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- 2027GLWN: Towards Provable Energy-Stable Overset Grid Methods Using Sub-Cell Summation-by-Parts Operators (2027). [DOI](#)
- 2027K: Acceleration of multicomponent multiple-precision arithmetic with branch-free algorithms and SIMD vectorization (2027). [DOI](#)
- 2026A: Semi-Lagrangian Discontinuous Galerkin Method with Adaptive Mesh Refinement for the Vlasov–Poisson System in 1X+3V (2026). [DOI](#)
- 2026A2: The Impact of the Generalized (r, q) Distributed Electrons on the Formation of Monotonic Shock Structures in an Anisotropic Plasma (2026). [DOI](#)
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- 2026CDGV: On p -Robust Convergence and Optimality of Adaptive FEM Driven by Equilibrated-Flux Estimators (2026). [DOI](#)
- 2026CH: A Comparative Study of Low-Dissipation Numerical Schemes for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026CHLW: COS(DG): A Convex Oscillation-Suppressing Framework for High-Order Discontinuous Galerkin Methods (2026). [DOI](#)
- 2026CKLS: A Parallel Basis Update and Galerkin Integrator for Tree Tensor Networks (2026). [DOI](#)
- 2026CLY: Semi-Lagrangian SAV method for Vlasov-Maxwell equations (2026). [DOI](#)
- 2026CSY: On the convergence of the discontinuous Galerkin scheme for Einstein-scalar equations (2026). [DOI](#)
- 2026CSZ: Unified Analysis of the Semidiscrete Discontinuous Galerkin Methods for 2-D Hyperbolic Equations on Cartesian Meshes Using $\mathbb{P}^k$ Elements: Optimal Error Estimates and Superconvergence (2026). [DOI](#)
- 2026CZQ: Fifth-order direct Arbitrary-Lagrangian-Eulerian Hermite WENO scheme for the compressible Euler equations (2026). [DOI](#)
- 2026DAKM: A polynomial approximation scheme for nonlinear model reduction by moment matching (2026). [DOI](#)
- 2026DBTT: A structure-preserving staggered semi-implicit four-split finite volume scheme for continuum mechanics on unstructured meshes (2026). [DOI](#)
- 2026DDP: A Reynolds- and Hartmann-Semi-robust Hybrid Method for Magnetohydrodynamics (2026). [DOI](#)
- 2026DFTW: A Deterministic Ionization Algorithm for the OSIRIS Particle-in-Cell Framework (2026). [DOI](#)
- 2026DGNP: Hierarchical Sparse-Grid Particle-in-Cell Method with Locally Adaptive Mesh Refinement (2026). [DOI](#)
- 2026DMN: High-Order Structure-Preserving SBP Finite Difference Methods for the Vlasov–Maxwell System on Matrix-Free GPUs (2026). [DOI](#)
- 2026FCSJ: Conservative Velocity Mappings for Discontinuous Galerkin Kinetics (2026). [DOI](#)
- 2026FJW: An energy-based discontinuous Galerkin method for the wave equation with nonsmooth solutions (2026). [DOI](#)
- 2026FMMM: Code-Verification Techniques for Particle-in-Cell Simulations with Direct Simulation Monte Carlo Collisions (2026). [DOI](#)
- 2026FMNV: Neural semi-Lagrangian method for high-dimensional advection-diffusion problems (2026). [DOI](#)
- 2026FW: Compact Finite-Average-Moment Schemes with Single-Step Oscillation Elimination for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026G3: Fully spectral scheme for the linear BGK equation on the whole space (2026). [DOI](#)
- 2026GAS: Kinetic-based regularization: Learning spatial derivatives and PDE applications (2026). [DOI](#)
- 2026GCYZ: Energy-stable compact exponential time differencing algorithm for phase field models (2026). [DOI](#)
- 2026GHJ: BEACONS: Bounded-Error, Algebraically-Composable Neural Solvers for Partial Differential Equations (2026). [DOI](#)
- 2026GHT: Adaptive mesh methods for hyperbolic conservation laws with bound-preserving flux limiters (2026). [DOI](#)
- 2026GILO: Why summation by parts is not enough (2026). [DOI](#)
- 2026GJZM: A Gegenbauer-Galerkin spectral scheme for three-dimensional incompressible Hall-magnetohydrodynamics system (2026). [DOI](#)
- 2026GR: Convergence of hyperbolic approximations to higher-order PDEs for smooth solutions (2026). [DOI](#)
- 2026GSC: A Stable Conservative Lagrange-Galerkin Method for Convection Equations Based on Curved-Element

- Mesh Intersection (2026). [DOI](#)
- 2026GT: Numerical instability of algebraic rootfinders (2026). [DOI](#)
 - 2026GT2: Invariant-Domain Preserving IMEX Schemes for the Nonequilibrium Gray Radiation-Hydrodynamics Equations Part I (2026). [DOI](#)
 - 2026GWSY: Do We Need Tensor Cores for Stencil Computations? (2026). [DOI](#)
 - 2026GZ: A Space-Time Galerkin Boundary Element Method for Aeroacoustic Scattering (2026). [DOI](#)
 - 2026H: Drifting to Boltzmann: Million-Fold Acceleration in Boltzmann Sampling with Force-Guided Drifting (2026). [DOI](#)
 - 2026H2: PDE Realization and Structure-Aware Solvers for a Compact Two-Stage Fourth-Order IMEX Method (2026). [DOI](#)
 - 2026HBKW: Novel approach to general curvilinear coordinates for plasma fluid applications (2026). [DOI](#)
 - 2026HGW: An Adaptive Integrating Factor Midpoint Method for Second Order Evolution Equations (2026). [DOI](#)
 - 2026HHLZ: Deep Picard Iteration for High-Dimensional Nonlinear PDEs (2026). [DOI](#)
 - 2026HLLS: An adaptive error estimator for stationary Navier-Stokes equations using variational physics informed neural networks (2026). [DOI](#)
 - 2026HM: A Priori Adaptive Numerical Methods for Estimating Blow-Up Times of Autonomous ODEs (2026). [DOI](#)
 - 2026HRT: A checkerboard-free, symmetry-preserving, conservative method for magnetohydrodynamics (2026). [DOI](#)
 - 2026HS: Local Time Integration for Friedrichs' Systems (2026). [DOI](#)
 - 2026HS2: Anchor-Based Function Extrapolation with Proven Bounds and Projection Guarantees (2026). [DOI](#)
 - 2026HVFO: Non-unitary time evolution via the Chebyshev expansion method (2026). [DOI](#)
 - 2026HY: An analytically-solvable, wave-form asymptotic-preserving and energy-conserving time-splitting scheme for Vlasov–Poisson equations in the quasi-neutral regime (2026). [DOI](#)
 - 2026HY2: Scaling Optimized Spectral Approximations on Unbounded Domains: The Generalized Hermite and Laguerre Methods (2026). [DOI](#)
 - 2026IDR: Mixed Hermite–Legendre spectral method for kinetic plasma simulations (2026). [DOI](#)
 - 2026J: An Upwind Summation-by-Part Scheme for the Non-Linear Boltzmann Equation (2026). [DOI](#)
 - 2026JHJ: A conservative discontinuous Galerkin algorithm for particle kinetics on smooth manifolds (2026). [DOI](#)
 - 2026JJPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
 - 2026JQZ: A Class of High Order Unconditional Maximum Principle Preserving Extended IFRK Schemes for Allen-Cahn Equation (2026). [DOI](#)
 - 2026KBK: A Taxonomy of Numerical Differentiation Methods (2026). [DOI](#)
 - 2026KLWB: A Hybrid Semi-Lagrangian Flow Mapping Approach for Vlasov Systems: Combining Iterative and Compositional Flow Maps (2026). [DOI](#)
 - 2026KSCP: Generalized Tadmor Conditions and Structure-Preserving Numerical Fluxes for the Compressible Flow of Real Gases (2026). [DOI](#)
 - 2026KSKB: Entropy stable high-order discontinuous Galerkin spectral-element methods on curvilinear, hybrid meshes (2026). [DOI](#)
 - 2026L: A Bivariate Spline Construction of Orthonormal Polynomials over Polygonal Domains and Its Applications to Quadrature (2026). [DOI](#)
 - 2026LJLH: FourierSpecNet: Neural collision operator approximation inspired by the Fourier spectral method for solving the Boltzmann equation (2026). [DOI](#)
 - 2026LLXZ: Efficient computation of the forcing term in Enskog Vlasov equation (2026). [DOI](#)
 - 2026LQ: A simple, high-order and compact WENO limiter based on control volume for spectral volume method (2026). [DOI](#)
 - 2026LSS: A Contour Integral-Based Algorithm for Computing Generalized Singular Values (2026). [DOI](#)
 - 2026LTTS: Comparative study of different quadrature methods for cut elements (2026). [DOI](#)
 - 2026LTWZ: A Class of Refined Implicit-Explicit Runge-Kutta Methods with Robust Time Adaptability and Unconditional Convergence for the Cahn-Hilliard Model (2026). [DOI](#)
 - 2026LV: Weak Scalability of time parallel Schwarz methods for parabolic optimal control problems (2026). [DOI](#)
 - 2026LVDN: Inverse initial data reconstruction for Maxwell's equations via time splitting method (2026). [DOI](#)
 - 2026LXWZ: Optimal convergence of local discontinuous Galerkin methods for convection–diffusion equations (2026). [DOI](#)

- 2026LZG: High-precision quadrature via local Fourier extension: analytic integration, uniform sampling, and correction for piecewise smooth integrands (2026). [DOI](#)
- 2026LZS: Increasingly high-order compact central schemes with a recursively Jacobian-free Cauchy-Kovalevskaya method (2026). [DOI](#)
- 2026MNM: A High Order Accurate and Provably Stable Fully Discrete Continuous Galerkin Framework on Summation-by-Parts Form for Advection-Diffusion Equations (2026). [DOI](#)
- 2026MR: Structure-preserving long-time simulations of turbulence in magnetised ideal fluids (2026). [DOI](#)
- 2026MRG: Entropy-Stable Discontinuous Spectral-Element Methods for the Spherical Shallow Water Equations in Covariant Form (2026). [DOI](#)
- 2026N: Differential Inversion of the Implicit Euler Method (2026). [DOI](#)
- 2026N3: Maximal Entropy Dissipation Numerical Scheme for Conservation Law Systems (2026). [DOI](#)
- 2026NO: Mixed Precision Iterative Refinement for Linear Inverse Problems (2026). [DOI](#)
- 2026OAGT: Numerical methods for quasi-stationary distributions (2026). [DOI](#)
- 2026P: Development of discontinuous Galerkin methods for hyperbolic systems that preserve a curl or a divergence constraint: The case of linear systems (2026). [DOI](#)
- 2026PN: Approximating sparse matrices and their functions using matrix-vector products (2026). [DOI](#)
- 2026QKLM: The tempered finite element method (2026). [DOI](#)
- 2026QNC: Entropy stable reduced order modeling of nonlinear conservation laws using discontinuous Galerkin methods (2026). [DOI](#)
- 2026QSW: A fast Fourier spectral method for wave kinetic equation (2026). [DOI](#)
- 2026R2: Numerical quadrature on bounded surfaces without parameterizations (2026). [DOI](#)
- 2026RARB: New Time Integrators and Capabilities in SUNDIALS Versions 6.2.0–7.4.0 (2026). [DOI](#)
- 2026RS: Micro-macro kinetic flux-vector splitting schemes for the multidimensional Boltzmann-ES-BGK equation (2026). [DOI](#)
- 2026RVKK: Adaptive hyperviscosity stabilisation for the RBF-FD method in solving advection-dominated transport equations (2026). [DOI](#)
- 2026S: High Order Lax-Wendroff Methods for Hyperbolic Systems (2026). [DOI](#)
- 2026S4: The State-Dependent Riccati Equation in Nonlinear Optimal Control: Analysis and Numerical Approximation (2026). [DOI](#)
- 2026S5: FastLSQ: Solving PDEs in One Shot via Fourier Features with Exact Analytical Derivatives (2026). [DOI](#)
- 2026S6: FFT-accelerated computation of the Dirichlet-to-Neumann map for inhomogeneous exterior Helmholtz problems using the method of fundamental solutions (2026). [DOI](#)
- 2026S7: Adaptive multigrid for high-order discontinuous Galerkin methods based on the full approximation scheme (2026). [DOI](#)
- 2026SDL: Half-explicit Runge–Kutta integrators for variational multiscale turbulence modeling: Toward higher-order accuracy in space and time (2026). [DOI](#)
- 2026SHJH: Constructing field-aligned coordinate systems for gyrokinetic simulations of tokamaks in X-point geometries (2026). [DOI](#)
- 2026SJKY: Multi-species Rosenbluth Fokker-Planck collision operator for discontinuous Galerkin method (2026). [DOI](#)
- 2026SNSK: A Discontinuous Galerkin Consistent Splitting Method for the Incompressible Navier–Stokes Equations (2026). [DOI](#)
- 2026SQHZ: An adaptive-rank approach with greedy sampling for multi-scale BGK equations (2026). [DOI](#)
- 2026SRHK: A splitting algorithm for a zonal quasineutral Vlasov model of strongly magnetized plasmas (2026). [DOI](#)
- 2026SSEH: Adaptive Sparse-grid Discontinuous Galerkin Approximations the Bhatnagar–Gross–Krook Model (2026). [DOI](#)
- 2026ST: Numerical Efficiency of Explicit Time Integrators for Phase-Field Models (2026). [DOI](#)
- 2026SV: Embedded Trefftz DG method for the Helmholtz equation (2026). [DOI](#)
- 2026SWJ: Projection enhanced triangular finite element method: a novel FEM approach for polygonal elements (2026). [DOI](#)
- 2026SYTN: Predictor-corrector method for solving anisotropic diffusion equation in magnetized plasmas (2026). [DOI](#)
- 2026TBA: A novel conditional formulation of the Vlasov–Ampère equations: a conservative, positivity, asymptotic

- and Gauss law preserving scheme (2026). [DOI](#)
- 2026TGRV: Numerical Analysis of a Coupled 3D-1D Transport Problem (2026). [DOI](#)
 - 2026TMS: An error control framework for computing the exponential of matrices arising from the finite element discretization (2026). [DOI](#)
 - 2026TSRC: A deterministic verification strategy for multidimensional gyrokinetic particle-in-cell algorithms using the method of manufactured solutions (2026). [DOI](#)
 - 2026VBGM: Local Cubic Spline Interpolation for Vlasov-Type Equations on a Multi-Patch Geometry (2026). [DOI](#)
 - 2026VC: On the choice of viscous discontinuous Galerkin discretization for entropy correction artificial viscosity methods (2026). [DOI](#)
 - 2026W2: Beyond Taylor: Divergence-Based Functional Expansions and Their Application to Numerical Integration (2026). [DOI](#)
 - 2026WFZ: Convergence of entropy-stable continuous summation-by-parts discretizations of symmetric hyperbolic conservation laws (2026). [DOI](#)
 - 2026WH2: Dynamical tensor train approximation for kinetic equations (2026). [DOI](#)
 - 2026WJPA: Accurate calculation of the gradients of the equilibrium poloidal flux in tokamaks (2026). [DOI](#)
 - 2026WKN: Numerical boundary flux functions that give provable bounds for nonlinear initial boundary value problems with open boundaries (2026). [DOI](#)
 - 2026WKRS: A meshless MUSCL method for the BGK-Boltzmann equation (2026). [DOI](#)
 - 2026WL: A unified high-resolution ODE framework for first-order methods (2026). [DOI](#)
 - 2026WLST: An Efficient Solver for Finite Element-based Constrained Transport in 3D Magnetohydrodynamics Applied to Magnetic Confinement Fusion (2026). [DOI](#)
 - 2026WWZ: Unconditionally Long-Time Stable Variable-Step Second-Order ETD Schemes for the 2D Periodic Incompressible NSE (2026). [DOI](#)
 - 2026X: A New Fractional Step Structure Preserving Method for The Landau-Lifshitz-Gilbert Equation (2026). [DOI](#)
 - 2026XL: On the Spectral Properties of Van Leer and AUSM Flux-Vector Splitting Schemes (2026). [DOI](#)
 - 2026XMY: An Efficient Energy Stable Structure Preserving Method for the Landau-Lifshitz Equation (2026). [DOI](#)
 - 2026YCW: A Fast Fourier Spectral Method for the Linearized Boltzmann Collision Operator (2026). [DOI](#)
 - 2026YF: An Entropy-Stable Oscillation-Eliminating DGSEM for the Euler Equations on Curvilinear Meshes (2026). [DOI](#)
 - 2026YM: On the Conjecture of Stability Preservation in Arbitrary-Order Adams-Bashforth-Type Integrators (2026). [DOI](#)
 - 2026ZLP: On energy consistency of intermediate states in HLL-type MHD Riemann solvers (2026). [DOI](#)
 - 2025ABGP: A nodally bound-preserving composite discontinuous Galerkin method on polytopic meshes (2025). [DOI](#)
 - 2025ADFM: Analysis of Floating-Point Matrix Multiplication Computed via Integer Arithmetic (2025). [DOI](#)
 - 2025AF: Enforcing Conservation Laws and Dissipation Inequalities Numerically via Auxiliary Variables (2025). [DOI](#)
 - 2025ART: L^2 and L^∞ rational approximation (2025). [DOI](#)
 - 2025BKKS: A hybrid Chebyshev-Tucker tensor format for approximation of multivariate functions (2025). [DOI](#)
 - 2025CGWZ: A Novel Preconditioning Framework for Solving Nonlinear PDEs based on Fenchel-Rockafellar Duality and Transformed Primal-Dual Techniques (2025). [DOI](#)
 - 2025CHLW: Asymptotic-preserving semi-Lagrangian discontinuous Galerkin schemes for the Boltzmann equation (2025). [DOI](#)
 - 2025CJV: The Dynamical Anatomy of Anderson Acceleration: From Adaptive Momentum to Variable-Mass ODEs (2025). [DOI](#)
 - 2025CKKS: Asymptotic-preserving conservative semi-Lagrangian discontinuous Galerkin schemes for the Vlasov-Poisson system in the quasi-neutral limit (2025). [DOI](#)
 - 2025CLLZ: A high-order energy-conserving semi-Lagrangian discontinuous Galerkin method for the Vlasov-Ampère system (2025). [DOI](#)
 - 2025CR: Interpretation of a Discrete de Rham method as a Finite Element System (2025). [DOI](#)
 - 2025DFHL: Mixed-precision algorithms for solving the Sylvester matrix equation (2025). [DOI](#)

- 2025DS: Matrix Factorizations with Uniformly Random Pivoting (2025). [DOI](#)
- 2025DUA: A time-space B-spline integrator for the Burgers' equation (2025). [DOI](#)
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- 2025G: A Method for Establishing Asymptotically Accurate Bounds for Extremal Roots of Eulerian Polynomials Using Polynomial Stability Preservers (2025). [DOI](#)
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- 2025H: An L-Stable Implicit Two-Stage Fourth-Order Temporal Discretization Scheme for Lax-Wendroff-Type Solvers Applied to Stiff Problems (2025). [DOI](#)
- 2025HL: A Generalized Alternating Anderson Acceleration Method (2025). [DOI](#)
- 2025HLS: On High-Order/Low-Order and Micro-Macro Methods for Implicit Time-Stepping of the BGK Model (2025). [DOI](#)
- 2025HMST: Structure-preserving finite-element approximations of the magnetic Euler-Poisson equations (2025). [DOI](#)
- 2025HS: Error Estimates for Gauss–Christoffel Quadrature under Reduced Regularity Conditions (2025). [DOI](#)
- 2025HYSB: An Overview of Meshfree Collocation Methods (2025). [DOI](#)
- 2025JHJ: A moment-conserving discontinuous Galerkin representation of the relativistic Maxwellian distribution (2025). [DOI](#)
- 2025K: A regret minimization approach to fixed-point iterations (2025). [DOI](#)
- 2025KMZ: A priori error analysis of the proximal Galerkin method (2025). [DOI](#)
- 2025L: Symplectic particle-in-cell methods for hybrid plasma models with Boltzmann electrons and space-charge effects (2025). [DOI](#)
- 2025LD: Structure-preserving schemes for nonlinear symmetric hyperbolic and thermodynamically compatible systems of partial differential equations (2025). [DOI](#)
- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)
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- 2025PGVA: Tensor-based multivariate function approximation: methods benchmarking and comparison (2025). [DOI](#)
- 2025PLGC: A scalable high-order multigrid-FFT Poisson solver for unbounded domains on adaptive multiresolution grids (2025). [DOI](#)
- 2025RAML: Efficient and Flexible Multirate Temporal Adaptivity (2025). [DOI](#)
- 2025RGHL: Efficient Solution of Fully Implicit Runge–Kutta Methods for Linear Wave Equations (2025). [DOI](#)
- 2025SETS: Explicit and Effectively Symmetric Runge-Kutta Methods (2025). [DOI](#)
- 2025SRHK: Convergence of Splitting Methods on Rotating Grids for the Magnetized Vlasov Equation (2025). [DOI](#)
- 2025SWRP: Moment-based adaptive time integration for thermal radiation transport (2025). [DOI](#)
- 2025TRBF: Adaptive Multidimensional Quadrature on Multi-GPU Systems (2025). [DOI](#)
- 2025VKCK: Optimal Transport, Timesteppers, Newton-Krylov Methods and Steady States of Collective Particle Dynamics (2025). [DOI](#)
- 2025VR: High-Order Meshfree Surface Integration, Including Singular Integrands (2025). [DOI](#)
- 2025YHR: An Explicit Energy-Conserving Particle Method for the Vlasov-Fokker-Planck Equation (2025). [DOI](#)
- 2025Z: Communication-Efficient and Memory-Aware Parallel Bootstrapping using MPI (2025). [DOI](#)
- 2025ZCSV: Numerical investigation of kinetic instabilities in BGK equilibria under collisional effects (2025). [DOI](#)
- 2025ZS: Optimal error estimates for a fully discrete, highly efficient decoupled scheme for the 2D/3D diffuse interface two-phase MHD flows (2025). [DOI](#)
- 2025ZSHC: A Semi-Lagrangian Adaptive Rank (SLAR) Method for High-Dimensional Vlasov Dynamics (2025).

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- 2025ZZS: A direct PinT algorithm for higher-order nonlinear time-evolution equations (2025). [DOI](#)
- 2024BC: Asymptotic Analysis of IMEX-RK Methods for ES-BGK Model at Navier–Stokes Level (2024). [DOI](#)
- 2024LJKL: An adaptive finite element multigrid solver using GPU acceleration (2024). [DOI](#)
- 2024LLW: A bi-fidelity method for the uncertain Vlasov-Poisson system near quasineutrality in an asymptotic-preserving particle-in-cell framework (2024). [DOI](#)
- 2024O: Parallelisation of partial differential equations via representation theory (2024). [DOI](#)
- 2024PS: A path-dependent PDE solver based on signature kernels (2024). [DOI](#)
- 2024SV: Quadratic projectable Runge-Kutta methods (2024). [DOI](#)
- 2023ATF: Temporal accuracy of FastRK3 (2023). [DOI](#)
- 2023T: Preconditioning for time-harmonic Maxwell’s equations using the Laguerre transform (2023). [DOI](#)
- 2022SKP: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part II: Nonlinearities and DAEs (2022). [DOI](#)
- 2022SKPS: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part I: The Linear Setting (2022). [DOI](#)
- 2020AF: A fast pressure-correction method for incompressible flows over curved walls (2020). [DOI](#)
- 2018MDL: Laguerre–Hermite pseudo-spectral velocity formulation of gyrokinetics (2018). [DOI](#)
- 2013LE: New velocity-space discretization for continuum kinetic calculations and Fokker–Planck collisions (2013). [DOI](#)
- 2009GKS: High Order Strong Stability Preserving Time Discretizations (2009). [DOI](#)
- 2005G: On High Order Strong Stability Preserving Runge–Kutta and Multi Step Time Discretizations (2005). [DOI](#)
- 2003LW: Comparison of Several Difference Schemes on 1D and 2D Test Problems for the Euler Equations (2003). [DOI](#)
- 2002S: Automatic control and adaptive time-stepping (2002). [DOI](#)
- 1999HW: Stiff differential equations solved by Radau methods (1999). [DOI](#)

Numerical Quadrature

- 2026HY2: Scaling Optimized Spectral Approximations on Unbounded Domains: The Generalized Hermite and Laguerre Methods (2026). [DOI](#)
- 2026LLXZ: Efficient computation of the forcing term in Enskog Vlasov equation (2026). [DOI](#)
- 2026LTTS: Comparative study of different quadrature methods for cut elements (2026). [DOI](#)
- 2026LZG: High-precision quadrature via local Fourier extension: analytic integration, uniform sampling, and correction for piecewise smooth integrands (2026). [DOI](#)
- 2026R2: Numerical quadrature on bounded surfaces without parameterizations (2026). [DOI](#)
- 2026W2: Beyond Taylor: Divergence-Based Functional Expansions and Their Application to Numerical Integration (2026). [DOI](#)
- 2025VCCC: Machine-precision energy conservative reduced models for Lagrangian hydrodynamics by quadrature methods (2025). [DOI](#)
- 2025VR: High-Order Meshfree Surface Integration, Including Singular Integrands (2025). [DOI](#)
- 2013LE: New velocity-space discretization for continuum kinetic calculations and Fokker–Planck collisions (2013). [DOI](#)

Finite Element Methods

- 2026AB: A conservative, discontinuous Galerkin, tracer transport scheme using compatible finite elements (2026). [DOI](#)
- 2026ABBG: Lethe 1.0: An open-source parallel high-order computational fluid dynamics software framework for single and multiphase flows (2026). [DOI](#)
- 2026ABL: Virtual finite element and hyperbolic problems: The PAMPA algorithm (2026). [DOI](#)
- 2026AFKP: A projection method for particle resampling (2026). [DOI](#)
- 2026AG: Finite Element Convergence Analysis For Wave Equations With Time-Dependent Coefficients (2026). [DOI](#)

- 2026BFKP: Fast solvers for the high-order FEM simplicial de Rham complex (2026). [DOI](#)
- 2026CBKS: JetSCI: A Hybrid JAX–PETSc Framework for Scalable Differentiable Simulation (2026). [DOI](#)
- 2026CDGV: On p -Robust Convergence and Optimality of Adaptive FEM Driven by Equilibrated-Flux Estimators (2026). [DOI](#)
- 2026CH2: Complexes from Complexes: Finite Element Complexes in Three Dimensions (2026). [DOI](#)
- 2026DDP: A Reynolds- and Hartmann-Semirobust Hybrid Method for Magnetohydrodynamics (2026). [DOI](#)
- 2026ECM: NektarIR: A Domain-Specific Compiler for High-Order Finite Element Operations on Heterogeneous Hardware (2026). [DOI](#)
- 2026GSC: A Stable Conservative Lagrange-Galerkin Method for Convection Equations Based on Curved-Element Mesh Intersection (2026). [DOI](#)
- 2026GT2: Invariant-Domain Preserving IMEX Schemes for the Nonequilibrium Gray Radiation-Hydrodynamics Equations Part I (2026). [DOI](#)
- 2026KRR: Efficient numerical strategies for entropy-regularized semi-discrete optimal transport (2026). [DOI](#)
- 2026LLR: Three-dimensional mesh adaptation in PFEM (2026). [DOI](#)
- 2026LMHL: TRIMEG-GKX: An electromagnetic gyrokinetic particle code with a piecewise field-aligned finite element method for micro- and macro-instability studies in tokamak core plasmas (2026). [DOI](#)
- 2026LMSH: A High-Order Piecewise Field-Aligned Triangular Finite Element Method for Electromagnetic Gyrokinetic Particle Simulations of Tokamak Plasmas with Open Field Lines (2026). [DOI](#)
- 2026LTTS: Comparative study of different quadrature methods for cut elements (2026). [DOI](#)
- 2026LYZC: A Strongly Discontinuous Boundary Condition Model With Discontinuous Galerkin Framework for Multiscale Electromagnetic Simulations Containing Imperfect Sliding Electrical Contact (2026). [DOI](#)
- 2026MNM: A High Order Accurate and Provably Stable Fully Discrete Continuous Galerkin Framework on Summation-by-Parts Form for Advection-Diffusion Equations (2026). [DOI](#)
- 2026QKLM: The tempered finite element method (2026). [DOI](#)
- 2026S6: FFT-accelerated computation of the Dirichlet-to-Neumann map for inhomogeneous exterior Helmholtz problems using the method of fundamental solutions (2026). [DOI](#)
- 2026S7: Adaptive multigrid for high-order discontinuous Galerkin methods based on the full approximation scheme (2026). [DOI](#)
- 2026SCL: Mass-lumping technique for fully explicit time integration of the shifted-basis variant of the 3D extended finite element method (2026). [DOI](#)
- 2026SDL: Half-explicit Runge–Kutta integrators for variational multiscale turbulence modeling: Toward higher-order accuracy in space and time (2026). [DOI](#)
- 2026SKIS: Field conserving adaptive mesh refinement (AMR) scheme on massively parallel adaptive octree meshes (2026). [DOI](#)
- 2026SWJ: Projection enhanced triangular finite element method: a novel FEM approach for polygonal elements (2026). [DOI](#)
- 2026TGRV: Numerical Analysis of a Coupled 3D-1D Transport Problem (2026). [DOI](#)
- 2026TKCD: Accelerating High-Order Finite Element Simulations at Extreme Scale with FP64 Tensor Cores (2026). [DOI](#)
- 2026TMS: An error control framework for computing the exponential of matrices arising from the finite element discretization (2026). [DOI](#)
- 2026WJPA: Accurate calculation of the gradients of the equilibrium poloidal flux in tokamaks (2026). [DOI](#)
- 2026WLST: An Efficient Solver for Finite Element-based Constrained Transport in 3D Magnetohydrodynamics Applied to Magnetic Confinement Fusion (2026). [DOI](#)
- 2026YF: An Entropy-Stable Oscillation-Eliminating DGSEM for the Euler Equations on Curvilinear Meshes (2026). [DOI](#)
- 2025ABGP: A nodally bound-preserving composite discontinuous Galerkin method on polytopic meshes (2025). [DOI](#)
- 2025CGWZ: A Novel Preconditioning Framework for Solving Nonlinear PDEs based on Fenchel-Rockafellar Duality and Transformed Primal-Dual Techniques (2025). [DOI](#)
- 2025CMPV: Mathematical modeling and sensitivity analysis of hypoxia-activated drugs (2025). [DOI](#)
- 2025CR: Interpretation of a Discrete de Rham method as a Finite Element System (2025). [DOI](#)
- 2025HMST: Structure-preserving finite-element approximations of the magnetic Euler-Poisson equations (2025). [DOI](#)

- 2025KM: Extending Irksome: Improvements in Automated Runge–Kutta Time Stepping for Finite Element Methods (2025). [DOI](#)
- 2025KMZ: A priori error analysis of the proximal Galerkin method (2025). [DOI](#)
- 2025RGHL: Efficient Solution of Fully Implicit Runge–Kutta Methods for Linear Wave Equations (2025). [DOI](#)
- 2025ZS: Optimal error estimates for a fully discrete, highly efficient decoupled scheme for the 2D/3D diffuse interface two-phase MHD flows (2025). [DOI](#)
- 2024AK: An Arbitrarily Higher-Order Time Implicit Scheme for Maxwell’s Equations (2024). [DOI](#)
- 2024LJKL: An adaptive finite element multigrid solver using GPU acceleration (2024). [DOI](#)
- 2023T: Preconditioning for time-harmonic Maxwell’s equations using the Laguerre transform (2023). [DOI](#)
- 2021FKM: Irksome: Automating Runge–Kutta Time-stepping for Finite Element Methods (2021). [DOI](#)

Computational Mechanics

- 2026AMS: Semi-explicit time discretization for linear thermo-poroelasticity (2026). [DOI](#)
- 2026BEAV: PENCO: A Physics–Energy–Numerics–Consistent Operator for 3D phase field modeling (2026). [DOI](#)
- 2026DBTT: A structure-preserving staggered semi-implicit four-split finite volume scheme for continuum mechanics on unstructured meshes (2026). [DOI](#)
- 2026FPBT: Electromagnetic Analysis of ITER Electron Cyclotron Emission Model (2026). [DOI](#)
- 2026GKSA: Design and mechanical analysis of the PRAGYA tokamak vacuum vessel (2026). [DOI](#)
- 2026LLR: Three-dimensional mesh adaptation in PFEM (2026). [DOI](#)
- 2026QKLM: The tempered finite element method (2026). [DOI](#)
- 2026RBCH: Automatic curved block structure and hexahedral mesh generation for hypersonic CFD (2026). [DOI](#)
- 2026SCL: Mass-lumping technique for fully explicit time integration of the shifted-basis variant of the 3D extended finite element method (2026). [DOI](#)
- 2026SWJ: Projection enhanced triangular finite element method: a novel FEM approach for polygonal elements (2026). [DOI](#)
- 2026ZL: Benchmarking and extending grid-based, meshfree, and physics-informed neural network methods for the generalized probability density evolution equation (2026). [DOI](#)

Mesh Generation

- 2027GLWN: Towards Provable Energy-Stable Overset Grid Methods Using Sub-Cell Summation-by-Parts Operators (2027). [DOI](#)
- 2026RBCH: Automatic curved block structure and hexahedral mesh generation for hypersonic CFD (2026). [DOI](#)
- 2026SHJH: Constructing field-aligned coordinate systems for gyrokinetic simulations of tokamaks in X-point geometries (2026). [DOI](#)

Finite Difference Methods

- 2027GLWN: Towards Provable Energy-Stable Overset Grid Methods Using Sub-Cell Summation-by-Parts Operators (2027). [DOI](#)
- 2026DMN: High-Order Structure-Preserving SBP Finite Difference Methods for the Vlasov–Maxwell System on Matrix-Free GPUs (2026). [DOI](#)
- 2026HBKW: Novel approach to general curvilinear coordinates for plasma fluid applications (2026). [DOI](#)
- 2026LZS: Increasingly high-order compact central schemes with a recursively Jacobian-free Cauchy–Kovalevskaya method (2026). [DOI](#)
- 2026MTWG: A GPU-accelerated matrix-free FAS multigrid solver for Navier–Stokes equations with memory-efficient implementations (2026). [DOI](#)
- 2026SYTN: Predictor-corrector method for solving anisotropic diffusion equation in magnetized plasmas (2026). [DOI](#)
- 2026TBA: A novel conditional formulation of the Vlasov–Ampère equations: a conservative, positivity, asymptotic and Gauss law preserving scheme (2026). [DOI](#)

Finite Volume Methods

- 2026ACKM: Dual formulation finite-volume methods on overlapping meshes for hyperbolic conservation laws (2026). [DOI](#)
- 2026AMC: Formulation of entropy-conservative discretizations for compressible flows of thermally perfect gases (2026). [DOI](#)
- 2026AR: On Affordable High-Order Entropy-Conservative/Stable and Well-Balanced Methods for Nonconservative Hyperbolic Systems (2026). [DOI](#)
- 2026BF: Monotone-based Numerical Schemes for Two-Dimensional Systems of Nonlocal Conservation Laws (2026). [DOI](#)
- 2026CH: A Comparative Study of Low-Dissipation Numerical Schemes for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026CZQ: Fifth-order direct Arbitrary-Lagrangian-Eulerian Hermite WENO scheme for the compressible Euler equations (2026). [DOI](#)
- 2026DBTT: A structure-preserving staggered semi-implicit four-split finite volume scheme for continuum mechanics on unstructured meshes (2026). [DOI](#)
- 2026FW: Compact Finite-Average-Moment Schemes with Single-Step Oscillation Elimination for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026FZHW: Fire: An open-source adaptive mesh refinement solver for supersonic reacting flows (2026). [DOI](#)
- 2026GHDL: Solving the Six-Dimensional Vlasov–Maxwell System with Active Flux and Splitting Methods (2026). [DOI](#)
- 2026GHT: Adaptive mesh methods for hyperbolic conservation laws with bound-preserving flux limiters (2026). [DOI](#)
- 2026K2: kalyпсо: A performance portable platform for compressible hydrodynamics simulations using adaptive mesh refinement (2026). [DOI](#)
- 2026LQ: A simple, high-order and compact WENO limiter based on control volume for spectral volume method (2026). [DOI](#)
- 2026LZS: Increasingly high-order compact central schemes with a recursively Jacobian-free Cauchy–Kovalevskaya method (2026). [DOI](#)
- 2026N2: A Flux-Correction Form of the Third-Order Edge-Based Scheme for a General Numerical Flux Function (2026). [DOI](#)
- 2026N3: Maximal Entropy Dissipation Numerical Scheme for Conservation Law Systems (2026). [DOI](#)
- 2026RS: Micro-macro kinetic flux-vector splitting schemes for the multidimensional Boltzmann-ES-BGK equation (2026). [DOI](#)
- 2026TCLH: Coupling of an External Circuit Model With a Plasma Fluid Model With a Time-Step Size Refinement (TSSR) Algorithm (2026). [DOI](#)
- 2026ZGZ: Predictor-corrector discrete unified gas kinetic scheme for Boltzmann transport models with non-conservative collision operators (2026). [DOI](#)
- 2026ZLP: On energy consistency of intermediate states in HLL-type MHD Riemann solvers (2026). [DOI](#)
- 2026ZW: Positivity-preserving implicit finite volume methods on unstructured grids for compressible flows (2026). [DOI](#)
- 2025GRD: On General and Complete Multidimensional Riemann Solvers for Nonlinear Systems of Hyperbolic Conservation Laws (2025). [DOI](#)
- 2025LD: Structure-preserving schemes for nonlinear symmetric hyperbolic and thermodynamically compatible systems of partial differential equations (2025). [DOI](#)
- 2025NH: Asymptotically compatible entropy-consistent discretization for a class of nonlocal conservation laws (2025). [DOI](#)
- 2025PE: Stable low diffusion flux splitting schemes on unstructured meshes (2025). [DOI](#)

Discontinuous Galerkin Methods

- 2026A: Semi-Lagrangian Discontinuous Galerkin Method with Adaptive Mesh Refinement for the Vlasov–Poisson System in 1X+3V (2026). [DOI](#)
- 2026AB: A conservative, discontinuous Galerkin, tracer transport scheme using compatible finite elements (2026).

[DOI](#)

- 2026AL: Robust PAMPA scheme in the DG formulation on unstructured triangular meshes: Bound preservation, oscillation elimination, and boundary conditions (2026). [DOI](#)
- 2026AR: On Affordable High-Order Entropy-Conservative/Stable and Well-Balanced Methods for Nonconservative Hyperbolic Systems (2026). [DOI](#)
- 2026BCSR: Automatic differentiation for performing the Cauchy–Kovalevskaya procedure in Lax–Wendroff type discretizations (2026). [DOI](#)
- 2026BR: Compact Runge-Kutta flux reconstruction methods for non-conservative hyperbolic equations (2026). [DOI](#)
- 2026BTG: Stability analysis of Arbitrary-Lagrangian-Eulerian ADER-DG methods on classical and degenerate spacetime geometries (2026). [DOI](#)
- 2026CHLW: COS(DG): A Convex Oscillation-Suppressing Framework for High-Order Discontinuous Galerkin Methods (2026). [DOI](#)
- 2026CSY: On the convergence of the discontinuous Galerkin scheme for Einstein-scalar equations (2026). [DOI](#)
- 2026CSZ: Unified Analysis of the Semidiscrete Discontinuous Galerkin Methods for 2-D Hyperbolic Equations on Cartesian Meshes Using $\mathbb{P}^k$ Elements: Optimal Error Estimates and Superconvergence (2026). [DOI](#)
- 2026FCSJ: Conservative Velocity Mappings for Discontinuous Galerkin Kinetics (2026). [DOI](#)
- 2026FJW: An energy-based discontinuous Galerkin method for the wave equation with nonsmooth solutions (2026). [DOI](#)
- 2026GK: Comparison of Structure-Preserving Methods for the Cahn–Hilliard–Navier–Stokes Equations (2026). [DOI](#)
- 2026HS: Local Time Integration for Friedrichs’ Systems (2026). [DOI](#)
- 2026HZCN: A Curvilinear Lagrangian discontinuous Galerkin method for resistive magneto-hydrodynamics (2026). [DOI](#)
- 2026J: An Upwind Summation-by-Part Scheme for the Non-Linear Boltzmann Equation (2026). [DOI](#)
- 2026JHJ: A conservative discontinuous Galerkin algorithm for particle kinetics on smooth manifolds (2026). [DOI](#)
- 2026JJPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
- 2026KSKB: Entropy stable high-order discontinuous Galerkin spectral-element methods on curvilinear, hybrid meshes (2026). [DOI](#)
- 2026LKCH: An unstructured block-based adaptive mesh refinement approach for discontinuous Galerkin method (2026). [DOI](#)
- 2026LMSZ: Efficient optimization-based invariant-domain-preserving limiters in solving gas dynamics equations (2026). [DOI](#)
- 2026LWY: High-order positivity-preserving and globally divergence-free discontinuous Galerkin methods for ideal magnetohydrodynamics with conditional energy conservation (2026). [DOI](#)
- 2026LXWZ: Optimal convergence of local discontinuous Galerkin methods for convection–diffusion equations (2026). [DOI](#)
- 2026LYZC: A Strongly Discontinuous Boundary Condition Model With Discontinuous Galerkin Framework for Multiscale Electromagnetic Simulations Containing Imperfect Sliding Electrical Contact (2026). [DOI](#)
- 2026MRG: Entropy-Stable Discontinuous Spectral-Element Methods for the Spherical Shallow Water Equations in Covariant Form (2026). [DOI](#)
- 2026OCP: A p-adaptive hybridizable discontinuous Galerkin spectral element method for electrostatic particle-in-cell simulations (2026). [DOI](#)
- 2026P: Development of discontinuous Galerkin methods for hyperbolic systems that preserve a curl or a divergence constraint: The case of linear systems (2026). [DOI](#)
- 2026QNC: Entropy stable reduced order modeling of nonlinear conservation laws using discontinuous Galerkin methods (2026). [DOI](#)
- 2026RCC: Efficient entropy-conserving/stable discontinuous Galerkin solution of the multicomponent compressible Euler equations (2026). [DOI](#)
- 2026S7: Adaptive multigrid for high-order discontinuous Galerkin methods based on the full approximation scheme (2026). [DOI](#)
- 2026SDLM: A well-balanced positivity-preserving discontinuous Galerkin method for shallow water models with variable density (2026). [DOI](#)
- 2026SH: Performance of a high-order MPI-Kokkos accelerated fluid solver (2026). [DOI](#)

- 2026SHJH: Constructing field-aligned coordinate systems for gyrokinetic simulations of tokamaks in X-point geometries (2026). [DOI](#)
- 2026SJKY: Multi-species Rosenbluth Fokker-Planck collision operator for discontinuous Galerkin method (2026). [DOI](#)
- 2026SNSK: A Discontinuous Galerkin Consistent Splitting Method for the Incompressible Navier–Stokes Equations (2026). [DOI](#)
- 2026SS: A multi-resolution limiter for the Runge–Kutta discontinuous Galerkin method (2026). [DOI](#)
- 2026SSEH: Adaptive Sparse-grid Discontinuous Galerkin Approximations the Bhatnagar–Gross–Krook Model (2026). [DOI](#)
- 2026SV: Embedded Trefftz DG method for the Helmholtz equation (2026). [DOI](#)
- 2026TGRV: Numerical Analysis of a Coupled 3D-1D Transport Problem (2026). [DOI](#)
- 2026VC: On the choice of viscous discontinuous Galerkin discretization for entropy correction artificial viscosity methods (2026). [DOI](#)
- 2026WEK: Adaptive mesh $h(p)$ -refinement of a discontinuous Bubnov–Galerkin isogeometric analysis spatial discretisation of the first-order form of the neutron transport equation with goal-based error measures and diffusion acceleration (2026). [DOI](#)
- 2026WKN: Numerical boundary flux functions that give provable bounds for nonlinear initial boundary value problems with open boundaries (2026). [DOI](#)
- 2026XM: Unified Interface Flux Evaluation in a General Discontinuous Galerkin Spectral Element Framework (2026). [DOI](#)
- 2026YF: An Entropy-Stable Oscillation-Eliminating DGSEM for the Euler Equations on Curvilinear Meshes (2026). [DOI](#)
- 2026ZL: Benchmarking and extending grid-based, meshfree, and physics-informed neural network methods for the generalized probability density evolution equation (2026). [DOI](#)
- 2026ZZHD: A subcell entropy-constrained limiter for entropy stable discontinuous Galerkin methods (2026). [DOI](#)
- 2025ABGP: A nodally bound-preserving composite discontinuous Galerkin method on polytopic meshes (2025). [DOI](#)
- 2025CHLW: Asymptotic-preserving semi-Lagrangian discontinuous Galerkin schemes for the Boltzmann equation (2025). [DOI](#)
- 2025CKKS: Asymptotic-preserving conservative semi-Lagrangian discontinuous Galerkin schemes for the Vlasov–Poisson system in the quasi-neutral limit (2025). [DOI](#)
- 2025CLLZ: A high-order energy-conserving semi-Lagrangian discontinuous Galerkin method for the Vlasov–Ampère system (2025). [DOI](#)
- 2025EDLN: An Asynchronous Many-Task Algorithm for Unstructured S_N Transport on Shared Memory Systems (2025). [DOI](#)
- 2025HLS: On High-Order/Low-Order and Micro-Macro Methods for Implicit Time-Stepping of the BGK Model (2025). [DOI](#)
- 2025JHJ: A moment-conserving discontinuous Galerkin representation of the relativistic Maxwellian distribution (2025). [DOI](#)
- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)
- 2022SKP: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part II: Nonlinearities and DAEs (2022). [DOI](#)
- 2022SKPS: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part I: The Linear Setting (2022). [DOI](#)
- 2020FBMH: Conservative discontinuous Galerkin scheme of a gyro-averaged Dougherty collision operator (2020). [DOI](#)
- 2020MHFF: Electromagnetic full- f gyrokinetics in the tokamak edge with discontinuous Galerkin methods (2020). [DOI](#)
- 2015SHH: A gyrokinetic one-dimensional scrape-off layer model of an edge-localized mode heat pulse (2015). [DOI](#)

Meshfree Methods

- 2026BKL: Compact LABFM: a framework for meshless methods with spectral-like resolving power (2026). [DOI](#)
- 2026FMNV: Neural semi-Lagrangian method for high-dimensional advection-diffusion problems (2026). [DOI](#)
- 2026LZLC: An implicit Lagrangian generalized finite difference method for flow problems (2026). [DOI](#)
- 2026R2: Numerical quadrature on bounded surfaces without parameterizations (2026). [DOI](#)
- 2026RVKK: Adaptive hyperviscosity stabilisation for the RBF-FD method in solving advection-dominated transport equations (2026). [DOI](#)
- 2026VRDK: Load Balanced Parallel Node Generation for Meshless Numerical Methods (2026). [DOI](#)
- 2026WKRS: A meshless MUSCL method for the BGK-Boltzmann equation (2026). [DOI](#)
- 2026ZL: Benchmarking and extending grid-based, meshfree, and physics-informed neural network methods for the generalized probability density evolution equation (2026). [DOI](#)
- 2025BKKS: A hybrid Chebyshev-Tucker tensor format for approximation of multivariate functions (2025). [DOI](#)
- 2025HYSB: An Overview of Meshfree Collocation Methods (2025). [DOI](#)
- 2025VR: High-Order Meshfree Surface Integration, Including Singular Integrands (2025). [DOI](#)

Adaptive Mesh Refinement

- 2026A: Semi-Lagrangian Discontinuous Galerkin Method with Adaptive Mesh Refinement for the Vlasov–Poisson System in 1X+3V (2026). [DOI](#)
- 2026ABBG: Lethé 1.0: An open-source parallel high-order computational fluid dynamics software framework for single and multiphase flows (2026). [DOI](#)
- 2026BB: Scalable Overset Computation Between a Forest-of-Octrees- and an Arbitrary Distributed Parallel Mesh (2026). [DOI](#)
- 2026CDGV: On p -Robust Convergence and Optimality of Adaptive FEM Driven by Equilibrated-Flux Estimators (2026). [DOI](#)
- 2026DGNP: Hierarchical Sparse-Grid Particle-in-Cell Method with Locally Adaptive Mesh Refinement (2026). [DOI](#)
- 2026FZHW: Fire: An open-source adaptive mesh refinement solver for supersonic reacting flows (2026). [DOI](#)
- 2026GHT: Adaptive mesh methods for hyperbolic conservation laws with bound-preserving flux limiters (2026). [DOI](#)
- 2026GK: Comparison of Structure-Preserving Methods for the Cahn–Hilliard–Navier–Stokes Equations (2026). [DOI](#)
- 2026HLLS: An adaptive error estimator for stationary Navier-Stokes equations using variational physics informed neural networks (2026). [DOI](#)
- 2026K2: kalypso: A performance portable platform for compressible hydrodynamics simulations using adaptive mesh refinement (2026). [DOI](#)
- 2026KHSB: A Morton-Type Space-Filling Curve for Pyramid Subdivision and Hybrid Adaptive Mesh Refinement (2026). [DOI](#)
- 2026LKCH: An unstructured block-based adaptive mesh refinement approach for discontinuous Galerkin method (2026). [DOI](#)
- 2026LLR: Three-dimensional mesh adaptation in PFEM (2026). [DOI](#)
- 2026OCP: A p -adaptive hybridizable discontinuous Galerkin spectral element method for electrostatic particle-in-cell simulations (2026). [DOI](#)
- 2026S7: Adaptive multigrid for high-order discontinuous Galerkin methods based on the full approximation scheme (2026). [DOI](#)
- 2026SKIS: Field conserving adaptive mesh refinement (AMR) scheme on massively parallel adaptive octree meshes (2026). [DOI](#)
- 2026SSEH: Adaptive Sparse-grid Discontinuous Galerkin Approximations the Bhatnagar–Gross–Krook Model (2026). [DOI](#)
- 2026WEK: Adaptive mesh $h(p)$ -refinement of a discontinuous Bubnov–Galerkin isogeometric analysis spatial discretisation of the first-order form of the neutron transport equation with goal-based error measures and diffusion acceleration (2026). [DOI](#)
- 2025HJP: Temporal Substepping Scheme for Magnetohydrodynamics with Cell-based Adaptive Mesh Refinement

on Staggered Grid (2025). [DOI](#)

- 2025PLGC: A scalable high-order multigrid-FFT Poisson solver for unbounded domains on adaptive multiresolution grids (2025). [DOI](#)

Time Integration

- 2026ACS: Exploring Exponential Runge–Kutta Methods: A Survey (2026). [DOI](#)
- 2026AMS: Semi-explicit time discretization for linear thermo-poroelasticity (2026). [DOI](#)
- 2026AVZ: Explicit stabilized implementation of singly diagonally implicit Runge–Kutta methods (2026). [DOI](#)
- 2026BB2: A Multirate Transient Simulation Technique for Circuits described by Integro-Differential Equations (2026). [DOI](#)
- 2026BCSR: Automatic differentiation for performing the Cauchy–Kovalevskaya procedure in Lax–Wendroff type discretizations (2026). [DOI](#)
- 2026BI: High Order Semi-implicit Rosenbrock type and Multistep Methods for Evolutionary Partial Differential Equations with Higher Order Derivatives (2026). [DOI](#)
- 2026BR: Compact Runge-Kutta flux reconstruction methods for non-conservative hyperbolic equations (2026). [DOI](#)
- 2026CKLS: A Parallel Basis Update and Galerkin Integrator for Tree Tensor Networks (2026). [DOI](#)
- 2026CLY: Semi-Lagrangian SAV method for Vlasov-Maxwell equations (2026). [DOI](#)
- 2026CMPT: A Lagrangian method for solving the spherical shallow water equations using power diagrams (2026). [DOI](#)
- 2026CZQ: Fifth-order direct Arbitrary-Lagrangian-Eulerian Hermite WENO scheme for the compressible Euler equations (2026). [DOI](#)
- 2026DBTT: A structure-preserving staggered semi-implicit four-split finite volume scheme for continuum mechanics on unstructured meshes (2026). [DOI](#)
- 2026FMNV: Neural semi-Lagrangian method for high-dimensional advection-diffusion problems (2026). [DOI](#)
- 2026FPKA: Structure Preservation Using Discrete Gradients in the Vlasov-Poisson-Landau System (2026). [DOI](#)
- 2026FRT: A modified Crank–Nicolson scheme for the Vlasov–Poisson system with a strong external magnetic field (2026). [DOI](#)
- 2026GCYZ: Energy-stable compact exponential time differencing algorithm for phase field models (2026). [DOI](#)
- 2026GHDL: Solving the Six-Dimensional Vlasov–Maxwell System with Active Flux and Splitting Methods (2026). [DOI](#)
- 2026GJZM: A Gegenbauer-Galerkin spectral scheme for three-dimensional incompressible Hall-magnetohydrodynamics system (2026). [DOI](#)
- 2026GR: Convergence of hyperbolic approximations to higher-order PDEs for smooth solutions (2026). [DOI](#)
- 2026GT2: Invariant-Domain Preserving IMEX Schemes for the Nonequilibrium Gray Radiation-Hydrodynamics Equations Part I (2026). [DOI](#)
- 2026GTCD: A second-order semi-Lagrangian backward differentiation formula with Picard iterations (PICSL) for the spherical shallow-water equations (2026). [DOI](#)
- 2026H2: PDE Realization and Structure-Aware Solvers for a Compact Two-Stage Fourth-Order IMEX Method (2026). [DOI](#)
- 2026HGW: An Adaptive Integrating Factor Midpoint Method for Second Order Evolution Equations (2026). [DOI](#)
- 2026HM: A Priori Adaptive Numerical Methods for Estimating Blow-Up Times of Autonomous ODEs (2026). [DOI](#)
- 2026HS: Local Time Integration for Friedrichs’ Systems (2026). [DOI](#)
- 2026HY: An analytically-solvable, wave-form asymptotic-preserving and energy-conserving time-splitting scheme for Vlasov–Poisson equations in the quasi-neutral regime (2026). [DOI](#)
- 2026HZCN: A Curvilinear Lagrangian discontinuous Galerkin method for resistive magneto-hydrodynamics (2026). [DOI](#)
- 2026JQZ: A Class of High Order Unconditional Maximum Principle Preserving Extended IFRK Schemes for Allen-Cahn Equation (2026). [DOI](#)
- 2026KLWB: A Hybrid Semi-Lagrangian Flow Mapping Approach for Vlasov Systems: Combining Iterative and Compositional Flow Maps (2026). [DOI](#)
- 2026LHD: jaxdae: A JAX-native Differentiable Solver for Differential-Algebraic Equations in Coupled Multi-

- physics (2026). [DOI](#)
- 2026LTWZ: A Class of Refined Implicit-Explicit Runge-Kutta Methods with Robust Time Adaptability and Unconditional Convergence for the Cahn-Hilliard Model (2026). [DOI](#)
 - 2026LYZR: Semi-implicit scheme for multi-species collision operators in Tokamak Plasma Simulations (2026). [DOI](#)
 - 2026LZLC: An implicit Lagrangian generalized finite difference method for flow problems (2026). [DOI](#)
 - 2026LZS: Increasingly high-order compact central schemes with a recursively Jacobian-free Cauchy-Kovalevskaya method (2026). [DOI](#)
 - 2026MC: Extended phase-space symplectic integration for electron dynamics (2026). [DOI](#)
 - 2026MHL: High-order exponential solver method for particle-in-cell simulations (2026). [DOI](#)
 - 2026MNM: A High Order Accurate and Provably Stable Fully Discrete Continuous Galerkin Framework on Summation-by-Parts Form for Advection-Diffusion Equations (2026). [DOI](#)
 - 2026MR: Structure-preserving long-time simulations of turbulence in magnetised ideal fluids (2026). [DOI](#)
 - 2026N: Differential Inversion of the Implicit Euler Method (2026). [DOI](#)
 - 2026NCE: A Low-Rank, High-Order Implicit-Explicit Integrator for Three-Dimensional Convection-Diffusion Equations (2026). [DOI](#)
 - 2026ODDE: STORM: Scrape-off layer turbulence in tokamak fusion reactors (2026). [DOI](#)
 - 2026P2: moljax: GPU-accelerated method of lines for stiff reaction-diffusion PDEs with FFT preconditioning (2026). [DOI](#)
 - 2026RARB: New Time Integrators and Capabilities in SUNDIALS Versions 6.2.0–7.4.0 (2026). [DOI](#)
 - 2026S: High Order Lax-Wendroff Methods for Hyperbolic Systems (2026). [DOI](#)
 - 2026S3: An Overview of Relativistic Particle Pushers and their Extension to Arbitrary Order Accuracy (2026). [DOI](#)
 - 2026SCL: Mass-lumping technique for fully explicit time integration of the shifted-basis variant of the 3D extended finite element method (2026). [DOI](#)
 - 2026SDL: Half-explicit Runge–Kutta integrators for variational multiscale turbulence modeling: Toward higher-order accuracy in space and time (2026). [DOI](#)
 - 2026SNSK: A Discontinuous Galerkin Consistent Splitting Method for the Incompressible Navier–Stokes Equations (2026). [DOI](#)
 - 2026SQHZ: An adaptive-rank approach with greedy sampling for multi-scale BGK equations (2026). [DOI](#)
 - 2026SRHK: A splitting algorithm for a zonal quasineutral Vlasov model of strongly magnetized plasmas (2026). [DOI](#)
 - 2026ST: Numerical Efficiency of Explicit Time Integrators for Phase-Field Models (2026). [DOI](#)
 - 2026TCLH: Coupling of an External Circuit Model With a Plasma Fluid Model With a Time-Step Size Refinement (TSSR) Algorithm (2026). [DOI](#)
 - 2026WLST: An Efficient Solver for Finite Element-based Constrained Transport in 3D Magnetohydrodynamics Applied to Magnetic Confinement Fusion (2026). [DOI](#)
 - 2026WWZ: Unconditionally Long-Time Stable Variable-Step Second-Order ETD Schemes for the 2D Periodic Incompressible NSE (2026). [DOI](#)
 - 2026X: A New Fractional Step Structure Preserving Method for The Landau-Lifshitz-Gilbert Equation (2026). [DOI](#)
 - 2026YM: On the Conjecture of Stability Preservation in Arbitrary-Order Adams-Bashforth-Type Integrators (2026). [DOI](#)
 - 2026ZGZ: Predictor-corrector discrete unified gas kinetic scheme for Boltzmann transport models with non-conservative collision operators (2026). [DOI](#)
 - 2026ZW: Positivity-preserving implicit finite volume methods on unstructured grids for compressible flows (2026). [DOI](#)
 - 2025AF: Enforcing Conservation Laws and Dissipation Inequalities Numerically via Auxiliary Variables (2025). [DOI](#)
 - 2025CHLW: Asymptotic-preserving semi-Lagrangian discontinuous Galerkin schemes for the Boltzmann equation (2025). [DOI](#)
 - 2025DDGG: Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror (2025). [DOI](#)
 - 2025DUA: A time-space B-spline integrator for the Burgers' equation (2025). [DOI](#)

- 2025H: An L-Stable Implicit Two-Stage Fourth-Order Temporal Discretization Scheme for Lax-Wendroff-Type Solvers Applied to Stiff Problems (2025). [DOI](#)
- 2025HJP: Temporal Substepping Scheme for Magnetohydrodynamics with Cell-based Adaptive Mesh Refinement on Staggered Grid (2025). [DOI](#)
- 2025HLS: On High-Order/Low-Order and Micro-Macro Methods for Implicit Time-Stepping of the BGK Model (2025). [DOI](#)
- 2025HMST: Structure-preserving finite-element approximations of the magnetic Euler-Poisson equations (2025). [DOI](#)
- 2025KM: Extending Irksome: Improvements in Automated Runge–Kutta Time Stepping for Finite Element Methods (2025). [DOI](#)
- 2025L: Symplectic particle-in-cell methods for hybrid plasma models with Boltzmann electrons and space-charge effects (2025). [DOI](#)
- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)
- 2025P: Multi-Order Runge-Kutta Methods or How to Numerically Solve Initial Value Problems of Any Order (2025). [DOI](#)
- 2025RAML: Efficient and Flexible Multirate Temporal Adaptivity (2025). [DOI](#)
- 2025RGHL: Efficient Solution of Fully Implicit Runge–Kutta Methods for Linear Wave Equations (2025). [DOI](#)
- 2025SETS: Explicit and Effectively Symmetric Runge-Kutta Methods (2025). [DOI](#)
- 2025SRHK: Convergence of Splitting Methods on Rotating Grids for the Magnetized Vlasov Equation (2025). [DOI](#)
- 2025SWRP: Moment-based adaptive time integration for thermal radiation transport (2025). [DOI](#)
- 2025YHR: An Explicit Energy-Conserving Particle Method for the Vlasov-Fokker-Planck Equation (2025). [DOI](#)
- 2025ZZS: A direct PinT algorithm for higher-order nonlinear time-evolution equations (2025). [DOI](#)
- 2024AK: An Arbitrarily Higher-Order Time Implicit Scheme for Maxwell’s Equations (2024). [DOI](#)
- 2024BC: Asymptotic Analysis of IMEX-RK Methods for ES-BGK Model at Navier–Stokes Level (2024). [DOI](#)
- 2024SV: Quadratic projectable Runge-Kutta methods (2024). [DOI](#)
- 2023ATF: Temporal accuracy of FastRK3 (2023). [DOI](#)
- 2023SKYI: Integrating Odeint Time Stepping into OpenFPM for Distributed and GPU Accelerated Numerical Solvers (2023). [DOI](#)
- 2022SKP: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part II: Nonlinearities and DAEs (2022). [DOI](#)
- 2022SKPS: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part I: The Linear Setting (2022). [DOI](#)
- 2021FKM: Irksome: Automating Runge–Kutta Time-stepping for Finite Element Methods (2021). [DOI](#)
- 2020AF: A fast pressure-correction method for incompressible flows over curved walls (2020). [DOI](#)
- 2009GKS: High Order Strong Stability Preserving Time Discretizations (2009). [DOI](#)
- 2005G: On High Order Strong Stability Preserving Runge–Kutta and Multi Step Time Discretizations (2005). [DOI](#)
- 2002S: Automatic control and adaptive time-stepping (2002). [DOI](#)
- 1999HW: Stiff differential equations solved by Radau methods (1999). [DOI](#)

Numerical Linear Algebra

- 2026ABBB: Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop (2026). [DOI](#)
- 2026AC: IrAA: Low-Rank Anderson Acceleration (2026). [DOI](#)
- 2026BBKL: Reduced rank extrapolation for multi-term Sylvester equations (2026). [DOI](#)
- 2026BDFR: The iterated Golub-Kahan-Tikhonov method (2026). [DOI](#)
- 2026BFKP: Fast solvers for the high-order FEM simplicial de Rham complex (2026). [DOI](#)
- 2026BMP: Momentum accelerated power iterations and the restarted Lanczos method (2026). [DOI](#)
- 2026CEMM: Does block size matter in randomized block Krylov low-rank approximation? (2026). [DOI](#)
- 2026CES: Fast GPU Linear Algebra via Compile Time Expression Fusion (2026). [DOI](#)
- 2026CHLZ: Preconditioning without a preconditioner using randomized block Krylov subspace methods (2026). [DOI](#)

- 2026CKLS: A Parallel Basis Update and Galerkin Integrator for Tree Tensor Networks (2026). [DOI](#)
- 2026DBCT: Scalable s -step Preconditioned Conjugate Gradient with Chebyshev Basis and Gauss-Seidel Gram Solve (2026). [DOI](#)
- 2026DEM: The matrix-vector complexity of $Ax = b$ (2026). [DOI](#)
- 2026G2: Unifying Optimization and Dynamics to Parallelize Sequential Computation: A Guide to Parallel Newton Methods for Breaking Sequential Bottlenecks (2026). [DOI](#)
- 2026GNY: Entrywise Approximation for Matrix Inversion and Linear Systems (2026). [DOI](#)
- 2026GT: Numerical instability of algebraic rootfinders (2026). [DOI](#)
- 2026H2: PDE Realization and Structure-Aware Solvers for a Compact Two-Stage Fourth-Order IMEX Method (2026). [DOI](#)
- 2026HVFO: Non-unitary time evolution via the Chebyshev expansion method (2026). [DOI](#)
- 2026HXLW: A Structure-Aware Irregular Blocking Method for Sparse LU Factorization (2026). [DOI](#)
- 2026HZYZ: IFNSO: Iteration-Free Newton-Schulz Orthogonalization (2026). [DOI](#)
- 2026KJWC: One-shot acceleration of transient PDE solvers via online-learned preconditioners (2026). [DOI](#)
- 2026KW: Everything is Vecchia: Unifying low-rank and sparse inverse Cholesky approximations (2026). [DOI](#)
- 2026LL: Parallel Sparse and Data-Sparse Factorization-Based Linear Solvers (2026). [DOI](#)
- 2026LSS: A Contour Integral-Based Algorithm for Computing Generalized Singular Values (2026). [DOI](#)
- 2026LYXB: Sub-Second Collisionless Gyrokinetic Eigenvalue Solutions via Orbit-Invariant Decomposition (2026). [DOI](#)
- 2026MTWG: A GPU-accelerated matrix-free FAS multigrid solver for Navier–Stokes equations with memory-efficient implementations (2026). [DOI](#)
- 2026N: Differential Inversion of the Implicit Euler Method (2026). [DOI](#)
- 2026N4: A 52-Addition, Rank-23 Scheme for Exact 3×3 Matrix Multiplication (2026). [DOI](#)
- 2026NO: Mixed Precision Iterative Refinement for Linear Inverse Problems (2026). [DOI](#)
- 2026P2: moljax: GPU-accelerated method of lines for stiff reaction-diffusion PDEs with FFT preconditioning (2026). [DOI](#)
- 2026PN: Approximating sparse matrices and their functions using matrix-vector products (2026). [DOI](#)
- 2026S7: Adaptive multigrid for high-order discontinuous Galerkin methods based on the full approximation scheme (2026). [DOI](#)
- 2026TMS: An error control framework for computing the exponential of matrices arising from the finite element discretization (2026). [DOI](#)
- 2026VPBA: Distributed Linear Solvers and Data Heterogeneity (2026). [DOI](#)
- 2026WEK: Adaptive mesh $h(p)$ -refinement of a discontinuous Bubnov–Galerkin isogeometric analysis spatial discretisation of the first-order form of the neutron transport equation with goal-based error measures and diffusion acceleration (2026). [DOI](#)
- 2026WLST: An Efficient Solver for Finite Element-based Constrained Transport in 3D Magnetohydrodynamics Applied to Magnetic Confinement Fusion (2026). [DOI](#)
- 2026WWQW: 2DESR: a two-dimensional Fourier-space gyrokinetic eigenvalue code for the ion-temperature-gradient modes in tokamaks (2026). [DOI](#)
- 2026YZLY: TokaGLINT: A Scalable GPU-Tailored Implicit Solver for Full 3D Tokamak Electromagnetic Simulations (2026). [DOI](#)
- 2025A: Formal Verification of COO to CSR Sparse Matrix Conversion (Invited Paper) (2025). [DOI](#)
- 2025A2: Stabilization of BiCGSTAB by the generalized residual cutting method (2025). [DOI](#)
- 2025ADFM: Analysis of Floating-Point Matrix Multiplication Computed via Integer Arithmetic (2025). [DOI](#)
- 2025BG: Slicing Is All You Need: Towards A Universal One-Sided Algorithm for Distributed Matrix Multiplication (2025). [DOI](#)
- 2025BKKS: A hybrid Chebyshev-Tucker tensor format for approximation of multivariate functions (2025). [DOI](#)
- 2025C: HYLW: Hybrid Parallel Sparse LU Factorization (2025). [DOI](#)
- 2025DFHL: Mixed-precision algorithms for solving the Sylvester matrix equation (2025). [DOI](#)
- 2025DS: Matrix Factorizations with Uniformly Random Pivoting (2025). [DOI](#)
- 2025EB: Global Asymptotic Rates Under Randomization: Gauss-Seidel and Kaczmarz (2025). [DOI](#)
- 2025HL: A Generalized Alternating Anderson Acceleration Method (2025). [DOI](#)
- 2025JMS: Harnessing Batched BLAS/LAPACK Kernels on GPUs for Parallel Solutions of Block Tridiagonal Systems (2025). [DOI](#)

- 2025KM: Extending Irksome: Improvements in Automated Runge–Kutta Time Stepping for Finite Element Methods (2025). [DOI](#)
- 2025RGHL: Efficient Solution of Fully Implicit Runge–Kutta Methods for Linear Wave Equations (2025). [DOI](#)
- 2025S3: Admissible and attainable convergence behavior with stagnation mirroring in restarted (block) GMRES (2025). [DOI](#)
- 2025SC: Armadillo: an Efficient Framework for Numerical Linear Algebra (2025). [DOI](#)
- 2025SSX: Even Faster Kernel Matrix Linear Algebra via Density Estimation (2025). [DOI](#)
- 2025TOMT: pyGinkgo: A Sparse Linear Algebra Operator Framework for Python (2025). [DOI](#)
- 2024K: Floating Point Compression of Hierarchical Matrix Formats and its Impact on Matrix-Vector Multiplication (2024). [DOI](#)
- 2023T: Preconditioning for time-harmonic Maxwell’s equations using the Laguerre transform (2023). [DOI](#)
- 2022SKP: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part II: Nonlinearities and DAEs (2022). [DOI](#)
- 2022SKPS: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part I: The Linear Setting (2022). [DOI](#)

Computational Algebra

- 2026CH2: Complexes from Complexes: Finite Element Complexes in Three Dimensions (2026). [DOI](#)
- 2026GT: Numerical instability of algebraic rootfinders (2026). [DOI](#)
- 2026N4: A 52-Addition, Rank-23 Scheme for Exact 3×3 Matrix Multiplication (2026). [DOI](#)
- 2026OPK: Quadratzation of Autonomous Partial Differential Equations: Theory and Algorithms (2026). [DOI](#)
- 2026TG: Object-oriented programming as a tool for constructing high-order quantum-kinetic BBGKY equations (2026). [DOI](#)
- 2025G: A Method for Establishing Asymptotically Accurate Bounds for Extremal Roots of Eulerian Polynomials Using Polynomial Stability Preservers (2025). [DOI](#)
- 2025GST: Fast and accurate computation of classical Gaussian quadratures (2025). [DOI](#)
- 2025SETS: Explicit and Effectively Symmetric Runge-Kutta Methods (2025). [DOI](#)

Representation Theory

- 2024O: Parallelisation of partial differential equations via representation theory (2024). [DOI](#)

Optimal Transport

- 2026CMPT: A Lagrangian method for solving the spherical shallow water equations using power diagrams (2026). [DOI](#)
- 2026KRR: Efficient numerical strategies for entropy-regularized semi-discrete optimal transport (2026). [DOI](#)
- 2026RF: Sampling Boltzmann distributions via normalizing flow approximation of transport maps (2026). [DOI](#)
- 2025VKCK: Optimal Transport, Timesteppers, Newton-Krylov Methods and Steady States of Collective Particle Dynamics (2025). [DOI](#)

Algorithms

- 2026ABBB: Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop (2026). [DOI](#)
- 2026AC: IrAA: Low-Rank Anderson Acceleration (2026). [DOI](#)
- 2026BB: Scalable Overset Computation Between a Forest-of-Octrees- and an Arbitrary Distributed Parallel Mesh (2026). [DOI](#)
- 2026BBKL: Reduced rank extrapolation for multi-term Sylvester equations (2026). [DOI](#)
- 2026BKSS: History-Independent Load Balancing (2026). [DOI](#)
- 2026BMP: Momentum accelerated power iterations and the restarted Lanczos method (2026). [DOI](#)
- 2026CDKM: Frontier Space-Time Algorithms Using Only Full Memory (2026). [DOI](#)
- 2026CEMM: Does block size matter in randomized block Krylov low-rank approximation? (2026). [DOI](#)

- 2026CHLZ: Preconditioning without a preconditioner using randomized block Krylov subspace methods (2026). [DOI](#)
- 2026CMPT: A Lagrangian method for solving the spherical shallow water equations using power diagrams (2026). [DOI](#)
- 2026DEM: The matrix-vector complexity of $Ax = b$ (2026). [DOI](#)
- 2026G: bsort: A theoretically efficient non-comparison-based sorting algorithm for integer and floating-point numbers (2026). [DOI](#)
- 2026GGT: PACO: A Fully Cache-Oblivious Parallel FFT with One Global Redistribution (2026). [DOI](#)
- 2026GNY: Entrywise Approximation for Matrix Inversion and Linear Systems (2026). [DOI](#)
- 2026HZYZ: IFNSO: Iteration-Free Newton-Schulz Orthogonalization (2026). [DOI](#)
- 2026KS: Resolving Conflicts with Grace: Dynamically Concurrent Universality (2026). [DOI](#)
- 2026KW: Everything is Vecchia: Unifying low-rank and sparse inverse Cholesky approximations (2026). [DOI](#)
- 2026L: A Bivariate Spline Construction of Orthonormal Polynomials over Polygonal Domains and Its Applications to Quadrature (2026). [DOI](#)
- 2026LSS: A Contour Integral-Based Algorithm for Computing Generalized Singular Values (2026). [DOI](#)
- 2026M: Parallelizing the Factorial Space: $3 \times$ SIMD Acceleration of the Steinhaus-Johnson-Trotter Algorithm via Dual-Lane AVX2 Execution (2026). [DOI](#)
- 2026N: Differential Inversion of the Implicit Euler Method (2026). [DOI](#)
- 2026N4: A 52-Addition, Rank-23 Scheme for Exact 3×3 Matrix Multiplication (2026). [DOI](#)
- 2026OAGT: Numerical methods for quasi-stationary distributions (2026). [DOI](#)
- 2026OFL: Space-efficient B-tree Implementation for Memory-Constrained Flash Embedded Devices (2026). [DOI](#)
- 2026OPK: Quadratzation of Autonomous Partial Differential Equations: Theory and Algorithms (2026). [DOI](#)
- 2026PN: Approximating sparse matrices and their functions using matrix-vector products (2026). [DOI](#)
- 2026R2: Numerical quadrature on bounded surfaces without parameterizations (2026). [DOI](#)
- 2026RBCH: Automatic curved block structure and hexahedral mesh generation for hypersonic CFD (2026). [DOI](#)
- 2026SQHZ: An adaptive-rank approach with greedy sampling for multi-scale BGK equations (2026). [DOI](#)
- 2026VPBA: Distributed Linear Solvers and Data Heterogeneity (2026). [DOI](#)
- 2026VRDK: Load Balanced Parallel Node Generation for Meshless Numerical Methods (2026). [DOI](#)
- 2025A2: Stabilization of BiCGSTAB by the generalized residual cutting method (2025). [DOI](#)
- 2025AB: Optimizing Allreduce Operations for Modern Heterogeneous Architectures with Multiple Processes per GPU (2025). [DOI](#)
- 2025ART: L^2 and L^∞ rational approximation (2025). [DOI](#)
- 2025CJV: The Dynamical Anatomy of Anderson Acceleration: From Adaptive Momentum to Variable-Mass ODEs (2025). [DOI](#)
- 2025DS: Matrix Factorizations with Uniformly Random Pivoting (2025). [DOI](#)
- 2025EB: Global Asymptotic Rates Under Randomization: Gauss-Seidel and Kaczmarz (2025). [DOI](#)
- 2025G: A Method for Establishing Asymptotically Accurate Bounds for Extremal Roots of Eulerian Polynomials Using Polynomial Stability Preservers (2025). [DOI](#)
- 2025GST: Fast and accurate computation of classical Gaussian quadratures (2025). [DOI](#)
- 2025HL: A Generalized Alternating Anderson Acceleration Method (2025). [DOI](#)
- 2025JM: Efficient Algorithms to Compute Closed Substrings (2025). [DOI](#)
- 2025K: A regret minimization approach to fixed-point iterations (2025). [DOI](#)
- 2025LMST: Tree-Like Shortcuttings of Trees (2025). [DOI](#)
- 2025MBK: A New Execution Model and Executor for Adaptively Optimizing the Performance of Parallel Algorithms Using HPX Runtime System (2025). [DOI](#)
- 2025RW: Lazy B-Trees (2025). [DOI](#)
- 2025S: Towards universally optimal sorting algorithms (2025). [DOI](#)
- 2025S3: Admissible and attainable convergence behavior with stagnation mirroring in restarted (block) GMRES (2025). [DOI](#)
- 2025SSX: Even Faster Kernel Matrix Linear Algebra via Density Estimation (2025). [DOI](#)
- 2025SZ: Prefix Sums via Kronecker Products (2025). [DOI](#)
- 2025Z: Communication-Efficient and Memory-Aware Parallel Bootstrapping using MPI (2025). [DOI](#)
- 2007CHPG: Collective communication: theory, practice, and experience (2007). [DOI](#)

Computational Complexity

- 2026CDKM: Frontier Space-Time Algorithms Using Only Full Memory (2026). [DOI](#)
- 2026GGT: PACO: A Fully Cache-Oblivious Parallel FFT with One Global Redistribution (2026). [DOI](#)
- 2026N4: A 52-Addition, Rank-23 Scheme for Exact 3×3 Matrix Multiplication (2026). [DOI](#)
- 2025JM: Efficient Algorithms to Compute Closed Substrings (2025). [DOI](#)

Optimization

- 2026BDJZ: Discovering new phases via computing second-order stationary states of Landau–Brazovskii model (2026). [DOI](#)
- 2026CFTB: A Gauss-Newton Method with No Additional PDE Solves Beyond Gradient Evaluation for Large-Scale PDE-Constrained Inverse Problems (2026). [DOI](#)
- 2026CHLZ: Preconditioning without a preconditioner using randomized block Krylov subspace methods (2026). [DOI](#)
- 2026CJLY: Control of a Uniformly Magnetized Plasma with External Electric Fields (2026). [DOI](#)
- 2026DEM: The matrix-vector complexity of $Ax = b$ (2026). [DOI](#)
- 2026FPPK: A Flexible and Differentiable Coil Proxy for Stellarator Equilibrium Optimization (2026). [DOI](#)
- 2026G2: Unifying Optimization and Dynamics to Parallelize Sequential Computation: A Guide to Parallel Newton Methods for Breaking Sequential Bottlenecks (2026). [DOI](#)
- 2026GSHJ: Designing A Buildable Optimized Stellarator to Confine Electron-Positron Plasmas (2026). [DOI](#)
- 2026GSJH: Stochastic single-stage stellarator optimization using fixed-boundary equilibria (2026). [DOI](#)
- 2026HZYZ: IFNSO: Iteration-Free Newton-Schulz Orthogonalization (2026). [DOI](#)
- 2026JFHZ: Robustness Optimization for Compact Free-Electron Laser Driven by Laser Wakefield Accelerators (2026). [DOI](#)
- 2026JKGH: Optimized laser wakefield acceleration: Generating stable, high-energy, monoenergetic electron beams and demonstrating extreme-ultraviolet free-electron lasers (2026). [DOI](#)
- 2026KLJH: How Does The Magnetic Gradient Scale Length Influence Complexity of Filamentary Coils in Stellarators? (2026). [DOI](#)
- 2026KRR: Efficient numerical strategies for entropy-regularized semi-discrete optimal transport (2026). [DOI](#)
- 2026LMSZ: Efficient optimization-based invariant-domain-preserving limiters in solving gas dynamics equations (2026). [DOI](#)
- 2026LV: Weak Scalability of time parallel Schwarz methods for parabolic optimal control problems (2026). [DOI](#)
- 2026LYVZ: Optimization of stellarator configurations combining omnigenity and piecewise omnigenity (2026). [DOI](#)
- 2026MHLH: Batch Bayesian Optimization of Attosecond Betatron Pulses from Laser Wakefield Acceleration (2026). [DOI](#)
- 2026N3: Maximal Entropy Dissipation Numerical Scheme for Conservation Law Systems (2026). [DOI](#)
- 2026OTWH: Sparse and low-rank kinetic distribution estimation (2026). [DOI](#)
- 2026PJCD: Deflation Techniques for Stellarator Equilibrium and Optimization (2026). [DOI](#)
- 2026QMLC: Optimization of laser-driven proton acceleration in a near-critical-density plasma (2026). [DOI](#)
- 2026RXPS: Critical gradient optimisation for quasi-isodynamic stellarators (2026). [DOI](#)
- 2026S4: The State-Dependent Riccati Equation in Nonlinear Optimal Control: Analysis and Numerical Approximation (2026). [DOI](#)
- 2026SNMD: Exploring the fusion power plant design space: comparative analysis of positive and negative triangularity tokamaks through optimization (2026). [DOI](#)
- 2026VBFP: Stellarator Island Divertor Shape Optimization for Reduced Peak Heat Fluxes (2026). [DOI](#)
- 2026VPBA: Distributed Linear Solvers and Data Heterogeneity (2026). [DOI](#)
- 2026WBBC: A tutorial on inversion-based shape control with design application to NSTX-U (2026). [DOI](#)
- 2026WL: A unified high-resolution ODE framework for first-order methods (2026). [DOI](#)
- 2026ZZHD: A subcell entropy-constrained limiter for entropy stable discontinuous Galerkin methods (2026). [DOI](#)
- 2025CJV: The Dynamical Anatomy of Anderson Acceleration: From Adaptive Momentum to Variable-Mass ODEs (2025). [DOI](#)

- 2025EB: Global Asymptotic Rates Under Randomization: Gauss-Seidel and Kaczmarz (2025). [DOI](#)
- 2025HL: A Generalized Alternating Anderson Acceleration Method (2025). [DOI](#)
- 2025K: A regret minimization approach to fixed-point iterations (2025). [DOI](#)

Uncertainty Quantification

- 2026CKLS: A Parallel Basis Update and Galerkin Integrator for Tree Tensor Networks (2026). [DOI](#)
- 2026GXPS: Real-Time Plasma State Prediction via FPGA-Accelerated Quantized Recurrent Probabilistic Neural Networks (2026). [DOI](#)
- 2026HS2: Anchor-Based Function Extrapolation with Proven Bounds and Projection Guarantees (2026). [DOI](#)
- 2026LHKD: Physics-Informed Drift Diagnosis for Laser-Plasma Accelerator Operations (2026). [DOI](#)
- 2026SGSH: A Predictive Formula for the H-Mode Electron Separatrix Density: Bridging Regression and Physics-Based Models across C-Mod, AUG and JET Tokamaks (2026). [DOI](#)
- 2026ZL: Benchmarking and extending grid-based, meshfree, and physics-informed neural network methods for the generalized probability density evolution equation (2026). [DOI](#)
- 2025CMPV: Mathematical modeling and sensitivity analysis of hypoxia-activated drugs (2025). [DOI](#)
- 2025UN: Nonlinear Gaussian process tomography with imposed non-negativity constraints on physical quantities for plasma diagnostics (2025). [DOI](#)
- 2025Z: Communication-Efficient and Memory-Aware Parallel Bootstrapping using MPI (2025). [DOI](#)
- 2024LLW: A bi-fidelity method for the uncertain Vlasov-Poisson system near quasineutrality in an asymptotic-preserving particle-in-cell framework (2024). [DOI](#)

Reduced Order Modeling

- 2026DAKM: A polynomial approximation scheme for nonlinear model reduction by moment matching (2026). [DOI](#)
- 2026NCE: A Low-Rank, High-Order Implicit-Explicit Integrator for Three-Dimensional Convection-Diffusion Equations (2026). [DOI](#)
- 2026NPSB: AMORE: Adaptive multi-output operator network for stiff chemical kinetics (2026). [DOI](#)
- 2026OC: Hamiltonian Hydrodynamic Reductions of One-Dimensional Vlasov Equations (2026). [DOI](#)
- 2026OPK: Quadraticization of Autonomous Partial Differential Equations: Theory and Algorithms (2026). [DOI](#)
- 2026QNC: Entropy stable reduced order modeling of nonlinear conservation laws using discontinuous Galerkin methods (2026). [DOI](#)
- 2026WH2: Dynamical tensor train approximation for kinetic equations (2026). [DOI](#)
- 2025HGR: The Hot-Electron Closure of the Moment-Based Gyrokinetic Plasma Model (2025). [DOI](#)
- 2025PGVA: Tensor-based multivariate function approximation: methods benchmarking and comparison (2025). [DOI](#)
- 2025VCCC: Machine-precision energy conservative reduced models for Lagrangian hydrodynamics by quadrature methods (2025). [DOI](#)

Computational Fluid Dynamics

- 2026AB: A conservative, discontinuous Galerkin, tracer transport scheme using compatible finite elements (2026). [DOI](#)
- 2026ABBG: Lethe 1.0: An open-source parallel high-order computational fluid dynamics software framework for single and multiphase flows (2026). [DOI](#)
- 2026ACKM: Dual formulation finite-volume methods on overlapping meshes for hyperbolic conservation laws (2026). [DOI](#)
- 2026AMC: Formulation of entropy-conservative discretizations for compressible flows of thermally perfect gases (2026). [DOI](#)
- 2026AMM: Energy Cascades in Rotating and Stratified Turbulence in Anisotropic Domains (2026). [DOI](#)
- 2026B4: Routes towards an effective AI in CFD: an epistemological and technical perspective (2026). [DOI](#)
- 2026BTG: Stability analysis of Arbitrary-Lagrangian-Eulerian ADER-DG methods on classical and degenerate spacetime geometries (2026). [DOI](#)

- 2026CC: Explosive combustion in a closed channel revisited (2026). [DOI](#)
- 2026CHLW: COS(DG): A Convex Oscillation-Suppressing Framework for High-Order Discontinuous Galerkin Methods (2026). [DOI](#)
- 2026CMPT: A Lagrangian method for solving the spherical shallow water equations using power diagrams (2026). [DOI](#)
- 2026CWML: A general kinematic theory of fluid-element rotation and intrinsic vorticity decompositions (2026). [DOI](#)
- 2026CZQ: Fifth-order direct Arbitrary-Lagrangian-Eulerian Hermite WENO scheme for the compressible Euler equations (2026). [DOI](#)
- 2026DBTT: A structure-preserving staggered semi-implicit four-split finite volume scheme for continuum mechanics on unstructured meshes (2026). [DOI](#)
- 2026DRM: A numerical strategy for the acceleration of resolved turbulence development in high Mach number turbulent boundary layers (2026). [DOI](#)
- 2026EAKU: Multi-Partner Project: Multi-GPU Performance Portability Analysis for CFD Simulations at Scale (2026). [DOI](#)
- 2026FW: Compact Finite-Average-Moment Schemes with Single-Step Oscillation Elimination for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026FZHW: Fire: An open-source adaptive mesh refinement solver for supersonic reacting flows (2026). [DOI](#)
- 2026GSC: A Stable Conservative Lagrange-Galerkin Method for Convection Equations Based on Curved-Element Mesh Intersection (2026). [DOI](#)
- 2026GTCD: A second-order semi-Lagrangian backward differentiation formula with Picard iterations (PICSL) for the spherical shallow-water equations (2026). [DOI](#)
- 2026GYWG: COREFL: An open-source GPU-accelerated high-fidelity solver for compressible reactive flows on generalized curvilinear coordinates (2026). [DOI](#)
- 2026GZ: A Space-Time Galerkin Boundary Element Method for Aeroacoustic Scattering (2026). [DOI](#)
- 2026HBKW: Novel approach to general curvilinear coordinates for plasma fluid applications (2026). [DOI](#)
- 2026HLLS: An adaptive error estimator for stationary Navier-Stokes equations using variational physics informed neural networks (2026). [DOI](#)
- 2026HRT: A checkerboard-free, symmetry-preserving, conservative method for magnetohydrodynamics (2026). [DOI](#)
- 2026JKGH: Optimized laser wakefield acceleration: Generating stable, high-energy, monoenergetic electron beams and demonstrating extreme-ultraviolet free-electron lasers (2026). [DOI](#)
- 2026K2: kalypso: A performance portable platform for compressible hydrodynamics simulations using adaptive mesh refinement (2026). [DOI](#)
- 2026KJWC: One-shot acceleration of transient PDE solvers via online-learned preconditioners (2026). [DOI](#)
- 2026KSCP: Generalized Tadmor Conditions and Structure-Preserving Numerical Fluxes for the Compressible Flow of Real Gases (2026). [DOI](#)
- 2026KSKB: Entropy stable high-order discontinuous Galerkin spectral-element methods on curvilinear, hybrid meshes (2026). [DOI](#)
- 2026LC: Multi-GPU acceleration of PALABOS fluid solver using C++ standard parallelism (2026). [DOI](#)
- 2026LG: An efficient discrete unified gas kinetic scheme for strongly inhomogeneous fluids at the nanoscale (2026). [DOI](#)
- 2026LLR: Three-dimensional mesh adaptation in PFEM (2026). [DOI](#)
- 2026LMSZ: Efficient optimization-based invariant-domain-preserving limiters in solving gas dynamics equations (2026). [DOI](#)
- 2026LZLC: An implicit Lagrangian generalized finite difference method for flow problems (2026). [DOI](#)
- 2026LZS: Increasingly high-order compact central schemes with a recursively Jacobian-free Cauchy-Kovalevskaya method (2026). [DOI](#)
- 2026MRG: Entropy-Stable Discontinuous Spectral-Element Methods for the Spherical Shallow Water Equations in Covariant Form (2026). [DOI](#)
- 2026MTWG: A GPU-accelerated matrix-free FAS multigrid solver for Navier–Stokes equations with memory-efficient implementations (2026). [DOI](#)
- 2026N2: A Flux-Correction Form of the Third-Order Edge-Based Scheme for a General Numerical Flux Function (2026). [DOI](#)

- 2026RBCH: Automatic curved block structure and hexahedral mesh generation for hypersonic CFD (2026). [DOI](#)
- 2026RCC: Efficient entropy-conserving/stable discontinuous Galerkin solution of the multicomponent compressible Euler equations (2026). [DOI](#)
- 2026SDL: Half-explicit Runge–Kutta integrators for variational multiscale turbulence modeling: Toward higher-order accuracy in space and time (2026). [DOI](#)
- 2026SDLM: A well-balanced positivity-preserving discontinuous Galerkin method for shallow water models with variable density (2026). [DOI](#)
- 2026SH: Performance of a high-order MPI-Kokkos accelerated fluid solver (2026). [DOI](#)
- 2026SNSK: A Discontinuous Galerkin Consistent Splitting Method for the Incompressible Navier–Stokes Equations (2026). [DOI](#)
- 2026SP: Lyapunov stability analysis of the chaotic flow past two square cylinders (2026). [DOI](#)
- 2026SS: A multi-resolution limiter for the Runge–Kutta discontinuous Galerkin method (2026). [DOI](#)
- 2026VLN: The inverse initial data problem for anisotropic Navier–Stokes equations via Legendre time reduction method (2026). [DOI](#)
- 2026WH: An effective implementation of the bidirectional buffer: Towards laminar and turbulent open-boundary flows (2026). [DOI](#)
- 2026WKRS: A meshless MUSCL method for the BGK-Boltzmann equation (2026). [DOI](#)
- 2026WRS: Unstable Magnetic Reconnection Self-Generates Turbulence (2026). [DOI](#)
- 2026WWZ: Unconditionally Long-Time Stable Variable-Step Second-Order ETD Schemes for the 2D Periodic Incompressible NSE (2026). [DOI](#)
- 2026XL: On the Spectral Properties of Van Leer and AUSM Flux-Vector Splitting Schemes (2026). [DOI](#)
- 2026YF: An Entropy-Stable Oscillation-Eliminating DGSEM for the Euler Equations on Curvilinear Meshes (2026). [DOI](#)
- 2026YLFP: Surfactant effects on gravity-capillary waves (2026). [DOI](#)
- 2026ZW: Positivity-preserving implicit finite volume methods on unstructured grids for compressible flows (2026). [DOI](#)
- 2026ZYH: A differentiable wall-modeled large-eddy simulation method for high-Reynolds-number wall-bounded turbulent flows (2026). [DOI](#)
- 2026ZZHD: A subcell entropy-constrained limiter for entropy stable discontinuous Galerkin methods (2026). [DOI](#)
- 2025AF: Enforcing Conservation Laws and Dissipation Inequalities Numerically via Auxiliary Variables (2025). [DOI](#)
- 2025GRD: On General and Complete Multidimensional Riemann Solvers for Nonlinear Systems of Hyperbolic Conservation Laws (2025). [DOI](#)
- 2025H: An L-Stable Implicit Two-Stage Fourth-Order Temporal Discretization Scheme for Lax-Wendroff-Type Solvers Applied to Stiff Problems (2025). [DOI](#)
- 2025PE: Stable low diffusion flux splitting schemes on unstructured meshes (2025). [DOI](#)
- 2025VCCC: Machine-precision energy conservative reduced models for Lagrangian hydrodynamics by quadrature methods (2025). [DOI](#)
- 2024BC: Asymptotic Analysis of IMEX-RK Methods for ES-BGK Model at Navier–Stokes Level (2024). [DOI](#)
- 2023ATF: Temporal accuracy of FastRK3 (2023). [DOI](#)
- 2022SKP: Fast Solution of Fully Implicit Runge–Kutta and Discontinuous Galerkin in Time for Numerical PDEs, Part II: Nonlinearities and DAEs (2022). [DOI](#)
- 2020AF: A fast pressure-correction method for incompressible flows over curved walls (2020). [DOI](#)
- 2003LW: Comparison of Several Difference Schemes on 1D and 2D Test Problems for the Euler Equations (2003). [DOI](#)

Reactive Flows

- 2026CC: Explosive combustion in a closed channel revisited (2026). [DOI](#)
- 2026FZHW: Fire: An open-source adaptive mesh refinement solver for supersonic reacting flows (2026). [DOI](#)
- 2026GYWG: COREFL: An open-source GPU-accelerated high-fidelity solver for compressible reactive flows on generalized curvilinear coordinates (2026). [DOI](#)
- 2026NPSB: AMORE: Adaptive multi-output operator network for stiff chemical kinetics (2026). [DOI](#)

Transport and Conservation Laws

- 2027GLWN: Towards Provable Energy-Stable Overset Grid Methods Using Sub-Cell Summation-by-Parts Operators (2027). [DOI](#)
- 2026AB: A conservative, discontinuous Galerkin, tracer transport scheme using compatible finite elements (2026). [DOI](#)
- 2026ABL: Virtual finite element and hyperbolic problems: The PAMPA algorithm (2026). [DOI](#)
- 2026ACKM: Dual formulation finite-volume methods on overlapping meshes for hyperbolic conservation laws (2026). [DOI](#)
- 2026AH: High-Order Moment Closure for Nonmagnetized Electrons in Partially Ionized Plasmas (2026). [DOI](#)
- 2026AL: Robust PAMPA scheme in the DG formulation on unstructured triangular meshes: Bound preservation, oscillation elimination, and boundary conditions (2026). [DOI](#)
- 2026AMC: Formulation of entropy-conservative discretizations for compressible flows of thermally perfect gases (2026). [DOI](#)
- 2026AR: On Affordable High-Order Entropy-Conservative/Stable and Well-Balanced Methods for Nonconservative Hyperbolic Systems (2026). [DOI](#)
- 2026BCSR: Automatic differentiation for performing the Cauchy–Kovalevskaya procedure in Lax–Wendroff type discretizations (2026). [DOI](#)
- 2026BF: Monotone-based Numerical Schemes for Two-Dimensional Systems of Nonlocal Conservation Laws (2026). [DOI](#)
- 2026BR: Compact Runge-Kutta flux reconstruction methods for non-conservative hyperbolic equations (2026). [DOI](#)
- 2026CC: Explosive combustion in a closed channel revisited (2026). [DOI](#)
- 2026CH: A Comparative Study of Low-Dissipation Numerical Schemes for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026CHLW: COS(DG): A Convex Oscillation-Suppressing Framework for High-Order Discontinuous Galerkin Methods (2026). [DOI](#)
- 2026CSZ: Unified Analysis of the Semidiscrete Discontinuous Galerkin Methods for 2-D Hyperbolic Equations on Cartesian Meshes Using $\mathbb{P}^k$ Elements: Optimal Error Estimates and Superconvergence (2026). [DOI](#)
- 2026CZQ: Fifth-order direct Arbitrary-Lagrangian-Eulerian Hermite WENO scheme for the compressible Euler equations (2026). [DOI](#)
- 2026FMNV: Neural semi-Lagrangian method for high-dimensional advection-diffusion problems (2026). [DOI](#)
- 2026FW: Compact Finite-Average-Moment Schemes with Single-Step Oscillation Elimination for Hyperbolic Conservation Laws (2026). [DOI](#)
- 2026GAS: Kinetic-based regularization: Learning spatial derivatives and PDE applications (2026). [DOI](#)
- 2026GHJ: BEACONS: Bounded-Error, Algebraically-Composable Neural Solvers for Partial Differential Equations (2026). [DOI](#)
- 2026GHT: Adaptive mesh methods for hyperbolic conservation laws with bound-preserving flux limiters (2026). [DOI](#)
- 2026GILO: Why summation by parts is not enough (2026). [DOI](#)
- 2026GSC: A Stable Conservative Lagrange-Galerkin Method for Convection Equations Based on Curved-Element Mesh Intersection (2026). [DOI](#)
- 2026GTCD: A second-order semi-Lagrangian backward differentiation formula with Picard iterations (PICSL) for the spherical shallow-water equations (2026). [DOI](#)
- 2026J: An Upwind Summation-by-Part Scheme for the Non-Linear Boltzmann Equation (2026). [DOI](#)
- 2026KSCP: Generalized Tadmor Conditions and Structure-Preserving Numerical Fluxes for the Compressible Flow of Real Gases (2026). [DOI](#)
- 2026KSKB: Entropy stable high-order discontinuous Galerkin spectral-element methods on curvilinear, hybrid meshes (2026). [DOI](#)
- 2026LHWP: Time-embedded convolutional neural networks for modeling plasma heat transport (2026). [DOI](#)
- 2026LKCH: An unstructured block-based adaptive mesh refinement approach for discontinuous Galerkin method (2026). [DOI](#)
- 2026LMSZ: Efficient optimization-based invariant-domain-preserving limiters in solving gas dynamics equations (2026). [DOI](#)

- 2026LQ: A simple, high-order and compact WENO limiter based on control volume for spectral volume method (2026). [DOI](#)
- 2026LRBG: Conservative Formulation of the Drift-Reduced Fluid Plasma Model (2026). [DOI](#)
- 2026LZS: Increasingly high-order compact central schemes with a recursively Jacobian-free Cauchy-Kovalevskaya method (2026). [DOI](#)
- 2026MRG: Entropy-Stable Discontinuous Spectral-Element Methods for the Spherical Shallow Water Equations in Covariant Form (2026). [DOI](#)
- 2026N2: A Flux-Correction Form of the Third-Order Edge-Based Scheme for a General Numerical Flux Function (2026). [DOI](#)
- 2026N3: Maximal Entropy Dissipation Numerical Scheme for Conservation Law Systems (2026). [DOI](#)
- 2026P: Development of discontinuous Galerkin methods for hyperbolic systems that preserve a curl or a divergence constraint: The case of linear systems (2026). [DOI](#)
- 2026QNC: Entropy stable reduced order modeling of nonlinear conservation laws using discontinuous Galerkin methods (2026). [DOI](#)
- 2026RCC: Efficient entropy-conserving/stable discontinuous Galerkin solution of the multicomponent compressible Euler equations (2026). [DOI](#)
- 2026RVKK: Adaptive hyperviscosity stabilisation for the RBF-FD method in solving advection-dominated transport equations (2026). [DOI](#)
- 2026S: High Order Lax-Wendroff Methods for Hyperbolic Systems (2026). [DOI](#)
- 2026SDLM: A well-balanced positivity-preserving discontinuous Galerkin method for shallow water models with variable density (2026). [DOI](#)
- 2026SQHZ: An adaptive-rank approach with greedy sampling for multi-scale BGK equations (2026). [DOI](#)
- 2026SS: A multi-resolution limiter for the Runge–Kutta discontinuous Galerkin method (2026). [DOI](#)
- 2026TBA: A novel conditional formulation of the Vlasov–Ampère equations: a conservative, positivity, asymptotic and Gauss law preserving scheme (2026). [DOI](#)
- 2026TGRV: Numerical Analysis of a Coupled 3D-1D Transport Problem (2026). [DOI](#)
- 2026VC: On the choice of viscous discontinuous Galerkin discretization for entropy correction artificial viscosity methods (2026). [DOI](#)
- 2026WFZ: Convergence of entropy-stable continuous summation-by-parts discretizations of symmetric hyperbolic conservation laws (2026). [DOI](#)
- 2026WKN: Numerical boundary flux functions that give provable bounds for nonlinear initial boundary value problems with open boundaries (2026). [DOI](#)
- 2026XL: On the Spectral Properties of Van Leer and AUSM Flux-Vector Splitting Schemes (2026). [DOI](#)
- 2026YF: An Entropy-Stable Oscillation-Eliminating DGSEM for the Euler Equations on Curvilinear Meshes (2026). [DOI](#)
- 2026ZGZ: Predictor-corrector discrete unified gas kinetic scheme for Boltzmann transport models with non-conservative collision operators (2026). [DOI](#)
- 2026ZW: Positivity-preserving implicit finite volume methods on unstructured grids for compressible flows (2026). [DOI](#)
- 2026ZZHD: A subcell entropy-constrained limiter for entropy stable discontinuous Galerkin methods (2026). [DOI](#)
- 2025AF: Enforcing Conservation Laws and Dissipation Inequalities Numerically via Auxiliary Variables (2025). [DOI](#)
- 2025CHLW: Asymptotic-preserving semi-Lagrangian discontinuous Galerkin schemes for the Boltzmann equation (2025). [DOI](#)
- 2025CKKS: Asymptotic-preserving conservative semi-Lagrangian discontinuous Galerkin schemes for the Vlasov-Poisson system in the quasi-neutral limit (2025). [DOI](#)
- 2025CLLZ: A high-order energy-conserving semi-Lagrangian discontinuous Galerkin method for the Vlasov-Ampère system (2025). [DOI](#)
- 2025EDLN: An Asynchronous Many-Task Algorithm for Unstructured S_N Transport on Shared Memory Systems (2025). [DOI](#)
- 2025GRD: On General and Complete Multidimensional Riemann Solvers for Nonlinear Systems of Hyperbolic Conservation Laws (2025). [DOI](#)
- 2025LD: Structure-preserving schemes for nonlinear symmetric hyperbolic and thermodynamically compatible

systems of partial differential equations (2025). [DOI](#)

- 2025NH: Asymptotically compatible entropy-consistent discretization for a class of nonlocal conservation laws (2025). [DOI](#)
- 2025PE: Stable low diffusion flux splitting schemes on unstructured meshes (2025). [DOI](#)
- 2025SWRP: Moment-based adaptive time integration for thermal radiation transport (2025). [DOI](#)
- 2009GKS: High Order Strong Stability Preserving Time Discretizations (2009). [DOI](#)
- 2005G: On High Order Strong Stability Preserving Runge–Kutta and Multi Step Time Discretizations (2005). [DOI](#)
- 2003LW: Comparison of Several Difference Schemes on 1D and 2D Test Problems for the Euler Equations (2003). [DOI](#)

Multiphysics

- 2026ABBG: Lethe 1.0: An open-source parallel high-order computational fluid dynamics software framework for single and multiphase flows (2026). [DOI](#)
- 2026AMS: Semi-explicit time discretization for linear thermo-poroelasticity (2026). [DOI](#)
- 2026FZHW: Fire: An open-source adaptive mesh refinement solver for supersonic reacting flows (2026). [DOI](#)
- 2026GK: Comparison of Structure-Preserving Methods for the Cahn–Hilliard–Navier–Stokes Equations (2026). [DOI](#)
- 2026LCLK: Discrete unified gas kinetic scheme for multiscale plasma simulations (2026). [DOI](#)
- 2026LHD: jaxdae: A JAX-native Differentiable Solver for Differential-Algebraic Equations in Coupled Multiphysics (2026). [DOI](#)
- 2026LYZC: A Strongly Discontinuous Boundary Condition Model With Discontinuous Galerkin Framework for Multiscale Electromagnetic Simulations Containing Imperfect Sliding Electrical Contact (2026). [DOI](#)
- 2026OUZI: Coupling Fluid Neutrals to Gyrokinetic Plasma Dynamics for Edge and SOL Turbulence Simulations (2026). [DOI](#)
- 2026RPRT: Thermal modeling of runaway electron induced damage in the SPARC tokamak (2026). [DOI](#)
- 2026TCLH: Coupling of an External Circuit Model With a Plasma Fluid Model With a Time-Step Size Refinement (TSSR) Algorithm (2026). [DOI](#)
- 2026WZJL: Numerical Modeling of Electromagnetic Wave Transmission Through a 2-D Magnetized Plasma Sheath (2026). [DOI](#)
- 2026YLFP: Surfactant effects on gravity-capillary waves (2026). [DOI](#)
- 2026ZCXZ: Three Dimensional Multiphysics Modelling of Helicon Wave Heating and Antenna Plasma Coupling for Boundary Density Control in Toroidal Fusion Plasmas (2026). [DOI](#)
- 2025CJBL: Case study of a differentiable heterogeneous multiphysics solver for a nuclear fusion application (2025). [DOI](#)
- 2025CMPV: Mathematical modeling and sensitivity analysis of hypoxia-activated drugs (2025). [DOI](#)
- 2025ZS: Optimal error estimates for a fully discrete, highly efficient decoupled scheme for the 2D/3D diffuse interface two-phase MHD flows (2025). [DOI](#)

Circuit Simulation

- 2026BB2: A Multirate Transient Simulation Technique for Circuits described by Integro-Differential Equations (2026). [DOI](#)
- 2026TCLH: Coupling of an External Circuit Model With a Plasma Fluid Model With a Time-Step Size Refinement (TSSR) Algorithm (2026). [DOI](#)

Biomedical Modeling

- 2025CMPV: Mathematical modeling and sensitivity analysis of hypoxia-activated drugs (2025). [DOI](#)

Plasma Physics

- 2026A: Semi-Lagrangian Discontinuous Galerkin Method with Adaptive Mesh Refinement for the Vlasov–Poisson System in 1X+3V (2026). [DOI](#)
- 2026A2: The Impact of the Generalized (r, q) Distributed Electrons on the Formation of Monotonic Shock Structures in an Anisotropic Plasma (2026). [DOI](#)
- 2026ABY: Helical core formation and MHD stability in ITER-scale plasmas with fusion-born alpha particles (2026). [DOI](#)
- 2026ADGM: Gyrokinetic simulation of the effect of transient fueling on plasma turbulence in ADITYA-U tokamak (2026). [DOI](#)
- 2026AFKP: A projection method for particle resampling (2026). [DOI](#)
- 2026AH: High-Order Moment Closure for Nonmagnetized Electrons in Partially Ionized Plasmas (2026). [DOI](#)
- 2026AKMI: Velocity-Space Signatures of Energy Transfer for Ion-Acoustic Instabilities (2026). [DOI](#)
- 2026ALGP: Energy Bunching from Subcycle Ionization Injection in Laser Wakefield Acceleration (2026). [DOI](#)
- 2026B: Anderson Localization of Ion-Temperature-Gradient Modes in Aperiodic Stellarators (2026). [DOI](#)
- 2026B3: Internal transport barriers in fluids and plasmas (2026). [DOI](#)
- 2026B5: Pfirsch-Schlüter Current (2026). [DOI](#)
- 2026B6: The toroidal flux and separatrix effects in tokamaks (2026). [DOI](#)
- 2026BC: Generalized flux-weighted boundary walls in kinetic models (2026). [DOI](#)
- 2026BC2: The machine learning approach to moment closure relations for plasma: a review (2026). [DOI](#)
- 2026BCPS: The Linearized Vlasov–Maxwell System as a Hamiltonian System (2026). [DOI](#)
- 2026BHJP: Capturing Secondary Kinetic Instabilities in Three-Dimensional Dayside Reconnection Using an Improved Gradient-Based Closure (2026). [DOI](#)
- 2026BKT: Fully kinetic 3-D simulations of high-beta plasma in a mirror trap (2026). [DOI](#)
- 2026BLAO: TokaMind: A Multi-Modal Transformer Foundation Model for Tokamak Plasma Dynamics (2026). [DOI](#)
- 2026BLBS: Longitudinally resolved measurement of energy-transfer efficiency in a plasma-wakefield accelerator (2026). [DOI](#)
- 2026BLGA: Study of Low-Frequency Core-Edge Coupling in a Tokamak: II. Spatial Channeling & Focusing In Antenna-Driven MHD (2026). [DOI](#)
- 2026BLPP: Tearing Stability Prediction Combining Toroidal Calculations With a Two-Fluid Slab Layer (2026). [DOI](#)
- 2026BMCC: Beam transverse dynamics in laser-plasma accelerators (2026). [DOI](#)
- 2026BP: A fast spectral particle method for the Landau equation (2026). [DOI](#)
- 2026BSSK: Accelerating the Particle-In-Cell code ECsim with OpenACC (2026). [DOI](#)
- 2026BTSL: Plasma wakefield dynamics of self-generated electron bunch trains (2026). [DOI](#)
- 2026CGAT: Experimental evidence of production of directional muons from a laser-wakefield accelerator (2026). [DOI](#)
- 2026CGSN: Plasma Confinement State Classification in Fusion Power Plants: Profile Reflectometer and Ensemble Diagnostics (2026). [DOI](#)
- 2026CHDW: Self-adaptive stabilization and quality boost for electron beams from all-optical plasma wakefield accelerators (2026). [DOI](#)
- 2026CJLY: Control of a Uniformly Magnetized Plasma with External Electric Fields (2026). [DOI](#)
- 2026CLY: Semi-Lagrangian SAV method for Vlasov-Maxwell equations (2026). [DOI](#)
- 2026CMSM: Kinetic equilibrium prediction at TCV using RAPTOR and FBT (2026). [DOI](#)
- 2026CSSM: Inertial-Range Suppression and Ponderomotive Density Cavitation in Broadband Sub-Alfvénic Turbulence under Plasma Sheet Boundary Layer Conditions (2026). [DOI](#)
- 2026CZGS: Detecting shearless phase-space transport barriers in global gyrokinetic turbulence simulations with test particle map models (2026). [DOI](#)
- 2026D: GPU Acceleration and Portability of the TRIMEG Code for Gyrokinetic Plasma Simulations Using OpenMP (2026). [DOI](#)
- 2026DFTW: A Deterministic Ionization Algorithm for the OSIRIS Particle-in-Cell Framework (2026). [DOI](#)
- 2026DGNP: Hierarchical Sparse-Grid Particle-in-Cell Method with Locally Adaptive Mesh Refinement (2026). [DOI](#)

- 2026DJUN: OpenEdge: A collaborative, open-source, multi-purpose direct simulation Monte Carlo for plasma simulation in magnetic fusion environments (2026). [DOI](#)
- 2026DM: Electron Acceleration by a Wakefield Excited in Plasma by a Profiled Train of Bunches Under Conditions of Their Greatest Stability and a Plateau on the Wakefield (2026). [DOI](#)
- 2026DMN: High-Order Structure-Preserving SBP Finite Difference Methods for the Vlasov–Maxwell System on Matrix-Free GPUs (2026). [DOI](#)
- 2026ELSD: Stellarator divertor design by optimising coils for surfaces with sharp corners (2026). [DOI](#)
- 2026F: Generalized Similarity Theory for Plasmas (2026). [DOI](#)
- 2026FABC: Efficient Acceleration of High-Quality GeV-Electron Bunches in a Hybrid Laser- and Beam-Driven Plasma Wakefield Accelerator (2026). [DOI](#)
- 2026FBAP: Plasma mixing driven by the collisionless Kelvin–Helmholtz instability: Insights from a fully kinetic simulation and density-based diagnostics (2026). [DOI](#)
- 2026FCSJ: Conservative Velocity Mappings for Discontinuous Galerkin Kinetics (2026). [DOI](#)
- 2026FFCR: Comparison of turbulent electron heat flux model predictions of the H-mode electron pedestal temperature profile across isotope mix and gas fuelling rate scans in JET with the Be/W wall (2026). [DOI](#)
- 2026FHDM: Gridless Quasistatic Model for Efficient Simulation of Plasma-based Accelerators (2026). [DOI](#)
- 2026FHWS: A HIP-Compatible Accelerator Backend for Fourier-Bessel Particle-in-Cell Simulations on CPU/DCU Heterogeneous Clusters (2026). [DOI](#)
- 2026FKO: Nonmodal growth and optimal perturbations in magnetohydrodynamic shear flows (2026). [DOI](#)
- 2026FMCH: A Coherent Structure Transport Model for Scrape-Off Layer Turbulence (2026). [DOI](#)
- 2026FMMM: Code-Verification Techniques for Particle-in-Cell Simulations with Direct Simulation Monte Carlo Collisions (2026). [DOI](#)
- 2026FPKA: Structure Preservation Using Discrete Gradients in the Vlasov-Poisson-Landau System (2026). [DOI](#)
- 2026FPPK: A Flexible and Differentiable Coil Proxy for Stellarator Equilibrium Optimization (2026). [DOI](#)
- 2026FRT: A modified Crank–Nicolson scheme for the Vlasov–Poisson system with a strong external magnetic field (2026). [DOI](#)
- 2026FT: An analytical optimization of plasma density profiles for downramp injection in laser wake-field acceleration (2026). [DOI](#)
- 2026FUSJ: Extending the gyrokinetic code GENE-X to stellarators (2026). [DOI](#)
- 2026GCSB: Runaway electron interactions with whistler waves in tokamak plasmas: energy-dependent transport scaling (2026). [DOI](#)
- 2026GDS: Plasma Wakes Driven by Compton Scattering: Nonlinear Regime and Particle Acceleration (2026). [DOI](#)
- 2026GHDL: Solving the Six-Dimensional Vlasov–Maxwell System with Active Flux and Splitting Methods (2026). [DOI](#)
- 2026GKSA: Design and mechanical analysis of the PRAGYA tokamak vacuum vessel (2026). [DOI](#)
- 2026GNP: Electric current dynamics in the stellarator coil winding surface model (2026). [DOI](#)
- 2026GS: Resonant amplification of zonal flow driven by the negative mass instability in tokamak plasma (2026). [DOI](#)
- 2026GSHJ: Designing A Buildable Optimized Stellarator to Confine Electron-Positron Plasmas (2026). [DOI](#)
- 2026GSJH: Stochastic single-stage stellarator optimization using fixed-boundary equilibria (2026). [DOI](#)
- 2026GSNE: High Field Diamond Magnetometry Towards Tokamak Diagnostics (2026). [DOI](#)
- 2026GVBC: Experimental Demonstration of Beam-Driven Wakefield Acceleration in Laser-Plasma Filament (2026). [DOI](#)
- 2026GXPS: Real-Time Plasma State Prediction via FPGA-Accelerated Quantized Recurrent Probabilistic Neural Networks (2026). [DOI](#)
- 2026HAGB: Laboratory observation of collective beam–plasma instabilities in a relativistic pair jet (2026). [DOI](#)
- 2026HBFH: Towards fully predictive gyrokinetic full- f simulations: validation and triangularity studies in TCV (2026). [DOI](#)
- 2026HBKW: Novel approach to general curvilinear coordinates for plasma fluid applications (2026). [DOI](#)
- 2026HFBH: High-throughput full- f gyrokinetics of the tokamak boundary (2026). [DOI](#)
- 2026HGR: The hot-electron closure of the moment-based gyrokinetic plasma model (2026). [DOI](#)
- 2026HNND: Relativistic magnetohydrodynamic wave excitation by laser pulse in a magnetized plasma (2026). [DOI](#)

- 2026HTSD: Real-time Tomography-based Bayesian Inference from TCV Bolometry Data (2026). [DOI](#)
- 2026HV: Flow-Regulated Suprathermal Particle Acceleration in Weakly Collisional Astrophysical Plasmas (2026). [DOI](#)
- 2026HY: An analytically-solvable, wave-form asymptotic-preserving and energy-conserving time-splitting scheme for Vlasov–Poisson equations in the quasi-neutral regime (2026). [DOI](#)
- 2026HYZ: Compressible Turbulence as a Source of Particle Beams and Ion Bernstein Waves in Collisionless Plasmas (2026). [DOI](#)
- 2026IDR: Mixed Hermite–Legendre spectral method for kinetic plasma simulations (2026). [DOI](#)
- 2026JFHZ: Robustness Optimization for Compact Free-Electron Laser Driven by Laser Wakefield Accelerators (2026). [DOI](#)
- 2026JGS: Magnetic Field Amplification and Particle Acceleration in Weakly Magnetized Trans-relativistic Electron-ion Shocks (2026). [DOI](#)
- 2026JHJ: A conservative discontinuous Galerkin algorithm for particle kinetics on smooth manifolds (2026). [DOI](#)
- 2026JPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
- 2026JKGH: Optimized laser wakefield acceleration: Generating stable, high-energy, monoenergetic electron beams and demonstrating extreme-ultraviolet free-electron lasers (2026). [DOI](#)
- 2026JT: On the evolution of a large-amplitude, weakly-collisional electron plasma wave (2026). [DOI](#)
- 2026K: Mercier–Cotsaftis and Grad–Shafranov equations for anisotropic plasma (2026). [DOI](#)
- 2026KDBK: Surrogate models study for laser-plasma accelerator electron source design through numerical optimisation (2026). [DOI](#)
- 2026KKLM: Physics-Informed Kolmogorov–Arnold Networks for Grad–Shafranov Tokamak Equilibria (2026). [DOI](#)
- 2026KLJH: How Does The Magnetic Gradient Scale Length Influence Complexity of Filamentary Coils in Stellarators? (2026). [DOI](#)
- 2026KLWB: A Hybrid Semi-Lagrangian Flow Mapping Approach for Vlasov Systems: Combining Iterative and Compositional Flow Maps (2026). [DOI](#)
- 2026KNB: Real-time time-dependent density functional theory for high-energy density physics (2026). [DOI](#)
- 2026KNTL: Understanding Carbon Sourcing and Transport Originating from the Helicon Antenna Surfaces During High-Power Helicon Discharge in DIII-D Tokamak (2026). [DOI](#)
- 2026KSG: T3FF: Toroidal 3-dimensional full- f full- k code for isothermal gyrofluid drift-Alfvén turbulence (2026). [DOI](#)
- 2026KSH: Effects of Turbulent Energy Exchange between Electrons and Ions on Global Temperature Profiles (2026). [DOI](#)
- 2026KT: Partially Ionized Plasma Physics and Technological Applications (2026). [DOI](#)
- 2026LBBK: Study of Low-Frequency Core-Edge Coupling in a Tokamak: I. Experimental Observation in KSTAR (2026). [DOI](#)
- 2026LCLK: Discrete unified gas kinetic scheme for multiscale plasma simulations (2026). [DOI](#)
- 2026LDGB: Simulating ultrarelativistic beam-plasma instabilities with a quasistatic particle-in-cell code (2026). [DOI](#)
- 2026LGST: First Electron Acceleration in a Tunable-Velocity Laser Wakefield (2026). [DOI](#)
- 2026LGTT: Probing flying-focus wakefields (2026). [DOI](#)
- 2026LHAH: Asymmetric self-injection of electrons in a laser wakefield accelerator via localized laser redshifting (2026). [DOI](#)
- 2026LHKD: Physics-Informed Drift Diagnosis for Laser-Plasma Accelerator Operations (2026). [DOI](#)
- 2026LHWP: Time-embedded convolutional neural networks for modeling plasma heat transport (2026). [DOI](#)
- 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
- 2026LLBZ: Hybrid Fluid–Kinetic Simulations of Resistive Instabilities in Runaway Electron Beams (2026). [DOI](#)
- 2026LLL: Role of ion acoustic instability in magnetic reconnection (2026). [DOI](#)
- 2026LMHL: TRIMEG-GKX: An electromagnetic gyrokinetic particle code with a piecewise field-aligned finite element method for micro- and macro-instability studies in tokamak core plasmas (2026). [DOI](#)
- 2026LMSH: A High-Order Piecewise Field-Aligned Triangular Finite Element Method for Electromagnetic Gyrokinetic Particle Simulations of Tokamak Plasmas with Open Field Lines (2026). [DOI](#)
- 2026LO: Numerical study of electron acceleration by microwave-driven plasma wakefields in rectangular waveg-

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- 2026LRBG: Conservative Formulation of the Drift-Reduced Fluid Plasma Model (2026). [DOI](#)
 - 2026LRDQ: Observation of stopping power reduction at strong ion-plasma coupling (2026). [DOI](#)
 - 2026LRGW: Long lasting plasma density structures utilizing tailored density profiles (2026). [DOI](#)
 - 2026LSGH: Laser-Plasma Based Seeded Free Electron Laser in the High-Gain Regime (2026). [DOI](#)
 - 2026LTGM: Expansion-Driven Self-Magnetization of High-Energy-Density Plasmas (2026). [DOI](#)
 - 2026LVCN: Theory of beam-driven nonlinear plasma wake and interior waves (2026). [DOI](#)
 - 2026LWY: High-order positivity-preserving and globally divergence-free discontinuous Galerkin methods for ideal magnetohydrodynamics with conditional energy conservation (2026). [DOI](#)
 - 2026LYVZ: Optimization of stellarator configurations combining omnigenity and piecewise omnigenity (2026). [DOI](#)
 - 2026LYXB: Sub-Second Collisionless Gyrokinetic Eigenvalue Solutions via Orbit-Invariant Decomposition (2026). [DOI](#)
 - 2026LYZR: Semi-implicit scheme for multi-species collision operators in Tokamak Plasma Simulations (2026). [DOI](#)
 - 2026LZZ: Near-Wall Pathways of Anomalous Electron Transport in Hall Thrusters Revealed by 3D PIC Simulations (2026). [DOI](#)
 - 2026MC: Extended phase-space symplectic integration for electron dynamics (2026). [DOI](#)
 - 2026MCCC: Non-dimensional confinement scaling in similar negative triangularity plasmas on the DIII-D and TCV tokamaks (2026). [DOI](#)
 - 2026MEVD: Superluminal Wave Activation at Relativistic Magnetized Shocks (2026). [DOI](#)
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 - 2026MHET: Experimental Evidence for Increased Particle Fluxes Due to a Change in Transport at the Separatrix near Density Limits on Alcator C-Mod (2026). [DOI](#)
 - 2026MHL: High-order exponential solver method for particle-in-cell simulations (2026). [DOI](#)
 - 2026MHLH: Batch Bayesian Optimization of Attosecond Betatron Pulses from Laser Wakefield Acceleration (2026). [DOI](#)
 - 2026MHSE: Empirical impact of near-separatrix plasma and neutral transport on the pedestal in the transition between EDA and ELMy H-modes on Alcator C-Mod (2026). [DOI](#)
 - 2026MK: Gravitational instability in partially ionized plasmas: A two-fluid approach (2026). [DOI](#)
 - 2026MKC: Proper Treatment of Energy and Momentum in Time-Modulated Plasmas (2026). [DOI](#)
 - 2026MLL: Suppressed Stiffness of energetic particle transport due to thermal plasma nonlinearity in tokamak plasmas (2026). [DOI](#)
 - 2026MMGK: Laser field reconstruction for the modeling of laser-plasma interaction in cylindrical geometry (2026). [DOI](#)
 - 2026MML: Parametric instabilities of the inhomogeneous near-SOL tokamak plasma, driven by the coupled effect of the high-harmonic fast wave and of the ion and electron temperature gradients, and anomalous heating of the near-SOL ions (2026). [DOI](#)
 - 2026MMWS: Laser wakefield acceleration in a capillary gas cell to produce high-quality GeV-scale electron beams (2026). [DOI](#)
 - 2026MRD: Comprehensive full- f drift-kinetic and δf gyrokinetic simulations of a linear plasma device based on the gyro-moment approach (2026). [DOI](#)
 - 2026MRHH: Core-edge integrated modeling of ARC: on the effect of impurity transport and detachment conditions (2026). [DOI](#)
 - 2026MSCM: Evidence of Nonlinear Coupling in the Edge Harmonic Oscillation Sustaining Quiescent High Confinement in a Tokamak Plasma (2026). [DOI](#)
 - 2026MZL: Effects of neoclassical toroidal viscosity on plasma flow evolution in the presence of resonant magnetic perturbation in a tokamak (2026). [DOI](#)
 - 2026NBBP: Nonlinear generation of global zonal structures in gyrokinetic simulations of TCV and ASDEX Upgrade magnetic configurations (2026). [DOI](#)
 - 2026NN: Reduced One-Fluid GENERIC Closure from Relativistic Moment Kinetics (2026). [DOI](#)
 - 2026NP: Theory of zonal flow growth and propagation in toroidal geometry (2026). [DOI](#)
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 - 2026OC: Hamiltonian Hydrodynamic Reductions of One-Dimensional Vlasov Equations (2026). [DOI](#)

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- 2026ODDE: STORM: Scrape-off layer turbulence in tokamak fusion reactors (2026). [DOI](#)
- 2026OMCL: Predicting core transport in ITER baseline discharges with neon injections (2026). [DOI](#)
- 2026OTWH: Sparse and low-rank kinetic distribution estimation (2026). [DOI](#)
- 2026OUZJ: Coupling Fluid Neutrals to Gyrokinetic Plasma Dynamics for Edge and SOL Turbulence Simulations (2026). [DOI](#)
- 2026PBLL: Coordinate-invariant flux-surface Fourier analysis in tokamaks (2026). [DOI](#)
- 2026PDD: Ion-Acoustic Shock Waves in Magnetized Plasma With Relativistically Degenerate Electrons and Positrons (2026). [DOI](#)
- 2026PJCD: Deflation Techniques for Stellarator Equilibrium and Optimization (2026). [DOI](#)
- 2026PLBE: Pedestal Origin and Extrapolation of High-Density Small Edge-Localised-Modes Peak Parallel Energy Fluence in ITER and SPARC (2026). [DOI](#)
- 2026PMAA: Dephasingless laser wakefield acceleration in a plasma waveguide (2026). [DOI](#)
- 2026PSHZ: On the mechanism of pedestal relaxation events: Insights gained by turbulence simulations with GRILLIX (2026). [DOI](#)
- 2026PSKA: Nonlinear Transport in Microwave-Generated Plasma (2026). [DOI](#)
- 2026QMLC: Optimization of laser-driven proton acceleration in a near-critical-density plasma (2026). [DOI](#)
- 2026R: Ions-electrons-states for the two-component Vlasov–Poisson equation (2026). [DOI](#)
- 2026R3: Bivariate incomplete-Bessel kernels for the first nonlinear Vlasov–Maxwell response (2026). [DOI](#)
- 2026RECB: Parallelized real-time physics codes for plasma control on DIII-D (2026). [DOI](#)
- 2026RFR: Variation of instability characteristics and resulting electron transport under external modulation in $E \times B$ plasmas (2026). [DOI](#)
- 2026RJOA: TokaMark: A Comprehensive Benchmark for MAST Tokamak Plasma Models (2026). [DOI](#)
- 2026RPRT: Thermal modeling of runaway electron induced damage in the SPARC tokamak (2026). [DOI](#)
- 2026RS2: Science Of Nuclear Fusion: Insights and Ideas (2026). [DOI](#)
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- 2026S2: Hybrid PIC-fluid model for numerical simulation of laser-plasma interaction (2026). [DOI](#)
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- 2026SBBM: First global gyrokinetic profile predictions of ITER burning plasma (2026). [DOI](#)
- 2026SCSS: Weibel Instability in Collisionless Plasmas Across Astrophysical and Laboratory Shocks (2026). [DOI](#)
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- 2026SGSH: A Predictive Formula for the H-Mode Electron Separatrix Density: Bridging Regression and Physics-Based Models across C-Mod, AUG and JET Tokamaks (2026). [DOI](#)
- 2026SHJH: Constructing field-aligned coordinate systems for gyrokinetic simulations of tokamaks in X-point geometries (2026). [DOI](#)
- 2026SJKY: Multi-species Rosenbluth Fokker-Planck collision operator for discontinuous Galerkin method (2026). [DOI](#)
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- 2026SWLA: Stable 50 MeV Beams from a 100 Hz Laser-Wakefield Accelerator Driven by an OPCPA Laser (2026). [DOI](#)
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- 2026TBA: A novel conditional formulation of the Vlasov–Ampère equations: a conservative, positivity, asymptotic and Gauss law preserving scheme (2026). [DOI](#)
- 2026TCLH: Coupling of an External Circuit Model With a Plasma Fluid Model With a Time-Step Size Refinement (TSSR) Algorithm (2026). [DOI](#)

- 2026TDAZ: Drift-reduced fluid modeling of rapidly rotating plasmas (2026). [DOI](#)
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- 2026TP: Neoclassical transport and profile prediction in transport barriers (2026). [DOI](#)
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- 2026VCG: Control of the bootstrap current in approximately quasi-axisymmetric magnetic fields (2026). [DOI](#)
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- 2026WBBC: A tutorial on inversion-based shape control with design application to NSTX-U (2026). [DOI](#)
- 2026WCMT: Integrating High Performance In-Memory Data Streaming and In-Situ Visualization in Hybrid MPI+OpenMP PIC MC Simulations Towards Exascale (2026). [DOI](#)
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- 2026YZLY: TokaGLINT: A Scalable GPU-Tailored Implicit Solver for Full 3D Tokamak Electromagnetic Simulations (2026). [DOI](#)
- 2026Z: Particle pinch in global tokamak edge simulations (2026). [DOI](#)
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- 2026ZCXY: Broadening of scrape-off layer width by turbulence spreading in the heavy hydrogen isotope plasmas (2026). [DOI](#)
- 2026ZCXZ: Three Dimensional Multiphysics Modelling of Helicon Wave Heating and Antenna Plasma Coupling for Boundary Density Control in Toroidal Fusion Plasmas (2026). [DOI](#)
- 2026ZD: The Properties of Plasma Sheath Containing the Primary Electrons with a Cairns Distribution (2026). [DOI](#)
- 2026ZHKB: Flux pumping and bifurcated relaxations of helical core in 3D magnetohydrodynamic modelling of ASDEX Upgrade plasmas (2026). [DOI](#)

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- 2026ZJZU: Fast Turbulence Phase Transition in a Flux-Driven Global Edge-SOL Simulation of a Tokamak Plasma (2026). [DOI](#)
- 2026ZMFA: Amplitude Modulation and Instability of Ion-Acoustic Solitons in Collisionless Plasmas With Tsallis–Gurevich Distributed Electrons (2026). [DOI](#)
- 2026ZPSR: Variability of MHD instabilities in benign termination of high-current runaway electron beams in the JET and DIII-D tokamaks (2026). [DOI](#)
- 2026ZRCE: Infrared Thermography in the Tokamak à Configuration Variable (2026). [DOI](#)
- 2026ZRZS: Triggers for Plasma Detachment Bifurcation in the Edge Divertor Region of Tokamaks (2026). [DOI](#)
- 2026ZUM: Loading Non-Maxwellian Velocity Distributions in Particle Simulations (2026). [DOI](#)
- 2026ZXLM: VEQ: a fast parametric Grad–Shafranov solver for fixed-boundary tokamak equilibria with flexible source profiles (2026). [DOI](#)
- 2026ZZLW: AlgoPlasma: Open Algorithms for Plasma Modeling (2026). [DOI](#)
- 2026ZZY: Three-dimensional particle-in-cell simulations on Hall thruster electron drift instability: influence of magnetic field on electron transport (2026). [DOI](#)
- 2025AA: Thermalization times and scaling laws for alpha particles and hydrogen isotopes in Tokamak plasmas (2025). [DOI](#)
- 2025ACLB: In situ axion generation and detection in laser-plasma wakefield interaction (2025). [DOI](#)
- 2025ALSE: Drive beam depletion with multi-Joule energy transfer in a plasma wakefield accelerator (2025). [DOI](#)
- 2025B: Electron Inertia and Magnetic Reconnection (2025). [DOI](#)
- 2025B2: Magnetic Field Line Chaos, Cantori, and Turnstiles in Toroidal Plasmas (2025). [DOI](#)
- 2025BBEG: Photon Accelerator in Magnetized Electron-Ion Plasma (2025). [DOI](#)
- 2025BOMG: π -PIC: a framework for modular particle-in-cell developments and simulations (2025). [DOI](#)
- 2025BPQM: Photon Phase-Space Dynamics in a Plasma Wakefield Accelerator (2025). [DOI](#)
- 2025BSBB: Study of the impact of fast ions on core turbulence at rational surfaces via global gyrokinetic simulations (2025). [DOI](#)
- 2025BSG: Operator Formalism for Laser-Plasma Wakefield Acceleration (2025). [DOI](#)
- 2025BSS: Phase-Space Shaping in Wakefield Accelerators due to Betatron Cooling (2025). [DOI](#)
- 2025CHHM: Measurement of Ion Acceleration and Diffusion in a Laser-Driven Magnetized Plasma (2025). [DOI](#)
- 2025CJBL: Case study of a differentiable heterogeneous multiphysics solver for a nuclear fusion application (2025). [DOI](#)
- 2025CKKS: Asymptotic-preserving conservative semi-Lagrangian discontinuous Galerkin schemes for the Vlasov-Poisson system in the quasi-neutral limit (2025). [DOI](#)
- 2025CLLZ: A high-order energy-conserving semi-Lagrangian discontinuous Galerkin method for the Vlasov-Ampère system (2025). [DOI](#)
- 2025COWC: Laser Amplification in e^- - μ^- -ion Plasmas (2025). [DOI](#)
- 2025DBML: Bayesian Optimization of Laser-Wakefield Acceleration via Spectral Pulse Shaping (2025). [DOI](#)
- 2025DDGG: Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror (2025). [DOI](#)
- 2025DHYD: Microturbulence Suppression by Alfvén Eigenmodes in the DIII-D Tokamak (2025). [DOI](#)
- 2025EGTL: Long Living Hot and Dense Plasma from Relativistic Laser-Nanowire Array Interaction (2025). [DOI](#)
- 2025EHBG: Combined Influence of Rotation and Scrape-Off Layer Drifts on Recycling Asymmetries in Tokamak Plasmas (2025). [DOI](#)
- 2025FKF: Unveiling a New β Scaling of the Tearing Instability in Weakly Collisional Plasmas (2025). [DOI](#)
- 2025GAST: MHD waves with mixed properties/Alfvén waves with pressure variations: a review (2025). [DOI](#)
- 2025GKST: High brightness multi-MeV photon source driven by a petawatt-scale laser wakefield accelerator (2025). [DOI](#)
- 2025GPSH: Computing Flux-Surface Shapes in Tokamaks and Stellarators (2025). [DOI](#)
- 2025HBCC: Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module (2025). [DOI](#)
- 2025HBS: Magnetohydrodynamic-guiding-center-particle-in-cell Method for Multiscale Plasma Kinetic Simulations (2025). [DOI](#)
- 2025HGR: The Hot-Electron Closure of the Moment-Based Gyrokinetic Plasma Model (2025). [DOI](#)
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- on Staggered Grid (2025). [DOI](#)
- 2025HMST: Structure-preserving finite-element approximations of the magnetic Euler-Poisson equations (2025). [DOI](#)
- 2025IMCR: Turbulence and Magnetic Reconnection in Relativistic Multispecies Plasmas (2025). [DOI](#)
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- 2025JHJ: A moment-conserving discontinuous Galerkin representation of the relativistic Maxwellian distribution (2025). [DOI](#)
- 2025L: Symplectic particle-in-cell methods for hybrid plasma models with Boltzmann electrons and space-charge effects (2025). [DOI](#)
- 2025L2: Magnetised turbulent plasmas as high-energy particle accelerators (2025). [DOI](#)
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- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)
- 2025LWZ: From filamentation to stratification: Instability dynamics in scissors-shaped relativistic beam-plasma system (2025). [DOI](#)
- 2025LZGY: Studying the properties of reconnection-driven turbulence (2025). [DOI](#)
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- 2025MATV: Optimization of the characteristics of a relativistic electron beam based on laser wake-field acceleration using a non-symmetric gas target profile (2025). [DOI](#)
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- 2025PGHS: GyroSwin: 5D Surrogates for Gyrokinetic Plasma Turbulence Simulations (2025). [DOI](#)
- 2025PHR: A New Approach to Compute Linear Landau Damping (2025). [DOI](#)
- 2025PSWR: Isotope Production in Fusion Systems (2025). [DOI](#)
- 2025RMBC: UK White Paper on Magnetic Reconnection (2025). [DOI](#)
- 2025S2: The Numerics of VMEC++ (2025). [DOI](#)
- 2025SA: Machine Learning-Integrated Hybrid Fluid-Kinetic Framework for Quantum Electrodynamical Laser Plasma Simulations (2025). [DOI](#)
- 2025SKDF: Overview of the Helios Design: A Practical Planar Coil Stellarator Fusion Power Plant (2025). [DOI](#)
- 2025SPH: Linear stability studies for a quasi-axisymmetric equilibrium including plasma flow, drift and viscosity effects (2025). [DOI](#)
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- 2025SRKH: Gyrokinetic Simulations of a Low Recycling Scrape-off Layer without a Lithium Target (2025). [DOI](#)
- 2025SVLN: Toward controlling electron beam charge with nanoparticle-assisted laser wakefield accelerators (2025). [DOI](#)
- 2025TBAB: T_i/T_e Dependence of Core Turbulence and Transport in DIII-D QH-Mode Plasmas (2025). [DOI](#)
- 2025TFSF: Reconstructions of electron-temperature profiles from EUROfusion pedestal database using turbulence models and machine learning (2025). [DOI](#)
- 2025UN: Nonlinear Gaussian process tomography with imposed non-negativity constraints on physical quantities for plasma diagnostics (2025). [DOI](#)
- 2025VGRB: Roadmap for warm dense matter physics (2025). [DOI](#)
- 2025WAFG: Particle-in-cell simulations of burning inertial confinement fusion capsule implosions (2025). [DOI](#)
- 2025WFTJ: Influence of ion motion in a resonantly driven wakefield accelerator (2025). [DOI](#)
- 2025WWTG: Comparison of filament properties in real-size GBS simulations and experiments of TCV-X21 (2025). [DOI](#)
- 2025WZLW: Achieving High Efficiency And Enhanced Beam Quality In Laser Wakefield Acceleration (2025). [DOI](#)
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- 2025ZCSV: Numerical investigation of kinetic instabilities in BGK equilibria under collisional effects (2025). [DOI](#)
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- 2025ZSRR: Divertor Detachment Characterization in Negative Triangularity Discharges in DIII-D via 2D Edge-Plasma Transport Modeling (2025). [DOI](#)
- 2024G: A coordinate-free expression of plasma theory (2024). [DOI](#)
- 2024HM: Rugged Magneto-Hydrodynamic Invariants in Weakly Collisional Plasma Turbulence: Two-Dimensional Hybrid Simulation Results (2024). [DOI](#)
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- 2021EMCW: Impact of magnetic islands on plasma flow and turbulence in W7-X (2021). [DOI](#)
- 2020FBMH: Conservative discontinuous Galerkin scheme of a gyro-averaged Dougherty collision operator (2020). [DOI](#)
- 2020MHF: Electromagnetic full- f gyrokinetics in the tokamak edge with discontinuous Galerkin methods (2020). [DOI](#)
- 2018MDL: Laguerre–Hermite pseudo-spectral velocity formulation of gyrokinetics (2018). [DOI](#)
- 2017NNSW: Isotope Effects on Trapped-Electron-Mode Driven Turbulence and Zonal Flows in Helical and Tokamak Plasmas (2017). [DOI](#)
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Accelerator Physics

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- 2026BLBS: Longitudinally resolved measurement of energy-transfer efficiency in a plasma-wakefield accelerator (2026). [DOI](#)
- 2026BMCC: Beam transverse dynamics in laser-plasma accelerators (2026). [DOI](#)
- 2026BTSL: Plasma wakefield dynamics of self-generated electron bunch trains (2026). [DOI](#)
- 2026CGAT: Experimental evidence of production of directional muons from a laser-wakefield accelerator (2026). [DOI](#)
- 2026CGG: A Scattered-Field Formulation for Coupled Geometric Wakefield and Space Charge Field Simulations in Particle Accelerators (2026). [DOI](#)
- 2026CHDW: Self-adaptive stabilization and quality boost for electron beams from all-optical plasma wakefield accelerators (2026). [DOI](#)
- 2026DM: Electron Acceleration by a Wakefield Excited in Plasma by a Profiled Train of Bunches Under Conditions of Their Greatest Stability and a Plateau on the Wakefield (2026). [DOI](#)
- 2026FABC: Efficient Acceleration of High-Quality GeV-Electron Bunches in a Hybrid Laser- and Beam-Driven Plasma Wakefield Accelerator (2026). [DOI](#)
- 2026FHDM: Gridless Quasistatic Model for Efficient Simulation of Plasma-based Accelerators (2026). [DOI](#)
- 2026FHWS: A HIP-Compatible Accelerator Backend for Fourier-Bessel Particle-in-Cell Simulations on CPU/DCU Heterogeneous Clusters (2026). [DOI](#)
- 2026FT: An analytical optimization of plasma density profiles for downramp injection in laser wake-field acceleration (2026). [DOI](#)
- 2026GDS: Plasma Wakes Driven by Compton Scattering: Nonlinear Regime and Particle Acceleration (2026). [DOI](#)

- 2026GVBC: Experimental Demonstration of Beam-Driven Wakefield Acceleration in Laser-Plasma Filament (2026). [DOI](#)
- 2026HAGB: Laboratory observation of collective beam–plasma instabilities in a relativistic pair jet (2026). [DOI](#)
- 2026JFHZ: Robustness Optimization for Compact Free-Electron Laser Driven by Laser Wakefield Accelerators (2026). [DOI](#)
- 2026JKGH: Optimized laser wakefield acceleration: Generating stable, high-energy, monoenergetic electron beams and demonstrating extreme-ultraviolet free-electron lasers (2026). [DOI](#)
- 2026KDBK: Surrogate models study for laser-plasma accelerator electron source design through numerical optimisation (2026). [DOI](#)
- 2026LDGB: Simulating ultrarelativistic beam-plasma instabilities with a quasistatic particle-in-cell code (2026). [DOI](#)
- 2026LGST: First Electron Acceleration in a Tunable-Velocity Laser Wakefield (2026). [DOI](#)
- 2026LGTT: Probing flying-focus wakefields (2026). [DOI](#)
- 2026LHAH: Asymmetric self-injection of electrons in a laser wakefield accelerator via localized laser redshifting (2026). [DOI](#)
- 2026LHKD: Physics-Informed Drift Diagnosis for Laser-Plasma Accelerator Operations (2026). [DOI](#)
- 2026LO: Numerical study of electron acceleration by microwave-driven plasma wakefields in rectangular waveguides (2026). [DOI](#)
- 2026LSGH: Laser-Plasma Based Seeded Free Electron Laser in the High-Gain Regime (2026). [DOI](#)
- 2026LVCN: Theory of beam-driven nonlinear plasma wake and interior waves (2026). [DOI](#)
- 2026MHL: High-order exponential solver method for particle-in-cell simulations (2026). [DOI](#)
- 2026MHLH: Batch Bayesian Optimization of Attosecond Betatron Pulses from Laser Wakefield Acceleration (2026). [DOI](#)
- 2026MMGK: Laser field reconstruction for the modeling of laser-plasma interaction in cylindrical geometry (2026). [DOI](#)
- 2026MMWS: Laser wakefield acceleration in a capillary gas cell to produce high-quality GeV-scale electron beams (2026). [DOI](#)
- 2026PMAA: Dephasingless laser wakefield acceleration in a plasma waveguide (2026). [DOI](#)
- 2026QMLC: Optimization of laser-driven proton acceleration in a near-critical-density plasma (2026). [DOI](#)
- 2026SWLA: Stable 50 MeV Beams from a 100 Hz Laser-Wakefield Accelerator Driven by an OPCPA Laser (2026). [DOI](#)
- 2026WXMx: Nanometer-scale pre-bunched electron beams generated from all-optical plasma-based acceleration (2026). [DOI](#)
- 2025ACLB: In situ axion generation and detection in laser-plasma wakefield interaction (2025). [DOI](#)
- 2025ALSE: Drive beam depletion with multi-Joule energy transfer in a plasma wakefield accelerator (2025). [DOI](#)
- 2025BBEG: Photon Accelerator in Magnetized Electron-Ion Plasma (2025). [DOI](#)
- 2025BPQM: Photon Phase-Space Dynamics in a Plasma Wakefield Accelerator (2025). [DOI](#)
- 2025BSG: Operator Formalism for Laser-Plasma Wakefield Acceleration (2025). [DOI](#)
- 2025BSS: Phase-Space Shaping in Wakefield Accelerators due to Betatron Cooling (2025). [DOI](#)
- 2025DBML: Bayesian Optimization of Laser-Wakefield Acceleration via Spectral Pulse Shaping (2025). [DOI](#)
- 2025GKST: High brightness multi-MeV photon source driven by a petawatt-scale laser wakefield accelerator (2025). [DOI](#)
- 2025LCDG: Beam-driven plasma-wakefield acceleration (2025). [DOI](#)
- 2025MATV: Optimization of the characteristics of a relativistic electron beam based on laser wake-field acceleration using a non-symmetric gas target profile (2025). [DOI](#)
- 2025SVLN: Toward controlling electron beam charge with nanoparticle-assisted laser wakefield accelerators (2025). [DOI](#)
- 2025WFTJ: Influence of ion motion in a resonantly driven wakefield accelerator (2025). [DOI](#)
- 2025WZLW: Achieving High Efficiency And Enhanced Beam Quality In Laser Wakefield Acceleration (2025). [DOI](#)

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- 2026ABY: Helical core formation and MHD stability in ITER-scale plasmas with fusion-born alpha particles (2026). [DOI](#)
- 2026ADGM: Gyrokinetic simulation of the effect of transient fueling on plasma turbulence in ADITYA-U tokamak (2026). [DOI](#)
- 2026B: Anderson Localization of Ion-Temperature-Gradient Modes in Aperiodic Stellarators (2026). [DOI](#)
- 2026B3: Internal transport barriers in fluids and plasmas (2026). [DOI](#)
- 2026B5: Pfirsch-Schlüter Current (2026). [DOI](#)
- 2026B6: The toroidal flux and separatrix effects in tokamaks (2026). [DOI](#)
- 2026BKT: Fully kinetic 3-D simulations of high-beta plasma in a mirror trap (2026). [DOI](#)
- 2026BLAO: TokaMind: A Multi-Modal Transformer Foundation Model for Tokamak Plasma Dynamics (2026). [DOI](#)
- 2026BLGA: Study of Low-Frequency Core-Edge Coupling in a Tokamak: II. Spatial Channeling & Focusing In Antenna-Driven MHD (2026). [DOI](#)
- 2026BLPP: Tearing Stability Prediction Combining Toroidal Calculations With a Two-Fluid Slab Layer (2026). [DOI](#)
- 2026CGSN: Plasma Confinement State Classification in Fusion Power Plants: Profile Reflectometer and Ensemble Diagnostics (2026). [DOI](#)
- 2026CMSM: Kinetic equilibrium prediction at TCV using RAPTOR and FBT (2026). [DOI](#)
- 2026CZGS: Detecting shearless phase-space transport barriers in global gyrokinetic turbulence simulations with test particle map models (2026). [DOI](#)
- 2026DJUN: OpenEdge: A collaborative, open-source, multi-purpose direct simulation Monte Carlo for plasma simulation in magnetic fusion environments (2026). [DOI](#)
- 2026ELSD: Stellarator divertor design by optimising coils for surfaces with sharp corners (2026). [DOI](#)
- 2026FFCR: Comparison of turbulent electron heat flux model predictions of the H-mode electron pedestal temperature profile across isotope mix and gas fuelling rate scans in JET with the Be/W wall (2026). [DOI](#)
- 2026FMCH: A Coherent Structure Transport Model for Scrape-Off Layer Turbulence (2026). [DOI](#)
- 2026FPBT: Electromagnetic Analysis of ITER Electron Cyclotron Emission Model (2026). [DOI](#)
- 2026FPPK: A Flexible and Differentiable Coil Proxy for Stellarator Equilibrium Optimization (2026). [DOI](#)
- 2026FUSJ: Extending the gyrokinetic code GENE-X to stellarators (2026). [DOI](#)
- 2026GCSB: Runaway electron interactions with whistler waves in tokamak plasmas: energy-dependent transport scaling (2026). [DOI](#)
- 2026GKSA: Design and mechanical analysis of the PRAGYA tokamak vacuum vessel (2026). [DOI](#)
- 2026GNP: Electric current dynamics in the stellarator coil winding surface model (2026). [DOI](#)
- 2026GS: Resonant amplification of zonal flow driven by the negative mass instability in tokamak plasma (2026). [DOI](#)
- 2026GSHJ: Designing A Buildable Optimized Stellarator to Confine Electron-Positron Plasmas (2026). [DOI](#)
- 2026GSJH: Stochastic single-stage stellarator optimization using fixed-boundary equilibria (2026). [DOI](#)
- 2026GSNE: High Field Diamond Magnetometry Towards Tokamak Diagnostics (2026). [DOI](#)
- 2026GXPS: Real-Time Plasma State Prediction via FPGA-Accelerated Quantized Recurrent Probabilistic Neural Networks (2026). [DOI](#)
- 2026HBFH: Towards fully predictive gyrokinetic full- f simulations: validation and triangularity studies in TCV (2026). [DOI](#)
- 2026HFBH: High-throughput full- f gyrokinetics of the tokamak boundary (2026). [DOI](#)
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- 2026KLJH: How Does The Magnetic Gradient Scale Length Influence Complexity of Filamentary Coils in Stellarators? (2026). [DOI](#)
- 2026KNL: Understanding Carbon Sourcing and Transport Originating from the Helicon Antenna Surfaces During High-Power Helicon Discharge in DIII-D Tokamak (2026). [DOI](#)
- 2026KSG: T3FF: Toroidal 3-dimensional full- f full- k code for isothermal gyrofluid drift-Alfvén turbulence

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- 2026KSH: Effects of Turbulent Energy Exchange between Electrons and Ions on Global Temperature Profiles (2026). [DOI](#)
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 - 2026LCHK: JOREK simulations of the X-point radiator formation and its movement in ASDEX Upgrade (2026). [DOI](#)
 - 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
 - 2026LLBZ: Hybrid Fluid–Kinetic Simulations of Resistive Instabilities in Runaway Electron Beams (2026). [DOI](#)
 - 2026LMHL: TRIMEG-GKX: An electromagnetic gyrokinetic particle code with a piecewise field-aligned finite element method for micro- and macro-instability studies in tokamak core plasmas (2026). [DOI](#)
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 - 2026LRDQ: Observation of stopping power reduction at strong ion-plasma coupling (2026). [DOI](#)
 - 2026LTGM: Expansion-Driven Self-Magnetization of High-Energy-Density Plasmas (2026). [DOI](#)
 - 2026LYVZ: Optimization of stellarator configurations combining omnigenity and piecewise omnigenity (2026). [DOI](#)
 - 2026LYZR: Semi-implicit scheme for multi-species collision operators in Tokamak Plasma Simulations (2026). [DOI](#)
 - 2026MCCC: Non-dimensional confinement scaling in similar negative triangularity plasmas on the DIII-D and TCV tokamaks (2026). [DOI](#)
 - 2026MHET: Experimental Evidence for Increased Particle Fluxes Due to a Change in Transport at the Separatrix near Density Limits on Alcator C-Mod (2026). [DOI](#)
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 - 2026MML: Parametric instabilities of the inhomogeneous near-SOL tokamak plasma, driven by the coupled effect of the high-harmonic fast wave and of the ion and electron temperature gradients, and anomalous heating of the near-SOL ions (2026). [DOI](#)
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 - 2026NHGN: PanoMHD: Multimodal Modelling of Plasma Dynamics towards Tokamak Control (2026). [DOI](#)
 - 2026NP: Theory of zonal flow growth and propagation in toroidal geometry (2026). [DOI](#)
 - 2026NPBM: Saturation of Magnetized Plasma Turbulence by Propagating Zonal Flows (2026). [DOI](#)
 - 2026ODDE: STORM: Scrape-off layer turbulence in tokamak fusion reactors (2026). [DOI](#)
 - 2026OMCL: Predicting core transport in ITER baseline discharges with neon injections (2026). [DOI](#)
 - 2026OUZJ: Coupling Fluid Neutrals to Gyrokinetic Plasma Dynamics for Edge and SOL Turbulence Simulations (2026). [DOI](#)
 - 2026PBLL: Coordinate-invariant flux-surface Fourier analysis in tokamaks (2026). [DOI](#)
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 - 2026PSHZ: On the mechanism of pedestal relaxation events: Insights gained by turbulence simulations with GRILLIX (2026). [DOI](#)
 - 2026RECB: Parallelized real-time physics codes for plasma control on DIII-D (2026). [DOI](#)
 - 2026RJOA: TokaMark: A Comprehensive Benchmark for MAST Tokamak Plasma Models (2026). [DOI](#)

- 2026RPRT: Thermal modeling of runaway electron induced damage in the SPARC tokamak (2026). [DOI](#)
- 2026RS2: Science Of Nuclear Fusion: Insights and Ideas (2026). [DOI](#)
- 2026RXPS: Critical gradient optimisation for quasi-isodynamic stellarators (2026). [DOI](#)
- 2026SBBM: First global gyrokinetic profile predictions of ITER burning plasma (2026). [DOI](#)
- 2026SGSH: A Predictive Formula for the H-Mode Electron Separatrix Density: Bridging Regression and Physics-Based Models across C-Mod, AUG and JET Tokamaks (2026). [DOI](#)
- 2026SHJH: Constructing field-aligned coordinate systems for gyrokinetic simulations of tokamaks in X-point geometries (2026). [DOI](#)
- 2026SJKY: Multi-species Rosenbluth Fokker-Planck collision operator for discontinuous Galerkin method (2026). [DOI](#)
- 2026SJNB: Compact quasisymmetric stellarators, a near axisymmetric theory (2026). [DOI](#)
- 2026SNMD: Exploring the fusion power plant design space: comparative analysis of positive and negative triangularity tokamaks through optimization (2026). [DOI](#)
- 2026TDAZ: Drift-reduced fluid modeling of rapidly rotating plasmas (2026). [DOI](#)
- 2026TP: Neoclassical transport and profile prediction in transport barriers (2026). [DOI](#)
- 2026TSRC: A deterministic verification strategy for multidimensional gyrokinetic particle-in-cell algorithms using the method of manufactured solutions (2026). [DOI](#)
- 2026VANV: Runaway Electron Generation in ITER Mitigated Disruptions with Improved Physics Models (2026). [DOI](#)
- 2026VBFP: Stellarator Island Divertor Shape Optimization for Reduced Peak Heat Fluxes (2026). [DOI](#)
- 2026VCG: Control of the bootstrap current in approximately quasi-axisymmetric magnetic fields (2026). [DOI](#)
- 2026VS: On the timescales of controlled termination of tokamak plasmas (2026). [DOI](#)
- 2026W: Electric Field Induced by Radial Redistribution of the Energetic Ion Pressure in a Fusion Plasma (2026). [DOI](#)
- 2026WBBC: A tutorial on inversion-based shape control with design application to NSTX-U (2026). [DOI](#)
- 2026WGL: Numerical Investigation of Plasma Vertical Growth Rate in Vertical Displacement Events on EAST (2026). [DOI](#)
- 2026WJPA: Accurate calculation of the gradients of the equilibrium poloidal flux in tokamaks (2026). [DOI](#)
- 2026WLST: An Efficient Solver for Finite Element-based Constrained Transport in 3D Magnetohydrodynamics Applied to Magnetic Confinement Fusion (2026). [DOI](#)
- 2026WLWC: Revisiting p-¹¹B Fusion: Updated Cross-sections, Reactivity, and Energy Balance (2026). [DOI](#)
- 2026WW: Effects of Geometric Curvature and Weak Magnetic Shear on the Ion-Temperature-Gradient Instability Near the Magnetic Axis in a Tokamak (2026). [DOI](#)
- 2026WWQW: 2DES: a two-dimensional Fourier-space gyrokinetic eigenvalue code for the ion-temperature-gradient modes in tokamaks (2026). [DOI](#)
- 2026XDZW: Turbulence-Driven Edge-Localized-Mode-Free High-Confinement Mode with Divertor Detachment in a Metal-Wall Tokamak (2026). [DOI](#)
- 2026XLDM: Development of a Reduced Multi-Fluid Equilibrium Model and Its Application to Proton-Boron Spherical Tokamaks (2026). [DOI](#)
- 2026YG: Feasibility of Negative Triangularity Equilibria in the SPARC Tokamak (2026). [DOI](#)
- 2026YT: Effect of eigenmode destabilisation by energetic ions on plasma rotation (2026). [DOI](#)
- 2026YZLY: TokaGLINT: A Scalable GPU-Tailored Implicit Solver for Full 3D Tokamak Electromagnetic Simulations (2026). [DOI](#)
- 2026Z: Particle pinch in global tokamak edge simulations (2026). [DOI](#)
- 2026ZCLB: Global linear drift-wave eigenmode structures on flux surfaces in stellarators: ion temperature gradient mode (2026). [DOI](#)
- 2026ZCXY: Broadening of scrape-off layer width by turbulence spreading in the heavy hydrogen isotope plasmas (2026). [DOI](#)
- 2026ZCXZ: Three Dimensional Multiphysics Modelling of Helicon Wave Heating and Antenna Plasma Coupling for Boundary Density Control in Toroidal Fusion Plasmas (2026). [DOI](#)
- 2026ZHKB: Flux pumping and bifurcated relaxations of helical core in 3D magnetohydrodynamic modelling of ASDEX Upgrade plasmas (2026). [DOI](#)
- 2026ZJZU: Fast Turbulence Phase Transition in a Flux-Driven Global Edge-SOL Simulation of a Tokamak Plasma (2026). [DOI](#)

- 2026ZPSR: Variability of MHD instabilities in benign termination of high-current runaway electron beams in the JET and DIII-D tokamaks (2026). [DOI](#)
- 2026ZRCE: Infrared Thermography in the Tokamak à Configuration Variable (2026). [DOI](#)
- 2026ZRZS: Triggers for Plasma Detachment Bifurcation in the Edge Divertor Region of Tokamaks (2026). [DOI](#)
- 2026ZXLM: VEQ: a fast parametric Grad–Shafranov solver for fixed-boundary tokamak equilibria with flexible source profiles (2026). [DOI](#)
- 2025AA: Thermalization times and scaling laws for alpha particles and hydrogen isotopes in Tokamak plasmas (2025). [DOI](#)
- 2025B2: Magnetic Field Line Chaos, Cantori, and Turnstiles in Toroidal Plasmas (2025). [DOI](#)
- 2025BSBB: Study of the impact of fast ions on core turbulence at rational surfaces via global gyrokinetic simulations (2025). [DOI](#)
- 2025DDGG: Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror (2025). [DOI](#)
- 2025DHYD: Microturbulence Suppression by Alfvén Eigenmodes in the DIII-D Tokamak (2025). [DOI](#)
- 2025EHBG: Combined Influence of Rotation and Scrape-Off Layer Drifts on Recycling Asymmetries in Tokamak Plasmas (2025). [DOI](#)
- 2025GPSH: Computing Flux-Surface Shapes in Tokamaks and Stellarators (2025). [DOI](#)
- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)
- 2025MRHH: Impurity peaking of SPARC H-modes: a sensitivity study on physics and engineering assumptions (2025). [DOI](#)
- 2025MSAK: Plasma position constrained free-boundary MHD equilibrium in Tokamaks using pyIPREQ (2025). [DOI](#)
- 2025PGHS: GyroSwin: 5D Surrogates for Gyrokinetic Plasma Turbulence Simulations (2025). [DOI](#)
- 2025PSWR: Isotope Production in Fusion Systems (2025). [DOI](#)
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- 2025SKDF: Overview of the Helios Design: A Practical Planar Coil Stellarator Fusion Power Plant (2025). [DOI](#)
- 2025SPH: Linear stability studies for a quasi-axisymmetric equilibrium including plasma flow, drift and viscosity effects (2025). [DOI](#)
- 2025SRKH: Gyrokinetic Simulations of a Low Recycling Scrape-off Layer without a Lithium Target (2025). [DOI](#)
- 2025TBAB: T_i/T_e Dependence of Core Turbulence and Transport in DIII-D QH-Mode Plasmas (2025). [DOI](#)
- 2025TFSF: Reconstructions of electron-temperature profiles from EUROfusion pedestal database using turbulence models and machine learning (2025). [DOI](#)
- 2025VGRB: Roadmap for warm dense matter physics (2025). [DOI](#)
- 2025WAFG: Particle-in-cell simulations of burning inertial confinement fusion capsule implosions (2025). [DOI](#)
- 2025WH: Continuing progress toward fusion energy breakeven and gain as measured against the Lawson criteria (2025). [DOI](#)
- 2025WWTG: Comparison of filament properties in real-size GBS simulations and experiments of TCV-X21 (2025). [DOI](#)
- 2025ZSRR: Divertor Detachment Characterization in Negative Triangularity Discharges in DIII-D via 2D Edge-Plasma Transport Modeling (2025). [DOI](#)
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- 2021EMCW: Impact of magnetic islands on plasma flow and turbulence in W7-X (2021). [DOI](#)
- 2020FBMH: Conservative discontinuous Galerkin scheme of a gyro-averaged Dougherty collision operator (2020). [DOI](#)
- 2020MHFF: Electromagnetic full- f gyrokinetics in the tokamak edge with discontinuous Galerkin methods (2020). [DOI](#)
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- 2026WTCJ: Multi-GPU Hybrid Particle-in-Cell Monte Carlo Simulations for Exascale Computing Systems (2026). [DOI](#)
- 2026WZJL: Numerical Modeling of Electromagnetic Wave Transmission Through a 2-D Magnetized Plasma Sheath (2026). [DOI](#)
- 2026XDZW: Turbulence-Driven Edge-Localized-Mode-Free High-Confinement Mode with Divertor Detachment in a Metal-Wall Tokamak (2026). [DOI](#)
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Electromagnetics

- 2026B6: The toroidal flux and separatrix effects in tokamaks (2026). [DOI](#)
- 2026BCPS: The Linearized Vlasov–Maxwell System as a Hamiltonian System (2026). [DOI](#)
- 2026CGG: A Scattered-Field Formulation for Coupled Geometric Wakefield and Space Charge Field Simulations in Particle Accelerators (2026). [DOI](#)
- 2026F: Generalized Similarity Theory for Plasmas (2026). [DOI](#)
- 2026FPBT: Electromagnetic Analysis of ITER Electron Cyclotron Emission Model (2026). [DOI](#)
- 2026HAGB: Laboratory observation of collective beam–plasma instabilities in a relativistic pair jet (2026). [DOI](#)
- 2026HS: Local Time Integration for Friedrichs’ Systems (2026). [DOI](#)
- 2026JJPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
- 2026LVDN: Inverse initial data reconstruction for Maxwell’s equations via time splitting method (2026). [DOI](#)
- 2026LYZC: A Strongly Discontinuous Boundary Condition Model With Discontinuous Galerkin Framework for Multiscale Electromagnetic Simulations Containing Imperfect Sliding Electrical Contact (2026). [DOI](#)
- 2026MHL: High-order exponential solver method for particle-in-cell simulations (2026). [DOI](#)
- 2026MKC: Proper Treatment of Energy and Momentum in Time-Modulated Plasmas (2026). [DOI](#)
- 2026NDLB: Particle-In-Cell Simulations of Quantum Plasmas (2026). [DOI](#)
- 2026PSKA: Nonlinear Transport in Microwave-Generated Plasma (2026). [DOI](#)
- 2026RFR: Variation of instability characteristics and resulting electron transport under external modulation in $E \times B$ plasmas (2026). [DOI](#)
- 2026SG: A Direct Sampling Method for Inverse Time-Dependent Electromagnetic Source Problems: Reconstruction of the Radiating Time and Spatial Support (2026). [DOI](#)
- 2026WZJL: Numerical Modeling of Electromagnetic Wave Transmission Through a 2-D Magnetized Plasma Sheath (2026). [DOI](#)
- 2026XMY: An Efficient Energy Stable Structure Preserving Method for the Landau–Lifshitz Equation (2026). [DOI](#)
- 2026YZLY: TokaGLINT: A Scalable GPU-Tailored Implicit Solver for Full 3D Tokamak Electromagnetic Simulations (2026). [DOI](#)
- 2026ZCXZ: Three Dimensional Multiphysics Modelling of Helicon Wave Heating and Antenna Plasma Coupling for Boundary Density Control in Toroidal Fusion Plasmas (2026). [DOI](#)
- 2025B: Electron Inertia and Magnetic Reconnection (2025). [DOI](#)
- 2025B2: Magnetic Field Line Chaos, Cantori, and Turnstiles in Toroidal Plasmas (2025). [DOI](#)
- 2024AK: An Arbitrarily Higher-Order Time Implicit Scheme for Maxwell’s Equations (2024). [DOI](#)
- 2023T: Preconditioning for time-harmonic Maxwell’s equations using the Laguerre transform (2023). [DOI](#)

Gyrokinetics

- 2026ADGM: Gyrokinetic simulation of the effect of transient fueling on plasma turbulence in ADITYA-U tokamak (2026). [DOI](#)
- 2026CZGS: Detecting shearless phase-space transport barriers in global gyrokinetic turbulence simulations with test particle map models (2026). [DOI](#)
- 2026D: GPU Acceleration and Portability of the TRIMEG Code for Gyrokinetic Plasma Simulations Using OpenMP (2026). [DOI](#)

- 2026FCSJ: Conservative Velocity Mappings for Discontinuous Galerkin Kinetics (2026). [DOI](#)
- 2026FUSJ: Extending the gyrokinetic code GENE-X to stellarators (2026). [DOI](#)
- 2026GS: Resonant amplification of zonal flow driven by the negative mass instability in tokamak plasma (2026). [DOI](#)
- 2026HBFH: Towards fully predictive gyrokinetic full- f simulations: validation and triangularity studies in TCV (2026). [DOI](#)
- 2026HFBH: High-throughput full- f gyrokinetics of the tokamak boundary (2026). [DOI](#)
- 2026HGR: The hot-electron closure of the moment-based gyrokinetic plasma model (2026). [DOI](#)
- 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
- 2026LMHL: TRIMEG-GKX: An electromagnetic gyrokinetic particle code with a piecewise field-aligned finite element method for micro- and macro-instability studies in tokamak core plasmas (2026). [DOI](#)
- 2026LMSH: A High-Order Piecewise Field-Aligned Triangular Finite Element Method for Electromagnetic Gyrokinetic Particle Simulations of Tokamak Plasmas with Open Field Lines (2026). [DOI](#)
- 2026LYXB: Sub-Second Collisionless Gyrokinetic Eigenvalue Solutions via Orbit-Invariant Decomposition (2026). [DOI](#)
- 2026LYZR: Semi-implicit scheme for multi-species collision operators in Tokamak Plasma Simulations (2026). [DOI](#)
- 2026MLL: Suppressed Stiffness of energetic particle transport due to thermal plasma nonlinearity in tokamak plasmas (2026). [DOI](#)
- 2026MRD: Comprehensive full- f drift-kinetic and δf gyrokinetic simulations of a linear plasma device based on the gyro-moment approach (2026). [DOI](#)
- 2026NBBP: Nonlinear generation of global zonal structures in gyrokinetic simulations of TCV and ASDEX Upgrade magnetic configurations (2026). [DOI](#)
- 2026NP: Theory of zonal flow growth and propagation in toroidal geometry (2026). [DOI](#)
- 2026NPBM: Saturation of Magnetized Plasma Turbulence by Propagating Zonal Flows (2026). [DOI](#)
- 2026OUZJ: Coupling Fluid Neutrals to Gyrokinetic Plasma Dynamics for Edge and SOL Turbulence Simulations (2026). [DOI](#)
- 2026RXPS: Critical gradient optimisation for quasi-isodynamic stellarators (2026). [DOI](#)
- 2026SBBM: First global gyrokinetic profile predictions of ITER burning plasma (2026). [DOI](#)
- 2026SHJH: Constructing field-aligned coordinate systems for gyrokinetic simulations of tokamaks in X-point geometries (2026). [DOI](#)
- 2026SLMW: ORB5X: a performance-portable global electromagnetic gyrokinetic PIC code in C++/Kokkos built using Agentic AI (2026). [DOI](#)
- 2026TOMP: Verification of a hybrid kinetic-gyrokinetic model using the advanced semi-Lagrange code ssV (2026). [DOI](#)
- 2026TSRC: A deterministic verification strategy for multidimensional gyrokinetic particle-in-cell algorithms using the method of manufactured solutions (2026). [DOI](#)
- 2026W: Electric Field Induced by Radial Redistribution of the Energetic Ion Pressure in a Fusion Plasma (2026). [DOI](#)
- 2026WW: Effects of Geometric Curvature and Weak Magnetic Shear on the Ion-Temperature-Gradient Instability Near the Magnetic Axis in a Tokamak (2026). [DOI](#)
- 2026WWQW: 2DESr: a two-dimensional Fourier-space gyrokinetic eigenvalue code for the ion-temperature-gradient modes in tokamaks (2026). [DOI](#)
- 2026XDZW: Turbulence-Driven Edge-Localized-Mode-Free High-Confinement Mode with Divertor Detachment in a Metal-Wall Tokamak (2026). [DOI](#)
- 2026YN: von Neumann Entropy of Phase Space Structures in Gyrokinetic Plasma Turbulence (2026). [DOI](#)
- 2026ZCLB: Global linear drift-wave eigenmode structures on flux surfaces in stellarators: ion temperature gradient mode (2026). [DOI](#)
- 2025BSBB: Study of the impact of fast ions on core turbulence at rational surfaces via global gyrokinetic simulations (2025). [DOI](#)
- 2025HGR: The Hot-Electron Closure of the Moment-Based Gyrokinetic Plasma Model (2025). [DOI](#)
- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)

- 2025PGHS: GyroSwin: 5D Surrogates for Gyrokinetic Plasma Turbulence Simulations (2025). [DOI](#)
- 2025SKDF: Overview of the Helios Design: A Practical Planar Coil Stellarator Fusion Power Plant (2025). [DOI](#)
- 2025SRKH: Gyrokinetic Simulations of a Low Recycling Scrape-off Layer without a Lithium Target (2025). [DOI](#)
- 2025TBAB: T_i/T_e Dependence of Core Turbulence and Transport in DIII-D QH-Mode Plasmas (2025). [DOI](#)
- 2022NH: Gyrokinetic Turbulent Transport Simulations on Steady Burning Condition in D-T-He Plasmas (2022). [DOI](#)
- 2020FBMH: Conservative discontinuous Galerkin scheme of a gyro-averaged Dougherty collision operator (2020). [DOI](#)
- 2020MHF: Electromagnetic full- f gyrokinetics in the tokamak edge with discontinuous Galerkin methods (2020). [DOI](#)
- 2018MDL: Laguerre–Hermite pseudo-spectral velocity formulation of gyrokinetics (2018). [DOI](#)
- 2017NNSW: Isotope Effects on Trapped-Electron-Mode Driven Turbulence and Zonal Flows in Helical and Tokamak Plasmas (2017). [DOI](#)
- 2015SHH: A gyrokinetic one-dimensional scrape-off layer model of an edge-localized mode heat pulse (2015). [DOI](#)
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- 2026A: Semi-Lagrangian Discontinuous Galerkin Method with Adaptive Mesh Refinement for the Vlasov–Poisson System in 1X+3V (2026). [DOI](#)
- 2026AFKP: A projection method for particle resampling (2026). [DOI](#)
- 2026AH: High-Order Moment Closure for Nonmagnetized Electrons in Partially Ionized Plasmas (2026). [DOI](#)
- 2026AKMI: Velocity-Space Signatures of Energy Transfer for Ion-Acoustic Instabilities (2026). [DOI](#)
- 2026B2: On Modeling and Solving the Boltzmann Equation (2026). [DOI](#)
- 2026BC: Generalized flux-weighted boundary walls in kinetic models (2026). [DOI](#)
- 2026BC2: The machine learning approach to moment closure relations for plasma: a review (2026). [DOI](#)
- 2026BCPS: The Linearized Vlasov–Maxwell System as a Hamiltonian System (2026). [DOI](#)
- 2026BHJP: Capturing Secondary Kinetic Instabilities in Three-Dimensional Dayside Reconnection Using an Improved Gradient-Based Closure (2026). [DOI](#)
- 2026BKT: Fully kinetic 3-D simulations of high-beta plasma in a mirror trap (2026). [DOI](#)
- 2026BP: A fast spectral particle method for the Landau equation (2026). [DOI](#)
- 2026CJLY: Control of a Uniformly Magnetized Plasma with External Electric Fields (2026). [DOI](#)
- 2026CLY: Semi-Lagrangian SAV method for Vlasov-Maxwell equations (2026). [DOI](#)
- 2026CZGS: Detecting shearless phase-space transport barriers in global gyrokinetic turbulence simulations with test particle map models (2026). [DOI](#)
- 2026DFTW: A Deterministic Ionization Algorithm for the OSIRIS Particle-in-Cell Framework (2026). [DOI](#)
- 2026DMN: High-Order Structure-Preserving SBP Finite Difference Methods for the Vlasov–Maxwell System on Matrix-Free GPUs (2026). [DOI](#)
- 2026F: Generalized Similarity Theory for Plasmas (2026). [DOI](#)
- 2026FBAP: Plasma mixing driven by the collisionless Kelvin–Helmholtz instability: Insights from a fully kinetic simulation and density-based diagnostics (2026). [DOI](#)
- 2026FCSJ: Conservative Velocity Mappings for Discontinuous Galerkin Kinetics (2026). [DOI](#)
- 2026FPKA: Structure Preservation Using Discrete Gradients in the Vlasov-Poisson-Landau System (2026). [DOI](#)
- 2026FRT: A modified Crank–Nicolson scheme for the Vlasov–Poisson system with a strong external magnetic field (2026). [DOI](#)
- 2026G3: Fully spectral scheme for the linear BGK equation on the whole space (2026). [DOI](#)
- 2026GHDL: Solving the Six-Dimensional Vlasov–Maxwell System with Active Flux and Splitting Methods (2026). [DOI](#)
- 2026GS: Resonant amplification of zonal flow driven by the negative mass instability in tokamak plasma (2026). [DOI](#)
- 2026HV: Flow-Regulated Suprathermal Particle Acceleration in Weakly Collisional Astrophysical Plasmas (2026). [DOI](#)

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- 2026HY: An analytically-solvable, wave-form asymptotic-preserving and energy-conserving time-splitting scheme for Vlasov–Poisson equations in the quasi-neutral regime (2026). [DOI](#)
- 2026IDR: Mixed Hermite–Legendre spectral method for kinetic plasma simulations (2026). [DOI](#)
- 2026J: An Upwind Summation-by-Part Scheme for the Non-Linear Boltzmann Equation (2026). [DOI](#)
- 2026JHJ: A conservative discontinuous Galerkin algorithm for particle kinetics on smooth manifolds (2026). [DOI](#)
- 2026JJPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
- 2026JT: On the evolution of a large-amplitude, weakly-collisional electron plasma wave (2026). [DOI](#)
- 2026KK: Relation of Exact Hydrodynamics to the Chapman–Enskog Series (2026). [DOI](#)
- 2026KLWB: A Hybrid Semi-Lagrangian Flow Mapping Approach for Vlasov Systems: Combining Iterative and Compositional Flow Maps (2026). [DOI](#)
- 2026KT: Partially Ionized Plasma Physics and Technological Applications (2026). [DOI](#)
- 2026LCLK: Discrete unified gas kinetic scheme for multiscale plasma simulations (2026). [DOI](#)
- 2026LDGB: Simulating ultrarelativistic beam-plasma instabilities with a quasistatic particle-in-cell code (2026). [DOI](#)
- 2026LG: An efficient discrete unified gas kinetic scheme for strongly inhomogeneous fluids at the nanoscale (2026). [DOI](#)
- 2026LG2: Transformation of the Maxwell-Boltzmann distribution to a power law (2026). [DOI](#)
- 2026LHWP: Time-embedded convolutional neural networks for modeling plasma heat transport (2026). [DOI](#)
- 2026LJLH: FourierSpecNet: Neural collision operator approximation inspired by the Fourier spectral method for solving the Boltzmann equation (2026). [DOI](#)
- 2026LLBZ: Hybrid Fluid–Kinetic Simulations of Resistive Instabilities in Runaway Electron Beams (2026). [DOI](#)
- 2026LLL: Role of ion acoustic instability in magnetic reconnection (2026). [DOI](#)
- 2026LLXZ: Efficient computation of the forcing term in Enskog Vlasov equation (2026). [DOI](#)
- 2026LTGM: Expansion-Driven Self-Magnetization of High-Energy-Density Plasmas (2026). [DOI](#)
- 2026LVCN: Theory of beam-driven nonlinear plasma wake and interior waves (2026). [DOI](#)
- 2026LYZR: Semi-implicit scheme for multi-species collision operators in Tokamak Plasma Simulations (2026). [DOI](#)
- 2026MF: Energy Partition in Collisionless Counterstreaming Plasmas (2026). [DOI](#)
- 2026MK: Gravitational instability in partially ionized plasmas: A two-fluid approach (2026). [DOI](#)
- 2026MLL: Suppressed Stiffness of energetic particle transport due to thermal plasma nonlinearity in tokamak plasmas (2026). [DOI](#)
- 2026MML: Parametric instabilities of the inhomogeneous near-SOL tokamak plasma, driven by the coupled effect of the high-harmonic fast wave and of the ion and electron temperature gradients, and anomalous heating of the near-SOL ions (2026). [DOI](#)
- 2026MRD: Comprehensive full- f drift-kinetic and δf gyrokinetic simulations of a linear plasma device based on the gyro-moment approach (2026). [DOI](#)
- 2026NN: Reduced One-Fluid GENERIC Closure from Relativistic Moment Kinetics (2026). [DOI](#)
- 2026OC: Hamiltonian Hydrodynamic Reductions of One-Dimensional Vlasov Equations (2026). [DOI](#)
- 2026OTWH: Sparse and low-rank kinetic distribution estimation (2026). [DOI](#)
- 2026OUZJ: Coupling Fluid Neutrals to Gyrokinetic Plasma Dynamics for Edge and SOL Turbulence Simulations (2026). [DOI](#)
- 2026QSW: A fast Fourier spectral method for wave kinetic equation (2026). [DOI](#)
- 2026R: Ions-electrons-states for the two-component Vlasov–Poisson equation (2026). [DOI](#)
- 2026R3: Bivariate incomplete-Bessel kernels for the first nonlinear Vlasov–Maxwell response (2026). [DOI](#)
- 2026RS: Micro-macro kinetic flux-vector splitting schemes for the multidimensional Boltzmann-ES-BGK equation (2026). [DOI](#)
- 2026SJKY: Multi-species Rosenbluth Fokker-Planck collision operator for discontinuous Galerkin method (2026). [DOI](#)
- 2026SQHZ: An adaptive-rank approach with greedy sampling for multi-scale BGK equations (2026). [DOI](#)
- 2026SRHK: A splitting algorithm for a zonal quasineutral Vlasov model of strongly magnetized plasmas (2026). [DOI](#)
- 2026SSEH: Adaptive Sparse-grid Discontinuous Galerkin Approximations the Bhatnagar–Gross–Krook Model (2026). [DOI](#)

- 2026TBA: A novel conditional formulation of the Vlasov–Ampère equations: a conservative, positivity, asymptotic and Gauss law preserving scheme (2026). [DOI](#)
- 2026TLF: Particle collisionality in scaled kinetic plasma simulations (2026). [DOI](#)
- 2026TOMP: Verification of a hybrid kinetic-gyrokinetic model using the advanced semi-Lagrange code ssV (2026). [DOI](#)
- 2026TP: Neoclassical transport and profile prediction in transport barriers (2026). [DOI](#)
- 2026VBGM: Local Cubic Spline Interpolation for Vlasov-Type Equations on a Multi-Patch Geometry (2026). [DOI](#)
- 2026WH2: Dynamical tensor train approximation for kinetic equations (2026). [DOI](#)
- 2026WKRS: A meshless MUSCL method for the BGK-Boltzmann equation (2026). [DOI](#)
- 2026YCW: A Fast Fourier Spectral Method for the Linearized Boltzmann Collision Operator (2026). [DOI](#)
- 2026YT: Effect of eigenmode destabilisation by energetic ions on plasma rotation (2026). [DOI](#)
- 2026ZD: The Properties of Plasma Sheath Containing the Primary Electrons with a Cairns Distribution (2026). [DOI](#)
- 2026ZGZ: Predictor-corrector discrete unified gas kinetic scheme for Boltzmann transport models with non-conservative collision operators (2026). [DOI](#)
- 2026ZHWD: Collisionless ion-electron energy exchange in magnetized shocks (2026). [DOI](#)
- 2026ZUM: Loading Non-Maxwellian Velocity Distributions in Particle Simulations (2026). [DOI](#)
- 2025AA: Thermalization times and scaling laws for alpha particles and hydrogen isotopes in Tokamak plasmas (2025). [DOI](#)
- 2025CHHM: Measurement of Ion Acceleration and Diffusion in a Laser-Driven Magnetized Plasma (2025). [DOI](#)
- 2025CHLW: Asymptotic-preserving semi-Lagrangian discontinuous Galerkin schemes for the Boltzmann equation (2025). [DOI](#)
- 2025CKKS: Asymptotic-preserving conservative semi-Lagrangian discontinuous Galerkin schemes for the Vlasov-Poisson system in the quasi-neutral limit (2025). [DOI](#)
- 2025CLLZ: A high-order energy-conserving semi-Lagrangian discontinuous Galerkin method for the Vlasov-Ampère system (2025). [DOI](#)
- 2025DDGG: Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror (2025). [DOI](#)
- 2025HLS: On High-Order/Low-Order and Micro-Macro Methods for Implicit Time-Stepping of the BGK Model (2025). [DOI](#)
- 2025IMCR: Turbulence and Magnetic Reconnection in Relativistic Multispecies Plasmas (2025). [DOI](#)
- 2025ISSS: Kinetic renormalization of auroral turbulence (2025). [DOI](#)
- 2025JHJ: A moment-conserving discontinuous Galerkin representation of the relativistic Maxwellian distribution (2025). [DOI](#)
- 2025L: Symplectic particle-in-cell methods for hybrid plasma models with Boltzmann electrons and space-charge effects (2025). [DOI](#)
- 2025LJHH: Axisymmetric gyrokinetic simulation of ASDEX-Upgrade scrape-off layer using a conservative implicit BGK collision operator (2025). [DOI](#)
- 2025LWZ: From filamentation to stratification: Instability dynamics in scissors-shaped relativistic beam-plasma system (2025). [DOI](#)
- 2025PHR: A New Approach to Compute Linear Landau Damping (2025). [DOI](#)
- 2025SRHK: Convergence of Splitting Methods on Rotating Grids for the Magnetized Vlasov Equation (2025). [DOI](#)
- 2025SRKH: Gyrokinetic Simulations of a Low Recycling Scrape-off Layer without a Lithium Target (2025). [DOI](#)
- 2025WAFG: Particle-in-cell simulations of burning inertial confinement fusion capsule implosions (2025). [DOI](#)
- 2025YHR: An Explicit Energy-Conserving Particle Method for the Vlasov-Fokker-Planck Equation (2025). [DOI](#)
- 2025ZCSV: Numerical investigation of kinetic instabilities in BGK equilibria under collisional effects (2025). [DOI](#)
- 2025ZSHC: A Semi-Lagrangian Adaptive Rank (SLAR) Method for High-Dimensional Vlasov Dynamics (2025). [DOI](#)
- 2024BC: Asymptotic Analysis of IMEX-RK Methods for ES-BGK Model at Navier–Stokes Level (2024). [DOI](#)
- 2024G: A coordinate-free expression of plasma theory (2024). [DOI](#)
- 2024HM: Rugged Magneto-Hydrodynamic Invariants in Weakly Collisional Plasma Turbulence: Two-

Dimensional Hybrid Simulation Results (2024). [DOI](#)

- 2024IMSC: Long-lived Equilibria in Kinetic Astrophysical Plasma Turbulence (2024). [DOI](#)
- 2024LLW: A bi-fidelity method for the uncertain Vlasov-Poisson system near quasineutrality in an asymptotic-preserving particle-in-cell framework (2024). [DOI](#)
- 2020FBMH: Conservative discontinuous Galerkin scheme of a gyro-averaged Dougherty collision operator (2020). [DOI](#)
- 2018MDL: Laguerre–Hermite pseudo-spectral velocity formulation of gyrokinetics (2018). [DOI](#)
- 2015SHH: A gyrokinetic one-dimensional scrape-off layer model of an edge-localized mode heat pulse (2015). [DOI](#)
- 2013LE: New velocity-space discretization for continuum kinetic calculations and Fokker–Planck collisions (2013). [DOI](#)
- 2007BH: Foundations of nonlinear gyrokinetic theory (2007). [DOI](#)

Statistical Mechanics

- 2026BC: Generalized flux-weighted boundary walls in kinetic models (2026). [DOI](#)
- 2026LG2: Transformation of the Maxwell-Boltzmann distribution to a power law (2026). [DOI](#)

Particle Methods

- 2026ABY: Helical core formation and MHD stability in ITER-scale plasmas with fusion-born alpha particles (2026). [DOI](#)
- 2026AFKP: A projection method for particle resampling (2026). [DOI](#)
- 2026ALGP: Energy Bunching from Subcycle Ionization Injection in Laser Wakefield Acceleration (2026). [DOI](#)
- 2026BC: Generalized flux-weighted boundary walls in kinetic models (2026). [DOI](#)
- 2026BCPS: The Linearized Vlasov–Maxwell System as a Hamiltonian System (2026). [DOI](#)
- 2026BKT: Fully kinetic 3-D simulations of high-beta plasma in a mirror trap (2026). [DOI](#)
- 2026BMCC: Beam transverse dynamics in laser-plasma accelerators (2026). [DOI](#)
- 2026BP: A fast spectral particle method for the Landau equation (2026). [DOI](#)
- 2026BSSK: Accelerating the Particle-In-Cell code ECsim with OpenACC (2026). [DOI](#)
- 2026BTSL: Plasma wakefield dynamics of self-generated electron bunch trains (2026). [DOI](#)
- 2026CGG: A Scattered-Field Formulation for Coupled Geometric Wakefield and Space Charge Field Simulations in Particle Accelerators (2026). [DOI](#)
- 2026CHDW: Self-adaptive stabilization and quality boost for electron beams from all-optical plasma wakefield accelerators (2026). [DOI](#)
- 2026CMPT: A Lagrangian method for solving the spherical shallow water equations using power diagrams (2026). [DOI](#)
- 2026CZGS: Detecting shearless phase-space transport barriers in global gyrokinetic turbulence simulations with test particle map models (2026). [DOI](#)
- 2026DFTW: A Deterministic Ionization Algorithm for the OSIRIS Particle-in-Cell Framework (2026). [DOI](#)
- 2026DGNP: Hierarchical Sparse-Grid Particle-in-Cell Method with Locally Adaptive Mesh Refinement (2026). [DOI](#)
- 2026DJUN: OpenEdge: A collaborative, open-source, multi-purpose direct simulation Monte Carlo for plasma simulation in magnetic fusion environments (2026). [DOI](#)
- 2026FBAP: Plasma mixing driven by the collisionless Kelvin–Helmholtz instability: Insights from a fully kinetic simulation and density-based diagnostics (2026). [DOI](#)
- 2026FHDM: Gridless Quasistatic Model for Efficient Simulation of Plasma-based Accelerators (2026). [DOI](#)
- 2026FHWS: A HIP-Compatible Accelerator Backend for Fourier-Bessel Particle-in-Cell Simulations on CPU/DCU Heterogeneous Clusters (2026). [DOI](#)
- 2026FMMM: Code-Verification Techniques for Particle-in-Cell Simulations with Direct Simulation Monte Carlo Collisions (2026). [DOI](#)
- 2026FPKA: Structure Preservation Using Discrete Gradients in the Vlasov-Poisson-Landau System (2026). [DOI](#)
- 2026FRT: A modified Crank–Nicolson scheme for the Vlasov–Poisson system with a strong external magnetic field (2026). [DOI](#)

- 2026GCSB: Runaway electron interactions with whistler waves in tokamak plasmas: energy-dependent transport scaling (2026). [DOI](#)
- 2026GDS: Plasma Wakes Driven by Compton Scattering: Nonlinear Regime and Particle Acceleration (2026). [DOI](#)
- 2026GVBC: Experimental Demonstration of Beam-Driven Wakefield Acceleration in Laser-Plasma Filament (2026). [DOI](#)
- 2026HAGB: Laboratory observation of collective beam–plasma instabilities in a relativistic pair jet (2026). [DOI](#)
- 2026HYZ: Compressible Turbulence as a Source of Particle Beams and Ion Bernstein Waves in Collisionless Plasmas (2026). [DOI](#)
- 2026JGS: Magnetic Field Amplification and Particle Acceleration in Weakly Magnetized Trans-relativistic Electron-ion Shocks (2026). [DOI](#)
- 2026JKGH: Optimized laser wakefield acceleration: Generating stable, high-energy, monoenergetic electron beams and demonstrating extreme-ultraviolet free-electron lasers (2026). [DOI](#)
- 2026KNTL: Understanding Carbon Sourcing and Transport Originating from the Helicon Antenna Surfaces During High-Power Helicon Discharge in DIII-D Tokamak (2026). [DOI](#)
- 2026LDGB: Simulating ultrarelativistic beam-plasma instabilities with a quasistatic particle-in-cell code (2026). [DOI](#)
- 2026LG2: Transformation of the Maxwell-Boltzmann distribution to a power law (2026). [DOI](#)
- 2026LHAH: Asymmetric self-injection of electrons in a laser wakefield accelerator via localized laser redshifting (2026). [DOI](#)
- 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
- 2026LLBZ: Hybrid Fluid–Kinetic Simulations of Resistive Instabilities in Runaway Electron Beams (2026). [DOI](#)
- 2026LLL: Role of ion acoustic instability in magnetic reconnection (2026). [DOI](#)
- 2026LLR: Three-dimensional mesh adaptation in PFEM (2026). [DOI](#)
- 2026LMHL: TRIMEG-GKX: An electromagnetic gyrokinetic particle code with a piecewise field-aligned finite element method for micro- and macro-instability studies in tokamak core plasmas (2026). [DOI](#)
- 2026LMSH: A High-Order Piecewise Field-Aligned Triangular Finite Element Method for Electromagnetic Gyrokinetic Particle Simulations of Tokamak Plasmas with Open Field Lines (2026). [DOI](#)
- 2026LO: Numerical study of electron acceleration by microwave-driven plasma wakefields in rectangular waveguides (2026). [DOI](#)
- 2026LRGW: Long lasting plasma density structures utilizing tailored density profiles (2026). [DOI](#)
- 2026LTGM: Expansion-Driven Self-Magnetization of High-Energy-Density Plasmas (2026). [DOI](#)
- 2026LZZ: Near-Wall Pathways of Anomalous Electron Transport in Hall Thrusters Revealed by 3D PIC Simulations (2026). [DOI](#)
- 2026MEVD: Superluminal Wave Activation at Relativistic Magnetized Shocks (2026). [DOI](#)
- 2026MF: Energy Partition in Collisionless Counterstreaming Plasmas (2026). [DOI](#)
- 2026MHL: High-order exponential solver method for particle-in-cell simulations (2026). [DOI](#)
- 2026MLL: Suppressed Stiffness of energetic particle transport due to thermal plasma nonlinearity in tokamak plasmas (2026). [DOI](#)
- 2026MMGK: Laser field reconstruction for the modeling of laser-plasma interaction in cylindrical geometry (2026). [DOI](#)
- 2026MMWS: Laser wakefield acceleration in a capillary gas cell to produce high-quality GeV-scale electron beams (2026). [DOI](#)
- 2026NDLB: Particle-In-Cell Simulations of Quantum Plasmas (2026). [DOI](#)
- 2026OCP: A p-adaptive hybridizable discontinuous Galerkin spectral element method for electrostatic particle-in-cell simulations (2026). [DOI](#)
- 2026PMAA: Dephasingless laser wakefield acceleration in a plasma waveguide (2026). [DOI](#)
- 2026RFR: Variation of instability characteristics and resulting electron transport under external modulation in $E \times B$ plasmas (2026). [DOI](#)
- 2026RLLW: Research on the advanced GPU-based Monte Carlo neutron transport methods (2026). [DOI](#)
- 2026S2: Hybrid PIC-fluid model for numerical simulation of laser-plasma interaction (2026). [DOI](#)
- 2026S3: An Overview of Relativistic Particle Pushers and their Extension to Arbitrary Order Accuracy (2026). [DOI](#)

- 2026SLMW: ORB5X: a performance-portable global electromagnetic gyrokinetic PIC code in C++/Kokkos built using Agentic AI (2026). [DOI](#)
- 2026STCR: The PLUTO code on GPUs: Offloading Lagrangian Particle methods (2026). [DOI](#)
- 2026SWLA: Stable 50 MeV Beams from a 100 Hz Laser-Wakefield Accelerator Driven by an OPCPA Laser (2026). [DOI](#)
- 2026TK: GPU-Native Compressed Neighbor Lists with a Space-Filling-Curve Data Layout (2026). [DOI](#)
- 2026TLF: Particle collisionality in scaled kinetic plasma simulations (2026). [DOI](#)
- 2026TSRC: A deterministic verification strategy for multidimensional gyrokinetic particle-in-cell algorithms using the method of manufactured solutions (2026). [DOI](#)
- 2026WCMT: Integrating High Performance In-Memory Data Streaming and In-Situ Visualization in Hybrid MPI+OpenMP PIC MC Simulations Towards Exascale (2026). [DOI](#)
- 2026WH: An effective implementation of the bidirectional buffer: Towards laminar and turbulent open-boundary flows (2026). [DOI](#)
- 2026WTCJ: Multi-GPU Hybrid Particle-in-Cell Monte Carlo Simulations for Exascale Computing Systems (2026). [DOI](#)
- 2026WXMx: Nanometer-scale pre-bunched electron beams generated from all-optical plasma-based acceleration (2026). [DOI](#)
- 2026YZLY: TokaGLINT: A Scalable GPU-Tailored Implicit Solver for Full 3D Tokamak Electromagnetic Simulations (2026). [DOI](#)
- 2026ZHWD: Collisionless ion-electron energy exchange in magnetized shocks (2026). [DOI](#)
- 2026ZUM: Loading Non-Maxwellian Velocity Distributions in Particle Simulations (2026). [DOI](#)
- 2026ZZLW: AlgoPlasma: Open Algorithms for Plasma Modeling (2026). [DOI](#)
- 2026ZZY: Three-dimensional particle-in-cell simulations on Hall thruster electron drift instability: influence of magnetic field on electron transport (2026). [DOI](#)
- 2025BOMG: π -PIC: a framework for modular particle-in-cell developments and simulations (2025). [DOI](#)
- 2025BSS: Phase-Space Shaping in Wakefield Accelerators due to Betatron Cooling (2025). [DOI](#)
- 2025COWC: Laser Amplification in e^- - μ^- -ion Plasmas (2025). [DOI](#)
- 2025DBML: Bayesian Optimization of Laser-Wakefield Acceleration via Spectral Pulse Shaping (2025). [DOI](#)
- 2025EGTL: Long Living Hot and Dense Plasma from Relativistic Laser-Nanowire Array Interaction (2025). [DOI](#)
- 2025HBCC: Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module (2025). [DOI](#)
- 2025HBS: Magnetohydrodynamic-guiding-center-particle-in-cell Method for Multiscale Plasma Kinetic Simulations (2025). [DOI](#)
- 2025IMCR: Turbulence and Magnetic Reconnection in Relativistic Multispecies Plasmas (2025). [DOI](#)
- 2025L: Symplectic particle-in-cell methods for hybrid plasma models with Boltzmann electrons and space-charge effects (2025). [DOI](#)
- 2025LWZ: From filamentation to stratification: Instability dynamics in scissors-shaped relativistic beam-plasma system (2025). [DOI](#)
- 2025LZL: Suppression of inverse magnetic energy transfer in collisionless marginally magnetized plasmas (2025). [DOI](#)
- 2025MJLT: JAX-in-Cell: A Differentiable Particle-in-Cell Code for Plasma Physics Applications (2025). [DOI](#)
- 2025SA: Machine Learning-Integrated Hybrid Fluid-Kinetic Framework for Quantum Electrodynamical Laser Plasma Simulations (2025). [DOI](#)
- 2025SVLN: Toward controlling electron beam charge with nanoparticle-assisted laser wakefield accelerators (2025). [DOI](#)
- 2025VKCK: Optimal Transport, Timesteppers, Newton-Krylov Methods and Steady States of Collective Particle Dynamics (2025). [DOI](#)
- 2025WAFG: Particle-in-cell simulations of burning inertial confinement fusion capsule implosions (2025). [DOI](#)
- 2025WFTJ: Influence of ion motion in a resonantly driven wakefield accelerator (2025). [DOI](#)
- 2025WZLW: Achieving High Efficiency And Enhanced Beam Quality In Laser Wakefield Acceleration (2025). [DOI](#)
- 2025YHR: An Explicit Energy-Conserving Particle Method for the Vlasov-Fokker-Planck Equation (2025). [DOI](#)
- 2024IMSC: Long-lived Equilibria in Kinetic Astrophysical Plasma Turbulence (2024). [DOI](#)
- 2024LLW: A bi-fidelity method for the uncertain Vlasov-Poisson system near quasineutrality in an asymptotic-

preserving particle-in-cell framework (2024). [DOI](#)

Magnetohydrodynamics

- 2026ABY: Helical core formation and MHD stability in ITER-scale plasmas with fusion-born alpha particles (2026). [DOI](#)
- 2026B6: The toroidal flux and separatrix effects in tokamaks (2026). [DOI](#)
- 2026BLGA: Study of Low-Frequency Core-Edge Coupling in a Tokamak: II. Spatial Channeling & Focusing In Antenna-Driven MHD (2026). [DOI](#)
- 2026BLPP: Tearing Stability Prediction Combining Toroidal Calculations With a Two-Fluid Slab Layer (2026). [DOI](#)
- 2026BR: Compact Runge-Kutta flux reconstruction methods for non-conservative hyperbolic equations (2026). [DOI](#)
- 2026CMSM: Kinetic equilibrium prediction at TCV using RAPTOR and FBT (2026). [DOI](#)
- 2026DDP: A Reynolds- and Hartmann-Semirobust Hybrid Method for Magnetohydrodynamics (2026). [DOI](#)
- 2026FKO: Nonmodal growth and optimal perturbations in magnetohydrodynamic shear flows (2026). [DOI](#)
- 2026GJZM: A Gegenbauer-Galerkin spectral scheme for three-dimensional incompressible Hall-magnetohydrodynamics system (2026). [DOI](#)
- 2026GSJH: Stochastic single-stage stellarator optimization using fixed-boundary equilibria (2026). [DOI](#)
- 2026HBKW: Novel approach to general curvilinear coordinates for plasma fluid applications (2026). [DOI](#)
- 2026HNND: Relativistic magnetohydrodynamic wave excitation by laser pulse in a magnetized plasma (2026). [DOI](#)
- 2026HRT: A checkerboard-free, symmetry-preserving, conservative method for magnetohydrodynamics (2026). [DOI](#)
- 2026HZCN: A Curvilinear Lagrangian discontinuous Galerkin method for resistive magneto-hydrodynamics (2026). [DOI](#)
- 2026K: Mercier–Cotsaftis and Grad–Shafranov equations for anisotropic plasma (2026). [DOI](#)
- 2026KKLM: Physics-Informed Kolmogorov–Arnold Networks for Grad–Shafranov Tokamak Equilibria (2026). [DOI](#)
- 2026LBBK: Study of Low-Frequency Core-Edge Coupling in a Tokamak: I. Experimental Observation in KSTAR (2026). [DOI](#)
- 2026LCHK: JOREK simulations of the X-point radiator formation and its movement in ASDEX Upgrade (2026). [DOI](#)
- 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
- 2026LLBZ: Hybrid Fluid–Kinetic Simulations of Resistive Instabilities in Runaway Electron Beams (2026). [DOI](#)
- 2026LWY: High-order positivity-preserving and globally divergence-free discontinuous Galerkin methods for ideal magnetohydrodynamics with conditional energy conservation (2026). [DOI](#)
- 2026MR: Structure-preserving long-time simulations of turbulence in magnetised ideal fluids (2026). [DOI](#)
- 2026MSCM: Evidence of Nonlinear Coupling in the Edge Harmonic Oscillation Sustaining Quiescent High Confinement in a Tokamak Plasma (2026). [DOI](#)
- 2026NHGN: PanoMHD: Multimodal Modelling of Plasma Dynamics towards Tokamak Control (2026). [DOI](#)
- 2026PBLL: Coordinate-invariant flux-surface Fourier analysis in tokamaks (2026). [DOI](#)
- 2026RECB: Parallelized real-time physics codes for plasma control on DIII-D (2026). [DOI](#)
- 2026SE: Global Non-axisymmetric Hall Instabilities in a Rotating Plasma (2026). [DOI](#)
- 2026SJNB: Compact quasiaxisymmetric stellarators, a near axisymmetric theory (2026). [DOI](#)
- 2026STCR: The PLUTO code on GPUs: Offloading Lagrangian Particle methods (2026). [DOI](#)
- 2026TDAZ: Drift-reduced fluid modeling of rapidly rotating plasmas (2026). [DOI](#)
- 2026WGL: Numerical Investigation of Plasma Vertical Growth Rate in Vertical Displacement Events on EAST (2026). [DOI](#)
- 2026WJPA: Accurate calculation of the gradients of the equilibrium poloidal flux in tokamaks (2026). [DOI](#)
- 2026WLST: An Efficient Solver for Finite Element-based Constrained Transport in 3D Magnetohydrodynamics Applied to Magnetic Confinement Fusion (2026). [DOI](#)
- 2026WRS: Unstable Magnetic Reconnection Self-Generates Turbulence (2026). [DOI](#)

- 2026XLDM: Development of a Reduced Multi-Fluid Equilibrium Model and Its Application to Proton-Boron Spherical Tokamaks (2026). [DOI](#)
- 2026YG: Feasibility of Negative Triangularity Equilibria in the SPARC Tokamak (2026). [DOI](#)
- 2026ZHKB: Flux pumping and bifurcated relaxations of helical core in 3D magnetohydrodynamic modelling of ASDEX Upgrade plasmas (2026). [DOI](#)
- 2026ZHWD: Collisionless ion-electron energy exchange in magnetized shocks (2026). [DOI](#)
- 2026ZLP: On energy consistency of intermediate states in HLL-type MHD Riemann solvers (2026). [DOI](#)
- 2026ZPSR: Variability of MHD instabilities in benign termination of high-current runaway electron beams in the JET and DIII-D tokamaks (2026). [DOI](#)
- 2026ZXLM: VEQ: a fast parametric Grad–Shafranov solver for fixed-boundary tokamak equilibria with flexible source profiles (2026). [DOI](#)
- 2025B: Electron Inertia and Magnetic Reconnection (2025). [DOI](#)
- 2025B2: Magnetic Field Line Chaos, Cantori, and Turnstiles in Toroidal Plasmas (2025). [DOI](#)
- 2025FKF: Unveiling a New β Scaling of the Tearing Instability in Weakly Collisional Plasmas (2025). [DOI](#)
- 2025GAST: MHD waves with mixed properties/Alfvén waves with pressure variations: a review (2025). [DOI](#)
- 2025GPSH: Computing Flux-Surface Shapes in Tokamaks and Stellarators (2025). [DOI](#)
- 2025HBS: Magnetohydrodynamic-guiding-center-particle-in-cell Method for Multiscale Plasma Kinetic Simulations (2025). [DOI](#)
- 2025HJP: Temporal Substepping Scheme for Magnetohydrodynamics with Cell-based Adaptive Mesh Refinement on Staggered Grid (2025). [DOI](#)
- 2025LZGY: Studying the properties of reconnection-driven turbulence (2025). [DOI](#)
- 2025MQPF: Wave topology in Hall magnetohydrodynamics (2025). [DOI](#)
- 2025MSAK: Plasma position constrained free-boundary MHD equilibrium in Tokamaks using pyIPREQ (2025). [DOI](#)
- 2025RMBC: UK White Paper on Magnetic Reconnection (2025). [DOI](#)
- 2025S2: The Numerics of VMEC++ (2025). [DOI](#)
- 2025SKDF: Overview of the Helios Design: A Practical Planar Coil Stellarator Fusion Power Plant (2025). [DOI](#)
- 2025SPH: Linear stability studies for a quasi-axisymmetric equilibrium including plasma flow, drift and viscosity effects (2025). [DOI](#)
- 2025ZS: Optimal error estimates for a fully discrete, highly efficient decoupled scheme for the 2D/3D diffuse interface two-phase MHD flows (2025). [DOI](#)
- 2024G: A coordinate-free expression of plasma theory (2024). [DOI](#)
- 2024HM: Rugged Magneto-Hydrodynamic Invariants in Weakly Collisional Plasma Turbulence: Two-Dimensional Hybrid Simulation Results (2024). [DOI](#)

Turbulence

- 2026ADGM: Gyrokinetic simulation of the effect of transient fueling on plasma turbulence in ADITYA-U tokamak (2026). [DOI](#)
- 2026AMM: Energy Cascades in Rotating and Stratified Turbulence in Anisotropic Domains (2026). [DOI](#)
- 2026B3: Internal transport barriers in fluids and plasmas (2026). [DOI](#)
- 2026CSSM: Inertial-Range Suppression and Ponderomotive Density Cavitation in Broadband Sub-Alfvénic Turbulence under Plasma Sheet Boundary Layer Conditions (2026). [DOI](#)
- 2026CZGS: Detecting shearless phase-space transport barriers in global gyrokinetic turbulence simulations with test particle map models (2026). [DOI](#)
- 2026DRM: A numerical strategy for the acceleration of resolved turbulence development in high Mach number turbulent boundary layers (2026). [DOI](#)
- 2026FFCR: Comparison of turbulent electron heat flux model predictions of the H-mode electron pedestal temperature profile across isotope mix and gas fuelling rate scans in JET with the Be/W wall (2026). [DOI](#)
- 2026FKO: Nonmodal growth and optimal perturbations in magnetohydrodynamic shear flows (2026). [DOI](#)
- 2026FMCH: A Coherent Structure Transport Model for Scrape-Off Layer Turbulence (2026). [DOI](#)
- 2026FUSJ: Extending the gyrokinetic code GENE-X to stellarators (2026). [DOI](#)
- 2026GS: Resonant amplification of zonal flow driven by the negative mass instability in tokamak plasma (2026). [DOI](#)

- 2026HBFH: Towards fully predictive gyrokinetic full- f simulations: validation and triangularity studies in TCV (2026). [DOI](#)
- 2026HFBH: High-throughput full- f gyrokinetics of the tokamak boundary (2026). [DOI](#)
- 2026HGR: The hot-electron closure of the moment-based gyrokinetic plasma model (2026). [DOI](#)
- 2026HRT: A checkerboard-free, symmetry-preserving, conservative method for magnetohydrodynamics (2026). [DOI](#)
- 2026HYZ: Compressible Turbulence as a Source of Particle Beams and Ion Bernstein Waves in Collisionless Plasmas (2026). [DOI](#)
- 2026KSG: T3FF: Toroidal 3-dimensional full- f full- k code for isothermal gyrofluid drift-Alfvén turbulence (2026). [DOI](#)
- 2026KSH: Effects of Turbulent Energy Exchange between Electrons and Ions on Global Temperature Profiles (2026). [DOI](#)
- 2026LRBG: Conservative Formulation of the Drift-Reduced Fluid Plasma Model (2026). [DOI](#)
- 2026MHET: Experimental Evidence for Increased Particle Fluxes Due to a Change in Transport at the Separatrix near Density Limits on Alcator C-Mod (2026). [DOI](#)
- 2026MHSE: Empirical impact of near-separatrix plasma and neutral transport on the pedestal in the transition between EDA and ELMy H-modes on Alcator C-Mod (2026). [DOI](#)
- 2026MML: Parametric instabilities of the inhomogeneous near-SOL tokamak plasma, driven by the coupled effect of the high-harmonic fast wave and of the ion and electron temperature gradients, and anomalous heating of the near-SOL ions (2026). [DOI](#)
- 2026MR: Structure-preserving long-time simulations of turbulence in magnetised ideal fluids (2026). [DOI](#)
- 2026MRD: Comprehensive full- f drift-kinetic and δf gyrokinetic simulations of a linear plasma device based on the gyro-moment approach (2026). [DOI](#)
- 2026MRHH: Core-edge integrated modeling of ARC: on the effect of impurity transport and detachment conditions (2026). [DOI](#)
- 2026NBBP: Nonlinear generation of global zonal structures in gyrokinetic simulations of TCV and ASDEX Upgrade magnetic configurations (2026). [DOI](#)
- 2026NP: Theory of zonal flow growth and propagation in toroidal geometry (2026). [DOI](#)
- 2026NPBM: Saturation of Magnetized Plasma Turbulence by Propagating Zonal Flows (2026). [DOI](#)
- 2026ODDE: STORM: Scrape-off layer turbulence in tokamak fusion reactors (2026). [DOI](#)
- 2026OMCL: Predicting core transport in ITER baseline discharges with neon injections (2026). [DOI](#)
- 2026PSHZ: On the mechanism of pedestal relaxation events: Insights gained by turbulence simulations with GRILLIX (2026). [DOI](#)
- 2026QSW: A fast Fourier spectral method for wave kinetic equation (2026). [DOI](#)
- 2026RXPS: Critical gradient optimisation for quasi-isodynamic stellarators (2026). [DOI](#)
- 2026SBBM: First global gyrokinetic profile predictions of ITER burning plasma (2026). [DOI](#)
- 2026SDL: Half-explicit Runge–Kutta integrators for variational multiscale turbulence modeling: Toward higher-order accuracy in space and time (2026). [DOI](#)
- 2026W: Electric Field Induced by Radial Redistribution of the Energetic Ion Pressure in a Fusion Plasma (2026). [DOI](#)
- 2026WH: An effective implementation of the bidirectional buffer: Towards laminar and turbulent open-boundary flows (2026). [DOI](#)
- 2026WRS: Unstable Magnetic Reconnection Self-Generates Turbulence (2026). [DOI](#)
- 2026WW: Effects of Geometric Curvature and Weak Magnetic Shear on the Ion-Temperature-Gradient Instability Near the Magnetic Axis in a Tokamak (2026). [DOI](#)
- 2026XDZW: Turbulence-Driven Edge-Localized-Mode-Free High-Confinement Mode with Divertor Detachment in a Metal-Wall Tokamak (2026). [DOI](#)
- 2026YN: von Neumann Entropy of Phase Space Structures in Gyrokinetic Plasma Turbulence (2026). [DOI](#)
- 2026Z: Particle pinch in global tokamak edge simulations (2026). [DOI](#)
- 2026ZCXY: Broadening of scrape-off layer width by turbulence spreading in the heavy hydrogen isotope plasmas (2026). [DOI](#)
- 2026ZJZU: Fast Turbulence Phase Transition in a Flux-Driven Global Edge-SOL Simulation of a Tokamak Plasma (2026). [DOI](#)
- 2026ZYH: A differentiable wall-modeled large-eddy simulation method for high-Reynolds-number wall-bounded

turbulent flows (2026). [DOI](#)

- 2026ZZY: Three-dimensional particle-in-cell simulations on Hall thruster electron drift instability: influence of magnetic field on electron transport (2026). [DOI](#)
- 2025BSBB: Study of the impact of fast ions on core turbulence at rational surfaces via global gyrokinetic simulations (2025). [DOI](#)
- 2025CHHM: Measurement of Ion Acceleration and Diffusion in a Laser-Driven Magnetized Plasma (2025). [DOI](#)
- 2025DHYD: Microturbulence Suppression by Alfvén Eigenmodes in the DIII-D Tokamak (2025). [DOI](#)
- 2025HGR: The Hot-Electron Closure of the Moment-Based Gyrokinetic Plasma Model (2025). [DOI](#)
- 2025IMCR: Turbulence and Magnetic Reconnection in Relativistic Multispecies Plasmas (2025). [DOI](#)
- 2025ISSS: Kinetic renormalization of auroral turbulence (2025). [DOI](#)
- 2025L2: Magnetised turbulent plasmas as high-energy particle accelerators (2025). [DOI](#)
- 2025LZGY: Studying the properties of reconnection-driven turbulence (2025). [DOI](#)
- 2025MRHH: Impurity peaking of SPARC H-modes: a sensitivity study on physics and engineering assumptions (2025). [DOI](#)
- 2025PGHS: GyroSwin: 5D Surrogates for Gyrokinetic Plasma Turbulence Simulations (2025). [DOI](#)
- 2025TBAB: T_i/T_e Dependence of Core Turbulence and Transport in DIII-D QH-Mode Plasmas (2025). [DOI](#)
- 2025TFSS: Reconstructions of electron-temperature profiles from EUROfusion pedestal database using turbulence models and machine learning (2025). [DOI](#)
- 2025WWTG: Comparison of filament properties in real-size GBS simulations and experiments of TCV-X21 (2025). [DOI](#)
- 2024HM: Rugged Magneto-Hydrodynamic Invariants in Weakly Collisional Plasma Turbulence: Two-Dimensional Hybrid Simulation Results (2024). [DOI](#)
- 2024IMSC: Long-lived Equilibria in Kinetic Astrophysical Plasma Turbulence (2024). [DOI](#)
- 2022NH: Gyrokinetic Turbulent Transport Simulations on Steady Burning Condition in D-T-He Plasmas (2022). [DOI](#)
- 2021EMCW: Impact of magnetic islands on plasma flow and turbulence in W7-X (2021). [DOI](#)
- 2020MHFF: Electromagnetic full- f gyrokinetics in the tokamak edge with discontinuous Galerkin methods (2020). [DOI](#)
- 2017NNSW: Isotope Effects on Trapped-Electron-Mode Driven Turbulence and Zonal Flows in Helical and Tokamak Plasmas (2017). [DOI](#)
- 2012NWS: Nonlinear Entropy Transfer via Zonal Flows in Gyrokinetic Plasma Turbulence (2012). [DOI](#)
- 2007GSAB: Global full- f gyrokinetic simulations of plasma turbulence (2007). [DOI](#)

Astrophysics

- 2026AKMI: Velocity-Space Signatures of Energy Transfer for Ion-Acoustic Instabilities (2026). [DOI](#)
- 2026AMM: Energy Cascades in Rotating and Stratified Turbulence in Anisotropic Domains (2026). [DOI](#)
- 2026CSSM: Inertial-Range Suppression and Ponderomotive Density Cavitation in Broadband Sub-Alfvénic Turbulence under Plasma Sheet Boundary Layer Conditions (2026). [DOI](#)
- 2026FKO: Nonmodal growth and optimal perturbations in magnetohydrodynamic shear flows (2026). [DOI](#)
- 2026GDS: Plasma Wakes Driven by Compton Scattering: Nonlinear Regime and Particle Acceleration (2026). [DOI](#)
- 2026GT2: Invariant-Domain Preserving IMEX Schemes for the Nonequilibrium Gray Radiation-Hydrodynamics Equations Part I (2026). [DOI](#)
- 2026HAGB: Laboratory observation of collective beam-plasma instabilities in a relativistic pair jet (2026). [DOI](#)
- 2026HV: Flow-Regulated Suprathermal Particle Acceleration in Weakly Collisional Astrophysical Plasmas (2026). [DOI](#)
- 2026HYZ: Compressible Turbulence as a Source of Particle Beams and Ion Bernstein Waves in Collisionless Plasmas (2026). [DOI](#)
- 2026JGS: Magnetic Field Amplification and Particle Acceleration in Weakly Magnetized Trans-relativistic Electron-ion Shocks (2026). [DOI](#)
- 2026JJPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
- 2026LLL: Role of ion acoustic instability in magnetic reconnection (2026). [DOI](#)
- 2026MEVD: Superluminal Wave Activation at Relativistic Magnetized Shocks (2026). [DOI](#)

- 2026MF: Energy Partition in Collisionless Counterstreaming Plasmas (2026). [DOI](#)
- 2026MK: Gravitational instability in partially ionized plasmas: A two-fluid approach (2026). [DOI](#)
- 2026PDD: Ion-Acoustic Shock Waves in Magnetized Plasma With Relativistically Degenerate Electrons and Positrons (2026). [DOI](#)
- 2026SCSS: Weibel Instability in Collisionless Plasmas Across Astrophysical and Laboratory Shocks (2026). [DOI](#)
- 2026SE: Global Non-axisymmetric Hall Instabilities in a Rotating Plasma (2026). [DOI](#)
- 2026STCR: The PLUTO code on GPUs: Offloading Lagrangian Particle methods (2026). [DOI](#)
- 2026TK: GPU-Native Compressed Neighbor Lists with a Space-Filling-Curve Data Layout (2026). [DOI](#)
- 2026ZHWD: Collisionless ion-electron energy exchange in magnetized shocks (2026). [DOI](#)
- 2025BBEG: Photon Accelerator in Magnetized Electron-Ion Plasma (2025). [DOI](#)
- 2025CHHM: Measurement of Ion Acceleration and Diffusion in a Laser-Driven Magnetized Plasma (2025). [DOI](#)
- 2025FKF: Unveiling a New β Scaling of the Tearing Instability in Weakly Collisional Plasmas (2025). [DOI](#)
- 2025GAST: MHD waves with mixed properties/Alfvén waves with pressure variations: a review (2025). [DOI](#)
- 2025HBCC: Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module (2025). [DOI](#)
- 2025HBS: Magnetohydrodynamic-guiding-center-particle-in-cell Method for Multiscale Plasma Kinetic Simulations (2025). [DOI](#)
- 2025IMCR: Turbulence and Magnetic Reconnection in Relativistic Multispecies Plasmas (2025). [DOI](#)
- 2025L2: Magnetised turbulent plasmas as high-energy particle accelerators (2025). [DOI](#)
- 2025LZGY: Studying the properties of reconnection-driven turbulence (2025). [DOI](#)
- 2025LZL: Suppression of inverse magnetic energy transfer in collisionless marginally magnetized plasmas (2025). [DOI](#)
- 2025RMBC: UK White Paper on Magnetic Reconnection (2025). [DOI](#)
- 2025VGRB: Roadmap for warm dense matter physics (2025). [DOI](#)
- 2024HM: Rugged Magneto-Hydrodynamic Invariants in Weakly Collisional Plasma Turbulence: Two-Dimensional Hybrid Simulation Results (2024). [DOI](#)
- 2024IMSC: Long-lived Equilibria in Kinetic Astrophysical Plasma Turbulence (2024). [DOI](#)

Numerical Relativity

- 2026CSY: On the convergence of the discontinuous Galerkin scheme for Einstein-scalar equations (2026). [DOI](#)

Nuclear Physics

- 2026RLLW: Research on the advanced GPU-based Monte Carlo neutron transport methods (2026). [DOI](#)
- 2026RS2: Science Of Nuclear Fusion: Insights and Ideas (2026). [DOI](#)
- 2026WEK: Adaptive mesh $h(p)$ -refinement of a discontinuous Bubnov–Galerkin isogeometric analysis spatial discretisation of the first-order form of the neutron transport equation with goal-based error measures and diffusion acceleration (2026). [DOI](#)
- 2026WLWC: Revisiting p-¹¹B Fusion: Updated Cross-sections, Reactivity, and Energy Balance (2026). [DOI](#)
- 2026ZGZ: Predictor-corrector discrete unified gas kinetic scheme for Boltzmann transport models with non-conservative collision operators (2026). [DOI](#)
- 2025AA: Thermalization times and scaling laws for alpha particles and hydrogen isotopes in Tokamak plasmas (2025). [DOI](#)

Quantum Physics

- 2026BKKG: H-NESSi: The hierarchical non-equilibrium systems simulation package (2026). [DOI](#)
- 2026GSNE: High Field Diamond Magnetometry Towards Tokamak Diagnostics (2026). [DOI](#)
- 2026HVFO: Non-unitary time evolution via the Chebyshev expansion method (2026). [DOI](#)
- 2026KNB: Real-time time-dependent density functional theory for high-energy density physics (2026). [DOI](#)
- 2026MC: Extended phase-space symplectic integration for electron dynamics (2026). [DOI](#)
- 2026NDLB: Particle-In-Cell Simulations of Quantum Plasmas (2026). [DOI](#)

- 2026TG: Object-oriented programming as a tool for constructing high-order quantum-kinetic BBGKY equations (2026). [DOI](#)
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- 2025KPPT: Probing the localization effects in Krylov basis (2025). [DOI](#)
- 2025SZ: Prefix Sums via Kronecker Products (2025). [DOI](#)

Condensed Matter Physics

- 2026BDJZ: Discovering new phases via computing second-order stationary states of Landau–Brazovskii model (2026). [DOI](#)
- 2026BKKG: H-NESSi: The hierarchical non-equilibrium systems simulation package (2026). [DOI](#)
- 2026GCYZ: Energy-stable compact exponential time differencing algorithm for phase field models (2026). [DOI](#)
- 2026NDLB: Particle-In-Cell Simulations of Quantum Plasmas (2026). [DOI](#)
- 2026ST: Numerical Efficiency of Explicit Time Integrators for Phase-Field Models (2026). [DOI](#)
- 2026X: A New Fractional Step Structure Preserving Method for The Landau-Lifshitz-Gilbert Equation (2026). [DOI](#)

Molecular Dynamics

- 2026H: Drifting to Boltzmann: Million-Fold Acceleration in Boltzmann Sampling with Force-Guided Drifting (2026). [DOI](#)
- 2026LRDQ: Observation of stopping power reduction at strong ion-plasma coupling (2026). [DOI](#)
- 2026MC: Extended phase-space symplectic integration for electron dynamics (2026). [DOI](#)
- 2026RF: Sampling Boltzmann distributions via normalizing flow approximation of transport maps (2026). [DOI](#)
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Machine Learning

- 2026B4: Routes towards an effective AI in CFD: an epistemological and technical perspective (2026). [DOI](#)
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- 2026BEAV: PENCO: A Physics–Energy–Numerics–Consistent Operator for 3D phase field modeling (2026). [DOI](#)
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- 2026KDBK: Surrogate models study for laser-plasma accelerator electron source design through numerical optimisation (2026). [DOI](#)
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- 2026LHWP: Time-embedded convolutional neural networks for modeling plasma heat transport (2026). [DOI](#)
- 2026LJLH: FourierSpecNet: Neural collision operator approximation inspired by the Fourier spectral method for solving the Boltzmann equation (2026). [DOI](#)
- 2026MHLH: Batch Bayesian Optimization of Attosecond Betatron Pulses from Laser Wakefield Acceleration (2026). [DOI](#)
- 2026NHGN: PanoMHD: Multimodal Modelling of Plasma Dynamics towards Tokamak Control (2026). [DOI](#)
- 2026NPSB: AMORE: Adaptive multi-output operator network for stiff chemical kinetics (2026). [DOI](#)
- 2026RF: Sampling Boltzmann distributions via normalizing flow approximation of transport maps (2026). [DOI](#)
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- 2026S5: FastLSQ: Solving PDEs in One Shot via Fourier Features with Exact Analytical Derivatives (2026). [DOI](#)
- 2026SC: Physics- and geometry-aware spatio-spectral graph neural operator for time-independent and time-dependent PDEs (2026). [DOI](#)
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- 2025MKK: Theoretical Foundations of GPU-Native Compilation for Rapid Code Iteration (2025). [DOI](#)
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- 2024PS: A path-dependent PDE solver based on signature kernels (2024). [DOI](#)

High-Performance Computing

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 - 2026EAKU: Multi-Partner Project: Multi-GPU Performance Portability Analysis for CFD Simulations at Scale (2026). [DOI](#)
 - 2026ECM: NektarIR: A Domain-Specific Compiler for High-Order Finite Element Operations on Heterogeneous Hardware (2026). [DOI](#)
 - 2026FCSJ: Conservative Velocity Mappings for Discontinuous Galerkin Kinetics (2026). [DOI](#)
 - 2026FHWS: A HIP-Compatible Accelerator Backend for Fourier-Bessel Particle-in-Cell Simulations on CPU/DCU Heterogeneous Clusters (2026). [DOI](#)
 - 2026GWSY: Do We Need Tensor Cores for Stencil Computations? (2026). [DOI](#)
 - 2026GYWG: COREFL: An open-source GPU-accelerated high-fidelity solver for compressible reactive flows on generalized curvilinear coordinates (2026). [DOI](#)
 - 2026HFBH: High-throughput full- f gyrokinetics of the tokamak boundary (2026). [DOI](#)
 - 2026HGMB: SpCCL: A Sparsity-Aware Collective Communication Library for GPU Platforms (2026). [DOI](#)
 - 2026HXLW: A Structure-Aware Irregular Blocking Method for Sparse LU Factorization (2026). [DOI](#)
 - 2026JJPH: Modeling of Relativistic Plasmas with a Conservative Discontinuous Galerkin Method (2026). [DOI](#)
 - 2026K2: kalypso: A performance portable platform for compressible hydrodynamics simulations using adaptive mesh refinement (2026). [DOI](#)
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 - 2026KNB: Real-time time-dependent density functional theory for high-energy density physics (2026). [DOI](#)
 - 2026KRR: Efficient numerical strategies for entropy-regularized semi-discrete optimal transport (2026). [DOI](#)
 - 2026LC: Multi-GPU acceleration of PALABOS fluid solver using C++ standard parallelism (2026). [DOI](#)
 - 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
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 - 2026P2: moljax: GPU-accelerated method of lines for stiff reaction-diffusion PDEs with FFT preconditioning (2026). [DOI](#)
 - 2026QSW: A fast Fourier spectral method for wave kinetic equation (2026). [DOI](#)
 - 2026RARB: New Time Integrators and Capabilities in SUNDIALS Versions 6.2.0–7.4.0 (2026). [DOI](#)
 - 2026RLLW: Research on the advanced GPU-based Monte Carlo neutron transport methods (2026). [DOI](#)
 - 2026S7: Adaptive multigrid for high-order discontinuous Galerkin methods based on the full approximation scheme (2026). [DOI](#)
 - 2026S8: The benefits and challenges of explicit memory management in OpenMP Target GPU offloading (2026). [DOI](#)
 - 2026SH: Performance of a high-order MPI-Kokkos accelerated fluid solver (2026). [DOI](#)

- 2026SKIS: Field conserving adaptive mesh refinement (AMR) scheme on massively parallel adaptive octree meshes (2026). [DOI](#)
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- 2026VRDK: Load Balanced Parallel Node Generation for Meshless Numerical Methods (2026). [DOI](#)
- 2026WCMT: Integrating High Performance In-Memory Data Streaming and In-Situ Visualization in Hybrid MPI+OpenMP PIC MC Simulations Towards Exascale (2026). [DOI](#)
- 2026WTCJ: Multi-GPU Hybrid Particle-in-Cell Monte Carlo Simulations for Exascale Computing Systems (2026). [DOI](#)
- 2026XM: Unified Interface Flux Evaluation in a General Discontinuous Galerkin Spectral Element Framework (2026). [DOI](#)
- 2026YCW: A Fast Fourier Spectral Method for the Linearized Boltzmann Collision Operator (2026). [DOI](#)
- 2026YZLY: TokaGLINT: A Scalable GPU-Tailored Implicit Solver for Full 3D Tokamak Electromagnetic Simulations (2026). [DOI](#)
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- 2025ADFM: Analysis of Floating-Point Matrix Multiplication Computed via Integer Arithmetic (2025). [DOI](#)
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- 2025C: HYLU: Hybrid Parallel Sparse LU Factorization (2025). [DOI](#)
- 2025DDGG: Semi-implicit continuum kinetic modeling of weakly collisional parallel transport in a magnetic mirror (2025). [DOI](#)
- 2025EDLN: An Asynchronous Many-Task Algorithm for Unstructured S_N Transport on Shared Memory Systems (2025). [DOI](#)
- 2025HBCC: Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module (2025). [DOI](#)
- 2025HBMG: GPU-Initiated Networking for NCCL (2025). [DOI](#)
- 2025HSBJ: Demystifying NCCL: An In-depth Analysis of GPU Communication Protocols and Algorithms (2025). [DOI](#)
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- 2025MKK: Theoretical Foundations of GPU-Native Compilation for Rapid Code Iteration (2025). [DOI](#)
- 2025PLGC: A scalable high-order multigrid-FFT Poisson solver for unbounded domains on adaptive multiresolution grids (2025). [DOI](#)
- 2025TOMT: pyGinkgo: A Sparse Linear Algebra Operator Framework for Python (2025). [DOI](#)
- 2025TRBF: Adaptive Multidimensional Quadrature on Multi-GPU Systems (2025). [DOI](#)
- 2025WMKK: cMPI: Using CXL Memory Sharing for MPI One-Sided and Two-Sided Inter-Node Communications (2025). [DOI](#)
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Parallel Computing

- 2027K: Acceleration of multicomponent multiple-precision arithmetic with branch-free algorithms and SIMD vectorization (2027). [DOI](#)
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- 2026DSMG: Scaling State-Space Models on Multiple GPUs with Tensor Parallelism (2026). [DOI](#)
- 2026GXPS: Real-Time Plasma State Prediction via FPGA-Accelerated Quantized Recurrent Probabilistic Neural Networks (2026). [DOI](#)
- 2026HGMB: SpCCL: A Sparsity-Aware Collective Communication Library for GPU Platforms (2026). [DOI](#)
- 2026KKB: Layout-Agnostic MPI Abstraction for Distributed Computing in Modern C++ (2026). [DOI](#)
- 2026KS: Resolving Conflicts with Grace: Dynamically Concurrent Universality (2026). [DOI](#)
- 2026OFL: Space-efficient B-tree Implementation for Memory-Constrained Flash Embedded Devices (2026). [DOI](#)
- 2026RECB: Parallelized real-time physics codes for plasma control on DIII-D (2026). [DOI](#)
- 2026S8: The benefits and challenges of explicit memory management in OpenMP Target GPU offloading (2026). [DOI](#)
- 2025AB: Optimizing Allreduce Operations for Modern Heterogeneous Architectures with Multiple Processes per GPU (2025). [DOI](#)
- 2025HBMG: GPU-Initiated Networking for NCCL (2025). [DOI](#)
- 2025HSBJ: Demystifying NCCL: An In-depth Analysis of GPU Communication Protocols and Algorithms (2025). [DOI](#)
- 2025MAC: Parallel Spawning Strategies for Dynamic-Aware MPI Applications (2025). [DOI](#)

- 2025MBK: A New Execution Model and Executor for Adaptively Optimizing the Performance of Parallel Algorithms Using HPX Runtime System (2025). [DOI](#)
- 2025MKK: Theoretical Foundations of GPU-Native Compilation for Rapid Code Iteration (2025). [DOI](#)
- 2025WMKK: cMPI: Using CXL Memory Sharing for MPI One-Sided and Two-Sided Inter-Node Communications (2025). [DOI](#)
- 2025YCT: CARMA: Collocation-Aware Resource Manager (2025). [DOI](#)
- 2024RBIW: An extension of C++ with memory-centric specifications for HPC to reduce memory footprints and streamline MPI development (2024). [DOI](#)
- 2024UTIS: The Landscape of GPU-Centric Communication (2024). [DOI](#)

Scientific Software

- 2026ABBG: Lethe 1.0: An open-source parallel high-order computational fluid dynamics software framework for single and multiphase flows (2026). [DOI](#)
- 2026BB: Scalable Overset Computation Between a Forest-of-Octrees- and an Arbitrary Distributed Parallel Mesh (2026). [DOI](#)
- 2026BKKG: H-NESSi: The hierarchical non-equilibrium systems simulation package (2026). [DOI](#)
- 2026CBKS: JetSCI: A Hybrid JAX–PETSc Framework for Scalable Differentiable Simulation (2026). [DOI](#)
- 2026CES: Fast GPU Linear Algebra via Compile Time Expression Fusion (2026). [DOI](#)
- 2026D: GPU Acceleration and Portability of the TRIMEG Code for Gyrokinetic Plasma Simulations Using OpenMP (2026). [DOI](#)
- 2026DFTW: A Deterministic Ionization Algorithm for the OSIRIS Particle-in-Cell Framework (2026). [DOI](#)
- 2026DJUN: OpenEdge: A collaborative, open-source, multi-purpose direct simulation Monte Carlo for plasma simulation in magnetic fusion environments (2026). [DOI](#)
- 2026DMN: High-Order Structure-Preserving SBP Finite Difference Methods for the Vlasov–Maxwell System on Matrix-Free GPUs (2026). [DOI](#)
- 2026EAKU: Multi-Partner Project: Multi-GPU Performance Portability Analysis for CFD Simulations at Scale (2026). [DOI](#)
- 2026ECM: NektarIR: A Domain-Specific Compiler for High-Order Finite Element Operations on Heterogeneous Hardware (2026). [DOI](#)
- 2026FHWS: A HIP-Compatible Accelerator Backend for Fourier-Bessel Particle-in-Cell Simulations on CPU/DCU Heterogeneous Clusters (2026). [DOI](#)
- 2026FMMM: Code-Verification Techniques for Particle-in-Cell Simulations with Direct Simulation Monte Carlo Collisions (2026). [DOI](#)
- 2026FPKA: Structure Preservation Using Discrete Gradients in the Vlasov-Poisson-Landau System (2026). [DOI](#)
- 2026FUSJ: Extending the gyrokinetic code GENE-X to stellarators (2026). [DOI](#)
- 2026FZHW: Fire: An open-source adaptive mesh refinement solver for supersonic reacting flows (2026). [DOI](#)
- 2026GYWG: COREFL: An open-source GPU-accelerated high-fidelity solver for compressible reactive flows on generalized curvilinear coordinates (2026). [DOI](#)
- 2026K2: kalypso: A performance portable platform for compressible hydrodynamics simulations using adaptive mesh refinement (2026). [DOI](#)
- 2026KBK: A Taxonomy of Numerical Differentiation Methods (2026). [DOI](#)
- 2026KJWC: One-shot acceleration of transient PDE solvers via online-learned preconditioners (2026). [DOI](#)
- 2026KSG: T3FF: Toroidal 3-dimensional full- f full- k code for isothermal gyrofluid drift-Alfvén turbulence (2026). [DOI](#)
- 2026KT: Partially Ionized Plasma Physics and Technological Applications (2026). [DOI](#)
- 2026LC: Multi-GPU acceleration of PALABOS fluid solver using C++ standard parallelism (2026). [DOI](#)
- 2026LDGB: Simulating ultrarelativistic beam-plasma instabilities with a quasistatic particle-in-cell code (2026). [DOI](#)
- 2026LHD: jaxdae: A JAX-native Differentiable Solver for Differential-Algebraic Equations in Coupled Multi-physics (2026). [DOI](#)
- 2026LJF: cuGMEC: A high-performance code for gyrokinetic-MHD hybrid simulation on GPUs with CUDA C++ (2026). [DOI](#)
- 2026LMHL: TRIMEG-GKX: An electromagnetic gyrokinetic particle code with a piecewise field-aligned finite

- element method for micro- and macro-instability studies in tokamak core plasmas (2026). [DOI](#)
- 2026LYXB: Sub-Second Collisionless Gyrokinetic Eigenvalue Solutions via Orbit-Invariant Decomposition (2026). [DOI](#)
 - 2026MHGR: Trilinos: Enabling Scientific Computing across Diverse Hardware Architectures at Scale (2026). [DOI](#)
 - 2026MTWG: A GPU-accelerated matrix-free FAS multigrid solver for Navier–Stokes equations with memory-efficient implementations (2026). [DOI](#)
 - 2026ODDE: STORM: Scrape-off layer turbulence in tokamak fusion reactors (2026). [DOI](#)
 - 2026P2: moljax: GPU-accelerated method of lines for stiff reaction-diffusion PDEs with FFT preconditioning (2026). [DOI](#)
 - 2026RARB: New Time Integrators and Capabilities in SUNDIALS Versions 6.2.0–7.4.0 (2026). [DOI](#)
 - 2026RECB: Parallelized real-time physics codes for plasma control on DIII-D (2026). [DOI](#)
 - 2026S2: Hybrid PIC-fluid model for numerical simulation of laser-plasma interaction (2026). [DOI](#)
 - 2026SH: Performance of a high-order MPI-Kokkos accelerated fluid solver (2026). [DOI](#)
 - 2026SLMW: ORB5X: a performance-portable global electromagnetic gyrokinetic PIC code in C++/Kokkos built using Agentic AI (2026). [DOI](#)
 - 2026TG: Object-oriented programming as a tool for constructing high-order quantum-kinetic BBGKY equations (2026). [DOI](#)
 - 2026TKCD: Accelerating High-Order Finite Element Simulations at Extreme Scale with FP64 Tensor Cores (2026). [DOI](#)
 - 2026TOMP: Verification of a hybrid kinetic-gyrokinetic model using the advanced semi-Lagrange code ssV (2026). [DOI](#)
 - 2026TSRC: A deterministic verification strategy for multidimensional gyrokinetic particle-in-cell algorithms using the method of manufactured solutions (2026). [DOI](#)
 - 2026VBGM: Local Cubic Spline Interpolation for Vlasov-Type Equations on a Multi-Patch Geometry (2026). [DOI](#)
 - 2026WTCJ: Multi-GPU Hybrid Particle-in-Cell Monte Carlo Simulations for Exascale Computing Systems (2026). [DOI](#)
 - 2026XM: Unified Interface Flux Evaluation in a General Discontinuous Galerkin Spectral Element Framework (2026). [DOI](#)
 - 2026ZXLM: VEQ: a fast parametric Grad–Shafranov solver for fixed-boundary tokamak equilibria with flexible source profiles (2026). [DOI](#)
 - 2026ZZLW: AlgoPlasma: Open Algorithms for Plasma Modeling (2026). [DOI](#)
 - 2025A: Formal Verification of COO to CSR Sparse Matrix Conversion (Invited Paper) (2025). [DOI](#)
 - 2025BOMG: π -PIC: a framework for modular particle-in-cell developments and simulations (2025). [DOI](#)
 - 2025C: HYL: Hybrid Parallel Sparse LU Factorization (2025). [DOI](#)
 - 2025CJBL: Case study of a differentiable heterogeneous multiphysics solver for a nuclear fusion application (2025). [DOI](#)
 - 2025GST: Fast and accurate computation of classical Gaussian quadratures (2025). [DOI](#)
 - 2025HBCC: Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module (2025). [DOI](#)
 - 2025KM: Extending Irksome: Improvements in Automated Runge–Kutta Time Stepping for Finite Element Methods (2025). [DOI](#)
 - 2025MBK: A New Execution Model and Executor for Adaptively Optimizing the Performance of Parallel Algorithms Using HPX Runtime System (2025). [DOI](#)
 - 2025MJLT: JAX-in-Cell: A Differentiable Particle-in-Cell Code for Plasma Physics Applications (2025). [DOI](#)
 - 2025MSAK: Plasma position constrained free-boundary MHD equilibrium in Tokamaks using pyIPREQ (2025). [DOI](#)
 - 2025S2: The Numerics of VMEC++ (2025). [DOI](#)
 - 2025SC: Armadillo: an Efficient Framework for Numerical Linear Algebra (2025). [DOI](#)
 - 2025TOMT: pyGinkgo: A Sparse Linear Algebra Operator Framework for Python (2025). [DOI](#)
 - 2023SKYI: Integrating Odeint Time Stepping into OpenFPM for Distributed and GPU Accelerated Numerical Solvers (2023). [DOI](#)
 - 2021FKM: Irksome: Automating Runge–Kutta Time-stepping for Finite Element Methods (2021). [DOI](#)

- 2007CHPG: Collective communication: theory, practice, and experience (2007). [DOI](#)
- 1999HW: Stiff differential equations solved by Radau methods (1999). [DOI](#)

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- 2026ACGS: Concepts for designing modern C++ interfaces for MPI (2026). [DOI](#)
- 2026BBCC: Growing Mathlib: maintenance of a large scale mathematical library (2026). [DOI](#)
- 2026KKB: Layout-Agnostic MPI Abstraction for Distributed Computing in Modern C++ (2026). [DOI](#)
- 2026LC: Multi-GPU acceleration of PALABOS fluid solver using C++ standard parallelism (2026). [DOI](#)
- 2026S8: The benefits and challenges of explicit memory management in OpenMP Target GPU offloading (2026). [DOI](#)
- 2026SLMW: ORB5X: a performance-portable global electromagnetic gyrokinetic PIC code in C++/Kokkos built using Agentic AI (2026). [DOI](#)
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- 2024RBIW: An extension of C++ with memory-centric specifications for HPC to reduce memory footprints and streamline MPI development (2024). [DOI](#)

Formal Methods

- 2026BBCC: Growing Mathlib: maintenance of a large scale mathematical library (2026). [DOI](#)
- 2026GHJ: BEACONS: Bounded-Error, Algebraically-Composable Neural Solvers for Partial Differential Equations (2026). [DOI](#)
- 2026KS: Resolving Conflicts with Grace: Dynamically Concurrent Universality (2026). [DOI](#)
- 2025A: Formal Verification of COO to CSR Sparse Matrix Conversion (Invited Paper) (2025). [DOI](#)

2027GLWN

ChatGPT (September 2026)

Summary

This paper introduces sub-cell summation-by-parts (SBP) operators to place overset-grid coupling on a provable conservation and energy-stability foundation. Ordinary SBP differentiation mimics integration by parts over a whole cell; the new operators satisfy corresponding identities on overlapping sub-cells, enabling a discrete analogue of recent continuous well-posedness estimates. The authors establish existence conditions, give constructive procedures, and combine the operators with simultaneous-approximation-term interface penalties. For fixed one-dimensional overset domains whose overlap does not change with time or refinement, the resulting semidiscretizations are conservative and energy stable. Experiments for linear advection, inviscid Burgers, Maxwell, and compressible Euler equations demonstrate high-order accuracy and substantially reduced long-time error growth relative to baseline overset coupling. The conditions are sufficient rather than necessary, and the proof is limited to fixed one-dimensional overlaps; multidimensional sub-cell operators, moving grids, and efficient implementations remain open problems.

Contributions

1. Defined SBP differentiation operators that reproduce integration by parts on prescribed sub-cells.
2. Proved existence criteria and supplied a procedure for constructing these operators.
3. Derived conservative, energy-stable overset coupling with sub-cell interface penalties.
4. Demonstrated high-order accuracy and improved long-time stability on four hyperbolic model classes.
5. Delineated the current guarantee to fixed one-dimensional overlaps and identified multidimensional extensions.

2027K

ChatGPT (July 2026)

Summary

This paper evaluates branch-free algorithms for fixed multiple-precision arithmetic represented by two, three, or four hardware floating-point components: double-double, triple-double, and quadruple-double precision. Error-free transformations and branch-free renormalization expose instruction-level parallelism to AVX2 on AMD EPYC and Neon on Qualcomm Snapdragon processors. Implementations are tested in real and complex Strassen matrix multiplication, real-coefficient polynomial evaluation at real and complex arguments, and a Durand–Kerner solver for a Chebyshev integration polynomial. The benchmarks preserve the intended significant-digit ranges while showing that branch elimination is precision- and architecture-dependent. It slows or does not improve double-double arithmetic, but typically accelerates triple- and quadruple-double work; real matrix multiplication gains average factors of about 1.2–1.5 and 1.4–2.0, respectively. Combining branch-free arithmetic with SIMD gives roughly 1.47–1.61 speedups in the EPYC Chebyshev test. A three-real-multiplication complex formulation also becomes increasingly advantageous at higher precision and larger sizes.

Contributions

1. Integrated branch-free triple- and quadruple-component arithmetic with AVX2 and Arm Neon vectorization.
2. Benchmarked real and complex matrix multiplication across x86 and Arm platforms with controlled numerical accuracy.
3. Quantified that branch-free conversion helps triple- and quadruple-double arithmetic but not double-double arithmetic.
4. Evaluated SIMD branch-free arithmetic in polynomial evaluation and a Durand–Kerner root solver.
5. Showed that three-real-multiplication complex arithmetic improves relative efficiency as precision and problem size grow.

2026A

ChatGPT (July 2026)

Summary

Semi-Lagrangian discontinuous Galerkin (SLDG) methods can take large time steps for kinetic transport, but their usual precomputed overlap matrices assume uniform cell widths. This paper extends SLDG to adaptively refined velocity meshes and from one to three velocity dimensions for the Vlasov–Poisson system. A hybrid sweep uses precomputed level-specific overlap matrices on conforming cells and evaluates generalized overlap integrals only at nonconforming refinement interfaces. A compressed-sparse-row pencil representation organizes dimensionally split sweeps, including weighted accumulation when coarse cells occur in multiple pencils. Tensor-product discontinuous Galerkin elements on PETSc hexahedral meshes provide the three-dimensional velocity discretization. A $1X + 3V$ Landau-damping test reproduces the expected damping behavior, conserves mass exactly, and shows convergence with polynomial degree and adaptive-refinement level. The work establishes the algorithmic machinery and baseline verification for adaptive high-dimensional SLDG; broader nonlinear plasma tests and performance/scalability characterization are not presented in the reported study.

Contributions

1. Extended semi-Lagrangian discontinuous Galerkin transport to adaptively refined, nonuniform velocity meshes.
2. Introduced fast conforming-cell sweeps and on-demand generalized overlap integration at refinement boundaries.
3. Designed a compressed-sparse-row pencil structure for dimensionally split adaptive sweeps.
4. Generalized the method to three velocity dimensions using tensor-product PETSc finite elements.
5. Verified damping rates, exact mass conservation, and convergence on a $1X + 3V$ Landau-damping problem.

2026A2

ChatGPT (September 2026)

Summary

This theoretical study examines small-amplitude ion-acoustic shocks in a magnetized electron–ion plasma with anisotropic warm ions and a generalized (r, q) electron distribution. Two spectral parameters describe nonthermal electron velocity profiles, while Chew–Goldberger–Low double-adiabatic relations distinguish ion pressures parallel and perpendicular to the magnetic field. Reductive perturbation yields a Korteweg–de Vries–Burgers equation balancing nonlinearity, dispersion, and viscous dissipation; a hyperbolic-tangent construction supplies monotonic travelling shock profiles. The derived phase speed increases with the parallel ion-to-electron temperature ratio and the field-aligned propagation direction cosine, but is independent of perpendicular ion temperature at this order. In the reported parameter scans, shock amplitude decreases with increasing spectral indices, either ion temperature ratio, or the parallel direction cosine, and increases with magnetic-field strength or viscosity. These trends follow the results section and figures; the conclusion contains conflicting statements about the spectral-index dependence. The findings characterize sensitivity within a reduced fluid model and have no experimental or observational validation.

Contributions

1. Combines viscous ion-fluid dynamics, anisotropic double-adiabatic pressure laws, and a generalized electron density response with admissibility conditions $q > 1$ and $q(r + 1) > 5/2$.
2. Derives the phase-speed relation $\lambda = \pm l_z \sqrt{1/a + 3\sigma_{\parallel}}$, where a is the linear electron-density response coefficient, l_z is the field-aligned direction cosine, and $\sigma_{\parallel}$ is the parallel ion-to-electron temperature ratio.
3. Obtains explicit nonlinear, dispersive, and dissipative coefficients for the Korteweg–de Vries–Burgers equation and constructs a quadratic hyperbolic-tangent travelling-shock solution.
4. Quantifies decreasing shock amplitudes with increasing spectral parameters and either ion temperature ratio in the reported scans, including comparisons across both signs of temperature anisotropy.
5. Characterizes the competing influences of propagation geometry, magnetic-field strength, and viscosity: the scanned shocks weaken with increasing l_z and strengthen with increasing ion-cyclotron-to-ion-plasma frequency ratio or viscosity.

2026AB

ChatGPT (July 2026)

Summary

The paper develops locally and globally conservative discontinuous Galerkin tracer transport for compatible finite-element dynamical cores. Rather than advecting a mixing ratio m , it advances tracer density ρm alongside dry density ρ after mapping both fields into a shared discontinuous transport space. Conservative Galerkin projection, recovery, and injection preserve tracer mass when m is either co-located with or vertically staggered from ρ on a Charney–Phillips grid. Recovered-space transport provides second-order accuracy for lowest-order elements, while embedded DG handles vertically continuous thermodynamic spaces. A consistency correction preserves spatially constant mixing ratios. A mean-mixing-ratio limiter blends the higher-order field with a mass-equivalent lowest-order projection to enforce non-negativity for co-located spaces. Gusto tests on spherical and vertical-slice transport, terminator toy chemistry, and a moist rising bubble show near-machine-precision mass conservation without degrading convergence, whereas advective-form controls accumulate larger mass errors. Extending the limiter to vertically staggered spaces and enforcing full monotonicity remain future work.

Contributions

1. Formulated DG transport for tracer density ρm , giving conservative evolution of dry density and tracer mass.
2. Defined conservative and consistent projection, recovery, and injection operators for density and mixing ratios in different spaces.

3. Combined recovered-space and embedded-DG mappings for co-located and vertically staggered tracer configurations.
4. Developed a mean-mixing-ratio limiter that preserves cell tracer mass while enforcing non-negativity for co-located spaces.
5. Validated conservation, consistency, accuracy, chemistry coupling, and moist dynamical-core behavior in Gusto tests.

2026ABBB

ChatGPT (July 2026)

Summary

This workshop report turns cross-disciplinary discussions between numerical linear algebra and theoretical computer science into precise research problems in matrix computation. For linear systems, it proposes parameterized PDE-derived benchmarks and asks for sharper correctness and finite-precision theories for multigrid, conjugate-gradient, sketch-and-project, GMRES, and preconditioning. Its eigenvalue section seeks deterministic substitutes for random pseudospectral shattering, near-optimal precision bounds for diagonalization, and better understanding of Ritz values and invariant subspaces. The low-rank section studies when greedy or randomized column selection and rank-revealing factorizations achieve the strong performance observed in practice, especially for structured matrices. The sketching section separates the weaker notion of subspace injection from embedding, asks whether injections suffice for more numerical tasks, and records near-sharp conjectures for sparse and rerandomized Hadamard maps. The final section extends the agenda to tensor-network approximation, eigenvector-dependent nonlinear eigenproblems from local Hamiltonians, and matrix-sign approximants constrained by multiplication count. Rather than claiming solutions, the report contributes a technically motivated, jointly negotiated agenda with concrete complexity, conditioning, approximation, and precision targets.

Contributions

1. Formulated a linear-solver agenda around scalable PDE benchmarks and unresolved correctness and convergence questions for multigrid and Krylov methods.
2. Specified eigenvalue challenges involving deterministic pseudospectral shattering, optimal-precision diagonalization, and Ritz and invariant-subspace guarantees.
3. Framed structural conditions under which column subset selection and rank-revealing factorizations should be provably quasi-optimal.
4. Developed open questions for subspace injections, optimal sparse sketches, and rerandomized Hadamard transforms.
5. Extended the program to tensor approximation, nonlinear Hamiltonian eigenproblems, and multiplication-budget-optimal matrix-sign polynomials.

2026ABBG

ChatGPT (July 2026)

Summary

This work presents a C++ finite-element framework for large-scale computational fluid dynamics, discrete-particle simulation, and coupled multiphysics. Built on deal.II, it combines stabilized continuous Galerkin formulations with distributed-memory parallelism, dynamic mesh adaptation, load balancing, and high-order isoparametric elements. Three fluid linear-algebra paths—monolithic matrix-based, block matrix-based, and matrix-free—support problems ranging from flexible tetrahedral meshes to hexahedral simulations exceeding one billion unknowns. The physics modules cover incompressible Newtonian and generalized-Newtonian flow, heat transfer, tracer transport, Volume-of-Fluid and Cahn–Hilliard two-phase models, and discrete element methods. Particle coupling supports unresolved CFD–DEM for millions of spheres and resolved immersed-boundary CFD–DEM for fewer complex particles, while a Nitsche immersed-boundary formulation handles moving geometries. A common parameter-file workflow and modular solver

architecture let users combine physics and extend models. Verification and application examples address turbulent Taylor–Couette flow, granular discharge, rising bubbles, fluidized beds, settling particles, and conjugate heat transfer, comparing against analytical, experimental, or reference solutions. Engineering maturity is supported by more than 450 automated tests and 74 documented examples with meshes and post-processing scripts. The framework therefore couples research-grade high-order numerics, scalable execution, multiphase breadth, and reproducible onboarding in one open-source platform.

Contributions

1. Delivers an open-source, parallel high-order finite-element framework for chemical and manufacturing flow simulations on the deal.II infrastructure.
2. Unifies single-phase CFD, heat transfer, fluid-fluid models, DEM, and resolved and unresolved CFD–DEM within a modular multiphysics architecture.
3. Provides matrix-based, block matrix-based, and matrix-free solver paths, with the latter demonstrated on problems exceeding one billion unknowns.
4. Integrates dynamic mesh adaptation for hexahedral CFD meshes and extends parallel load balancing to DEM and coupled CFD–DEM workflows.
5. Establishes reproducibility and extensibility through more than 450 automated tests, 74 documented examples, and validations against analytical, experimental, and reference results.

2026ABL

ChatGPT (July 2026)

Summary

The PAMPA algorithm extends active-flux and residual-distribution ideas to hyperbolic problems on arbitrary polygonal control volumes using virtual finite element concepts. The method keeps point values on element boundaries and cell averages as degrees of freedom, but avoids explicitly constructing local polynomial bases; gradients are instead discretized from virtual-element degrees of freedom. This produces a globally continuous, formally third-order approximation with a diagonal-mass-like update structure suited to polygonal meshes. For nonlinear systems and discontinuities, the paper adds stabilization through either a MOOD-type a posteriori strategy or a monolithic convex limiter that blends high- and low-order schemes while preserving invariant-domain constraints. Numerical tests include scalar problems, acoustics, and the two-dimensional Euler equations on triangular, quadrilateral, and agglomerated polygonal meshes. The results support accuracy and robustness on complex meshes, while the authors identify higher-than-third-order extensions and mesh adaptation as future work.

Contributions

1. Generalized active-flux-style point-average methods to arbitrary polygonal meshes through virtual finite element ideas.
2. Built the PAMPA discretization without explicit local polynomial basis functions.
3. Combined globally continuous point values with cell averages for third-order hyperbolic schemes.
4. Added MOOD and bound-preserving convex limiting strategies for shocks and invariant domains.
5. Demonstrated the method on scalar, acoustic, and Euler benchmarks over multiple mesh types.

2026ABP

ChatGPT (September 2026)

Summary

This paper extends Galley’s doubled-variable initial-value action into a canonical Hamiltonian framework for smooth nonconservative systems. Two copies of each coordinate are coupled through an antisymmetric term K ; identical initial data and final-time equality conditions make the variational problem well posed, while uniqueness places both on-shell

paths on the physical-limit submanifold. A Legendre transform produces Hamiltonian dynamics on a doubled, $4N$ -dimensional symplectic phase space. Liouville volume is preserved globally even when the physical $2N$ -dimensional slice contracts, as for a damped oscillator. Terms beyond first order in virtual variables do not affect physical motion, yielding the exact linear Lie-form Hamiltonian

$$A = \pi_- \cdot V(q_+, \pi_+, t) - q_- \cdot G(q_+, \pi_+, t).$$

A canonical gauge freedom shifts physical momentum by an arbitrary regular function without changing trajectories. Consequently, a smooth second-order initial-value problem $\ddot{q} = U(q, \dot{q}, t)$ can be embedded by choosing any locally invertible momentum map $P(q, \dot{q}, t)$. Earth–Moon tides, an adiabatic piston, and a charged particle illustrate the construction.

Contributions

1. Clarified the boundary conditions and uniqueness assumptions that produce the physical-limit trajectory.
2. Derived the doubled Hamiltonian system and its extended Poisson bracket through an ordinary Legendre transform.
3. Proved that higher-order virtual terms are dynamically irrelevant and obtained equivalent linear Lie forms.
4. Identified canonical momentum-gauge transformations that preserve physical trajectories.
5. Constructed explicit Lagrangian and Hamiltonian embeddings for general smooth second-order initial-value problems and applied them to three physical examples.

2026ABY

ChatGPT (July 2026)

Summary

This study uses nonlinear MHD–particle-in-cell simulations to examine how fusion-born alpha particles affect a helical core, the saturated $m/n = 1/1$ kink or quasi-interchange state expected in low-shear ITER-scale plasmas with $q \gtrsim 1$. Increasing alpha pressure enlarges the magnetic-axis displacement. At ITER-relevant central alpha pressure, $\beta_\alpha(0) \lesssim 1\%$, the alpha contribution behaves approximately like additional total MHD pressure and does not notably flatten the pressure profiles, so the helical core retains omnigenity. Beyond this range, displacement continues growing until an observed limit near $\beta_\alpha(0) = 3\%$ in the reference case, where bulk and alpha profiles partially flatten. The simulations also reveal a secondary resistive pressure-driven instability localized in the compressed-flux region after core formation. Its coherent spectrum of short-wavelength harmonics may promote magnetic chaos, but the authors caution that the present model cannot represent this mode adequately enough for predictive confinement conclusions.

Contributions

1. Quantified alpha-pressure effects on helical-core displacement in ITER-scale MHD–PIC simulations.
2. Showed that ITER-relevant alpha pressure can be approximated as part of total MHD pressure.
3. Found preserved pressure profiles and omnigenity for $\beta_\alpha(0) \lesssim 1\%$.
4. Identified displacement saturation and partial profile flattening near $\beta_\alpha(0) = 3\%$ in the reference case.
5. Characterized a secondary resistive instability that may generate magnetic chaos after core formation.

2026AC

Claude (September 2026)

Summary

The authors propose low-rank Anderson acceleration (lrAA) for nonlinear matrix equations $G(X) = X$, $X \in \mathbb{R}^{m \times n}$, from local-stencil discretizations of two-dimensional PDEs with approximately low-rank solutions, targeting $O(m + n)$ rather than $O(mn)$ cost and storage. As background, Anderson acceleration (AA) combines the previous $\min(k, \hat{m}) + 1$ iterates and residuals with weights minimizing a linearized residual. lrAA keeps iterates, function values, and differences

as truncated-SVD factors, solves the least-squares problem on them, and schedules the truncation tolerance as $\epsilon_G = \theta \rho_k$, with ρ_k an approximate residual norm, so ranks grow gradually. Nonlinear terms are compressed by Cross-DEIM, a new adaptive cross approximation (reconstruction from selected rows and columns) alternating discrete-empirical-interpolation index selection with stabilized cross approximation until both the update and an error indicator meet the tolerance. No convergence analysis is given. Tests cover matrix approximation, Laplace’s equation, and Bratu, Monge–Ampère, Allen–Cahn, and level-set advection problems; for Laplace, unpreconditioned Bratu, and Monge–Ampère, intermediate ranks stay near the converged rank, and for Laplace lrAA needs fewer iterations than full-rank AA. The authors note that Cross-DEIM can fail on adversarial matrices and its cost depends on initialization under slow singular-value decay, that accurate least squares need a tiny rounding tolerance, and that finer meshes show some rank overshoot.

Contributions

1. Designed Cross-DEIM (Algorithm 3.1) to merge new QDEIM (pivoted-QR DEIM of Drmač and Gugercin) row and column indices from the current singular vectors into the previous index sets, add one random index to a set that fails to grow, drop rows and columns whose pivoted-QR diagonal entries fall below 10^{-12} , stop once the change between consecutive approximations and the indicator $\min(\eta_1(1 + \eta_2), \eta_2(1 + \eta_1))S_{\min}$, built on estimates of Donello et al. with $S_{\min}$ the smallest current singular value, both fall below ϵ , and truncate to the smallest rank whose discarded squared singular values sum below ϵ^2 ; its stabilized cross step (Algorithm 3.3), a version of that of Donello et al., solves $Q(\mathcal{I}, :)W = R$ with Q from a pivoted QR of $C = G(:, \mathcal{J})$, or the transposed system, and then takes the SVD of W .
2. Solved the lrAA least-squares problem (Algorithm 2.4) through column-pivoted QR factorizations of the stacked factors $[U_1, \dots, U_s]$ and $[V_1, \dots, V_s]$ of the residual differences with right-hand side $\text{vec}(Q_1^T U_B S_B V_B^T Q_2)$, rounded residual differences at $\epsilon_F = 10^{-12}$ because larger values caused ill-conditioning and more iterations, recommended windows $\hat{m}$ of 3–5 because larger ones raise both iteration counts and least-squares cost, and replaced the rounding of the AA update by Cross-DEIM warm-started with the singular vectors of X_k , with every operation other than Cross-DEIM costing $O((m + n)r^2)$.
3. Benchmarked Cross-DEIM against the truncated SVD and adaptive cross approximation (ACA) with partial pivoting: over 100 random rank-one initializations for the 500×500 matrix $(|x_i + y_j|/2)^5$, sampled from a C^4 function with a nonsmooth feature along $x + y = 0$, ACA missed tolerances below about 10^{-3} and departed from the SVD above rank 10 while Cross-DEIM stayed close to the SVD, and for solid-body-rotated anisotropic Gaussian and $(|x + y|/2)^5$ matrices of size 500×500 at tolerance 10^{-2} , warm starts roughly halved the Cross-DEIM iterations and maximum intermediate ranks.
4. For Laplace’s equation with $m = n = 31$ and $\hat{m} = 5$, $\theta = 1$ ended at rank 5 against rank 10 of the truncated direct solution, $\theta = 0.5$ raised the rank stepwise to 10 in about 330 iterations, and a fixed tolerance 10^{-10} drove the rank to nearly full early and needed about 520 iterations; at $m = n = 63$ lrAA took about a quarter of the iterations of full-rank AA, and an exponential-sum preconditioner acting only on the factors gave convergence on 1023^2 and 4096^2 grids in about ten iterations.
5. Solved the Monge–Ampère equation with $f = 1/\sqrt{x^2 + y^2}$ and exact solution $u = \frac{2\sqrt{2}}{3}(x^2 + y^2)^{3/4}$ through the relaxed map $G = X + 0.9(H - X)$ built on Method 1 of Benamou et al. with $\theta = 0.25$: on 21, 61, 101, and 221 points per direction lrAA needed 109, 287, 443, and 675 iterations (final ranks 13, 18, 20, and 26) at tolerance 10^{-10} , against 1083 to 107388 Gauss–Seidel iterations reported by Benamou et al. at the stricter 10^{-14} , and 7 to 13 iterations with ranks 4 to 8 at the truncation-error-guided tolerance $0.01h_x$.

Comments

- Section 2.2: “Lines 9 to 11 find the solution to the least squares minimization problem using Algorithm 2.4 and update the solution as a linear combination of previous G_k . The rounding of sums of low-rank matrices used in line 9 is described in Algorithm 2.3”, but in Algorithm 2.2 line 9 is $\hat{m}_k = \min(\hat{m}, k)$, the rounding $\mathcal{T}_{\epsilon_F, r_{\max}}^{\text{round}}$ is applied in line 10, and the update is line 12, while the section’s references to lines 4, 7, 12, and 13 match the box; both references are evidently shifted by one line.
- Sections 2.2, 4.2.1, and 4.2.2: Section 2.2 states “In the computations in this paper we take $\theta \approx 0.1 - 0.5$ ”, and Section 4.2.1 finds $0.1 \leq \theta \leq 0.5$ effective while $\theta = 1$ “does not lead to a robust scheduling”, but the Bratu parameters in Section 4.2.2 are “TOL = 10^{-6} , $\hat{m} = 5$, $\theta = 0.9$ ”; unresolved.

- Algorithms 2.3, 2.4, and 3.3: with $[Q_1, R_1, \Pi_1] = \text{qr}(U)$ and $[Q_2, R_2, \Pi_2] = \text{qr}(V)$ in both 2.3 and 2.4, Algorithm 2.3 truncates the core $R_1 \Pi_1 S \Pi_2^T R_2^T$ while Algorithm 2.4 uses $a_k = \text{vec}(R_1 \Pi_1^T D_k \Pi_2 R_2^T)$, which differ for general permutations; the factorization $C P_C = Q R_C$ in line 4 of Algorithm 3.3 gives $U = Q_1 R_1 \Pi_1^T$ and makes Algorithm 2.4 consistent, but line 16 of the same algorithm, $r_C = \text{diag}(P_C^T R_C P_C)$, follows the opposite orientation, mapping pivoted diagonal entries back to the original columns only for involutive permutations; the intended convention is unresolved.
- Section 3: “for IrAA applications tested in this paper, we observe that the range of the Cross-DEIM iteration number is from 2 to 4”, but the per-step averages in Fig. 18 (level-set advection) are about 5, and up to about 6, for the nonlinear evaluation on the 256- and 512-point meshes, and those in Fig. 16 (Allen–Cahn) fall to about 1.3–1.6 for the rounding in early steps, so individual Cross-DEIM counts outside 2–4 occur.
- Algorithm 3.1, lines 26–30: the loop that removes redundant columns runs “for $l = 1, 2, \dots, |\mathcal{J}_k|$ ” while testing $(r_C)_l$ and removing “element l from $\mathcal{J}_k$ ” (“Remove redundant columns in $C = G(:, \mathcal{J}_k)$ ”), although $r_C = \text{diag}(P_C^T R_C P_C)$ of Algorithm 3.3 has one entry per column of C ; the evident intended bound is $|\mathcal{J}_k|$, mirroring the row loop of lines 21–25.
- Algorithm 3.3: line 1 gives $\mathcal{J}, \mathcal{I}$ “with k and l elements, respectively”, but line 3 gives $C = G(:, \mathcal{J}) \in \mathbb{R}^{m \times k}$ and $R = G(\mathcal{I}, :) \in \mathbb{R}^{l \times n}$, which require $|\mathcal{J}| = k$ and $|\mathcal{I}| = l$; only this reading makes “if $k \leq l$ then Solve $Q(\mathcal{J}, :)W = R$ ” a square or overdetermined solve (and likewise the else branch with $Z(\mathcal{I}, :)$), so line 1 evidently has k and l swapped.
- Section 4.2.1: the Fig. 8 computations are described as “computations with $n = m = 32$ ”, but the Fig. 8 caption states “Here all results are for $n = m = 31$ ”, and the scheduled run in Fig. 8 reproduces the rank history and the roughly 330 iterations of the $\theta = 0.5$, $m = n = 31$ panel of Fig. 5; the evident intended value is 31.
- Section 4.2.2: the cumulative average of the Cross-DEIM maximum intermediate rank is said to be “on par with the final rank (that is 10) for the unpreconditioned case”, but in Fig. 11 the unpreconditioned rank history ends at 11 and the singular-value panel shows eleven singular values of the unpreconditioned IrAA solution; unresolved.
- Section 4.2.2: “The unpreconditioned IrAA gives a monotonically increasing intermediate rank”, but the unpreconditioned rank history in Fig. 11 falls from 8 to 7 for roughly iterations 335–430, after isolated drops to 7 near iterations 275–315, before rising to 9, 10, and 11.
- Section 4.2.3, definitions below Eq. (4.3): “ $2a_2 = u_{i,j+1} + u_{i,j-1}$ ” repeats $u_{i,j+1}$ and omits $u_{i,j-1}$, so $2(a_2 - u_{i,j})/h^2$ would not approximate u_{yy} , unlike the symmetric $2a_1 = u_{i+1,j} + u_{i-1,j}$ and the “second order finite difference scheme” invoked for the tolerance $0.01h_x$; the evident intended definition is $2a_2 = u_{i,j+1} + u_{i,j-1}$.
- Section 4.2.3, Eq. (4.3): the radicand $(a_1 - a_2)^2 + \frac{1}{4}(a_3 - a_4)^2 - h^4 f_{i,j}$ has the wrong sign on $h^4 f_{i,j}$ for the stated equation $u_{xx}u_{yy} - u_{xy}^2 = f$: with $u_{xx} \approx 2(a_1 - u_{i,j})/h^2$, $u_{yy} \approx 2(a_2 - u_{i,j})/h^2$, and $u_{xy} \approx (a_3 - a_4)/(2h^2)$, the smaller root of $4(a_1 - u)(a_2 - u) - \frac{1}{4}(a_3 - a_4)^2 = h^4 f_{i,j}$ has radicand $(a_1 - a_2)^2 + \frac{1}{4}(a_3 - a_4)^2 + h^4 f_{i,j}$, and for $u = (x^2 + y^2)/2$, where $f = 1$, the printed radicand is $-h^4 < 0$ whereas $+h^4 f_{i,j}$ returns u exactly; the evident intended sign is $+$.
- Section 4.2.5: the initial data $\phi(x, y, 0) = 1/(1 + e^{(r_c - 0.15)/\nu})$ satisfy $\phi \leq \frac{1}{2}$, with equality only on $r_c = 0.15$, so the “ $\phi = 0.1, 0.5, 0.9$ level sets at times 0.0, 0.5, and 1” of the text and Fig. 17 cannot include $\phi = 0.9$ at time 0, and $\phi = 0.5$, said to define the interface, is a ridge; the evident intended profile is $1/(1 + e^{(r_c - 0.15)/\nu})$, which decreases from about 0.95 at the centre.
- Section 4.2.5 and Fig. 17: under the flow “ $u = 0.5, v = 0.5$, until time $t = 0.5$ ” the profile centred at $(0.75, 0.75)$ is translated by $(0.25, 0.25)$ to $(1, 1) \equiv (0, 0)$ of the periodic unit square, but the middle row of Fig. 17, titled “Solution at time 0.50”, shows the contours centred at $(0.5, 0.5)$ on all four meshes, while the top and bottom rows are centred at $(0.75, 0.75)$; unresolved.

2026ACGS

ChatGPT (July 2026)

Summary

Modern C++ MPI wrappers must reconcile language-level types, ownership, and deterministic lifetimes with MPI’s opaque handles, global state, and underspecified mutation rules. Rather than proposing another concrete binding, this paper develops design guidance for a future high-level interface. It examines RAII representations of MPI objects, the

consequences of `const` for communicators, compile-time mapping from C++ types to MPI datatypes, and contiguous, sized, typed buffer concepts. Moving buffers into request objects can prevent invalid access during nonblocking communication, while compile-time checks, layered assertions, exceptions, and `std::expected` separate usage errors from recoverable failures. The discussion also covers serialization, datatype pools, and sensible defaults. It identifies specification gaps, especially datatype lifetimes under sessions and callback state, that obstruct safe automated resource management. The result is a conceptual blueprint, not a standardized API or performance evaluation.

Contributions

1. Organized modern MPI C++ interface design around object models, types, lifetimes, ownership, and communication buffers.
2. Proposed C++ concepts for contiguous, sized, typed buffers with compile-time datatype checking.
3. Showed how move semantics can enforce exclusive buffer ownership during nonblocking operations.
4. Developed an idiomatic error-handling hierarchy spanning static checks, optional runtime assertions, exceptions, and `std::expected`.
5. Identified MPI specification ambiguities that must be resolved before robust language-level resource management is possible.

2026ACKM

ChatGPT (July 2026)

Summary

Conservative formulations select physically relevant weak solutions of hyperbolic conservation laws, whereas primitive-variable formulations can simplify wave treatment but may converge to incorrect shock solutions. The paper combines both representations in second-order dual-formulation finite-volume methods for one and two dimensions. Cell averages of conserved and primitive variables evolve on overlapping staggered meshes. A path-conservative central-upwind scheme advances the nonconservative primitive system, and those values provide simple fluxes for the conservative update. Because the two evolutions are otherwise independent, a conservative post-processing step couples them after each timestep, suppresses spurious oscillations, and recovers the correct weak solution. The two-dimensional construction uses separate staggered primitive grids in each coordinate direction and dimension-by-dimension post-processing; several updates can be parallelized. Euler-equation tests recover second-order convergence for a smooth vortex and handle shock tubes, near-vacuum rarefactions, shock–turbulence interaction, blast waves, and multidimensional Riemann problems. In one-dimensional comparisons, the method generally resolves features at least as well as, and often better than, a central scheme on overlapping cells. The current approach remains second order: extending it requires a uniformly accurate higher-order conservative post-processing procedure.

Contributions

1. Introduced second-order dual-formulation finite-volume methods that evolve conservative and primitive variables on overlapping staggered meshes.
2. Coupled a path-conservative central-upwind primitive update to a conservative update with unusually simple numerical fluxes.
3. Designed conservative post-processing that suppresses oscillations and prevents convergence to nonphysical weak solutions.
4. Extended the framework from one dimension to two using directional primitive grids and dimension-by-dimension coupling.
5. Verified second-order accuracy and robust shock resolution across eight Euler benchmarks, including near-vacuum and multidimensional flows.

2026ACS

ChatGPT (July 2026)

Summary

This survey presents exponential Runge–Kutta methods as schemes for semilinear problems $u' = Au + g(t, u)$ that propagate the stiff linear component through matrix exponentials and approximate the nonlinear variation-of-constants integral with Runge–Kutta-like stages and φ -functions. It develops the basic construction, consistency, stability, and order framework before tracing more than sixty years of work from early exponential formulas through Krylov and contour methods, stiff order conditions, exponential Rosenbrock methods, parallel and multirate schemes, delay equations, and recent structure-preserving applications. The review emphasizes that practical performance depends on efficiently computing matrix-function actions and on whether stiffness is concentrated in the linear operator. Three deliberately didactic examples compare second- and fourth-order exponential, classical Runge–Kutta, and Rosenbrock schemes on a scalar stiff problem, a Duffing oscillator, and a pseudospectral delay-equation system. The experiments illustrate substantial stability and accuracy advantages in suitable stiff settings but are not a comprehensive benchmark. The numerical section covers explicit exponential methods; implicit variants and broader robustness studies remain open.

Contributions

1. Gave a self-contained derivation of exponential Runge–Kutta schemes from variation of constants and explained the role of matrix exponentials and φ -functions.
2. Reconstructed the field’s historical development from the late 1950s through modern stiff-order, high-order, parallel, and multirate methods.
3. Synthesized practical matrix-function techniques, including scaling and squaring, Krylov, interpolation, and contour-integral approaches.
4. Surveyed applications and theory for partial differential equations, delay equations, advection–diffusion–reaction systems, order reduction, and structure preservation.
5. Supplied three didactic comparisons against classical Runge–Kutta and Rosenbrock methods, clarifying when explicit exponential schemes gain stability or accuracy and when overhead dominates.

2026ADGM

ChatGPT (July 2026)

Summary

Experiments and global electrostatic gyrokinetic simulations examine why short hydrogen gas puffs improve confinement in ADITYA-U. A puff raises line-averaged density within about 1 ms and electron temperature after 3–4 ms, while flattening the density profile mainly at normalized radius $0.3 < \rho < 0.8$. GTC simulations use measured pre- and post-puff profiles, an IPREQ equilibrium, and kinetic passing and trapped electrons. Trapped-electron modes (TEMs) dominate both cases, but after fueling their peak shifts outward from $\psi/\psi_X \approx 0.5$ to 0.75, their poloidal mode number falls from about 160 to 120, and their radial structure retreats from the core. Nonlinear simulations consequently show lower particle and heat transport and curtailed inward turbulence propagation; saturated electron and ion particle diffusivities fall by about 84% and 94%, respectively. The results support profile relaxation, rather than turbulence spreading, as the mechanism linking transient fueling to the rapid core-temperature rise. They establish neutral fueling as an active turbulence-control mechanism in this discharge, although the simulations compare measured states rather than predicting the neutral-plasma profile evolution.

Contributions

1. Connected the post-puff mid-radius density flattening to reduced trapped-electron-mode drive.
2. Reproduced the pre- and post-puff regimes with global GTC simulations using measured profiles and equilibrium geometry.
3. Quantified the outward shift, localization, and reduced mode number of the dominant TEM after fueling.

4. Measured large reductions in saturated particle diffusivities and suppressed inward turbulence propagation.
5. Provided a transport explanation for the observed millisecond-scale core-temperature increase.

2026AFKP

ChatGPT (July 2026)

Summary

This paper formulates particle resampling as a projection through a finite-element function space. Particle weights are first deposited into the continuum basis; a pseudoinverse particle mass matrix then transfers that representation to a freely selected replacement particle set. Because new particle locations and counts are decoupled from the continuum mesh, both discretizations can in principle adapt independently. The projection exactly preserves every moment representable by the finite-element polynomial space, including kinetic energy when quadratic velocity elements are used, and leaves the continuum field solve unchanged. A matrix-free LSQR implementation in PETSc-PIC is tested on $1X + 1V$ Vlasov–Poisson two-stream instability and linear and nonlinear Landau damping. Resampling restores well-populated phase-space grids, suppresses particle noise, and sustains coherent dynamics to long times while preserving energy to solver tolerance. However, excessive resampling can smooth filaments, change vortex-merger timing, or trigger oscillatory instability. Tests use structured static grids and remap mainly to the original particle layout; stability analysis, adaptive unstructured meshes, optimized high-dimensional operators, and physically realistic $2X + 3V$ cases remain future work.

Contributions

1. Expressed conservative resampling as finite-element deposition followed by pseudoinverse projection.
2. Proved preservation of all moments represented by the chosen polynomial basis.
3. Decoupled replacement-particle placement and count from the continuum mesh.
4. Implemented a matrix-free PETSc-PIC solver with practical oversampling conditions.
5. Quantified noise reduction, conservation, and long-time behavior in kinetic benchmarks.

2026AG

ChatGPT (July 2026)

Summary

This paper establishes finite-element error estimates for second-order hyperbolic equations whose material coefficients vary in both space and time, including the sign-changing gain/loss term induced by temporal modulation. The central device is a time-dependent Ritz-like projection that incorporates this first-order term and permits control of the projection error and its time derivatives. Combined with a duality argument and energy estimates, it yields the optimal semidiscrete bound $\|u - U_h\|_{L^\infty(0,T;H^1)} \leq Ch^{r-1}$ for conforming polynomial spaces under stated regularity assumptions. One-dimensional experiments with mass-lumped quadratic elements and leapfrog or leapfrog–Crank–Nicolson time stepping recover second-order H^1 and third-order L^2 convergence for smooth time-modulated density, bulk modulus, and gain/loss coefficients. A final high-contrast example models 50 time-modulated subwavelength resonators and exhibits strong, spatially localized, exponentially growing wave amplification. Thus, the work both closes a convergence-theory gap for non-autonomous wave operators and demonstrates behavior relevant to actively modulated metamaterials.

Contributions

1. Formulated a finite-element convergence framework for wave equations with fully space-time-dependent coefficients and gain/loss terms.
2. Introduced a time-dependent Ritz-like projection adapted to amplification and attenuation in the medium.
3. Proved optimal H^1 convergence for the semidiscrete Galerkin approximation and bounds for projection time derivatives.
4. Numerically confirmed the predicted H^1 and L^2 rates with mass-lumped quadratic finite elements.

5. Demonstrated localized exponential field growth in a chain of time-modulated subwavelength resonators.

2026AH

ChatGPT (July 2026)

Summary

This paper derives a nonlinear fourteen-moment electron closure for rarefied, nonmagnetized, partially ionized plasmas where strong fields and finite electron drift invalidate linearized transport theory. A Hermitian velocity-distribution expansion evolves density, momentum, the anisotropic pressure tensor, contracted heat flux, and scalar kurtosis. Collision moments are integrated analytically at finite Mach number for elastic and inelastic electron-neutral interactions, electron-impact ionization, and electron-electron and electron-ion Coulomb collisions. The resulting terms couple moments of different tensorial rank, breaking the linear-regime Curie symmetry; examples include pressure-anisotropy effects on momentum and inelastic rates and kurtosis effects on heat-flux relaxation. In homogeneous argon calculations over 10–1000 Td, the nonlinear model agrees substantially better with Monte Carlo collision results than linear 13- or 14-moment models, particularly above 100 Td and before runaway. It also captures high-order moments and high-energy-distribution depletion missed by linear closures.

Contributions

1. Formulated a nonlinear 14-moment model for density, velocity, anisotropic pressure, heat flux, and scalar kurtosis.
2. Integrated finite-drift nonlinear collision moments analytically for elastic, inelastic, ionization, and Coulomb processes.
3. Identified cross-tensorial collisional couplings absent from linear closures.
4. Quantified large anisotropy- and kurtosis-dependent corrections to collision rates.
5. Demonstrated substantially better agreement with Monte Carlo kinetics than linear closures through 1000 Td.

2026AKMI

ChatGPT (July 2026)

Summary

The paper identifies velocity-space signatures of the ion-ion-acoustic instability that could be sought in single-spacecraft measurements of the young solar wind. Fully kinetic one-dimensional Vlasov–Poisson simulations model core protons, drifting beam protons, and electrons over parameter ranges motivated by Parker Solar Probe observations. Simulated frequencies and growth rates agree with linear Landau theory. Global energy accounting shows that the proton beam drives the instability and transfers most of its lost energy to core protons; electron energization is small relative to the electrons’ initial energy. Field-particle correlations separate reversible oscillatory exchange from secular transfer and localize the latter near the ion-acoustic resonance velocity: beam protons lose energy, core protons gain it, and electrons mainly show oscillatory exchange. Increasing electron temperature, beam density, or appropriate beam drift strengthens growth and transfer. Probe position affects beam and electron signatures through ion-hole structure but little affects the core signature. Present spacecraft cadence is too slow to resolve the predicted millisecond-scale evolution.

Contributions

1. Validated fully kinetic instability simulations against linear-theory frequencies and growth rates across three parameter scans.
2. Established that beam-proton energy primarily heats core protons, with the field acting mainly as an intermediary.
3. Identified distinct resonant field-particle-correlation signatures for beam protons, core protons, and electrons.
4. Quantified how electron temperature, beam density, drift speed, and sampling position alter energy-transfer signatures.
5. Defined measurement-cadence requirements by showing that current Parker Solar Probe data cannot resolve the predicted signatures.

2026AL

ChatGPT (September 2026)

Summary

Abgrall and Liu extend the point-average-moment polynomial-interpreted (PAMPA) method for hyperbolic conservation laws on unstructured triangles. A discontinuous Galerkin (DG) formulation followed by projection produces a globally continuous solution representation without a global mass-matrix solve, while retaining local conservation. The formulation clarifies how boundary fluxes enter updates of both interior and boundary degrees of freedom. Convex blending with low-order updates preserves admissible bounds; a rotationally invariant damping measure adds oscillation control near strong shocks. A truncation-error analysis predicts third-order accuracy for smooth solutions, which the reported convergence tests support. Numerical examples cover scalar conservation laws and compressible Euler flow. Entropy properties, error estimates, and extension beyond triangular meshes remain open in this work.

Contributions

1. Formulated a family of PAMPA updates through DG residuals and projection onto continuous polynomials.
2. Incorporated boundary conditions consistently into residuals for interior and boundary degrees of freedom.
3. Combined bound-preserving and oscillation-eliminating parameters through convex blending.
4. Analyzed truncation error to explain third-order smooth-solution accuracy on triangular meshes.
5. Demonstrated bound preservation and shock control on scalar and Euler benchmark problems.

2026ALGP

ChatGPT (July 2026)

Summary

Experiments with the 9 fs, multiterawatt LUCID laser provide the first evidence that carrier-envelope-phase evolution can bunch laser-wakefield electrons into a comb of narrow energy peaks. Ionization injection from nitrogen K-shell electrons in a helium-nitrogen target occurs near threshold within a narrow density window around $1.5 \times 10^{19} \text{ cm}^{-3}$. As the carrier-envelope phase slips during propagation, successive optical half-cycles enter and leave the trapping window at discrete positions. Their different acceleration lengths generate peaks separated by about 3 MeV, corresponding to an effective field near 70 GV m^{-1} . Neighboring peaks have alternating angular chirps, as expected from half-cycle injection. Particle-in-cell simulations reproduce discrete injection, the comb, alternating angles, and approximately 6% relative spacing. The mechanism offers waveform-synchronized, potentially attosecond injection and could be scaled toward higher energies and repetition rates.

Contributions

1. Reported the first experimental observation of carrier-envelope-phase-driven energy bunching in laser wakefield acceleration.
2. Resolved regularly spaced quasimonoenergetic peaks in a narrow near-threshold density window.
3. Connected the roughly 3 MeV spacing to half-cycle-separated injection and a 70 GV m^{-1} effective field.
4. Measured alternating angular chirps between neighboring bunches as an independent half-cycle signature.
5. Reproduced discrete injection, bunching, angular tilt, and relative spacing with particle-in-cell simulations.

2026AMC

ChatGPT (July 2026)

Summary

This paper derives a locally conservative spatial discretization of the compressible Euler equations for thermally perfect gases, whose heat capacities vary with temperature. The numerical flux exactly conserves discrete thermodynamic

entropy, mass, momentum, and total energy while preserving kinetic energy. By exploiting the thermally perfect equation of state, the formulation removes singular terms that arise in a generic real-gas entropy-conservative flux; only logarithmic means remain. The framework accommodates different caloric models and multicomponent nonreacting mixtures, extends to arbitrary spatial order, and yields an asymptotically entropy-conservative hierarchy through Taylor expansion. Tests with periodic jets, a three-dimensional Taylor–Green vortex, and a shock tube verify conservation and show improved kinetic-energy evolution and small-scale statistics relative to comparable entropy-conservative schemes. Local Lax–Friedrichs dissipation supplies shock-capturing robustness when discontinuities are present.

Contributions

1. Derived an exactly entropy-conservative flux for thermally perfect-gas Euler equations.
2. Preserved primary invariants and kinetic energy within the same locally conservative discretization.
3. Removed generic real-gas flux singularities by using the structure of thermally perfect equations of state.
4. Extended the construction to high order, multicomponent mixtures, and asymptotically entropy-conservative approximations.
5. Demonstrated improved robustness and kinetic-energy fidelity on two- and three-dimensional flow tests.

2026AMM

ChatGPT (July 2026)

Summary

This study uses direct numerical simulations to determine when fully three-dimensional rotating, stably stratified turbulence develops an inverse energy cascade in a thin periodic domain. Thirty $6144 \times 6144 \times 192$ simulations at $Re = 500$ vary Rossby and Froude numbers while fixing the aspect ratio and forcing scale. The resulting phase diagram shows that rotation promotes upscale transfer whereas increasing stratification suppresses it; when both effects are strong, the transition occurs near $Ro/Fr \approx 80$ for the chosen geometry. Every detected inverse cascade is bidirectional, with some injected energy still transferred to dissipative small scales. Spectral projections distinguish quasi-geostrophic modes from gravity-inertial waves and show how rotation channels energy toward vertically invariant modes that support upscale transfer, while waves contribute to the forward cascade. Several inverse-cascading cases occupy parameter ranges relevant to planetary atmospheres, demonstrating physical plausibility without claiming that the simplified simulations reproduce atmospheric dynamics.

Contributions

1. Mapped inverse-cascade behavior across 30 high-resolution simulations spanning a broad Rossby-Froude parameter grid.
2. Established how rotation, stratification, and their ratio control the inverse-transfer boundary for the selected geometry.
3. Showed that every observed inverse cascade is split, retaining simultaneous transfer toward dissipative scales.
4. Identified vertically invariant rotational modes as the upscale channel and gravity-inertial modes as an important forward-transfer channel.
5. Demonstrated inverse cascades in non-quasi-geostrophic regimes compatible with selected planetary-atmosphere parameters.

2026AMS

ChatGPT (July 2026)

Summary

Linear thermo-poroelasticity couples solid displacement, fluid pressure, and temperature, making fully implicit time steps expensive. This paper introduces half-decoupled and fully decoupled first-order schemes that solve the three component equations sequentially. Selected coupling terms are lagged by one time step, allowing the methods to be

interpreted as implicit Euler discretizations of related delay equations. The authors prove an abstract first-order error estimate for parabolic delay problems and transfer it to the coupled system. Convergence is guaranteed by weak-coupling conditions expressed only through material and operator constants, rather than a time-step restriction. Finite-element experiments on geothermal-type and manufactured problems confirm first-order convergence when these computable conditions hold, explore their sharpness, and compare the methods with stabilized iterative splittings. The new schemes avoid inner coupling iterations and larger monolithic matrices, but their guarantee depends on sufficiently weak physical coupling.

Contributions

1. Introduced partially and fully sequential time-stepping schemes for linear thermo-poroelasticity.
2. Reduced each time step to separate mechanics, flow, and heat solves without inner iterations.
3. Recast the lagged schemes as implicit Euler methods for parabolic delay equations.
4. Proved first-order convergence under a priori material-parameter coupling conditions.
5. Numerically assessed convergence, condition sharpness, and performance relative to existing decouplings.

2026AR

ChatGPT (July 2026)

Summary

Entropy-stable discretizations improve robustness for hyperbolic systems, but constructions for nonconservative products often require path integrals or problem-specific derivations. The paper recasts Tadmor-type entropy conditions algebraically and proves necessary and sufficient conditions for entropy conservation or stability without requiring a convex entropy. It proposes four concrete fluctuation-based semi-discretizations of nonconservative terms and fully characterizes their entropy properties for three-point finite-volume methods. These forms enable algorithmic construction of affordable fluxes without evaluating path integrals. Applications recover existing constructions and derive new schemes for systems including variable-coefficient advection, shallow-water models, ideal magnetohydrodynamics, and compressible Euler equations written with internal energy. The paper then extends the finite-volume construction to high-order summation-by-parts operators on Cartesian and curvilinear meshes, carrying well-balanced properties from the underlying finite-volume method under stated conditions. Numerical tests verify accuracy, robustness, entropy behavior, and preservation of selected steady states. The results provide a general construction framework, while each application still requires compatible entropy variables, fluctuations, and dissipation.

Contributions

1. Proved algebraic necessary-and-sufficient entropy conditions that also apply to nonconvex functionals.
2. Characterized four practical fluctuation forms for nonconservative three-point finite-volume schemes.
3. Enabled algorithmic construction of entropy-conservative and entropy-stable fluxes without path-integral evaluation.
4. Extended the framework to multidimensional high-order summation-by-parts methods on Cartesian and curved meshes.
5. Recovered existing constructions and derived new entropy-preserving schemes for representative hyperbolic systems.

2026AVZ

ChatGPT (September 2026)

Summary

This paper computes singly diagonally implicit Runge–Kutta (SDIRK) stages through an explicit stabilized iteration. It rewrites the implicit stage equations as the steady state of an auxiliary system, then applies a partitioned Runge–Kutta–Chebyshev iteration. The partition isolates diagonal diffusion from off-diagonal diffusion and advection,

avoiding large Newton systems; stiff reactions require only local Jacobian inversions. For linear diffusion, the authors prove convergence independently of spatial dimension and a cost of $O(\sqrt{\Delta t}/\Delta x)$ diffusion evaluations per SDIRK step on a grid of spacing Δx . For linear advection–diffusion, they prove convergence under stated spectral and stability conditions. An adaptive fourth-order implementation, exSDIRK4, is tested on one- and two-dimensional Brusselator problems. Compared with second-order PIROCK, it uses more evaluations at loose tolerances but becomes more efficient as accuracy demands increase. The analysis does not extend directly to fully implicit Runge–Kutta methods.

Contributions

1. Recast SDIRK stage equations as a steady-state problem solvable by explicit stabilized iteration.
2. Split the stage operator so diffusion drives Chebyshev stabilization, while stiff reactions use spatially local Jacobian inversions.
3. Proved dimension-independent convergence for linear diffusion and the stated $O(\sqrt{\Delta t}/\Delta x)$ diffusion-evaluation cost.
4. Established conditional convergence for linear advection–diffusion and analyzed the iteration’s diffusion–advection stability region.
5. Built an adaptive fourth-order implementation and demonstrated its cost advantage over PIROCK at tight tolerances on stiff Brusselator tests.

2026B

Claude (September 2026)

Summary

The author studies how the aperiodic geometry that a field line samples in a stellarator affects the structure of ion-temperature-gradient (ITG) eigenmodes. Extending a reduced fluid drift-wave model with an ion temperature gradient and expanding about the drift-wave-resonant branch $\Omega = \omega/\omega_{*e} = 1 + \Lambda$ gives a Hill equation $\phi'' + \epsilon_n(1 + \eta_i)K(\theta)\phi = -\Lambda\phi$ in the ballooning angle, where K is the normalized magnetic drift, $\epsilon_n = L_n/R$, and $\eta_i = L_n/L_{Ti}$. Because the line samples incommensurate Boozer harmonics, K is modeled as $\cos\theta + \lambda\cos(\alpha\theta + \delta)$ with irrational α . Projecting onto Wannier functions of the lowest Mathieu band gives the Aubry–André–Harper (AAH) model, whose eigenstates all localize above unit effective coupling. For the continuous equation, localization is diagnosed by a positive transfer-matrix Lyapunov exponent at $\Lambda = 0$. With $\lambda = 1$, $\alpha \approx 2.14$, and $\epsilon_n = 0.12$, the exponent becomes persistently positive above $\eta_i^* \approx 2.3$, against 2.8 for the periodic case, where positivity marks a Floquet gap instead; the onset is non-monotonic because the spectrum is a Cantor set. The result concerns mode structure on the drift-wave branch, not the ITG branch proper, at weak magnetic shear. The author states that its effect on transport, even in sign, and any link to ion-temperature clamping are unsettled.

Contributions

1. Derived the two-fluid equation $\phi'' + [\Omega(\Omega - 1) + \epsilon_d(\theta)(\Omega + \eta_i)]\phi = 0$ from adiabatic electrons, $E \times B$ advection of the ion temperature, and ion continuity and parallel momentum, and bounded its drift-wave-resonant reduction by $\epsilon_n(1 + \eta_i) \lesssim 1$.
2. Related the incommensurate wavenumber ratio to Boozer harmonics through field-line wavenumbers $k_{m,n} = m - n/\iota$, so that α is irrational for irrational rotational transform unless both harmonics belong to one family, and argued that field-period modularity does not restore periodicity along a line.
3. Expressed the AAH coupling as $\lambda_{\text{AAH}} = \epsilon_n(1 + \eta_i)\lambda F_\alpha/(2|t|)$ through the lowest-band Wannier hopping amplitude t and a form factor F_α of the Wannier probability density.
4. Reported Lyapunov exponents converged in period number, integrator tolerance, and phase δ (within 0.5%), with isolated positive windows below onset, and predicted that the localized mode’s position along the line varies with field-line label.
5. Estimated that finite-Larmor-radius confinement from global shear competes with the aperiodic mechanism near $\hat{s} \approx 0.1$, placing W7-X near that bound, and set out opposing suppression and enhancement arguments that prevent any conclusion about clamping.

Comments

- Section 2.1, last paragraph: “The genuine breakdown is the limit $L_n \rightarrow \infty$ at fixed L_{T_i} ... that limit is approached with $\epsilon_n \rightarrow 0$ and $\eta_i \rightarrow \infty$ such that A remains finite”, but with $\epsilon_n = L_n/R$ and $\eta_i = L_n/L_{T_i}$ as defined in the same section, $L_n \rightarrow \infty$ gives $\epsilon_n \rightarrow \infty$ and $A = \epsilon_n(1 + \eta_i) \rightarrow \infty$.
- Fig. 2 caption: the aperiodic model is said to use $\alpha = 2 + \sqrt{2}/10 \simeq 2.14$ “(the legend rounds this to 2.1)”, but the legend reads “Aperiodic: $\lambda = 1, \alpha \simeq 2.14$ ”.
- Section 5, item 3: “Both values carry an uncertainty of order 0.1 arising from the Cantor structure of the spectrum”, but the second value, 2.8, belongs to the periodic ($\lambda = 0$) case, which Sections 2.2 and 3.1 describe as a Mathieu problem with ordinary bands and Floquet gaps, and Section 3.1 attaches the uncertainty of order 0.1 only to $\eta_i^* \approx 2.3$; the Cantor-structure uncertainty evidently applies to the aperiodic value only.
- Appendix B, Eq. (22): $t = -\int_0^1 E_0(k)e^{2\pi i k} dk \approx [E_0(k=0) - E_0(k=\frac{1}{2})]/4$, but keeping only the fundamental harmonic, $E_0(k) = \bar{E} + e_1 \cos 2\pi k$, the integral gives $t = -e_1/2$ while $[E_0(0) - E_0(\frac{1}{2})]/4 = +e_1/2$, so the approximation and the band width $W = E_0(0) - E_0(\frac{1}{2})$ used after it have the opposite sign; Eqs. (26)–(27) are unaffected because they use $|t|$.

2026B2

ChatGPT (July 2026)

Summary

This review connects linear Boltzmann transport across radiative transfer, neutron transport, and rarefied-gas dynamics through the Analytical Discrete Ordinates (ADO) method. Angular quadrature converts the integro-differential transport equation into coupled first-order equations; ADO solves their homogeneous part through a reduced-order eigensystem and supplies explicit spatial dependence useful for accurate direct and inverse calculations. The article develops the slab problem, general boundary conditions, anisotropic scattering, and particular solutions, then shows how transverse integration and nodal balance extend the construction to two-dimensional Cartesian geometry. It surveys tailored quadratures, including Legendre–Chebyshev product rules, and reports that ADO nodal schemes can outperform comparison methods on coarse meshes while supporting spatial and angular error studies. Applications include reactor shielding and multiplication factors, radiative transfer, source or coefficient reconstruction, and kinetic models for microflows. The review closes by identifying large linear systems, joint spatial-angular error analysis, and multidimensional optical-tomography inversion as open directions rather than presenting ADO as a finished universal solver.

Contributions

1. Unified photon, neutron, and rarefied-gas linear Boltzmann models under a common discrete-ordinates viewpoint.
2. Derived the one-dimensional ADO solution using half-range quadrature and a reduced-order eigenvalue problem with general boundary and particular solutions.
3. Extended ADO to two-dimensional Cartesian transport through transverse integration and an analytical nodal formulation.
4. Synthesized quadrature design and error studies, including product quadratures and Richardson-based spatial convergence estimates.
5. Connected explicit ADO spatial formulas to shielding, criticality, inverse transport, and linearized-Boltzmann kinetic problems while identifying concrete open problems.

2026B3

ChatGPT (July 2026)

Summary

This review compares internal interface layers in neutral fluids with internal transport barriers in fusion plasmas. It proposes a conjectural common cyclic process linking exact coherent states or magnetic islands, Reynolds-stress-driven zonal flows, and interface layers or transport barriers. A mini-review traces fluid interfaces separating uniform momentum, temperature, and concentration zones. Reduced mean-momentum, plasma zonal-flow, quasi-geostrophic potential-vorticity, and scalar-transport balances provide common diagnostics for flow generation and transport suppression. Evidence is strongest for coherent-structure-to-zonal-flow and zonal-flow-to-barrier links; feedback from barriers to coherent structures remains poorly established, especially in fluids. The roadmap calls for stress-budget, flux-suppression, potential-vorticity-staircase, and topography-aware closure tests, along with toroidal fluid experiments and comparative fluid-plasma devices. The proposed cycle is explicitly presented as a framework to test, not as established universality.

Contributions

1. Organized fluid internal interface layers and plasma transport barriers within one comparative framework.
2. Proposed a three-element common cycle connecting coherent structures, zonal flows, and barriers.
3. Mapped Reynolds-stress and potential-vorticity-flux descriptions onto shared mean-flow rectification physics.
4. Separated well-supported causal links from the unresolved barrier-to-coherent-structure feedback.
5. Defined cross-disciplinary closure tests and experiments capable of supporting or falsifying the framework.

2026B4

ChatGPT (July 2026)

Summary

This perspective asks what would make artificial intelligence effective enough to complement or replace mathematical algorithms in computational fluid dynamics. Rather than cataloguing applications, it derives four pillars of effectiveness from the historical success of mathematics: symmetry and invariance, separation of scales, sparsity, and semantic significance. It maps each pillar to current techniques, including physics-aware losses and equivariant architectures, multiscale and multifidelity learning, neural operators, sparse identification and compressed representations, and methods that embed interpretable scientific structure. Because empirical accuracy alone does not supply guarantees analogous to mathematical proof, the article adds credibility as a practical requirement, emphasizing robustness, uncertainty quantification, representative data, and explicit model doubt. Generative and foundation models may constitute a paradigm shift: they can incorporate some pillars indirectly, but their overparameterization, data demands, and emergent semantics may require a revised epistemological framework. The perspective concludes that physics-structured AI can become integral to hybrid numerical tools, while universality-oriented models require further scrutiny.

Contributions

1. Framed AI effectiveness in computational fluid dynamics as an epistemological problem beyond benchmark accuracy.
2. Derived four pillars: symmetry, scale separation, sparsity, and semantic significance.
3. Connected each pillar to concrete AI architectures and workflows used in fluid mechanics.
4. Added robustness, uncertainty-aware doubt, and credibility as requirements for scientific adoption.
5. Explained why generative and foundation models may require revised pillars of effectiveness.

2026B5

ChatGPT (September 2026)

Summary

The Pfirsch-Schlüter current flows along magnetic field lines to keep the current density divergence free in a toroidal plasma with a pressure gradient. It can distort magnetic surfaces and limit the pressure attainable in a stellarator. Boozer derives a condition for confining this current to a half or whole stellarator period: $v_s = \int_0^{\ell_s} d\ell/B$ must be the same for every field line on a magnetic surface, where ℓ_s is its length through the segment. The integral also measures flux-tube volume per unit magnetic flux. Because $v_s = \sigma_s \ell_s^2/I_s$ and σ_s varies weakly between field lines, equal segment lengths closely approximate the exact condition. Using magnetic-field profiles from a published quasi-isodynamic design, the paper estimates accuracy of about four parts per thousand for a half period and two parts per thousand for a full period. It connects equal lengths to stronger inboard helical shaping, achievable with modular coils or potentially a helical central column.

Contributions

1. Derived the parallel-current balance and separated the Pfirsch-Schlüter current from the divergence-free net current.
2. Identified field-line independence of $\int_0^{\ell_s} d\ell/B$ as the exact segment-localization condition.
3. Interpreted that integral as flux-tube volume per unit magnetic flux.
4. Related the exact condition to segment length through $v_s = \sigma_s \ell_s^2/I_s$ and bounded the effect of magnetic-field variation.
5. Estimated the length criterion's accuracy for a published quasi-isodynamic configuration and explained the resulting inboard coil-shaping requirement.

2026B6

Claude (October 2026)

Summary

Boozer argues that tokamak analysis should use the toroidal flux ψ_t enclosed by a magnetic surface as radial coordinate rather than the internal poloidal flux $\psi_p^{in} = \Psi_p^{ax}(t) - \psi_p$, because Faraday's law fixes the slippage of poloidal relative to toroidal flux, $\partial\psi_p(\psi_t, t)/\partial t = V_\ell$, and because a divertor separatrix makes poloidal-flux definitions of the edge safety factor q_{95} and internal inductance ℓ_i depend on the current profile. In axisymmetry $q = \sigma G/I$, with I and G the toroidal and poloidal currents and $\sigma(\psi_t)$ a shape function of the surface geometry. A two-wire model, the plasma current and an equal divertor-coil current at $y = \pm a$ in a slab periodic in $z = R\varphi$, is solved with Jacobi elliptic functions: the surfaces are $k^2 = r^2 r_e^2/a^4$, $\sigma = k^2 K(k)a^2/(2\pi R^2)$, $\psi_t = a^2 B_\varphi [E(k) - (1 - k^2)K(k)]$, and $\sigma = (a^2/\pi R^2)(\psi_t/\Psi_t)\{1 + \Pi_\sigma\}$ with Π_σ rising from zero to about 0.7 near the separatrix. For a flat current channel, 95% of the poloidal flux lies at 81% to 89% of the toroidal flux depending on channel width, which a fixed toroidal-flux fraction such as 85% would remove. The optimal edge fraction is left open.

Contributions

1. Derived from the contravariant and covariant forms (14)–(15) of $\vec{B}$ in $(\psi_t, \theta, \varphi)$ coordinates, and from their generalizations (17)–(18) to angles θ_* , φ_* with $\varphi = \varphi_* + \nu$, that $\iota\sigma = I/G$ and hence $q = \sigma G/I$ (28)–(29), where σ is the ratio of the J_*^{-1} -weighted surface averages of $(\partial\vec{x}/\partial\theta_*)^2$ and $(\partial\vec{x}/\partial\varphi_*)^2$ (27); for two-to-one elliptical surfaces with triangular shaping (33) this gives $\sigma = 2.5r^2/R^2 + 0.36r^4/(a^2 R^2)$ against r^2/R^2 for circular surfaces (36).
2. Built the two-wire divertor model with poloidal flux $\psi_p^{in} = (\mu_0 R I_p/2) \ln k^2$, $k^2 = (r_x/a)^4 + 2(r_x/a)^2 \cos(2\theta_x) + 1$ in cylindrical coordinates about the X-point (46)–(50), regularized the logarithmic singularity at the axis by a flat current channel of radius r_I , giving $\psi_p^{in}(k) = \mu_0 R I_p \ln(\sqrt{e}k/k_I)$ for $k \geq k_I = 2r_I/a$ (55), (57), and located the q_{95} surface at $k_{95p} = (k_I/\sqrt{e})^{0.05}$ (58).

3. Parametrized the surfaces by $\cos(2\theta_x) = -\text{dn}(u)$, $r_x^2/a^2 = \text{dn}(u) + k \text{cn}(u)$, $\theta_* = 2\pi u/4K(k)$ (60)–(66), showed $(\partial\bar{x}/\partial u)^2 = k^2 r_x^2/4$ (81), and obtained $\sigma(k) = k^2 K(k)a^2/(2\pi R^2)$ (84), so that $\sigma/\sigma_c = 2K(k)/\pi$ relative to circular surfaces (86) and $\iota = \mu_0 R I_p / (B_\varphi a^2 k^2 K)$ (89), which reproduces $\mu_0 R I_p / (2\pi B_\varphi r^2)$ as $k \rightarrow 0$.
4. Integrated $d\psi_p^{in}/\iota$ to $\psi_t(k) = a^2 B_\varphi [E(k) - (1 - k^2)K(k)]$ (94), so $\Psi_t = a^2 B_\varphi$ is the flux inside the separatrix, and wrote $\Pi_\sigma = \frac{1}{2}k^2 K/[E - (1 - k^2)K] - 1$ (96), with the fit $\Pi_\sigma = 0.2019x^{1.1426}/(1 - x)^{0.4176}$, $x = \psi_t/\Psi_t$ (97), stated to be accurate to within 4%.
5. Using $(\psi_t/\Psi_t)^{(I)} = \pi k_I^2/4$ for the channel width, found that the 95% poloidal-flux surface sits at ψ_t^{p95}/Ψ_t between about 0.81 and 0.89 (Fig. 6), so $(\psi_t^{p95}/\Psi_t)\{1 + \Pi_\sigma\}$ varies between about 1.05 and 1.33 with the current profile, leaving σ , q_{95} , and $\ell_i = \int_0^1 I(s)q(1)/(I(1)q(s)) ds$ (11) correspondingly uncertain; proposed fixing the edge at a toroidal-flux fraction such as 0.85, noted that the pedestal bootstrap current of a separatrix-bounded plasma lowers ℓ_i at fixed edge q , and suggested data-driven and linear tearing and kink stability studies to identify disruption-avoidance parameters.

Comments

- Section IV A, Eq. (51): $\psi_p^{in}(r) = 2\pi\mu_0 R \int_0^r B_\theta dr$, but with $B_\theta = \mu_0 I_p r / (2\pi r_I^2)$ inside and $\mu_0 I_p / (2\pi r)$ outside the current channel the stated results (52)–(53), $\mu_0 R I_p [\frac{1}{2} + \ln(r/r_I)]$, follow from $2\pi R \int_0^r B_\theta dr$, consistent with (41)–(44); the factor μ_0 in (51) is evidently extraneous.

2026BB

ChatGPT (July 2026)

Summary

This paper develops a distributed algorithm for one-way overset coupling between an adaptively refined forest-of-octrees mesh and an unrelated parallel mesh represented by query points. Globally replicated partition boundaries in Morton order let every process determine exact candidate owners without communication. Nonblocking point-to-point exchange then routes queries only to relevant forest processes, where a local top-down search locates leaves and evaluates application-defined data before returning results. Smooth per-tree mappings support curved multiblock domains, while opaque query objects and callbacks generalize the method beyond points. The same machinery enables refinement near mesh intersections and weighted repartitioning based on query workload. Two- and three-dimensional tests show near-linear behavior in balanced cases through 12,288 processes; a deliberately imbalanced curved-arc problem scales poorly until space-filling-curve repartitioning reduces the maximum per-process query load. An implementation in ForestClaw couples atmospheric and ionospheric solvers in a larger geophysics application. The algorithm is intrinsically one-directional, though applying it twice can couple two forest meshes.

Contributions

1. Formulated exact, communication-free owner searches from global forest partition boundaries.
2. Combined selective nonblocking routing with local mapped-octree searches for arbitrary query sets.
3. Designed callback-based query objects that support interpolation, geometric tests, and other remote data requests.
4. Reused overset queries to drive intersection refinement and workload-aware repartitioning.
5. Demonstrated balanced scalability to 12,288 processes and integration in a multimesh geophysics simulation.

2026BB2

Claude (September 2026)

Summary

The multirate PDE technique simulates radio-frequency (RF) circuits by lifting the circuit differential-algebraic equation (DAE) to a PDE in a slow envelope time τ and a 2π -periodic fast variable $\hat{t}$, whose solution along a characteristic curve

recovers the DAE solution. Distributed devices such as transmission lines and filters are characterized instead by impulse responses h or transfer functions H , which add a convolution integral to the DAE. The authors extend the multirate PDE to such integro-differential algebraic equations through a bivariate convolution, acyclic in τ and cyclic in $\hat{t}$, with a multivariate impulse response $\hat{h}(\tau, \hat{t})$. Requiring the DAE and PDE solutions to coincide along the characteristics yields pointwise equalities for $\hat{h}$. For a constant carrier frequency ω_0 , $\hat{h}(\tau, \hat{t}) = h(\tau) \delta(\hat{t} - \omega_0 \tau)$ satisfies them, so the convolution runs along the characteristics. A Fourier series in $\hat{t}$ turns it into convolutions of each envelope $\hat{x}_k$ with the modulated impulse response $h(\tau) e^{-jk\omega_0 \tau}$, whose transform $H(\omega + k\omega_0)$ is lowpass and can be discretized coarsely. The $k = 1$ term matches the equivalent complex baseband (ECB) method of communication engineering without requiring a Hilbert transform. The letter is purely theoretical, treats only the bivariate case, and reports no numerical experiments.

Contributions

1. Formulated a multirate integro-PDE for RF circuits whose convolution term is a bivariate convolution with a multivariate impulse response, required to reproduce the ordinary DAE convolution along every characteristic curve.
2. Derived, for a general instantaneous frequency $\omega(\tau)$, pointwise equalities between the multivariate and ordinary impulse responses that link the integro-DAE and integro-PDE solutions along the characteristics.
3. Showed for a fixed carrier frequency that the impulse response $h(\tau) \delta(\hat{t} - \omega_0 \tau)$, with a periodic Dirac distribution, satisfies these equalities and reduces the output to a univariate convolution along each characteristic.
4. Derived a harmonic-wise form in which each Fourier envelope is convolved with a frequency-shifted impulse response, and proved it equivalent to the Dirac form.
5. Identified the $k = 1$ envelope and the shifted transfer function $H(\omega + \omega_0)$ with the ECB signal and baseband transfer function, making the method compatible with coupled circuit and communication-system simulation.

Comments

- §II, Eq. (2), and §III, Eq. (5): the multirate PDE is printed as $\frac{\partial}{\partial \tau} q(\hat{x}(\tau, \hat{t})) + \frac{d(\omega(\tau)\tau)}{d\tau} q(\hat{x}(\tau, \hat{t})) + i(\hat{x}(\tau, \hat{t})) = \hat{s}(\tau, \hat{t})$, with no derivative in $\hat{t}$ in the second term, which conflicts with the stated characteristic curves $(\tau, \hat{t}) = (t, \omega(t)t + \Theta)$ along which the PDE and DAE solutions coincide; the intended term is $\frac{d(\omega(\tau)\tau)}{d\tau} \frac{\partial}{\partial \hat{t}} q(\hat{x}(\tau, \hat{t}))$.
- §III and Conclusion: Theorem 3.2 is introduced with “A necessary condition on $\hat{h}$ is derived next” and is called a necessary condition in the Conclusion, but the theorem states that the DAE and PDE solutions coincide “if” equalities (8) and (9) hold, a sufficient condition, and the preceding text announces “sufficient conditions”; which is intended is unresolved.
- §IV, Eq. (13): the integral in $y_\Theta(t) = \hat{y}(t, \omega_0 t + \Theta) = \int_{-\infty}^{\tau} h(t - \tau') \hat{x}(\tau', \omega_0 \tau' + \Theta) d\tau'$ has upper limit τ , although $\tau = t$ on this characteristic and the corresponding integrals in the proof of Theorem 3.2 end at t ; the intended limit is t .
- Proof of Theorem 5.2: the final line is $h(\tau) \sum_k \delta(\omega_0 \tau - \hat{t})$, whose summand does not depend on k , whereas the theorem states $h(\tau) \delta(\omega_0 \tau - \hat{t})$ with the periodic Dirac distribution; the theorem’s statement is the evident intended result.

2026BBCC

ChatGPT (July 2026)

Summary

Mathlib’s rapid growth creates software-engineering and social bottlenecks beyond the correctness guarantees supplied by the Lean proof checker. Drawing on a 1.9-million-line community library, the paper documents mechanisms that keep changes manageable without overwhelming maintainers. A deprecation system preserves renamed declarations and modules temporarily while issuing actionable migration warnings. Syntax and environment linters automate style, import-structure, redundancy, documentation, and maintainability checks, giving contributors immediate feedback and reserving reviewer attention for mathematical and architectural questions. Conscious dependency design, profiling, and specialized data structures control compilation and elaboration costs. Explicit metrics and issue-tracking practices make technical debt visible rather than allowing temporary exceptions to become permanent. Mandatory but decentralized

code review spreads knowledge and onboards contributors; bots expose import and debt changes, suggest reviewers, and implement routine maintainer actions. A custom dashboard groups pull requests by state and accumulated review time, supporting systematic triage at scale. The account is primarily qualitative rather than a controlled evaluation, but it turns Mathlib’s operational experience into reusable practices for other large formal libraries and open-source projects.

Contributions

1. Documented a deprecation workflow that permits frequent refactoring without abruptly breaking downstream formalizations.
2. Classified semantic linters that automate local feedback and global coherence checks across a large Lean library.
3. Explained library-design and profiling practices used to contain compilation and elaboration costs.
4. Established measurable technical-debt tracking integrated into issue and pull-request workflows.
5. Presented scalable review and triage automation, including reviewer suggestions, status summaries, and a sortable dashboard.

2026BBKL

ChatGPT (July 2026)

Summary

This preprint applies reduced rank extrapolation (RRE) to splitting-based fixed-point solvers for generalized multi-term Sylvester equations, including Lyapunov-plus-positive problems. For small dense equations, cycling RRE forms extrapolants from windows of matrix iterates and can improve the asymptotic reduction factor by suppressing dominant iteration modes. For large sparse equations, the authors extend RRE to nonsymmetric low-rank factors and embed it in a nonstationary outer iteration whose inner Sylvester solves use dynamically adjusted tolerances. Rank truncation controls storage, and factored residual estimators avoid assembling dense residual matrices. Convergence results cover exact stationary and inexact nonstationary processes. Dense and low-rank experiments using ADI or extended Krylov inner solvers show that suitable RRE windows reduce outer iterations, runtime, and often memory, although performance depends on window size and extrapolation can increase intermediate ranks.

Contributions

1. Extended cycling RRE to fixed-point iterations for multi-term Sylvester equations.
2. Derived convergence results for exact stationary and inexact nonstationary iterations.
3. Generalized RRE to sequences of nonsymmetric low-rank matrix approximations.
4. Integrated adaptive inner tolerances, rank truncation, and cheap residual estimation.
5. Demonstrated iteration, runtime, and storage reductions in dense and large-scale tests.

2026BC

Claude (September 2026)

Summary

Particle simulations with thermal walls commonly reinject particles with flux-weighted Maxwellian momenta, which earlier simulations showed to thermalize an ideal gas whereas half-Maxwellian sampling does not. The authors consider non-interacting particles in a one-dimensional external potential U that vanishes at both walls, reinjected with momentum density proportional to $p^n e^{-p^2/(2T)}$ for integer n : $n = 0$ is the half-Maxwellian, $n = 1$ the mass flux, and $n = 3$ the energy flux. Equating the incoming phase-space flux at the walls to this rule and applying Jeans’ theorem along trajectories that reach the walls gives the stationary distribution $f_n \propto H^{(n-1)/2} e^{-H/T}$ for any U , where H is the single-particle energy; it is Maxwell–Boltzmann only for $n = 1$. For $U = 2 \cos(q/2)$ on $[-\pi, \pi]$, a dimensionless coronal-loop potential, closed-form density and kinetic-temperature profiles show, for $n = 0$, a logarithmically diverging density and logarithmically vanishing temperature at the walls; for $n = 1$, an isothermal Boltzmann profile; and for $n = 3$, a non-monotonic density with temperature falling from $3T$ at the walls. Fourth-order symplectic simulations

with 2^{16} particles at $T = 3$ reproduce all profiles. The authors state that self-consistent mean fields make the problem nonlinear and that trapped regions disconnected from the walls require additional information.

Contributions

1. Derived the normalization $A_{T,n} = 2^{(1-n)/2} T^{-(n+1)/2} / \Gamma(\frac{n+1}{2})$ and inverse-transform samplers, the absolute value of a Box–Muller Gaussian for $n = 0$, $p = \sqrt{-2T \ln(1-r)}$ for $n = 1$, and $p = \sqrt{2T[-1 - W_{-1}(-(1-r)/e)]}$ for $n = 3$, with the lower Lambert branch W_{-1} computed by a branch-safeguarded Halley iteration started from branch-point and $x \rightarrow 0^-$ asymptotic guesses.
2. Expressed the $n = 0$ moments through modified Bessel functions, $n_0 \propto e^{-U/(2T)} K_0(U/(2T))$ and $T_0 = U[K_1(U/(2T))/K_0(U/(2T)) - 1]$, and expanded both about the domain center $q = 0$ to show opposite-curvature quadratic behavior without linear terms, a density minimum and a temperature maximum there.
3. Obtained $n_3 \propto (U + T/2)e^{-U/T}$ and $T_3 = T(U + \frac{3}{2}T)/(U + \frac{1}{2}T)$ for energy-flux injection, attributing the non-monotonic density to competition between the energy prefactor H and the Boltzmann factor.
4. Initialized every run in unit-temperature equilibrium by von Neumann rejection sampling, integrated with a fixed step 10^{-2} , declared stationarity once the mean kinetic energy fluctuated about a constant, and compared time-averaged profiles with the analytic curves on the half-domain $[-\pi, 0]$.
5. Offered a theoretical explanation of the earlier numerical finding that only mass-flux sampling thermalizes an ideal gas, extended it to arbitrary n and general potentials, argued that it also holds when particles can escape, and proposed energy-flux injection ($n = 3$) as an alternative model of heating pulses in the authors’ two-component coronal-plasma simulations.

Comments

- §1: long-range interactions are described as “mediated by two-body potentials decaying slower than $1/r^{-d}$ at large distances r ”, but $1/r^{-d} = r^d$ grows with r ; the evident intended bound is $1/r^d$.
- §2 and Fig. 1 caption: the threshold x_{th} that switches between the W_{-1} initial guesses Eq. (12) and Eq. (13) “is chosen empirically to ensure continuity of the initial guess”, but at the reported $x_{\text{th}} \simeq -0.30$ Eq. (12) gives -1.39 and Eq. (13) gives -1.73 (with $W_{-1}(-0.30) \approx -1.78$), a jump of 0.34 , nearly the largest gap between the two guesses for $x \in [-1/e, -0.16]$, and they coincide only near $x \approx -0.16$; the intended threshold is unresolved.

2026BC2

Claude (September 2026)

Summary

This review surveys machine-learning (ML) closures for plasma fluid models, which truncate the velocity-moment hierarchy of the collisionless Vlasov equation by expressing the highest retained moment through lower-order ones. It first summarizes analytic closures, namely adiabatic models extending the Chew–Goldberger–Low closure, Landau-fluid closures descended from the Hammett–Perkins (HP) closure, and gyrofluid models, and then introduces neural networks, physics-informed neural networks (PINNs), neural operators, and equation discovery by sparse or symbolic regression. The twenty ML studies in its summary table are grouped by training data (analytic closures or kinetic simulations) or as equation discovery, and by whether the closure was tested offline against reference data or online inside a time-evolving fluid solver; only five were tested online. The authors conclude that most work consists of one-dimensional proof-of-concept studies, mainly of Landau damping; that a Fourier neural operator (FNO) closure run online reproduced linear and non-linear Landau damping better than HP; and that off-diagonal pressure-tensor components are poorly captured by both neural-network and sparse-regression methods. Stated open problems are extrapolation beyond the training parameter range, numerical stability in multidimensional large-scale simulations, handling different phenomena and boundary conditions without retraining, and the cost of kinetic training data.

Contributions

1. Tabulated the analytic adiabatic, Landau-fluid, and gyrofluid closures by closure class, target regime, and limitation, and identified three limitations they share: finite-Larmor-radius orderings that break at sub-gyroradius scales

and weak guide field, Padé fits to the linear response that miss the suppressed phase mixing of turbulence, and near-Maxwellian expansions that fail in reconnection exhausts, shocks, and driven turbulence.

2. Used the offline/online distinction to single out the singular-value-decomposition closure of Shukla et al., trained at 3 of 35 parameter points with about 8% online error, whose online run far from its training point jumped spuriously to a higher transport level that offline scoring could not detect, and noted that the online neural-network closures of Wang et al., Huang et al., and Joglekar & Thomas all treat only one-dimensional Landau damping.
3. Contrasted the methods' practical limitations: PINNs that learn the whole fluid system (Qin et al.) give an implicit closure that cannot be moved into an independent solver, neural operators need more data and do not guarantee accuracy at finer resolutions, and whether the FNO's 10.8% error against 66.7% for a multilayer perceptron (Wei et al.) reflects the Fourier representation or model capacity is left open until operators are tested on closures not naturally expressed in Fourier space.
4. Identified that general symbolic regression and uncertainty quantification have not yet been applied to plasma closures, and that learned closures are rarely checked for energy and momentum conservation, Galilean or Lorentz invariance, entropy production, hyperbolicity of the closed system, and spurious growing modes.
5. Recommended a two-stage workflow in which a neural network first sets an accuracy baseline and sparse regression, or symbolic regression when the library fails, then seeks an interpretable form (after Ingelsten et al.), together with symmetry-based data augmentation, end-to-end training through differentiable solvers, locally selected specialized closures, pooled community kinetic data, and satellite observations as training data.

Comments

- Fig. 1 caption: the figure is “Reproduced from Tóth et al. (2016), J. Geophys. Res. Space Phys. 122, 10,336–10,363, Fig. 2”, but the reference list gives Tóth et al. (2016) as the Ganymede-magnetosphere article in volume 121 (2), 1273–1293, so the cited source of the figure is unresolved.
- Section 1.2.3: the small parameter ϵ is described as, among others, “the ratio of perpendicular fluctuation scale to the Larmor radius”, which conflicts with the ordering given in the same passage, $\epsilon \sim k_{\parallel}/k_{\perp}$ and $k_{\perp}\rho \sim 1$; the formula indicates that the intended ratio is that of the perpendicular to the parallel fluctuation scale.
- Section 1.2.3: Passot et al. (2018) are said to assume “anisotropic turbulence $k_{\perp} \ll k_{\parallel}$ ”, which reverses the ordering $k_{\parallel} \ll k_{\perp}$ stated earlier in the same section for the gyrokinetic models from which gyrofluid models are derived; the intended reading is evidently $k_{\parallel} \ll k_{\perp}$.
- Table 1: the caption defines “5-moment (scalar pressure)”, as does Section 1.2, yet Chew et al. (1956), Abraham-Shrauner (1967, 1968, 1973), and Le et al. (2009) are classed as “5-moment” although their own entries and Eqs. (1.18)–(1.21), (1.23), and (1.24) describe separate pressures $p_{\parallel}$ and $p_{\perp}$.
- Section 1.3: Maulik et al. (2020) is cited with Wang et al. (2020) for surrogate closures “integrated ... into fluid simulations, demonstrating that neural network closures can be coupled to fluid solvers and run stably in time”, but Section 3.1 states that this study “was conducted entirely offline” and Table 2 lists it as offline.
- Fig. 6 caption: “Orange circles denote training-set error and blue triangles denote test-set error”, but the legend assigns both a circle and a triangle to each set, orange to “train set” and blue to “test sets”, and the plot switches from circles to triangles beyond five terms, so colour, not shape, distinguishes the sets.
- Section 3.3.3: “only Donaghy & Germaschewski (2023) has applied sparse regression to a two-dimensional problem” conflicts with Section 3.3.1 and Table 2, where McGrae-Menge et al. (2026) apply symmetry-augmented sparse regression to “two-dimensional PIC simulations of magnetic reconnection”.
- Table 2: Liu et al. (2022) is listed under “Section 3.2: Kinetic simulation training data” with target “Implicit fluid closure”, but the study is discussed in Section 3.3.2 as PINN-based equation discovery that yields the explicit heat-flux closure of Eq. (3.9), whereas Section 3.4 states that the table is organised by subsection.
- Section 4.1: “Ma et al. (2020) showed that several architectures could reproduce the non-linear Hammett-Perkins heat flux”, but the HP closure of Eqs. (1.25) and (3.1) is linear in $\bar{T}$ and Table 1 calls it “limited to linear, electrostatic regime”; Section 1.3 describes the same result as learning “the non-local Hammett-Perkins closure”, evidently the intended word.

2026BCPS

ChatGPT (July 2026)

Summary

The authors derive a noncanonical Hamiltonian formulation of the Vlasov-Maxwell system linearized about a Maxwellian distribution and consistent background fields. An energy-Casimir construction yields a quadratic Hamiltonian and Poisson bracket, then a curvilinear transformation makes the model usable in tokamak- and stellarator-like geometries. GEMPIC discretization combines particles for the perturbed distribution with finite-element exterior calculus fields in an exact de Rham sequence, preserving the Hamiltonian structure semidiscretely. A Poisson splitting provides tractable time integrators for Maxwell, coupling, spatial-advection, and Lorentz substeps, implemented in STRUPHY. Weak Landau damping agrees with control-variate and direct-delta- f models at short times, while the geometric method maintains solver-level energy conservation and avoids their long-time noise growth. In magnetized tests it reproduces the analytical Bernstein-wave spectrum, though particle discretization introduces identifiable Case-van Kampen modes at hybrid-frequency harmonics.

Contributions

1. Derived a quadratic Hamiltonian and noncanonical Poisson structure for Maxwellian-background linearized Vlasov-Maxwell dynamics.
2. Extended the Hamiltonian formulation to curvilinear geometry.
3. Produced a GEMPIC particle-FEEC semidiscretization that preserves geometric constraints.
4. Derived Poisson-split substep integrators and implemented them in STRUPHY.
5. Demonstrated superior long-time Landau-damping stability and recovery of the Bernstein-wave spectrum.

2026BCSR

ChatGPT (July 2026)

Summary

This paper replaces finite-difference approximations in the Cauchy–Kovalevskaya predictor of high-order Lax–Wendroff flux-reconstruction and discontinuous-Galerkin schemes with forward automatic differentiation (AD). A recursive derivative-bundle identity evaluates each temporal flux derivative as a directional derivative of the preceding one, exactly reproducing the multivariate chain rule without forming flux Jacobians or writing order-specific finite-difference formulas. The authors implement point-wise AD with Enzyme.jl and element-wise AD with TaylorDiff.jl. Because AD evaluates fluxes only at the admissible numerical state, it avoids the inadmissible perturbed states that make approximate Lax–Wendroff fail for isentropic Euler and relativistic-hydrodynamics Riemann problems. Tests recover the expected order through sixth order, preserve admissibility, and handle a Mach-2000 jet. Across Intel Xeon and Apple M3 benchmarks, AD is at least competitive with the finite-difference predictor; tested speedups reach 1.61, although whether point-wise or element-wise differentiation is faster depends on hardware. The result is a Jacobian-free, problem-independent predictor whose implementation is uniform across orders.

Contributions

1. Derived a recursive automatic-differentiation formula for exact Cauchy–Kovalevskaya temporal flux derivatives without explicit Jacobian tensors.
2. Implemented point-wise Enzyme.jl and element-wise TaylorDiff.jl variants in Tenkai.jl and TrixiLW.jl.
3. Showed why AD avoids predictor-stage positivity corrections by never evaluating fluxes at perturbed inadmissible states.
4. Demonstrated robustness on demanding Euler and relativistic-hydrodynamics problems while recovering the expected orders through sixth order.
5. Benchmarked both AD variants on Intel and Apple CPUs, finding speedups as high as 1.61 with architecture-dependent variant rankings.

2026BDFR

ChatGPT (July 2026)

Summary

This paper analyzes iterated Golub–Kahan–Tikhonov regularization for large linear inverse problems obtained by discretizing ill-posed Hilbert-space operator equations. Partial Golub–Kahan bidiagonalization projects the problem into a small Krylov subspace, where iterated Tikhonov regularization is applied. The analysis simultaneously tracks data noise, operator discretization, and projection error. Under source conditions and balanced projection and data errors, it obtains $O(\delta^{2i/(2i+1)})$ convergence, improving the $O(\delta^{2/3})$ saturation of ordinary Tikhonov as the iteration count i increases. Dominant work remains Golub–Kahan matrix-vector products and is essentially independent of i . Two regularization-parameter strategies are given, including an alternative permitting smaller Krylov spaces. Deblurring and tomography experiments show gains over non-iterated Golub–Kahan–Tikhonov and, for nonsymmetric cases, iterated Arnoldi–Tikhonov.

Contributions

1. Defined iterated Tikhonov regularization on a partial Golub–Kahan reduction.
2. Jointly analyzed noise, discretization, Krylov-projection, and regularization errors.
3. Proved convergence rates up to $O(\delta^{2i/(2i+1)})$ under the stated assumptions.
4. Developed two parameter-selection rules, including one that can use a smaller Krylov subspace.
5. Demonstrated improved image-restoration and rectangular-tomography results against non-iterated and Arnoldi-based alternatives.

2026BDJZ

ChatGPT (July 2026)

Summary

The paper treats equilibrium-phase discovery in the Landau–Brazovskii model as computation of second-order stationary points, thereby distinguishing local minima from saddle points that can trap energy-decreasing first-order schemes. After Fourier pseudospectral discretization, the authors introduce an implicit–explicit trust-region method whose outer iteration is proved to approach second-order stationarity. Its nonconvex trust-region subproblem is solved globally by exploiting complementary Hessian structure: the interaction operator is diagonal in reciprocal space, while the nonlinear bulk term is diagonal in physical space, enabling FFT-based operations. Numerical comparisons with Bregman proximal-gradient, semi-implicit, SAV, IEQ, and exponential-time-differencing methods show that first-order methods can retain negative Hessian eigenvalues, whereas IMEX-TR reaches nonnegative curvature. The method also escapes converged saddles when applied as post-processing. It discovers cubic FDDD as a stable Landau–Brazovskii phase and maps a previously missing thermodynamic stability region without altering established neighboring regions.

Contributions

1. Reframed stable Landau–Brazovskii phase computation around second-order rather than merely first-order stationarity.
2. Introduced IMEX-TR and proved convergence of its outer iteration to second-order stationary points.
3. Provided a globally convergent trust-region subproblem solver using negative curvature and FFT-exploitable Hessian structure.
4. Showed that IMEX-TR avoids or escapes saddles where representative first-order methods stagnate.
5. Identified the cubic FDDD phase as stable and added its stability region to the model’s phase diagram.

2026BEAV

ChatGPT (July 2026)

Summary

PENCO is a physics-guided neural-operator framework for three-dimensional phase-field evolution. It can use either FNO-4D or a multi-head neural operator and augments data fitting with symmetric Gauss–Lobatto PDE residuals, consistency with a semi-implicit numerical teacher, low-frequency Fourier anchoring, free-energy dissipation, and epoch-dependent loss weights. Tests cover Allen–Cahn, Cahn–Hilliard, Swift–Hohenberg, phase-field-crystal, and molecular-beam-epitaxy equations on periodic cubes. Across in-distribution rollouts, the hybrid models reduce long-horizon drift and generally outperform purely data-driven operators; they also generalize to spherical, star-shaped, and toroidal initial conditions. The pure-physics model slightly outperforms PENCO for out-of-distribution Cahn–Hilliard evolution, where strict conservation dominates. Experiments use 32^3 grids, two operator layers, and substantially shorter training than the original two-dimensional MHNO study. Current demonstrations are limited to periodic domains and phase-field systems; fracture, crack growth, nonperiodic boundaries, and realistic multiphysics geometries remain future work.

Contributions

1. Unified physics, energy, and numerical-scheme consistency in a neural operator.
2. Added symmetric Gauss–Lobatto residual enforcement for stiff phase-field partial differential equations.
3. Enforced semi-implicit teacher consistency, spectral anchoring, and energy dissipation.
4. Evaluated five three-dimensional phase-field systems under fixed data budgets.
5. Demonstrated long-horizon and unseen-geometry improvements over FNO-4D and MHNO.

2026BF

ChatGPT (July 2026)

Summary

The paper builds first-order finite-volume schemes for weakly coupled two-dimensional systems whose flux depends on convolutions of all state variables. It approximates nonlocal terms at cell interfaces so standard monotone local fluxes, including Lax–Friedrichs- and Godunov-type choices, can be reused even though the complete nonlocal update is not monotone in every argument. Under regularity, CFL, and bounded-variation assumptions, the authors prove positivity and maximum principles, L^1 and bounded-variation estimates, existence and uniqueness of the weak entropy solution, convergence of the approximations, and a Kuznetsov-type $O(\sqrt{\Delta t})$ error estimate. Tests cover a scalar nonlocal model and intersecting pedestrian streams with an obstacle. At $N = 800$, errors against an $N = 3200$ Godunov reference are 0.801 for the earlier Lax–Friedrichs scheme, 0.267 for the proposed lower-viscosity variant, and 0.156 for Godunov; only Godunov approaches first-order observed convergence. The analysis is first-order and assumption-dependent; higher-order extensions are deferred.

Contributions

1. Reduced broad nonlocal fluxes to interface-local forms usable with monotone flux functions.
2. Established sufficient conditions for convergence to a unique weak entropy solution.
3. Proved positivity, maximum, L^1 , bounded-variation, and time-regularity estimates.
4. Derived an $O(\sqrt{\Delta t})$ error bound for the scheme class.
5. Demonstrated reduced diffusion and near-first-order Godunov behavior in two applications.

2026BFBK

ChatGPT (July 2026)

Summary

High-order finite elements on triangles and tetrahedra can make local Riesz-map solves scale prohibitively with polynomial degree p . This paper constructs alternative bases for the same simplicial de Rham spaces using integral moments against numerically generated polynomials orthogonal in two inner products. The resulting cell matrices have weak interior–interface and interior–interior couplings. Neglecting selected couplings yields sparse auxiliary operators, and combining the split with Pavarino, Arnold–Falk–Winther, and Hiptmair patch decompositions produces p -robust preconditioners in the targeted stiffness-dominated regime. In three dimensions, setup falls from the naive $O(p^9)$ cost to $O(p^6)$, with $O(p^4)$ storage and lower application cost. Incomplete Cholesky factorizations preserve the static-condensation sparsity pattern. Numerical tests across $H(\text{grad})$, $H(\text{curl})$, and $H(\text{div})$ confirm bounded iteration counts in the predicted regimes. Augmented-Lagrangian block preconditioners then extend the solvers to Hodge Laplacians. Robustness for strongly mass-dominated parameter choices is not generally guaranteed.

Contributions

1. Constructed high-order simplicial de Rham bases with provably weak interior coupling.
2. Derived interior–interface splittings compatible with standard patch-space decompositions.
3. Reduced three-dimensional setup complexity from $O(p^9)$ to $O(p^6)$.
4. Built sparse incomplete-Cholesky patch solvers with favorable degree scaling.
5. Applied the Riesz-map solvers to augmented-Lagrangian preconditioning of Hodge Laplacians.

2026BHJP

ChatGPT (July 2026)

Summary

Three-dimensional Gkeyll simulations revisit asymmetric dayside magnetic reconnection modeled on the 16 October 2015 Magnetospheric Multiscale diffusion-region crossing. The two-fluid ten-moment model evolves electron inertia and the full pressure tensor, but replaces a local heat-flux relaxation with a gradient-based closure. A symmetric vertex discretization and limiter enforce heat flow from hot to cold regions. Unlike the local closure, the improved model develops lower-hybrid drift instability within about five ion cyclotron periods, followed by a drift-kink mode, turbulence, magnetic islands, and flux ropes. It also produces sharper cross-field density transitions and a reconnection rate whose post-peak decline resembles kinetic simulations. A companion Vlasov comparison captures the same major instability sequence, although the kinetic kink grows substantially faster and contains additional small-scale modes. The closure still generates overly extensive temperature anisotropy and agyrotropy, indicating that restricted cross-field diffusion or a hybrid isotropizing term is needed before global magnetospheric use.

Contributions

1. Implemented a symmetric, limited gradient heat-flux closure in the Gkeyll ten-moment model.
2. Recovered lower-hybrid drift and secondary drift-kink instabilities absent with the local closure.
3. Connected instability saturation to turbulence, secondary magnetic islands, flux ropes, and declining reconnection rate.
4. Compared fluid and Vlasov calculations to distinguish captured instability physics from kinetic timing and scale differences.
5. Diagnosed excessive pressure-tensor anisotropy and proposed closure terms that restrict cross-field heat flow and restore upstream isotropy.

2026BI

ChatGPT (July 2026)

Summary

The paper develops linearly implicit time integrators for one-dimensional evolutionary partial differential equations with second-, third-, and fourth-order spatial derivatives, avoiding both Newton solves and the explicit stability restriction $\Delta t = O(\Delta x^k)$. It derives semi-implicit Rosenbrock-type methods, including order conditions through third order and L-stable four-stage schemes sharing one Jacobian matrix per step. It then repairs the instability of earlier explicit-predictor multistep formulas by using a semi-implicit predictor followed by backward-differentiation-formula corrections; a p th-order method generally needs p or $p + 1$ linear solves per step. Coupled to high-order finite differences and WENO convection discretizations, the methods attain their designed orders on nonlinear diffusion, convection–diffusion, compacton/KdV-type dispersion, and biharmonic diffusion. The second-, third-, and fourth-order predictor–corrector schemes converge at approximately their designed orders with Δt proportional to Δx . Evidence is limited to periodic, one-dimensional manufactured or analytic tests.

Contributions

1. Converted nonlinear stiff spatial terms into linear stage solves without Newton iteration.
2. Derived third-order conditions and L-stable semi-implicit Rosenbrock schemes.
3. Replaced an unstable explicit multistep predictor with a semi-implicit formulation.
4. Related predictor order and correction count to accuracy and linear-solve cost.
5. Demonstrated designed convergence for second- through fourth-derivative equations.

2026BKKG

Claude (September 2026)

Summary

The authors present H-NESSi, an open-source C++ library solving the Kadanoff–Baym equations (KBE), integro-differential equations of motion for two-time nonequilibrium Green’s functions. Earlier solvers on dense two-time grids cost $\mathcal{O}(T^3)$ operations and $\mathcal{O}(T^2)$ memory for T timesteps. Extending an earlier hierarchical off-diagonal low-rank (HODLR) scheme to multiorbital systems, H-NESSi partitions the lower-triangular retarded and lesser components into blocks stored as truncated singular value decompositions (TSVD) with tolerance ε_{SVD} , updated row by row during high-order multistep integration, and samples imaginary-time functions on sparse discrete Lehmann representation nodes. The interface mirrors the NESSi package, users implement self-energies, and an MPI class redistributes data between time points and momenta. For a driven superconductor in dynamical mean-field theory, retained ranks grow as N^α with block size N ($\alpha \approx 0.5$ superconducting, ≈ 0 normal), the Dyson solver scales approximately as $T^{2+\alpha}$ against T^3 for NESSi, and SVD updates as $T^{2+2\alpha}$. For the two-dimensional Hubbard model in the second Born approximation, ranks saturate, integration scales as $N_t^{2.05}$, and equal memory allows about three times longer propagation. The authors state that physical accuracy depends on the user-provided self-energy, that $\varepsilon_{\text{SVD}} > 10^{-5}$ can amplify errors, and that integrating the density matrix along the diagonal can diverge in long steady states.

Contributions

1. Kept the current row and the previous k rows of G^R and $G^<$, needed by the high-order integrators, uncompressed, compression thus lagging timestepping by $k + 1$ steps, with read-write Eigen maps only there and read-only access elsewhere costing $\mathcal{O}(N_S^\varepsilon)$ operations per value reconstructed from a TSVD block, left the Matsubara and mixed components uncompressed with imaginary time on DLR nodes, advised $N_t/2^{N_l} \approx 10$ for the number of levels N_l , and compressed the N_o^2 orbital components separately, offline tests on simple multiorbital systems indicating no significant gain from combining orbital and time indices.
2. Integrated the density matrix along the diagonal, $\partial_t \rho(t)|_{t=T} = \xi i \partial_t G^<(t, T) + \xi i \partial_t G^<(T, t)$, at no extra cost (RHO_DIAGONAL): for a continuously driven two-band, two-site Hubbard model with $\varepsilon_{\text{SVD}} = 10^{-10}$ and

$h = 0.025$ to 0.0125 , $\text{Re Tr}[\rho(t) - \rho^M]$ stayed constant at about 10^{-10} to 10^{-7} , whereas horizontal integration let it grow to about 3×10^{-5} to 3×10^{-3} and gave $\text{Im Tr} \rho(t)$ up to about 2×10^{-5} , a non-Hermiticity that forcing a Hermitian mean-field Hamiltonian would not remove.

3. Used MPI_Alltoallv in the mpi_comm class to pass from buffers holding all t' at local momenta to buffers holding all momenta at a local subset of t' and DLR nodes, reassigning t' points at every timestep because their number grows, so that the time-local second-Born self-energy is evaluated in real space via FFTW at $\mathcal{O}(N_k \log N_k)$ rather than $\mathcal{O}(N_k^2)$ cost, with $G_{(k_x, k_y)} = G_{(k_x, -k_y)}$ reducing L^2 to $L(L/2 + 1)$ momenta and one dyson and dlr_info instance per OpenMP thread because neither class is thread-safe.
4. Fitted, for dynamical mean-field runs with $U = 2$ and $\varepsilon_{\text{SVD}} = 10^{-6}$ at $\beta = 18$ (superconducting) and $\beta = 1$ (normal), log-log slopes of cumulative wall time of 2.38 and 2.06 for the timestep solve and 3.03 and 2.12 for the SVD updates, against 3.03 for NESSi, and, for the Hubbard lattice with $L = 128$, $\beta = 50$, $U = 0.5$, and $\varepsilon_{\text{SVD}} = 10^{-12}$, propagated $N_t = 6289$ time points in the memory that limits NESSi to $N_t = 2048$.
5. Measured nearly linear MPI speedup up to 32 ranks on one node and OpenMP speedup near $N_{th}^{0.95}$ up to 16 threads for $L = 48$, $N_t = 2048$, and, for $L = 64$, $N_t = 16384$, found the fewest CPU hours on 2048 CPUs at 8 threads per rank (up to about 1.6 times more otherwise) with about 1 TB of memory, maximum ε -ranks ranging over the Brillouin zone from about 20 to 33 at $\beta = 1$ and 40 to 50 at $\beta = 5$, and memory savings near 100 for the largest blocks at $\beta = 5$.

Comments

- Abstract, Section 4.7, and Fig. 4(a): the abstract claims for the driven-superconductor and two-dimensional Hubbard benchmarks “asymptotic time complexity significantly below the cubic scaling of uncompressed approaches”, but in the superconducting state the TSVD update step follows, per Section 4.7, “the expected scaling of $T^{2+2\alpha}$ ” with “ $\alpha \approx 0.5$ for the superconducting state”, that is T^3 ; Fig. 4(a) fits it with slope 3.03, as for NESSi, and at the last step its cumulative time (about 60 h) exceeds that of the timestep solve (about 18 h, slope 2.38), so in the driven-superconductor benchmark only the Dyson solver and the normal state are shown to be sub-cubic; unresolved.
- Sections 2.4 and 3: the Integration class “supports discretization error scaling as h^{k+1} , where $1 \leq k \leq 5$ is a user-specified parameter”, and k is called “the integration order” (Sections 3.1 and 4.1), which gives five integrators, of first to fifth order in k or second to sixth order in the error exponent, but Section 3 lists “first- to sixth-order integrators implemented in the integration class”; unresolved.
- Section 4.3, Eqs. (23)–(24): at $t' = 0$ and $\tau = 0$ the second contour arguments of the two components are the adjacent points t_{0+} and $t_{0|}$ (Table 1, Fig. 1), both after every real-time point, so $G^{\uparrow, HF}(t, 0)$ must equal $G^{<, HF}(t, 0)$, but (23) gives $i \exp[-i\epsilon(0|)t] \rho^M$ and (24) gives $\xi i \exp[-i\epsilon(0|)t] \rho^M$, which differ in sign for fermions ($\xi = -1$), the statistics of the example; since $\rho(t) = \xi i G^{<}(t, t)$ (Section 3.2) implies $G^{<}(t, t) = -\xi i \rho$, which (23) matches for fermions, the evident intended prefactor in (24) is $-\xi i$.
- Section 5.2, Eqs. (32)–(33): writing $\Sigma(z, z') = U^2 G(z, z') G(z, z') G(z', z)$ as $-U^2$ times the Bubble2 product of G with the Bubble1 product of G and G , the rules of Section 3.3 reproduce $\Sigma^{<}$ of (33) but, through $C^M(\tau) = -A^M(\tau) B^M(-\tau)$ and $C^M(\tau) = -A^M(\tau) B^M(\tau)$, give $\Sigma^M(\tau) = -U^2 G^M(\tau) G^M(\tau) G^M(-\tau)$, opposite in sign to (32), $\Sigma_r^M(\tau) = U^2 G_r^M(\tau) G_r^M(\tau) G_{-r}^M(-\tau)$; with $G^M(-\tau) = \xi G^M(\beta - \tau)$ (Section 3.1) the derived form equals $U^2 G^M(\tau) G^M(\tau) G^M(\beta - \tau)$ for fermions, which Sigma_mat in Appendix B implements for the analogous bubble with GM_reversed, so (32) evidently intends $U^2 G_r^M(\tau) G_r^M(\tau) G_{-r}^M(\beta - \tau)$, equivalently a leading minus sign.
- Section 5.2, Eq. (34): $\Sigma_r^{\uparrow}(t, \tau) = U^2 G_r^{\uparrow}(t, \tau) G_r^{\uparrow}(t, \tau) G_{-r}^{\uparrow}(\tau, t)$ attaches the right-mixed symbol to the arguments (t, τ) and the left-mixed symbol to (τ, t) , the reverse of Table 1, where $G^{\uparrow}(t, \tau)$ is left-mixed and $G^{\uparrow}(\tau, t)$ right-mixed; since H-NESSi stores and propagates the left-mixed component (Section 3.1, Eq. (7)) and the product that gives (33) yields $\Sigma^{\uparrow}(t, \tau) = U^2 G^{\uparrow}(t, \tau) G^{\uparrow}(t, \tau) G^{\uparrow}(\tau, t)$, as Sigma_tstp in Appendix B implements, that is evidently the intended equation.
- Fig. 10 caption: the figure is described as “Performance scaling of the hybrid MPI/OpenMP implementation for a fixed total number of CPUs ($N_{CPU} = 2048$)”, and panel (c) as showing CPU hours “for two different simulation lengths”, but the legend of panel (c) labels the second series “ $N_t = 8192$, $N_{CPU} = 512$ ”; unresolved.
- Appendix A, Eq. (A.10): in $\omega_i = \sum_{j=0}^i b_{k+1-i}$ the summand does not depend on the summation index j , so the sum reduces to $(i+1)b_{k+1-i}$ rather than a cumulative sum of Adams–Moulton weights (for $k = 2$, with

$b_3, b_2, b_1 = 5/12, 8/12, -1/12$, it gives $\omega_1 = 16/12$ and $\omega_2 = -3/12$ instead of $13/12$ and 1); the evident intended summand is b_{k+1-j} .

2026BKL

ChatGPT (July 2026)

Summary

This paper develops a compact, implicit form of the local anisotropic basis function method (LABFM) for high-order meshless differentiation on irregular point clouds. Lele-type implicit stencils are selected locally, and their coefficients are optimized over two-dimensional wavenumber space to improve resolving power while encouraging diagonal dominance and a well-conditioned global sparse system. Second- and fourth-order gradient and Laplacian operators resolve substantially more of the admissible spectrum than explicit LABFM; some fourth-order Laplacians exhibit spectral-like behavior. Convergence tests confirm lower differentiation errors, although compact gradient operators remain unstable for purely advective problems and may require filtering. In viscous Burgers simulations, compact schemes reduce early, advection-dominated errors by as much as an order of magnitude. For Poisson problems on a domain with a curved internal boundary, the largest compact stencils also reduce error by about an order of magnitude with nearly no additional solve cost. The study is a proof of concept, and coefficient forms and stencil selection remain open optimization questions.

Contributions

1. Formulated compact LABFM operators using local implicit Lele-type stencils on unstructured node sets.
2. Designed a coefficient-selection procedure that balances wavenumber resolution, diagonal dominance, and conditioning.
3. Quantified spectral-like resolving behavior for second- and fourth-order gradient and Laplacian operators.
4. Established convergence and stability characteristics, including the remaining instability of compact advection operators.
5. Demonstrated up to order-of-magnitude error reductions for viscous Burgers and curved-domain Poisson problems.

2026BKSS

ChatGPT (July 2026)

Summary

The paper constructs a strongly history-independent, fully dynamic two-choice balls-and-bins allocation. Its representation depends only on the current ball set and fixed randomness, so snapshots reveal no update history. For at most m balls on n bins, the final algorithm guarantees maximum load $m/n + O(1)$ with high probability while requiring only $O(\log \log(m/n))$ expected ball movements per insertion or deletion. The construction begins with a canonically ordered history-independent greedy baseline, then develops Slice and Spread, which repeatedly removes excess balls above carefully chosen thresholds and reallocates them using fresh second-choice randomness. A history-independent cuckoo-hashing-style cleanup converts small cumulative overload into constant maximum overload. The result improves even the prior unrestricted fully dynamic trade-off by combining constant overload with $o(m/n)$ expected recourse.

Contributions

1. Formalized strongly history-independent dynamic two-choice allocation through unique representability from the current set and random tape.
2. Analyzed canonical-order history-independent greedy, including its recourse behavior over a broad parameter range.
3. Introduced Slice and Spread, whose thresholded rounds progressively smooth loads while retaining low update sensitivity.

4. Developed a history-independent cuckoo-hashing cleanup that turns low cumulative overload into $O(1)$ maximum overload.
5. Proved maximum load $m/n + O(1)$ with high probability and $O(\log \log(m/n))$ expected recourse per operation.

2026BKT

ChatGPT (September 2026)

Summary

Three-dimensional particle-in-cell simulations examine high-pressure plasma formation and stability in a compact magnetic mirror, treating both ions and electrons kinetically. An energy- and charge-conserving semi-implicit model follows continuous Maxwellian plasma injection into an initially empty trap. A radial electric field develops alongside the longitudinal ambipolar potential, redistributing azimuthal currents so that electrons carry almost all the current supporting equilibrium. Electron cooling and temperature anisotropy in the magnetic expanders also modify the escaping ion flow relative to an isothermal-electron model. As pressure rises, an azimuthal $m = 1$ flute instability displaces the plasma toward the wall before β , the ratio of plasma pressure to vacuum magnetic pressure, reaches unity. Neither conducting ends nor conducting cylindrical or square sidewalls stabilize this mode in the simulated regime. The authors tentatively associate this failure with rapid ion motion and strong electrostatic fluctuations that disrupt particle attachment to magnetic field lines. The reduced mass ratio $m_i/m_e = 100$, collisionless treatment, rapid model injection, and omission of secondary electron emission limit direct experimental extrapolation.

Contributions

1. Simulated compact high-beta mirror-plasma formation with fully kinetic ions and electrons using a semi-implicit scheme conserving global energy and local charge while resolving electron gyromotion.
2. Quantified how the radial electric field suppresses ion azimuthal current and enhances electron current, with force-balance analysis identifying a significant magnetic-curvature contribution.
3. Resolved electron cooling and anisotropy in magnetic expanders, reducing the ambipolar potential drop and ion outflow speed relative to an isothermal-electron nozzle model.
4. Identified the pressure-limiting $m = 1$ flute instability, whose displacement grows at a rate comparable to the inverse ion transit time while higher azimuthal modes saturate at smaller amplitudes.
5. Tested conducting cylindrical and square sidewalls and found no stabilization in the fast-instability regime, documenting plasma displacement relative to weakly distorted magnetic field lines.

2026BLAO

ChatGPT (July 2026)

Summary

TokaMind is an open-source multi-modal transformer pretrained on public MAST tokamak diagnostics spanning time series, two-dimensional profiles, and videos with different sampling rates and missing signals. Fixed-basis DCT3D embeddings, masked attention and losses, a shared transformer, modality heads, and per-target adapters permit partial parameter reuse across changing input and output schemas. Pretraining uses 16 diagnostics, five actuators, and 34 targets; the Base and Tiny models contain 6.93 million and 3.69 million parameters. On the 14-task TokaMark benchmark, fine-tuned Base achieves overall normalized root-mean-square error 0.212, compared with 0.249 for the strongest benchmark baseline and 0.229 for matched-epoch training from scratch. It wins 13 of 14 tasks and 52 of 57 signal outputs; pretraining helps most for long-horizon and high-dimensional equilibrium tasks. Evaluation is confined to MAST, does not establish cross-device transfer, and leaves rapidly varying and long-horizon signals challenging.

Contributions

1. Introduced an open-source foundation model for heterogeneous tokamak diagnostics.
2. Provided schema-flexible masking, tokenization, and adapters for missing or changing signals.

3. Used lightweight DCT3D embeddings that outperform the tested learned codecs.
4. Improved overall TokaMark error to 0.212 and won 13 of 14 tasks.
5. Showed that pretraining transfers most clearly to long-horizon and high-dimensional objectives.

2026BLBS

Claude (September 2026)

Summary

In beam-driven plasma-wakefield accelerators, the efficiency of energy transfer from the wake to a trailing bunch is usually measured with an electron spectrometer, which is invasive and averages over the whole plasma stage. At DESY's FLASHForward facility, the authors instead use the excess light emitted by a 195-mm argon discharge capillary after the beams pass, recorded by a side-view camera. The light yield rises with driver energy loss and falls when a trailing bunch extracts energy, so a linear response curve calibrated on driver-only shots converts light into remaining wake energy and hence efficiency. At a high-efficiency working point, the spectrometer gave $34 \pm 2\%$ wake-to-trailing-bunch efficiency and $2.2 \pm 0.1\%$ driver-to-trailing-bunch efficiency. Light-based and spectrometer efficiencies are linearly correlated, but the light-based values are higher, which the authors attribute mainly to trailing-bunch charge loss. Dividing the image into 3-mm segments yields longitudinally resolved efficiencies: the final segment agrees much better with the spectrometer, and an efficiency decline along the density plateau, removed by collimating the bunch tail, is consistent with in-plasma charge loss. Segment calibration assumes local and integrated driver energy losses are proportional; simulations indicate resulting errors of a few percentage points. The light-emission mechanism itself is not fully understood.

Contributions

1. Demonstrated excess plasma-emission light as a noninvasive, in situ measure of the energy remaining in a plasma wake, including its reduction when a trailing bunch extracts energy.
2. Formulated an efficiency measurement from a calibrated light-versus-wake-energy response curve, with a shot-to-shot driver energy-loss estimate combining light and spectrometer data.
3. Reached a working point with 2.2% driver-to-trailing-bunch efficiency, compared with 0.9% at a previous working point, mainly because a longer plasma quadrupled the trailing-bunch energy gain.
4. Measured the local energy-transfer efficiency along the plasma, revealing a decline that tail collimation removes and that the downstream spectrometer cannot detect.
5. Quantified with HiPACE++ simulations the systematic error of the segment-calibration assumption, reaching 8 percentage points only for an unrealistically high driver emittance.

Comments

- §V, first paragraph: the stated best fit $\eta_p = 0.75 \eta_s$, with bounds $\eta_p = 0.65 \eta_s$ and $\eta_p = 0.81 \eta_s$, implies $\eta_p < \eta_s$. The preceding sentence (points lie above the line $\eta_p = \eta_s$, and for $\eta_p > 30\%$ the spectrometer value η_s is on average 9.7 percentage points lower) and Fig. 3 (η_p on the vertical axis, η_s on the horizontal axis, points above the diagonal) both show $\eta_p > \eta_s$. The intended relations are most likely $\eta_s = 0.75 \eta_p$, $0.65 \eta_p$, and $0.81 \eta_p$.

2026BLGA

ChatGPT (September 2026)

Summary

Motivated by synchronized core and edge oscillations in KSTAR double-peaked fishbones, this study investigates radial coupling using visco-resistive, single-fluid magnetohydrodynamic simulations with MEGA. Modified KSTAR equilibria contain nearly flat safety-factor profiles that produce low-frequency Alfvén-continuum plateaus, while a spatially localized internal antenna supplies periodic drive. An antenna near the edge excites a coherent core-localized

quasi-mode, demonstrating that a local response need not require local drive. Comparisons between central and off-axis receiving plateaus show stronger inward than outward excitation, consistent with energy concentration toward smaller plasma volumes. The core also responds below its continuum-plateau frequency, suggesting a mechanism that could accommodate frequency chirping without an exactly matching continuum. Artificially suppressing pressure and density fluctuations changes the response but preserves remote excitation. Analysis of transients distinguishes rapid magnetoacoustic propagation, poloidal Alfvén-wave spreading, radial structure formation, and slower amplitude saturation. These results establish a possible MHD coupling mechanism; its relevance to experimental fishbones remains conditional because energetic-particle kinetics and differential plasma rotation are omitted, and chirping is not simulated self-consistently.

Contributions

1. Demonstrated coherent core Alfvénic quasi-modes excited by antennas at distant radii on deliberately constructed low-frequency continuum plateaus.
2. Used two receiving plateaus and an antenna-position scan to establish stronger inward than outward excitation, supporting a volumetric-focusing interpretation.
3. Found substantial central responses to edge drive at 4.5 and 6.75 kHz below an approximately 10 kHz continuum plateau, including configurations without a $q = 1$ surface.
4. Verified that remote core excitation survives artificial suppression of density and pressure perturbations, although polarization, continuum frequencies, and amplitudes change.
5. Resolved distinct propagation, radial structure formation, and amplitude saturation stages, with dissipation scans distinguishing the settling time from the final core response.

2026BLPP

ChatGPT (July 2026)

Summary

The paper develops a rapid classical tearing-mode stability workflow by asymptotically matching two complementary calculations. STRIDE supplies the ideal-MHD toroidal outer-region index Δ' , including shaping, while SLAYER solves a two-fluid drift-MHD slab model for the resistive inner-layer response $\Delta(\gamma, \omega)$. A toroidal proxy for thermal-conduction-modified Glasser stabilization enters through $\Delta_{\text{eff}} = \Delta' - \Delta_{\text{crit}}$. SLAYER smoothly connects four analytic growth-rate regimes, and the combined workflow agrees with the TJ code in a low- β benchmark. At inverse aspect ratio above about 0.45, STRIDE+SLAYER and TJ diverge as their geometric approximations approach different ideal limits. Application to a shaped, DIII-D-like synthetic H-mode equilibrium tracks 2/1 and 3/1 growth rates as the toroidal-current profile broadens. The model is fast and interpretable but omits inner-layer pressure gradients, parallel flow, rational-surface coupling, and resistive-wall effects; experimental validation of Δ_{crit} remains necessary.

Contributions

1. Coupled toroidal STRIDE outer-region calculations to the two-fluid SLAYER resistive layer.
2. Solved jointly for tearing-mode growth rate and rotation frequency using an effective stability index.
3. Incorporated a toroidal proxy for thermal-conduction-modified Glasser stabilization.
4. Validated the workflow against four analytic regimes and TJ scans in plasma β and inverse aspect ratio.
5. Demonstrated stable calculations for multiple rational surfaces in a shaped H-mode-like tokamak equilibrium.

2026BMCC

ChatGPT (September 2026)

Summary

This paper develops a unified description of transverse electron-beam dynamics across a laser-plasma accelerator's density up-ramp, plateau, down-ramp, and neighboring transport lines. The transverse focusing gradient K controls the

evolution of the Twiss parameters and emittance. For a constant- K plateau, the authors derive formulas for chromatic betatron decoherence, the distance to saturation, and the saturated emittance; particle-in-cell and transfer-matrix simulations validate these predictions. Matching $\alpha = 0$ and $\beta = 1/\sqrt{K}$ at a plasma-section entrance suppresses emittance growth, and a matched beam can remain matched as K subsequently varies without requiring an adiabatic ramp. Density ramps chiefly ease coupling to conventional optics by reducing the Twiss parameter $\gamma = (1 + \alpha^2)/\beta$ before the beam reaches a transport line, where chromatic growth can otherwise exceed that inside the plasma by orders of magnitude. Simulations of linear, exponential, and Gaussian ramps show that ramp length matters more than detailed shape and quantify practical mismatch tolerances. The results support joint optimization of ramps, plasma plateaus, and transport optics in multistage accelerators.

Contributions

1. Identified the focusing gradient K as the organizing parameter for transverse dynamics throughout a realistic plasma stage.
2. Derived consistent expressions for betatron decoherence length and saturated emittance from energy spread and initial Twiss mismatch.
3. Validated the analytical model with FBPIC, WAKE-T, and TRACEWIN simulations.
4. Showed that entrance matching preserves emittance under subsequent variations of K independently of adiabaticity.
5. Established ramp-design guidelines and mismatch tolerances that reduce γ and limit chromatic emittance growth in adjoining transport lines.

2026BMP

ChatGPT (July 2026)

Summary

This paper compares momentum-accelerated power iteration with restarted Lanczos for dominant eigenpairs of symmetric or Hermitian matrices using matrix-vector products as the common cost. Both methods apply polynomial filters whose convergence is governed by Chebyshev-polynomial ratios, but their suppression of subdominant eigenmodes differs: momentum has a nearly uniform average rate with oscillations, whereas restarted Lanczos has quasi-periodic mode-wise behavior. The analysis derives a critical Krylov subspace size m_c , determined by relative spectral-gap parameters, below which static or dynamic momentum converges faster than restarted Lanczos with the same product budget. Numerical examples confirm the predicted crossover. The authors then alternate restarted Lanczos cycles with momentum steps, interpreting momentum as a polynomial preconditioner that damps the peaks in Lanczos's mode-wise convergence. Tests on synthetic spectra and eight SuiteSparse matrices show that the hybrid can reduce products and improve attainable residuals, especially for smaller Krylov spaces, though it is not uniformly best. Because momentum needs no orthogonalization or auxiliary projected eigenproblem, the comparison also exposes memory advantages beyond iteration counts.

Contributions

1. Put momentum power iteration and restarted Lanczos in a common Chebyshev-polynomial convergence framework.
2. Derived spectral-gap-dependent crossover criteria for their matrix-vector-product efficiency.
3. Characterized uniform-versus-quasi-periodic damping of subdominant eigenmodes.
4. Introduced momentum-preconditioned restarted Lanczos as an alternating polynomial-filter scheme.
5. Tested the theory and hybrid method on synthetic spectra and eight sparse benchmark matrices.

Summary

The authors propose a deterministic particle method for the spatially homogeneous Landau equation with per-step cost $O(N + M^d \log M)$ for N particles and M Fourier modes per dimension at fixed transform tolerance. As background, the equation is a continuity equation with a velocity field $U[f]$, which blob methods evaluate pairwise at $O(N^2)$ cost. Writing $U[f] = -J(f, f)/f$ with J the collision flux, they reconstruct the density as the Fourier truncation $P_M f^N$ of the particle measure on the torus $[-L, L]^d$ by a type-1 nonuniform FFT; flux modes follow by FFT convolution with precomputed Landau kernel modes, and a type-2 transform evaluates flux and floored density at the particles. For smooth f supported in $B(0, R)$, $L > 2R$, and Sobolev index $s > d/2 + 1$, a consistency theorem bounds the velocity-field error on $\{f > \kappa\}$ by $C(\kappa^{-1}M^{-(s-d/2-1)} + \kappa^{-2}\|P_M f^N - f\|_{C^1})$, separating spectral truncation from density reconstruction. In two-dimensional Maxwellian-molecule tests, relative L^2 errors on the BKW solution reach order 10^{-7} , against about 10^{-2} for the blob method, with runtime sub-linear in N ; errors stagnate in N when M is too small, and too few particles for M cause instability, which Gaussian filtering removes at an accuracy cost. Only mass is conserved exactly; trajectory convergence, low-density regions, and three-dimensional Coulomb interactions remain open.

Contributions

1. Derived the flux modes $(\hat{J}_M)_k = \sum_m \hat{f}_m \hat{f}_{k-m} \hat{\beta}(k-m, m)$ with $\hat{\beta}(l, m) = (l-m)D(m) - (l-m)I(m)$, where $D(m) = \frac{i\pi}{L} \int \Psi(q) e^{-i\pi m \cdot q/L} dq$ and $I_{pq}(m)$ is the same integral weighted by $\mu_p \mu_q$, with $\Psi = |q|^{\gamma+2}$ and $\mu = q/|q|$, the kernel modes of the Pareschi–Russo–Toscani spectral method; identified $P_M f^N$ as a kernel density estimate with the unit-mass but sign-changing Dirichlet kernel $\psi_M = (2L)^{-d} \sum_k e^{i\pi k \cdot x/L}$, handled by the denominator $\max\{P_M f^N, f\}$, related it to the Fourier-in-cell approach, and found a drift–diffusion splitting based on a spectral approximation of $\nabla f/f$ less stable.
2. Proved that the periodic flux equals the whole-space flux on $[-L, L]^d$ for data supported in $B(0, R)$ when $L > 2R$, the stability bounds $\|J_{\text{per}}(f, g)\|_{C^0} \leq C\|g\|_{C^1}\|f\|_{C^1}$ and $\|J_{\text{per}}(f, g)\|_{H^s} \leq C\|g\|_{H^{s+1}}\|f\|_{H^{s+1}}$ for $s > d/2$ from the integrability of $|q|^{\gamma+2}$ for $\gamma > -d-2$ and the algebra property of H^s , truncation errors of order $M^{-(s-d/2)}$ in C^0 and $M^{-(s-1-d/2)}$ in C^1 , and the flux error $CM^{-(s-d/2-1)}\|f\|_{H^s}^2$ (Proposition 3.3), with the theorem further assuming $\|P_M f^N - f\|_{C^0} \leq \kappa/2$, $\|P_M f^N - f\|_{H^s} \leq 1$, and a floor $\underline{f} \leq \kappa/2$, which is then inactive on $\{f > \kappa\}$.
3. On the BKW solution with ring-shaped $f(0, v) = |v|^2 e^{-|v|^2}/\pi$ and the collision operator scaled by $1/16$, showed at $t = 0$, with 100^2 particles on $[-10, 10]^2$ and $L = 10$, the error of $P_M f^N$ falling to about 10^{-10} at $M = 64$ and that of U_M worsening as κ decreases from 10^{-1} to 10^{-5} , and at $t = 8$, with particles on $[-5, 5]^2$ and RK4 with $\Delta t = 10^{-3}$, errors of order 10^{-5} for $M = 48$ that do not improve beyond $N = 60^2$, errors for $M = 32$ that grow from $N = 40^2$ to 60^2 , and errors for $M = 64$ falling to order 10^{-7} at $N = 120^2$, while $M = 128$ was less accurate than $M = 64$ at both $N = 100^2$ and 200^2 .
4. Timed MATLAB R2026a runs on an Apple M2 MacBook Air with FINUFFT at tolerance 10^{-13} and floor 10^{-13} , finding log–log runtime slopes in N of 0.42, 0.44, and 0.62 for $M = 32, 48$, and 64 against roughly quadratic growth for the blob method (Euler, $\Delta t = 10^{-2}$, $\varepsilon = 0.64h^2$), which the spectral method undercut from $N > 3 \cdot 10^3$ despite RK4 with $\Delta t = 10^{-3}$, and tracked momentum errors near 10^{-8} and energy errors below 10^{-5} for $M = 64$, $N = 100^2$, whereas the blob method conserves momentum exactly at the fully discrete level.
5. Stabilized the under-resolved case $M = 128$, $N = 80^2$, whose unfiltered error grows to about 10^{-1} by $t = 8$, with the Gaussian filter $(\hat{\psi})_k = \exp(-\pi^2 \varepsilon |k|^2/(2L^2))$, tested for $\varepsilon \in \{0.1, 0.3, 0.64\}h^2$, which stabilized the run at a significant loss of accuracy, with $\varepsilon = 0.64h^2$ matching the blob accuracy, and reproduced Trubnikov’s decay $\Delta T(t) = \Delta T(0)e^{-t/2}$ from $T_1(0) = 0.75$, $T_2(0) = 0.5$ with final errors 5×10^{-7} ($M = 48$, $N = 80^2$, $\Delta t = 10^{-2}$), 4×10^{-8} ($M = 64$, $N = 100^2$, $\Delta t = 10^{-3}$), and 10^{-2} for the blob method, and the relaxation of $0.9\mathcal{M}(v; (0, 0), 1) + 0.1\mathcal{M}(v; (3, 0), 0.1)$ toward the Maxwellian with mean $(0.3, 0)$ and temperature 1.315 with 140^2 particles on $[-6, 6]^2$ and $M = 64$.

Comments

- Section 4 and Fig. 3: errors and diagnostics are said to use “a fixed grid made of 120×120 points over $[-L, L]^2$ ”, with $L = 10$, and “This grid is shared across all simulations and baseline algorithms for better comparison”, but the Fig. 3 caption states “The error is computed here on a grid of 200×200 points over $[-10, 10]^2$ ”, and Fig. 4 reuses these runs (“same as Figure 3a”); unresolved.
- Section 4.1, “Spectral accuracy at $t = 0$ ”: the computed quantities are listed as “the truncated Fourier approximations $f^M = \mathcal{P}_M f$, $J_M(f, f)$, and $U_M[f^N]$ ”, the projections of the exact density used in the proof of Proposition 3.3, but the Fig. 2 panels are titled f_M^N and $J_M(f_M^N, f_M^N)$, the caption refers to the Fourier KDE f_M^N and to $J_M(\mathcal{P}_M f^N, \mathcal{P}_M f^N)$, the following sentences to f_M^N and $J_M(f_M^N, f_M^N)$, and the preceding paragraph motivates the test by the accuracy of $\mathcal{P}_M f^N$ on the particle initialization grid; the evident intended quantities are $f_M^N = \mathcal{P}_M f^N$ and $J_M(f_M^N, f_M^N)$.

2026BR

ChatGPT (September 2026)

Summary

This paper extends compact Runge–Kutta flux reconstruction (cRKFR) to hyperbolic systems with stiff source terms and non-conservative products, which cannot generally be expressed as flux divergences. An implicit–explicit time discretization treats source terms implicitly at each solution point while keeping advection explicit and requiring only one inter-element numerical flux evaluation per time step. Interface fluxes represent non-conservative terms; a first-order finite-volume method on element subcells is blended with the high-order method near nonsmooth solutions. Flux and subcell limiters enforce physical admissibility, such as positive density and pressure, under the paper’s assumptions on the low-order scheme and admissible set. Tests cover stiff scalar equations, reactive Euler flow, the ten-moment model, shear shallow water flow, GLM magnetohydrodynamics, and multi-ion magnetohydrodynamics. Smooth-wave and manufactured-solution tests recover the expected accuracy, while shock and instability tests show robust behavior. The authors identify path-conservative and well-balanced extensions as open work.

Contributions

1. Extended the time-averaged cRKFR formulation to stiff sources through point-local implicit–explicit Runge–Kutta updates.
2. Formulated interface numerical fluxes for non-conservative products while retaining one inter-element flux computation per time step.
3. Built a first-order subcell finite-volume scheme and blended it with high-order cRKFR for nonsmooth solutions.
4. Combined interface flux limiting and additional subcell blending to preserve admissibility at solution points under stated assumptions.
5. Verified accuracy and robustness across stiff reactive, shallow-water, single-ion MHD, and multi-ion MHD tests.

2026BSLW

ChatGPT (July 2026)

Summary

This paper co-designs an MPI extension and HPE Slingshot 11 implementation that removes the CPU from the fast path of two-sided GPU communication while retaining familiar MPI semantics. Persistent requests separate setup from repeated communication; `MPI_Match` permanently resolves partners before execution, and an `MPI_Queue` orders matched request starts and waits with GPU-stream kernels. The prototype maps these abstractions to libfabric deferred work queues, triggered operations, and GPU-accessible counters while preserving receiver-buffer readiness. A Cabana `StreamHalo` implementation shows that gather and scatter exchanges can enqueue receives, packing, sends, completion waits, and unpacking without host synchronization. On Frontier and Tuolumne, stream-triggered ping-pong lowers latency by as much as 50% for small and medium messages, although the untuned prototype loses to production MPI for

large messages and some small non-ready sends. A CabanaGhost strong-scaling benchmark reaches 8,192 Frontier GPUs and improves maximum mean speedup by 28% for the large problem. Source code, run scripts, raw measurements, and analysis are released.

Contributions

1. Designed a CPU-free GPU communication API around MPI persistent operations, permanent matching, and stream-associated queues.
2. Preserved two-sided matching and remote-buffer-readiness semantics outside the GPU–NIC critical path.
3. Implemented the API on Slingshot 11 using libfabric deferred work queues, triggered operations, and counters.
4. Integrated the abstraction into Cabana halo exchanges with only limited changes to conventional MPI structure.
5. Demonstrated up to 50% lower ping-pong latency and 28% higher maximum strong-scaling speedup on leadership-class systems.

2026BSSK

ChatGPT (July 2026)

Summary

This paper ports the energy-conserving semi-implicit particle-in-cell code ECsim to GPUs with OpenACC directives while preserving its MPI decomposition and production code structure. The particle mover and moment-gathering kernels are offloaded after restructuring loops, managing device-resident data, and addressing atomics and particle-to-cell access patterns. Numerical comparisons confirm agreement with the CPU implementation. On Leonardo Booster nodes, the accelerated version achieves about a fivefold reduction in time to solution and a threefold reduction in energy use relative to the CPU baseline. Cross-generation tests show that ECsim’s memory-bound kernels benefit from newer bandwidth and that the GH200 unified-memory design reduces data-movement penalties; newer atomic hardware particularly improves moment gathering. Strong scaling retains roughly 70% efficiency through 64 GPUs, and weak scaling retains about 78% through 1,024 GPUs. Comparisons are deliberately limited to the original ECsim implementation because other PIC codes use different algorithms and accuracy-cost tradeoffs.

Contributions

1. Ported ECsim’s dominant particle kernels to GPUs with minimally invasive OpenACC directives.
2. Preserved the code’s semi-implicit, machine-precision energy-conserving formulation.
3. Validated GPU results against the original CPU implementation.
4. Quantified time-to-solution and energy-to-solution gains across NVIDIA GPU generations.
5. Demonstrated strong and weak scaling to 64 and 1,024 GPUs, respectively.

2026BTG

ChatGPT (July 2026)

Summary

The paper develops a von Neumann stability analysis for explicit and implicit arbitrary-Lagrangian–Eulerian ADER discontinuous Galerkin schemes for hyperbolic equations. On classical one-dimensional spacetime meshes, it derives amplification-matrix criteria, clarifies the limited validity of commonly used CFL values at higher polynomial degrees, and bounds admissible mesh velocities. It then models topology-changing moving meshes by inserting degenerate sliver spacetime elements that have zero spatial size at time-step endpoints but nonzero spacetime volume. Conservative predictor and corrector formulations exchange fluxes through these slivers without projection or reconstruction. The analysis and numerical studies show that slivers do not alter the explicit scheme’s classical CFL restrictions. The implicit scheme is stable throughout the tested parameter range, with a proof for a representative sliver configuration. Linear-advection convergence tests recover the expected consistency order for both formulations. The analysis is

restricted to scalar one-dimensional model problems, so nonlinear systems and multidimensional topology changes remain future work.

Contributions

1. Formulated explicit and implicit direct ALE ADER-DG schemes in a common spacetime framework.
2. Derived Fourier amplification-matrix stability criteria for the explicit method.
3. Showed that conventional higher-order CFL values can be weakly unstable and derived mesh-velocity restrictions.
4. Extended conservative predictor and corrector formulations to degenerate sliver elements representing mesh topology changes.
5. Established unchanged explicit CFL behavior with slivers, proved an implicit stability case, and verified the expected consistency orders.

2026BTSL

Claude (September 2026)

Summary

Downramp injection in laser wakefield accelerators can trap electrons in several plasma periods behind the driver. In the authors' experiments, a $6 \times 10^{18} \text{ W cm}^{-2}$ driver, focused 1.5–2.5 mm beyond a hydrodynamic optical-field-ionization (HOFI) shock made by a line-focused heater beam in a helium jet at $1\text{--}2 \times 10^{18} \text{ cm}^{-3}$, yielded spectra with 2 to 7 quasi-monoenergetic peaks whose energies rise linearly with peak index. Particle-in-cell simulations, initialized with a FLASH density profile whose roughly 2 μm gradient is much shorter than the 33 μm plasma wavelength, reproduce the trains. Electrons are injected at nearly the same phase at the back of each period of a nearly sinusoidal wake; as the laser intensity rises toward focus, relativistic lengthening enlarges every period by about the same amount, so the already relativistic bunches shift toward the bubble centers by multiples of a constant offset and sample linearly weaker fields. The shift relaxes as the laser defocuses. Narrow peaks further require energy compression, a phase-space rotation as beam loading reverses the field gradient along each bunch, completed just before the jet exit; 0.5 mm focus shifts alter it. Excessive lengthening loses later bunches. The authors propose the trains as a single-shot probe of relativistic lengthening.

Contributions

1. Quantified the experimental periodicity by linear fits of peak energy versus index for shots with 3 to 7 peaks, giving spacings of 31 to 86 MeV with R^2 from 0.988 to 0.999, and relative FWHM spreads of individual peaks as low as 1.9%.
2. Tracked the distance Δz between each bunch's center of mass and its bubble rear for four successive periods, showing that it grows with period index, peaks at the laser focus 2 mm after the shock, and returns close to the injection phase as the laser defocuses.
3. Showed that the linear dependence of the mean Δz on period index persists after removing the laser-centroid motion from density-induced dephasing and the beam-centroid motion from transverse betatron oscillations, excluding both as the origin of the shift.
4. Found that the peak accelerating field is nearly the same in successive periods, with a slight beam-loading reduction smaller than the phase-shift effect, and that most energy gain occurs during lengthening, so the mean energy gained in each bubble represents the accelerating potential produced by the lengthening.
5. Showed that the relative RMS energy spread of the trailing bunches peaks as the laser begins to defocus and then falls to a minimum near 5 mm after the shock with little energy gain, whereas before compression the energy distributions of successive bunches overlap and would give a continuous spectrum.

Comments

- Fig. 4 caption: panel (b) is said to show “the raw data (solid lines), data referenced to the laser centroid motion (squares), and data referenced to the beam COM motion arising from transverse betatron oscillations (circles)”, but the panel contains no squares, and its legend marks laser-subtracted data with filled circles and beam-subtracted

data with diamonds, as does the text (“The circles denote data in the laser-referenced frame ... while diamonds denote data with the beam centroid motion ... removed”); the legend and text give the intended reading.

- Fig. 2(c): the five-peak series (“5 peaks, $\Delta E = 63$ MeV”) is described as peak energies “extracted from (b)”, but its markers lie at about 150, 214, 272, 335, and 404 MeV, whereas the only five-peak spectrum in Fig. 2(b), which matches the fourth Lanex image in Fig. 2(a), peaks at about 157, 197, 249, 305, and 366 MeV, whose linear fit gives a spacing near 53 MeV; the markers of the other four series agree with their spectra within a few MeV, so the source of the five-peak data is unresolved.
- Fig. 5(e): the vertical-axis ticks labeled 0, 0.1, and 0.3 are equally spaced, which a linear axis does not allow; as in Fig. 5(d), where “1.3” marks the midpoint of 0 to 2.5, the middle tick likely marks 0.15 rounded to one decimal, so the largest plotted spread, of B4, is near 0.29 rather than the 0.19 implied by the 0.1 label.

2026CBKS

ChatGPT (July 2026)

Summary

JetSCI combines JAX’s automatic differentiation and GPU-vectorized local finite-element kernels with PETSc’s sparse solvers, preconditioners, and MPI distribution. For heterogeneous micromechanics problems, JAX differentiates material-batched residual kernels to form dense element Jacobians, which are assembled into a sparse global operator. A GPU-resident interoperability path uses DLPack, CuPy, ctypes, and `petsc4py` to expose device pointers to PETSc without staging through host memory; solutions return to JAX for Newton iteration. Component-level tests compare sparse matrix–vector products, JAX and PETSc Krylov solvers, matrix-free and explicit JAX paths, direct solvers, preconditioners, and one- versus two-GPU execution. PETSc-backed paths are faster and remain convergent at sizes where several JAX or CuPy baselines fail, while two-GPU execution becomes beneficial only after communication overhead is amortized. The study is limited to a two-material benchmark and modest GPU counts; distributed matrix and vector construction is less developed than the solve phase.

Contributions

1. Designed a hybrid differentiable-simulation architecture that assigns local differentiated kernels to JAX and global sparse solves to PETSc.
2. Built element-batched automatic Jacobian generation while preserving sparse finite-element assembly.
3. Implemented a low-copy GPU bridge through DLPack, CuPy, ctypes, and PETSc’s native construction routines.
4. Exposed PETSc solver, preconditioner, and MPI options inside a JAX-driven Newton workflow.
5. Evaluated solver robustness, preconditioning, matrix representations, and multi-GPU crossover behavior on heterogeneous micromechanics systems.

2026CC

ChatGPT (July 2026)

Summary

This study explains deflagration-to-detonation transition in a short closed channel containing stoichiometric hydrogen-air at knocking-relevant conditions of 10 atm and 1100 K. An analytical low-Mach-number solution for a planar flame approaching inert end-gas develops a formal finite-time temperature singularity, but reactions in the end-gas determine the physical transition. During autoignition induction, even weak heat release raises the compressed end-gas temperature. Because the induction delay is far more temperature-sensitive than the laminar flame speed, the reaction time scale collapses, invalidating the low-Mach approximation. The resulting flame acceleration emits compression waves that steepen into shocks and reinforce the flame through thermal feedback, destroying its laminar structure and initiating detonation before bulk autoignition. One-dimensional detailed-chemistry simulations support this mechanism: selectively suppressing small induction-period heat release delays or eliminates detonation without changing the flame

kinetics. The mechanism differs from long-tube flame acceleration because it originates in thermally sensitive end-gas induction near autoignition.

Contributions

1. Derived an analytical low-Mach solution for planar flame propagation and pressure growth in a closed channel.
2. Identified induction-period end-gas heat release as the trigger for breakdown of the low-Mach approximation.
3. Explained detonation onset through a thermal feedback loop between flame acceleration and compression waves.
4. Used selectively modified detailed chemistry to isolate the sensitivity of transition timing to weak induction heat release.
5. Distinguished this knocking-relevant mechanism from deflagration-to-detonation transition in long tubes under normal conditions.

2026CDGV

ChatGPT (July 2026)

Summary

The paper proves polynomial-degree-robust convergence properties for an h -adaptive conforming finite element method for the Poisson equation in two or three dimensions. Its vertex-patch algorithm uses equilibrated-flux error indicators, Dörfler marking, local residual liftings, and a bounded sequence of newest-vertex bisections. An a posteriori computable criterion controls error contraction; when satisfied, the contraction factor is independent of the fixed polynomial degree p . The authors establish discrete reliability with a p -independent constant using a recent robust projector. For piecewise-polynomial right-hand sides, this combines with robust estimator efficiency to give a p -independent upper threshold for the Dörfler parameter and optimal algebraic convergence rates, although rate-dependent constants and conditional contraction robustness remain. Tests on L-shaped and cross-shaped domains for $p = 1, \dots, 4$ attain the expected $\text{DoF}^{-p/2}$ behavior. The computable lower-bound constants stay below 1.6 and generally require only one local refinement, supporting the criterion's practicality.

Contributions

1. Designed a fixed- p , vertex-based adaptive FEM driven directly by equilibrated-flux estimators.
2. Derived an a posteriori verifiable condition for p -robust contraction at every adaptive step.
3. Proved p -robust discrete reliability using a polynomial-degree-robust projector.
4. Established optimal algebraic rates under a p -independent Dörfler-marking threshold.
5. Confirmed optimal rates, robust estimator effectivity, and small contraction constants numerically.

2026CDKM

ChatGPT (July 2026)

Summary

Catalytic computation gives an algorithm a large read–write memory initially filled with arbitrary data, provided that the memory is restored exactly afterward. The paper asks whether such full memory can reproduce leading polynomial-time, sublinear-space algorithms while retaining only logarithmic clean workspace. It answers affirmatively for directed s – t connectivity, edit distance, longest common subsequence, and discrete Fréchet distance. For connectivity, it combines a long/short-path decomposition with reversible flow propagation. Representative vertices are generated from pairwise-independent hashes sampled through an explicit expander walk, reducing the random seed to logarithmic length. For grid problems, the authors propagate weighted path sums, encode additive path costs multiplicatively, and reconstruct large integers from Chinese-remainder residues. These techniques produce polynomial-time algorithms with $O(\log n)$ ordinary workspace and sublinear catalytic space matching the established $n/2^{\Theta(\sqrt{\log n})}$ frontier. The work also supplies the first such sublinear-space result for discrete Fréchet distance under its stated metric assumptions.

Contributions

1. Gave deterministic polynomial-time directed-connectivity algorithms with logarithmic clean workspace and frontier-level sublinear catalytic space.
2. Gave a one-sided-error randomized connectivity variant using only logarithmically many random bits.
3. Built representative graph subsets from pairwise-independent hashes and explicit expander walks without storing the subset in clean memory.
4. Developed catalytic weighted-reachability algorithms for directed grid graphs using reversible flow propagation and Chinese-remainder representations.
5. Applied the grid framework to exact edit distance, longest common subsequence, and discrete Fréchet distance.

2026CEMM

ChatGPT (July 2026)

Summary

Randomized block Krylov iteration computes a rank- k matrix approximation by repeatedly multiplying a target matrix by a random starting block of b vectors. Previous near-optimal theory covered only $b = 1$ and $b = k$; bounds for intermediate blocks contained a $b(k - b)$ factor and could require $O(k^2)$ matrix–vector products, despite intermediate block sizes often performing best in practice. This paper closes that gap. For every $1 \leq b \leq k$, it proves that randomized block Krylov iteration obtains a $(1 + \varepsilon)$ -factor approximation to the best rank- k approximation using $\tilde{O}(k/\sqrt{\varepsilon})$ matrix–vector products, with both Frobenius- and spectral-norm guarantees and associated singular-vector capture. The proof reduces the approximation analysis to conditioning of a square random block Krylov matrix. New lower bounds on its minimum singular value combine Gaussian anti-concentration, Vandermonde structure, and a non-sparsification argument; the conditioning results may also inform sparse linear-system algorithms. The theorem supplies theoretical support for practical intermediate block sizes without claiming that all choices have equal wall-clock cost. Remaining questions include sharper singular-gap dependence, removing condition-number terms, finite-precision behavior, and sparse random starting blocks.

Contributions

1. Proved near-optimal randomized block Krylov guarantees for every block size $1 \leq b \leq k$.
2. Removed the prior worst-case $b(k - b)$ dependence from matrix–vector-product complexity.
3. Established $(1 + \varepsilon)$ low-rank approximation guarantees in Frobenius and spectral norms.
4. Reduced the analysis to quantitative conditioning of square random block Krylov matrices.
5. Developed minimum-singular-value bounds using anti-concentration, Vandermonde structure, and non-sparsification.

2026CES

ChatGPT (July 2026)

Summary

This paper describes Bandicoot, a C++ GPU linear-algebra library whose Armadillo-compatible interface is intended to ease migration from CPU code. Expression templates encode a user’s operation as a compile-time abstract syntax tree. Template specializations traverse that tree to fill reusable skeleton kernels with generated argument lists, indexing expressions, bounds checks, and element operations, thereby fusing compound expressions into one GPU kernel without a runtime expression optimizer or JIT source generator. The implementation separates the high-level API and optimizer from CUDA and OpenCL backends; further backends remain incomplete. Benchmarks on an NVIDIA RTX 4090 compare 32-bit, mostly $10,000 \times 10,000$ expressions with several array and machine-learning toolkits after warm-up. Bandicoot is fastest or tied in most reported cases and sustains approximately the measured memory-bandwidth ceiling while adding 2–16 matrices, whereas competitors degrade for longer expressions. These experiments exclude warm-up

and cover one GPU, precision, and workload suite, so they establish competitive kernel execution rather than universal application performance.

Contributions

1. Presented an Armadillo-compatible C++ interface for GPU linear algebra with CUDA and OpenCL backends.
2. Represented compound expressions as compile-time types and generated fused GPU kernel source from their syntax trees.
3. Reused backend-independent skeleton kernels to reduce specialized-kernel implementation and maintenance.
4. Eliminated runtime expression analysis and JIT source-generation infrastructure while retaining device-time kernel compilation.
5. Benchmarked compound expressions and demonstrated sustained measured memory bandwidth for sums of up to sixteen matrices.

2026CFTB

ChatGPT (July 2026)

Summary

Gauss–Newton methods can converge rapidly on PDE-constrained inverse problems, but their Jacobian–vector products normally require additional forward or adjoint PDE solves. This paper proposes gradient-only Gauss–Newton for objectives formed from sums of per-experiment losses. Each loss ϕ_i is written as $\rho_i^2/2$, where $\rho_i = \sqrt{2\phi_i}$. The Jacobian row $\nabla \rho_i = \nabla \phi_i / \sqrt{2\phi_i}$ is therefore assembled from function values and gradients already computed for the full objective. Combined with a regularizer Hessian, this gives a low-rank Gauss–Newton approximation without extra PDE solves. Under compact-sublevel, Lipschitz-gradient, uniformly positive regularizer-Hessian, and Wolfe-line-search assumptions, the gradient norm converges to zero. Acoustic full-waveform-inversion tests compare the method with nonlinear conjugate gradient, L-BFGS, and Gauss–Newton CG under equal PDE-solve budgets. It is competitive for uniform acquisition and strongest for realistic uneven receiver coverage. Severe rank deficiency in the approximate Jacobian makes the method approach gradient descent.

Contributions

1. Reformulated summed PDE-constrained losses so a Gauss–Newton Jacobian can be assembled from per-loss values and gradients.
2. Eliminated sensitivity and adjoint PDE solves beyond those already required for gradient evaluation.
3. Exploited the low-rank structure and regularization through an experiment-space linear system.
4. Proved global convergence to first-order stationarity under the stated regularity and line-search assumptions.
5. Demonstrated competitive full-waveform-inversion performance under fixed PDE-solve budgets and identified rank deficiency as a limiting case.

2026CGAT

ChatGPT (July 2026)

Summary

The authors report experimental evidence for directional, high-energy muons produced by a petawatt laser-wakefield accelerator. A high-charge GeV electron beam struck a 2 cm lead converter, where Bethe-Heitler pair production generated muons amid much larger electron, positron, and photon backgrounds. Two dipole magnets selected and transported charged particles through a shielded path to a Timepix3 detector, whose pixel tracks recorded both deposited energy and arrival time. Track-level likelihood analysis, compared with Monte Carlo modeling, gave a $(99.1 \pm 0.5)\%$ confidence that at least one detected event was a muon rather than noise. Measurements with thinner converters served as controls and produced likelihood distributions consistent with leptonic background. Simulations reproduced the

observed signal and support scaling to higher-charge, higher-energy drivers. For a 10 Hz petawatt system, the study projects that suitable transport and shielding could deliver roughly 10^4 muons per second onto square-centimeter areas. This establishes a compact route toward directed laser-driven muon beams for radiography and other applications.

Contributions

1. Produced experimental evidence of directional high-energy muons from a laser-wakefield accelerator.
2. Combined magnetic transport, heavy shielding, and pixelated track detection to isolate muons from intense secondary-particle noise.
3. Quantified muon detection at $(99.1 \pm 0.5)\%$ confidence using track likelihoods and numerical modeling.
4. Used thinner-converter control shots to show that the corresponding tracks were consistent with background.
5. Projected a path to approximately 10^4 guided muons per second with a 10 Hz petawatt laser.

2026CGG

Claude (September 2026)

Summary

The authors propose self-consistent beam-dynamics simulations including geometric wakefields without interpolating particle currents onto a mesh. In a scattered-field formulation, the total field is the sum of an incident field, taken as the beam's free-space space-charge field (SCF), and a scattered (wake) field that solves the source-free Maxwell equations with $n \times E_s = -n \times E_i$ on perfectly conducting walls. Discretized with the Finite Integration Technique (FIT) on a window moving at the speed of light, this boundary condition becomes an equivalent magnetic current, with a boundary-conformal variant for curved walls. The SCF comes from a rest-frame Green's-function solver at the particles and, at the walls, from a fast multipole method (FMM) or retarded Liénard–Wiechert (LW) fields; the wakefield uses a dispersion-free split-operator scheme. For a rigid 15 MeV bunch in uniform pipes, the steady-state field matches the analytic solution and converges at second order in rectangular pipes, and in a circular pipe the conformal current is far more accurate than a staircase one. In the seven-cell SuperKEKB RF photo-gun, wakefields raise the exit energy spread by about 14%, consistently with full electromagnetic particle-in-cell (CST) simulations, while needing about 8 GB and under 1.5 h instead of 160 GB and about 10 h.

Contributions

1. Split the beam-driven Maxwell problem into space-charge and wakefield subproblems coupled only through a magnetic boundary current, so that each uses its own mesh, time step (with integer sub-cycling of the SCF solver), and specialized code.
2. Introduced a boundary-conformal magnetic current $(\mathbf{CR}_l - \mathbf{R}_A \mathbf{C})\hat{\mathbf{e}}_i$ built from cut-edge and cut-face ratios, which reduces to the staircase current on mesh-aligned walls.
3. Computed incident boundary voltages with an asymmetric source–target FMM that treats the cathode with image charges, and alternatively with LW fields whose retarded times are found by bisection over stored trajectories.
4. Decomposed the steady-state space-charge wake in a pipe into free-space SCF and scattered field, showing that the scattered field compensates the SCF, and simulated a dual-bunch corrugated waveguide with beam loading, giving witness gradients of 5.2 and 4.71 MV/m against measured 4.93 and 5.02 MV/m.
5. Showed that retardation delays the gun wakefield, so the quasistatic FMM model locally overestimates the energy spread near irises and systematically overestimates the energy loss, while the final energy spread nearly agrees and mean-energy differences remain at the few-per-mille level.

Comments

- Section 4.1, Eq. (16): the back-transformed fields are written $B_x = -\sqrt{\gamma_0^2 - 1} E'_y$ and $B_y = \sqrt{\gamma_0^2 - 1} E'_x$, which are dimensionally inconsistent in the SI units used throughout ($D = \varepsilon_0 E$, and $B = \frac{1}{c} n_p \times E$ in Eq. (17)); the evident intended reading divides both by c .
- Section 6.3, Figs. 4 and 5: Fig. 5 is introduced as the decomposition of the steady-state solution for the $w = 100$ mm, $h = 15$ mm pipe shown in Fig. 4, but its total field peaks near 143 V/m, whereas Fig. 4 peaks near 162 V/m;

Eqs. (19)–(20) with the Section 6.2 bunch give about 163 V/m for $h = 15$ mm and 144 V/m for $h = 10$ mm, so Fig. 5 appears to use the $h = 10$ mm pipe of Fig. 6, which neither the text nor the caption states.

- Section 6.4, Fig. 7: the text states that “the second order convergence is fully recovered when the conformal boundary approach introduced in Section 3.3 is used” (and the Conclusion calls it “optimally convergent”), but the conformal errors in Fig. 7 fall from about 4×10^{-3} at $\sigma_z/\Delta = 4$ to about 4×10^{-4} at 24, an average rate near 1.3, and remain above the plotted second-order guide line; unresolved.

2026CGSN

ChatGPT (July 2026)

Summary

Fusion pilot plants will have little diagnostic port space and harsh operating conditions, motivating confinement-state inference from reactor-compatible microwave diagnostics. Using 260 DIII-D shots with profile-reflectometer data, the authors construct 8,102 time-slice samples and represent each density profile by a cubic spline sampled at ten radial locations concentrated near the plasma edge. A gradient-boosted classifier distinguishes L- from H-mode with 97% mean test accuracy under shot-level train/test splitting. SHAP analysis confirms that edge and pedestal information dominates the prediction. The paper then combines this density-based classifier with an earlier electron-cyclotron-emission temperature classifier. Distances to k-means centers in each diagnostic’s training feature space provide confidence weights for a probability ensemble, which reaches 99.2% accuracy across randomized shot splits. A chronological sliding-window test lowers accuracies to 93.1% for reflectometry and 96.0% for the ensemble, showing useful robustness but also distribution drift and a likely need for periodic recalibration.

Contributions

1. Constructed a fusion-power-plant-relevant H-mode classifier from profile-reflectometer density profiles, including a spline representation that handles incomplete core penetration.
2. Used shot-level rather than snapshot-level splitting to reduce temporal leakage from correlated time slices.
3. Achieved 97% mean test accuracy, with 98% precision, recall, and F1 score, across 100 shot reshuffles.
4. Introduced a confidence-weighted reflectometry and electron-cyclotron-emission ensemble based on distance to training-data clusters, reaching 99.2% accuracy.
5. Quantified chronological distribution drift and identified periodic recalibration as a likely deployment requirement.

2026CH

ChatGPT (July 2026)

Summary

The paper benchmarks four low-dissipation alternatives to the Harten–Lax–van Leer (HLL) flux for compressible Euler equations: HLLC, Toro–Vazquez flux splitting, low-dissipation central-upwind (LDCU), and local-characteristic-decomposition central-upwind (LCDCU). Each flux is embedded in first-, second-, third-, and fifth-order finite-volume and finite-difference constructions, with high order supplied by reconstruction and alternative WENO machinery. Fourteen one- and two-dimensional experiments probe smooth convergence, contact and shear resolution, shock interactions, multidimensional instabilities, robustness, and CPU-normalized efficiency. All four alternatives reduce contact-wave smearing relative to HLL and usually have similar dissipation; case-dependent differences are smaller than the gain from moving from first to second order. Third- and fifth-order variants give further benefits for complex multidimensional structure and can be more efficient at a target resolution. The study therefore does not identify a universal winner: characteristic decomposition, antidiffusion, shock behavior, and implementation cost determine the preferred solver for a given flow.

Contributions

1. Placed HLLC, Toro–Vazquez splitting, LDCU, and LCDCU in one consistent HLL-referenced comparison.

2. Built first-, second-, third-, and fifth-order versions in both finite-volume and finite-difference frameworks.
3. Verified designed orders on smooth problems and tested robustness and resolution on a broad one- and two-dimensional Euler benchmark suite.
4. Showed that all four low-dissipation schemes resolve contacts and jets better than HLL while their mutual differences remain problem dependent.
5. Compared CPU-normalized solutions to separate formal order and flux cost from raw grid-resolution effects.

2026CH2

Claude (September 2026)

Summary

Finite element Hessian, elasticity, and divdiv complexes in three dimensions have so far been built case by case. Arnold and Hu derive the continuous complexes with the Bernstein–Gelfand–Gelfand (BGG) construction from stacked de Rham complexes whose Sobolev smoothness matches, but finite element spaces conforming only in $H(\text{curl})$ or $H(\text{div})$ lack the tangential or normal continuity that the connecting maps require. The authors introduce two reduction operations on short exact sequences, a tilde operation that restricts the target space and takes its preimage and a hat operation that restricts the domain through an auxiliary operator, plus an inverse-hat augmentation, each preserving exactness under stated conditions. Applied to their smooth finite element de Rham complexes on tetrahedral meshes, whose vertex, edge, and face smoothness is set by vectors $r = (r^v, r^e, r^f)$, the BGG construction yields exact finite element Hessian, elasticity, and divdiv complexes for families of smoothness vectors and degrees, with bubble counterparts. Degrees of freedom for the new tensor spaces come from tangential–normal decompositions, face trace complexes, and two-dimensional analogues. Special cases recover several earlier complexes, but those on Alfeld splits, which correspond to $r^0 = (2, 1, 1)$, lie outside the framework, and no discretization of a specific PDE is analysed.

Contributions

1. Formulated tilde, hat, and inverse-hat operations on short exact sequences, with dimension-count criteria for exactness, that make the BGG construction applicable to finite element spaces with mismatched continuity.
2. Derived exact finite element Hessian complexes, with a variant allowing inequality constraints between smoothness vectors, that recover an earlier finite element Hessian complex for $r^0 = (4, 2, 1)$ and $k \geq 7$.
3. Derived exact finite element elasticity complexes for $r^0 \geq (2, 1, 0)$, including a smoother variant beginning in H^2 , by combining BGG with inverse hat operations.
4. Derived exact finite element divdiv complexes and relaxed variants for $r^0 \geq (1, 0, 0)$, recovering two earlier finite element divdiv complexes.
5. Constructed unisolvent degrees of freedom for $H(\text{curl}; \mathbb{S})$, $H(\text{inc}; \mathbb{S})$, and $H(\text{sym curl}; \mathbb{T})$ elements from tangential–normal decompositions, trace complexes, and polynomial bubble complexes.

Comments

- §4.2, proof of Lemma 4.6, Eq. (35): the identity is printed as $\dim \mathbb{V}_{k-1}^{\text{div}}(r; \mathbb{M}) - \dim \mathbb{V}_{k-1}^{\text{div}}(r; \mathbb{T}) = \dim \mathbb{V}_k^{L^2}(r)$, but the difference between $\mathbb{M}$ - and $\mathbb{T}$ -valued spaces of degree $k-1$ is the trace part of degree $k-1$, and the proof of Lemma 6.2 uses (35) with right-hand side $\dim \mathbb{V}_{k-1}^{L^2}(r_2)$; the intended subscript is $k-1$.
- §5.3, Lemma 5.20: the lemma assumes only that $(r_2, r_2 \ominus 1, k)$ is div stable, but its proof states “We do need the div stability $\text{div } \mathbb{V}_{k+2}^{\text{div}}(r_0) = \mathbb{V}_{k+1}^{L^2}(r_1)$ ”, which appears neither in the lemma nor in (55) or Theorem 5.21; whether it follows from the stated hypotheses is unresolved.
- §6.2, DoF (67i), and the proof of Lemma 6.5: the face moments of $\Pi_f(\text{inc } \tau)n$ are taken against $\mathbb{B}_{k-1}^{\text{div}}(f; r_2)/\text{RT}(f)$, and the proof claims that these moments vanish for $q \in \text{RT}(f)$ when $\tau|_e = 0$ and $(\text{curl } \tau)^\top|_e = 0$, but the analogous $H(\text{div}; \mathbb{S})$ DoF (32d) uses $\text{RM}(f)$, and for $\tau = (xy(1-x-y-z))^3 e_1 \otimes e_1$, which satisfies both edge conditions on the face $z = 0$, the moment against the dilation field in $\text{RT}(f)$ equals $1/16800$ while those against constants and rotations vanish; the intended quotient is evidently $\text{RM}(f)$.
- §6.3, after Eq. (82): the count is justified by “ $\text{div div } \mathbb{B}_k^{\text{div div}^+}(r_2; \mathbb{T}) = \mathbb{B}_{k-2}(r_3)/\text{RT}$ ”, whereas (82) itself uses $\mathbb{B}_k^{\text{div div}^+}(r_2; \mathbb{S})$ and Lemma 4.18 states the result for $\mathbb{S}$ -valued bubbles with quotient $\mathbb{P}_1(T)$; the count is unaffected

because both quotients have dimension 4.

2026CHDW

ChatGPT (July 2026)

Summary

This simulation study proposes a hybrid laser-wakefield-acceleration to plasma-wakefield-acceleration stage in which a plasma photocathode converts fluctuating driver beams into lower-charge electron bunches with improved quality and reproducibility. Quasi-three-dimensional particle-in-cell simulations show that witness charge is governed by the photocathode laser and helium density, becoming effectively independent of driver charge once trapping is complete. Relativistic driver velocity makes early witness energy gain weakly sensitive to large changes in driver mean energy and energy spread; suitable extraction before depletion yields approximately 1% energy stability and sub-1% projected witness energy spread over broad driver variations. Release near the symmetric field-minimum region of the blowout provides tolerance to timing and pointing jitter. When driver charge changes, the trapping position shifts within the wake so that accelerating-gradient variations are partially self-compensated. The stage simultaneously transforms brightness, energy, and stability, producing ultralow-emittance, multi-kA witness beams, but sacrifices charge: the demonstrated driver-to-witness charge ratio is about 1%.

Contributions

1. Demonstrated that plasma-photocathode witness charge decouples from broad driver-charge fluctuations above full trapping.
2. Showed that dephasing-free propagation suppresses witness-energy sensitivity to driver energy and spread before depletion.
3. Quantified a regime with roughly 1% energy stability, sub-1% spread, nanometre-scale emittance, and multi-kA current.
4. Established timing and pointing tolerance from injection near the symmetric minimum of the blowout potential.
5. Identified a trapping-position shift that mitigates accelerating-gradient changes caused by driver-density jitter.

2026CHLW

ChatGPT (July 2026)

Summary

COS(DG) is a compact, nonintrusive oscillation-control wrapper for high-order DG discretizations of hyperbolic conservation laws. Once per time step, it convexly blends each DG polynomial with its cell average using entropy-induced distances that are locally scale- and evolution-invariant. The method needs only immediate neighbors, no derivatives or characteristic decomposition, and works identically for modal and nodal formulations on general meshes. Analysis proves optimal convergence for smooth solutions, L^2 nonexpansiveness, and inheritance of entropy stability from an entropy-stable base scheme; a finite convex covering argument supports problem-independent oscillation suppression. A symmetry-preserving entropy sensor localizes the operation, and convex scaling integrates naturally with the Zhang-Shu bound-preserving limiter. Extensive one- and two-dimensional advection, Burgers, and Euler benchmarks, including comparisons with nine TVB-, WENO-, MP-, OFDG-, and OEDG-type approaches, show robust shock capture, fine-structure resolution, scale-independent parameter behavior, and low overhead.

Contributions

1. Introduced a characteristic-free, entropy-guided convex oscillation-suppression framework for arbitrary-order DG methods.
2. Proved optimal accuracy, L^2 nonexpansiveness, and inherited entropy stability.
3. Established local scale and evolution invariance with problem-independent suppression guarantees.
4. Added a symmetry-preserving entropy criterion and unified Zhang-Shu bound preservation.

5. Demonstrated high resolution and robustness against nine established controls on scalar and Euler shock problems.

2026CHLZ

ChatGPT (July 2026)

Summary

The paper proposes augmented block conjugate gradients for a single symmetric positive-definite system: append a Gaussian random block to the right-hand side and solve in their joint block Krylov space. Although no preconditioner is formed, a deterministic subspace-containment theorem shows that the method implicitly contains the iterates of preconditioned conjugate gradients for a broad family of Nyström deflation preconditioners. Consequently, after equal numbers of matrix loads, augmented block conjugate gradients is never worse than the corresponding Nyström-preconditioned method in exact arithmetic. Probabilistic bounds relate convergence to the condition number remaining after dominant eigenvalues are removed, and new guarantees for deeper Krylov Nyström preconditioners follow along the way. Shift invariance lets one block-Lanczos factorization solve many $(A + \mu I)x = b$ systems, making the approach attractive for an entire ridge-regression regularization path. Integrating these shifted solves also gives guarantees for generating multiple Gaussian samples via matrix square roots. Experiments show substantial improvements when loading the matrix dominates cost, while the authors explicitly note extra arithmetic, storage, and finite-precision issues for block Lanczos.

Contributions

1. Introduced randomized augmented block conjugate gradients using Gaussian auxiliary right-hand sides as implicit spectral information.
2. Proved Krylov-subspace containment and deterministic same-matrix-load dominance over a broad class of Nyström-preconditioned methods.
3. Derived probabilistic convergence and matrix-vector complexity bounds through deeper-Krylov Nyström-preconditioning estimates.
4. Exploited shift invariance to evaluate many regularization parameters from one block-Lanczos run for ridge-regression paths.
5. Extended the analysis to matrix-square-root iterations for multiple Gaussian samples and verified the gains experimentally.

2026CJLY

ChatGPT (July 2026)

Summary

This paper studies stabilization of the magnetized Vlasov–Poisson system by an imposed electric field. Laplace–Fourier analysis yields a dispersion relation and a Penrose-type criterion for Bernstein modes perpendicular to a uniform magnetic field. Gaussian equilibria exhibit persistent but bounded oscillations, whereas a ring-shaped equilibrium supports the Dory–Guest–Harris instability. The authors design controls by modifying or eliminating unstable dispersion roots. One strategy chooses the external potential to cancel the self-consistent field, recovering magnetized free streaming exactly; for separable Gaussian data, non-Bernstein electric-energy modes then decay while Bernstein modes remain periodic at the gyrofrequency. A second root-placement strategy suppresses selected unstable modes without full cancellation. Particle simulations agree with the linear predictions and show stabilization persisting in tested nonlinear cases. The approach provides an analytically guided control principle, but the geometry and field are idealized; extension to drift-kinetic and gyrokinetic regimes is left open.

Contributions

1. Derived the magnetized dispersion relation and a Bernstein-mode Penrose stability condition.
2. Characterized stable Gaussian and unstable Dory–Guest–Harris equilibria within one framework.

3. Formulated electric-field controls through unstable-root elimination and placement.
4. Proved bounded controlled energy behavior for the exact free-streaming strategy.
5. Demonstrated suppression of unstable modes in linear and tested nonlinear simulations.

2026CKLS

ChatGPT (July 2026)

Summary

This work extends the parallel basis-update-and-Galerkin integrator for dynamical low-rank matrix equations to Tucker tensors and general tree tensor networks. During each time step, differential equations for all leaf bases and connecting tensors are evolved concurrently; only hierarchical orthogonalization remains sequential. The method adapts tensor ranks, avoids backward time integration, and omits the augmented-rank connecting-tensor update required by an earlier rank-adaptive scheme. The authors prove a first-order global error bound whose constants are independent of small singular values, establishing robustness in regimes where conventional integrators become restrictive. Tests on long-range quantum spin dynamics and uncertain radiative-transfer benchmarks reproduce the predicted first-order behavior and agree with reference solutions. Even in serial implementations, removing augmented-rank coefficient updates reduces runtimes, including from 30 to 12 seconds for the plane-source case and from 105 to 72 seconds for the line-source case. The additional speedup available from genuinely parallel execution remains hardware-dependent and was not measured.

Contributions

1. Extended a robust parallel low-rank matrix integrator to Tucker tensors and arbitrary tree tensor networks.
2. Enabled concurrent evolution of every basis and connecting tensor within a time step.
3. Established a first-order error bound independent of small matricization singular values.
4. Combined rank adaptivity with forward-only evolution and avoided augmented-rank connecting-tensor updates.
5. Demonstrated accuracy and serial runtime reductions on quantum and uncertain radiative-transfer problems.

2026CLY

ChatGPT (July 2026)

Summary

This paper constructs a semi-Lagrangian scalar auxiliary variable (SAV) method for the Vlasov-Ampere and Vlasov-Maxwell equations. Standard Eulerian or semi-Lagrangian splittings often preserve either Gauss's law or total energy, but not both, because the Ampere and transport pieces are separated. The authors introduce an auxiliary scalar variable that reformulates the energy while retaining a splitting compatible with semi-Lagrangian transport. The resulting method preserves Gauss's law without solving the constraint equation and preserves a modified total energy, while keeping an explicit update structure. The paper develops the method first for Vlasov-Ampere, then for 1d-2v and full Vlasov-Maxwell systems, and tests it on standard plasma benchmarks. Numerical comparisons show stable behavior and improved invariant preservation relative to methods that enforce only one structure. Remaining work includes non-Fourier spatial discretizations for complex geometries, Maxwell-step CFL improvements, and possible PIC adaptations.

Contributions

1. Introduced an SAV reformulation for Hamiltonian Vlasov-Ampere and Vlasov-Maxwell dynamics.
2. Designed a splitting that preserves Gauss's law while supporting semi-Lagrangian transport.
3. Preserved a reformulated total energy without requiring a costly implicit nonlinear solve.
4. Extended the construction from Vlasov-Ampere to 1d-2v and full Vlasov-Maxwell models.
5. Demonstrated the method on numerical tests against invariant-preserving alternatives.

2026CMPT

ChatGPT (July 2026)

Summary

Caplan et al. develop a Lagrangian particle method for the spherical shallow-water equations in which particles are represented by weighted spherical Voronoi, or power, cells. A halfspace-clipping construction and spherical-quadtree neighbor search make the tessellation practical at very large scale. Cell geometry supplies discrete gradient and divergence operators, a semi-implicit update relaxes the gravity-wave time-step restriction, and semi-discrete optimal transport adjusts cell weights to enforce each particle's target mass. Williamson and Galewsky benchmarks show approximately first-order convergence, momentum and energy behavior comparable to a recent Lagrangian solver, and broadly consistent height fields relative to an Eulerian TRiSK solver. The method avoids polar singularities by working in three-dimensional Cartesian coordinates, but its current linear solves and repeated optimal-transport iterations make it 12–20 times slower than the comparison solver, and some test solutions remain noticeably smoothed.

Contributions

1. Introduced a power-cell Lagrangian discretization of the spherical shallow-water equations.
2. Developed spherical halfspace-clipping and quadtree algorithms that compute a 100-million-site Voronoi diagram in under two minutes.
3. Combined power-cell differential operators with semi-implicit time integration without artificial viscosity or particle merging.
4. Enforced local particle mass through a Newton solution of a semi-discrete optimal-transport problem.
5. Validated four standard spherical-flow benchmarks and quantified accuracy, conservation, and runtime against an Eulerian solver.

2026CMSM

Claude (September 2026)

Summary

The authors couple predict-first simulations of the 1.5D transport code RAPTOR with the free-boundary inverse equilibrium solver FBT into a Kinetic Equilibrium Prediction (KEP) workflow for preparing full discharges of the TCV tokamak. FBT computes the feedforward poloidal-field (PF) coil currents producing a target shape, but is usually given simple polynomial pressure-gradient $p'(\psi)$ and current-function $TT'(\psi)$ profiles tuned by operators. Here RAPTOR predicts these profiles from the pulse schedule, given an assumed confinement factor $H^{98(y,2)}$ and line-averaged density, using an empirical gradient-based transport model with feedback-controlled pedestal gradients, L–H transitions from the Martin power-threshold scaling, and a separate negative-triangularity (NT) parametrization. FBT takes the RAPTOR profiles, rescaling p' to keep the plasma current, and returns its equilibria to RAPTOR; this loose iteration converges in a few steps. With measured density, electron stored energy is predicted within 20% in essentially all of 211 shots, and ion energy in most of 148. Experimentally, KEP-prepared coil currents improved X-point alignment in a lower-single-null H-mode and held a snowflake-divertor NT L-mode shape near target longer. The authors state that predicting density from gas programming remains delicate, that X-point height and radiation limit H-mode prediction, and that FBT neglects plasma dynamics and vessel currents.

Contributions

1. Rescaled the Martin threshold by a divertor factor $f_{\text{div}}(s) = 1 + 1/(1 + e^s)$ of the signed normalized separation s of the two X-point flux surfaces, giving 1 for favorable and 2 for unfavorable ion ∇B drift and 1.5 for a connected double null, with prohibitive values for limited and NT plasmas and an alternative f_{div}^* that lowers the threshold near $s = 0$.
2. Extended the gradient-based model with an NT confinement state, triggered when $\min(\delta_{up}, \delta_{bot}) < -0.2$ with a 100 ms linear ramp and using fixed parameters selected to match 41 NT shots, and with an ion heat equation

whose pedestal T_i/T_e follows an ad hoc density law and which receives 60% of the absorbed NBI power, as an alternative to a T_i/T_e scaling that runs 10–15% faster.

3. Reduced the runtime through schedule-based non-uniform time steps ($dt_{min} \leq 0.5$ ms in transients, up to 1 ms elsewhere) and C code generation (a factor 2), reaching 2.3 ± 0.38 min per second of discharge, and made FBT robust by freezing the plasma domain after 20 unconverged iterations, so that all FBT equilibria converged given a converged RAPTOR input.
4. For 48 of 51 single-null shots with a classifier-detected H-mode, RAPTOR placed the L–H transition within a 100 ms window, the other 3 being ohmic shots without a clear pedestal; spurious H-mode predictions in 21 ohmic shots were attributed to high X-point height, and three programmed double-null shots transitioned late, showing no clear favorable double-null effect on TCV.
5. Found KEP corrections to feedforward PF coil currents mostly of tens of amperes and occasionally several hundred, as in the H-mode phase of #81882, showed that imposing only RAPTOR's β_{pol} and q_A on polynomial profiles leaves differences in plasma and coil flux, and found that the predicted l_{i3} drops at the L–H transition as in an interpretative LIUQE–ASTRA kinetic reconstruction, opposite to the diamagnetic-loop-constrained magnetic reconstruction.

Comments

- Section 2.1, “Ion temperature”: the separatrix condition “ $T_i|_{sep} = \gamma T_e|_{ped}$ ”, such that the ion temperature gradient follows $\mu_{T_i} = \gamma \mu_{T_e}$ ” conflicts with the definition $\gamma = T_i/T_e|_{ped}$, since with the linear pedestal profile of Eq. (2.4), $T_\sigma|_{ped} = T_\sigma|_{sep} + \mu_\sigma(1 - \hat{\rho}_{ped})$, it gives $T_i|_{ped} = \gamma(2T_e|_{ped} - T_e|_{sep}) \neq \gamma T_e|_{ped}$; the consistent reading is $T_i|_{sep} = \gamma T_e|_{sep}$.
- Section 3, discussion of #82274: “having a slightly LSN, favorable $s \simeq +0.5$, would have led to a factor 1.25 instead of 1.75” (with $s \simeq -0.5$, as Fig. 10(c) shows), but Eq. (2.8) with $\alpha_f = 1$, $\alpha_u = 2$, and $G(s) = 1/(1 + e^s)$ gives $f_{div}(0.5) \approx 1.38$ and $f_{div}(-0.5) \approx 1.62$, as does the solid curve of Fig. 9; 1.25 and 1.75 correspond to $|s| \approx 1.1$; unresolved.
- Section 4.1: the text says the schedule was tested “with the new FBT-RAPTOR result (#83940)”, but the Fig. 13 caption and the Fig. 14 caption and legend identify the FBT-RAPTOR-prepared shot as #83740; the evident intended reading is #83740.
- Fig. 16 caption: the initial FBT run is said to be “used for the preparation of shot #83926” and the FBT-RAPTOR result “used to prepare #83946”, whereas Section 4.2 and the Fig. 17 caption assign #83946 (with #83947 and #83950) to the initial FBT programming and #83926 (with #83978) to the FBT-RAPTOR correction; the caption’s shot numbers appear swapped.
- Fig. 17 caption: #83946, #83947, and #83950 are described as “prepared with an initial FBT equilibrium with high β_{pol} and l_{i3} ”, but Section 4.2 calls them “the shots prepared with low- β_{pol} and low- l_{i3} ”, the Fig. 16 caption gives the initial FBT run “a low β_{pol} and l_{i3} ”, and Figs. 16(a)–(b) show the initial-FBT β_{pol} (about 0.05) and l_{i3} (about 0.86) below the FBT-RAPTOR values in the NT phase; “high” evidently should read “low”.
- Fig. 16 caption: LIUQE-MER is said to be displayed in (a)–(b), but the legends of panels (a) and (b) label the black curve “LIUQE KER”; unresolved.

2026CSSM

ChatGPT (July 2026)

Summary

This study numerically examines ponderomotive density structuring by broadband kinetic Alfvén-wave turbulence under plasma-sheet-boundary-layer conditions. A two-dimensional pseudospectral solver advances coupled modified nonlinear Schrödinger and magnetosonic equations on a 256×256 grid with $\beta \sim 0.1$ – 0.3 . The undamped calculation conserves energy within 0.085% through normalized time 40, while a scalar Landau-damping run loses about 4% of magnetic energy. The reported nonlinearity parameter remains near 0.25, indicating a sub-Alfvénic regime. Magnetic filaments and co-located density cavities emerge within several wave periods; density perturbations reach $|\delta n/n_0| \sim 0.3$ – 0.5 , although the domain-wide intensity–density correlation is only -0.022 . Spectra transition rapidly from injection to dissipation without an extended inertial range, which the authors associate with moderate magnetic Reynolds number.

Interpretation is limited by the two-dimensional fluid-envelope model, parallel-only acoustic response, scalar damping, fixed ion temperature, and absence of self-consistent wave–particle resonances.

Contributions

1. Simulated broadband, rather than single-packet, kinetic Alfvén-wave forcing with a coupled MNLS–magnetosonic model.
2. Documented early formation and persistence of intermittent magnetic filaments and density cavities at ion-scale widths.
3. Quantified numerical conservation, damping losses, and sub-Alfvénic nonlinearity over the full run.
4. Reported a compressed injection-to-dissipation spectrum and interpreted it through moderate-Reynolds-number scale separation.
5. Compared the simulated structures with plasma-sheet observations while explicitly identifying missing three-dimensional and kinetic physics.

2026CSY

ChatGPT (July 2026)

Summary

This paper proves stability and convergence results for discontinuous Galerkin discretizations of spherically symmetric Einstein-scalar equations in Bondi coordinates. The setting is chosen as a mathematically tractable nonlinear-relativity model that still includes large initial data leading to black-hole formation. The authors formulate the DG scheme, prove L^2 stability for arbitrary L^2 initial data, and establish a priori estimates for a broad class of exact solutions. Those estimates include behavior up to $r = 0$ and provide event-horizon formation information. For polynomial degree $k \geq 1$, the paper proves an optimal $O(h^{k+1})$ error estimate, and it separately treats the P^0 case. Numerical experiments confirm that the theoretical convergence rates are sharp. The analysis gives rigorous support for high-order DG methods in a nonlinear general-relativistic model where much prior numerical-relativity stability work has been empirical or linearized.

Contributions

1. Formulated a DG method for spherically symmetric Einstein-scalar equations in Bondi coordinates.
2. Proved L^2 stability for arbitrary L^2 initial data.
3. Established a priori estimates for large-data solutions that form black holes.
4. Proved optimal convergence for high-order DG approximations and handled the P^0 case separately.
5. Confirmed the sharpness of the error estimates through numerical experiments.

2026CSZ

Claude (September 2026)

Summary

Discontinuous Galerkin (DG) methods with total-degree $\mathbb{P}^k$ elements on Cartesian meshes converge numerically at the optimal L^2 rate $\mathcal{O}(h^{k+1})$, but with $(k+1)(k+2)/2$ rather than $(k+1)^2$ unknowns per cell this had been proved only for constant coefficients or for small k . The authors analyse semidiscrete upwind $\mathbb{P}^k$ DG for $u_t + (\alpha u)_x + (\beta u)_y = 0$ and $u_t + f(u)_x + g(u)_y = 0$ with periodic boundaries, assuming uniform meshes and wind components α, β or f', g' that do not change sign. A special projection is built from the truncated $\mathbb{Q}^k$ Radau expansion of the exact solution by keeping terms of index sum at most $k+1$ and removing their degree- $(k+1)$ part. Correction functions, defined by DG problems with cell-averaged coefficients and estimated through differences of expansion coefficients on the uniform mesh, then make the DG bilinear form of the error higher order. The L^2 error is $\mathcal{O}(h^{k+1})$, for all $k \geq 0$ in the linear case; for $k \geq 1$ and $u \in H^{k+3}$, cell-average and downwind-edge-average errors are $\mathcal{O}(h^{k+2})$ in the linear case and

$\mathcal{O}(h^{\min(2k, k+2)})$ in the nonlinear case, which relies on an a priori L^∞ error assumption justified for $k \geq 1$. Experiments with $k \leq 3$ also show cell-average rates $2k + 1$ for $k = 1, 2$, and loss of superconvergence on randomly perturbed meshes.

Contributions

1. Proved optimal L^2 convergence of upwind $\mathbb{P}^k$ DG on uniform Cartesian meshes for every $k \geq 0$ for linear variable-coefficient equations and, under an a priori error assumption justified for $k \geq 1$, for nonlinear conservation laws, which the authors present as answering an open question.
2. Isolated the degree- $(k + 1)$ Radau terms that $\mathbb{P}^k$ cannot represent into a single term ζ_0 , and proved well-posedness of the operators that generate the correction functions ζ_1 and ζ_2 from it.
3. Derived, for uniform meshes, estimates of mixed differences of the expansion coefficients of the exact solution between neighbouring cells, which provide the extra powers of h in the error analysis.
4. Established superconvergence of cell-average and downwind-edge-average errors for $\mathbb{P}^k$ DG, which the authors describe as the first superconvergence results for $\mathbb{P}^k$ DG methods.
5. Observed in constant-coefficient, variable-coefficient, and nonlinear tests that cell-average errors reach the $\mathcal{O}(h^{2k+1})$ rate known for $\mathbb{Q}^k$ when $k = 1, 2$, whereas perturbing the mesh nodes by up to a quarter of the mesh size reduces all three errors to order $k + 1$.

Comments

- §1: the introduction states that the optimal order $\mathcal{O}(h^{k+1})$ “for DG method solving both linear and nonlinear equations holds true when $\mathbb{P}^k$, $k \geq 0$ elements” are used, but the nonlinear analysis rests on the a priori assumption (3.3), which §3, including Remark 3.2, justifies only “for $k \geq 1$ ”.
- §2.3, Eq. (2.26): the proof of Theorem 2.5 uses “ $(Q_3 u, v)_\tau = 0$, $\forall v \in V_h$ ”, but $Q_3 u$ collects the terms $u_{p,q}(L_{i,p} - L_{i,p-1})(x)(L_{j,q} - L_{j,q-1})(y)$ with $p, q \leq k$ and $p + q \geq k + 2$, so for $k = 2$ the term $p = q = 2$ has nonzero inner product with $L_{i,1}(x)L_{j,1}(y) \in V_h$; the half of (2.26) for ζ_0 holds, and whether the bound (2.25) survives without the $Q_3 u$ half is not addressed.
- §2.3: the estimate of J_τ relies on “ $\pi_1^x \pi_1^y \alpha v_x \in V_h$, $\pi_1^x \pi_1^y \beta v_y \in V_h$ for $v \in V_h$ ”, but $\pi_1^x \pi_1^y \alpha$ is bilinear and v_x has degree up to $k - 1$, so the product can have degree $k + 1$ and in general lies outside $V_h = \mathbb{P}^k$.
- §3, Eq. (3.2), and $A(w, v)$: the interface terms are printed as $\hat{f}(u_h)v^-|_{i+\frac{1}{2},y} - \hat{f}(u_h)v^-|_{i-\frac{1}{2},y}$ and $\tilde{g}(u_h)v^-|_{x,j+\frac{1}{2}} - \tilde{g}(u_h)v^+|_{x,j-\frac{1}{2}}$, whereas the linear scheme and $\mathcal{H}_\tau$ use the traces v^+ at $x_{i-\frac{1}{2}}$ and $y_{j-\frac{1}{2}}$; the evident intended terms are $v^+|_{i-\frac{1}{2},y}$ and $v^+|_{x,j-\frac{1}{2}}$.
- §4, Example 3: the nonlinear results are introduced with “Tables 1–2 present errors and the corresponding convergence rates”, but Tables 1–2 hold the constant-coefficient results of Example 1; the nonlinear results are in Tables 5–6.
- Table 6: the caption reads “Errors and convergence rates for the nonlinear equation in uniform meshes”, identical to Table 5, although Example 3 describes the second table as the nonuniform-mesh results and Table 6 shows no superconvergence (for example, the e_c rate for $k = 1$ falls to 1.82); the intended caption is nonuniform meshes.
- §5: the conclusion states that superconvergence of cell and downwind-edge averages “with an order of $\mathcal{O}(h^{k+2})$ ” is derived for linear and nonlinear equations, but Theorem 3.1, Eq. (3.6), gives only $\mathcal{O}(h^{\min(2k, k+2)})$ for the nonlinear case, which is $\mathcal{O}(h^2)$, no better than optimal, for $k = 1$.

2026CWML

ChatGPT (July 2026)

Summary

This paper develops a unified kinematic theory separating rigid rotation from spin or shear contributions to vorticity in three-dimensional flow. Direction-dependent decompositions are derived from the motion of material line and surface elements, whose angular velocities contain complementary strain-induced terms around the mean $\omega/2$. The surface-element spin is shown to equal the relative vorticity in the generalized Caswell formula, linking it directly to surface shear stress. Analogous decompositions are constructed for streamlines and streamsurfaces. In the real-Schur representation

of the velocity-gradient tensor, two intrinsic vorticity decompositions distinguish positive- and negative-spin cases associated with Liutex–shear and complementary anti-KKB mechanisms. The authors prove that direction-dependent modes on the rotation-axis-normal plane are bounded by these intrinsic extrema. Simple shear, point and Burgers vortices, vortex-sheet instability, triaxial strain, and Saturn’s polar flow illustrate how the framework distinguishes orbital rotation, vortex structure, and shear.

Contributions

1. Derived direction-dependent vorticity decompositions from material line- and surface-element rotation.
2. Proved complementary relations between orthogonal line and surface angular velocities.
3. Identified surface-element spin with relative vorticity and surface shear stress.
4. Formulated two invariant decompositions describing KKB and anti-KKB rotational mechanisms.
5. Established bounds linking streamline-based rotation modes to intrinsic vorticity extrema.

2026CZGS

ChatGPT (July 2026)

Summary

The paper shows that weak-shear regions at extrema of zonal $E \times B$ flow can help confine turbulence rather than merely weaken shear suppression. Global total- f XGC gyrokinetic particle-in-cell simulations exhibit zonal jets with nonzero flow curvature, steep ion gyrocenter-density gradients, and converging avalanche activity. The authors isolate quasi-coherent fluctuations near a jet and reduce gyrokinetic test-particle motion to a planar nontwist map. Poincaré sections identify shearless invariant tori, while signatures in the simulated ion distribution connect the reduced model to the self-consistent turbulence. A transmissivity analysis finds that these structures reduce radial transport by about a factor of two in trapped-particle phase space. Avalanches can nevertheless cross the partial barrier through eddy-detachment events that create warm- and cold-core-ring-like structures without fully destroying it. The results extend shearless-transport-barrier concepts from idealized maps to high-fidelity gyrokinetic turbulence and suggest relevance to staircase, pedestal, and turbulence-spreading dynamics.

Contributions

1. Identified robust shearless transport barriers around zonal jets in global XGC turbulence.
2. Derived a planar nontwist map from gyrokinetic test-particle dynamics and measured fluctuations.
3. Connected shearless invariant tori in the map to structures in the simulated ion distribution.
4. Measured roughly twofold transport reduction in trapped-particle phase space.
5. Explained intermittent barrier crossing through avalanche-driven eddy detachment and ring formation.

2026CZQ

ChatGPT (July 2026)

Summary

This paper develops a fifth-order direct arbitrary-Lagrangian-Eulerian Hermite WENO finite-volume method for one- and two-dimensional compressible Euler equations. Compact HWENO reconstruction combines cell averages and first moments, permits prescribed positive linear weights on unequal moving-mesh stencils, and evaluates smoothness indicators by midpoint quadrature. A nodal Lagrangian velocity solver plus mesh regularization controls distortion, while a modified strong-stability-preserving Runge–Kutta discretization updates moments and geometric terms so the fully discrete method satisfies the geometric conservation law. Accuracy and constant-state tests verify fifth-order convergence and geometric conservation. Shock, vortex, Sedov, Noh, and multimedium gas-water tests show high resolution and robustness without remeshing, remapping, or extra interface equations. Relative to ALE-WENO, ALE-HWENO attains smaller errors, uses compact stencils, requires fewer regularization steps, better retains Lagrangian mesh features, and can reduce CPU time. Extension to three dimensions and magnetohydrodynamics remains future work.

Contributions

1. Formulated a fifth-order direct ALE finite-volume method using compact HWENO reconstruction of cell averages and first moments.
2. Avoided geometry-dependent linear-weight solves on unequal moving-mesh stencils and used midpoint-quadrature smoothness indicators.
3. Modified Runge–Kutta moment and geometric updates to satisfy the geometric conservation law.
4. Coupled nodal Lagrangian velocities with mesh regularization to handle distorted meshes and strong discontinuities robustly.
5. Demonstrated one- and two-dimensional multimedium capability, fewer mesh regularizations, and lower reported costs than ALE-WENO.

2026D

ChatGPT (July 2026)

Summary

This master’s thesis ports the computationally dominant kernels of TRIMEG, a Fortran particle-in-cell gyrokinetic code on triangular meshes, to GPUs with OpenMP target offloading. Particle pushing, pullback, density calculation, and grid-to-particle operations were restructured to accommodate incomplete compiler support for object-oriented Fortran and polymorphism. A unified implementation runs on AMD and NVIDIA GPUs, although compiler versions, memory models, and unified shared memory requirements limit practical portability. Kernel profiling, launch-configuration searches, single-node comparisons, and multi-node strong-scaling tests show substantial acceleration for offloaded workloads and characterize contention when multiple MPI ranks share a GPU. The hybrid MPI–OpenMP design scales across GPU nodes under suitable rank placement, but whole-application gains depend on CPU resources and remaining non-offloaded work. Correctness is checked with Cyclone and TCV-X21 ion-temperature-gradient benchmarks: GPU and CPU versions reproduce comparable field-energy growth rates and two-dimensional mode structures. The work therefore provides both an accelerated TRIMEG implementation and detailed evidence about current OpenMP-offloading constraints for complex Fortran applications.

Contributions

1. Ported TRIMEG’s particle and grid-interaction kernels to OpenMP GPU offloading.
2. Produced a common implementation for AMD and NVIDIA accelerator platforms.
3. Documented compiler failures and code transformations required for object-oriented Fortran offloading.
4. Measured kernel speedups, launch sensitivity, MPI-rank contention, and multi-node scaling.
5. Verified the accelerated code with Cyclone and TCV-X21 ion-temperature-gradient simulations.

2026DAKM

Claude (September 2026)

Summary

The authors propose a numerical method for the invariance equation of nonlinear moment matching, a model-reduction approach in which a high-dimensional system $\dot{x} = f(x, u)$, $y = h(x)$ is driven by a low-dimensional signal generator $\dot{\omega} = s(\omega)$, $u = \ell(\omega)$. In the background theory, if the generator is neutrally stable and the system is locally exponentially stable, a mapping $x = \pi(\omega)$ solving the n coupled first-order PDEs $\frac{\partial \pi}{\partial \omega} s(\omega) = f(\pi(\omega), \ell(\omega))$, $\pi(0) = 0$, describes the steady-state response, and $h(\pi(r))$ gives the output of a reduced-order model (ROM) of dimension d . Each component of π is expanded in monomials of total degree 1 to M on a box Ω , and the Galerkin residual equations are solved by Newton’s method, in the least-squares sense when the Jacobian is singular. The scheme recovers the exact solution of a four-state cart-pendulum problem with residuals near 10^{-9} . For a nonlinear resistor-inductor ladder with $n = 1000$ driven by a linear or a Van der Pol oscillator ($d = 2$), residuals decrease with M , and two-state ROMs reproduce the steady-state output with relative RMS errors of 4.5×10^{-3} and 4.1×10^{-2} at $M = 8$. The authors note that accuracy

degrades on larger domains and with stronger generator nonlinearity, that tensorized quadrature cost grows exponentially in d , and that higher-dimensional generators remain future work.

Contributions

1. Exploited the ladder structure by precomputing the exact tensors $[\mathbf{N}]_{ijk} = \int_{\Omega} \phi_i \phi_j \phi_k$ and $[\mathbf{O}]_{ijkl} = \int_{\Omega} \phi_i \phi_j \phi_k \phi_l$, giving a block-tridiagonal Jacobian with diagonal blocks $\mathbf{A} + 2\kappa\mathbf{M} + \mathcal{N}(\mathbf{c}_i) + \mathcal{O}(\mathbf{c}_i)$ and off-diagonal blocks $-\mathbf{M}$, so that switching from the linear to the Van der Pol generator only requires reassembling $\mathbf{A}$.
2. Chose the free ROM input maps $\bar{g}(r, u) = (0, c)^\top u$ for the linear oscillator and $(0, \mu(1 - r_1^2) + c)^\top u$ for the Van der Pol generator, whose linearizations $\begin{bmatrix} 0 & a \\ -a & -c \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ -1 & -c \end{bmatrix}$ are Hurwitz for every $c > 0$.
3. Reported, for $n = 1000$ on $\Omega = [-0.6, 0.6]^2$ and $M = 2$ to 8 ($N = 5$ to 44 basis functions per component), residuals falling from 3.4×10^{-4} to 6.5×10^{-10} for the linear oscillator and from 7.4×10^{-3} to 8.1×10^{-7} for the Van der Pol generator, with three Newton iterations from $\mathbf{c}^0 = \mathbf{0}$ in every Van der Pol case.
4. Showed that at $M = 6$ enlarging $\Omega = [-a, a]^2$ from $a = 0.6$ to 1.2 raises the residual on $[-0.6, 0.6]^2$ for $\mu = 0.25, 1$, and 2 , most strongly for $\mu = 2$ (from about 4×10^{-5} to about 50), which the authors attribute to the cubic term $\mu\omega_1^2\omega_2$ amplifying coefficient errors near the boundary.
5. At reduced dimension 2 , the Van der Pol ROM outperforms proper orthogonal decomposition ($\epsilon_{\text{rel}} = 5.7 \times 10^{-2}$) and balanced truncation of the linearization (1.2×10^{-1}), and simulates to $t = 50$ in 0.0120 s against 0.4243 s for the full model, at an offline cost of 8423 s for $M = 8$.

Comments

- Section 3.2, Eqs. (3.19)–(3.20): the basis is defined as the tensor product of the one-dimensional basis $\varphi_M(\omega) = (\omega, \omega^2, \dots, \omega^M)$, which has no constant term, so every product in (3.20) contains each ω_j at least to the first power and pure powers such as ω_1 are excluded (for $d = 2$, $M = 2$ only $\omega_1\omega_2$ remains); this conflicts with the multi-index set $\Lambda_M = \{\nu : 1 \leq |\nu| \leq M\}$, the count $N = \binom{M+d}{M} - 1$, Table 3 ($N = 5, 14, 27, 44$), and the Test 1 solution $(\omega_1, k\omega_1, \omega_2, k\omega_2)^\top$, so the evident intended reading includes the constant in the one-dimensional factors and removes only the constant product.
- Sections 4.3 and 5, Eq. (4.15): Section 2.1 states “In the rest of the paper, we shall assume that assumptions (A1) and (A2) hold”, and Section 5.1 relies on the resulting guarantee that π exists near the origin, but the Van der Pol generator with $\mu > 0$ linearizes at the origin to $S = \begin{bmatrix} 0 & 1 \\ -1 & \mu \end{bmatrix}$, whose eigenvalues have real part $\mu/2 > 0$, so its origin is unstable and (A1) fails (Section 2 notes that neutral stability implies purely imaginary eigenvalues of S), leaving Proposition 2.7 without its hypotheses; unresolved.
- Section 4.1: the statement in Section 2.1 that (A1) and (A2) hold in the rest of the paper conflicts with the cart-pendulum dynamics, whose Jacobian $\partial f/\partial x$ at the origin has $\partial f_3/\partial x_1 = a_1 > 0$ and eigenvalues $\pm\sqrt{a_1}$ and 0 (double), so (A2), which requires all eigenvalues to have negative real part, fails; unresolved.
- Section 5.4, Table 3: the text states that “The Jacobian JF is stored sparsely, so the time complexity scales linearly in n and quadratically in N ”, but the total times in Table 3 rise from 1.92 s at $N = 5$ to 8423 s at $N = 44$, a factor of about 4400 for an 8.8 -fold increase in N (quadratic scaling would give about 77), roughly as N^4 , in line with the $O(N^4)$ tensor $\mathbf{O}$ named later in the same section; unresolved.

2026DBCT

ChatGPT (July 2026)

Summary

The paper develops a communication-avoiding s -step preconditioned conjugate-gradient method for distributed GPU systems. It builds each block Krylov basis with Chebyshev polynomials and solves the resulting small Gram systems approximately using forward Gauss-Seidel sweeps, reducing global synchronizations while retaining practical numerical stability. A moment representation explains why the Chebyshev Gram matrix remains reasonably conditioned for moderate block sizes, and the equivalence between forward Gauss-Seidel and modified Gram-Schmidt provides a

structural justification for the inner iteration. A performance model makes explicit the trade-off between fewer global reductions and extra local arithmetic. Multi-GPU tests on three-dimensional Poisson systems show improved strong scaling relative to classical CG. Weak-scaling experiments with algebraic multigrid exceed four billion unknowns; at 512 GPUs, $s = 2, 3, 4$ reduce time to solution, with $s = 4$ giving the best tested balance. The authors limit these conclusions to the explored scales and propose adaptive Gram-solve accuracy and communication-reducing preconditioners for larger systems.

Contributions

1. Combined a Chebyshev block basis with iterative forward Gauss-Seidel Gram solves in s -step PCG.
2. Related Gram conditioning to a moment structure and the inner solve to modified Gram-Schmidt.
3. Derived a performance model balancing synchronization savings against additional local work.
4. Implemented and evaluated the method on NVIDIA A100 and H100 multi-GPU systems.
5. Demonstrated convergence beyond four billion unknowns and weak-scaling gains at 512 GPUs for moderate s .

2026DBTT

Claude (September 2026)

Summary

The Godunov–Peshkov–Romenski (GPR) model describes fluids and solids by one first-order hyperbolic system for density, momentum, energy, a distortion field A , and a thermal impulse J , whose stiff relaxation sources recover the compressible Navier–Stokes–Fourier equations as the relaxation times $\tau_1, \tau_2 \rightarrow 0$. Its acoustic, shear, and heat waves make explicit schemes costly. Extending an earlier Cartesian four-split scheme to unstructured triangles, the authors split the system into convective, temperature, mechanical, and pressure subsystems. Only convection is explicit, so the time step depends on material velocity alone. The implicit stages reduce to symmetric positive definite scalar wave equations for temperature and pressure, solved by conjugate gradients, and a nonsymmetric vector wave equation for the nodal velocity, solved by GMRES. Momentum, A , and J live in cells and density, energy, pressure, and temperature at vertices; corner-normal discrete operators satisfying discrete curl–gradient and divergence–curl identities keep A and J curl-free when sources are absent or linear. Formal expansions show consistency with the Navier–Stokes stress and Fourier heat flux, and the pressure step recovers the low-Mach limit. Tests include Taylor–Green vortex, Riemann, shear-flow, lid-driven-cavity, solid-rotor, and explosion problems. Rescaling A to enforce its determinant constraint, applied only in viscous regimes, breaks the curl-free property.

Contributions

1. Extended the four-split semi-implicit GPR scheme to unstructured triangular meshes, which the authors describe as the first all-Mach-number GPR solver on unstructured meshes that respects both discrete vector-calculus identities.
2. Built a vertex-staggered collocation with mutually adjoint corner-normal gradient, divergence, and curl operators and a compatible final update of the distortion field and thermal impulse that starts from the old time level.
3. Reduced the mechanical subsystem, written for the trace-free metric tensor, to a single vector wave equation for the nodal velocity by eliminating the metric tensor and thermal impulse through semi-implicit stress approximations.
4. Showed by Chapman–Enskog-type expansions that the discrete scheme yields the Navier–Stokes stress with viscosity $\mu = \rho_0 c_s^2 \tau_1 / 6$ as $\tau_1 \rightarrow 0$ and a consistent Fourier heat flux as $\tau_2 \rightarrow 0$.
5. Verified second-order convergence of density errors in the acoustic Mach number down to $\text{Ma} = 5 \times 10^{-4}$ and curl errors of A and J at machine precision in a solid-rotor test.

Comments

- §2.2, Eq. (17c): the primitive-variable temperature subsystem is written as $\frac{c_v}{T c_h^2} \frac{\partial T}{\partial t} + \frac{\partial J_k}{\partial x_k} = 0$, but eliminating $\mathcal{E}$ from (16b)–(16c) with ρ , $\mathbf{v}$, and $\mathbf{A}$ fixed, using $\mathcal{E}_1 = \rho c_v T$ (from (3) and $T = \partial_{\rho S} \mathcal{E}$), $\mathcal{E}_4 = \frac{1}{2} c_h^2 \rho J_i J_i$ (3), $q_k = \rho c_h^2 T J_k$ (5), and $\theta_2 = \rho c_h^2 \tau_2$ (7), gives the right-hand side $J_k J_k / (\tau_2 T)$, the relaxation heating that

corresponds to $\beta_k \beta_k / (T \theta_2)$ in (2); the discrete form (45b) also omits it. Whether the term was dropped deliberately is unresolved.

2026DDP

ChatGPT (September 2026)

Summary

The authors introduce and analyze a hybrid discretization for unsteady incompressible magnetohydrodynamics on convex domains. Velocity, magnetic field, and their associated pressures are approximated by element and face unknowns inspired by Hybrid High-Order methods, with Raviart–Thomas–N’ed’elec element spaces for vector fields. Carefully designed reconstructions and skew-symmetric nonlinear couplings yield energy stability and convection-semirobust a priori estimates: for sufficiently regular solutions, velocity and magnetic-field errors do not scale with the inverse viscosity or magnetic diffusivity. The analysis also predicts a higher asymptotic order when diffusion dominates, characterized through local Reynolds- and Hartmann-type numbers. Because no interelement penalty terms are required, the method retains a compact stencil, and static condensation can eliminate element unknowns from the global algebraic system. Manufactured-solution experiments confirm the predicted convergence regimes and parameter robustness, while a magnetically driven lid-cavity example illustrates the coupled flow behavior. The results apply under the stated regularity and convex-domain assumptions.

Contributions

1. Developed a hybrid high-order-type method for all velocity, magnetic, and pressure variables in incompressible MHD.
2. Proved energy stability and convection-semirobust error bounds independent of inverse diffusion coefficients.
3. Identified improved convergence orders in the asymptotic diffusion-dominated regime.
4. Achieved compact coupling without interelement penalty terms and enabled static condensation.
5. Confirmed theoretical rates and parameter robustness through comprehensive numerical experiments.

2026DEM

ChatGPT (July 2026)

Summary

This paper determines worst-case matrix–vector query complexity for approximately solving general linear systems under randomized adaptive algorithms. For two-sided access to A and $A^\top$, it proves a lower bound of essentially $\frac{1}{4}\kappa \log(1/\varepsilon)$ products, within a factor of four of conjugate gradient on the normal equations. The proof reduces a sufficiently accurate linear solver to estimating $\text{tr}(A^{-1})$ and combines this with polynomial inapproximability of $1/x$ on $[-\kappa, -1] \cup [1, \kappa]$. For transpose-free one-sided access, a hidden-Haar argument shows that an orthogonal, perfectly conditioned hard instance needs at least $\lceil n/2 \rceil$ products for residual $1/2$ and all n products for residual $0.2/\sqrt{n}$. Extensions give the sharp $\Omega(\sqrt{\kappa} \log(1/\varepsilon))$ symmetric-positive-definite dependence and a fine-grained $\Omega(k + \kappa_k \log(1/\varepsilon))$ bound. These are worst-case oracle results; structured matrices, preconditioners, and typical GMRES behavior may be much better.

Contributions

1. Proved an explicit randomized two-sided lower bound proportional to $\kappa \log(1/\varepsilon)$.
2. Proved dimension-scale one-sided lower bounds even for orthogonal matrices.
3. Established a hidden-Haar theorem for information concealed from adaptive queries.
4. Connected linear solving to inverse-trace estimation and polynomial inapproximability.
5. Extended the framework to sharp positive-definite and fine-grained bounds.

2026DFTW

ChatGPT (July 2026)

Summary

The paper extends OSIRIS with a deterministic particle-grid model for field and electron-impact ionization. Electron macroparticles deposit MRBEB-based ionization-rate, energy-loss, and secondary-momentum contributions to grids; coupled charge-state rate equations are advanced deterministically, and electrons and ions are injected with conserved charge and modeled momentum. The implementation supports immobile neutral media and mobile ion macroparticles and augments tunneling field ionization with barrier-suppression corrections. Verification against analytical rate equations, stopping-power predictions, and momentum-transfer formulas covers several elements and charge states. Comparisons with Smilei and EPOCH show agreement after accounting for cross-section choices and expose inaccuracies in a multiple-ionization-per-timestep Monte Carlo treatment. Relative to particle-pair Monte Carlo schemes, the grid method reduces ionization-rate error by up to two orders of magnitude at moderate particles per cell while retaining linear runtime scaling. The current implementation is slower and omits recombination, excited-state kinetics, ion-frame Lorentz transforms, and secondary-electron angular scattering.

Contributions

1. Introduced a deterministic particle-grid collisional-ionization algorithm for immobile media and mobile ion macroparticles in OSIRIS.
2. Integrated MRBEB cross sections, energy exchange, momentum transfer, and tunneling-to-barrier-suppression field-ionization rates.
3. Advanced coupled charge-state populations deterministically while injecting ionized particles with charge conservation and numerical safeguards.
4. Verified rates, stopping power, and secondary momentum and benchmarked the implementation against Smilei and EPOCH.
5. Demonstrated linear particle-count scaling and substantially faster error reduction than Monte Carlo sampling at moderate particle counts.

2026DGNP

ChatGPT (September 2026)

Summary

This paper extends hierarchical sparse-grid particle-in-cell methods for the Vlasov–Poisson system to resolve localized plasma structures while retaining reduced particle noise. An energy-based B-spline approximation space prioritizes accuracy in the H^1 seminorm, which measures electric-field error through the potential gradient. Generalized sparse-grid spaces enrich the hierarchy toward full-grid resolution. Local adaptation selects basis functions using normalized density coefficients, called hierarchical surpluses. An incremental algorithm evaluates candidate subspaces with Schur complements and reused Cholesky factorizations, avoiding a solve over the complete enriched space. Two-dimensional manufactured-solution and diocotron-instability tests distinguish spatial bias from statistical error and demonstrate improved density reconstruction with fewer active nodes. In one manufactured-solution benchmark, the adaptive method uses approximately 17 times fewer particles and 45 times fewer nodes than standard PIC, with roughly fivefold shorter execution time. The results support local refinement as a way to balance sparse-grid noise reduction against representation error; three-dimensional validation remains future work.

Contributions

1. Introduces an energy-based hierarchical approximation space with $O(h^{-1})$ degrees of freedom, targeting H^1 accuracy without the conventional sparse-grid dimension factor $|\log h|^{d-1}$.
2. Defines generalized sparse-grid spaces that add hierarchical levels, supplying additional basis functions for structures poorly resolved by conventional sparse grids.

3. Develops a local adaptation criterion based on normalized density surpluses, retaining significant basis functions and removing insignificant degrees of freedom.
4. Derives an incremental refinement algorithm using Schur-complement reduction and reusable Cholesky factors to assess candidate subspaces without solving the complete generalized mass system.
5. Demonstrates restored second-order spatial convergence for a challenging diocotron initial density and substantial particle, node, and runtime reductions in separate two-dimensional accuracy benchmarks.

2026DJUN

ChatGPT (July 2026)

Summary

OpenEdge is an open-source DSMC framework for plasma-material interaction and charged-particle transport in magnetic-fusion environments. Rather than replacing SPARTA, it builds domain-specific fusion capabilities on SPARTA's DSMC engine, adding charged-particle transport, Coulomb collisions, ionization and recombination, sputtering and reflection models, sheath effects, and coupling to plasma backgrounds. The implementation uses object-oriented C++ with a Python interface and Kokkos-enabled parallelism for modern high-performance platforms. Verification and validation examples test collision operators, ionization, sheath behavior, impurity transport, and quantitative comparisons with WEST facility data. The paper positions OpenEdge as a modular tool for whole-device and edge-plasma studies where kinetic impurity and surface physics matter. Current limitations are also clear: predictive use depends on external background plasma data and ongoing development of dynamic surface evolution, re-erosion, evaporation, charging, and changing-field models.

Contributions

1. Introduced OpenEdge as a fusion-oriented DSMC framework built on the SPARTA engine.
2. Added PMI physics modules for collisions, ionization, recombination, reflection, sputtering, and sheath effects.
3. Provided a C++/Python software architecture with Kokkos-based parallel execution support.
4. Verified key physics components and compared impurity-transport behavior with WEST observations.
5. Defined development paths for dynamic surfaces, deposited layers, lithium droplets, and evolving plasma backgrounds.

2026DM

ChatGPT (July 2026)

Summary

This paper analyzes plasma wakefield acceleration driven by profiled trains of triangular electron bunches. The central result is that the conditions for forming plateau-like accelerating or decelerating wakefields coincide with the conditions for placing bunches in their most stable focusing regions. Analytical formulas determine bunch spacing and peak-density ratios for triangular and rectangular precursors, and show that the density ratio becomes universal from the third bunch onward. The authors also analyze driver-witness pairs, finding that the accelerating field can be restored after each pair, which theoretically permits indefinitely repeated bunch trains. Quasi-static 2.5D PIC simulations with LCODE verify the predicted bunch parameters and plateau fields. Ion mobility breaks the ideal infinite-train picture by shifting ions and eventually introducing inhomogeneity, but the simulations indicate that short stable trains with uniform acceleration remain feasible. The work provides design rules for stable low-charge bunch-train plasma accelerators.

Contributions

1. Derived plateau-formation conditions for triangular bunch trains with different precursor shapes.
2. Showed that plateau conditions coincide with high-stability focusing-force placement.
3. Identified a universal peak-density ratio for bunches beginning with the third bunch.
4. Verified the analytical design rules using quasi-static 2.5D PIC simulations.

5. Quantified how ion mobility limits infinite trains while preserving feasible short stable trains.

2026DMN

ChatGPT (September 2026)

Summary

This paper develops high-order summation-by-parts finite differences for the two-dimensional, two-velocity Vlasov–Maxwell system. Tensor-product operators retain the one-dimensional SBP structure, while a Lax–Friedrichs split upwind operator suppresses filamentation instabilities that destabilize the central scheme. The semidiscrete method conserves mass and momentum, propagates Gauss’s law, and makes the distribution’s discrete L^2 norm nonincreasing; exact energy conservation applies to the central variant, while upwinding adds controlled dissipation. A matrix-free CUDA implementation stores only local stencil coefficients, flattens the four-dimensional distribution, and uses atomic moment accumulation instead of assembled global matrices. Manufactured solutions recover second-, fourth-, and sixth-order convergence. Diocotron, Weibel, and two-species Kelvin–Helmholtz simulations demonstrate the method on a single NVIDIA L40 GPU, including a reported case with about 420 million degrees of freedom. Strong magnetic fields remain costly under the explicit time integrator, and extending the approach to 3D3V will require implicit treatment and multi-GPU decomposition.

Contributions

1. Constructed tensor-product SBP discretizations for the 2D2V Vlasov–Maxwell equations.
2. Introduced split upwind stabilization with provable mass, momentum, Gauss-law, and norm properties.
3. Implemented the scheme as matrix-free CUDA kernels without assembled multidimensional operators.
4. Verified second-, fourth-, and sixth-order accuracy using manufactured solutions.
5. Demonstrated large single-GPU diocotron, Weibel, and Kelvin–Helmholtz simulations and documented scaling limits.

2026DRM

ChatGPT (July 2026)

Summary

This paper develops a compressible dynamic forcing strategy that accelerates the transition from an upstream Reynolds-averaged Navier–Stokes solution to wall-modelled large-eddy simulation on curvilinear grids. The method amplifies wall-normal velocity fluctuations supplied by a synthetic turbulence generator, with density and forcing scalings derived from Morkovin’s hypothesis and the Van Driest transformation. The generator can operate as an inlet boundary condition or as a volumetric source embedded inside the domain. In a Mach-2 zero-pressure-gradient boundary layer spanning $1000 \leq Re_\tau \leq 2500$, the combined method recovers skin friction within 5% of the Coles–Fernholz correlation after roughly 7–8 inlet boundary-layer thicknesses and shortens adaptation distance by as much as 70% relative to synthetic turbulence alone. A Mach-4.14 rounded-base calculation shows that resolved upstream turbulence is essential when the separated mixing layer has $M_c \approx 1$: the method reproduces experimental shock, recirculation, mixing-layer, and base-pressure behavior that a RANS-inlet hybrid calculation misses.

Contributions

1. Formulated compressible dynamic forcing with Morkovin and Van Driest scaling.
2. Supported synthetic turbulence as either an inlet condition or an embedded source.
3. Reduced Mach-2 boundary-layer adaptation distance by up to 70% without downstream artifacts.
4. Showed that hypersonic $M_c \approx 1$ mixing layers require resolved upstream turbulence.
5. Validated the embedded RANS-to-WMLES strategy against rounded-base experiments.

2026DSMG

ChatGPT (July 2026)

Summary

Selective state-space models avoid attention’s quadratic sequence cost, but their recurrent state updates, local convolutions, and packed mixer parameters make conventional Transformer tensor parallelism inefficient. This paper develops a state-space-model-specific multi-GPU inference design. It caches each layer’s recurrent state after prefill so decode does not reprocess the prompt, shards channels so convolution and state updates remain local, and explicitly distributes packed quantities such as Δ , B , C , A , and D according to their computational roles. Communication is restricted mainly to output aggregation, where optional FP32-to-FP16 all-reduce reduces bandwidth. Experiments cover Mamba, Mamba-2, Falcon-Mamba, and Zamba on NVIDIA A6000 PCIe and A100 NVLink systems. Tensor parallelism supports prompts two to four times longer at fixed batch size and substantially improves throughput over single-GPU inference, particularly for long contexts. Quantized communication provides further gains but slightly changes token rankings, making it an explicit accuracy–throughput trade-off.

Contributions

1. Introduced a distributed, sharded state-space-model cache that carries recurrent state from prefill into decode without extra communication.
2. Designed channel-wise tensor partitioning that keeps convolution, gating, and recurrent state updates GPU-local.
3. Separated packed mixer fields and placed channel-dependent, token-dependent, and replicated parameters according to their data dependencies.
4. Added optional FP32-to-FP16 quantized all-reduce at the remaining synchronization boundary.
5. Evaluated the design across four pure and hybrid state-space architectures on PCIe- and NVLink-connected GPU systems.

2026EAKU

ChatGPT (July 2026)

Summary

This paper studies performance portability in SOD2D, a Fortran spectral-element solver using OpenACC and MPI for compressible, turbulence-resolving computational fluid dynamics. The authors profile convection and diffusion kernels and explore a design space spanning FP32/FP64 arithmetic, unified or split kernels, four memory-access variants, NVIDIA V100/A100 GPUs, an AMD MI250X compute die, and weak scaling on LUMI. The same optimization does not transfer reliably across vendors, precisions, or scales. Removing explicit preloading usually beats the baseline, prefetching improves preloaded variants in selected cases, and regularizing accesses can lose performance because its auxiliary matrix is 18 times larger. Kernel splitting helps AMD in FP32 but hurts NVIDIA, with the trend reversing in FP64. At scale, the optimal channel-flow configuration changes with GPU count: applying the 32-GPU optimum at 64 GPUs can cost 23.8% throughput in FP32 and 8.3% in FP64. Small-scale or single-platform measurements are therefore inadequate predictors of multi-GPU performance, motivating architecture-, precision-, workload-, and scale-aware tuning.

Contributions

1. Defined a performance-portability design space spanning application, memory, compiler, hardware, precision, and scale.
2. Identified convection as the dominant hotspot on both NVIDIA and AMD platforms.
3. Evaluated kernel splitting, preloading removal, cache prefetching, and regularized access on three GPU architectures.
4. Showed that optimization effects reverse across vendor and precision combinations.
5. Quantified throughput penalties from extrapolating a 32-GPU optimum to 64-GPU channel-flow simulations.

2026ECM

ChatGPT (September 2026)

Summary

NektarIR is a domain-specific compiler for high-order spectral/*hp* finite-element operators used in computational fluid dynamics. Its custom MLIR dialect represents elemental operations and records element shape, basis or quadrature, and data layout before lowering the operations through reusable MLIR dialects to CPU or GPU code. The paper implements sum-factorized Helmholtz kernels on three-dimensional hexahedral and tetrahedral elements. CPU code vectorizes across elements; GPU code explores one thread per element and one thread per expansion coefficient. Benchmarks against hand-written Nektar++ kernels show higher throughput for NektarIR on the tested AMD EPYC CPUs, but lower throughput on the NVIDIA H100 for large problems, particularly for coefficient-threaded tetrahedral kernels. Measured generation and compilation overhead is under one second for the tested kernels. The work demonstrates a route to generating hardware-specific kernels from one operator description while identifying GPU register pressure and other performance gaps as remaining work.

Contributions

1. Defined an MLIR dialect whose block type records element shape, expansion or quadrature data, and memory layout for spectral/*hp* operators.
2. Lowered composed Helmholtz operations to fused, sum-factorized elemental kernels through domain-specific transformations and standard MLIR dialects.
3. Implemented CPU vectorization across elements and two GPU threading strategies, including loop coalescing for triangular and tetrahedral iteration spaces.
4. Measured sub-second kernel lowering and just-in-time compilation on the tested CPU and GPU configurations.
5. Compared generated and hand-written Nektar++ Helmholtz kernels, finding higher CPU throughput but unresolved GPU performance deficits.

2026ELSD

ChatGPT (July 2026)

Summary

Stellarator divertors must guide exhaust along three-dimensional magnetic fields, where separatrices often acquire broad stochastic layers. This paper designs an LHD-like helical divertor using optimized modular coils rather than continuous helical coils. The central device is a rotating, lemon-shaped target plasma surface with toroidally continuous sharp corners. Standard coil optimization minimizes the normalized normal magnetic field on this surface, causing its corners to become X-lines of a clean separatrix while also constraining coil length, spacing, and curvature. The resulting modular coils produce the desired topology with substantially less chaos than the Large Helical Device, showing that strong stochasticity is not intrinsic to helical divertors. A weighted squared-flux objective emphasizes accuracy near the sharp corners and improves field-line behavior. The authors also introduce manifold optimization, which directly controls stable and unstable manifolds without minimizing normal field and can further reduce chaos. That configuration is sensitive to coil perturbations, however, and the target boundary was not optimized simultaneously for confinement, stability, quasisymmetry, or quasi-isodynamicity. The work therefore establishes a new divertor-design route while identifying robustness and core-edge co-optimization as unresolved requirements.

Contributions

1. Produced the first LHD-like helical divertor design based on optimized modular coils.
2. Used toroidally continuous sharp target-surface corners to create X-lines and a clean separatrix.
3. Demonstrated that a helical divertor can have substantially less magnetic chaos than LHD.
4. Added weighted squared-flux optimization to concentrate field accuracy near diverting corners.
5. Introduced manifold optimization as a direct separatrix-control objective and quantified its perturbation sensitivity.

2026F

ChatGPT (September 2026)

Summary

The paper derives plasma similarity transformations from kinetic equations coupled to Maxwell's equations and tests them with particle-in-cell and fluid simulations. For a system enlarged by a factor k , the collisionless scaling multiplies lengths and times by k , fields by k^{-1} , and charged-particle distribution functions and densities by k^{-2} , while leaving velocities, species masses, and charges unchanged. The same transformation accommodates electromagnetic and relativistic dynamics. In weakly ionized plasmas, neutral density scales as k^{-1} so that charge-neutral collisions remain compatible. Simulations compare two-stream instabilities, electron diodes, and radio-frequency discharges with standing-wave effects under matched scaling conditions. Highly ionized, quasineutral plasmas require a different density scaling, k^{-1} , when charge-charge collisions dominate and an ambipolar field relation replaces Poisson's equation. Fluid microdischarge calculations show violations between the weakly and strongly ionized limits due to gas heating and chemistry. A temperature-dependent correction reduces these discrepancies but is presented as less rigorous than the limiting scalings. Thus, the results establish conditional similarity relations, rather than a single exact scaling across arbitrary plasma regimes.

Contributions

1. Derives a common space-time, field, and distribution-function scaling for collisionless kinetic equations coupled to the full Maxwell system.
2. Shows compatibility with relativistic electron dynamics because the velocity and Lorentz factor remain invariant.
3. Demonstrates scaled agreement in particle-in-cell simulations of two-stream evolution, electron diodes, and electromagnetic radio-frequency discharges.
4. Distinguishes charge-neutral-dominated density scaling from the charge-charge-dominated scaling obtained under a quasineutral ambipolar approximation.
5. Proposes and tests a neutral-temperature correction that reduces microdischarge similarity violations during the transition between weak and strong ionization.

2026FABC

ChatGPT (July 2026)

Summary

This experiment combines a laser wakefield accelerator with a second, beam-driven plasma stage: the first-stage electron bunch drives the second wake, while a density down-ramp injects a new witness bunch internally. Operating near depletion of the driver simultaneously increases witness energy and improves beam quality relative to the laser-only output. The best witnesses reach 1.3 GeV, with 38–53 pC charge in representative cases, roughly 0.1 mrad divergence at the spectral peak, and few-percent energy spread. The energy transformer ratio approaches two. From measured driver and witness spectra, about 20% of the initial integrated driver energy transfers to the witness, while a lower-bound estimate assigns at least 39% of the wake energy extracted from the driver to the witness. Particle-in-cell modeling supports the injection and depletion interpretation. The authors report simultaneous high depletion and high plasma-to-witness efficiency beyond previous plasma-wakefield experiments, while noting that future application studies need complete phase-space measurements, including current and emittance.

Contributions

1. Demonstrated a compact laser-driven source feeding an internally injected beam-driven plasma stage.
2. Operated the second stage near driver depletion while retaining narrow, low-divergence witness bunches.
3. Reached record 1.3 GeV witness energy for this hybrid scheme.
4. Measured an energy transformer ratio near two and about 20% driver-to-witness efficiency.
5. Connected the observed injection, energy gain, and depletion with particle-in-cell simulations.

2026FBAP

ChatGPT (July 2026)

Summary

High-resolution two-dimensional particle-in-cell simulations examine how the nonlinear collisionless Kelvin–Helmholtz instability mixes plasmas across a magnetized shear layer. Particle-origin labels directly measure interpenetration, while a complementary density-based tracer derived from labeled-population densities localizes the same process. Mixing rises during vortex merging but remains inefficient and spatially localized: values near 30% occur only in narrow interfaces, detached parcels, and reconnection regions. Ions mix more than electrons because their weaker magnetization permits more cross-field motion; electrons remain largely tied to magnetic field lines. The growth of mixed regions coincides spatially and temporally with current intensification and clusters of magnetic X-points, quantitatively associating localized reconnection with transport. Larger-domain tests preserve this localization. The conclusions are limited to a symmetric, two-dimensional configuration with reduced ion-to-electron mass ratio and periodic boundaries; realistic asymmetry and three-dimensional secondary instabilities may increase cumulative transport.

Contributions

1. Combined particle-origin labeling with a density-based tracer to quantify genuine plasma interpenetration.
2. Demonstrated that nonlinear Kelvin–Helmholtz mixing remains confined to narrow interfaces and detached plasma parcels.
3. Quantified species-dependent transport, finding stronger ion mixing and tightly constrained electron mixing.
4. Linked enhanced mixing to the temporal growth and spatial clustering of magnetic-reconnection signatures.
5. Tested domain-size sensitivity and identified dimensionality, mass-ratio, symmetry, and boundary-condition limitations.

2026FCSJ

ChatGPT (July 2026)

Summary

Continuum kinetic solvers can waste most velocity-space degrees of freedom where the particle distribution is negligible. This work introduces nonuniform velocity coordinates into a discontinuous Galerkin full- f gyrokinetic formulation. User-defined mappings concentrate resolution in physically important regions while a continuous intercell representation of the mapping and carefully constructed numerical fluxes preserve the conservative structure. Collisionless dynamics conserve particle number and total energy to machine precision apart from temporal-discretization error; a hybrid polynomial basis also enables the collision operator to conserve particles, momentum, and energy exactly. Tests cover isolated collisionless and collisional behavior and increasingly realistic one-, two-, and three-dimensional Gkeyll simulations of high-temperature-superconducting mirrors, an ASDEX Upgrade scrape-off layer, and LAPD turbulence. The mapped grids reproduce corresponding uniform-grid results using 6–60 times fewer velocity-space cells and achieve 22–60-fold wall-clock speedups. Very coarse grids can nevertheless produce nonphysical negative parallel temperatures, motivating future positivity enforcement and more flexible mappings.

Contributions

1. Formulated a mapped-coordinate discontinuous Galerkin velocity discretization for full- f gyrokinetics.
2. Designed continuous discrete mappings and numerical fluxes that preserve particles and energy in collisionless dynamics.
3. Introduced a hybrid basis giving the mapped collision operator exact particle, momentum, and energy conservation.
4. Verified accuracy and conservation across collisionless, collisional, and coupled tests in one to three spatial dimensions.
5. Reproduced uniform-grid applications with 6–60 times fewer velocity cells and 22–60-fold speedups.

2026FFCR

ChatGPT (September 2026)

Summary

This study tests an electron-temperature-gradient (ETG) turbulent heat-flux model against JET H-mode pedestals spanning deuterium–tritium isotope mixtures and gas-fuelling rates. Using measured density profiles and separatrix boundary conditions, the model predicts the electron temperature at the density-pedestal top within about 20% across the scans. This supports slab-ETG turbulence, whose threshold depends on $\eta_e = L_{n_e}/L_{T_e}$, as the dominant heat-transport mechanism in the steep-density-gradient region. The model nevertheless underpredicts the fitted temperature-pedestal top by as much as a factor of two because that location lies farther inward, where the density gradient is weak and the adopted heat-flux scaling overpredicts transport. The isotope trend arises indirectly through isotope-dependent density profiles rather than an ion-mass dependence in the ETG scaling. Higher fuelling raises $n_{e,\text{sep}}/n_{e,\text{ped}}$ and lowers the predicted temperature, consistent with experiment. A combined density-pedestal, ETG, and EPED model predicts pedestal density and temperature generally within roughly 10%, although its pedestal-width trend disagrees with measurements.

Contributions

1. Validated the ETG heat-flux model across JET isotope-mix and gas-fuelling scans rather than a single operating condition.
2. Found density-pedestal-top temperatures within about 20% of measurements throughout the dataset.
3. Showed that applying the steep-gradient scaling inward can underpredict the fitted pedestal-top temperature by more than a factor of two.
4. Explained the modeled isotope dependence through measured density-profile changes and the separatrix-to-pedestal density ratio.
5. Combined density, ETG, and EPED constraints into a full pedestal model with density and temperature predictions generally within about 10%.

2026FHDM

ChatGPT (July 2026)

Summary

This paper generalizes a gridless quasistatic algorithm for axisymmetric plasma wakes and implements it in Wake-T. Background plasma macroparticles are advanced on radial trajectories, while fields are reconstructed directly from enclosed charge and current rather than deposited onto a fixed plasma grid. The formulation supports arbitrary radial density profiles, multiple mobile plasma species, nonuniform sampling, laser drivers, and particle beams. Independent adaptive grids remain available for the laser envelope and beam pushers, allowing very narrow witness beams to be resolved without globally refining the plasma representation. Benchmarks against FBPIC and HiPACE++ cover laser- and beam-driven wakes, ion motion, beam loading, and tailored local refinement. In the reported laser case, comparable convergence requires roughly 40 times less combined resolution and reduces runtime from 9.8 GPU-hours on an A100 to about seven minutes on one CPU core. The approach assumes axial symmetry and the quasistatic approximation, so asymmetric and rapidly evolving phenomena remain outside its scope.

Contributions

1. Generalized gridless quasistatic wake modeling to arbitrary profiles and multiple plasma species.
2. Coupled the model in Wake-T to laser-envelope and particle-beam solvers.
3. Enabled independent local refinement around narrow beams without a global plasma grid.
4. Benchmarked laser wakes, beam wakes, ion motion, and beam loading against established codes.
5. Demonstrated large resolution and runtime reductions for converged axisymmetric simulations.

2026FHWS

ChatGPT (September 2026)

Summary

The authors replace FBPIC’s Numba CUDA-dependent accelerator kernels with C/C++ kernels called through CuPy RawKernel, enabling its quasi-cylindrical Fourier–Bessel particle-in-cell simulations on HIP-compatible DCUs. The Python interface, device-resident arrays, and MPI domain decomposition remain in place. Linear-wakefield and laser-wakefield-acceleration tests reproduce the original backend’s field behavior, with reported relative L^2 field differences no greater than 5.19×10^{-4} . On an NVIDIA V100, the new backend is 1.32–1.54 times faster for three tested workloads. Device-specific block tuning reduces DCU step time by 24.3%. Four DCUs provide a 1.88-fold strong-scaling speedup and 2.72-fold aggregate weak-scaling throughput, about 68% efficiency; eight-DCU scaling is limited by communication and synchronization. The evidence supports portability and performance on the tested CUDA and DCU systems, with larger-cluster scaling still constrained.

Contributions

1. Ported FBPIC’s dominant particle and field kernels to a HIP-compatible CuPy RawKernel backend.
2. Preserved the existing Python workflow, device-resident data, and MPI decomposition.
3. Checked numerical agreement with the original code on wakefield benchmarks.
4. Measured single-device performance on V100 and tuned DCU thread-block configurations.
5. Profiled multi-DCU strong and weak scaling and identified inter-node communication costs.

2026FJW

ChatGPT (July 2026)

Summary

Fu, Jiang, and Wang construct an oscillation-free energy-based discontinuous Galerkin method for second-order wave equations with nonsmooth solutions. Standard EDG fluxes and derivative-based damping cannot move piecewise-constant data whose jumps coincide with element interfaces, so the authors add a solution-jump penalty alongside projection-based adaptive damping. The penalty restores the correct wave propagation, while damping suppresses Gibbs-type oscillations; their mesh scaling preserves high-order behavior in smooth regions. Semi-discrete energy stability and a priori error estimates are proved in one and multiple dimensions for general numerical fluxes. Tests with smooth linear and semilinear waves attain optimal convergence for alternating and Sommerfeld fluxes, while nonsmooth linear, Dirac-source, Sine–Gordon, and Klein–Gordon problems show accurate wave speeds and non-oscillatory profiles. The analysis is restricted mainly to Cartesian meshes and periodic conditions, and nonlinear experiments without exact solutions use fine-grid finite differences as references.

Contributions

1. Extended energy-based discontinuous Galerkin methods to nonsmooth second-order wave solutions using complementary damping and jump-penalty terms.
2. Identified why derivative-only formulations can freeze discontinuous piecewise-constant initial data.
3. Established semi-discrete energy dissipation for general flux parameters in one and multiple dimensions.
4. Derived a priori energy-norm error estimates showing that stabilization retains high-order smooth-solution accuracy.
5. Demonstrated robustness on linear discontinuities, distributional sources, and nonlinear Sine–Gordon and Klein–Gordon equations in one and two dimensions.

2026FKO

ChatGPT (September 2026)

Summary

The authors compute finite-time optimal growth of small three-dimensional perturbations to an incompressible magnetohydrodynamic tanh shear layer with a uniform streamwise magnetic field, finite viscosity, and finite resistivity. Singular-value analysis of the linear propagator shows that magnetic tension can suppress Kelvin–Helmholtz eigenmodes while still allowing large transient energy amplification. In sufficiently long domains, the marginal case reaches gain above 10^4 , and even a strongly sub-Alfvénic case with $M_A = 0.1$ reaches about 10^3 . Besides the hydrodynamic Orr and lift-up mechanisms, magnetic shear production creates a magnetic analogue of lift-up: cross-stream field perturbations generate streamwise magnetic energy. Optimal perturbations at short times are inherently three-dimensional, while stable-flow optima at longer times approach streamwise-uniform structures with finite spanwise variation. After peak growth, kinetic and magnetic energy exchange through standing Alfvén waves. The results question field-strength bounds inferred solely from Kelvin–Helmholtz stability and show that two-dimensional simulations can omit dominant transient-growth pathways. Reynolds and magnetic Reynolds numbers are fixed at 250, so their scaling remains unresolved.

Contributions

1. Quantified finite-time optimal perturbation growth in a resistive, viscous MHD shear layer with a streamwise field.
2. Found energy gains above 10^4 despite Kelvin–Helmholtz stability and about 10^3 at $M_A = 0.1$.
3. Identified magnetic shear production as a magnetic lift-up mechanism analogous to the dynamo omega effect.
4. Showed that dominant transient perturbations can be three-dimensional or nearly streamwise uniform, outside standard two-dimensional analyses.
5. Connected the post-growth kinetic–magnetic energy exchange to standing Alfvén waves and the quarter-Alfvén-crossing-time optimum.

2026FMCH

ChatGPT (July 2026)

Summary

The paper introduces Coherent Structure Transport (CST), a reduced analytic model of scrape-off-layer blobs coupled to full-orbit deuterium-ion tracking in GEMX. A SOLPS-ITER solution supplies realistic axisymmetric electric and magnetic fields in DIII-D X-point geometry; time-dependent blob potentials are superimposed to study divertor ion heat-flux profiles without an expensive first-principles electron-turbulence calculation. Magnetic-only calculations recover the expected inverse-poloidal-field scaling of the heat-load width. Adding the SOLPS-ITER electric field substantially changes ion orbits and produces a broadened profile with a secondary peak outside the strike point. Blob turbulence further lowers the peak heat flux and broadens the profile approximately linearly with fluctuation amplitude; a few-percent normalized density fluctuation can roughly double the width. For a DIII-D wide-pedestal quiescent-H-mode case, the combined model predicts a 2.9 mm width consistent with experiment and XGC1. The fast model currently treats ions only and requires future sheath, electron, neutral-collision, and broader parameter-scan work.

Contributions

1. Defined the CST analytic blob-field model and coupled it to GEMX full-orbit ion tracking in realistic X-point geometry.
2. Integrated SOLPS-ITER equilibrium electric fields and showed that they alter ion drifts and generate a secondary divertor heat-flux peak.
3. Recovered the expected $\lambda_q \propto B_p^{-1}$ magnetic-only scaling and compared it with Eich and Goldston expectations.
4. Quantified heat-load broadening with blob amplitude and size, finding near-linear amplitude dependence but modest size dependence at fixed packing fraction.

5. Produced a fast DIII-D estimate, $\lambda_q = 2.9$ mm, comparable to experiment and XGC1 while identifying missing physics.

2026FMMM

ChatGPT (July 2026)

Summary

The paper develops a method-of-manufactured-solutions framework for three-dimensional particle-in-cell simulations with direct-simulation Monte Carlo binary elastic collisions. Rather than evolving manufactured particle weights, it manufactures separable phase-space distributions and uses inverse cumulative distributions to define exact particle positions and velocities, avoiding negative weights and interference with weight-dependent collision logic. Collision effects are verified by averaging independent stochastic outcomes and deriving an analytic velocity source from manufactured cross sections and angular anisotropy. Error models couple spatial and temporal discretization, finite-particle sampling, and collision statistics. Across uncoupled, one-way-coupled, and fully coupled tests, particle and potential errors recover second-order convergence with one run per discretization; isolated collision error behaves as $O(h^2)$ in the maximum norm and approximately $O(h^{5/2})$ in the L^2 norm. Complementary scattering-angle tests expose an injected collision error that trajectory metrics miss, while another injected error shows the reverse. Charged-particle Coulomb collision extensions are outside scope.

Contributions

1. Constructed exact manufactured particle trajectories by inverse sampling without changing weights.
2. Derived an analytic manufactured collision source from averaged stochastic outcomes.
3. Derived refinement rates for coupled discretization, sampling, and collision errors.
4. Recovered expected second-order convergence across multiple three-dimensional coupling cases.
5. Showed that trajectory and scattering-angle metrics detect complementary collision bugs.

2026FMNV

ChatGPT (July 2026)

Summary

Grid-based solvers for advection–diffusion equations suffer exponential growth in cost with dimension, while explicit Eulerian methods also face stability-limited timesteps. The proposed neural semi-Lagrangian method represents the spatial and parametric solution with a neural network and advances it sequentially in time. After fitting the initial condition, each step traces backward characteristics and optimizes a new network to match the transported previous solution; stochastic characteristic sampling extends the construction to diffusion. The method is mesh-free, highly parallel on GPUs, and has no timestep restriction for stability. Rough estimates separate characteristic-integration, neural approximation, optimization, and sampling errors, and the implementation adds natural-gradient preconditioning, adaptive sampling, and boundary treatments. Tests cover parametric transport, Vlasov dynamics, two- and three-dimensional level-set deformation, advection–diffusion through eight dimensions, and a Vlasov–Poisson proof of concept. Compared with PINNs, neural Galerkin methods, and classical semi-Lagrangian schemes, it maintains competitive accuracy at lower cost and becomes increasingly advantageous in high dimensions. Its main weakness is repeated nonconvex optimization: training becomes harder during long-time, fine-structure evolution, and the available analysis gives only rough rather than rigorous convergence guarantees.

Contributions

1. Replaced grid interpolation in semi-Lagrangian time stepping with projection onto a neural-network approximation space.
2. Extended the mesh-free construction from advection to diffusion through stochastic backward characteristics.
3. Derived an error decomposition covering characteristic, approximation, optimization, and sampling errors.

4. Demonstrated GPU-parallel solutions for parametric, kinetic, level-set, and eight-dimensional advection–diffusion problems.
5. Showed favorable high-dimensional accuracy and cost against PINN, neural Galerkin, and classical semi-Lagrangian baselines.

2026FPBT

ChatGPT (September 2026)

Summary

This study compares electromagnetic load models for mirrors in the ITER electron cyclotron emission diagnostic during a prescribed severe plasma disruption. The diagnostic’s copper mirrors experience eddy-current forces and torques that must be supplied to subsequent structural analysis. A simplified ANSYS Maxwell model uses six Helmholtz coil pairs to impose locally uniform magnetic fields and their time derivatives. The authors compare this approximation with an existing global APDL model and a new transient Maxwell model covering a 20° ITER sector with the Bay3 components. For the plasma-facing mirrors, global-model torques are approximately 40–50% of the simplified-model values because the latter combines conservative field-variation envelopes. Conversely, the global Maxwell model predicts peak net forces about five times larger than the simplified model. The study therefore recommends global transient modeling to capture spatial field variations and obtain loads consistent with the APDL reference. The comparisons concern one disruption scenario and electromagnetic loading; combined thermal, nuclear, seismic, and structural qualification remains subsequent work.

Contributions

1. Constructed a local Helmholtz-coil model that independently prescribes background magnetic fields and field-variation rates in three directions for ITER ECE load estimation.
2. Developed a transient 20° global Maxwell sector model incorporating the Bay3 mirrors for the same prescribed disruption used in the APDL reference.
3. Showed that conservative local field-variation envelopes overestimate plasma-facing mirror torques relative to both global modeling approaches.
4. Identified an opposing force discrepancy: global Maxwell peak forces are about five times the local-model predictions, so torque conservatism does not imply conservative force estimates.
5. Established a comparison-based workflow for supplying component electromagnetic loads to later structural assessments, favoring the global Maxwell model for the ECE system.

2026FPKA

ChatGPT (July 2026)

Summary

The paper develops particle-in-cell discretizations and discrete-gradient time integrators for both the Hamiltonian Vlasov-Poisson subsystem and dissipative Landau collisions, viewing their combination as a metriplectic system. Spatial bracket discretizations preserve the relevant degeneracies, while the time-discrete constructions conserve mass, momentum, and energy and maintain entropy-production monotonicity up to nonlinear-solver tolerances. PETSc implementations are tested on collisionless Landau damping and electron-positron thermal equilibration. In damping tests, discrete gradients match symplectic accuracy and outperform RK4 in preserving invariants, but implicit solves cost roughly 130% more per step than symplectic integration and 160% more than RK4. In equilibration, the integrator preserves momentum to $O(10^{-13})$, kinetic energy to $O(10^{-12})$, and monotone entropy until equilibrium-scale solver fluctuations. The work supplies an open-source, cross-verified structure-preserving Landau implementation while identifying nonlinear-solver efficiency as the main limitation.

Contributions

1. Placed Vlasov-Poisson and Landau particle discretizations in one metriplectic structure-preserving framework.
2. Derived a discrete-gradient Landau integrator that conserves moments and enforces entropy monotonicity.
3. Avoided Gonzalez-gradient singularities near equilibrium through an alternative discrete-gradient choice.
4. Implemented Hamiltonian and collisional integrators in PETSc with nonlinear-solver tolerance studies.
5. Quantified conservation, entropy, and runtime trade-offs on Landau damping and two-species equilibration.

2026FPPK

ChatGPT (July 2026)

Summary

Stellarator design usually separates plasma-equilibrium optimization from coil design, which can yield attractive plasmas that require impractical coils. This work turns QUADCOIL's winding-surface optimization into a differentiable coil-complexity proxy embedded inside equilibrium optimization. The resulting quasi-single-stage method retains only equilibrium degrees of freedom while representing filament coils, permanent magnets, or dipole arrays and supporting realistic quadratic objectives and constraints. The authors add adjoint differentiation, a JAX-based implementation, and a differentiable winding-surface generator that removes self-intersections while following the plasma boundary. Two studies validate the proxy after discarding the winding-surface solution and cold-starting higher-fidelity coil optimization. For a MUSE-like configuration, the optimized equilibrium needs about 34% fewer permanent magnets at comparable field accuracy. For an ARIES-CS-like configuration, subsequent filament-coil optimization reduces both peak and root-mean-square Lorentz forces by about 30% while preserving geometric constraints. The method provides an inexpensive bridge between conventional two-stage and full single-stage stellarator co-design.

Contributions

1. Formulated a flexible QUADCOIL-based quasi-single-stage proxy that embeds coil optimization inside equilibrium optimization without explicit coil degrees of freedom.
2. Derived an adjoint differentiation route and implemented end-to-end differentiation in JAX for constrained winding-surface subproblems.
3. Introduced a differentiable winding-surface generator that removes self-intersections and follows changing plasma boundaries.
4. Demonstrated a MUSE-like permanent-magnet design using 34.2% fewer magnets at matched magnetic-field error.
5. Demonstrated an ARIES-CS-like filament-coil design with roughly 30% lower peak and rms forces while retaining geometric constraints.

2026FRT

ChatGPT (July 2026)

Summary

This work proposes an asymptotic-preserving modified Crank–Nicolson particle pusher for two-dimensional Vlasov–Poisson dynamics under a strong, spatially varying magnetic field of fixed direction. Standard Crank–Nicolson variants lose consistency when the Larmor scale is unresolved. The new formulation augments particle position and velocity with kinetic energy and an effective magnetic-gradient force, allowing time steps independent of the small magnetization parameter while filtering fast gyromotion. Under boundedness assumptions, the authors prove convergence at fixed step size to a second-order discrete guiding-center system and second-order behavior of limiting energy and magnetic moment. Particle-in-cell tests for single trajectories, diocotron instability, and merging D-shaped vortices reproduce slow drift, confinement, energy exchange, and asymptotic reference densities over long times. The method is substantially more accurate than the compared IMEX scheme but costs more than twice as much in reported tests. Its analysis assumes

stability rather than proving it, positivity of the auxiliary energy is not guaranteed, large steps do not resolve the physical gyrofrequency, and extension to three-dimensional curvature and parallel motion remains open.

Contributions

1. Diagnosed the strong-field inconsistency of conventional Crank–Nicolson particle pushers.
2. Added kinetic energy and a magnetic-gradient force to restore uniform consistency.
3. Proved convergence to a second-order discrete guiding-center limit.
4. Filtered unresolved fast scales without a magnetization-dependent time-step restriction.
5. Validated long-time behavior in self-consistent particle-in-cell plasma tests.

2026FT

ChatGPT (July 2026)

Summary

This paper develops an analytical design procedure for tailoring plasma density downramps in laser wake-field acceleration. The model starts from an improved fully relativistic plane description and links the initial cold plasma density to the laser pulse profile. The goal is to control partial wave breaking, trigger electron self-injection at the downramp, and increase the acceleration of small injected bunches during the early acceleration phase. The analysis yields a multi-step optimization workflow rather than only a numerical scan. Particle-in-cell simulations using the same input data are then used as an independent check, and the paper reports excellent agreement between the analytical predictions and simulated behavior. The result is a reduced-model tool for designing downramp injection stages before running expensive kinetic simulations. Its scope is deliberately early-stage acceleration in an idealized geometry, so the paper is best read as a controlled density-profile optimization method rather than a full accelerator-system design.

Contributions

1. Built an analytical procedure for optimizing plasma density profiles for downramp injection.
2. Connected the density tailoring problem to a fully relativistic plane model of laser wake-field acceleration.
3. Targeted controlled partial wave breaking and self-injection of small electron bunches.
4. Compared the optimized profiles against particle-in-cell simulations with matched inputs.
5. Provided a design workflow that can reduce reliance on brute-force kinetic parameter scans.

2026FUSJ

ChatGPT (September 2026)

Summary

GENE-X is extended from axisymmetric tokamaks to non-axisymmetric stellarators while retaining its full- f gyrokinetic model, which evolves the complete particle distribution. The flux-coordinate independent approach uses local field-line tracing between poloidal grids and remains usable across magnetic islands and the scrape-off layer, where field lines intersect material boundaries. Analytic Dommaschk vacuum fields, plane-dependent geometry and meshes, and masked array operations remove axisymmetry assumptions in GENE-X and its shared PARALLAX infrastructure. Manufactured-solution tests recover second-order convergence of the distribution function and electromagnetic fields in three-dimensional geometry. A source-free, low-pressure simulation covers the magnetic axis through a non-segmented island divertor. It develops large edge fluctuations, gradient-localized turbulent transport, and a radial electric-field well. Comparable perpendicular ion and electron heat input to the scrape-off layer, combined with weaker parallel ion losses, explains ion-temperature accumulation there. This is a proof of capability rather than experimental validation: the calculation uses vacuum geometry, simplified Dirichlet wall conditions, one stellarator field period, and long-wavelength and small-field-pitch approximations.

Contributions

1. Enabled global Eulerian gyrokinetic simulations spanning the stellarator core, edge, and island-divertor scrape-off layer within one tokamak/stellarator codebase.
2. Implemented and verified analytic three-dimensional vacuum magnetic fields and an approximate flux-surface label for mesh construction, initialization, and diagnostics.
3. Generalized PARALLAX and GENE-X data structures to plane-dependent meshes, field-line maps, and elliptic solvers, with masks distinguishing active, boundary, and filler points.
4. Verified second-order convergence in three-dimensional geometry by refining space, velocity space, and time in manufactured-solution tests, while retaining axisymmetric verification results.
5. Demonstrated self-consistent edge turbulence and related scrape-off-layer ion heating to the imbalance between cross-field heat supply and species-dependent parallel exhaust.

2026FW

Claude (September 2026)

Summary

High-order WENO and Hermite WENO (HWENO) shock-capturing schemes spend much of their cost on wide stencils, stagewise nonlinear weights, and characteristic decompositions. The authors propose finite-average-moment (FAM) schemes that evolve only the P^1 moment state, cell averages and scaled first moments, with a DG-style weak form and SSP-RK3, and obtain orders $k = 3, 4, 5, 6$ from fixed linear, polynomially exact reconstructions on compact stencils: three cells in one dimension and a 3×3 patch on uniform Cartesian grids in two. Nonlinear stabilization is a single oscillation-elimination (OE) step after the final Runge–Kutta stage, an exact exponential damping of the first moments driven by interface jumps of the reconstruction, which preserves cell averages and changes smooth solutions only by $\mathcal{O}(\Delta x^{k+1})$. For linear advection without OE, Fourier analysis gives a k th-order physical mode, a strongly damped spurious mode, and $(k + 1)$ th-order cell-average superconvergence. Smooth Burgers and Euler tests confirm these orders, and Buckley–Leverett and one- and two-dimensional Euler shock problems, reconstructed componentwise in conservative variables, show no visible oscillations; strong-shock tests need a positivity-preserving limiter. FAM5 and FAM6 run faster than MR-WENO5, HWENO-U, and OE-HWENO, with time ratios down to 0.10, whereas FAM3 is slower than MR-WENO3. No nonlinear stability theory is given.

Contributions

1. Decoupled formal order from the number of evolved unknowns: two per component per cell in one dimension and three in two, with order supplied by paired auxiliary polynomials, including incomplete quartic and sixth-degree ones in two dimensions.
2. Proved unique solvability and polynomial exactness of the compact reconstruction systems, and cell-average preservation, nonexpansive damping, and accuracy preservation of the one- and two-dimensional OE steps.
3. Derived sampled admissible CFL numbers 0.40, 0.44, 0.58, and 0.56 for $k = 3, 4, 5, 6$ from the SSP-RK3 amplification matrix of the linear backbone.
4. Introduced a Type II damping coefficient normalized by an unnormalized local sum, which reduces undershoots and preserves small structures relative to the globally normalized Type I in a multiscale Lax problem.
5. Attributed 77–85% of HWENO-U and OE-HWENO runtimes to stagewise reconstruction and showed that local characteristic variables increased FAM5 runtime by factors of 1.25 and 1.80 without visible change in the displayed densities.

Comments

- Abstract: “the FAM schemes deliver significantly lower complete-run wall-clock times compared to” MR-WENO, HWENO-U, and OE-HWENO, but Table 8 gives FAM3 versus MR-WENO3 time ratios of 1.42 (Sedov) and 1.61 (LeBlanc), a matched-order pair that §4 lists among the emphasized comparisons; the claim holds only for the FAM5 and FAM6 comparisons.

- Fig. 2: the caption describes the “ L^1 error of the density cell averages versus wall-clock time for the 1D smooth Euler test”, but the plotted error ranges match the Burgers errors of Table 5, not the Euler errors of Table 6; MR-WENO3 runs from about 10^{-3} to $10^{-5.3}$, as in Table 5 (9.10×10^{-4} to 4.85×10^{-6}) rather than Table 6 (4.63×10^{-3} to 1.12×10^{-4}), and FAM6 starts near $10^{-5.8}$ (Table 5: 1.64×10^{-6} ; Table 6: 9.84×10^{-9}), so the error axes repeat Fig. 1; the intended Euler data are unresolved.
- Table 14, $k = 6$: Appendix L defines the FAM6 reconstruction as $\sum_{\ell=0}^{20} c_{\ell} \phi_{i,j}^{(\ell)}$, discarding c_{21} and c_{22} of the auxiliary polynomial Q_6 , and (2.12) takes jumps of the reconstruction, but the printed $\mathbf{A}_6$, $\mathbf{B}_6$, and $\mathbf{C}_6$ equal the face-center jumps of the unprojected Q_6 computed from the printed coefficients $c_0, \dots, c_{22}$, whereas the rows $k = 3, 4, 5$ correspond to the projected polynomials; which jumps the computations use is unresolved.

2026FZHW

ChatGPT (July 2026)

Summary

This work extends the cell-based ECOGEN adaptive-mesh framework into an open-source C++ solver for multiscale supersonic reacting flows. It solves the compressible, multicomponent reactive Navier–Stokes equations using thermally perfect species, mixture-averaged transport, and detailed finite-rate chemistry supplied through Cantera. A MUSCL finite-volume discretization reconstructs interface states; a multicomponent HLLC Riemann solver with low-Mach correction improves inviscid-flux accuracy for detonations, while a skewness-corrected central scheme handles viscous fluxes. Hydrodynamics advances with second-order Runge–Kutta integration, and operator-split chemical source terms use Cantera solvers of up to sixth order. To concentrate resolution on physically important structures, the authors design a three-stage AMR strategy that separately addresses inert-flow features, flame fronts, and coupled shock–flame regions, including secondary shocks missed by simple gradient criteria. They also derive a theoretical AMR-to-uniform-grid speedup ratio from advancement costs and show agreement with measured speedups. Validation spans one-, two-, and three-dimensional convection, diffusion, planar flames, inert and reacting shock–bubble interactions, planar detonations, and cellular detonation structures. The result is a validated GPLv3 tool for studying shock-induced ignition, Richtmyer–Meshkov dynamics, supersonic mixing, and combustion across widely separated length scales.

Contributions

1. Extends ECOGEN into a GPLv3 solver for compressible multicomponent reacting flows with detailed thermochemistry and transport through Cantera.
2. Introduces a multicomponent HLLC solver with low-Mach correction that improves detonation simulations over the conventional HLLC formulation.
3. Develops a three-stage AMR strategy specialized for inert-flow structures, flame fronts, and coupled shock–flame interactions.
4. Derives an AMR speedup model from advancement costs and validates its theoretical predictions against measured speedups.
5. Validates accuracy and efficiency across inert and reacting one-, two-, and three-dimensional benchmarks, including shock–bubble interactions and cellular detonations.

2026G

ChatGPT (July 2026)

Summary

This paper presents bsort, an in-place, non-comparison sorting algorithm for unsigned integers, two’s-complement signed integers, and IEEE-754 floating-point values. It extends binary quicksort’s bitwise partitioning with sign-aware direction changes. Floating-point values are ordered hierarchically by sign, exponent, and mantissa; negative partitions reverse the required directions. The paper specifies positions for infinities, signed zeros, and NaNs, placing NaNs at the array extremes according to sign and mantissa payload and ordering -0 before $+0$. Correctness is proved for the integer

and floating-point procedures. For n elements of word width w , bsort takes $O(wn)$ time and $O(w)$ recursion space, so fixed machine widths give linear scaling in n . C and C++ benchmarks show competitiveness with standard-library hybrid sorts for small element widths, but performance degrades relative to them as word width grows and depends on data distribution.

Contributions

1. Unified in-place bitwise sorting for unsigned integers, signed integers, and floating-point values.
2. Derived sign-aware partition rules and a sign–exponent–mantissa hierarchy for IEEE-754 data.
3. Defined deterministic placement of infinities, signed zeros, and signed NaN payloads.
4. Proved $O(wn)$ running time and $O(w)$ auxiliary space for w -bit elements.
5. Benchmarked C and C++ implementations against optimized standard-library sorting routines across types and input distributions.

2026G2

ChatGPT (July 2026)

Summary

This dissertation studies how state-space computations, including nonlinear recurrent networks, sampling chains, and explicit dynamical solvers, can be parallelized across sequence length. Their recurrences are recast as nonlinear systems and solved with Newton-like iterations whose structured linear subproblems admit parallel associative scans. Quasi-Newton variants reduce state-dimension cost and memory, while trust-region variants use parallel Kalman filtering to improve robustness when undamped iterations fail. The theoretical analysis unifies Newton, Picard, Jacobi, and related fixed-point methods through their Jacobian approximations and derives linear rates governed by approximation error and stability. For nonlinear state-space models, dynamical predictability connects the merit function’s conditioning to the largest Lyapunov exponent: stable dynamics can converge in $O(\log T)$ iterations, while unstable dynamics may remain effectively sequential. Experiments on sequence models and sampling problems demonstrate speed, memory, and stability tradeoffs. The results provide both algorithms and a diagnostic for when time parallelism is worthwhile.

Contributions

1. Developed scan-based quasi-Newton methods with improved state-dimension scaling and memory use.
2. Constructed trust-region parallel Newton methods using parallel Kalman filtering.
3. Unified Newton, Picard, Jacobi, and other fixed-point iterations through Jacobian approximation.
4. Derived convergence rates linking optimization conditioning to dynamical stability.
5. Identified the largest Lyapunov exponent as a criterion for profitable sequence parallelization.

2026G3

Claude (September 2026)

Summary

The author designs a fully spectral discretization of the linearized Bhatnagar–Gross–Krook (BGK) equation $\partial_t f + v \partial_x f - \partial_x \phi \partial_v f = \mathcal{L}(f)$ on $\mathbb{R}_x \times \mathbb{R}_v$, with a confining even polynomial potential ϕ and a collision operator conserving mass, momentum, and energy. The perturbation of equilibrium is expanded in Hermite polynomials in velocity and in polynomials orthonormal with respect to $\rho = e^{-\phi}$ in space, truncated at degrees K and N , and advanced by implicit Euler, without domain truncation or artificial boundary conditions. For $N \geq \deg \phi$, the semi-discrete and fully discrete schemes conserve mass and total energy and, for the harmonic potential, keep zero the four further moments associated with oscillatory equilibria. Adapting the continuous entropy method of Carrapatoso et al., the author proves the semi-discrete scheme hypocoercive: although collisions damp only Hermite modes of degree three and higher, solutions decay exponentially in the weighted L^2 norm, with constants depending on N and ϕ . Experiments show N -independent decay rates, with a fast-then-slow relaxation for a double-well potential. The author notes that the constants could

degenerate as N grows unless certain projected operator norms stay bounded, which holds for the harmonic potential and is conjectured in general, and leaves open the closure choice and the two-phase relaxation.

Contributions

1. Applied the projection $\Pi_{\mathbb{P}_N}$ in the Hermite-moment hierarchy $\partial_t C_k = \sqrt{k+1} \partial_x^* C_{k+1} - \sqrt{k} \partial_x C_{k-1} - \delta_{k \geq 3} C_k$ only after the weighted adjoint $\partial_x^* = -\partial_x + \partial_x \phi$, because polynomials are stable under ∂_x but not under multiplication by $\partial_x \phi$, and evaluated the entries $\int \tilde{P}_r' \tilde{P}_n \rho dx$, which vanish for $r \leq n$, as $\int \tilde{P}_r \tilde{P}_n \partial_x \phi \rho dx$ from the recurrence, in a sparse block-tridiagonal iteration matrix.
2. Built an entropy $\mathcal{H}_1$ adding to $\|\tilde{h}\|^2$ four cross terms with Ω^{-1} , $\Omega = \partial_x^* \partial_x + 1$, weighted by $\varepsilon, \varepsilon^{3/2}, \varepsilon^{7/4}, \varepsilon^{15/8}$, which suffices for the harmonic potential, and $\mathcal{H}_2 = \mathcal{H}_1 - \varepsilon^{62/32} (b'b + c'c'')$ otherwise, with $b = \langle \tilde{C}_1 \rangle$ and $c = \langle \tilde{C}_2 \rangle$ the spatial means of the momentum and kinetic-energy moments, and with N entering only through a constant K_N bounding four projected operators on $\mathbb{P}_N$.
3. Transferred the decay to the implicit Euler iterates on the invariant-free subspace through a Lyapunov norm, $\|u_i\|_P^2 \leq (1 + 2\alpha\Delta t)^{-i} \|u_0\|_P^2$, which tends to $e^{-2\alpha T} \|u_0\|_P^2$ at fixed $T = I\Delta t$ as $I \rightarrow \infty$, with α possibly depending on N .
4. Computed the three-term recurrence coefficients of the orthonormal polynomials by iterating Freud's equation, which is ill-conditioned and diverges at large index, up to $K_0 = 10$ and by the Deift et al. asymptotic expansion beyond, after showing on three quartic potentials that the expansion is inaccurate at small index for the non-convex $x^4 - 2x^2$.
5. Reported, with $K = 20$ and $N = 15$ or 30 , log-scale decay slopes of -0.14 for the harmonic potential and of -0.29 for $t \leq 20$ then -0.05 for the normalized double well $(x-1)^2(x+1)^2$, and harmonic-case norms $\|\tilde{C}_0\|$ and $\|\tilde{C}_1\|$ oscillating with a common period shifted by half a period.

Comments

- Section 1.1: the oscillatory modes $f_+ = (x \cos t + v \sin t)\mathcal{M}$ and $f_- = (v \cos t - x \sin t)\mathcal{M}$ do not satisfy the first equation of (1.3) for the harmonic potential ($\partial_x \phi = x$), which leaves $2(v \cos t - x \sin t)\mathcal{M}$ and $-2(x \cos t + v \sin t)\mathcal{M}$, and f_+ gives $\frac{d}{dt} \langle rx \rangle = -\langle m \rangle$, contrary to $\frac{d}{dt} \langle rx \rangle = \langle m \rangle$ in the proof of Proposition 2.1; $g_{\pm}$ do satisfy (1.3), so the evident intended modes are $(x \cos t - v \sin t)\mathcal{M}$ and $(v \cos t + x \sin t)\mathcal{M}$.
- Section 2.2, proof of Proposition 2.1: part 2 states “It was already proven above that” $\frac{d}{dt} \langle r\phi \rangle = -\langle m \partial_x \phi \rangle$, but part 1 derives $\frac{d}{dt} \langle r\phi \rangle = \langle m \partial_x \phi \rangle$, and only the latter sign gives the next line $\frac{d}{dt} (\frac{1}{\sqrt{2}} \langle e \rangle - \langle r\phi \rangle) = -2\langle mx \rangle$; the recalled sign would give zero.
- Section 3.1, Eqs. (3.2) and (3.5): the initial data are written $\tilde{C}_{k,n}(0) = \int h(0, x, v) \tilde{P}_n(x) \tilde{H}_n(v) \mathcal{M} dx dv$ and $\tilde{C}_k(0, x) = \Pi_{\mathbb{P}_N} \int h(0, x, v) \mu(v) dv$, so neither depends on k , contrary to $C_{k,n} = \langle C_k, \tilde{P}_n \rangle$ with C_k the Hermite coefficients of (2.1); the evident intended weight is $\tilde{H}_k(v)$ in both.
- Section 4.1, Eq. (4.1): with $\alpha_{r,n} := \int \partial_x P_r P_n \rho dx$, the coefficients $\langle \tilde{P}_r, \partial_x \tilde{P}_n \rangle$ and $\langle \tilde{P}_n, \partial_x \tilde{P}_r \rangle$ of (3.2) are $\alpha_{n,r}$ and $\alpha_{r,n}$, but (4.1), stated as inherited from (3.2) and equivalent to (4.2), writes $\sum_r \alpha_{r,n} \tilde{C}_{k+1,n}^{i+1}$ and $\sum_r \alpha_{n,r} \tilde{C}_{k-1,r}^{i+1}$; the evident intended terms are $\sum_r \alpha_{n,r} \tilde{C}_{k+1,r}^{i+1}$ and $\sum_r \alpha_{r,n} \tilde{C}_{k-1,r}^{i+1}$.
- Section 4.3: the reduced system is written $\frac{d}{dt} u = Au$ with A “ α -coercive” in $\|\cdot\|_P$, and its implicit Euler step as $u_{i+1} + \Delta t A u_{i+1} = u_i$; implicit Euler for $\frac{d}{dt} u = Au$ is $u_{i+1} - \Delta t A u_{i+1} = u_i$, and a coercive A would make $\frac{d}{dt} u = Au$ grow rather than be exponentially stable, so the evident intended system is $\frac{d}{dt} u = -Au$.
- Section 5.2, Eq. (5.27) and Lemma 5.6: z is defined with the term $-\sqrt{2}c(t)b\xi_{\phi}(x)$, and Lemma 5.6 inverts it as $\tilde{r} = -b'x + c''\xi_2/(2\sqrt{2}) + \sqrt{2}c(t) + z$, whereas (5.28), the proof of Lemma 5.8, and Lemma 5.11 ($\tilde{C}_0 = \sqrt{2}\xi_{\phi}c + S$) use $z = \tilde{r} + b'x - c''\xi_2/(2\sqrt{2}) - \sqrt{2}c\xi_{\phi}$, which is evidently intended.
- Section 5.2, proof of Lemma 5.9: it starts by “multiplying Equation (5.23) by $\partial_x \rho$ ” and later injects the expression for b “into (5.24)”, but (5.23) and (5.24) are inequalities in the proof of Lemma 5.5; the computations use the master equation (5.29) and its consequence (5.30), from which (5.31) follows.
- Appendix A: ϕ is expanded as $\phi(x) = \sum_{i=0}^m \gamma_i x^{2i}$ with leading coefficient γ_m , as also used in the proof of Lemma 5.9, so γ_2 multiplies x^4 for a quartic, but the quartic evaluation of (A.2) and Freud's equation (A.3), $k+1 = 2\gamma_2 a_k^2 + 4\gamma_4 a_k^2 (a_{k-1}^2 + a_k^2 + a_{k+1}^2)$, use γ_2 and γ_4 as the coefficients of x^2 and x^4 , the reading under

which (A.3) follows from (A.1).

2026GAS

ChatGPT (July 2026)

Summary

This work extends kinetic-based regularization (KBR), a localized one-parameter kernel regression method, to spatial-derivative estimation. An improved correction makes the local quadratic fit second-order accurate away from training points. An explicit scheme extracts derivatives from closed-form fitted coefficients; an implicit scheme perturbs a query point and solves a small local system. On one-dimensional Camel and Rastrigin functions, explicit extraction is more stable for clean unseen data, while implicit extraction is more robust to noise. Clean-data tests outperform a differential-neural-network baseline and compare favorably with nonuniform finite differences. A preliminary two-dimensional example recovers gradients and Hessians. KBR is also inserted into conservative solvers through interface-flux prediction for inviscid Burgers and Sod shock-tube problems. MacCormack- and Roe-based variants remain stable and preserve conservative structure, although the Burgers solution exhibits Gibbs oscillations and irregular high-dimensional point clouds remain future work.

Contributions

1. Corrected KBR's local quadratic reconstruction to attain the intended accuracy at unseen query points.
2. Derived explicit and implicit procedures for extracting gradients and second derivatives.
3. Characterized complementary behavior: stronger clean-data stability for the explicit scheme and greater noise robustness for the implicit scheme.
4. Demonstrated preliminary two-dimensional gradient and Hessian recovery.
5. Integrated KBR interface-flux predictions into conservative MacCormack and Roe shock-capturing solvers.

2026GCSB

Claude (September 2026)

Summary

The authors couple the full-wave solver AORSA, which computes a whistler eigenmode at 200 MHz and toroidal mode number $n = 35$ in a DIII-D equilibrium, with the full-orbit code KORC to study pitch-angle scattering of runaway electrons (REs) in 3D tokamak geometry beyond quasilinear diffusion. The coupling is one-way: 10,240 REs with energies uniform in 1–20 MeV and initial pitch angle 10° first relax without waves for 1 ms and then evolve for 2 ms in the prescribed fields, with no damping mechanism included. After the waves are switched on, the mean and variance of the pitch-angle distribution increase monotonically, skewness and kurtosis rise transiently, a high-pitch-angle tail develops, and the number of confined REs decays linearly. Pitch-angle increments over one wave period have Gaussian cores and Laplace (exponential) tails. The variance of pitch-angle displacements scales as t^α with α depending on initial energy: sub-diffusive for 1–3 MeV, diffusive only in a few bins, and super-diffusive for most energies between 4 and 17 MeV. The authors conclude that the dynamics are intermittent and not captured by quasilinear theory over these time scales, and caution that the finite simulation time may not reveal the asymptotic scaling.

Contributions

1. Coupled AORSA full-wave whistler fields with KORC full-orbit tracking, capturing eigenmodes with many coupled poloidal harmonics and a range of local parallel wavenumbers that slab and single-mode resonance models omit.
2. Measured Laplace tail scales of 1 and 0.65 for positive and negative pitch-angle kicks at $1\text{--}5^\circ$ and $8\text{--}10$ MeV, and a symmetric scale of 0.75 at $6\text{--}16^\circ$ and $8\text{--}9$ MeV, as input for kick-type reduced transport models.
3. Resolved the transport exponent both in broad initial-energy groups and in 1 MeV bins, the latter alternating between diffusive and super-diffusive behavior above 12 MeV.

4. Found that 1–5 MeV REs roughly double their mean kinetic energy, 10–15 MeV REs lose energy on average, and 15–20 MeV REs keep a nearly constant mean energy, with a power-law positive tail (exponent about 1.39) in normalized energy changes.
5. Argued that the Laplace statistics and anomalous variance scaling contradict the small, uncorrelated kicks assumed by quasilinear theory, and linked the pitch-angle increase to possible RE mitigation through enhanced synchrotron losses.

Comments

- Section IV, Eq. (5): the kurtosis is written $\gamma_{2,\eta} = \langle (\eta - \mu_\eta)^4 - 3 \rangle / \sigma_\eta^4$, which subtracts a pure number inside an average of a quantity in degrees⁴; the text calls it the normalized fourth central moment, and the values near 3 in Fig. 2(d) before the waves are switched on indicate $\langle (\eta - \mu_\eta)^4 \rangle / \sigma_\eta^4$.
- Section IV, Fig. 4: the text reports “a prompt loss of about 700 particles”, but Fig. 4 drops from 10,240 to about 9,350 confined particles at the start, a loss of about 900.
- Section IV, Fig. 5(b): the text assigns $\lambda = 1$ to the positive tail and $\lambda = 0.65$ to the negative tail, but the legend labels the blue dashed line on the positive tail “ $\lambda=0.65$ ” and gives “ $\lambda=1.0$ ” a red dashed style shared with the Gaussian; the slope of the positive-tail line corresponds to $\lambda \approx 1$, so the legend labels appear swapped.
- Fig. 5 caption: panel (b) is described as the distribution “on log-log scale”, but it plots $\log_{10}$ of the PDF against a linear $\Delta\eta$ axis that includes negative values, i.e. on a semi-log scale.
- Section VI, Fig. 9: the text lists four groups, (a) 1–5, (b) 5–10, (c) 10–15, and (d) 15–20 MeV, but Fig. 9 labels only (a) 1–5, (b) 10–15, and (c) 15–20 MeV; its fourth, unlabelled plot starts at 7.5 MeV and is evidently the 5–10 MeV group.
- Section VI: “those with $K_0 > 10$ MeV exhibit modest energy gain of approximately 1 MeV” conflicts with the next sentences, which state that the 10–15 MeV group loses energy and the 15–20 MeV group stays nearly constant, as Figs. 9(b) and 9(c) show; the unlabelled 5–10 MeV plot rises by about 1 MeV, so the evident intended reading is $K_0 = 5$ –10 MeV.

2026GCYZ

ChatGPT (July 2026)

Summary

This paper develops a second-order compact exponential time-differencing scheme, cETD2, for three-dimensional phase-field models. The method integrates the stiff linear differential operators exactly and approximates nonlinear local-free-energy and elastic-interaction terms by polynomials. A carefully chosen linear splitting controls those nonlinearities and, under explicit parameter conditions, preserves the original discrete free-energy decay rather than a modified energy. The authors prove the temporal convergence orders and energy stability of the first- and second-order cETD schemes. Their parallel software uses compact central differences, tensor contractions implemented with DGEMM, MPI data exchange, and configurable modules for Allen–Cahn, Cahn–Hilliard, anisotropic interfacial energy, elastic strain energy, and boundary conditions. Numerical tests confirm the expected temporal orders and energy decay. Large three-dimensional titanium $\beta \rightarrow \omega$ and $\beta \rightarrow \alpha$ transformation simulations, including a 2048^3 grid, demonstrate long-time microstructure evolution with anisotropic interfaces and elastic effects.

Contributions

1. Constructed first- and second-order compact exponential time-differencing schemes for general phase-field systems.
2. Derived splitting conditions that preserve the original discrete energy-dissipation law.
3. Proved temporal convergence and error estimates for cETD1 and cETD2.
4. Implemented modular parallel software covering Allen–Cahn, Cahn–Hilliard, anisotropy, elasticity, and multiple boundary treatments.
5. Verified the method on convergent, energy-decaying titanium phase-transformation simulations up to 2048^3 grid points.

2026GDS

ChatGPT (July 2026)

Summary

This study develops nonlinear theory and particle-in-cell simulations for plasma wakes driven by Compton scattering of an incoherent high-frequency photon burst, rather than by a coherent laser's ponderomotive force. The analysis extends earlier linear theory to asymmetric and nonresonant drivers and derives a relativistic fluid description in which sufficiently energetic bursts approach the cold wave-breaking field. A perfectly collimated photon driver propagates at c , enabling trapped electrons to remain phase-locked; practical acceleration is then limited by photon depletion. Angular divergence lowers the driver's ensemble speed and therefore the wake phase velocity, restoring a dephasing limit. One- and two-dimensional OSIRIS simulations validate the predicted wake amplitudes, relativistic wavelength increase, trapping, depletion scaling, and divergence dependence. Finite-width drivers create transverse structures unlike laser wakefields: stochastic Compton momentum transfer produces a persistent current and DC magnetic field that focuses relativistic electrons throughout the wake. Current laboratory photon sources cannot reach the nonlinear regime, whereas intense compact-object radiation propagating through tenuous astrophysical plasma may do so.

Contributions

1. Extended linear Compton-wake theory to asymmetric and long nonresonant photon bursts.
2. Derived and validated a nonlinear relativistic fluid model reaching the wave-breaking regime.
3. Established electron trapping and phase-locking conditions and depletion- and dephasing-limited energy scalings.
4. Revealed a persistent DC magnetic component that provides transverse focusing in two dimensions.
5. Assessed laboratory and astrophysical regimes for nonlinear Compton-driven wake acceleration.

2026GGT

ChatGPT (September 2026)

Summary

PACO is a parallel fast Fourier transform that simultaneously targets cache-efficient local computation and minimal global data movement. It performs a local FFT, one global permutation, and a second local FFT. Recursive local stages partition transform and batch dimensions without knowing cache parameters. Rather than materializing each transpose-like layout induced by the recursion, PACO defers these layouts, composes them into a base- b digit reversal, and fuses that permutation with the ownership redistribution already required between transform dimensions. Under an exact base- b slab decomposition in a hybrid ideal-cache/Bulk Synchronous Parallel model, an N -point transform on p processors has optimal maximum work $\Theta((N/p) \log N)$, optimal cache complexity $\Theta((N/(pB))(1 + \log_M N))$, and exactly one global redistribution. Every source-destination pair transfers N/p^2 elements, and the output uses factor-swapped slab ownership. The analysis assumes bounded power-of-two radix, exact divisibility, homogeneous parameters, no replication, and one-dimensional slab distributions.

Contributions

1. Designed a fully cache-oblivious distributed FFT with a single global redistribution round.
2. Represented recursive layout changes lazily and composed them into one digit-reversal permutation.
3. Fused local-layout permutation with the unavoidable global ownership change.
4. Proved optimal per-processor work and ideal-cache complexity under the stated decomposition model.
5. Established balanced pairwise communication and optimal migration volume for the prescribed ownership contract.

2026GHDL

ChatGPT (September 2026)

Summary

This paper constructs a second-order solver for the six-dimensional Vlasov–Maxwell system by combining Active Flux, operator splitting, and a finite-difference time-domain Maxwell update. Active Flux augments finite-volume cell averages with interface point values, using their compact local evolution to obtain high-order fluxes. Splitting the phase-space transport into one-dimensional advections keeps the stencil and implementation manageable despite the six-dimensional distribution function. The resulting method conserves particle number through its flux formulation and couples charge and current moments to electromagnetic fields. Against the semi-Lagrangian Positive and Flux-Conservative method, the compact Active Flux stencil introduces markedly less numerical dissipation and reduced directional anisotropy. Standard kinetic tests, including Landau damping, two-stream instability, and electromagnetic plasma phenomena, reproduce benchmark behavior at comparable or lower resolution while reducing computational cost. The present discretization does not explicitly enforce higher-moment conservation or positivity in every regime, which the authors identify as targets for further development.

Contributions

1. Formulated an Active Flux discretization for fully six-dimensional collisionless Vlasov transport.
2. Used operator splitting to reduce the phase-space update to efficient one-dimensional advection solves.
3. Coupled the kinetic update to finite-difference time-domain Maxwell equations.
4. Showed lower dissipation and anisotropy than a semi-Lagrangian positive flux-conservative benchmark.
5. Validated important electrostatic and electromagnetic kinetic phenomena at reduced computational cost.

2026GHJ

ChatGPT (July 2026)

Summary

BEACONS treats neural PDE surrogates as numerical methods that should carry explicit error guarantees. For homogeneous first-order hyperbolic conservation laws, the method of characteristics predicts solution smoothness and shock formation; combined with shallow-network approximation theory, this yields worst-case L^∞ bounds that extend beyond the training interval. To handle discontinuities, the framework decomposes a solution into simpler functions and composes separately trained shallow networks. Its key inequality bounds composition error by $e_f + Le_g$, allowing a smooth outer function with Lipschitz constant L to suppress error from a discontinuous inner approximation. A Racket domain-specific language specifies PDEs and architectures, generates C code for training and inference, and produces executable proof certificates through IEEE-754-aware symbolic rewriting. Deterministic tests train on the first third of simulations and extrapolate through the remainder for one- and two-dimensional advection, inviscid Burgers, and compressible Euler problems. Against equal-sized fully connected networks, BEACONS generally better preserves waves and conserved quantities, has lower L^2 and L^∞ errors, and stays within its certified worst-case bounds. The evidence is limited to selected benchmarks; moreover, Euler certificates are conditional because their reference solver is not formally verified, unlike the advection and Burgers solvers.

Contributions

1. Combined characteristic analysis with neural approximation theory to derive extrapolatory worst-case L^∞ bounds for hyperbolic PDE solutions.
2. Established an algebraic composition rule that propagates layerwise approximation errors and motivates bounded-error deep architectures for discontinuous solutions.
3. Implemented a Racket-based specification, C code-generation, training, and validation pipeline for BEACONS solvers.

4. Generated machine-checkable error certificates using a terminating symbolic rewrite system designed to respect IEEE-754 floating-point algebra.
5. Demonstrated improved extrapolation, conservation, and error behavior over equal-sized fully connected networks on one- and two-dimensional advection, Burgers, and Euler benchmarks.

2026GHT

ChatGPT (July 2026)

Summary

The paper develops third-order bound-preserving finite-volume methods for one-dimensional hyperbolic conservation laws on adaptive moving meshes. A high-order update is decomposed into conservative subcell updates; whenever a high-order subcell state leaves the admissible set, it is blended with a bound-preserving first-order counterpart. This produces an inexpensive flux limiter whose admissibility restriction depends only on the first-order subcell schemes. The authors also preserve the discrete geometric conservation law and prove that the limiter retains third-order spatial accuracy under a mild local CFL bound of $1/6$. The framework is extended from scalar laws to the compressible Euler equations and a nonconservative five-equation two-medium-flow model. Advection, Burgers, extreme Euler, and gas–liquid shock-tube tests show exact uniform-flow preservation, suppression of nonphysical states, and sharper resolution than comparable uniform meshes.

Contributions

1. Derived a conservative subcell decomposition for high-order finite-volume updates on time-dependent moving meshes.
2. Converted admissibility enforcement into an inexpensive flux blend between high- and first-order subcell schemes.
3. Established bound-preserving CFL conditions that depend solely on the first-order subcell operators.
4. Proved third-order accuracy preservation under the local CFL restriction $\lambda_j \alpha \leq 1/6$ while satisfying the discrete geometric conservation law.
5. Extended and validated the construction for Euler flow and the five-equation gas–liquid model, including an extreme case in which the unlimited method fails.

2026GILO

ChatGPT (July 2026)

Summary

Summation-by-parts identities provide discrete conservation and energy estimates, but this paper shows that satisfying the formal SBP definition does not ensure an accurate differentiation operator. Counterexamples for linear advection exhibit conservative, energy-stable function-space SBP operators with severe errors and poor convergence because unresolved modes remain in the differentiation matrix’s nullspace. The authors make nullspace consistency and approximation outside the exactness space explicit design requirements. They then extend an optimization-based construction of function-space SBP operators in two ways. A prescribed sparse block or band structure reduces the differentiation-matrix optimization from $O(N^2)$ to $O(Nb)$ unknowns and can restore useful nullspace and error behavior for global operators. For local operators, regularization minimizes derivative errors on an augmented function set when exactness on the larger space is impossible. Linear-advection and manufactured two-dimensional Euler tests show sparse operators outperforming dense counterparts and regularized operators outperforming unregularized ones. The framework applies to arbitrary one-dimensional nodes and finite-dimensional function spaces, with higher-dimensional use through tensor products.

Contributions

1. Demonstrated that algebraic SBP exactness, positive norm, conservation, and energy stability can coexist with poor approximation and convergence.

2. Identified unresolved nullspace modes and failure of nullspace consistency as the central structural cause of inaccurate solutions.
3. Formulated augmented-basis derivative-error minimization as an additional accuracy criterion beyond formal SBP exactness.
4. Introduced a sparse parameterization that reduces optimization unknowns from $O(N^2)$ to $O(Nb)$ and makes global function-space SBP operators more practical.
5. Validated sparse and regularized constructions on one-dimensional advection and manufactured two-dimensional compressible Euler problems.

2026GJZM

ChatGPT (July 2026)

Summary

The authors construct a linear, second-order scheme for three-dimensional incompressible Hall magnetohydrodynamics on $\mathbb{R}^3$, avoiding artificial domain truncation. Mapped Gegenbauer functions provide global spatial approximation, while a non-zero auxiliary variable linearizes nonlinear advection, Lorentz-force, and Hall terms. A Lagrange multiplier enforces the magnetic divergence constraint weakly, and a projection method decouples velocity and pressure through Stokes solves. Time-stepping-varying Crank–Nicolson discretization supports adaptive steps and preserves an unconditional discrete energy-dissipation law. Manufactured-solution tests show exponential spatial convergence and second-order temporal accuracy; adaptive stepping tracks rapid early energy decay while reducing CPU time. Energy remains monotone for time steps as large as one and across viscosity, diffusivity, and Hall-parameter tests. Simulations of magnetic X - and O -points further reproduce Hall-driven current-layer intensification, with current magnitude increasing as the Hall parameter grows.

Contributions

1. Formulated a Gegenbauer–Galerkin discretization for the full unbounded three-dimensional domain.
2. Combined an auxiliary variable, magnetic Lagrange multiplier, and projection method into a linear algorithm.
3. Proved weak magnetic divergence preservation, second-order accuracy, and unconditional energy stability.
4. Demonstrated exponential spatial convergence and effective adaptive time stepping numerically.
5. Recovered Hall-induced current-density intensification during magnetic-null evolution.

2026GK

ChatGPT (July 2026)

Summary

This paper develops discontinuous Galerkin discretizations for the Cahn–Hilliard and coupled Cahn–Hilliard–Navier–Stokes equations with degenerate mobility. The new SIPGD-L and SWIPD-L schemes alter the penalty mobility flux using, respectively, an interface maximum and a harmonic average, providing tunable control of coercivity and stability. A coercivity result is established for the generalized mobility-dependent trilinear form. Combined with an Eyre splitting, Zhang–Shu limiter, and divergence-free velocity postprocessing, the methods are designed to retain mass conservation, energy dissipation, and phase-field bounds. Manufactured-solution studies recover optimal L^2 and H^1 convergence in two and three dimensions. Merging-droplet and rotating-droplet tests show numerical structure preservation, while hp -adaptivity reduces degrees of freedom and computational complexity relative to h -adaptivity without materially degrading accuracy. Rigorous discrete structure-preservation results for SWIPD-L beyond coercivity remain future work.

Contributions

1. Introduced SIPGD-L and SWIPD-L with interface-maximum and harmonic-average mobility fluxes.
2. Proved coercivity of the generalized mobility-dependent DG trilinear form under an explicit penalty-flux condition.

3. Combined the schemes with Zhang–Shu limiting and hp -adaptive indicators while retaining numerical mass and energy structure.
4. Demonstrated optimal L^2 and H^1 convergence in two and three dimensions across four DG variants.
5. Showed that hp -adaptive droplet simulations preserve bounds and reduce degrees of freedom relative to h -adaptivity.

2026GKSA

ChatGPT (July 2026)

Summary

The paper presents the vacuum-vessel design and three-dimensional finite-element assessment for PRAGYA, a privately developed low-aspect-ratio tokamak with major radius about 0.4 m, minor radius above 0.18 m, plasma current up to 25 kA, and toroidal field 0.1 T. The SS304L vessel is split by a toroidal electrical break to reduce eddy currents and uses double O-ring seals, multiple diagnostic and pumping ports, internal stiffeners, and supported pumping to reach approximately 10^{-6} mbar. A mesh-refinement study motivates a roughly 1.012-million-element model and selection of a 6 mm wall. Under atmospheric pressure and self-weight, the model predicts about 110 MPa maximum von Mises stress and 0.5 mm maximum deformation. Baking at 150 °C produces about 280 MPa peak secondary stress, judged acceptable under the adopted criteria. Linear buckling analyses find the support leg and vessel assembly stable for the modeled loads. Electromagnetic loading during normal operation and disruptions is deferred.

Contributions

1. Specified an integrated vacuum-vessel design for the compact PRAGYA tokamak and its operational geometry.
2. Incorporated a toroidal electrical break, double O-ring seals, ports, stiffeners, supports, and pumping interfaces.
3. Used mesh refinement and thickness studies to select a converged model and a 6 mm SS304L wall.
4. Quantified stress and deformation under atmospheric pressure, self-weight, and 150 °C baking.
5. Assessed linear buckling margins for the support leg and complete vessel assembly while defining the electromagnetic-load limitation.

2026GNP

ChatGPT (July 2026)

Summary

The paper studies field-line dynamics of divergence-free surface currents used to represent stellarator coils. On smooth toroidal winding surfaces, generic currents satisfy a dichotomy: a nowhere-vanishing current has either dense trajectories or periodic trajectories with a common winding type, whereas a current with zeros contains both centre and saddle regions and has only finitely many nonperiodic connecting trajectories. For cylindrical coil surfaces, oppositely directed currents on the two boundary circles generically force centre and saddle regions; nearly all remaining trajectories are periodic. Imposing steady-state Maxwell and Ohmic assumptions makes the current both divergence- and curl-free, hence a harmonic Neumann field. On a cylinder this space is one-dimensional, the nonzero field has no zeros, and every trajectory is a closed poloidal orbit, excluding centre and saddle regions. Additional recurrence and Poincaré–Bendixson arguments show that almost every orbit of a general divergence-free cylindrical current is periodic. These results explain observed current patterns and identify assumptions under which coil contours simplify.

Contributions

1. Proved a generic toroidal-current dichotomy between nowhere-vanishing dense or periodic dynamics and dynamics with centre and saddle regions.
2. Classified exceptional toroidal trajectories as singularities or finitely many saddle connections.
3. Showed that opposite boundary-current orientations on a cylindrical surface generically force centre and saddle regions.

4. Derived harmonic Neumann currents from steady-state Maxwell and linear isotropic Ohmic assumptions.
5. Proved that nonzero harmonic cylindrical currents consist entirely of closed, noncontractible poloidal field lines.

2026GNY

ChatGPT (July 2026)

Summary

Entries of inverses of diagonally dominant matrices encode quantities such as random-walk escape probabilities and hitting times, which may be exponentially small. Normwise error bounds therefore do not provide useful relative accuracy for every entry without high bit complexity. This paper studies multiplicative, entrywise $\exp(\varepsilon)$ approximation for row diagonally dominant L -matrices (RDDL) and symmetric diagonally dominant M -matrices (SDDM). For floating-point RDDL inputs, it gives a cubic-time inversion algorithm whose running time is condition-number independent and conditionally optimal under the all-pairs shortest-paths conjecture. For fixed-point inputs, a threshold-decay framework yields SDDM linear solves in $\tilde{O}(m\sqrt{n})$ bit operations and inversion in $\tilde{O}(mn)$. For directed RDDL matrices, the authors obtain a linear solve in $\tilde{O}(mn^{1+o(1)})$ and inversion algorithms with costs $\tilde{O}(n^{\omega+1/2})$ or $\tilde{O}(mn^{3/2+o(1)})$. The analysis combines Schur complements, random-walk interpretations, and graph shortcutters, and transfers the guarantees to all-pairs escape probabilities and hitting times. The results separate arithmetic-operation speed from the precision cost required for entrywise accuracy.

Contributions

1. Formulated and analyzed multiplicative entrywise approximation for inverses and solutions of RDDL and SDDM systems.
2. Gave a condition-number-independent cubic-time algorithm for floating-point RDDL inversion.
3. Proved conditional optimality of the floating-point algorithm through a reduction from all-pairs shortest paths.
4. Developed fixed-point SDDM algorithms requiring $\tilde{O}(m\sqrt{n})$ bit operations for one solve and $\tilde{O}(mn)$ for inversion.
5. Derived subcubic directed-RDDL algorithms and corresponding guarantees for escape probabilities and hitting times.

2026GR

ChatGPT (July 2026)

Summary

This paper rigorously justifies first-order hyperbolic relaxations of several higher-order partial differential equations. Using relative-energy estimates, it compares weak entropy solutions of the relaxation systems with smooth solutions of the limiting equations and proves convergence as the relaxation parameter vanishes. The analysis treats mixed space–time derivatives typified by Benjamin–Bona–Mahony equations and purely spatial derivatives of arbitrary odd or even order, covering Korteweg–de Vries, KdV–Burgers, Gardner, Kawahara, biharmonic, and Kuramoto–Sivashinsky models. The authors manage the degenerating convexity of the relaxation energy and, where existing hyperbolizations lack a usable energy, propose alternative energy-compatible systems. Structure-preserving summation-by-parts spatial discretizations with additive Runge–Kutta time integration reproduce the predicted asymptotic rates. Numerically, auxiliary derivative variables often converge as rapidly as the primary solution, a stronger behavior left as a conjectured analytical refinement.

Contributions

1. Applied a unified relative-energy framework requiring only weak entropy solutions of the hyperbolizations and a smooth limiting solution.
2. Proved convergence for BBM-type equations despite degenerating convexity of the relative energy.
3. Extended the construction to broad odd- and even-order spatial PDE classes.

4. Replaced an unsuitable Kuramoto–Sivashinsky hyperbolization with an energy-compatible approximation and proved convergence.
5. Supplied reproducible SBP and additive-Runge–Kutta experiments confirming the theoretical rates.

2026GS

ChatGPT (September 2026)

Summary

This work investigates how energetic ions can promote internal transport barriers through kinetic $E \times B$ staircase structures in tokamak plasmas. Using a reduced Vlasov model averaged over cyclotron and bounce or transit motion, the authors connect staircase formation to negative mass instability, which bunches particles when their precession frequency decreases with radial action. Linear analysis and nonlinear TERESA simulations identify propagating staircases as analogues of Bernstein–Greene–Kruskal equilibria: coherent structures sustained by particle trapping. Trapped-particle modes drive their growth through autoresonance, while pairwise merging produces a dominant toroidal $n = 1$ structure alongside a zonal $n = 0$ component. With a stronger energetic-ion population, phase synchronization and resonant energy transfer amplify low-frequency zonal flow. The staircase subsequently disappears as its sustaining resonance is lost, leaving suppressed turbulence and an internal transport barrier. A separate simulation distinguishes initial energetic-ion–trapped-electron resonance from later staircase-mediated amplification. The mechanism is demonstrated within the reduced electrostatic model; the energetic-particle threshold is not determined analytically, and the correspondence with Kuramoto synchronization is a formal analogy.

Contributions

1. Extended negative mass instability to a tokamak particle-mode model, deriving a dispersion relation connecting instability and negative frequency shift to radially decreasing precession frequency.
2. Identified propagating kinetic staircases as BGK-like structures sustained by particle trapping, with phase mixing smoothing internal filaments and a separate Vlasov–Poisson calculation clarifying the phase-space-hole analogy.
3. Demonstrated formation of a dominant toroidal $n = 1$ staircase through pairwise merging of smaller structures, alongside an axisymmetric zonal-flow component.
4. Showed staircase-mediated zonal-flow amplification and turbulence suppression with increased energetic-ion density, followed by staircase destruction and internal transport barrier formation.
5. Distinguished early energetic-ion–trapped-electron resonance from later staircase-mediated amplification in a dedicated simulation, using frequency sidebands to support the autoresonance interpretation.

2026GSC

ChatGPT (July 2026)

Summary

This work extends conservative Lagrange–Galerkin convection schemes to quadratic finite elements, where characteristic transport curves element sides and invalidates straight-element intersection formulas. It develops three supermesh algorithms: Single Straight-sided Triangles (SST), Division in Straight-sided Triangles (DST), and Division in Curved Triangles (DCT). These compute transported-mesh integrals to machine accuracy, restoring mass conservation and unconditional time-step stability. Numerical tests on uniform and adaptively refined meshes show DCT retains projection-like L^2 accuracy while reducing mass error by as much as ten orders of magnitude; SST and DST match this on uniform grids but can lose accuracy under local refinement. DCT is best for very low-error refined solutions, DST for intermediate errors, and SST when larger errors are acceptable. A dynamic side-correction criterion reduces the added supermesh cost, and all intersection operations are element-local and readily parallelizable.

Contributions

1. Extended conservative Lagrange–Galerkin mesh intersection to transported quadratic curved elements.

2. Formulated SST, DST, and genuinely curved DCT supermesh algorithms.
3. Achieved machine-precision transported integrals, mass conservation, and stability without a CFL restriction.
4. Established accuracy-cost regimes and up to ten orders of mass-error improvement on uniform and adaptive meshes.
5. Introduced a dynamic correction-selection criterion and identified elementwise parallelism.

2026GSHJ

ChatGPT (July 2026)

Summary

The study develops plasma equilibria and high-temperature-superconducting coils for EPOS, a compact stellarator intended to confine electron-positron plasmas long enough for collective behavior and cyclotron cooling. EPOS targets a 2 T quasi-axisymmetric field, a roughly 10 L plasma, low rotational transform, and special weave-lane coils for positron injection. The optimization combines VMEC equilibria and finite-build coils in SIMSOPT, simultaneously penalizing quasisymmetry error, magnetic-field mismatch, coil spacing, curvature, and REBCO torsional and binormal strain. Gaussian-process coil perturbations make robustness to manufacturing error part of the objective. Eight candidates span 16-19 cm major radii and standard-to-weave-lane current ratios of three or four. All keep strain below the conservative 0.2% limit, and several tolerate millimeter-scale perturbations. Guiding-center calculations lose at most 10% of 5 eV particles over two seconds when initialized near the axis, but edge initialization loses 40-55%. Targeted refinement of the C4 R19 candidate produces convex coils, adequate spacing, low field error, and no visible islands. With 1 mm coil perturbations, however, its mean 0.5 s particle loss rises from 12.7% to about 30%, so engineering verification remains essential.

Contributions

1. Translated EPOS confinement, injection, size, field-strength, and REBCO winding constraints into a coupled stellarator objective function.
2. Combined stochastic and single-stage optimization of plasma boundaries and finite-build coils to account for manufacturing deviations during design.
3. Produced eight compact candidates whose coil strains remain below 0.2% while retaining useful quasisymmetry and magnetic-surface accuracy.
4. Quantified radial-injection and coil-error sensitivity with guiding-center particle-loss simulations over experimentally relevant times.
5. Refined the C4 R19 candidate to convex, separated, lower-current-ratio coils with low field error and intact nested flux surfaces.

2026GSJH

ChatGPT (July 2026)

Summary

This work combines single-stage stellarator design, which optimizes plasma and coils together, with stochastic coil optimization over ensembles of randomly perturbed coils. A fixed-boundary VMEC equilibrium is coupled to the coils through squared magnetic flux, allowing the objective to favor broad optima that tolerate manufacturing and assembly errors. Warm-started quasi-axisymmetric and quasi-helically symmetric configurations are compared with deterministic single-stage and stochastic stage-II designs using field error, quasisymmetry, engineering constraints, and guiding-center particle loss. The stochastic single-stage method escapes local minima that stall deterministic optimization and attains robustness comparable to stochastic stage II while improving optimized equilibrium quasisymmetry and squared flux. Gains are modest for the quasi-axisymmetric case, but the quasi-helical design has about threefold lower quasisymmetry error at 1.5 mm perturbations; its perturbed particle loss rises by 44%, compared with 430% for the deterministic design. Free-boundary optimization remains a proposed improvement.

Contributions

1. Unified stochastic coil perturbations with simultaneous fixed-boundary plasma-and-coil optimization.
2. Showed that perturbation-amplitude increases need not degrade optimization of the other physics objectives.
3. Benchmarked the method on quasi-axisymmetric and quasi-helically symmetric warm-start configurations.
4. Quantified tolerance using perturbed quasisymmetry, squared flux, and ensembles of guiding-center particle-loss simulations.
5. Demonstrated that optimizing robustness can be more valuable than pursuing vanishing unperturbed field error.

2026GSNE

ChatGPT (July 2026)

Summary

This paper tests whether nitrogen-vacancy centres in diamond can provide steady-state magnetic diagnostics in the tesla-scale fields expected inside future tokamaks. The authors build a fibre-coupled, millimetre-scale ensemble sensor using continuous-wave optically detected magnetic resonance and microwave excitation up to tens of gigahertz. They demonstrate operation at bias fields up to 1.2 T and compare non-degenerate orientations, which retain multiple resonances for vector information, with a near- $\langle 111 \rangle$ alignment optimized for single-axis sensitivity. Across 10–150 Hz, measured unshielded sensitivities are approximately $240\text{--}600\text{ nT}/\sqrt{\text{Hz}}$ for non-degenerate alignments and $110 \pm 20\text{ nT}/\sqrt{\text{Hz}}$ near $\langle 111 \rangle$. The high-field noise is dominated by technical laser noise rather than an intrinsic nitrogen-vacancy limit. These sensitivities may be sufficient for tokamak control, but deployment still requires resonance tracking, high dynamic range, radiation- and temperature-qualified fibres and microwave lines, and validation under DEMO-level neutron and gamma exposure.

Contributions

1. Demonstrated fibre-coupled ensemble nitrogen-vacancy magnetometry at fields up to 1.2 T.
2. Measured sub-microtesla sensitivities over 10–150 Hz for two field-alignment regimes.
3. Quantified the high-field sensitivity penalty and identified laser noise as the present limit.
4. Related field alignment to resonance visibility, sensitive-axis geometry, and vector-magnetometry capability.
5. Defined tokamak-engineering requirements for tracking, dynamic range, remote delivery, temperature tolerance, and radiation qualification.

2026GT

ChatGPT (July 2026)

Summary

Many global solvers for multivariate polynomial systems replace the original problem with univariate rootfinding or generalized eigenvalue subproblems. The paper shows that popular algebraic constructions can make these intermediate problems exponentially worse conditioned as the number of variables grows, even when the target root is comparatively well conditioned. It treats Gröbner-basis elimination, rational univariate representations, multiparameter eigenproblems solved by operator determinants, normal-form multiplication matrices, and Macaulay resultants, extending earlier analyses of hidden-variable resultants. Explicit polynomial systems quantify the condition-number amplification and show that changing polynomial bases or variable order does not remove several of the failures. New formulas connect operator-determinant conditioning to the original Jacobian and singular values of determinantal representations, and connect multiplication-matrix and Macaulay conditioning to Lagrange interpolants represented through a determinant. Numerical experiments reproduce the predicted loss of accuracy for bivariate and higher-dimensional examples. The results diagnose instability in common algorithmic variants rather than prove that every possible algebraic formulation is unstable: unusually chosen projections, bases, or determinantal representations remain potential escape routes, and homotopy-continuation methods are outside the analysis.

Contributions

1. Unified the stability analysis of major algebraic multidimensional rootfinders by comparing each constructed subproblem with the conditioning of the original root.
2. Constructed polynomial systems for which Gröbner elimination and bounded linear rational-univariate projections incur condition-number growth exponential in dimension.
3. Derived a condition-number formula for operator-determinant generalized eigenproblems and used it to expose exponential amplification in standard multiparameter formulations.
4. Characterized multiplication-matrix and Macaulay-resultant conditioning through Jacobians and Lagrange interpolants, yielding new exponential-instability examples.
5. Confirmed the theoretical scaling in numerical experiments while identifying the precise assumptions and possible routes around each instability mechanism.

2026GT2

ChatGPT (September 2026)

Summary

The paper develops a first-order implicit–explicit method for nonequilibrium gray radiation hydrodynamics that preserves the model’s physically admissible set. The coupled compressible-fluid and radiation system is split into two hyperbolic subsystems and one stiff parabolic radiation-diffusion subsystem. Graph-based finite element updates and derived local wave-speed bounds make each hyperbolic stage invariant-domain preserving. The parabolic stage uses backward Euler, a Picard iteration, and local Newton solves for radiation energy and material temperature; its admissibility proof holds under mild equation-of-state and spatial-discretization assumptions and does not depend on the Picard stopping tolerance. The composed approximation is shown to be consistent, conservative, invariant-domain preserving, and first-order accurate. Numerical experiments confirm convergence and robustness, including regimes with strongly contrasted opacity and high Mach number. The construction is deliberately a low-order foundation for a subsequent high-order method, where limiting can blend high-order states toward these admissible low-order updates.

Contributions

1. Introduced a three-part split of gray radiation hydrodynamics into two hyperbolic and one parabolic subsystem.
2. Derived the maximum wave speed needed for invariant-domain preservation in the second hyperbolic stage.
3. Constructed conservative graph-based finite element updates that preserve the admissible thermodynamic domain.
4. Designed a robust implicit radiation solve using Picard iteration and local Newton updates.
5. Proved first-order consistency, conservation, and admissibility and confirmed the predicted convergence numerically.

2026GTCD

ChatGPT (July 2026)

Summary

The paper introduces PICSL, a fully implicit semi-Lagrangian integrator that applies BDF2 consistently to spherical shallow-water dynamics and trajectory equations, solving the coupled nonlinear system by Picard iteration with convergence-based stopping. Analysis establishes second-order temporal accuracy, A-stability for the dynamics, absence of the spurious BDF2 resonance that affects centered Crank–Nicolson semi-Lagrangian schemes, and favorable Rossby- and gravity-wave dispersion. The time step is not Courant-limited by the dynamics, although fixed-point trajectory convergence requires a Lipschitz condition in regions with strong velocity gradients. Steady geostrophic flow, Rossby–Haurwitz waves, isolated topography, barotropic instability, and realistic-orography tests confirm second-order convergence, suppress terrain-induced noise without off-centering or filtering, and generally improve energy and potential-enstrophy preservation relative to centered or off-centered CNSL. Costs are comparable in most cases, but

semi-Lagrangian advection is more expensive because PICS� traces midpoint and departure trajectories. Evaluation in three-dimensional operational GEM configurations remains future work.

Contributions

1. Defined a fully implicit BDF2 semi-Lagrangian scheme with Picard nonlinear solves and tolerance-based stopping.
2. Applied BDF2 consistently to governing dynamics and kinematic trajectories, removing a source of semi-Lagrangian inconsistency.
3. Established second-order accuracy, A-stability, and absence of spurious BDF2 resonance without off-centering or filtering.
4. Derived Rossby- and gravity-wave dispersion behavior and the trajectory Lipschitz convergence restriction.
5. Demonstrated reduced terrain noise and improved invariant preservation on spherical shallow-water benchmarks at costs comparable to CNSL in most cases.

2026GVBC

ChatGPT (July 2026)

Summary

The paper reports a proof-of-principle beam-driven plasma-wakefield accelerator in a plasma filament produced by self-guided femtosecond laser propagation through low-pressure nitrogen. The filament supplies a narrow, reproducible plasma channel without a high-current discharge, relativistic ionizing beam, or terawatt-class ionization laser. In a 3-cm capillary with an approximately $8 \times 10^{14} \text{ cm}^{-3}$ plasma plateau, a driver-witness electron-bunch pair generates wakefields that raise the witness energy by about 8 MeV to 104.5 ± 0.4 MeV, corresponding to an average gradient near 266 MV/m, while preserving charge and relative energy spread. Across 100 consecutive shots, acceleration occurs in 95% of filament shots with less than 0.5% rms energy jitter. A matched discharge-plasma comparison succeeds in 75% of shots with about 1.3% jitter. Particle-in-cell simulations reproduce the measured energy changes and yield an accelerating field near 260 MV/m. The approach offers tunable channel geometry and low deposited energy, motivating high-repetition-rate compact accelerator stages.

Contributions

1. Demonstrated beam-driven wakefield acceleration in a self-generated laser-plasma filament.
2. Measured an approximately 8 MeV witness gain over 3 cm, corresponding to about 266 MV/m.
3. Showed reproducible operation over 100 shots with 95% acceleration success and below 0.5% rms energy jitter.
4. Benchmarked the filament against a discharge plasma, finding higher success and roughly threefold lower jitter.
5. Used particle-in-cell simulations to reproduce the measured wakefield and energy-transfer behavior.

2026GWSY

ChatGPT (July 2026)

Summary

This paper explains when matrix-multiply Tensor Cores can accelerate stencil computations despite their reputation as memory-bound kernels. Mapping stencils to fixed-size matrix operations introduces zero-padding sparsity and, under kernel fusion, redundant arithmetic that ordinary roofline models omit. The authors incorporate both penalties into an enhanced model for CUDA Cores, dense Tensor Cores, and sparse Tensor Cores, then classify four combinations of memory- and compute-bound execution. Tensor Cores provide no benefit when the CUDA implementation remains memory-bound; their profitable region instead occurs after temporal fusion makes CUDA execution compute-bound and specialized throughput outweighs transformation overhead. Sparse Tensor Cores enlarge this region by exploiting structured zeros. Measurements on an NVIDIA A100 GPU across DRStencil, EBISU, ConvStencil, and SPIDER support the model: selected cases range from a 27.12% slowdown outside the predicted region to 7.73-fold and 6.64-fold

speedups inside it, while sparse execution gives 3.06-fold over its dense counterpart in one tested configuration. Cache reuse and halo overlap explain small model deviations.

Contributions

1. Quantified sparse-padding and kernel-fusion redundancy introduced by stencil-to-matrix transformations.
2. Extended roofline analysis to compare effective CUDA-Core and Tensor-Core stencil throughput.
3. Derived four bottleneck regimes and an analytical sweet spot for profitable Tensor-Core use.
4. Showed how structured-sparse Tensor Cores broaden the predicted acceleration region.
5. Validated positive, neutral, and negative predictions on an NVIDIA A100 across four optimized stencil implementations.

2026GXPS

Claude (September 2026)

Summary

The authors deploy on an FPGA a recurrent probabilistic neural network (RPNN) that predicts the next DIII-D tokamak plasma state, for model-predictive-control-style plasma control, which needs low, deterministic latency. As background, the RPNN of Char et al. encodes a 27-dimensional state (7 scalars and 20 principal-component coefficients of 6 profiles) and 13 actuators, given as levels and commanded changes, through an MLP, a gated recurrent unit (GRU), and a residual-MLP decoder with mean and log-variance heads, and bootstrap ensembles add epistemic uncertainty. Replacing LayerNorm by BatchNorm for hls4ml compatibility, the authors retrain on 19,648 chronologically split shots at 20 ms cadence, shrink the model by 94.9% to 175,628 parameters (+1.9% one-step test MSE), and apply 16-bit quantization-aware training in QKeras (a further +2.8% MSE and -4.8% explained variance). Synthesized with hls4ml, Vitis HLS, and Vivado for a Xilinx Alveo U50, the kernel uses at most 65% of any resource and takes 276 cycles, 2.64 μ s, per inference at 104.7 MHz. Free-running ensembles track a held-out shot, but errors compound in closed loop. Timing comes from post-implementation analysis because no card was available, and training does not penalize rollout error growth.

Contributions

1. Validated the BatchNorm substitution on retrained checkpoints, with validation MSE 0.02622 against 0.02728 (-3.9%) and explained variance (EV) 0.5166 against 0.4967, and test MSE 0.02375 against 0.02407 and EV 0.4243 against 0.4038, trained by a two-stage curriculum (MSE training of the whole model, then fine-tuning of only the log-variance head and variance-pinning bounds by a heteroscedastic negative log-likelihood with a 10^{-3} penalty on wide pinning intervals; AdamW, learning rate 3×10^{-4} , batch 512, early-stopping patience 250 epochs, first two timesteps excluded from the loss) on 17,676/982/990 training/validation/test shots.
2. Ablated 25 reductions of the decoder width, residual-block count, GRU width, and backbone width from the 3.45M-parameter model, 74.2% of which sits in the decoder: on the validation split the four smallest backbones worsened MSE monotonically from +12.4% at 73,228 parameters to +35.5% at 22,508, and on the test split removing all residual blocks raised MSE by 26.9–54.4%, whereas the selected hid128 gru64 dec128 b1 cost +1.9% MSE and -2.7% EV and its two- and three-block variants, 19–38% larger, were no better.
3. Quantized weights, activations, and biases to ap_fixed<16,6> after preliminary sweeps in which ap_fixed<8,3> and ap_fixed<6,1> lost substantially more accuracy, showed that naive post-training quantization destroys the model (validation MSE 1.469, EV -13.57), and reached test MSE 0.02487 and EV 0.3932 after quantization-aware fine-tuning, +4.7% MSE and -7.3% EV relative to the full-size floating-point baseline.
4. Made the GRU hidden state explicit kernel I/O, streaming 117 input and 118 output 32-bit words per inference with zeros supplied at shot start, left the normalizer and variance pinning outside the dataflow region (175,414 of 175,628 parameters synthesized), and used reuse factor 64, streaming I/O, ap_fixed<32,20> accumulators, and an ap_fixed<27,17> input type, giving post-route utilization of 35.24% LUT, 17.79% FF, 65.03% DSP, and 48.32% BRAM, a 122-cycle (1.17 μ s) initiation interval, and batch-1 latencies about 230 and 411 times below the 608 μ s and 1,085 μ s of the same model in TensorFlow on an 8-core AMD EPYC 7763 and an NVIDIA A100.

5. Evaluated closed-loop behavior: in a 225-step replay of test shot 193369 with 15-member ensembles, per-channel increment clamping engaged in 1.3% (baseline) and 2.0% (quantized) of updates; over horizons, against a persistence model scoring EV 0.96 at one step and crossing zero near 99 steps, the quantized ensemble trailed the baseline by 0.51 EV after 100 steps in sample mode (-0.41 against $+0.10$); and of three ensemble-sampling schemes, Sample-TS1 (member resampled and Gaussian sampled every step) kept 90%-interval coverage within 0.81–0.90 (baseline) and 0.84–0.92 (quantized) with miscalibration areas at most 0.08 and 0.10 through 200 steps, against coverage near 0.27–0.30 for Mean-TS1 and over-coverage of 0.97–0.99 for Sample-TSInf.

Comments

- Section III B: “After fine-tuning, the 16-bit quantized model lands within $+2.8\%$ MSE / -4.8% EV of the FP32 reference on the held-out test split (Table VI)”, but Table VI is the architecture-size ablation; the quantized results (MSE 0.02487 and EV 0.3932 against 0.02420 and 0.4128) are in Table VIII.
- Section III A: “every configuration down to roughly 93% parameter reduction stays within $\pm 5\%$ of the full-size baseline’s MSE”, but Table V lists gru64 dec64 b0, at -86.6% parameters, with $+6.3\%$ MSE, the value the same sentence gives as the largest loss above 73,228 parameters.
- Section IV B: “We therefore screen each member in both prediction modes. Figure 2 reports results for these ensembles”, but Section IV A states for the Fig. 2 replay that “Each ensemble comprises its full set of fifteen members” and “No stability screening was applied”, and the Fig. 2 caption lists “all 15 bootstrap members”; only Figs. 3 and 4 use four screened members.
- Section V B: “the deployed FPGA kernel’s $2.64\ \mu\text{s}$ is $\approx 240\times$ faster than the same architecture on the CPU”, but Table XII gives $608\ \mu\text{s}$ for the ablated architecture on the CPU at batch size 1, and $608/2.64 \approx 230$; the A100 ratio, $1085/2.64 \approx 411$, matches the stated “ $\approx 410\times$ ”.

2026GYWG

ChatGPT (July 2026)

Summary

COREFL is an open-source C++/CUDA solver for high-fidelity compressible reactive-flow simulations on structured generalized curvilinear grids. It solves multicomponent compressible Navier–Stokes equations using a hybrid seventh-order WENO-Z/linear-upwind discretization, mixture-averaged transport, and balanced splitting for stiff chemical kinetics. CUDA-aware MPI supports multi-GPU execution, while static polymorphism and GPU-oriented multidimensional data structures preserve extensibility without virtual-dispatch overhead. Validation spans formal accuracy tests, inert and reactive shock tubes, oblique detonation, supersonic and hypersonic boundary layers, and a reacting turbulent mixing layer. Reported results reproduce reference shocks, heat transfer, turbulence statistics, and reacting-flow behavior. Kernel tuning reduces selected execution times by 17–35%; weak scaling remains near 90% through 16 A100 GPUs, and a reactive case shows more than 800-fold speedup over one core of an in-house CPU solver. Deployment currently depends on NVIDIA CUDA and CUDA-aware MPI.

Contributions

1. Released an Apache-2.0 C++/CUDA/MPI solver for compressible multicomponent reactive-flow direct and large-eddy simulations.
2. Integrated hybrid seventh-order WENO-Z/upwind convection, detailed mixture transport, and balanced chemical-kinetics splitting.
3. Used compile-time polymorphism to extend physical models with little runtime overhead or code duplication.
4. Designed and profiled GPU-specific data layouts and kernels, obtaining 17–35% reductions in selected kernel times.
5. Demonstrated broad physical validation, near-90% weak-scaling efficiency through 16 GPUs, and over 800-fold reactive-case acceleration relative to a CPU core.

2026GZ

ChatGPT (July 2026)

Summary

This paper develops a space-time Galerkin time-domain boundary element method for predicting how aircraft surfaces scatter and shield noise from transient, rotating, or broadband sources. The formulation enforces sound-hard boundary conditions and incorporates uniform mean flow through coordinate transformations. Unlike collocation TDBEMs, its Galerkin discretization is unconditionally stable, requires no tuned stabilization parameter, and can represent thin bodies with a single element layer. The principal numerical obstacle—singular double integration over space and time—is handled by decomposing triangle intersections with light-cone boundaries, analytically integrating the radial coordinate, and reducing the remaining inner calculation to one-dimensional quadrature. Sphere, disk, and plane tests cover curved and sharp-edged geometries, closed and thin surfaces, harmonic and transient sources, and media with and without mean flow; all agree closely with analytical solutions, with joint refinement showing second-order accuracy. Coupling the method to RANS/Ffowcs Williams–Hawkings propeller data reproduces measured amplification and shielding trends for trailing-edge-mounted propellers, with most installation-effect discrepancies below 5 dB.

Contributions

1. Formulated an unconditionally stable space-time Galerkin TDBEM for aeroacoustic scattering without tuned stabilization parameters.
2. Incorporated uniform mean-flow effects through variable transformations without altering the boundary-integral machinery.
3. Reduced singular double space-time integrals to efficient decomposition-based quadrature with analytical radial integration.
4. Validated the method across three analytical cases spanning curved, sharp-edged, closed, thin, stationary-flow, and mean-flow configurations.
5. Coupled the solver to CFD-derived propeller noise and reproduced experimental scattering and shielding trends over plate positions and sizes.

2026H

ChatGPT (July 2026)

Summary

This paper adapts one-step Drifting Models to molecular conformations and uses force labels to shift generation from biased training data toward the Boltzmann distribution. For Gaussian kernels, a Drifting Score Identity expresses attraction as a kernel-weighted score average; the force–score relation then gives a Boltzmann force identity. Force-Interpolated Drifting blends Cartesian displacements with force directions, while Force-Aligned Kernel changes neighbor weights. On a biased 3,000-frame MD17 ethanol subset, force interpolation works best in Cartesian coordinates, whereas kernel reweighting works best in pairwise-distance space because it preserves realizable molecular geometry. The latter reaches distance-histogram total-variation distance 0.089, bond mean absolute error 0.006 Å, and 100% bond stability. Generation is one-step and millisecond-scale. The title’s million-fold acceleration is an extrapolation across reported baselines and hardware rather than a direct same-hardware benchmark.

Contributions

1. Established Drifting Score and Drifting Force identities connecting data attraction, score functions, and molecular forces.
2. Introduced force-interpolated and force-aligned-kernel guidance mechanisms.
3. Showed how force guidance moves the drifting equilibrium toward, though not generally exactly to, the Boltzmann distribution.
4. Identified a representation-dependent reversal in which guidance mechanism performs best.

5. Achieved low distributional error and perfect tested bond stability on the biased MD17 ethanol experiment.

2026H2

ChatGPT (September 2026)

Summary

This paper realizes a compact two-stage, fourth-order, two-derivative implicit–explicit method for stiff partial differential equations and develops solvers for its widened implicit stages. A second-order ADER/Cauchy–Kowalevski local evolution supplies sufficient derivative information for the fourth-order outer composition, but derivatives must follow the full split field: differentiating explicit and implicit terms separately creates a leading Lie-bracket defect and can reduce noncommuting problems to first order. The fully discrete analysis requires reconstruction order $q \geq 3$ when $\Delta t = O(\Delta x)$ and permits asymmetric inexact-solve residuals, $O(\Delta t^3)$ at the midpoint and $O(\Delta t^5)$ at the endpoint. Matrix-free stage actions retain mixed terms exactly, while quadratic, shifted, Fourier, multilevel, semilinear, and source-local preconditioners approximate dominant stiff physics. Analysis explains exact quadratic cancellation, diffusion-mode clustering, and an $O(\varepsilon/\Delta x)$ correction for Jin–Xin relaxation. Ablations verify the order and tolerance claims, and Brusselator tests show favorable error-versus-runtime behavior over a useful regime. A stiff-front benchmark also shows that the time method does not cure underresolved shock–source inconsistency.

Contributions

1. Established that second-order local evolution suffices for the inherited fourth-order temporal composition.
2. Derived the Lie-bracket consistency defect caused by differentiating split fields separately.
3. Proved fully discrete reconstruction requirements and asymmetric residual tolerances for inexact stages.
4. Developed matrix-free, structure-aware preconditioners for diffusion-, reaction-, and relaxation-dominated systems.
5. Validated regime-dependent efficiency while isolating the separate limitation from underresolved shock–source coupling.

2026HAGB

ChatGPT (July 2026)

Summary

The Fireball experiment at CERN’s HiRadMat facility sends a 440 GeV, 3×10^{11} -proton bunch through graphite–tantalum to create an approximately charge-neutral electron–positron beam with mean Lorentz factor about 500 and peak pair density $5 \times 10^{11} \text{ cm}^{-3}$, which then traverses an RF argon plasma. A calibrated 532 nm Faraday-rotation probe detects no magnetic signal when the plasma is off but a clear signal when it is on. The fitted peak path-integrated field is $14.8 \pm 0.6 \text{ mT mm}$, consistent within systematic uncertainty with three-dimensional OSIRIS particle-in-cell simulations predicting $18.8 \pm 0.3 \text{ mT mm}$. The inferred spatially averaged instability growth rate, 0.40 ns^{-1} , agrees with the simulated 0.50 ns^{-1} . A homogeneous oblique-mode estimate gives 2.5 ns^{-1} , indicating that finite radius, energy spread, and geometry suppress growth by about a factor of five. Astrophysical extrapolation remains model-dependent.

Contributions

1. Produced and characterized an ultra-relativistic, approximately neutral laboratory pair jet.
2. Implemented an impulse-response-calibrated Faraday diagnostic for sub-nanosecond fields.
3. Observed plasma-dependent magnetic amplification rather than a free-propagation current field.
4. Quantitatively matched experimental field amplitude and growth rate with 3D simulations.
5. Constrained kinetic and geometric suppression missing from homogeneous growth theory.

2026HBFH

ChatGPT (July 2026)

Summary

This work advances predictive full- f , global, long-wavelength gyrokinetic simulation of tokamak edge and scrape-off-layer turbulence using magnetic geometry, heating power, and total particle inventory rather than prescribed plasma profiles. An adaptive Gkeyll source controls injected energy and models recycling with zero net particle addition, allowing profiles and turbulence to relax self-consistently. For TCV discharge #65125, simulated electron density and temperature agree reasonably with Thomson-scattering and Langmuir-probe measurements, while the model reproduces scrape-off-layer blobs, a self-organized radial electric field, and consistent power accounting. Comparing positive- and negative-triangularity discharges, the simulations find higher core density, about 20% stronger outer-edge $E \times B$ shear, reduced mid-to-high-wavenumber fluctuations, and modestly redistributed exhaust power under negative triangularity. Linear analysis indicates trapped-electron-mode-driven, electron-dominated heat transport.

Contributions

1. Implemented an adaptive energy and recycling source for profile-predictive full- f Gkeyll simulations.
2. Evolved edge and scrape-off-layer turbulence without imposing density or temperature profiles.
3. Validated predicted kinetic profiles and turbulence features against TCV #65125 diagnostics.
4. Quantified triangularity-related changes in flow shear, fluctuation spectra, and power partition.
5. Linked electron-dominated turbulent heat flux to trapped-electron-mode activity and identified missing physics.

2026HBKW

ChatGPT (September 2026)

Summary

This paper reformulates plasma-fluid equations in general nonorthogonal curvilinear coordinates so that geometric source terms need not appear explicitly in the nonlinear kinetic stress. Density, momentum, and internal energy are rescaled by the coordinate Jacobian, making conserved mass, kinetic energy, and internal energy quadratic under ordinary Euclidean inner products. For a resistive-magnetohydrodynamic testbed, antisymmetric first derivatives and elementary dot- and cross-product identities are sufficient to derive discrete mass, cyclic-coordinate angular-momentum, and total-energy conservation. The ALMA implementation generates Fortran operators from symbolic Python and uses centered finite differences with a staggered curl-curl operator. Shercliff-flow tests show second-order convergence and large accuracy gains from grid stretching; an Orszag–Tang vortex on a distorted grid retains total energy to about 12 digits and magnetic solenoidality to about 11 digits. The centered scheme can oscillate near sharp gradients, and computing metrics with the same finite differences does not guarantee an exact discrete geometric conservation law, which the authors identify as future work.

Contributions

1. Recast the plasma-fluid hierarchy in Jacobian-weighted variables that conceal explicit geometric forces.
2. Derived coordinate-independent discrete conservation conditions based on antisymmetry and vector identities.
3. Applied the formulation to resistive magnetohydrodynamics with compatible viscosity and Ohmic heating.
4. Implemented symbolic-to-Fortran curvilinear operators in the ALMA finite-difference code.
5. Verified convergence and conservation on Shercliff flow and distorted-grid Orszag–Tang tests.

2026HFBH

ChatGPT (July 2026)

Summary

Using Gkeyll, the authors run 256 concurrent, unsupervised full- f global electrostatic gyrokinetic simulations of a TCV-inspired boundary spanning the closed-field-line edge and open-field-line scrape-off layer. The scan combines eight triangularities, eight elongations, and four injected powers and evolves each case beyond turbulence relaxation. Shaping effects depend strongly on power: at low power, positive triangularity produces a relatively hot scrape-off layer, whereas at high power negative triangularity develops stronger edge ion-temperature gradients and better confinement. The authors attribute the low-power hot scrape-off layer to trapped-ion banana geometry: triangularity changes the field-line distance between turning points and the high-field-side limiter, altering exposure to cold recycled neutrals. Transport fingerprints place most cases in the ITG/TEM regime, while local linear Gyacomio calculations identify trapped-electron modes as the most unstable electrostatic modes. The approximately 75 TB open ensemble supports transport-model benchmarking and data-driven modeling. The model omits electromagnetic fluctuations, higher-order finite-Larmor-radius effects, and detailed neutral and sheath physics.

Contributions

1. Demonstrated an unattended, concurrent 256-case full- f boundary gyrokinetic parameter scan.
2. Quantified how triangularity affects scrape-off-layer temperature and edge gradients differently across heating powers.
3. Proposed a trapped-ion arc-length mechanism for the hot low-power positive-triangularity scrape-off layer.
4. Combined nonlinear transport fingerprints with local linear calculations to diagnose the ITG/TEM regime.
5. Released an approximately 75 TB simulation ensemble for transport benchmarks and data-driven modeling.

2026HGMB

Claude (September 2026)

Summary

The authors present SpCCL, a fork of the NCCLX GPU collective library whose all-gather, reduce-scatter, and all-reduce send only the nonzeros of unstructured sparse payloads, mainly at 1–99% sparsity, where reductions cause fill-in (densification of partial sums). As background, flat-index COO doubles the bytes per nonzero and reaches dense size at 50% sparsity, whereas a bitvector costs 3.125% of fp32 data. Their format, Pici, adds to a bitvector one prefix sum per 64×64 virtual tile, about 3.15% overhead in total, with warp-per-tile CUDA kernels that derive column offsets from popcounts and warp shuffles. The ring collectives compress once in all-gather and at every reduce-scatter step, and send dense data once the measured or extrapolated sparsity drops to empirically tuned thresholds that differ between intra- and inter-node links and for the all-gather phase of all-reduce. On Perlmutter A100 GPUs, at 99% sparsity, speedups over NCCL reach $5.25\times$ (all-gather), $2.5\times$ (reduce-scatter), and $2.66\times$ (all-reduce), SparCML is uncompetitive even with dense NCCL, and gradient-pruned data-parallel training is up to 26% faster end to end. The authors note slowdowns for small, near-latency-bound slices, reduced computation–communication overlap from using more channels, offline threshold tuning, and an application study that does not exhaust sparsification strategies.

Contributions

1. Laid out Pici as a bitvector B whose 64-bit words each hold one column of a row-major 64×64 tile, an index array I of exclusive per-tile prefix sums ($k = 4096$) closed by the total nonzero count, and values A ordered by tile, column, and row, an index overhead of about 0.78% of the bitvector against 3.125% for the rank structures poppy and pasta and 50% for Coruscant; in compression each lane of a 32-thread warp scans columns l and $l + 32$ for coalesced loads and the tile count is reduced in five shuffles, a block-local prefix sum suffices because each CUDA block compresses an independent segment, and compaction gives each lane the adjacent columns $2l$ and

- $2l + 1$, with popcounts, a warp exclusive prefix sum, and find-first-set bit searches that decompression and a scatter-add variant reuse.
2. Extended the Hockney model $T = \alpha + n\beta$ for ring all-gather to $T = (p - 1)(\alpha + \beta n'/p) + T_{\text{comp}}(n/p) + T_{\text{decomp}}(n'(p - 1)/p)$ with $n' = n(1 - \sigma + \omega)$ and $\omega = 0.0315$, extrapolated post-reduction sparsity as $\sigma_{\text{next}} = \sigma_{\text{prev}}\sigma_0$ for independent uniform nonzero positions, also across dense-send steps and recalibrated at each compression, and, from threshold sweeps in steps of 0.1, switched to dense at sparsity ≤ 0.6 intra-node, ≤ 0.5 inter-node, and ≤ 0.1 at the all-gather phase of all-reduce for Pici (0.8, 0.7, and 0.5 for COO), separating link types because a node's four 25 GB/s Slingshot 11 NICs only match the four 25 GB/s NVLink links of one GPU pair.
 3. Compressed and decompressed at slice granularity inside NCCLX's `genericOp` for the Simple protocol, prepending a 48-byte header with format, size, and component offsets, enlarged the FIFO buffers to hold a worst-case compressed slice, reused the application send buffer as accumulation scratch and the send FIFO as a staging buffer so that scatter-adds of intra-node data read local memory instead of making uncoalesced reads over NVLink, and raised the channel count, 64 channels being best in 72.5% of sparse all-gather cases and 32 otherwise; in an all-reduce ablation at 16 GPUs, 512 MiB, and 95% sparsity, going from 8 to 64 channels gave 3.80 $\times$, reduce-scatter thresholds 1.19 $\times$, the all-gather threshold 1.08 $\times$, and Pici instead of COO 1.37 $\times$, only the final configuration beating NCCLX.
 4. Showed on 16 GPUs with 2048 MiB buffers that Pici all-reduce is 1.19 $\times$ faster than COO at 99% sparsity and 1.33 $\times$ at 95%, where COO is slower than dense, while COO all-gather stays competitive down to about 90% sparsity owing to its simpler decompression, and found all-gather speedups at a fixed 32 MiB per-GPU input to grow with GPU count, 3.22 $\times$, 3.94 $\times$, 4.51 $\times$, and 4.88 $\times$ at 99% sparsity on 8 to 64 GPUs, whereas at 256 MiB on 64 GPUs all-reduce and reduce-scatter drop to 0.81 $\times$ and 0.77 $\times$, which the authors attribute to dense 64 KiB slices near the Hockney crossover $\alpha/\beta \approx 60.8$ KiB for 2.49 μ s latency and 25 GB/s.
 5. Pruned data-parallel gradients in Megatron-LM to 99% sparsity by sampling-based top- k , estimating the threshold from 0.1% of the gradients, with leaky error feedback in a Triton kernel, for a GPT-2 XL-based 1.5B model on 40 GB nodes and a StarCoder2-3B-based 3.3B model cut to 28 layers on 80 GB nodes, trained on WikiText-103 with sequence length 512 and micro-batch 2; SpCCL was at least 13% faster on eight 40 GB GPUs and 22% on 64 80 GB GPUs, peaking at 16% and 26% on 16 GPUs, and on 32 GPUs over 80 iterations all-reduce was over 2 $\times$ faster without error feedback, pruning cost 1.5 s and 3 s, all-reduce time fell by over 40% in every case, and error feedback added overhead to both pruning and all-reduce.

2026HGR

Claude (September 2026)

Summary

Gyromoment (GM) models expand the local δf gyrokinetic ion distribution in Hermite–Laguerre moments and need a closure when few moments are kept. For electrostatic flux-tube turbulence with adiabatic electrons, the authors expand the gyroaverage kernels $\bar{K}_n = (\tau \ell_\perp)^n e^{-\tau \ell_\perp} / n!$ for small ion-to-electron temperature ratio τ , truncate at moment degree $p + 2j \leq 2$ (density, parallel velocity, and parallel and perpendicular temperatures), and keep $\mathcal{O}(\tau)$ terms only in the density equation, an ad hoc mixed-order truncation. After a gyrocentre-to-particle transformation and rescaling of fields and gradients with τ , this hot-electron-limit (HEL) system reproduces the Z-pinch fluid model of Ivanov et al.; an empirical factor $c_f = 4$ matches the Dougherty collisionality to their Landau-based one. Gyacommo runs at $\tau = 10^{-3}$ reproduce Ivanov's linear growth rates, two-dimensional heat fluxes with low-collisionality blow-up, and three-dimensional heat-flux plateaus versus parallel box length. In Cyclone Base Case s - α geometry, compared with $\tau = 1$ gyrokinetic runs, the closure overpredicts peak growth rates (about 0.9 versus 0.25) and heat flux but preserves the temporal dynamics; it shows no Dimits shift and misses gyrokinetic transport suppression. The authors attribute the weak zonal-flow regulation to the missing higher-order moments.

Contributions

1. Showed that Z-pinch ITG growth rates become independent of τ for $\tau \lesssim 10^{-2}$ and coincide for 4 and 9 GMs, whereas removing N_i^{20} or N_i^{01} changes them, identifying the four-moment set as closed in the hot-electron limit.

2. Traced the residual linear mismatch with Ivanov's growth rates to higher-order τ terms of Gyacomo's Dougherty operator, using an eigenvalue solver of the HEL–GM system that agrees better with Ivanov at finite collisionality and matches the gyrokinetic runs exactly without collisions.
3. Found that lowering the collisionality by 20% from $\chi = 0.2$ at $\kappa_T = 1.2$ destroys the zonal flows and makes the heat flux blow up while barely changing linear growth rates, attributing the blow-up to the nonlinear saturation mechanism and noting that gyrokinetic studies report no such state.
4. Related the three-dimensional heat-flux decrease with parallel domain length to eddy self-interaction through periodic parallel boundaries and parallel decorrelation, and showed that stronger gradients excite slab ITG modes whose short parallel scales make the saturated flux sensitive to parallel resolution.
5. Compared Z-pinch and s - α geometries at identical parameters, finding lower transport in the Z-pinch despite larger growth rates for both models, and proposed that uniform curvature drives one coherent zonal-flow mode whereas z -dependent tokamak curvature favours competing zonal scales.

Comments

- §2, after Eq. (2.1): the non-adiabatic part of the distribution function is defined as $h_i = g_i - \langle \phi \rangle$, whose Hermite–Laguerre moments are, by Eq. (2.6), $N_i^{pj} - \hat{K}_j \phi \delta_{p0}$, whereas Eq. (2.11) defines the non-adiabatic GMs as $n_i^{pj} = N_i^{pj} + (q_i/\tau) \hat{K}_j \phi \delta_{p0}$, which the expansions (2.20)–(2.22), including the discussed τ^{-1} term, use; the sign and the factor q_i/τ differ, and (2.11) is the form the derivation relies on.
- §2.1: the evolved wavenumbers are given as $k_x = 2\pi m N_x/L_x$ and $k_y = 2\pi n N_y/L_y$ with $n = 1, \dots, N_y/2$; with the §3 values $L_y = 150$ and $N_y = 256$ the smallest printed k_y would be $2\pi \cdot 256/150 \approx 10.7$, far above the unstable range $k_y \rho_s \lesssim 2.5$ of Figs. 1 and 3, and excluding $n = 0$ would remove the $k_y = 0$ component required by the $\bar{\phi}_{yz}$ term of Eq. (2.16) and by the zonal flows of Fig. 7(a); the evident intended modes are $k_x = 2\pi m/L_x$ and $k_y = 2\pi n/L_y$ with $n = 0, \dots, N_y/2$.
- §2.2, Eqs. (2.28)–(2.29): applying Eq. (2.10), the rule that yields the printed $-C_{\parallel}^B T_{\perp}^*$ in Eq. (2.27), gives the parallel terms $\sqrt{\tau}[\sqrt{3}(\hat{C}_{\parallel} - C_{\parallel}^B)q_{\parallel}^* + \sqrt{2}\hat{C}_{\parallel}u_{\parallel}^* - \sqrt{2}C_{\parallel}^B q_{\perp}^*]$ for $(p, j) = (2, 0)$ and $\sqrt{\tau}[(\hat{C}_{\parallel} - 2C_{\parallel}^B)q_{\perp}^* + C_{\parallel}^B u_{\parallel}^*]$ for $(p, j) = (0, 1)$, but the printed equations lack $-\sqrt{2}\tau C_{\parallel}^B q_{\perp}^*$ and $-2\sqrt{\tau} C_{\parallel}^B q_{\perp}^*$; these terms are discarded in Eqs. (2.32)–(2.33) and vanish in the Z-pinch, so the closed model is unaffected.
- §2.2, Eq. (2.29): Eq. (2.15) gives $\mathcal{D}_N^{01} = R_N i k_y \hat{K}_1 \phi = \tau R_N i k_y \ell_{\perp} \phi + \mathcal{O}(\tau^2)$ and Eq. (2.14) gives $\mathcal{D}_T^{01} = R_T i k_y (2\hat{K}_1 - 2\hat{K}_2 - \hat{K}_0) \phi = R_T i k_y (-1 + 3\tau \ell_{\perp}) \phi + \mathcal{O}(\tau^2)$, so the $\mathcal{O}(\tau)$ gradient factor is $(3R_T + R_N)$ rather than the printed $(R_T + 3R_N)$; likewise Eq. (2.9) with $L_1^2 = L_0 - 2L_1 + 2L_2$ and $\hat{K}_0 = 1 - \tau \ell_{\perp} + \dots$ gives $\tau\{\ell_{\perp} \phi, n^* - 3T_{\perp}^* + 2P_{\perp}^{*}\}$ rather than the printed $-2T_{\perp}^*$; both terms are dropped in Eq. (2.33).
- §2.2, Eq. (2.30): the polarization factor is printed as $1 - 2(\ell_{\perp} - \tau \ell_{\perp}^2)$, but Eq. (2.16) with $q_i = 1$, $\Gamma_0 = \sum_n \hat{K}_n^2$, and Eqs. (2.17)–(2.18) give $\sum_n \hat{K}_n^2 = 1 - 2\tau \ell_{\perp} + 3\tau^2 \ell_{\perp}^2 + \mathcal{O}(\tau^3)$ and hence $1 + 2\ell_{\perp} - 3\tau \ell_{\perp}^2$; the linear term has the wrong sign for $\ell_{\perp} = k_{\perp}^2/2$, the convention of the same equation's right-hand side $n^* + \tau \ell_{\perp} (T_{\perp}^* - n^*)$, and the printed $\tau \ell_{\perp}^2$ coefficient corresponds to keeping only $\hat{K}_0^2$.
- §2.3, Eq. (2.38): the gyroradius vector is defined as $\rho = \sqrt{2\tau w_{\perp}} \hat{b}$, parallel to the magnetic field, which contradicts the properties $\langle \rho \rangle = 0$ and $\langle \rho \rho \rangle = \tau w_{\perp} \mathbf{I}_{\perp}$ stated in the same paragraph; the intended vector is perpendicular to $\hat{b}$ and rotates with the gyroangle.
- §3: the Gyacomo gradient is said to be scaled as “ $R_T = \tau \kappa_T$ ”, which conflicts with the rescaling $\kappa_T = \tau R_T/2$ of §2.3, with “despite an increasing temperature gradient R_T ” in the same paragraph as τ decreases, and with §5, where $\kappa_T = 3.5$ “corresponds to $R_T = 7$ for $\tau = 1$ ”, as does the Fig. 13 axis label “ $R_T, 2 \times \kappa_T$ ”; the intended scaling is $R_T = 2\kappa_T/\tau$.
- §3 and Fig. 5: §3 states that “Figure 5 compares the heat fluxes obtained in the HEGS with the results from Ivanov et al. (2022)”, and §2.3 says c_f was calibrated against Ivanov, Schekochihin & Dorland (2022) “(see figure 5)”, but the Fig. 5 caption attributes the reference data to Ivanov et al. (2020), as §6 does for the $c_f = 4$ calibration; the source of the two-dimensional reference heat fluxes is unresolved.
- §4 and Fig. 8: the caption gives $L_{\parallel} = 4$ (top) and $L_{\parallel} = 128$ (bottom), whereas the text describing the figure gives “ $L_{\parallel} = 8$ ” for the short domain and “ $L_{\parallel} = 32$ ” for the longer one; the plotted ranges $z/L_B \approx [-2, 2]$ and $[-32, 32]$, with $L_{\parallel} = 2L_z$ (§3) as confirmed by Fig. 9, whose $L_{\parallel} = 32$ panels span $z/L_B \approx [-8, 8]$, correspond to $L_{\parallel} = 8$ and $L_{\parallel} = 128$.
- §4: the weak- and strong-turbulence comparison is introduced as “ $\kappa_T = 0.38$ and $\kappa_T = 0.8$ ”, but the Fig. 9

caption gives $\kappa_T = 0.36$ for the weak case, the value used for the weak-turbulence runs in §3 and Fig. 8; the intended value is 0.36.

2026HGW

ChatGPT (July 2026)

Summary

This paper develops computable a posteriori error control for the integrating factor midpoint method applied to linear second-order evolution equations, including wave-type problems $u'' + Au = f$ with a positive self-adjoint operator A . A continuous piecewise-linear approximation first yields residual-based bounds that are suboptimal in time. The authors then construct a continuous piecewise-quadratic reconstruction and use an energy argument to recover optimal-order bounds in L^∞ -in-time and energy-in-space norms for both displacement and velocity. Because the estimators depend on the discrete solution and problem data, they support an adaptive step-size algorithm without requiring specialized evaluation of matrix-exponential actions. Four numerical examples confirm second-order convergence of the reconstructed estimator and show that the controller refines near localized temporal features and coarsens elsewhere. The analysis assumes a linear problem whose first-order operator generates a bounded C_0 semigroup; extensions to nonlinear equations are not established.

Contributions

1. Derived residual-based a posteriori bounds for an integrating factor midpoint discretization of second-order evolution equations.
2. Identified the loss of optimal order caused by the direct piecewise-linear approximation.
3. Introduced a quadratic time reconstruction that restores optimal-order error estimates.
4. Converted the reliable estimator into an adaptive time-step acceptance and update strategy.
5. Demonstrated convergence and localized refinement on smooth and sharply varying wave-type tests.

2026HHLZ

ChatGPT (July 2026)

Summary

Deep Picard Iteration solves high-dimensional semilinear and fully nonlinear parabolic PDEs by converting each Picard step into supervised regression for solution values and gradients. Feynman–Kac sampling supplies value labels and Bismut–Elworthy–Li formulas supply gradient labels. The authors prove that naive Malliavin-weight gradient estimators have infinite variance and remove their singular components with control variates, yielding finite-variance training targets. Warm-started neural networks then regress the updated functions, with data generation separable across samples. On 100-dimensional Burgers-type, Hamilton–Jacobi–Bellman, and fully nonlinear equations, the method is more accurate and less sensitive to the gradient-loss weight than the tested PINN, Deep BSDE, diffusion-DBSDE, and DBDP baselines, especially for stronger nonlinearities and longer horizons. A ten-dimensional reverse-SDE example recovers a multimodal target distribution, while the corresponding 100-dimensional gradient remains insufficient for accurate sampling. Experiments use synthetic problems with known solutions on one GPU; convergence conditions for fully nonlinear Picard iteration, boundary handling, training-distribution design, and demonstrated parallel scaling remain open.

Contributions

1. Recast nonlinear high-dimensional PDE solution as sequential value-and-gradient regression.
2. Combined Feynman–Kac and Bismut–Elworthy–Li labels in a practical neural method.
3. Proved infinite variance of naive gradient estimators and derived finite-variance controls.
4. Improved accuracy and hyperparameter robustness across several 100-dimensional benchmarks.
5. Demonstrated modest gradient-label overhead and embarrassingly parallel data generation.

2026HLLS

ChatGPT (July 2026)

Summary

This paper develops an adaptive variational physics-informed neural network (VPINN) method for stationary Navier-Stokes equations. Its a posteriori estimator combines projection errors in the PDE terms, data oscillation, VPINN loss, and element and interelement residuals. Under regularity and proximity assumptions, the authors prove that the estimator bounds the velocity H^1 and pressure L^2 error from above and is locally efficient up to projection and data-oscillation terms. Two-dimensional tests use Kovasznay flow and a singular solution on an L-shaped domain. The estimator tracks the true error, identifies steep-gradient and singular regions, and drives Dörfler mesh refinement. With comparable quadrature-point counts, adaptive meshes produce smaller errors and estimator values than uniform refinement. On very fine meshes the estimator can stagnate, which the authors associate with mesh-dependent local efficiency and difficulty optimizing the nonconvex VPINN loss.

Contributions

1. Constructed a locally computable four-part a posteriori estimator for stationary Navier-Stokes VPINNs.
2. Proved global reliability for the combined velocity H^1 and pressure L^2 error near a regular solution.
3. Established local efficiency estimates for volume residuals, divergence defects, flux jumps, and VPINN loss.
4. Integrated the estimator with Dörfler marking to adapt the VPINN test-space mesh.
5. Demonstrated improved accuracy over uniform refinement for steep-gradient and singular benchmark solutions.

2026HM

ChatGPT (July 2026)

Summary

This preprint develops a priori adaptive algorithms for estimating finite blow-up times of autonomous ordinary differential equations. Rather than following an unbounded solution indefinitely, each method selects a tolerance-dependent radius whose hitting time approximates the true blow-up time, then applies forward Euler steps scaled by the sensitivity of that auxiliary hitting time. Separate algorithms cover scalar and multidimensional systems under stated growth, regularity, and monotonicity assumptions. Both attain $O(\epsilon)$ blow-up-time error at $O(\epsilon^{-1})$ computational cost. For representative scalar problems, the analysis shows that uniform first-order stepping incurs at least an additional logarithmic cost factor. Numerical tests include polynomial, exponential, and logarithmically modified growth laws, uncoupled and coupled systems, and a semidiscretized reaction–diffusion equation. Experiments recover the predicted rates and illustrate extensions to higher-order integrators and some problems outside the main assumptions.

Contributions

1. Recast blow-up-time estimation as a tolerance-controlled auxiliary hitting-time problem.
2. Designed sensitivity-weighted adaptive algorithms for scalar and multidimensional autonomous ODEs.
3. Proved $O(\epsilon)$ error with $O(\epsilon^{-1})$ work for both algorithms.
4. Quantified the asymptotic efficiency gain over uniform first-order time stepping.
5. Validated the theory across diverse ODEs and a semidiscretized reaction–diffusion model.

2026HNND

ChatGPT (July 2026)

Summary

The paper develops an analytic fluid model for modulational instability when an intense electromagnetic pulse drives relativistic electron motion in a magnetized plasma. Ions are treated as an immobile neutralizing background, and

perturbation analysis of the electron magnetohydrodynamic equations reduces the coupled electromagnetic and plasma-wave dynamics to a nonlinear Schrödinger equation (NLSE). The dispersion and nonlinear coefficients determine whether envelope perturbations grow, and the authors derive an explicit maximum growth rate in terms of plasma parameters and wave intensity. Special parameter limits yield electron-driven modes analogous to Alfvén and magnetoacoustic waves. Interpreting the inverse-scattering solution, the work treats solitons as nonlinear normal modes accompanied by dispersive wave trains. A Bogoliubov–Mitropolsky perturbation analysis then separates real and imaginary coefficient effects: a phenomenological nonlinear Landau-damping term conserves wave quanta while accelerating an initially stationary soliton, whereas a complex growth/damping term controls amplitude through competition between linear growth and nonlinear stabilization. Because nonlinear Landau damping is introduced phenomenologically rather than derived from kinetic theory, that part of the model is a scoped fluid approximation and requires future kinetic justification.

Contributions

1. Reduced relativistic electron-fluid dynamics in a magnetized plasma to an NLSE for the laser-driven envelope.
2. Derived the modulational-instability criterion and an analytical maximum growth rate.
3. Identified limiting electron modes with Alfvén-like and magnetoacoustic-like behavior.
4. Interpreted the NLSE solitons as nonlinear normal modes within the inverse-scattering solution.
5. Distinguished phenomenological nonlinear Landau damping from complex growth/damping effects on soliton motion and amplitude.

2026HRT

ChatGPT (July 2026)

Summary

Collocated-grid solvers are convenient for complex magnetohydrodynamic (MHD) geometries, but their pressure and electric-potential Poisson equations can generate checkerboard modes, while standard stabilization introduces unwanted dissipation. For low magnetic Reynolds number MHD, this paper combines a symmetry-preserving discretization with runtime checkerboard detection and negative-feedback predictor fields. A volume-weighted adjoint relation between face-to-cell and cell-to-face interpolation retains the discrete operator symmetries responsible for conserving mass, momentum, kinetic energy, and charge density. The feedback activates only as checkerboarding develops, balancing suppression against numerical dissipation. A laminar MHD Taylor–Green vortex shows that the dynamic predictor avoids checkerboarding on skew meshes while remaining minimally dissipative on uniform meshes. Three-dimensional turbulent duct-flow tests with insulating and conducting walls show that non-symmetry-preserving alternatives can distort statistics or spuriously laminarize flow on skew grids. The benefit of dynamically adjusting the electric-potential predictor was less conclusive and requires tests with stronger charge-density fluctuations.

Contributions

1. Combined symmetry-preserving MHD operators with adaptive checkerboard suppression on collocated grids.
2. Derived volume-weighted adjoint interpolation that retains discrete conservation properties.
3. Introduced a runtime checkerboard metric and negative-feedback predictor mechanism.
4. Demonstrated low-dissipation behavior on uniform and skew Taylor–Green vortex meshes.
5. Produced accurate turbulent duct-flow results where a comparison method distorted or laminarized the flow.

2026HS

ChatGPT (July 2026)

Summary

This paper constructs and analyzes local time-integration methods for discontinuous Galerkin discretizations of two-field Friedrichs systems, including Maxwell and acoustic wave equations. These problems become locally stiff when only a small mesh region is highly refined or has large wave speed. The method applies leapfrog integration to the nonstiff

degrees of freedom and modifies the evolution locally through a general filter function. Polynomial Chebyshev filters yield fully explicit local time stepping, while a rational filter recovers a known locally implicit method. The authors establish sufficient filter conditions for stability under a CFL restriction independent of the fine mesh region and prove second-order convergence in time with optimal spatial order. Maxwell-equation experiments validate the estimates and show that the explicit leapfrog-Chebyshev method is most efficient when refinement is confined to a small portion of the mesh.

Contributions

1. Formulated a leapfrog-based local time-integration family for central-flux discontinuous Galerkin discretizations of two-field Friedrichs systems.
2. Unified fully explicit local time stepping and locally implicit integration through a common filter-function framework.
3. Constructed Chebyshev polynomial filters that remove the fine-region restriction from the global CFL condition.
4. Proved stability, second-order temporal convergence, and optimal spatial convergence under verifiable filter assumptions.
5. Showed on Maxwell problems that the explicit method is efficient when stiffness is confined to a small refined region.

2026HS2

ChatGPT (July 2026)

Summary

This paper recasts function extrapolation as projection onto a certified feasible set. An anchor is an auxiliary function with a bound on its distance from the unknown target over the extrapolation domain Ξ ; one or more anchor balls therefore constrain admissible predictions directly where data are absent. Projecting any baseline predictor onto such a set is proved not to increase its Ξ -error and to improve it strictly when the predictor lies outside the set, with quantitative improvement bounds. To construct certificates from in-domain error, the authors derive a tight spectral amplification constant from the largest eigenvalue of a Ξ -Gram matrix and a more numerically robust inner-domain bound. They also model error directions probabilistically to obtain less conservative, confidence-controlled radii. Polynomial, geomagnetic-field, spherical-harmonic, and nonlinear-oscillator experiments illustrate the guarantees and often reduce extrapolation error. The guarantees remain conditional on valid anchor radii; probabilistic certificates additionally depend on the assumed directional error model.

Contributions

1. Formulated model-independent extrapolation as projection onto intersections of anchor-defined feasible balls over Ξ .
2. Proved non-worsening and strict-improvement results, including explicit bounds based on the predictor's distance from a feasible ball.
3. Derived a Rayleigh–Ritz-tight spectral condition number relating in-domain and extrapolation errors.
4. Developed an inner-domain certificate for improved numerical robustness and probabilistic radii with prescribed confidence.
5. Tested simple and data-derived anchors across synthetic, geomagnetic, manifold, and differential-equation examples.

2026HTSD

ChatGPT (July 2026)

Summary

This paper converts Gaussian Bayesian bolometric tomography into a real-time diagnostic for TCV. Because radiated power over any chosen plasma region is a weighted sum of emissivity pixels, its posterior estimate can be written as a linear combination of bolometer measurements. The expensive coefficients are precomputed from the planned FBT magnetic equilibria, fixed noise and prior hyperparameters, and detector selection; no learned mapping or emissivity-profile training set is required. The method simultaneously estimates total, core, divertor, and main-chamber radiation with Bayesian credible intervals. Tests cover 50 discharges across lower- and upper-single-null, negative-triangularity, X-point-target, and long-legged configurations. Channel selections based on recent failure history yield mean correlations of 0.970–0.994 with offline tomography and region-wise mean RMSEs near 10.5–12.6 kW. Excluding several failure-prone channels would have avoided bad data in about 97% of the campaign. Coefficient generation takes roughly 10 s per equilibrium, comfortably within TCV’s inter-discharge interval; integration into feedback control remains future work.

Contributions

1. Derived real-time regional radiated-power estimators as linear combinations of bolometer signals.
2. Proved equivalence to conventional Gaussian-posterior tomographic estimates under the stated model assumptions.
3. Incorporated discharge-specific magnetic geometry through coefficients precomputed from FBT equilibria.
4. Validated total and regional estimates over 50 discharges spanning five magnetic configurations.
5. Developed and quantified detector-failure strategies suitable for routine real-time deployment.

2026HV

Claude (September 2026)

Summary

Suprathermal tails beyond a Maxwellian core are common in weakly collisional space plasmas, but how time-dependent bulk flows shape them is less studied. The authors solve a one-dimensional, spatially periodic Fokker–Planck equation with spatial advection, velocity-space diffusion representing wave–particle interactions, Bhatnagar–Gross–Krook (BGK) collisional relaxation, and a phenomenological drift $A_{\text{eff}} = (v - v_{st})/\tau_0$ relative to a streaming velocity v_{st} , an anti-drag representing energization that outpaces relaxation toward the flow frame. The streaming velocity is constant or grows or decays exponentially over $5\tau_0$, and the suprathermal fraction ζ counts particles with $|v| \geq 5v_{th}$. For constant $v_{st} \leq 0.1v_{th}$, broadening is nearly isotropic, driven by advection and diffusion, whereas for $v_{st} \gtrsim 0.5v_{th}$ streaming-driven acceleration dominates and the distributions become asymmetric; ζ grows on a timescale of order τ_0 and saturates at a level that increases with v_{st} . For time-dependent flows, early acceleration follows the instantaneous streaming speed, but saturated fractions barely differ from the constant flow with the same time average, so the authors conclude that the time-averaged streaming velocity controls the suprathermal yield. They regard the drift as a coarse-grained closure rather than a microscopic force, use BGK instead of a Coulomb collision operator, and call the description simplified, one-dimensional, and isotropic.

Contributions

1. Wrote the drift coefficient as the difference of an energization rate and a rate of relaxation toward the flow frame, $\tau_0^{-1} = \tau_{acc}^{-1} - \tau_{rel}^{-1}$, so that it acts as anti-drag for $\tau_{acc} < \tau_{rel}$ and as drag otherwise, and justified its linear dependence on $v - v_{st}$ as the leading term of an expansion about the streaming frame.
2. Modeled stochastic acceleration with $D(v) \propto v^{(s-1)/3} \exp(-v/v_{max})$, taking $s = 2$ as baseline and $s = 3/2$ for comparison from whistler-turbulence transport models with a cutoff at $v_{max} = 10v_{th}$, and chose $\tau_{coll} = 10\tau_0$, with $100\tau_0$ as a check, from a solar-wind comparison of dynamical and electron–electron collision times.

3. Discretized the equation with Crank–Nicolson time stepping and second-order central differences, evaluating the drift as the central difference of the flux $J_j = A_{\text{eff},j} f_j$ and the diffusion term in expanded product form averaged over two time levels, with steps of 10^{-3} times the domain size, v_{max} , and τ_0 .
4. Found for constant streaming velocities from 0 to $2v_{th}$ that the suprathermal fraction at $t = 5\tau_0$ increases approximately linearly with v_{st} and is nearly unchanged for $s = 3/2$ or $\tau_{\text{coll}} = 100\tau_0$, with small parameter-dependent deviations only for $v_{st} \lesssim 0.5v_{th}$, where diffusion matters more early on.
5. Compared flows that change tenfold over $T = 5\tau_0$ with $\Gamma_{st} = \ln 10/T$, finding that the decelerating flow produces suprathermal particles fastest while the accelerating flow first broadens isotropically before the trend reverses, and argued that static-force treatments suffice for long-term yields whereas transient bursts require time-dependent flows.

Comments

- §2.2, Eq. (6): the diffusion term enters as $+\partial_v(D\partial_v f/\tau_0^2)$ on the left-hand side of an equation set to zero, so that $\partial_t f = -\partial_v(D\partial_v f)/\tau_0^2 + \dots$, which for a positive diffusion coefficient is backward diffusion, whereas the conservation form used for the drift, $+\partial_v(A_{\text{eff}}f)$, requires the diffusive flux $-D\partial_v f/\tau_0^2$ and the text describes this term as stochastic acceleration that forms suprathermal tails; the intended sign is evidently negative.
- §2.2, Eq. (6): the collision term is printed as $\partial_v[v(f - f_{eq})/\tau_{\text{coll}}]$, a velocity-space divergence, on which the text relies when it states that on integration over velocity the “collisional relaxation terms reduce to boundary contributions”, but the same section defines the BGK term as $-(f - f_{eq})/\tau_{\text{coll}}$, which is not a divergence; which form was solved is unresolved.
- §2.2, Eq. (7): $D(v)$ is written in terms of v , but the velocity domain is $[-v_{\text{max}}, v_{\text{max}}]$ (Eq. (23)); for $v < 0$, $(v/v_{th})^{(s-1)/3}$ is not real for $s = 3/2$ and $\exp(-v/v_{\text{max}})$ exceeds unity, so the cutoff enhances rather than suppresses diffusion at large negative velocities, contrary to the stated role of v_{max} as the velocity “beyond which the efficiency of resonant wave-particle interactions is suppressed”; whether $|v|$ is intended is unresolved.
- §2.2, Eq. (9): evaluating the printed expression in Gaussian units with the stated $n_e = 5 \text{ cm}^{-3}$, $T_e = 10^5 \text{ K}$, and $\ln \Lambda = 10$ gives $\tau_{ee} \approx 1.7 \times 10^5 \text{ s}$ rather than $3 \times 10^4 \text{ s}$, and hence $\tau_{ee}/\tau_{\text{dyn}} \approx 87$ rather than 15 with Eq. (8); the weak-collisionality conclusion is unaffected, and the source of the discrepancy is unresolved.
- §2.3 and §3: with $\Gamma_{st}T = \ln 10$ and the initial speeds shown in Fig. 2, $v_{i,1} = 0.1v_{th}$ and $v_{i,2} = v_{th}$, Eqs. (13) and (14) both give $\langle v_{st} \rangle = 0.9v_{th}/\ln 10 \approx 0.39v_{th}$, but §3 and the Fig. 5 caption state $\langle v_{st} \rangle \approx 0.41v_{th}$, and the Model 3 line in Fig. 2 lies at about $0.41v_{th}$, so the plotted Model 3 differs from the time average that Eq. (12) prescribes; which value was simulated is unresolved.
- §2.1 and Figs. 3 and 5: the drift $A_{\text{eff}} = +(v - v_{st})/\tau_0$ pushes particles away from v_{st} (“outward transport in velocity space”, Fig. 1(c)), so for $v_{st} = 2v_{th}$ it drives the roughly 98% of the initial Maxwellian with $v < v_{st}$ toward negative v , and for $v_{st} = 0$ it alone would stretch the distribution by $e^{t/\tau_0} \approx 20$ by $t = 3\tau_0$; yet Figs. 3 and 5 show the distributions shifting toward positive v , with tails enhanced at positive and reduced at negative v , which §3 attributes to the streaming drift, and Fig. 3(a) shows only modest broadening for $v_{st} = 0$; unresolved.
- Figs. 3 and 4: at $t/\tau_0 = 3$, Fig. 4(a) gives $\zeta \approx 0.06$ for $v_{st} = 0$ and $\zeta \approx 0.21$ for $v_{st} = 2v_{th}$, but in Fig. 3(a) the $v_{st} = 0$ distribution, whose total is about 5 (initial peak about 2), falls below the 10^{-2} axis floor near $|v| = 5.5v_{th}$, which bounds ζ by about 0.02 even if $f = 10^{-2}$ out to v_{max} , and the $v_{st} = 2v_{th}$ curve integrates to $\zeta \approx 0.08$; Figs. 5(a) and 6 at $t/\tau_0 = 2$ disagree similarly, with bounds of 0.02–0.03 from the distributions against 0.03–0.07; unresolved.

2026HVFO

ChatGPT (July 2026)

Summary

This paper extends Chebyshev-polynomial time propagation from Hermitian Hamiltonians with spectra in $[-1, 1]$ to arbitrary non-Hermitian matrices with complex spectra. Because the Chebyshev series for the exponential converges over the whole complex plane, the restriction is numerical rather than analytic. Outside $[-1, 1]$, Chebyshev polynomials grow exponentially, so cancellation with decreasing Bessel coefficients exposes floating-point rounding error. Using Bernstein ellipses, the authors derive a large- $t\rho$ bound, $\tilde{\Delta} < 2et\rho \exp(t\rho/2)$, and invert it to obtain a safe spectral

radius for a prescribed tolerance and time step. A Jordan-form argument extends the expansion to diagonalizable and nondiagonalizable square matrices, while a matrix-vector recurrence avoids forming the propagator. Simulations of Gaussian wave packets in the Hatano-Nelson model under periodic and open boundaries agree with analytic evolution when the step size is chosen from the spectral geometry. Stable non-unitary propagation is therefore achievable by reducing the time step as the enclosing Bernstein ellipse grows.

Contributions

1. Extended the Chebyshev expansion of the exponential to arbitrary non-Hermitian square matrices.
2. Identified floating-point rounding as the practical limitation outside the conventional spectral interval.
3. Derived a rounding-error bound using the Bernstein ellipse enclosing the complex spectrum.
4. Provided a tolerance criterion for choosing a safe spectral radius or time step.
5. Validated matrix-vector propagation on the Hatano-Nelson model for open and periodic boundaries.

2026HXHW

ChatGPT (July 2026)

Summary

Regular two-dimensional blocking wastes parallelism in sparse LU factorization when fill is concentrated near the diagonal and lower-right region. This paper introduces a diagonal-block feature that measures cumulative nonzero distribution locally and globally, then uses it to choose unequal block sizes. Dense regions receive fine blocks and sparse regions coarse blocks, balancing work both among blocks at one dependency-tree level and across levels. The method is integrated into a PanguLU-style numerical factorization and evaluated on NVIDIA A100 GPUs. Across the test matrices, it reaches geometric-mean speedups of $1.50\times$ over PanguLU and $3.32\times$ over SuperLU_DIST on one GPU, and $1.40\times$ and $3.84\times$, respectively, on four GPUs. Benefits are largest for strongly irregular fill patterns; nearly uniform diagonal distributions can show little gain. Irregular metadata and lookups increase preprocessing cost, which the authors accept because numerical factorization dominates total time.

Contributions

1. Defined a diagonal-block feature that captures local sparse-fill distribution.
2. Designed an irregular blocking rule with fine dense blocks and coarse sparse blocks.
3. Balanced numerical work within and across dependency-tree levels.
4. Demonstrated geometric-mean speedups on one and four NVIDIA A100 GPUs.
5. Characterized when irregularity yields gains and when preprocessing overhead or uniform structure limits them.

2026HY

ChatGPT (July 2026)

Summary

Vlasov–Poisson solutions in the quasi-neutral regime oscillate on the small plasma period, making standard explicit and implicit schemes expensive or inaccurate. This paper introduces analytically solvable asymptotic-preserving (ASAP) splitting schemes by reformulating Vlasov–Poisson as a constrained Vlasov–moment–Ampère system. Moment equations reduce the nonlinear field subproblem to an analytically solvable ordinary differential equation, and characteristics then give the distribution function; both split subproblems therefore have no temporal discretization error. The schemes preserve the plasma frequency, temporal waveforms, and quasi-neutral limit. At the time-discrete level they also conserve global mass and energy, preserve a maximum principle and Casimir invariants, and retain time-reversal symmetry for Vlasov–Ampère equations. Experiments up to $3D3V$ reproduce highly oscillatory dynamics with steps substantially larger than the plasma period and outperform classical Hamiltonian splitting. Rigorous error bounds showing dependence on the Debye length remain future work.

Contributions

1. Reformulated Vlasov–Poisson into constrained subproblems that admit analytical time evolution.
2. Constructed splitting schemes that preserve plasma frequency, waveform, and the quasi-neutral asymptotic limit.
3. Established discrete conservation of mass and energy together with maximum-principle and Casimir preservation.
4. Extended the construction to Vlasov–Ampère equations while retaining time-reversal symmetry.
5. Demonstrated accurate large-step simulations, including a six-dimensional phase-space problem.

2026HY2

ChatGPT (July 2026)

Summary

This paper develops a bandwidth-based error theory for scaled generalized Hermite approximations on the real line and generalized Laguerre approximations on the half-line. The projection error is decomposed into spatial truncation, frequency truncation, and exponentially small spectral components, making the scale parameter interpretable as a balance between physical-space coverage and Fourier-space resolution. A transformation between Laguerre and Hermite systems transfers the principal estimate to both bases. The framework extends to derivatives, interpolation, Gaussian quadrature, and Galerkin discretization of a half-line elliptic problem. It predicts behavior missed by classical finite-smoothness estimates: suitable scaling can turn subgeometric convergence into geometric convergence, double algebraic order, explain root-exponential pre-asymptotics, and restore competitive unbounded-domain quadrature accuracy. Numerical experiments support these predictions. The comparison also overturns a simple decay-matching intuition: for a class of Gaussian-decaying analytic functions, two spliced half-line Laguerre expansions are at least competitive with, and can converge faster than, one Hermite expansion, though general superiority remains conjectural.

Contributions

1. Decomposed scaled Hermite and Laguerre projection error into spatial, frequency, and spectral terms analogous to a sampling-bandwidth criterion.
2. Established an equivalence that transfers generalized Hermite estimates to generalized Laguerre approximation.
3. Derived scale-selection rules predicting geometric, root-exponential, and improved algebraic convergence rates.
4. Extended the analysis to derivatives, interpolation, Gaussian quadrature, and a Laguerre–Galerkin half-line model problem.
5. Proved a class-specific comparison in which two spliced Laguerre bases match or outperform one Hermite basis and validated the predicted rates numerically.

2026HYZ

ChatGPT (July 2026)

Summary

High-resolution VPIC simulations follow decaying compressible fast-wave turbulence from magnetohydrodynamic to sub-ion scales in a collisionless plasma. At large scales, angle-dependent transit-time damping transfers compressive-wave energy to particles, creating suprathermal electrons and field-aligned proton beams. The proton resonant velocity grows with plasma beta; realistic-mass-ratio estimates give about $1.3V_A$ at $\beta = 0.5$ and $2.4V_A$ at $\beta = 2$, offering a mechanism for super-Alfvénic beams observed by Parker Solar Probe. At sub-ion scales, quasi-perpendicular fast-mode fluctuations couple to multiple ion Bernstein branches whose dispersion, quasi-electrostatic polarization, and compressibility agree with a plasma-dispersion solver. These waves provide a channel for perpendicular energization and Kappa-like proton tails. At higher beta, stronger transit-time damping may suppress the cascade, while beam-driven instabilities provide an alternative Bernstein-wave source.

Contributions

1. Demonstrated production of suprathermal electrons and proton beams by transit-time damping of compressible turbulence.
2. Connected plasma beta quantitatively to super-Alfvénic proton-beam drift speeds.
3. Resolved multiple ion Bernstein branches generated through fast-wave evolution at sub-ion scales.
4. Verified the waves through their dispersion, electrostatic character, and compressibility.
5. Linked ion Bernstein waves to perpendicular proton energization and proposed beta-dependent competing excitation pathways.

2026HZCN

ChatGPT (July 2026)

Summary

The paper develops a cell-centered curvilinear Lagrangian discontinuous Galerkin method for two-dimensional resistive magnetohydrodynamics. An implicit-explicit time integrator advances hydrodynamic advection explicitly and magnetic diffusion implicitly. Conservative variables use Taylor bases on reference elements, while a locally divergence-free magnetic basis is carried to curved physical cells by a Piola transformation. A multidimensional approximate Riemann solver supplies nodal velocities and forces, and basis-function deformation moves the curvilinear mesh with the material. Tests include Taylor–Green flow, MHD wave families, smooth and discontinuous-resistivity diffusion, coupled advection–diffusion, a rotor, and a magnetized blast. Smooth problems approach third-order accuracy, discontinuous tests capture the expected field and electric-field structure, and curved cells remain robust under severe rotor deformation. Oscillations persist near a slow shock, and a deliberately non-solenoidal diffusion test requires relaxing the divergence-free representation. The method is limited to two dimensions, with three-dimensional bases and hexahedral meshes left for future work; robustness is demonstrated numerically without a complete stability proof.

Contributions

1. Constructed a curvilinear cell-centered Lagrangian DG scheme for resistive MHD.
2. Split hyperbolic hydrodynamics and magnetic diffusion in an IMEX integrator.
3. Preserved element-local $\nabla \cdot \mathbf{B} = 0$ through divergence-free bases and Piola mapping.
4. Combined moving curved meshes, Taylor bases, nodal Riemann solving, and limiting.
5. Demonstrated high-order accuracy and robustness across smooth and shock-dominated tests.

2026HZYZ

ChatGPT (July 2026)

Summary

The paper proposes IFNSO, a single-pass polynomial approximation to the orthogonal polar factor intended to replace repeated Newton–Schulz steps in Muon-style optimization. After transposing a tall matrix and scaling it using $\|AA^\top\|_F^{1/2}$, the method constructs powers of $I - AA^\top / \|AA^\top\|_F$ whose exponents grow exponentially. Learnable coefficients combine these terms in a polynomial designed to approach one near unit singular values, with the final coefficient constrained from a selected extremum. Repeated squaring generates high powers using short-dimension matrix products, reducing expensive long-dimension products from N to one. With polynomial depth $L = 14$, synthetic $128 \times \{128, 512, 1024\}$ tests report orthogonality error 0.040 and increasingly favorable operation counts as width grows. A small MNIST/Muon experiment reports 98.87% accuracy and the fastest loss convergence among four baselines. Evidence remains preliminary: testing uses one RTX 4050, random small matrices, and MNIST, while oscillations, slower convergence near one, and high cost for very wide matrices remain.

Contributions

1. Recast repeated Newton–Schulz compositions as one learnable polynomial for approximate polar-factor orthogonalization.
2. Selected exponentially growing polynomial powers to cover a broad singular-value range with relatively few terms.
3. Introduced a constrained coefficient optimization that fixes the terminal coefficient and stabilizes the remaining coefficients.
4. Used one long-dimension Gram product followed by repeated squaring and aggregation in the shorter dimension.
5. Reported lower error and operation counts on synthetic matrices plus slightly higher MNIST accuracy than the tested baselines.

2026IDR

Claude (September 2026)

Summary

The authors propose a mixed velocity discretization of the 1D–1V Vlasov–Poisson equations with immobile ions. The distribution is split as $f = f_0 + \delta f$: the near-Maxwellian part f_0 is expanded in asymmetrically weighted (AW) Hermite functions, and the non-Maxwellian part δf in Legendre polynomials on a finite interval $[v_a, v_b]$. Requiring the Vlasov derivative $d\delta f/dt$ to be orthogonal to the retained Hermite modes leaves the Hermite equations unchanged and lets the last Hermite coefficient drive every Legendre coefficient, so information passes from f_0 to δf once filamentation reaches the Hermite truncation. The semi-discrete system conserves mass and energy for odd, and momentum for even, Hermite resolution N_H (with symmetric bounds and zero shift), and all three for an unbounded Legendre domain; removing the coupling to the first three Legendre modes restores exact conservation in general. With central differences in space and implicit-midpoint time stepping, the mixed method is more accurate than Hermite-only or Legendre-only runs with equal velocity degrees of freedom and comparable runtime for the two-stream and bump-on-tail instabilities. The authors state that it gains nothing when the Legendre part spans the whole velocity domain, as in linear advection, where recurrence is set by the better expansion.

Contributions

1. Derived the coupled system Eqs. (23)–(25), whose Legendre equations carry the source $-\frac{\alpha}{v_b - v_a} \mathcal{J}_{N_H, m} \sqrt{N_H/2} [\partial_x - \frac{2}{\alpha^2} \partial_x \phi] C_{N_H-1}$ with $\mathcal{J}_{N_H, m} = \int_{v_a}^{v_b} \psi_{N_H} \xi_m dv$, together with a weak Dirichlet penalty and an artificial Lenard–Bernstein-type collision operator acting only on modes of order at least 3.
2. Showed by Hermite parity that, for $u = (v_a + v_b)/2$, $\mathcal{J}_{N_H, 0}$ and $\mathcal{J}_{N_H, 2}$ vanish for odd N_H and $\mathcal{J}_{N_H, 1}$ for even N_H , and confirmed this by quadrature for $-v_a = v_b = 2, 4, 6$, where the nonvanishing integrals shrink by many orders of magnitude as the interval widens.
3. In linear advection with $N_H = N_L = 100$, showed the Legendre part compensating Hermite recurrence from $t \approx 10$ until it recurs itself near $t \approx 18$, and that the alternative closure $C_{N_H} = \sum_m B_m \mathcal{J}_{N_H, m}$ suppresses the nonphysical back-propagation of the Hermite cascade without delaying recurrence.
4. For the two-stream instability, found that enforcing $\mathcal{J}_{N_H, 0} = \mathcal{J}_{N_H, 1} = \mathcal{J}_{N_H, 2} = 0$ changes the distribution by only 0.06% ($N_H = 100$) and 0.28% ($N_H = 101$), and that $N_H = 85$, $N_L = 171$ on $[-2.5, 2.5]$ reproduces the phase-space vortex of a 60,000-point finite-difference reference.
5. For the bump-on-tail instability with Legendre bounds $[4, 15]$, used the weak bulk–beam coupling to justify strong Hermite damping ($\nu_H = 10$), found that the conservation constraint changes the solution by only $6 \times 10^{-8}\%$ ($N_H = 16$) and $3 \times 10^{-8}\%$ ($N_H = 17$), and reproduced the reference beam filamentation with $N_H = 16$, $N_L = 112$.

Comments

- Section 5: “When the Legendre velocity domain is finite (but symmetric), the conservation of total mass and energy (momentum) is satisfied if N_H is even (odd)” reverses the parity derived in Eqs. (35)–(37), where mass

and energy are conserved for odd N_H and momentum for even N_H , and shown in Fig. 7, where $N_H = 101$ conserves mass and energy and $N_H = 100$ conserves momentum; the evident intended reading is odd (even).

- Section 2.4: the text states that “each coefficient of order n interacts with its nearest neighbors ($n \pm 1$), except that the highest Hermite coefficient couples explicitly to all Legendre coefficients”, but the term $\partial_x \phi \sum_{i=0}^{m-1} \sigma_{m,i} B_i$ in Eq. (24) couples each B_m to every lower-order B_i with $m - i$ odd, and Fig. 1, cited in the same sentence, draws such long-range arcs (for example from B_0 to B_3).

2026J

ChatGPT (July 2026)

Summary

This paper couples flux-vector splitting with diagonal-norm upwind summation-by-parts (USBP) discontinuous Galerkin operators in physical space, a fast Fourier spectral treatment of the full Boltzmann collision operator, and first-order implicit-explicit time stepping. The semidiscrete formulation is proved locally and globally conservative for mass, momentum, and energy; unlike entropy-stable DG, it does not guarantee entropy decay but avoids costly two-point entropy-stable fluxes. Its fully discrete vanishing-Knudsen limit becomes an explicit flux-split Euler scheme. One-dimensional accuracy, shock propagation, Couette flow, Sod, and Lax tests span continuum through rarefied regimes, including over 85 million degrees of freedom and 50 billion operations per step. Results show high-order accuracy, shock robustness without limiters in most tests, close agreement with DSMC, classical DG, and entropy-stable DG references, and no extra flux-computation overhead.

Contributions

1. Introduced the first USBP scheme for the nonlinear inhomogeneous full Boltzmann equation using velocity-sign flux splitting.
2. Constructed compatible USBP, quadrature, and dissipation operators and proved discrete conservation of collision invariants.
3. Coupled USBP transport to fast spectral collision evaluation and iteration-free IMEX stepping across Knudsen regimes.
4. Established consistency of the fully discrete kinetic scheme with a flux-split Euler method in the vanishing-Knudsen limit.
5. Demonstrated high-order accuracy, low artificial dissipation, and mostly limiter-free shock stability on five benchmark families.

2026JFHZ

ChatGPT (July 2026)

Summary

The paper presents a start-to-end robustness optimization of a compact free-electron laser driven by a laser wakefield accelerator. Simulated laser and plasma fluctuations—laser-energy jitter, focal-position displacement from wavefront distortion, and shock-front-position variation—are propagated into electron-beam distributions and through the transport line and undulator. A covariance-matrix-adaptation evolution strategy optimizes beamline settings against these fluctuating inputs rather than a single nominal beam. The resulting configuration maintains matching between the electron beam and radiation field and produces approximately 25-nm radiation above 1 μJ over twice the assumed RMS fluctuation ranges. Gain-length and Pierce-parameter analysis explains cases that do not saturate within the selected 4.5-m undulator and indicates that a longer undulator could recover further output. Separate tests retain mean radiation energy above 1 μJ for pointing deviations up to 1 mrad.

Contributions

1. Built a start-to-end model connecting laser and plasma jitter to free-electron-laser output.

2. Quantified electron-beam degradation from three experimentally relevant wakefield-accelerator instabilities.
3. Applied CMA-ES to optimize the transport and undulator configuration for robustness.
4. Demonstrated greater than 1 μJ , roughly 25-nm output across twice-RMS variations.
5. Evaluated pointing-jitter tolerance and related losses to gain length, Pierce parameter, and undulator length.

2026JGS

ChatGPT (July 2026)

Summary

Long-duration two-dimensional OSIRIS particle-in-cell simulations map how magnetization controls quasi-parallel, trans-relativistic electron–ion shocks with Lorentz factor about two. Across $\sigma = 10^{-4.5}$ – 10^{-3} , the upstream precursor changes from Weibel or filamentation dominated at $\sigma \lesssim 10^{-4}$ to Bell dominated at $\sigma \gtrsim 10^{-3}$. Bell turbulence reaches larger magnetic-energy fractions and supports Bohm-like ion acceleration, $E_{\text{max},i} \propto t$; Weibel-scale turbulence yields $E_{\text{max}} \propto t^{1/2}$. Ions retain substantial nonthermal energy fractions, roughly 0.1–0.2, throughout. Electron behavior changes sharply: Bell-generated transverse fields suppress injection, whereas Weibel-dominated shocks heat electrons toward ion equipartition and place comparable fractions of flow energy into nonthermal ions and electrons. Supplementary one-dimensional simulations explain why high-current Bell amplification is capped by the ambient field at low magnetization, allowing Weibel modes to win, while a feedback model identifies low-current Bell and electron–ion Weibel states as steady configurations. The results connect shock microphysics to cosmic-ray acceleration and synchrotron emission in several astrophysical transients.

Contributions

1. Established the Bell-to-Weibel precursor transition between $\sigma \gtrsim 10^{-3}$ and $\sigma \lesssim 10^{-4}$ with long-duration particle-in-cell simulations.
2. Quantified regime-dependent magnetic-field amplification in the upstream shock region.
3. Measured ion acceleration efficiencies of roughly 0.1–0.2 and separated Bohm-like from Weibel-limited energy growth.
4. Demonstrated magnetization-controlled electron injection and energy partition between Bell- and Weibel-dominated shocks.
5. Used saturation simulations and a cosmic-ray-current feedback model to explain the stable precursor configurations.

2026JHJ

ChatGPT (September 2026)

Summary

This paper develops a discontinuous Galerkin solver for nonrelativistic particle kinetics on smooth, time-independent manifolds. Hamiltonian formulations encode curved geometry in either canonical or noncanonical coordinates. With a continuous discrete Hamiltonian, the collisionless scheme conserves particle number and energy; the noncanonical form also requires a continuous Poisson tensor. In canonical coordinates, central fluxes conserve the phase-space L^2 norm, while upwind fluxes make it decay. An implicit Bhatnagar–Gross–Krook (BGK) collision step, which relaxes the distribution toward local equilibrium, uses iterative moment corrections to preserve density, momentum, and energy. Shock and Kelvin–Helmholtz tests on flat, spherical, and hyperbolic geometries, including a rotating sphere, demonstrate the method. In the strongly collisional limit, the shock solution approaches the curved-manifold Euler result. The noncanonical momentum representation can reduce geometry-induced resolution demands, although the paper does not establish the same L^2 stability for it. A general relativistic extension remains future work.

Contributions

1. Derived canonical and noncanonical Hamiltonian kinetic equations for particle motion on smooth manifolds.

2. Constructed a discontinuous Galerkin discretization with discrete particle-number and energy conservation under stated continuity conditions.
3. Proved L^2 conservation for central fluxes and monotone L^2 decay for upwind fluxes in the canonical collisionless scheme.
4. Coupled an implicit BGK operator to iterative corrections that preserve its density, momentum, and energy moments.
5. Demonstrated curved-surface shocks, Kelvin–Helmholtz instability, rotation, and recovery of the Euler limit at high collisionality.

2026JJPH

ChatGPT (July 2026)

Summary

This work introduces a continuum modal discontinuous Galerkin solver for the relativistic Vlasov–Maxwell equations in Gkeyll. A mixed nodal–modal representation of the Lorentz factor and current eliminates weak-form aliasing, proving semidiscrete energy conservation and L^2 stability, while independently mapped four-velocity coordinates span large energy ranges without particle-in-cell sampling noise. In a pulsar-inspired pair discharge, density grows by about 10^3 and particles reach $\gamma \sim 5 \times 10^3$; Gkeyll screens the electric field and resolves a smooth distribution over four decades in four-velocity. A $2x3v$ reconnection run at magnetization $\sigma = 1$ recovers the expected local $dN/d\gamma \propto \gamma^{-4}$ spectrum. Linear instability tests match warm relativistic theory, and temporal energy errors scale as Δt^3 . The method is expensive: the reconnection example costs about 5×10^5 CPU-hours. It presently lacks exact Gauss-law enforcement, strict positivity, photons, and self-consistent pair production.

Contributions

1. Constructed an energy-conserving, L^2 -stable relativistic Vlasov–Maxwell DG method.
2. Introduced mapped four-velocity grids with conservation-compatible geometric factors.
3. Resolved a noise-free pair discharge across large density and velocity ranges.
4. Matched relativistic instability theory and verified temporal energy convergence.
5. Recovered spatially local nonthermal spectra in relativistic magnetic reconnection.

2026JKGH

ChatGPT (July 2026)

Summary

This study demonstrates extreme-ultraviolet free-electron-laser amplification driven by a compact laser wakefield accelerator. A 0.8 J laser, shock-front injection, a stabilized supersonic gas jet, and laser-wavefront filtering generate reproducible monoenergetic electron beams approaching 400 MeV with sub-1% energy spread. The aperture mask halves wavefront fluctuations and improves electron pointing, while a nozzle stilling chamber suppresses gas-density and shock-position variation. Controlled beam loading and wakefield phase rotation minimize energy spread near a 20 pC bunch charge. Start-to-end particle-in-cell, beam-transport, and FEL simulations reproduce the measured diagnostics. Transporting the beam through a two-meter undulator produces radiation near 40 nm with approximately twentyfold nonlinear gain; spectral broadening, K -dependence, and suppression after emittance degradation distinguish amplification from spontaneous emission. Gain remains fluctuation-sensitive, especially to laser and beam pointing stability, but the experiment establishes a route toward compact femtosecond XUV and x-ray light sources.

Contributions

1. Demonstrated approximately twentyfold XUV FEL gain from a laser-wakefield-accelerated beam.
2. Stabilized injection with a sharp density down-ramp and a long-stilling-chamber gas nozzle.
3. Improved laser wavefront and electron pointing using a simple movable aperture mask.

4. Tuned beam loading and phase rotation to obtain narrow-spread monoenergetic beams.
5. Validated the accelerator-to-undulator experiment with a start-to-end simulation chain.

2026JQZ

ChatGPT (July 2026)

Summary

This paper develops explicit extended integrating factor Runge-Kutta schemes of fourth through sixth order for the Allen-Cahn equation that preserve its maximum bound principle without a time-step restriction. The framework relaxes earlier constraints on exponential coefficients and constructs a target-order method recursively from a method one order lower. Rather than solving increasingly complicated Runge-Kutta order conditions directly, it interprets the local truncation requirement through Duhamel's formula and uses Gauss quadrature to determine suitable stages and coefficients. The authors prove maximum-bound preservation and L^∞ -error estimates for the resulting eIFRK4-6 schemes. Two- and three-dimensional tests confirm their formal temporal orders, bound preservation, and favorable energy behavior. In long-time phase-field simulations with large steps, eIFRK6 remains substantially more accurate and stable than lower-order stabilized IFRK schemes, although unconditional discrete energy dissipation is not established.

Contributions

1. Introduced a relaxed extended-IFRK framework for high-order explicit schemes with unconditional maximum-bound preservation.
2. Gave a recursive construction in which a lower-order method supplies nonlinear values at quadrature nodes for the target-order method.
3. Connected Runge-Kutta order requirements to Duhamel-formula local error and used Gauss quadrature to derive eIFRK4-6 formulas.
4. Proved unconditional maximum-bound preservation and optimal temporal L^∞ -error estimates for orders four through six.
5. Demonstrated formal convergence, bound preservation, and robust long-time behavior in two- and three-dimensional phase-field tests.

2026JT

ChatGPT (July 2026)

Summary

Using 1,600 one-dimensional, one-velocity-dimensional Vlasov–Poisson–Fokker–Planck simulations, this paper resolves the evolution of large-amplitude electron plasma waves when the bounce frequency greatly exceeds the electron–electron collision frequency. The dynamics separate into three phases: rapid collisionless trapping and plateau formation; a long quasi-steady detrapping phase governed by collisions and wave–particle interactions; and a return to near-linear Landau damping after the distribution becomes nearly Maxwellian. During the middle phase, collisions counterintuitively strengthen the nonlinear frequency downshift, which can approximately double because the Fokker–Planck contribution to susceptibility is only partly opposed by Vlasov dynamics. Fits quantify the initial shift, shift-growth rate, effective damping, phase lifetime, and final damping across wave amplitude, wavenumber, and collision frequency. The measured middle-phase damping is approximately linear in collision frequency but about an order of magnitude above Zakharov–Karpman theory and scales as $\omega_B^{-1.48}$ rather than ω_B^{-3} .

Contributions

1. Identified a three-phase kinetic evolution for weakly collisional, large-amplitude electron plasma waves.
2. Isolated the Fokker–Planck susceptibility contribution that enhances the nonlinear frequency downshift during detrapping.
3. Characterized the long-lived quasi-steady phase with empirical damping, frequency-shift, and lifetime laws.

4. Quantified where Zakharov–Karpman damping theory agrees with and departs from the simulations.
5. Reduced the simulated behavior to phase-specific dispersion relations suitable for piecewise wave-evolution models.

2026K

ChatGPT (July 2026)

Summary

This review clarifies the history, terminology, and mathematical scope of axisymmetric equilibrium equations for plasmas with anisotropic pressure. Starting from the canonical Grad–Shafranov equation for scalar pressure, it derives the generalized form for the tensor $\mathbf{P} = p_{\perp} \mathbf{I} + (p_{\parallel} - p_{\perp}) \mathbf{b}\mathbf{b}$. The anisotropy factor $\sigma = 1 - \mu_0(p_{\parallel} - p_{\perp})/B^2$ and modified current function $F = \sigma i_p$ yield the divergence-form Mercier–Cotsaftis equation; importantly, F is a magnetic-surface function even though σ and i_p need not be. The review connects ellipticity and well-posedness conditions to firehose and mirror stability, traces priority to Mercier and Cotsaftis’s 1961 work, and documents the equation’s later rediscovery and inconsistent naming. For mirror traps without poloidal current, it gives a reduced equation and argues that it can describe intermediate diamagnetic-bubble states, while fully developed diamagnetic traps instead require a nonlocal integro-differential model. The author recommends “Mercier–Cotsaftis equation” as the least ambiguous name for the anisotropic equilibrium equation.

Contributions

1. Reconstructed the historical priority and proliferation of names for the anisotropic equilibrium equation.
2. Presented parallel derivations of the canonical Grad–Shafranov and generalized Mercier–Cotsaftis equations.
3. Explained why $F = \sigma i_p$ remains a flux-surface function under pressure anisotropy.
4. Related ellipticity and solvability conditions to firehose, mirror, and Mercier stability criteria.
5. Delimited when the reduced equation applies to mirror and diamagnetic traps and when nonlocal physics changes the equation type.

2026K2

Claude (September 2026)

Summary

The author presents kalypsso, a block-structured adaptive mesh refinement (AMR) framework for compressible flow on CPUs and GPUs. The MPI-parallel, CPU-only forest-of-octrees library p4est refines, coarsens, 2:1-balances (neighbours differ by at most one level), and repartitions the mesh on the host, while solver data, one block of cells per leaf with an optional ghost halo, stay on the device for Kokkos kernels. Each AMR cycle downloads one refinement flag per leaf and uploads a hash table of 64-bit keys (tree, Morton index, level, outside-domain bits), from which kernels compute neighbour keys without searching; halos move by GPU-aware MPI. Finite-volume solvers for Euler (second-order MUSCL–Hancock), five-equation two-phase flow with THINC interface sharpening, and constrained-transport MHD are tested on shock-tube, Riemann, triple-point, shock–bubble, and blast-wave problems, the shock–bubble run matching measured wave speeds within 10%. On one GH200, 2D time-to-solution is shortest with 16^2 blocks on GPU and 8^2 on CPU, although GPU cell-update rates rise with block size, and 3D weak scaling retains about 80% node-to-node efficiency on 512 GPUs. As limitations, the author notes a single global time step, an AMR cycle taking up to 56.5% of GPU run time with 4^2 blocks, rectangular domains only, and no communication–computation overlap.

Contributions

1. Laid out the default 3D key as 15 tree-id bits (5 per direction, so up to 32^3 octrees), 42 Morton-index bits, 4 level bits, and $d = 3$ outside-status bits, and padded rectangular domains with outside blocks that reuse the key of their periodic image inside the box plus bits marking the directions of the faces through which they touch it, so

that outside blocks, filled from the boundary condition, are otherwise handled like MPI ghost blocks and kernels access all neighbours alike.

2. Kept device hash tables for both the old and the new mesh so that a `Kokkos::parallel_scan` computes the memory offsets of the remapping, and let the block-data class wrapping a `Kokkos::View` reallocate only when a resize exceeds a capacity held 20% above the logical size (instead of the STL's doubling), since reallocating at every AMR cycle made runs on the CUDA backend about 20% slower; CPU backends showed no gain.
3. Measured on one GH200 (Table 1, 4096² equivalent) best 2D AMR times of 41.2 s on the GPU and 164.7 s on the 72 ARM cores, against 109.1 s and 861.8 s for the uniform-grid code `euler_kokkos` with the same scheme, with GPU rates rising from 406 to 2678 million cell updates/s between 4² and 512² blocks, and in 3D (512³, Table 2) found 8³ blocks best on both devices, at 151.9 s and 1407.0 s against 306.0 s and 2121.3 s.
4. Weak-scaled a replicated blast wave with one tree per GPU on up to 64 A100 and 512 GH200 GPUs; in 2D (16² blocks, levels 4–9) every run ended with 11.56% of the finest uniform grid's cells (7.7 million per process, about 4×10^9 on 512 GPUs), the AMR cycle every 20 steps took 11% of run time, and MPI communication took about 15% within one node, 33% at 8 processes, and 39% at 512.
5. Implemented constrained transport with face-averaged magnetic fields in a dedicated staggered array layout and HLLD fluxes, replacing the coarse EMF at nonconforming edges by the mean of the two fine EMFs as in AthenaK and prolongating with the Tóth–Roe operator, which kept $|\nabla \cdot \mathbf{B}|$ at machine precision (10^{-15}) at conforming edges and at most 10^{-11} near nonconforming faces; with this costlier solver the GPU AMR-cycle share fell below 10% for blocks above 8×8 (Table 4).

Comments

- Abstract: with AMR activated and “comparable hardware resources”, “a consistent $\times 5$ CPU to GPU time to solution speed-up is obtained”, but Section 5.2 reports “a $\times 4$ speedup” between the best 2D times (164.7 s and 41.2 s in Table 1), the best 3D times in Table 2 give $1407.0/151.9 \approx 9.3$, and at equal block size the AMR runs give CPU-to-GPU time ratios of 3.1 to 6.6 in Table 1 and 8.0 to 9.3 in Table 2; unresolved.
- Section 4.3: “The ratio between the smallest and the largest block can be as high as 2^{5+14} ”, counting the 5 tree-id bits per direction, and “the maximum ratio between largest and smallest mesh cell can be larger than 1×10^6 ” once the cells of the leaf blocks are included, but the octrees lie on a regular Cartesian grid (the brick connectivity of Section 2) and every leaf carries the same block of cells, so block sizes, and hence cell sizes, differ by at most 2^{14} (14 Morton bits per direction; 2^{13} with $l \in [0, 13]$ in the Fig. 3 caption); both figures match instead the ratios of the domain extent, up to 2^5 octrees per direction, to the finest block and cell.
- Fig. 8 caption: the solutions are “obtained at final time $t_f = 0.6$ ”, but Section 5.2 states that “Fig. 8(b) to (f) correspond to the final fluid state at time $t_f = 0.8$ ”, and Table 1, which reports these simulations, runs them “to $t_f = 0.8$ ”; the evident intended value is 0.8.
- Table 1, 512² row: this run has no AMR and a fixed 16.77 M cells, yet its CPU entries (282 Mcu/s, 1155.9 s) imply 3.26×10^{11} cell updates, whereas its GPU entries (2678 Mcu/s, 139.8 s) and both `euler_kokkos` entries for the same uniform 4096² problem to $t_f = 0.8$ imply $3.74\text{--}3.75 \times 10^{11}$; the stated time would require about 324 Mcu/s; unresolved.
- Section 6.1: the mixture internal energy is defined as $\rho e = \sum_{i=0,1} \rho_i e_i$ next to the mixture density $\rho = \sum_{i=0,1} \rho_i \alpha_i$, but without volume fractions it counts the energy of a phase absent from a cell ($\alpha_i = 0$) and does not reduce to ρe when $e_0 = e_1 = e$; the evident intended definition, consistent with the phasic densities $\alpha_i \rho_i$ of Eq. (6), is $\rho e = \sum_i \alpha_i \rho_i e_i$.
- Section 6.1.2: region 2 of the triple point, filled with the second fluid ($\alpha_0 = 0, \alpha_1 = 1$), is given “ $\gamma_0 = 1.4$ ”, whereas regions 1 and 3, filled with the first fluid, have $\gamma_0 = 1.5$ and Section 6.1 calls the γ_i material constants; the evident intended value is $\gamma_1 = 1.4$.
- Section 6.1.3, Eq. (7): the post-shock state (1.6571, 0, −114.51, 0, 159060, 1.0) is said to follow from the Rankine–Hugoniot relations for the Mach 1.22 shock, but with the listed pre-shock air state (1.1839, 101325) and $\gamma_{\text{air}} = 1.4$ these give density 1.6295 and speed 115.5 m/s (only the pressure agrees), and $\alpha_0 = 0.28 \rho_{\text{bubble}} / \rho_{\text{air}}$ with $\rho_{\text{bubble}} = 0.0614 + 0.1579$ gives 0.0519 rather than 0.051; all three stated values follow instead from a pre-shock density of 1.204; unresolved.
- Section 6.2, Eq. (11): the update is written for $B_{x,i+1/2,j,k}^{n+1}$ but uses the EMFs $E_{z,i-1/2,j\pm 1/2,k}$ and $E_{y,i-1/2,j,k\pm 1/2}$ on the edges of face $i - 1/2$, and the preceding sentence announces “the explicit time evolution of $B_{x,i-1/2,j,k}^n$ ”;

the evident intended face index on the left-hand side is $i - 1/2$.

- Section 6.2, Eq. (12): $p = 0.1$ if $r < 0.1$ and $p = 10.0$ if $r \geq 0.1$, but the text states that “a sphere of radius $r_c = 0.1$ is at high pressure $P = 10.0$ while the background pressure is $P = .1$ ”, as the thermal-pressure colour bar of Fig. 17(b), spanning 0.091 to 1.0, also indicates; the two cases of Eq. (12) are evidently swapped.
- Section 6.2: Eq. (12) sets $\mathbf{B} = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]$, whereas the following text gives “ $B_x = 0.5$, $B_y = -0.5$ and $B_z = 0$ ”, which differs in magnitude and in the sign of B_y ; unresolved.
- Section 6.2: the blast wave is said to have “a moderately low plasma beta of $\beta = \frac{2P}{B^2} = 0.02$ ”, but with the background pressure $P = 0.1$ this formula gives $\beta = 0.2$ for the field of Eq. (12) ($B^2 = 1$) and $\beta = 0.4$ for the text’s $B_x = 0.5$, $B_y = -0.5$ ($B^2 = 0.5$); unresolved.
- Section 6.2: the text says that in Fig. 17 “The fluid density at final time $t_f = 0.15$ is shown on the left, and the adaptive mesh levels on the right. We use a block size of 16×16 and AMR levels in range $\llbracket 2, 6 \rrbracket$ ”, but the Fig. 17 caption gives $t_f = 0.2$, six panels, “Block size is 8^2 ” and “ $l \in \llbracket 2, 8 \rrbracket$ ”, consistent with the level colour bar of Fig. 17(c), which reaches 8; the final time is unresolved.
- Section 6.2: with both optima at 8^2 blocks, AMR is said to bring “a $\times 4$ speedup in the CPU case and $\times 2.66$ speedup in the GPU case” over the full Cartesian resolution, but the 8^2 row of Table 4 against the euler_kokkos row (165.8 s and 23.8 s against 801.3 s and 67.4 s) gives 4.8 and 2.8, whereas 2.66 matches the 16^2 row ($67.4/25.2 \approx 2.67$); unresolved.
- Section 7: small blocks are said to be “usually associated to higher sparsity index, hence smaller mesh for a given space resolution”, but Section 4.4 defines lower sparsity indices as meshes with fewer cells, and Section 5.2 and Table 1 show smaller blocks giving lower indices (10.6% at 4^2 against 43.3% at 64^2 at final time); the evident intended word is “lower”.

2026KBK

Claude (September 2026)

Summary

This review sorts numerical differentiation into five situations with recommended methods. Automatic differentiation suits fixed analytic relationships, not evolving simulated fields, except as the gradient engine of differentiable simulators. For noiseless simulation data, finite differences are simple, Fourier and Chebyshev spectral methods are highly accurate but require periodicity or special sampling, and finite elements handle irregular domains and weak derivatives at higher cost. Noise, amplified as $\mathcal{O}(\Delta x^{-\nu})$ by finite differences of the ν th derivative, is handled through its band separation from the signal: spectral low-pass filtering for periodic data, Kalman filtering and smoothing with robust and nonlinear variants when a model exists, and otherwise smoothing methods such as polynomial, spline, basis, wavelet, total-variation, naive-model Kalman, and local linear-dynamics fits for the ill-posed model-free problem. Hyperparameters are tuned by minimizing a fidelity-plus-total-variation loss, or a Huber variant for outliers, whose weight follows a bandlimit heuristic. Methods from the authors’ PyNumDiff package are compared on six simulated signals across outliers, noise type and scale, step size, and bandlimit, with 52 seeds. Most perform similarly, so auxiliary properties decide; the authors recommend RTSDiff, a Rauch–Tung–Striebel smoother, as the default. No method can separate noise that shares the signal’s character.

Contributions

1. Provided decision maps linking analytic structure, simulation complexity, noise, model availability, periodicity, and sampling to preferred differentiation algorithms and their error behavior.
2. Presented Kalman filtering and smoothing as a derivative-estimation framework, including its Bayesian interpretation, forward–backward smoothing, robust generalizations for outliers and non-Gaussian noise, and nonlinear filtering.
3. Generalized the fidelity–smoothness hyperparameter loss to a Huber form scaled by the median absolute deviation, which greatly improved all methods in the presence of outliers.
4. Found that accuracy is governed mainly by problem difficulty, with LinearDiff and RTSDiff best in average RMSE rank, RobustDiff and TVRDiff least biased, and sliding-window fits degrading least at large step sizes.

5. Released PyNumDiff with the comparison notebooks, and described how finite differences, splines, and Kalman smoothing extend to irregularly spaced samples.

Comments

- §6.1, Eq. (6.3): the second line contains the factor $(\mathbb{I} + \mathbf{K}\mathbf{C})^\top$, which conflicts with the first line of the same equation, whose transposed factor is $((\mathbb{I} - \mathbf{K}\mathbf{C})(\mathbf{A}\mathbf{e}_{n-1} + \mathbf{w}_n) - \mathbf{K}\mathbf{v}_n)^\top$, and with Eq. (6.5), which uses $(\mathbb{I} - \mathbf{K}\mathbf{C})^\top$; the intended factor is $(\mathbb{I} - \mathbf{K}\mathbf{C})^\top$.
- §6.2: the measurement and state densities are written as $\propto e^{\frac{1}{2}\|\mathbf{R}^{-1/2}(\mathbf{y}_n - \mathbf{C}\mathbf{x}_n)\|_2^2}$ and $\propto e^{\frac{1}{2}\|\mathbf{P}_{n|n-1}^{-1/2}(\mathbf{x}_n - \hat{\mathbf{x}}_{n|n-1})\|_2^2}$, without the minus sign that Gaussian densities and Eq. (6.8), obtained by taking the negative logarithm and minimizing, require; the intended exponents are $-\frac{1}{2}\|\cdot\|_2^2$.
- §6.5: the nonlinear form of the cruise-control model of §6.1.1 gives the third state row as $-f_r x_1 - mg u_0 + \frac{k_i}{\Delta t^2} x_2 + \frac{k_p}{\Delta t} u_1$, the fourth as $x_4 + \Delta t u_1$, and the measurement as $[x_1]_{n-1}$, whereas Eq. (6.6) gives $-(f_r + \frac{k_p}{\Delta t})x_1 + \frac{k_i}{\Delta t^2}x_2 - mg u_0 + \frac{k_p}{\Delta t}u_1$, $x_3 - \Delta t x_1 + \Delta t u_1$, and $\mathbf{C} = [1 \ 0 \ 0 \ 0]$, that is, position x_0 ; Eq. (6.6) makes these the evident intended forms.
- §7.1, Eq. (7.2): the bias metric, described as “the correlation coefficient between estimate error, $\hat{\mathbf{x}} - \dot{\mathbf{x}}$, and the actual derivative”, is printed as $R^2 = \text{cov}(\hat{\mathbf{x}} - \dot{\mathbf{x}}, \dot{\mathbf{x}}) / (\text{var}(\hat{\mathbf{x}} - \dot{\mathbf{x}}) \text{var}(\dot{\mathbf{x}}))$, which is neither the correlation coefficient nor its square and is not dimensionless; the label R^2 and the nonnegative values plotted in Figs. 57–61 suggest the squared covariance in the numerator, but the paper does not state it.
- §7.3: with first-order-FD($\mathbf{y}$) = $[y_1 - y_0, y_2 - y_1, y_3 - y_2, y_3 - y_2] / \Delta t$, the stated result $\text{cumsum}(\text{first-order-FD}(\mathbf{y})) \cdot \Delta t = [y_1, y_2, y_3, y_3] - y_0$ is wrong in its last entry, which is $2y_3 - y_2 - y_0$.
- §7.4.3, Eq. (7.8): the smoothing penalty is printed as $\lambda \int_{t_0}^{t_{N-1}} S''(u; \theta) du$, whereas the substituted form is $\lambda \int (\sum_j \alpha_j B_j''(u))^2 du$, written as $\lambda \alpha^\top \mathbf{R} \alpha$ in Eq. (7.11); the intended integrand is $(S'')^2$.
- Fig. 38 caption: each cubic basis function is said to be nonzero on “a number of between-knot intervals at most equal to the spline’s degree”, that is, three, but the plotted B_4 spans four intervals, as cubic B-splines do; the intended bound is the degree plus one.
- §7.7: the naive-model measurement is written as the vector $\mathbf{y}_n = [y_n, 0, 0]^\top$, which conflicts with $\mathbf{C} = [1 \ 0 \ 0]$, for which $\mathbf{C}\mathbf{x}_n$ is a scalar, and with the statement that $\mathbf{R}$ is 1×1 ; the intended measurement is the scalar y_n .
- §7.9.2: the study is said to explore “12 methods”, and Table 4 has 12 rows, but the legends of Figs. 57–61 list and plot 14 methods, including ButterDiff and SmoothAccelTVR as well as KernelDiff and TVRDiff; the intended count is unresolved.
- §8.3: the block equations give $\dot{\mathbf{M}}_2 = \mathcal{M}_1 \mathbf{M}_2 + \mathcal{M}_3 \mathbf{M}_3$, whereas the block product on the preceding line yields $\mathcal{M}_1 \mathbf{M}_2 + \mathcal{M}_2 \mathbf{M}_3$, which the following lines use; the intended term is $\mathcal{M}_2 \mathbf{M}_3$.

2026KDBK

ChatGPT (July 2026)

Summary

This study replaces expensive particle-in-cell scans of a laser-plasma injector with machine-learning surrogate models for rapid design and optimization. Using more than 3,000 simulations spanning laser and plasma-target parameters, the authors compare multilayer perceptrons, decision-tree models, and Gaussian processes for predicting electron-beam energy, charge, energy spread, and emittance. A compact multilayer perceptron trained on a 500-simulation random scan achieved $R^2 = 0.93$ and evaluated configurations roughly 10^7 times faster than particle-in-cell simulation. The resulting surrogate supports random search, Bayesian optimization, and multi-objective Bayesian optimization, enabling the discovery of target working points and stable operating regions that would be impractical to map directly. The analysis also emphasizes a central limitation: predictions degrade where training samples are sparse and at the low end of several beam outputs. Uniform coverage and targeted additional simulations are therefore important for reliable design. The models can be updated with new simulation or experimental data and could serve as fast electron-source components in start-to-end accelerator studies and feedback-control systems.

Contributions

1. Built laser-plasma-injector surrogates from a large particle-in-cell simulation data set.
2. Compared multilayer perceptron, decision-tree, and Gaussian-process models across four beam-quality outputs.
3. Achieved $R^2 = 0.93$ with a compact model trained on 500 simulations and approximately 10^7 -fold faster evaluation.
4. Used surrogates with single- and multi-objective optimization to locate target working points and stability regions.
5. Quantified the dependence of predictive reliability on training-data coverage and identified sparse-region limitations.

2026KHSB

ChatGPT (July 2026)

Summary

Forest-of-refinement-trees adaptive mesh refinement uses space-filling curves to distribute dynamic meshes cheaply, but mixed tetrahedral–hexahedral meshes require pyramids as conforming bridge elements. This paper supplies the missing pyramidal primitive for the open-source t8code framework. A pyramid is recursively divided into six pyramids and four tetrahedra in a construction that commutes with cubic refinement. Encoding each descendant by its enclosing subcube and one of two pyramid orientations yields a Morton-type index, which the authors prove is a valid space-filling-curve index. They derive parent, child, same-tree neighbor, inter-tree neighbor, and face transformations despite heterogeneous descendants and triangular and quadrilateral faces. At forest level, they generalize uniform construction, adaptation, partitioning, and ghost exchange, including load balancing for root shapes with different refinement counts. Large-scale tests show behavior comparable to existing element types and near-ideal scaling for central operations. A hybrid airplane-like mesh demonstrates the complete method with every supported three-dimensional element shape.

Contributions

1. Defined a commutative pyramidal refinement into six pyramids and four tetrahedra using two pyramid orientations.
2. Constructed and proved validity of a Morton-type pyramidal space-filling-curve index.
3. Developed the complete set of low-level parent, child, face-neighbor, and cross-tree transformation algorithms.
4. Generalized forest construction and uniform partition bounds to mixed root shapes with unequal descendant counts.
5. Implemented the design in t8code and demonstrated scalable adaptation, partitioning, construction, and ghost exchange.

2026KJWC

ChatGPT (July 2026)

Summary

PCGBandit accelerates transient PDE simulations without pretrained models or additional simulations by learning from linear solves generated during the run itself. Each preconditioner choice is treated as an arm in a nonstationary adversarial-bandit problem, with solve wall time as feedback. A reduced-variance Tsallis-INF variant adaptively selects among 33 incomplete-Cholesky, thresholded incomplete-Cholesky, and multigrid configurations; its learning update costs $O(d)$, independent of matrix size. A fallback to IC(0) preserves solver convergence when a selected configuration reaches its iteration limit. Integrated into OpenFOAM, the method is evaluated on six fluid, multiphase, and magnetohydrodynamic simulations. It is optimal or near-optimal across the tested tasks, exceeds both static baselines by more than fourfold for some linear-solve workloads, and usually breaks even within roughly 10^3 solves, often within 10^2 . Deterministic operation-count feedback remains within 1.2 times the wall-clock-feedback result, improving reproducibility. The simplest direct-numerical-simulation case receives no net benefit.

Contributions

1. Cast within-simulation preconditioner selection as an adversarial-bandit problem requiring no training runs.
2. Developed reduced-variance PCGBandit with $O(d)$ learning cost and worst-case $O(\sqrt{dT})$ regret.
3. Added adaptive selection across 33 OpenFOAM preconditioner configurations.
4. Preserved convergence through an IC(0) backstop and deterministic cost feedback.
5. Measured wall-clock and break-even behavior on six fluid and MHD simulations.

2026KK

ChatGPT (September 2026)

Summary

This paper relates the Chapman–Enskog asymptotic expansion of kinetic equations to exact hydrodynamic closure on slow spectral modes. For linearized kinetic dynamics, the authors show that the Chapman–Enskog series agrees locally, to every order as the Knudsen number tends to zero, with the eigenvalue and invariant-manifold expansion defining the slow spectral closure. An explicit one-dimensional linear BGK model makes the distinction between local asymptotics and global hydrodynamics concrete. Its slow diffusion eigenvalue has an exact closed form and defines the reduced dynamics for arbitrary Knudsen number up to the relevant spectral continuation, whereas the corresponding Chapman–Enskog power series diverges at every nonzero Knudsen number. The divergence is tied to a critical wave number where the hydrodynamic eigenvalue encounters the kinetic spectrum. Finite Burnett and super-Burnett truncations can therefore develop artifacts such as sign changes and Bobylev instability even though the exact spectral closure remains meaningful. The result clarifies why asymptotic agreement near equilibrium does not imply globally convergent hydrodynamic equations.

Contributions

1. Proved all-order local equivalence between Chapman–Enskog expansion and slow spectral closure.
2. Expressed hydrodynamic reduction through slow eigenvalues and invariant spectral manifolds.
3. Derived an exact diffusion-type closure for a one-dimensional linear BGK model.
4. Showed that its Chapman–Enskog series diverges for every nonzero Knudsen number.
5. Connected divergence and higher-order instability to the critical wave number of the hydrodynamic eigenvalue.

2026KKB

ChatGPT (July 2026)

Summary

The paper presents Noarr-MPI, a modern C++ abstraction that integrates MPI communication with Noarr’s first-class descriptions of memory layouts and traversals. Ordinary MPI exposes untyped buffers and requires programmers to construct derived datatypes manually for noncontiguous layouts. Noarr-MPI instead maps compile-time Noarr structures to MPI datatypes, checks transferred element types, and can transform data automatically when sender and receiver layouts differ. Its interface also composes with Noarr traversers, so distributed communication can share layout and traversal descriptions with OpenMP or CUDA computation. A distributed GEMM case study exercises row-major, column-major, tiled, and other layout choices while keeping the algorithmic structure largely independent of physical storage. Benchmarks compare Noarr-MPI with handwritten MPI using `mdspan`, `Boost.MPI`, and `KokkosComm` on up to eight cluster nodes. Noarr-MPI performs comparably to handwritten MPI and other modern bindings in most configurations, and is fastest in the four most efficient large-problem configurations, though several layout combinations remain outliers. The result is a type-checked interface that reduces datatype and layout-conversion code without imposing a systematic performance penalty in the tested kernel.

Contributions

1. Introduced a layout-agnostic, type-checked C++ abstraction over MPI communication.
2. Derived MPI datatypes automatically from Noarr structure and traversal descriptions.
3. Supported communication between differing physical layouts through automatic transformation.
4. Demonstrated the interface with a distributed, layout-agnostic GEMM implementation.
5. Showed performance broadly comparable to handwritten MPI and modern C++ MPI bindings across tested layouts and sizes.

2026KKLM

Claude (September 2026)

Summary

The authors solve the fixed-boundary Grad–Shafranov (GS) equation for tokamak equilibria with physics-informed Kolmogorov–Arnold networks (KANs), which place learnable univariate functions on network edges, here FastKANs built from Gaussian radial basis functions. The loss combines the GS residual at about 500 interior collocation points with a Dirichlet penalty at about 390 boundary points of an ITER-like lower single-null shape, and training uses AdamW followed by the self-scaled Broyden (SSBroyden) quasi-Newton method. The profile amplitudes (P_1, H_1) are updated in alternation with the network to meet target toroidal beta and plasma current, and the magnetic axis is tracked by damped Newton iterations. For linear Solov’ev profiles, KANs and parameter-matched multilayer perceptrons (MLPs) reach comparable losses, the MLPs faster, and a KAN reproduces an analytic single-null solution with relative error below 10^{-3} in the core. For nonlinear L-mode profiles, only KANs with guided training converge: homotopy continuation from Solov’ev profiles in all cases, and transfer learning from a pretrained Solov’ev network when the amplitudes are fixed. For an H-mode pressure-pedestal profile, different training strategies converge to distinct equilibria, indicating multiple solution branches. The authors note that MLPs with tuned hyperparameters might outperform KANs.

Contributions

1. Combined FastKANs, SSBroyden refinement, and guided training into a mesh-free GS solver that identifies (P_1, H_1) from beta and current constraints, with an alternative constraint on the on-axis safety factor from an analytic formula.
2. Introduced a homotopy curriculum that blends the Solov’ev and target profile derivatives $dP/d\psi$ and $dH/d\psi$, which converged in every nonlinear case considered and needs no stored pretrained network.
3. Built a collocation point generator for data-prescribed boundaries using Fourier-fitted boundary parametrizations, k-d-tree removal of clusters, potential-based void filling, and X-point refinement.
4. Constructed pedestal equilibria with edge current peaks and negative edge magnetic shear, finding that homotopy and transfer learning with parameter discovery reach similar beta and current but different H_1 and on-axis safety factor, and that unguided training with frozen amplitudes converges to a flatter-pressure branch.
5. Gave parameter-count formulas for FastKANs and MLPs to match model sizes and an ablation showing that unguided training and MLPs fail for the nonlinear L-mode profiles, whereas MLPs with frozen pedestal amplitudes converge toward the unguided branch.

Comments

- Section 3.2, Eq. (17): the outer sum of the Kolmogorov–Arnold representation of $f(x_1, \dots, x_d)$ runs to $2n + 1$, but n is not defined there; the representation for d variables evidently intends $2d + 1$.
- Section 5.1.2, Table 3 and Fig. 7: Table 3 gives the Solov’ev FastKAN runtime as “01m 50s”, while the Fig. 7 legend for the same comparison reads “FastKAN (L:2, N:16) runtime: 01m 36s” (the MLP runtimes agree in both); which value is correct is unresolved.
- Section 5.1.3, Fig. 9: Section 5.1.1 states that the hyperparameters “used in this and subsequent experiments are summarized in Table 1”, where the initial AdamW learning rate is 5×10^{-4} , but the Fig. 9 title reads “MLP lrs: [5e-03, 5e-01] | FastKAN lrs: [5e-03, 5e-01]”; unresolved.

- Section 5.2, text after Eq. (41): the integration constants are said to follow from “the boundary requirements at the plasma edge, $P(\psi = \psi_a) = 0$ and $F(\psi = \psi_a) = \sqrt{2H(\psi = \psi_a)} = 1$ ”, but Eq. (7) defines ψ_a as the flux on the magnetic axis, where Eq. (38) gives $P = P_0 \neq 0$; Eqs. (40)–(41) follow from these conditions at the boundary $\psi = \psi_b$ ($\psi_n = 1$), the evident intended reading.
- Section 5.3, Table 5 (Case B): the text says the frozen P_1 and H_1 “are set to the values identified by the homotopy method”, (0.354, 0.109), as listed in every other Case B row, but the MLP₂ row lists (0.355, 0.110).

2026KLJH

ChatGPT (July 2026)

Summary

The study tests whether the minimum magnetic-gradient scale length $\min(L_{\nabla B})$ on a stellarator’s last closed flux surface predicts engineering difficulty for discrete filamentary coils. Across 3,027 single-stage QUASR configurations, $\min(L_{\nabla B})$ correlates with minimum coil–surface distance, and its location is usually near the closest coil point; the second-gradient metric $L_{\nabla\nabla B}$ is even more spatially correlated. The authors then optimize 42 quasihelically symmetric equilibria over $\min(L_{\nabla B})$ and use continuation to design progressively longer coils. Larger scale length generally permits greater coil–surface and coil–coil separation and lower normal-field error, provided enough coil length is available. At short coil lengths the relationship breaks down in a high-ripple regime. Alpha-particle tracing exposes a trade-off: increasing $\min(L_{\nabla B})$ reduces coil-ripple loss but worsens quasisymmetry, producing an intermediate optimum. A third ensemble of finite- β random-boundary equilibria retains moderate correlations, showing robustness but also that scale length alone cannot determine coil distance; its location and global boundary shape matter.

Contributions

1. Established filament-coil correlations between $\min(L_{\nabla B})$ and coil–plasma spacing across three datasets.
2. Quantified spatial coincidence between magnetic-scale-length minima and closest coil locations in QUASR.
3. Demonstrated improvement through staged equilibrium optimization and continuation-based coil design.
4. Identified insufficient coil length and high ripple as a regime where the proxy fails.
5. Exposed the confinement trade-off between quasisymmetry error and coil-ripple reduction.

2026KLWB

ChatGPT (July 2026)

Summary

This paper combines Numerical Flow Iteration (NuFI) with the Characteristic Mapping Method (CMM) for the 1D+1V Vlasov–Poisson system. NuFI supplies symplectic, conservative local evolution through a compact electric-field representation, while periodically stored CMM submaps limit the growing cost of repeatedly tracing characteristics. The hybrid reconstructs the distribution from a composed backward map and permits the map grid to be much coarser than the sampling grid. Landau-damping and two-stream-instability experiments reproduce expected dynamics and resolve fine phase-space filamentation. Relative to classical CMM, the method improves mass, energy, and norm behavior; relative to plain NuFI, remapping sharply reduces runtime, with the best tested balance near 20 NuFI steps per remap. Compared with a cubic semi-Lagrangian predictor–corrector, flow mapping preserves the L^2 and maximum norms and momentum substantially better, although the baseline conserves mass better.

Contributions

1. Introduced a hybrid CMM–NuFI backward-flow-map algorithm for Vlasov–Poisson dynamics.
2. Combined symplectic NuFI local stepping with CMM submap composition for cheaper long-time propagation.
3. Separated coarse map storage from fine distribution sampling to provide high effective phase-space resolution.
4. Derived memory and computational-cost models exposing the remapping-frequency trade-off.
5. Validated conservation, accuracy, fine-structure resolution, and runtime against three baseline schemes.

2026KNB

ChatGPT (July 2026)

Summary

This review presents real-time time-dependent density functional theory (TDDFT) as a unified first-principles tool for electronic response in warm dense and high-energy-density matter. Propagating finite-temperature Kohn–Sham orbitals after a controlled perturbation captures collective and single-particle behavior, including free-free, bound-free, and bound-bound processes without an imposed state decomposition. The paper gives practical workflows for extracting dynamic structure factors relevant to x-ray Thomson scattering, frequency-dependent conductivity and optical properties, and nonlinear electronic stopping power. It emphasizes convergence requirements involving supercell size, time step, trajectories, pseudopotentials, and spectral broadening. The same versatility creates substantial cost and uncertainty: exchange-correlation approximations, frozen cores, finite cells, and extreme frequencies can bias predictions, while high temperature and high atomic number demand many states and fine resolution. The review positions TDDFT both as a predictive method and as a benchmark for cheaper response models.

Contributions

1. Unified linear and nonlinear high-energy-density response calculations within one real-time TDDFT framework.
2. Provided practical procedures for dynamic structure factors, conductivity, optical properties, and stopping power.
3. Explained how the method captures collective, free-free, bound-free, and bound-bound dynamics self-consistently.
4. Catalogued numerical sensitivities from cells, trajectories, time propagation, pseudopotentials, and frequency resolution.
5. Identified accuracy and cost limits that motivate improved functionals, heterogeneous computing, and reduced models.

2026KNTL

ChatGPT (July 2026)

Summary

This study applies the STRIPE framework to carbon erosion and impurity transport around DIII-D’s high-power helicon antenna. STRIPE combines SOLPS-ITER edge-plasma backgrounds, COMSOL radio-frequency sheath fields, RustBCA sputtering yields, and GITR/GITRm full-orbit transport. Two H-mode discharges compare antenna–plasma gaps of roughly 7 cm and 4 cm. Modeled rectified sheath potentials reach 1–5 kV and localize near the antenna bottom, where field lines meet surfaces at grazing incidence. Carbon self-sputtering dominates; sheath-accelerated deuterium contributes about 1% of gross erosion. The small-gap case produces 7.2×10^{20} eroded particles per second, redeposits about 12.3%, and sends 58% into the confined plasma, whereas the large-gap case yields much less erosion, about 4% redeposition, and 35% core penetration. Total helicon-sourced carbon remains small relative to the background, consistent with present graphite-wall observations. Quantitative sheath amplitudes remain uncertain because current models underresolve short-wavelength slow waves and are not validated for nearly tangential fields, a concern for future high- Z walls.

Contributions

1. Coupled edge plasma, full-wave sheath, sputtering, and full-orbit impurity models in one antenna analysis.
2. Predicted kilovolt rectified sheaths localized by antenna geometry and grazing magnetic incidence.
3. Quantified carbon self-sputtering and the smaller sheath-accelerated deuterium contribution.
4. Compared erosion, redeposition, and core penetration for two antenna–plasma gaps.
5. Identified sheath-model and slow-wave-resolution limitations relevant to high- Z plasma-facing materials.

2026KRR

ChatGPT (July 2026)

Summary

The paper presents a scalable framework for entropy-regularized semi-discrete optimal transport from a finite-element source measure to discrete target points. Its L-BFGS dual solver accelerates dense Gibbs-kernel evaluations through adaptive, error-controlled distance truncation and R-tree neighbor queries. Coarse-to-fine hierarchies for both the source mesh and target measure provide effective initial guesses, while regularization scheduling improves convergence toward small entropy parameters. The combined strategy addresses complex geometries and PDE-derived densities without explicitly constructing Laguerre cells. Benchmarks show a 175-fold multilevel speedup in a demanding case, up to 29.5% acceleration from regularization scheduling, and 9.27-fold strong-scaling speedup on 15 nodes. Applications compute Wasserstein barycenters, register three-dimensional vascular geometries, and perform density-aware blue-noise sampling on a sphere with geodesic cost. The authors provide an open-source C++ implementation, SemiDiscreteOT, based on deal.II with hybrid MPI and OpenMP parallelism.

Contributions

1. Developed a finite-element framework for large-scale regularized semi-discrete optimal transport.
2. Combined error-controlled kernel truncation with R-tree spatial queries.
3. Unified source and target multilevel hierarchies with regularization scheduling.
4. Demonstrated 175-fold multilevel acceleration and scalable hybrid parallel execution.
5. Validated vascular and manifold applications in an open-source deal.II implementation.

2026KS

ChatGPT (July 2026)

Summary

Universal constructions usually serialize operations through consensus whenever they might conflict in any state, even when they commute in the state actually reached. The paper formalizes dynamic concurrency: an implementation uses a strong synchronization primitive only when an operation can conflict with concurrent operations in a possible object state during its execution. Its construction records operations in a shared acyclic dependency graph and tracks active operations with snapshots. An operation that commutes with every relevant subset of active operations can be committed directly; otherwise, a consensus-based conflict-resolution path orders it. A commit-adopt-like mechanism prevents operations committed through the two paths from bypassing one another. The authors prove linearizability, wait-freedom, and dynamic concurrency for arbitrary sequential objects. This is a possibility result rather than a performance result: conflict-free operations take two write-read rounds, but checking commutativity may require exponential local work in the number of concurrent operations, so avoiding consensus may cost more than it saves.

Contributions

1. Defined dynamic concurrency for arbitrary sequential objects using state-dependent operation commutativity.
2. Designed a universal construction that uses strong synchronization only when a dynamic conflict is detected.
3. Represented operation dependencies with an acyclic graph whose topological orders are equivalent linearizations.
4. Coordinated direct and consensus-based commits so conflicting operations cannot bypass one another.
5. Proved linearizability, wait-freedom, and dynamic concurrency while identifying the cost of exhaustive conflict detection.

2026KSCP

ChatGPT (July 2026)

Summary

Tadmor’s entropy-conservative flux condition applies to convex conserved secondary quantities, but real-gas simulations also need non-convex or non-conservative structures such as kinetic energy. This paper derives continuous and discrete compatibility conditions for secondary balance laws that may contain work terms. Singular Hessians and non-injective gradients introduce a necessary property called null-consistency: admissible work terms must vanish along null directions for the generalized flux condition to be well posed. Discrete-gradient operators then provide systematic jump expansions even when thermodynamic functions are supplied by an arbitrary equation of state rather than closed formulas. Applying the framework to the Euler equations yields KEEP-DG, an entropy-conserving and kinetic-energy-consistent flux for general equations of state that removes a singularity in an earlier construction. Supercritical and transcritical tests, including turbulent flow with detailed thermodynamics and transport, remain stable and conservative to machine precision on the tested grids. The theory is developed for smooth solutions; pressure-equilibrium preservation is left for future work.

Contributions

1. Generalized continuous and discrete Tadmor compatibility conditions beyond convex entropy pairs.
2. Identified null-consistency as necessary for well-posed secondary-structure-preserving fluxes.
3. Used discrete gradients to construct fluxes for implicitly specified real-gas thermodynamics.
4. Derived the singularity-free KEEP-DG flux for arbitrary equations of state.
5. Demonstrated robust, machine-precision conservation in supercritical and transcritical flow tests.

2026KSG

ChatGPT (July 2026)

Summary

T3FF is an open-source C++ code for isothermal, three-dimensional electromagnetic gyrofluid simulations of tokamak-edge drift–Alfvén turbulence. Its full- f formulation permits finite-amplitude density fluctuations, while its full- k polarization retains second-order Padé finite-Larmor-radius effects beyond the usual long-wavelength approximation. The present version uses a field-aligned flux tube in simplified circular toroidal geometry with resistive parallel dynamics. Perpendicular advection is discretized with an Arakawa scheme, time evolution uses Adams–Bashforth stepping, and polarization, Ampère, and gyroaveraging equations are handled by iterative and FFT-based solvers with OpenMP parallelism. Verification includes recovery of a published edge-turbulence benchmark in the delta- f limit, parallel and perpendicular grid-convergence studies, and sound- and Alfvén-wave tests. Electromagnetic beta scans reproduce transport reduction from magnetic-flutter damping followed by transition to a resistive ballooning regime. The paper is a code-reference study; steep-gradient non-Boussinesq applications, scrape-off-layer boundaries, and a thermal six-moment extension remain future work.

Contributions

1. Introduced a three-dimensional full- f , full- k electromagnetic gyrofluid model and reference implementation.
2. Combined field-aligned toroidal geometry with consistent finite-Larmor-radius polarization and resistive parallel dynamics.
3. Documented the finite-difference, Arakawa, FFT, iterative-solver, and OpenMP implementation.
4. Verified delta- f benchmark recovery, spatial convergence, and sound- and Alfvén-wave dynamics.
5. Demonstrated beta-dependent magnetic-flutter damping and the transition to resistive ballooning transport.

2026KSH

ChatGPT (July 2026)

Summary

This work incorporates a reduced model of microturbulence-driven electron–ion energy exchange into the one-dimensional steady-state transport solver GOTRESS and evaluates its effect on global tokamak temperature profiles. The exchange is estimated from turbulent particle and heat fluxes, then compared with Coulomb exchange for DIII-D discharge 128913, an electron-heated DIII-D-like case, and ITER and SPARC scenarios. For the measured DIII-D case, turbulent exchange is only a few percent of total power and negligibly changes the profiles. Under strongly unbalanced electron heating, trapped-electron-mode turbulence transfers roughly 1 MW to ions, exceeds collisional transfer by more than a factor of two, and raises the central ion temperature by about 0.55 keV. In ITER and SPARC, ion-to-electron turbulent transfer is largely offset by enhanced collisional transfer, leaving profiles nearly unchanged. The results identify strongly unequal species heating, especially during startup, as the regime where turbulent exchange requires explicit treatment.

Contributions

1. Integrated a low-cost turbulent electron–ion exchange model into GOTRESS for self-consistent steady-state global profile prediction.
2. Showed that turbulent exchange contributes only a few percent of total power and negligibly changes profiles for DIII-D discharge 128913.
3. Demonstrated that turbulent transfer exceeds collisional transfer by more than twofold and adds about 1 MW to ions in a strongly electron-heated case.
4. Quantified central-temperature shifts of -0.13 keV for electrons and $+0.55$ keV for ions under unbalanced heating.
5. Established that compensation by collisional exchange weakens the effect in ITER and SPARC steady states while identifying startup as a priority regime.

2026KSKB

ChatGPT (July 2026)

Summary

This work extends entropy-stable discontinuous Galerkin spectral-element methods to arbitrary-order curvilinear hybrid meshes containing quadrilaterals and triangles in two dimensions and hexahedra, prisms, tetrahedra, and pyramids in three. Collapsed-coordinate tensor products retain sum-factorization efficiency; Legendre–Gauss collocation, generalized summation-by-parts operators, entropy projection, and two-point fluxes yield free-stream-preserving entropy-conservative or entropy-stable hyperbolic discretizations. A lifting formulation handles viscous terms. To avoid the severe explicit time-step restriction associated with collapsed nodal coordinates, the method evolves modal degrees of freedom and applies a memory-efficient weight-adjusted inverse mass matrix. Analyses and experiments establish free-stream preservation, grid and polynomial convergence, and entropy behavior on hybrid curvilinear meshes. Taylor–Green vortex, NACA 0012, and NASA Common Research Model simulations demonstrate application-scale compressible-flow capability. The entropy proof is restricted to the hyperbolic operator, curved-pyramid convergence is not guaranteed, and the Common Research Model test does not resolve the boundary layer.

Contributions

1. Constructed an arbitrary-order entropy-stable DG spectral-element method for curvilinear meshes mixing all standard two- and three-dimensional element shapes.
2. Used collapsed-coordinate tensor products to retain sum-factorization structure instead of dense multidimensional summation-by-parts costs.
3. Combined Legendre–Gauss collocation, generalized summation by parts, entropy projection, and two-point fluxes to establish free-stream and hyperbolic entropy properties.

4. Evolved modal unknowns with a weight-adjusted inverse mass approximation to alleviate collapsed-grid time-step restrictions.
5. Validated convergence and entropy behavior and demonstrated NACA 0012 and 216,718-element NASA Common Research Model simulations.

2026KT

ChatGPT (July 2026)

Summary

Kaganovich and Tendler review partially ionized plasma physics with emphasis on low-temperature technological plasmas. The paper contrasts partially ionized systems with fully ionized plasmas by stressing the role of neutrals, molecular reactions, and non-Maxwellian electron energy distribution functions obtained from kinetic Boltzmann analysis. Examples include non-monotonic distributions in argon–NF₃ afterglows, explosive cold-electron generation in capacitive discharges, hysteresis in inductively coupled plasmas, spokes in Penning discharges, and anomalous transport in Hall thrusters. The review also discusses modern particle-in-cell, fluid, and hybrid simulation tools used for plasma reactor design, etching, deposition, semiconductor processing, and emerging quantum-device manufacturing. Its conclusion emphasizes that practical reactor optimization needs hybrid kinetic-fluid models: fluid solvers are fast but miss kinetic effects, while fully kinetic models remain too expensive for routine engineering sweeps.

Contributions

1. Reviewed the physics distinguishing partially ionized from fully ionized plasmas.
2. Emphasized non-Maxwellian electron energy distributions in reactive gas discharges.
3. Connected self-organization phenomena to afterglow, capacitive, inductive, Penning, and Hall systems.
4. Surveyed modern PIC, fluid, and hybrid modeling for technological plasma applications.
5. Argued for hybrid and machine-learning-accelerated simulation workflows in reactor design.

2026KW

ChatGPT (July 2026)

Summary

This paper unifies partial pivoted Cholesky and Vecchia approximations for large positive-semidefinite matrices. Its central theorem shows that adding a Vecchia approximation of a partial-Cholesky residual gives exactly a Vecchia approximation of the original matrix with an augmented inverse-Cholesky sparsity pattern. The hybrid construction needs $O(rn)$ target-entry accesses rather than $O(r^2n)$ in the stated regime. The authors extend Vecchia optimality in the Kaporin condition number to positive-semidefinite targets and derive bounds connecting that quantity to linear solves and determinant estimation. They compare adaptive search, randomized and greedy pivots, nearest neighbors, and orthogonal matching pursuit. Experiments use 22 machine-learning datasets, each with 20,000 observations, and Gaussian kernel matrices at three regularization levels. Partial Cholesky plus Vecchia generally outperforms earlier Cholesky-derived preconditioners, but none of the tested methods solves half of the most nearly singular cases.

Contributions

1. Proved that partial Cholesky plus a Vecchia residual approximation equals one augmented-pattern Vecchia approximation.
2. Reduced target-matrix entry accesses from $O(r^2n)$ to $O(rn)$ in the analyzed regime.
3. Extended Kaporin-condition-number optimality from positive-definite to positive-semidefinite target matrices.
4. Derived Kaporin-based guarantees for linear solves and determinant estimates.
5. Compared pivot and sparsity strategies on 22 large kernel datasets and documented their near-singular limitations.

2026L

ChatGPT (July 2026)

Summary

Lai develops computational algorithms for constructing orthogonal and orthonormal polynomials over arbitrary polygonal domains in $\mathbb{R}^2$ using bivariate spline representations. A mature MATLAB spline-space implementation supplies exact polynomial representations over triangulated polygonal regions. Two algorithms are described: one constructs orthonormal polynomials up to degree d , and the other constructs degree- $d + 1$ orthonormal polynomials in the complement of $\mathbb{P}_d$. Numerical examples for degrees 1–5 visualize polynomial structures and zero curves, including evidence against Gauss quadrature on centrally symmetric polygonal domains. The paper then uses odd- and even-degree orthogonal polynomials to reduce integration to residual quadratic or linear parts, motivating new quadrature rules. Polynomial interpolation further extends these rules to efficient high-precision quadrature on diverse polygonal domains, with possible extensions to weights, nonpolygonal approximations, Sobolev bases, wavelets, and higher dimensions.

Contributions

1. Constructed orthonormal polynomial bases over arbitrary polygonal domains via bivariate splines.
2. Provided complementary algorithms for degree-limited and next-degree orthogonal spaces.
3. Used numerical zero-curve evidence to test Gauss-quadrature possibilities.
4. Derived polynomial-reduction strategies for odd- and even-degree integration.
5. Built high-precision quadrature schemes for polygonal domains from the constructed polynomials.

2026LBBK

ChatGPT (September 2026)

Summary

This study characterizes double-peaked fishbones: chirping plasma oscillations with synchronized core and edge activity separated by weaker intermediate fluctuations. A survey of approximately 3000 events in 40 KSTAR discharges relates magnetic fluctuation strength to operating conditions, finding stronger events at higher normalized beta and lower edge safety factor. Events are weakest when applied magnetic perturbations degrade confinement and strongest when optimized perturbations enhance it. Hilbert-transform analysis separates electron-temperature and magnetic signals into amplitude envelopes and phases. As fishbones strengthen, edge temperature envelopes become more correlated with magnetic fluctuations, while core correlations change comparatively little. Phase measurements near the outer midplane show that edge temperature oscillations generally lead core oscillations; the weakest events remain inconclusive because edge signals approach background levels. A case with an L-mode-like temperature profile retains two correlation peaks, showing that steep H-mode edge gradients are not required. Rare radial phase reversals also occur within the edge component. The observations constrain theories of long-range coupling and support investigating an active edge contribution, but do not establish that edge activity causes the core response.

Contributions

1. Characterized approximately 3000 double-peaked fishbones across 40 discharges and related their magnetic fluctuation strength to normalized beta, edge safety factor, and externally modified confinement.
2. Separated signal envelopes from phases and showed that edge-temperature correlations with magnetic fluctuations increase with fishbone strength while core correlations remain comparatively insensitive.
3. Identified an edge-leading phase relation near the outer midplane using complementary phase diagnostics and a survey of 28 discharges, with explicit limits for weak edge signals.
4. Observed two correlation peaks with an L-mode-like edge temperature profile, showing that steep H-mode gradients alone cannot explain the double-peaked structure.
5. Documented rare approximately π phase reversals between neighboring edge channels during strong fishbones, distinguishing internal structure of the outer component from an additional radial activity peak.

2026LC

ChatGPT (July 2026)

Summary

The paper presents a GPU and multi-GPU redesign of the Palabos lattice-Boltzmann framework using C++17 standard parallel algorithms. A new AcceleratedLattice preserves the established CPU-facing interface while replacing the CPU array-of-structures and virtual-polymorphism design with a GPU-oriented structure-of-arrays layout, thread-safe collision-streaming, compile-time kernels, and integer-tag dispatch. Shared MPI infrastructure and explicit bridges between CPU and GPU containers support hybrid execution and incremental porting instead of a complete rewrite. Operations use standard algorithms such as `for_each`, `transform_reduce`, and `exclusive_scan`, avoiding vendor-specific language extensions. Taylor-Green vortex, lid-driven cavity, and Berea-sandstone tests agree with reference solutions. On A100 GPUs, tested configurations reach roughly 75–85% of the memory-bandwidth performance model, comparable to CUDA/OpenCL-native LBM codes. Four-GPU tests achieve 80–90% weak-scaling efficiency and approximately 65–80% strong-scaling efficiency despite lacking communication-computation overlap.

Contributions

1. Designed a data-oriented GPU lattice compatible with Palabos' existing CPU interface.
2. Used C++17 standard parallel algorithms and templates for accelerator kernels.
3. Enabled hybrid CPU/GPU interoperability and incremental application migration.
4. Validated three-dimensional transitional, cavity, and porous-media flows.
5. Demonstrated near-bandwidth-limited single-GPU performance and strong multi-GPU scaling.

2026LCHK

ChatGPT (July 2026)

Summary

This study uses axisymmetric JOREK simulations to resolve the formation and vertical dynamics of an X-point radiator (XPR) in ASDEX Upgrade geometry. Reduced-MHD evolution of the deuterium plasma is two-way coupled to full- f kinetic deuterium neutrals and nitrogen impurities. Starting from discharge 38773, the model follows divertor detachment, high-field-side high-density formation, and establishment of a quasi-stationary radiating mantle 6.8 cm above the X-point. Increasing nitrogen seeding raises the radiation and ionization fronts until the XPR becomes a cold, recombining MARFE-like state; removing seeding reverses this sequence, moves the radiator downward, depletes neutrals, and ultimately destroys it. The reference XPR has a 1–2 eV core, electron density near $3 \times 10^{20} \text{ m}^{-3}$, and 10–15% impurity concentration. Including impurity-background collisions localizes the radiation poloidally, improving qualitative agreement with experiment and SOLPS-ITER. The simulations establish a mechanistic two-dimensional baseline for future three-dimensional studies of XPR-MHD interaction.

Contributions

1. Built a wall-extended JOREK model coupling reduced MHD to kinetic deuterium neutrals and nitrogen impurities for an ASDEX Upgrade discharge.
2. Reproduced the progression from attached divertor through detachment and high-field-side high-density formation to an X-point radiator.
3. Characterized a quasi-stationary radiator 6.8 cm above the X-point, including its temperature, density, impurity, and front structure.
4. Resolved upward motion and MARFE formation under increased nitrogen seeding and downward retreat and loss when seeding is removed.
5. Showed that impurity collisions compress nitrogen into a more poloidally localized radiating mantle closer to experimental morphology.

2026LCLK

ChatGPT (July 2026)

Summary

This paper develops a discrete unified gas kinetic scheme (DUGKS) for the multispecies BGK–Vlasov–Maxwell system across collisional, collisionless, quasi-neutral, and strongly magnetized plasma regimes. The method couples transport and BGK collisions semi-implicitly, couples distribution moments to reformulated Maxwell equations, and advances velocity-space acceleration with an unsplit forward semi-Lagrangian step. These choices remove stability restrictions associated with short collision times, small normalized Debye lengths, and small Larmor radii without requiring nonlinear solves. The authors show that the discretization is asymptotic preserving: as the controlling parameters vanish, it recovers consistent limits including ideal magnetohydrodynamics, two-fluid equations, and the collisionless Vlasov model. Numerical tests spanning electrostatic and electromagnetic problems confirm the predicted limiting behavior, stability, and accuracy. The approach retains an explicit-like computational structure, but its present BGK collision model and fixed uniform velocity grid limit physical fidelity and efficiency for Coulomb collisions or broad velocity distributions.

Contributions

1. Formulated a single DUGKS for multispecies BGK–Vlasov–Maxwell dynamics across fluid and kinetic plasma regimes.
2. Removed collision-scale stiffness by semi-implicitly coupling particle transport with BGK relaxation.
3. Combined macroscopic moments with reformulated Maxwell equations to remain stable in the quasi-neutral limit.
4. Used an unsplit forward semi-Lagrangian velocity update to avoid force-induced CFL restrictions in strongly magnetized regimes.
5. Established and numerically verified asymptotic recovery of ideal-MHD, two-fluid, and collisionless Vlasov limits.

2026LDGB

ChatGPT (September 2026)

Summary

QuaSSis is a two-dimensional, three-velocity-component quasistatic particle-in-cell code for dilute ultrarelativistic beams propagating through denser plasmas. It separates slow beam evolution from the rapid plasma response and extends this approach to the oblique two-stream instability. A finite-difference boundary scheme supports transversely periodic systems by evolving boundary field values before solving the interior equations. Benchmarks with Gaussian electron beams and transversely uniform electron–positron beams reproduce CALDER simulations of density modulation, field spectra, instability growth, and saturation. The reported comparisons reduce computational consumption from thousands of core-hours to a few core-hours, rather than demonstrating equivalent wall-clock speedups. Regularly spaced macroparticles with randomized weights allow the initial instability seed to be adjusted without increasing particle count. Analytical density- and current-noise estimates explain this control and identify a floor imposed by finite transverse momentum spread. Lower initial noise delays saturation without changing its level in the tested cases. Validation concerns the oblique-instability regime with effectively immobile background ions; it does not establish accuracy for arbitrary beam-plasma dynamics.

Contributions

1. Demonstrated quasistatic modeling of the oblique two-stream instability with QuaSSis, including finite-temperature plasma equations and separate beam and plasma advances.
2. Detailed a finite-difference treatment of transverse periodic boundaries that resolves the otherwise undetermined boundary values of the quasistatic Poisson problems.

3. Reproduced full-PIC beam structures, spectra, growth, and saturation for Gaussian and transversely uniform beams, while reducing the reported core-hour costs by roughly three orders of magnitude.
4. Introduced ordered macroparticle positions with randomized weights to control instability seeding at fixed particle count and computational cost.
5. Derived noise scalings for multiple particle shape orders and dimensions, explaining both delayed saturation at lower noise and the limiting contribution of transverse momentum fluctuations.

2026LG

ChatGPT (July 2026)

Summary

The paper accelerates a discrete unified gas kinetic scheme for strongly inhomogeneous nanoscale fluids whose kinetic model contains costly nonlocal-gradient, weighted-density, and mean-field-potential integrals. It replaces direct integral evaluation with finite-difference nonlocal gradients, adaptive volume averaging, and coarser distance-dependent sampling of rapidly decaying attractive interactions. These changes reduce complexity from $O(NN_\sigma)$ to $O(N)$, where N is the number of physical cells and N_σ is the integration-domain size. Static density structures and force-driven slit flows agree with Monte Carlo, molecular-dynamics, and original-scheme results, while two-dimensional tests achieve speedups of up to two orders of magnitude. The efficient scheme enables previously impractical two-dimensional applications. Pressure-driven channel simulations show that pressure and body-force driving are generally inequivalent at the nanoscale because the streamwise density profile is nonlinear. Square-duct simulations reveal wall adsorption, corner density peaks, and weakening of the Knudsen minimum as the duct narrows.

Contributions

1. Reduced nonlocal kinetic-scheme complexity from $O(NN_\sigma)$ to $O(N)$.
2. Replaced integral nonlocal gradients with an efficient finite-difference construction.
3. Added adaptive volume averaging and distance-coarsened mean-field evaluation.
4. Matched reference methods while attaining up to two orders of magnitude speedup.
5. Resolved inequivalent nanoscale driving and square-duct corner density accumulation.

2026LG2

Claude (September 2026)

Summary

The authors ask when a Maxwell–Boltzmann (MB) energy distribution becomes a power law (PL), using direct-simulation Monte Carlo of spatially uniform hard-sphere gases with elastic binary collisions, relativistic kinematics, and acceptance weighted by the covariant flux factor; the code reproduces MB, Maxwell–Jüttner, and two-species equipartition equilibria. By Euler’s homogeneous-function theorem a scale-free function is a PL. If one injection evolves self-similarly, $dN/dE = g(t)f(E/E_0)$ with characteristic energy $E_0(t)$, superposing steady injections yields an index independent of f : $\Gamma = a/b - 1$ for $N_p \propto t^a$ and $E_0 \propto t^b$, $\alpha/\beta - 1$ for exponential growth rates, and -1 for linear N_p with exponential E_0 . Light particles injected steadily into a heavy background give $\Gamma = -0.5, 0.5$, and 0 for cross sections $\sigma \propto E^\delta$ with $\delta = 0, -1, -\frac{1}{2}$, as predicted from $E_0 \propto t^{2/(1-2\delta)}$, and $\Gamma = -1$ once ultrarelativistic. Cascades of fast particles through a thermal bath give $\Gamma = -2.5$ (nonrelativistic) and -2 (ultrarelativistic). The authors conclude that far-from-equilibrium initial conditions, scale-free dynamics, and a scale-free open boundary condition suffice. The PL holds only at intermediate energies, away from the initial, final, and rest-mass scales; the self-similar shapes and the nonrelativistic cascade index are not derived analytically.

Contributions

1. Showed that mixed time dependences, an exponential N_p with a PL E_0 or a PL N_p with $a \neq 1$ and an exponential E_0 , give no PL, so a self-similar distribution need not superpose into a scale-free one.

2. Derived from light–heavy reflection kinematics a constant absolute gain $\Delta E_{kl} \simeq 2m_l v_h^2$ per scattering in the nonrelativistic limit and a constant fractional gain $2\gamma_h^2 \beta_h^2$ in the relativistic limit, implying $E_0 \propto t^2$ and exponential growth, and estimated about 230 scatterings per particle for equipartition at mass ratio 1836, against 480 in the simulation.
3. Found self-similar shapes $f(x) \propto \sqrt{x} \exp(-x^\beta)$ with fitted $\beta = 1.4 \pm 0.1$ and 0.98 ± 0.06 for $\delta = -1$ and $-\frac{1}{2}$ and $\beta = \frac{1}{2}$ for $\delta = 0$, suggesting $\beta \simeq -\delta + \frac{1}{2}$, so a steeper decline of the cross section steepens the cutoff while its amplitude only rescales time.
4. Kept the exponentially growing cascade tractable by randomly subsampling the tracked population, for example to 10^4 particles whenever it reaches 10^5 , while following the true population size, which reaches about 10^{20} , and showed that energy conservation gives the self-similar scaling $(E_k, F) \rightarrow (sE_k, F/s^2)$ and, through $\alpha + \beta = 0$, the ultrarelativistic index -2 .
5. Showed in Appendix C that energy injection into a finite thermal gas yields an MB core, shifted to higher temperature at high injection, with a sharp transition to a $\Gamma \simeq -2.5$ tail rather than a κ distribution, and in Appendix B that relativistic light and nonrelativistic heavy particles in equilibrium share T but differ in $\langle E_k \rangle$ by up to a factor of 2.

Comments

- §IV A and Figs. 3 and 5: the constant-cross-section acceleration run uses light ($m = 1$) and heavy ($M = 10^6$) particles with “the same initial $\gamma\beta = 0.001$ ”, which gives mean kinetic energies 5×10^{-7} and 0.5 (Fig. 2, with $m_1 = 1$ and $\gamma\beta = 10^{-3}$, indeed places the light particles at 5×10^{-7}), but the initial light-particle MB in Figs. 3 and 5 peaks near 2×10^{-9} and the background MB lies near 10^{-3} – 10^{-2} , as the text also states (“At $E_k < 10^{-8}$ the DF is set by the initial MB”, PL at $10^{-9} \ll E_k \ll 10^{-3}$); these energies correspond to $\gamma\beta \approx 10^{-4}$ (means 5×10^{-9} and 5×10^{-3}), not 10^{-3} ; which value was used is unresolved.
- §II D, Eqs. (38)–(39): substituting $E_0 = E/x$ into the factor $E_0^{-\alpha-3/2}$ of Eq. (37) gives $(E/x)^{-\alpha-3/2}$, not the printed $(Ex)^{-\alpha-3/2}$, so the integrand of Eq. (39) should be $x^{\alpha-1/2}e^{-x}$ rather than $x^{-\alpha-7/2}e^{-x}$; the printed integrand diverges as $x \rightarrow 0$ for every $\alpha > -5/2$, contradicting the statement that the integral “converges to a definite integral at $0 - \infty$ ”, whereas the corrected one converges for $\alpha > -\frac{1}{2}$; the $E^{-1/2}$ scaling of Eq. (40) is unaffected.
- §IV A 4, Eq. (51), and Fig. 9(b): the exponential self-similarity variable is defined as $x = \exp(E/E_0)$, as also in §II B and §VI, so $x \geq 1$ for $E, E_0 > 0$, but Fig. 9(b), described as the three DFs “transformed according to Eq. (51)”, plots $f(x)$ over $10^{-10} \lesssim x \lesssim 30$, a range the printed transformation cannot produce; which variable is plotted is unresolved.
- Fig. 10 caption: the thermal background is given as “ $\langle E_k \rangle = 5 \times 10^{-8}$ ”, but §IV B 1 gives $\langle E_k \rangle = 5 \times 10^{-9}$, §IV B 2 repeats 5×10^{-9} “as in the nonrelativistic simulation”, and the “Background energy” marker in Fig. 10 lies near 5×10^{-9} ; the intended value is 5×10^{-9} .
- Fig. 10 caption: the nonrelativistic cascade is said to demonstrate “the formation of a PL DF when the system dynamics leads to $N_p(t) \propto e^{\alpha t}$ and $E_0(t) \propto e^{\beta t}$ ”, but §IV B assigns this exponential evolution to the relativistic regime, the Fig. 11 caption states that the growth “switches to a PL growth in the nonrelativistic regime”, the Fig. 10 legend values $\langle E_k \rangle = 5.4 \times 10^{-2}$, 6.2×10^{-3} , and 5.0×10^{-4} at $t = 5, 20$, and 75 give logarithmic decay rates 0.144 and 0.046 rather than a constant rate, and exponential evolution with conserved energy gives $\Gamma = -2$ by the §IV B 2 argument with Eq. (19), not the plotted -2.5 ; the exponential description applies only to the ultrarelativistic cascade of Fig. 11.
- Appendix B: at $T = 10^{11}$ K, with “ $kT \simeq 10$ MeV”, the electrons are said to be relativistic “with a mean $\gamma \simeq 20$ ”, but the same appendix states that “in MJ $\langle E_k \rangle = 3 \times k_B T$ ” [Eq. (48)], which gives a mean $\gamma \approx 59$ for $kT = 10$ MeV and 51 for the exact $kT = 8.6$ MeV; the quoted 20 is closer to $kT/m_e c^2 \approx 17$ – 20 ; unresolved.
- Appendix C: the injection rate is $Q = ne_i/\Delta t$ with $n = 10$ and $e_i = 1$, and “ Δt in the simulations is varied between 10^{-2} and 1 , always shorter than t_{th} ” ($t_{\text{th}} \sim 10$), which gives $10 \leq Q \leq 10^3$, but the simulations and Fig. 13(a) include $Q = 1$, which requires $\Delta t = 10 \sim t_{\text{th}}$ for the stated n and e_i ; unresolved.

2026LGST

ChatGPT (September 2026)

Summary

The authors experimentally demonstrate relativistic electron acceleration in a laser wakefield driven by a flying focus, whose intensity maximum moves at an adjustable velocity. An axiparabola, a mirror with an extended focal region, focuses a laser pulse into a helium–nitrogen gas jet, while adjustable pulse-front curvature controls the timing across the beam. Measurements over twenty shots at each of three settings show that faster wakefields produce higher electron cutoff energies, increasing from approximately 350 to 400 MeV. Optical propagation and particle-in-cell simulations reproduce the energy shift and show reduced electron dephasing, the loss of accelerating phase caused by electrons outrunning the wake. A simplified analytical model predicts the same trend but overestimates the energies. The experiment establishes that this structured wakefield can sustain relativistic acceleration and provides evidence for partial dephasing mitigation. Fully dephasingless operation has not yet been demonstrated and requires improved control and measurement of wake velocity inside plasma.

Contributions

1. Demonstrates electron acceleration to several hundred MeV, with charges of tens of picocoulombs, using an axiparabola and adjustable pulse-front curvature.
2. Controls electron cutoff energy through wake-velocity tuning, observing an approximately 50 MeV increase between the slowest and fastest settings over twenty shots per setting.
3. Establishes statistically distinct maximum-energy distributions despite substantial charge fluctuations, including $p = 0.0004$ between the fastest and intermediate settings.
4. Reproduces an approximately 45 MeV energy separation in particle-in-cell simulations and resolves changes in electron position within the wake that support reduced dephasing.
5. Develops a simplified bubble-regime model linking the wake velocity controlled by pulse-front curvature to longitudinal momentum gain and the observed ordering of cutoff energies.

2026LGTT

ChatGPT (July 2026)

Summary

Flying-focus laser pulses combine spatiotemporal coupling with an axiparabola to create a quasi-Bessel intensity peak whose axial velocity can be tuned, potentially preventing electrons from outrunning a laser wakefield. This work directly images the resulting plasma structures with femtosecond relativistic electron microscopy and reproduces them using linked optical, particle-in-cell, and probe-beam simulations. Repeated measurements establish that the observed V-shaped, off-axis wake structure is stable rather than a single-shot artifact. Density scans show that increasing plasma density shortens the wake period, strengthens and nonlinearizes the longitudinal field, splits the on-axis driver, and changes the phase relation between on- and off-axis wakefields; the simulated field rises from 11 to 99 GV/m across the tested range. Ionization materially changes the structure in helium compared with a preionized plasma, whereas adding a small nitrogen fraction has little structural effect despite its relevance to electron injection. Simulations also show a sharp difference between focusing at the start of the gas plateau and focusing farther inside it. These results validate a diagnostic-and-simulation workflow for optimizing flying-focus wakefields, but the study characterizes wake structure rather than demonstrating dephasing-free electron acceleration.

Contributions

1. Directly imaged stable flying-focus wakefield structures with femtosecond relativistic electron microscopy.
2. Reproduced the measured structures with coupled optical, particle-in-cell, and probe simulations.
3. Quantified density-dependent changes in wake period, field strength, nonlinearity, and off-axis phase.
4. Distinguished the structural effects of preionization, hydrogen, helium, and helium–nitrogen targets.

5. Identified the focal-line position relative to the gas plateau as an important control parameter.

2026LHAH

Claude (September 2026)

Summary

In laser wakefield acceleration with tightly focused, near-petawatt pulses in plasma of density near 10^{18} cm^{-3} , self-focusing and self-phase modulation distort the laser envelope before electrons are self-injected. The authors build a fast quasistatic model: cold relativistic fluid equations for plasma electrons, with immobile ions, are solved on a cylindrical grid moving at c ; the laser is divided into longitudinal slices whose Gaussian spot radii follow an envelope equation; and the local frequency follows from the refractive-index gradient. Group-velocity dispersion and energy depletion are neglected. For $a_0 < 1$ the redshift grows linearly with density and quadratically with a_0 , as the authors' analytic estimate predicts. For $a_0 > 1$ it becomes localized within the pulse, and for oversized spots the distance before the maximum local wavelength exceeds 1.5 times the carrier wavelength is shorter than the self-guiding and pump-depletion lengths. Experiments with the 4.3 J, 21 fs HF-PW laser at ELI ALPS show larger divergence and pointing of electrons below 200 MeV along the polarization axis. FBPIC simulations reproduce the localized redshifting where the model predicts it and show that the resulting single-cycle waveform drives an asymmetric bubble and off-axis injection. An envelope-solver PIC simulation lacks the asymmetry below 200 MeV.

Contributions

1. Developed a fluid-envelope slicing model that evolves the local laser frequency and spot radius self-consistently with the plasma response, as a quick tool valid over about one to two Rayleigh lengths.
2. Derived an analytic weak-pump redshift estimate and confirmed its density and amplitude scalings numerically, with stronger redshift for shorter pulses up to an optimum.
3. Computed average and maximum redshift versus density and a_0 , and the redshift-limited acceleration length for matched and oversized spots, placing the onset of the redshift-dominated regime near 10^{18} cm^{-3} at $0.8 \mu\text{m}$ wavelength.
4. Linked localized redshifting to a single-cycle waveform, bubble asymmetry, and off-axis injection using PIC velocity maps that separate the symmetric and asymmetric parts of the transverse electron velocity.
5. Showed with an exponential-propagator PIC code that full-field simulations inject more off-axis charge below 200 MeV than envelope-solver simulations, which the authors attribute to the combined effect of bubble expansion and undulation on trapping.

Comments

- §III, Eq. (10): the first-order expansion of Eq. (9) gives $R_a \approx \frac{1}{2} c \frac{\partial \gamma}{\partial \xi} \frac{n_{e0}}{n_{c0}} t$, and with $\gamma = \sqrt{1 + |\vec{p}|^2 / (m_e c)^2 + a^2 / 2}$, $\partial \gamma / \partial \xi \approx \frac{1}{4} \partial a^2 / \partial \xi$, so the coefficient is $\frac{1}{8}$ rather than the printed $\frac{1}{4}$; the stated linear and quadratic scalings are unaffected, and where the factor of 2 arises is unresolved.
- §IV B: the text states that Eq. (7) with $K = -1$ and $n_e \approx n_{e0}$ gives $w_{sg} \approx \sqrt{a_0} \lambda_p / \pi$, that is, $k_p w = 2\sqrt{a_0}$, for $a_0 \gg 1$, but evaluating Eq. (7) as printed with $n_e = n_{e0}$ and $w = w_0$ gives $k_p w \approx 3.6, 3.9$, and 4.7 for $a_0 = 4, 10$, and 20 , rather than $4.0, 6.3$, and 8.9 ; unresolved.
- §IV B and §VI: the onset density $n_0 = 10^{18} \text{ cm}^{-3}$ at $\lambda_0 = 0.8 \mu\text{m}$ is restated “in other words” as $n_0 / n_c > 10^{-3}$ in §IV B but as $n_{e0} > 5 \times 10^{-4} n_{c0}$ in §VI; since $n_c \approx 1.74 \times 10^{21} \text{ cm}^{-3}$ at $0.8 \mu\text{m}$, 10^{18} cm^{-3} is $5.7 \times 10^{-4} n_c$, so the §VI value is the consistent one.
- §V and Fig. 7: the text states that “the electron’s angular-energy distribution in the orthogonal plane was not measured”, where the same paragraph identifies the orthogonal plane with Fig. 7(c), the θ_y plane perpendicular to the x polarization, but the Fig. 7 caption describes panels (d) and (e) as “Example spectra from experiments in the plane perpendicular to the laser’s polarization plane”; the unmeasured plane is likely the polarization plane.
- §V: the PIC laser has “FWHM pulse duration is 21 fs ($t_L = 18$ fs)”, but §II defines $\tau_L = ct_L$ as “the FWHM of the intensity envelope”; since $21 / \sqrt{2 \ln 2} \approx 17.8$ fs parallels “15 μm ($w_0 = 12.7 \mu\text{m}$)” in the same sentence, t_L in §V is evidently the $1/e$ field half-duration rather than the FWHM.

2026LHD

ChatGPT (September 2026)

Summary

This paper presents `jaxdae`, a JAX-native solver for differential-algebraic equations (DAEs), which combine differential evolution with constraints that must hold at each time. It combines adaptive BDF, Radau, and Rosenbrock integration with Pantelides index reduction, dummy derivatives, and acausal assembly of coupled physical components. The adaptive BDF gradient uses a custom reverse rule: it records the accepted time grid, freezes that grid, and re-solves a variable-step BDF-2 problem for its adjoint. This avoids differentiating discontinuous step-acceptance decisions; the gradient is the adjoint of the replay problem, not exactly of the forward adaptive solve. BDF and fixed-step Rosenbrock paths support JAX differentiation, whereas Radau is forward-only. Order tests, cross-solver comparisons, gradient checks, and chemical, nuclear, and power-system examples evaluate the implementation. On measured GPU benchmarks, compiled batched gradients remain nearly flat in wall time through batch size 1,000 for small models; comparison runs for the PyTorch counterpart reach only batch size 10. The work leaves distributed multi-GPU execution and wider event-adjoint verification open.

Contributions

1. Integrated implicit DAE solvers, structural index reduction, and acausal multiphysics assembly in JAX.
2. Implemented a frozen-grid BDF-2 replay adjoint for adaptive solves that composes with `jax.grad`, `vmap`, and `jit`.
3. Verified fixed-step BDF orders one through five and checked DAE gradients against finite differences and independent references.
4. Demonstrated batched GPU gradient throughput and parameter-inference workflows in chemical and nuclear models.
5. Documented the replay-adjoint approximation and the present limits of Radau differentiation, event handling, and multi-GPU execution.

2026LHKD

Claude (September 2026)

Summary

Laser-plasma accelerator beams drift over an operating shift, while the interaction-point conditions responsible are not measured. The authors model operation as a latent state-space system whose hidden state is the normalized laser amplitude a_0 , normalized plasma density $\tilde{n}_e$, and residual chirp C , estimated shot by shot from centroid energy, energy spread, and charge with an extended Kalman filter. A closed-form emission model, built from relativistic self-focusing, blowout-regime energy scaling, and chirp-dependent downramp injection, maps the state to observables; its Jacobian localizes which variable moved. Separately, hardware-derived transition models linking laser-head temperature, humidity, and room temperature to a_0 , C , and $\tilde{n}_e$ compete against a random-walk null by predictive likelihood, with profiled coupling gains and a BIC penalty. Two innovation screens and a four-condition gate decide whether to recommend an intervention. In synthetic sessions generated from the same model family, the responsible mechanism family is recovered whenever its energy signature is visible, humidity drift is unidentifiable because $\partial E / \partial C = 0$ at the operating point, and attribution is limited by environmental excitation rather than session length. The authors stress that the emission model bounds the achievable accuracy and that validation on measured data remains open.

Contributions

1. Formulated laser-plasma accelerator operation as a three-variable physical latent state-space model whose separate emission and transition models split drift diagnosis into localization and attribution.
2. Constructed a differentiable reduced-physics emission map in which chirp enters energy spread and charge at first order, making compressor drift distinguishable from laser-energy and gas-density drift.

3. Made the candidate transition models memoryless so that each score is an exact log marginal likelihood, and derived a family-margin evidence measure growing with shot count and with the squared difference of mechanism energy signatures.
4. Designed a lag-1 autocorrelation and normalized-innovation-squared screen that lowered the session-level false-alarm rate under a correct null from 85.0% to 0.01%, feeding a gate that issues falsifiable recovery predictions.
5. Showed that deliberate 6 °C laser-head dithering yields decisive attribution that ambient drift never provides, that margins stay decisive up to twice the baseline observation noise, and that the full causal candidate excludes the null without identifying a subsystem.

2026LHWP

ChatGPT (July 2026)

Summary

This paper introduces a time-embedded convolutional neural network (TCNN) for one-dimensional nonlocal electron heat transport in plasmas. Trained on fully kinetic OSIRIS particle-in-cell simulations, the model jointly predicts normalized heat flux and the nonlocality measure $\lambda_{\text{free}}/L_T$. Time-indexed convolutions represent evolving transport rather than imposing the quasistationary kernel assumed by the Luciani–Mora–Virmont model and the authors’ earlier LINN surrogate. Across collisionalities, TCNN most clearly improves predictions in strongly nonlocal, rapidly evolving regimes; local quasistatic cases remain better suited to a possible TCNN–LINN hybrid. Architecture variants show that convolutional depth and latent spatial projection materially improve nonlocal predictions. Tests with perturbed inputs show sublinear error growth as noise increases. The current isobaric one-dimensional training data and expensive multidimensional kinetic data limit immediate deployment, but the architecture provides a stable, physically structured surrogate for future radiation-hydrodynamics coupling.

Contributions

1. Designed a time-embedded convolutional architecture for dynamic nonlocal plasma heat transport.
2. Jointly predicted heat flux and a physically coupled nonlocality parameter.
3. Trained and tested the surrogate on fully kinetic particle-in-cell simulations across collisional regimes.
4. Used architecture variants to isolate the roles of convolutional depth and latent spatial projection.
5. Demonstrated robustness to noisy inputs and clarified extensions toward multidimensional transport models.

2026LJF

Claude (September 2026)

Summary

The authors present cuGMEC, a gyrokinetic–magnetohydrodynamic (MHD) hybrid code written from scratch in CUDA C++ for GPUs to simulate Alfvén eigenmodes driven by energetic particles (EPs) in tokamaks. Relative to the earlier CPU code GMEC, the fluid model adds thermal-ion finite-Larmor-radius (FLR) effects to the gyrokinetic vorticity equation through a Padé approximant, a parallel electric field in the generalized Ohm’s law, an electron continuity equation with an isothermal closure, and all nonlinear terms; thermal ions and EPs are advanced by gyrokinetic δf particle-in-cell (PIC) and enter through their perturbed pressures. Five-point central differences and fourth-order Runge–Kutta are applied in Boozer, field-aligned, and shifted-metric coordinates built from DESC equilibria. Multi-GPU runs partition the poloidal direction, exchange ghost cells through NCCL and MPI, all-reduce particle pressures, and solve the gyrokinetic Poisson equation with cuDSS. Benchmarks against theory and other codes cover tearing modes, toroidal and reversed-shear Alfvén eigenmodes, FLR radiative damping, nonlinear multi- n coupling, and cyclone-base-case ion-temperature-gradient and kinetic ballooning modes; trapped-electron modes are outside the model because electrons are fluid. On A800 GPUs, strong scaling to 32 GPUs retains 81.7% (MHD) and 72.0% (full model) efficiency, and one GPU replaces about 21.9 Xeon Gold 6348 CPUs.

Contributions

1. Diagnosed an odd–even decoupling instability along field lines caused by central differencing of $b_0 \cdot \nabla$ on a collocated grid and removed it by staggering only the y grid, on which $\delta A_{\parallel}$ and $\delta J_{\parallel}$ are evolved.
2. Replaced fully expanded stencils (25 coefficients for $\delta J_{\parallel}$) with Mathematica-generated compressed coefficients multiplying retained derivatives (6 coefficients), gaining about 1.5–1.75 $\times$ in double precision, and fixed N_z to a half or full warp so that interpolation along z can use warp-shuffle instructions.
3. Accelerated PIC field gathering by storing the eight gathered fields contiguously per grid point (3.0–4.25 $\times$ on A800, 2.0–3.5 $\times$ on RTX 4090) and by radix-sorting particles by cell every 20 steps to improve cache reuse.
4. Verified the FLR radiative damping scaling, with fitted slope and intercept within 4.8% and 8.5% of theory, and nonlinear mode generation in which the $n = 0, 12, 18$ growth rates are 1.95, 1.93, and 2.98 times that of $n = 6$.
5. Provided per-step timing tables versus particles per cell and gyro-average points for single- and multi-mode grids, and a GPU-to-CPU cost equivalence of 17.4 $\times$ for MHD and 22.5 $\times$ for PIC.

Comments

- Section 3.2, Fig. 3: the text states that “the shuffle instruction provides further improvement of about 1.5–2.0 $\times$ on top of coefficient compression”, but in the double-precision panels of Fig. 3 compression plus shuffle exceeds compression alone by only about 1.04–1.09 $\times$ (RTX 4090, L40) and 1.24–1.40 $\times$ (V100, A800); the stated range matches only the single-precision panels.
- Section 3.4, Fig. 4: the text states “Speedups of 2.0 $\times$ –3.0 $\times$ and 5.0 $\times$ –8.0 $\times$ are achieved on the A800 and RTX 4090”, but the RTX 4090 double-precision sorting curves lie at about 1.0–1.9 $\times$, and the claim that the two optimizations together “achieve a speedup of approximately 10.0 $\times$ –25.0 $\times$ for the PIC component” exceeds the products of the rearrangement and sorting speedups in Fig. 4 at matching precision and gyro-average points, which stay below about 16 $\times$; unresolved.
- Section 4.4, text after Eq. (26): “The background ions are hydrogen with $\rho_i/a = 125$ ” conflicts with the stated $T_{\text{ref}} = 2.2$ keV, $B_0 = 2$ T, and $a = 0.36R_0$ with $R_0 = 0.835$ m, which give a gyroradius of a few millimetres and $a/\rho_i \approx 125$; the evident intended reading is $a/\rho_i = 125$, the ratio used in Section 4.2.

2026LJLH

ChatGPT (July 2026)

Summary

FourierSpecNet is a learned Boltzmann collision operator whose architecture mirrors the separable convolution structure of a fast Fourier spectral solver. Learnable coefficients occupy a fixed truncated Fourier band, so the parameter count does not grow with velocity-grid resolution and one trained model can evaluate unseen grids. Training targets are spectral collision operators for Gaussians, Gaussian mixtures, and perturbed Gaussians. A consistency result bounds learned-operator error by approximation and spectral-truncation terms and gives convergence as bandwidth and discretization are refined under kernel-mode decay assumptions. Experiments cover Maxwellian, hard-sphere, inelastic, and three-dimensional velocity cases. Predictions track analytic or spectral references, approximately preserve physical moments, and retain similar errors across tested resolutions. At large grids, repeated inference is substantially faster than the reference fast spectral implementation, excluding training cost.

Contributions

1. Designed a neural architecture that reproduces the separable convolution structure of a fast spectral collision operator.
2. Made the learned parameters independent of grid resolution, enabling zero-shot super-resolution.
3. Proved a consistency result relating learned-operator error to coefficient approximation and spectral truncation.
4. Validated the method for elastic, inelastic, and three-dimensional velocity-space problems.
5. Demonstrated approximate physical-moment preservation and up to roughly 72 $\times$ inference speedup at the largest tested grid.

2026LKCH

ChatGPT (July 2026)

Summary

This paper proposes an adaptive mesh refinement strategy for discontinuous Galerkin solvers of conservation laws. The method uses structured DG grids inside recursively refinable blocks, while allowing the root blocks to have unstructured connectivity represented as a forest. This combines simple local data layouts with the geometric flexibility needed for complex domains. Guard cells provide inter-block communication, and L^2 projection transfers solution data across nonconforming refinement interfaces so flux calculations remain local to structured block grids. Smooth boundary fidelity is maintained by projecting refined boundary nodes onto analytic surfaces or reconstructed NURBS surfaces. The FSmesh implementation includes MPI parallelism, dynamic load balancing through space-filling curves or graph partitioning, and 2D and 3D benchmark tests. Results show expected convergence for smooth problems, effective shock and vortex tracking, and good parallel efficiency. The method targets DG users who need AMR without giving up complex geometry support.

Contributions

1. Developed a forest-of-blocks AMR structure with structured interiors and unstructured root-block topology.
2. Used guard cells and L^2 projection to simplify conservative DG communication across refinement levels.
3. Added analytic and NURBS-based boundary projection to preserve smooth geometry during refinement.
4. Implemented MPI load balancing with space-filling-curve and graph-partitioning options.
5. Verified convergence, dynamic refinement, and parallel performance on 2D and 3D conservation-law benchmarks.

2026LL

ChatGPT (July 2026)

Summary

This review explains how factorization-based solvers for large sparse or compressible dense systems can remain robust on distributed CPU and GPU machines. It first frames sparse LU and multifrontal elimination through dependency DAGs and elimination trees, then examines communication- and synchronization-avoiding execution: MPI one-sided queues, three-dimensional process grids, replicated computation, and communication lower bounds. GPU sections emphasize irregular batching and fine-grained task and data parallelism. The second half surveys data-sparse formats—BLR, HSS, HODLR, $\mathcal{H}$, $\mathcal{H}^2$, butterfly, and related hierarchical factorizations—that lower arithmetic and memory complexity when off-diagonal blocks are compressible. Embedding these formats into multifrontal sparse solvers can approach linear complexity, often as a Krylov preconditioner. The chapter compares distributed packages and closes with open problems in GPU-resident preprocessing, symmetric-indefinite distributed solvers, stability, portability, and automatic compression decisions.

Contributions

1. Gave a unified DAG and elimination-tree account of parallel supernodal and multifrontal sparse factorization and triangular solution.
2. Synthesized latency- and bandwidth-reduction techniques, including one-sided and three-dimensional communication-avoiding algorithms with explicit cost models.
3. Explained how irregular batched kernels and fine-grained scheduling expose GPU parallelism hidden by small, uneven sparse blocks.
4. Systematized hierarchical low-rank and butterfly formats, their construction and factorization, distributed layouts, and stability trade-offs.
5. Showed how structure sparsity and data sparsity combine in multifrontal solvers, surveyed distributed software, and identified major open problems.

2026LLBZ

ChatGPT (July 2026)

Summary

The authors couple relativistic guiding-center, full- f particle-in-cell runaway electrons to JOREK's reduced-MHD background plasma, creating an efficient hybrid fluid-kinetic model for slowly evolving resistive instabilities. Axisymmetric force balance, runaway drift-orbit shifts, and linear $(2, 1)$ tearing-mode behavior agree with analytical theory, full-orbit JOREK, M3D-C1, and eigenvalue results. Runaway electrons create a narrow current sublayer, and their current gradient adds a mode-frequency component absent from Ohmic plasmas. Nonlinearly, runaway-driven magnetic islands saturate wider than Ohmic islands: the width ratio is about 1.7 for $\Delta' \approx 0.2$ and exceeds 2 for $\Delta' \approx 2$, where topology becomes more complex. Magnetic energy grows strongly while bulk-plasma kinetic energy and reconnection flows remain weaker because perturbed runaway current drives reconnection without the usual Lorentz-force acceleration of the fluid. The framework targets disruption and benign runaway-beam termination studies.

Contributions

1. Implemented relativistic guiding-center full- f runaway electrons coupled self-consistently to JOREK reduced MHD.
2. Verified equilibrium and drift-orbit shifts against analytical predictions and full-orbit calculations.
3. Reproduced linear tearing-mode growth and frequency benchmarks, including the narrow runaway-current sublayer.
4. Quantified magnetic-island saturation substantially wider than in otherwise identical Ohmic cases.
5. Demonstrated magnetic-fluid decoupling that produces stronger islands but weaker reconnection flows.

2026LLL

ChatGPT (July 2026)

Summary

The authors use two-dimensional first-principles particle-in-cell simulations to determine how ion-acoustic instability affects collisionless magnetic reconnection. Harris-sheet runs compare upstream ion-to-electron temperature ratios $T_i/T_e = 1/50, 1/10$, and 1. In the cold-ion cases, electron-ion drift in the diffusion region exceeds the ion-acoustic threshold, producing wave activity consistent with linear Vlasov-Poisson theory and nonlinear ion phase-space structures. The instability heats outflowing ions until $T_i \approx T_e/10$ and causes local momentum fluctuations, but the diffusion-region-averaged anomalous terms are nearly zero. Peak reconnection rates vary by less than a factor of two across the temperature ratios, and anomalous resistivity remains negligible, contrary to estimates based on initially supercritical or steady-state assumptions. Ion-acoustic turbulence therefore redistributes reconnection energy mainly through ion heating rather than enabling Petschek-like reconnection. Scaling arguments suggest that analogous instability-driven heating may operate at sub-ion reconnection sites in low- T_i/T_e solar-wind turbulence.

Contributions

1. Demonstrated self-consistent ion-acoustic instability in cold-ion reconnection diffusion regions.
2. Matched instability onset and wave scales to local linear kinetic theory.
3. Showed that the instability heats ions to approximately one-tenth of the electron temperature.
4. Found negligible diffusion-region-averaged anomalous resistivity and little change in reconnection rate.
5. Derived conditions connecting the mechanism to sub-ion-scale solar-wind reconnection.

2026LLR

ChatGPT (July 2026)

Summary

This work addresses geometric robustness in the three-dimensional particle finite element method (PFEM) for chaotic free-surface flows. Instead of reconstructing the fluid domain with a classical α -shape, it advects the boundary and uses winding-number tests to classify remeshed tetrahedra, improving the treatment of topology changes and mass conservation. Adaptation proceeds in two stages: edge splitting refines the free surface while preserving its shape, then circumcenter insertion refines the interior according to a prescribed size field. Solid and fluid geometries are decoupled through inside/outside queries, segment–triangle intersections, and point projections, which also support slip boundaries. Dam-break comparisons reproduce measured and published pressure histories, while tire-splash and point-cloud-derived “triangle soup” examples demonstrate operation on complex, poor-quality, and non-watertight inputs. The resulting framework makes PFEM remeshing locally adaptive and tolerant of industrial geometry defects, although sliver control and parallel or GPU acceleration remain future work.

Contributions

1. Replaced α -shape domain recovery with advected-boundary winding-number classification.
2. Designed separate surface edge-splitting and volume circumcenter-insertion adaptation stages.
3. Reduced topology-change mass errors while preserving free-surface geometry.
4. Decoupled solid and fluid meshes through a small set of robust geometric queries.
5. Validated the approach on dam-break, tire-splash, and non-watertight-geometry simulations.

2026LLXZ

ChatGPT (July 2026)

Summary

Li et al. accelerate forcing-term evaluation in the Enskog–Vlasov–Shakhov kinetic equation for dense, nonequilibrium fluids. They expand the molecular distribution in Hermite polynomials in a locally shifted and temperature-scaled velocity coordinate, so the Maxwellian is recovered exactly at zeroth order and higher coefficients represent nonequilibrium corrections. A matching shifted-and-scaled Gauss–Hermite quadrature reduces the number of discrete velocities, while a component-wise recurrence avoids full tensor manipulation and supports arbitrary expansion order. Fourier, Couette, and Poiseuille flows validate surface-confinement behavior; equilibrium coexistence and evaporation tests assess phase transitions and interfacial Knudsen layers. The method matches high-resolution finite-difference references with far fewer velocity nodes and larger time steps, reaching a reported 566-fold speedup in a two-dimensional Poiseuille case. An adaptive expansion raises order near interfaces and vapor regions while retaining low order in near-equilibrium liquid.

Contributions

1. Reformulated the Enskog–Vlasov forcing term through a locally shifted-and-scaled Hermite expansion.
2. Derived component-wise arbitrary-order coefficient formulas that avoid expensive tensor operations.
3. Coupled the expansion to shifted-and-scaled Gauss–Hermite quadrature, reducing velocity nodes by one to two orders of magnitude.
4. Introduced a local adaptive-order criterion that concentrates nonequilibrium corrections near interfaces and Knudsen layers.
5. Validated surface-confined, phase-equilibrium, and evaporative flows, including up to 566-fold speedup over fourth-order velocity-space finite differences.

2026LMHL

ChatGPT (July 2026)

Summary

TRIMEG-GKX is presented as an electromagnetic gyrokinetic particle-in-cell code for tokamak core plasma simulations. The code combines object-oriented multi-species modeling, a mixed-variable/pullback electromagnetic formulation, and a high-order piecewise field-aligned finite element method using cubic B-splines. Its field-aligned discretization reduces grid requirements along magnetic-field directions, while particle decomposition with domain cloning replaces traditional domain decomposition for the demonstrated use cases. The paper emphasizes filter- and buffer-free treatment of the strong gyrocenter equations, plus cache optimization for particle-field interpolation that yields a reported tenfold speed-up in typical single-harmonic simulations. Benchmarks cover energetic-particle-driven Alfvén eigenmodes, reversed-shear Alfvén eigenmodes, ion-temperature-gradient modes, and kinetic ballooning modes in ad hoc and experimentally reconstructed ASDEX Upgrade, TCV, and JET equilibria. The work establishes the code’s numerical scope and benchmark behavior; nonlinear physics validation, open-field-line configurations, and collisions are left as future development paths.

Contributions

1. Described TRIMEG-GKX as an electromagnetic, multi-species gyrokinetic particle code for core tokamak instabilities.
2. Integrated a high-order piecewise field-aligned finite element discretization into the particle-in-cell workflow.
3. Implemented mixed-variable/pullback electromagnetic dynamics without relying on Fourier filters or numerical buffers.
4. Reported cache and parallelization choices that improve particle-field interpolation and multi-harmonic simulations.
5. Benchmarked Alfvénic, ion-temperature-gradient, and kinetic-ballooning cases in idealized and reconstructed equilibria.

2026LMSH

ChatGPT (July 2026)

Summary

The authors extend TRIMEG-C1 with a high-order, C^1 -continuous, piecewise field-aligned finite-element method for global electromagnetic gyrokinetic particle-in-cell simulations spanning tokamak core, separatrix, and open-field-line regions. Locally field-aligned basis functions combine with unstructured triangular poloidal meshes in cylindrical coordinates, preserving parallel accuracy without the distortion and separatrix singularity of globally field-aligned coordinates. The formulation supports δf and full- f models and couples mixed-variable electromagnetic equations, generalized pullback mitigation of cancellation, and an iterative Ampère solver with analytically controlled skin-depth terms. Linear simulations of TCV-X21 reproduce expected electromagnetic ion-temperature-gradient and kinetic-ballooning-mode behavior. Scans over toroidal mode number, plasma β , and normalized machine size capture the ITG-to-KBM transition and shifts in dominant modes. Multi-mode nonlinear runs demonstrate three-dimensional electromagnetic capability without toroidal filtering.

Contributions

1. Extended piecewise field-aligned finite elements to whole-volume tokamak geometry with open field lines.
2. Combined high-order C^1 triangular elements with local field alignment to avoid global-coordinate singularities.
3. Integrated mixed-variable and generalized pullback schemes for electromagnetic cancellation control.
4. Validated ion-temperature-gradient and kinetic-ballooning physics through TCV-X21 parameter scans.
5. Demonstrated multi-mode three-dimensional electromagnetic gyrokinetic particle simulations without toroidal filtering.

2026LMSZ

ChatGPT (July 2026)

Summary

The paper constructs optimization-based postprocessors that keep high-order gas-dynamics solutions inside the convex invariant domain of positive density and internal energy while retaining global conservation. An explicit projection onto the admissible Euler-state set enables Davis–Yin splitting for the ℓ_2 limiter; Douglas–Rachford splitting, with an inner Davis–Yin solve, handles the ℓ_1 limiter. In discontinuous Galerkin schemes, the optimization first repairs cell averages and a Zhang–Shu scaling limiter then repairs quadrature values. Manufactured tests retain optimal high-order convergence. Sedov blast-wave and Mach-2000 astrophysical-jet tests remain robust, locate shocks correctly, and work with both strong-stability-preserving and non-strong-stability-preserving Runge–Kutta schemes. The ℓ_2 method is generally cheaper, whereas the ℓ_1 limiter is activated less often in the jet test. The approach preserves global, but not cell-local, conservation.

Contributions

1. Derived an efficient explicit projection onto the gas-dynamics invariant domain.
2. Formulated a Davis–Yin ℓ_2 invariant-domain limiter.
3. Formulated a nested Douglas–Rachford/Davis–Yin ℓ_1 limiter.
4. Combined cell-average optimization with Zhang–Shu pointwise scaling for high-order discontinuous Galerkin methods.
5. Verified accuracy and robustness on manufactured solutions, Sedov blast waves, and Mach-2000 jets.

2026LO

ChatGPT (July 2026)

Summary

Three-dimensional particle-in-cell calculations assess externally injected electron acceleration by microwave-driven wakefields in a low-density-plasma-filled rectangular waveguide. A reduced model first maps injection phase and initial velocity, showing that useful acceleration requires placement in the first accelerating bucket and a speed close to the microwave pulse’s group velocity so electrons remain phase synchronized. Under the studied parameters, optimized electrons gain roughly 10^2 keV over meter-scale propagation, whereas small phase offsets sharply reduce the gain and can cause net deceleration. Finite-bunch test-particle simulations then expose effects omitted by the one-dimensional estimate: the transverse electric field of the waveguide’s TE_{10} mode deforms the bunch anisotropically and lowers acceleration efficiency. Fully self-consistent simulations show that moderate bunch charge leaves the wake largely intact and retains comparable net gain, but space charge broadens the bunch longitudinally and increases transverse distortion. The work therefore identifies synchronization and beam-quality constraints, rather than only attainable energy gain, for evaluating this compact accelerator concept.

Contributions

1. Mapped acceleration versus electron injection phase and initial velocity in the microwave-driven wake.
2. Identified near-group-velocity injection into the first accelerating bucket as the condition for sustained phase synchronization.
3. Predicted order-100 keV energy gain over meter-scale interaction lengths for optimized injection.
4. Quantified anisotropic bunch deformation caused by the TE_{10} mode’s transverse electric field.
5. Showed self-consistently that space charge preserves moderate energy gain while degrading longitudinal and transverse beam quality.

2026LQ

ChatGPT (July 2026)

Summary

The spectral-volume method obtains a high-order polynomial in each spectral volume from control-volume averages, but discontinuities require limiting to suppress spurious oscillations. This paper develops a control-volume simple WENO limiter that combines the original high-order reconstruction with linear polynomials formed from a target control volume and its neighbors using nonlinear weights. The limited polynomial may be discontinuous inside a spectral volume, so numerical fluxes are used at troubled-control-volume and spectral-volume boundaries, while direct physical-flux evaluation remains possible where the reconstructed solution is continuous. The construction uses only a few linear candidate polynomials, preserves the compact stencil of the spectral-volume method, and retains control-volume resolution. Tests on one- and two-dimensional scalar conservation laws and Euler systems recover the designed convergence rates for smooth solutions and resolve shocks, blast waves, Riemann problems, a forward-facing step, and double-Mach reflection without unstable oscillations.

Contributions

1. Introduced a control-volume-based simple WENO limiter for the spectral-volume method.
2. Combined the original high-order reconstruction with local linear candidates through nonlinear weights.
3. Supplied consistent numerical-flux treatment for newly discontinuous internal control-volume interfaces.
4. Preserved compactness and control-volume resolution with a small set of simple candidate polynomials.
5. Demonstrated high-order accuracy and robust shock resolution on one- and two-dimensional scalar and system tests.

2026LRBG

ChatGPT (July 2026)

Summary

Drift-reduced fluid models remove fast gyrofrequency dynamics from collisional plasma simulations, but common reductions can obscure exact conservation laws. This paper derives a conservative formulation by analytically inverting, without a perturbative truncation, the implicit relation that defines polarization velocity through the time derivative of the electric field. The inversion supplies a transport equation for leading-order perpendicular momentum and closes the reduced fluid system in arbitrary magnetic geometry. Electromagnetic fluctuations are retained rather than imposing the electrostatic limit. Starting from the species equations and Maxwell equations, the authors establish exact conservation of mass, charge, energy, and the relevant momentum components, including exchanges between particles and fields. The construction is independent of the chosen fluid closure and can therefore be combined with Braginskii two-fluid or more general multispecies closures. Unlike variational reductions based on an expanded guiding-center Lagrangian, the method operates directly on the fluid equations and makes exact conservation the organizing constraint. Its implications for long-time plasma-turbulence calculations and conservative numerical solvers remain to be assessed computationally.

Contributions

1. Analytically inverted the implicit polarization-velocity relation without a perturbative expansion.
2. Derived a closed conservative drift-reduced system in arbitrary magnetic geometry.
3. Retained electromagnetic fluctuations beyond the electrostatic approximation.
4. Proved exact mass, charge, energy, and momentum conservation for the reduced equations.
5. Made the formulation independent of the fluid closure, enabling two-fluid and multispecies applications.

2026LRDQ

Claude (September 2026)

Summary

The authors measure carbon-ion stopping in dense plasma at strong ion–plasma coupling, where linear-response and binary-collision theories are expected to fail. Coupling is quantified by $\mathcal{Z} = Z_{eff}/(n_0\lambda_D^3)$, the effective ion charge over the electron number in a Debye sphere; as background, weak coupling ($\mathcal{Z} \ll 1$) gives $dE/dx \propto Z_{eff}^2$. Laser-accelerated ions pass a magnetic dipole with slits that selects quasi-monoenergetic pulses of equal momentum-to-charge ratio, separated by time of flight. Incident 7.00 MeV (583 keV/u) C^{5+} ions cross a 1 mm foam heated by hohlraum X-rays to $T_e = 17 \pm 1$ eV and $n_e = (4.0 \pm 0.3) \times 10^{20} \text{ cm}^{-3}$, while earlier C^{6+} ions probe the cold foam, where their 1.51 MeV loss agrees with SRIM. A Thomson parabola spectrometer resolves exit energy and charge state, so Z_{eff} need not be assumed. The plasma loss is 1.92 MeV at an estimated $\mathcal{Z} \approx 3.8$. Hybrid charge-state and stopping simulations show that T-Matrix, Bethe–Bloch, Vlasov, Brown–Preston–Singleton, Li–Petrasso, and Zwicky’s combined models overestimate the loss and underestimate the mean charge, whereas classical molecular dynamics with quantum corrections agrees within uncertainties. Only C^{6+} and C^{5+} are analysed, as later ions meet expanded plasma of unknown state; the authors call for slower, more highly charged ions and self-consistent first-principles calculations.

Contributions

1. Drove target normal sheath acceleration with a 131 J, 759 fs laser focused to a 20 μm spot on a 15 μm CH-coated copper foil, trimmed the spectrum with a 0.28 T dipole between 500 μm entrance and exit slits, and heated a porous tricellulose-acetate foam ($C_9H_{16}O_8$, 2 mg/cm³, 1 mm) through a gold hohlraum irradiated by a 147 J, 2 ns beam, so that on the Fig. 2 time-of-flight axis C^{6+} reaches the foam before the ns-laser trigger, C^{5+} inside the 10 ns window before significant expansion with an interaction time of about 0.26 ns, and the 4.49 MeV C^{4+} and 2.47 MeV C^{3+} pulses after expansion.
2. Converted the magnetic (X) and electric (Y) Thomson-parabola deflections recorded on CR39 into spectra fitted by Gaussians, counting only the unsaturated high- and low-energy tails of C^{5+} and C^{4+} , whose central tracks saturate the detector, and found that C^{6+} leaves the cold foam as C^{5+} and C^{6+} centred at 8.45 MeV while C^{5+} leaves the plasma spread over C^{3+} – C^{6+} centred at 5.08 MeV, with an energy-loss error bar of about ± 0.4 MeV in Fig. 4(a) from statistics, fitting, and spectrometer resolution.
3. Solved charge-state rate equations with Coulomb ionization by bound and free target electrons, capture of bound and free electrons, three-body recombination, and excited states through target-density-effect modifications, coupled to each free-electron stopping model with bound-electron stopping from the Bethe–Bloch terms, over 20 loops across the target; charge equilibrium is reached within about 0.01 mm in every step, so $Z_{eff}(x) = Z_{eq}(v(x))$, C^{1+} – C^{3+} fractions stay about seven orders of magnitude lower, and models with stronger stopping give lower mean charges because slower ions favour recombination.
4. For a representative 6 MeV C^{5+} ion, the mean of incident and exit energies, evaluated the Brown–Preston–Singleton projectile–plasma coupling $g^2 = \sum_b \beta_b^2 (Z_{eff} Z_b / 4\pi)^2 \kappa_b^2$ of Eq. (1), with Debye wavenumbers $\kappa_b = \sqrt{\beta_b e_b^2 n_b}$ and $\beta_b = 1/T_b$, as $g \sim 2.1$ and Zwicky’s $\langle \eta \rangle = Z_{eff} e^2 / (4\pi\epsilon_0 \hbar \langle v_r \rangle)$ of Eq. (2) as about 1.1, violating $g \ll 1$ and $\langle \eta \rangle \ll 1$, although the plasma itself is ideal with electron coupling $\Gamma_{ee} = e^2 / (a_e k_B T) \sim 0.09$ and degeneracy $\Theta = k_B T / E_F \sim 100$.
5. Attributed the remaining excess of Zwicky’s combined model, which adds strong-coupling corrections to linear response, to its linear treatment of static and dynamic collective screening, and built the agreeing prediction from the classical molecular-dynamics stopping of Grabowski et al. averaged over the differing quantum corrections of Brown et al. and of Issanova et al., whose deviation sets its uncertainty, motivated by a closest-approach distance comparable to the electron de Broglie wavelength; it gives a mean exit charge of about 4.31 against the measured 4.34 and 4.13–4.21 for the other models in Fig. 4(b).

Comments

- Fig. 2 and its discussion: the central energies of the incident C^{3+} , C^{4+} , C^{5+} , and C^{6+} ions are given as “2.47 MeV, 4.49 MeV, 7.00 MeV, and 10.01 MeV”, but the cold-foam loss is obtained from a downshift “from 9.96

MeV to 8.45 MeV”, and in Fig. 2(a) the Gaussian fit to the incident C^{6+} spectrum peaks, and the dashed C^{6+} reference line of Figs. 2(a)–(b) lies, at about 9.96 MeV, whereas the other three fits peak at 2.47, 4.49, and 7.00 MeV; the stated 1.51 MeV cold-foam loss rests on 9.96 MeV.

- Figs. 3(b) and 4(a): the Fig. 3(b) remaining energies at the 1 mm exit, about 3.93 MeV (T-Matrix), 3.97 MeV (Bethe–Bloch), 4.35 MeV (Vlasov), 4.39–4.46 MeV (BPS, LP, Combined), and 4.76 MeV (MD+Quantum), imply losses from 7.00 MeV of about 3.07, 3.03, 2.65, 2.54–2.61, and 2.24 MeV, but Fig. 4(a) plots about 3.26, 3.21, 2.75, 2.62–2.68, and 2.29 MeV, although the exit average charge states of Fig. 3(b) match those of Fig. 4(b) to within about 0.005; unresolved.

2026LRGW

ChatGPT (July 2026)

Summary

One-dimensional fully kinetic SMILEI simulations show that linear or parabolic density gradients can replace laser-frequency chirping in autoresonant plasma beat-wave excitation. Spatial variation of the plasma frequency compensates the nonlinear wavelength shift, preserving phase locking and driving waves beyond the Rosenbluth–Liu limit. For normalized laser amplitudes $a_{1,2} = 0.1$ and 0.2 , saturation reaches about $0.60E_0$ and $0.95E_0$, respectively; the classical limit is exceeded for $a > 0.06$. Saturated amplitude is nearly independent of gradient length, while dephasing length scales linearly with gradient length and drive amplitude. In a steep parabolic profile, two counter-propagating laser pairs create a two-phase quasiperiodic density lattice that persists after the lasers depart; an 800 nm, $7 \times 10^{17} \text{ cm}^{-3}$ example lasts about 2.1 ps. The collisionless, cold-electron, one-dimensional model omits multidimensional bowing and filamentation that may degrade the structure.

Contributions

1. Demonstrated chirp-free autoresonance using tailored density rather than time-varying laser frequency.
2. Derived and validated density compensation for the nonlinear phase shift.
3. Found nearly gradient-independent saturation and linear dephasing-length scaling.
4. Drove fields beyond the Rosenbluth–Liu limit and near wave breaking at moderate intensity.
5. Produced a long-lived two-phase quasiperiodic plasma lattice with four unchirped lasers.

2026LSGH

Claude (September 2026)

Summary

The authors report the first seeded free electron laser (FEL) driven by a laser-plasma accelerator (LPA) in the high-gain regime, where radiation grows exponentially along the undulator. As background, LPA-driven FELs had lased in self-amplified spontaneous emission, which amplifies the beam’s own undulator radiation, and, seeded by an external coherent pulse, only in the low-gain regime. At HZDR, a 188.8 MeV, 400 pC beam with 7% RMS energy spread and peak spectral charge density above 10 pC/MeV is energy-chirped by a decompressing chicane and enters a 1.98 m undulator with a 268.2 nm third-harmonic seed, whose wavelength chirp, set by fused-silica rods, gives durations of 295, 473, and 1186 fs FWHM at constant energy. The FEL, red-shifted near 273 nm, interferes with the seed. GENESIS simulations matching the measured spatio-spectral images show exponential growth over the last 0.6 m with gain lengths of 0.55–0.68 m, deemed independent of seed duration, and about threefold amplification; shortening the seed fourfold raises exit peak power and measured FEL spectral intensity about fourfold. The authors report pulse-energy fluctuations up to 100% and measurement uncertainties above 30%, and note that saturation, predicted at 360 MW after 6 m, lies beyond the single undulator module.

Contributions

1. Drove the LPA with the 150-TW arm of DRACO in a tailored self-truncated ionization-induced injection scheme, whose energy, divergence, and charge varied shot to shot by up to 35% while their means stayed within 2% without drift over four eight-hour days, and transported the beam through a quadrupole triplet against chromatic emittance growth, a four-dipole chicane with $R_{56} = 2.0$ mm lowering the slice energy spread at the cost of peak current, and four quadrupoles for chromatic sliding focusing into a 99-period undulator with $\lambda_u = 20$ mm and $K_u = 2.31$, with the 2.55 μm RMS bunch length and (2.0, 1.5) mm mrad emittances obtained by matching simulated to measured spatio-spectral distributions.
2. Modeled the seed as $\lambda_{\text{seed}}(t) = \lambda_{\text{seed}-0} + (t - \tau)/D_\lambda$ from a Fourier-limited 29.3 fs pulse dispersed by at least 9.9 mm of transmissive optics plus fused-silica rods of 10 or 50 mm with group-velocity dispersion -4945.5 ps/(nm km) at 270 nm, and, because the resulting $D_\lambda = -0.049, -0.098$, and -0.296 ps/nm poorly reproduced the FEL wavelength versus delay, added a residual -0.025 ps/nm attributed to the infrared driver laser, giving $-0.074, -0.123$, and -0.321 ps/nm and the three seed durations.
3. Simulated the FEL by transporting a 6D Gaussian distribution of 10^6 particles with BETA through a chromatic matching lattice for 188.8 MeV into GENESIS with a seed described in amplitude and phase, Fourier transforming the exit field at $x = 0$ into a (y, λ) distribution back-propagated over 10 cm, and adjusting the seed-beam delay to the measured spectral centroid, which reproduced the interference-fringe spacing and curvature; fits of $\hat{P}_{\text{FEL}} \propto e^{z/L_g}$ between $z = 1.4$ and 2.0 m gave 0.68, 0.61, and 0.55 m for the 1186, 473, and 295 fs seeds.
4. Calibrated absolute pulse energies with the LPA off by splitting the transported seed 60%/40% between a power meter and the spectrometer over two series of ten shots, giving 9×10^{-5} pJ per count, opened the entrance slit to 2 mm, integrated the seed in an ellipse of 100 by 50 pixel radii and the FEL above a manually set wavelength within ± 15 pixels of its vertical position, rescaled each shot to the mean seed energy of 33 nJ assuming linear scaling, and found a double-Gaussian fit within 30% overall and 6% for the shortest seed, attributing the larger model deviation for the longest seed to mistuned transport on that day.
5. Scanned the seed-beam delay by $\pm 800, \pm 350$, and ± 230 fs for the 1186, 473, and 295 fs seeds, taking the FEL central wavelength and intensity from Gaussian fits to Fourier-filtered spectral line profiles beyond a threshold wavelength; the wavelength rose linearly with delay at a rate set by the seed chirp, as predicted from the single resonance point $\lambda_R(t_0) = \lambda_{\text{seed}}(t_0)$ plus the undulator red shift with $L_{\text{eff}} = 1.0$ m, and the intensity peaked at an optimal delay within a window that narrowed for shorter seeds, which also widened the seed-FEL interference fringes.

Comments

- Fig. 1: the caption assigns “(green) $D_{\lambda 1}$, (blue) $D_{\lambda 2}$ and (red) $D_{\lambda 3}$ ”, with $D_{\lambda 1} = -0.321$, $D_{\lambda 2} = -0.123$, and $D_{\lambda 3} = -0.074$ ps/nm, but the lines in panel (b) are labeled $\lambda_{\text{seed}}(t, D_{\lambda 0})$ (green), $\lambda_{\text{seed}}(t, D_{\lambda 1})$ (blue), and $\lambda_{\text{seed}}(t, D_{\lambda 2})$ (red); their slopes, shallowest for green and steepest for red, follow the caption’s values, so the panel labels are evidently shifted down by one index.
- Abstract, main text, and Fig. 3(b): the normalized energies E_{FEL}^* (dots) of Fig. 3(b) are said to exhibit “a maximum of 45 nJ using the shortest seed duration”, and the abstract states that 45 nJ pulse energies were achieved “in good agreement with quantitative numerical simulations”, but in the logarithmic Fig. 3(b) the 295 fs measurement lies near 13 nJ, the top of its RMS error bar near 24 nJ, and the simulated star near 12 nJ, so neither the plotted measurements nor the simulations reach 45 nJ; whether 45 nJ is an unlogged single-shot value is unresolved.
- Table I and Fig. 3: Table I lists the seed energy $E_{\text{seed}} = 250$ nJ among the FEL experiment parameters, but the numerical modeling of Fig. 3 uses “an input seed energy of 33 nJ”, and the measured seed energies E_{seed} extracted from the spectrometer images have mean $\langle E_{\text{seed}} \rangle = 33$ nJ, while the captions of Figs. 2 and 3 refer to Table I for the other parameters; where in the beamline 250 nJ applies is not stated; unresolved.
- Appendix A: “ $D_\lambda = \text{GVD}^* \times d \times 10^6$ ”, with D_λ in ps/nm, d in mm, and $\text{GVD}^* = -4945.5$ ps/(nm.km), gives about -4.9×10^{10} ps/nm for $d = 9.9$ mm, whereas Table II lists -0.049 ps/nm $= -4945.5 \times 9.9 \times 10^{-6}$, and likewise -0.098 and -0.296 ps/nm for 19.9 and 59.9 mm; the evident intended factor is 10^{-6} , the mm-to-km conversion.
- Appendix A, Eq. (A1): after dispersion “the seed pulse duration is increased up to” $\Delta T_{\text{seed}} = \Delta T_{\text{seed}-i} + d \times \Delta \lambda \times \text{GVD}^*$, but the stated $\text{GVD}^* = -4945.5$ ps/(nm.km) makes the increment negative, giving about -147 fs from the Fourier-limited 29.3 fs for $d = 9.9$ mm and the 3.6 nm seed bandwidth of Table I, whereas the Table II durations 205.56, 383.60, and 1095.75 fs equal $29.3 \text{ fs} + d \Delta \lambda |\text{GVD}^*|$ for $d = 9.9, 19.9$, and 59.9 mm; the

evident intended increment uses $|GVD^*|$.

- Table II caption: the seed properties are a “Calculation using Eq. (B1)”, but Appendix B contains no equation, and Appendix A states that “According to those analytical calculations”, which culminate in Eq. (A1), “the seed should exhibit the parameters summarized in Table II”; the evident intended reference is Eq. (A1).

2026LSS

ChatGPT (July 2026)

Summary

This paper develops contour-integral algorithms for computing selected interior singular values of a matrix and generalized singular values of a matrix pair. Although these quantities can be represented as eigenvalues of a Hermitian Jordan–Wielandt matrix or pencil, directly applying FEAST ignores its paired spectral structure and can be sensitive to the initial subspace. The authors analyze four structured spectral-projector choices and construct FEAST-SVD and FEAST-GSVD schemes. Their robust strategy augments the first trial subspace using projectors associated with positive and negative contours, retaining information that a single positive-contour projection may cancel or discard; later iterations revert to the cheaper single-contour update. A Monte Carlo trace estimator approximates the number of values in the requested interval, while Rayleigh–Ritz extraction, residual tests, and soft locking complete the solver. Experiments on twelve SuiteSparse matrices show that the methods typically recover all requested singular or generalized singular values in three or four iterations with high accuracy. The approach also refines low-precision initial solutions and may support spectral slicing or mixed-precision algorithms.

Contributions

1. Recast partial SVD and GSVD as structured contour-integral eigenproblems on Jordan–Wielandt matrices and pencils.
2. Analyzed four projection strategies and their sensitivity to cancellation and the initial subspace.
3. Introduced a first-iteration two-contour augmentation that improves robustness and accelerates later iterations.
4. Developed FEAST-SVD and FEAST-GSVD algorithms with trace estimation, Rayleigh–Ritz extraction, residual stopping, and soft locking.
5. Demonstrated rapid recovery on twelve sparse test matrices and successful refinement of low-precision starting solutions.

2026LTGM

ChatGPT (July 2026)

Summary

This study uses collisional particle-in-cell simulations with laser ray tracing to model laser ablation, plasma expansion, and self-magnetization self-consistently. Planar geometry suppresses Biermann-battery generation and isolates anisotropy-driven fields. Above roughly $4 \times 10^{13} \text{ W/cm}^2$, expansion preferentially cools electrons along the target normal, producing the positive temperature anisotropy needed for an expansion-driven electron Weibel instability. One- and two-dimensional simulations generate transverse magnetic filaments of order 50 T, magnetic energy near one percent of thermal energy, plasma $\beta \sim 100$, and Hall parameter $\omega_{ce}\tau_e > 1$. A self-similar expansion model explains how flow-driven anisotropy persists against collisional isotropization. The authors also derive a dimensionless threshold Γ from laser intensity, wavelength, and target properties; it separates magnetized from unmagnetized cases in their limited scan. Comparisons with artificially suppressed magnetic forces show about 20% temperature changes and reduced axial heat transport, demonstrating that the generated fields feed back on the expansion.

Contributions

1. Simulated laser ablation, expansion, collisions, and magnetogenesis in one self-consistent kinetic framework.
2. Identified an intensity threshold for expansion-driven Weibel self-magnetization.

3. Distinguished expansion-driven from temperature-gradient-driven anisotropy through field polarization and anisotropy sign.
4. Derived analytical anisotropy evolution and a simple laser–target threshold parameter Γ .
5. Demonstrated that self-generated fields magnetize electrons and modify plasma heat transport.

2026LTTS

ChatGPT (July 2026)

Summary

This study reviews and benchmarks open-source quadrature software for finite elements cut by implicitly or explicitly represented boundaries. A common MATLAB interface exercises Algoim, BoSSS, FCMLab, Gridap, mlhp, ngxfem, Nutils, QuESo, and related tools on two- and three-dimensional domain integrals, using matched geometries and polynomial test functions to compare accuracy, efficiency, robustness, and sensitivity to requested quadrature order. All tested codes are usable and robust for the selected cases, but their error–cost behavior differs substantially. Tessellation-based approaches introduce geometry approximation error that commonly limits convergence to second order; methods preserving curved boundaries or higher-order approximations can attain higher-order convergence. Increasing the nominal quadrature order does not necessarily reduce error when geometric approximation dominates, so total quadrature-point count is a more informative cost measure than an input order alone. The accompanying test environment and reusable quadrature-point outputs support reproducible comparisons. The review also identifies a missing open-source capability: volume quadrature for parametric boundary representations needed in CAD-based workflows.

Contributions

1. Cataloged open-source cut-element quadrature implementations and classified their boundary-representation strategies.
2. Built reproducible two- and three-dimensional benchmarks with both implicit and explicit geometry descriptions.
3. Compared accuracy against total quadrature-point cost rather than relying only on nominal integration order.
4. Demonstrated how tessellated geometry errors cap convergence while higher-order boundary representations permit higher accuracy.
5. Released a common MATLAB test interface and identified parametric-boundary volume quadrature as an unresolved software gap.

2026LTWZ

ChatGPT (July 2026)

Summary

The paper introduces refined implicit-explicit Runge-Kutta (R-IERK) schemes for the Cahn-Hilliard equation whose differentiation matrices and average dissipation rates do not depend on the spatial mesh or time-step size. Using discrete orthogonal convolution kernels and an updated time-space error splitting, the authors prove uniform boundedness of internal stage solutions without assuming global Lipschitz continuity of the quartic bulk nonlinearity. This permits proofs of the original, rather than modified, energy dissipation law at every stage and unconditional L^2 convergence. Parameterized second- and third-order schemes are constructed by optimizing average dissipation, including methods whose implicit part has an explicit first stage and need not be algebraically stable. Fourier pseudospectral experiments confirm the designed accuracy and show stable energy curves under much larger adaptive steps than comparison IERK methods, substantially reducing time levels and CPU cost.

Contributions

1. Defined R-IERK methods with mesh- and step-independent differentiation matrices and average dissipation rates.
2. Proved uniform boundedness of all internal stages without global Lipschitz continuity of the nonlinear bulk term.
3. Established the original discrete energy dissipation law at the internal stages.

4. Proved unconditional L^2 convergence with full $O(\tau^p + h^m)$ accuracy under the stated positivity conditions.
5. Constructed adaptive second- and third-order schemes that use far fewer time levels than fixed-step comparisons.

2026LV

ChatGPT (July 2026)

Summary

Parabolic optimal-control discretizations produce coupled forward–backward systems that cannot be advanced by ordinary sequential time stepping and become expensive at scale. This paper studies a time-parallel Schwarz domain-decomposition method and asks whether convergence remains effective as the number of fixed-size time subintervals grows. Two complementary analyses characterize the iteration matrix. A problem-tailored matrix norm gives a spectral-radius bound strictly below one and independent of the subinterval count. Block Toeplitz theory then locates every eigenvalue nonasymptotically within the same bound and characterizes the limiting spectrum as the count tends to infinity. Together these results establish weak scalability under the paper’s assumptions. Scalar-mode, heat-control, and periodic advection–diffusion experiments show that the bounds are sharp and iteration counts remain essentially independent of the interval count, including a discretization with more than eight million state-and-adjoint unknowns. The analysis provides a basis for future multilevel methods and high-performance implementations, which are not developed here.

Contributions

1. Formulated a weak-scalability analysis for time-parallel Schwarz iteration on parabolic optimal-control systems.
2. Constructed a tailored matrix norm yielding an interval-count-independent spectral-radius bound below one.
3. Used block Toeplitz theory to locate all eigenvalues nonasymptotically in the complex plane.
4. Characterized the asymptotic convergence spectrum as the number of time subintervals grows.
5. Confirmed numerically that the estimates sharply predict the observed convergence behavior.

2026LVCN

ChatGPT (September 2026)

Summary

The paper develops a one-dimensional cold, collisionless theory for nonlinear plasma waves driven by a rigid charged-particle beam. In a coordinate moving with the beam, the relativistic fluid and Poisson equations reduce to an energy integral that applies to either beam-charge sign and arbitrary driver velocity up to wave breaking. The resulting analytic expressions give electric-field amplitude, wavelength, peak electron density, and Lorentz factor for both the wake behind the beam and a distinct wave inside it. These interior waves exist for normalized beam charge density $\rho_b > -1/2$; unlike resonantly driven wakes, their properties do not depend on driver length. For dense positive drivers their wavelength contracts as $\lambda_{\text{int}}/\lambda_{pe} \sim (2/\rho_b)^{1/2}/\pi$, permitting localized, short-wavelength structures. Choosing an integer number of interior wavelengths can eliminate the trailing wake. Numerical solutions closely match the analytic results, including the mildly nonlinear regime where earlier asymptotic formulas fail. The model assumes a wide, non-evolving beam, immobile ions, and one-dimensional electrostatic dynamics.

Contributions

1. Derived nonlinear wake and interior-wave solutions valid for either beam-charge sign and arbitrary driver velocity up to wave breaking.
2. Identified the interior-wave existence condition $\rho_b > -1/2$ and calculated its amplitude, wavelength, and peak density.
3. Showed that interior waves are insensitive to driver length and can be isolated by suppressing the trailing wake.
4. Obtained the positive-driver short-wavelength scaling $\lambda_{\text{int}}/\lambda_{pe} \sim (2/\rho_b)^{1/2}/\pi$.
5. Verified the analytic predictions numerically and derived wave-breaking bounds for wakefield acceleration.

2026LVDN

ChatGPT (July 2026)

Summary

This paper reconstructs an unknown initial electric field in the time-domain Maxwell system from noisy lateral Cauchy data. The method expands time dependence in a Legendre polynomial–exponential basis, converting the original three-space-plus-time inverse problem into a coupled sequence of three-dimensional spatial equations. This time splitting avoids requiring the unknown initial time derivative of the electric field. Because the available data leave the reconstruction underdetermined, the authors select a minimum-norm solution in combined space–time Sobolev norms and prove convergence under noisy measurements. Three-dimensional tests recover multiple field geometries with 10% data noise, producing relative errors of about 8%–16%. Perfectly matched layers permit longer observation intervals without boundary-reflection artifacts. A comparison with Tikhonov optimization gives similar reconstructed shapes, but the proposed approach solves one reduced linear system rather than repeatedly advancing the full four-dimensional Maxwell problem inside an outer optimization loop.

Contributions

1. Reduced a time-dependent Maxwell initial-data inverse problem to coupled three-dimensional spatial equations.
2. Removed the need to prescribe the generally unavailable initial electric-field time derivative.
3. Defined a minimum-norm solution for the underdetermined reconstruction problem.
4. Proved convergence and stability with respect to noise-contaminated boundary measurements.
5. Demonstrated robust noisy reconstructions and lower computational structure than Tikhonov optimization.

2026LWY

ChatGPT (September 2026)

Summary

High-order ideal-magnetohydrodynamics schemes must keep density and pressure positive while maintaining $\nabla \cdot \mathbf{B} = 0$. Standard positivity limiters can disrupt continuity of the normal magnetic field. The authors instead construct a globally divergence-free discontinuous Galerkin method that leaves the magnetic field untouched and corrects total energy when needed for admissibility. They add an entropy-Hessian-based convex oscillation-suppression procedure, derive a positivity condition using an optimal convex decomposition, and offer two energy corrections: constrained optimization and closed-form scaling. Conservation is conditional because rare cells where the constraint is infeasible require a fallback energy adjustment. The paper proves positivity under a specified CFL restriction and tests high-order accuracy and robustness on strong shocks, low- β plasmas, and other demanding two-dimensional benchmarks. Reported global energy errors are near 10^{-12} , although strict local energy conservation is lost in fallback cells.

Contributions

1. Reconciled positivity limiting with globally divergence-free magnetic-field reconstruction.
2. Introduced entropy-Hessian convex oscillation suppression near discontinuities.
3. Developed optimization-based and closed-form energy corrections that are conservative when feasible.
4. Proved positivity under a mild CFL condition from an optimal convex decomposition.
5. Quantified rare fallback events and global energy errors in extreme MHD benchmarks.

2026LXWZ

ChatGPT (July 2026)

Summary

The paper resolves a discrepancy between theoretical and observed polynomial-degree convergence of a one-dimensional hp local discontinuous Galerkin method for convection–diffusion equations with singular solutions. Its central device is a fractional-regularity framework in which endpoint singularities are characterized through Caputo derivatives, while exact Legendre-coefficient formulas yield sharp Gauss–Radau projection estimates. These estimates produce explicit h - and p -version a priori bounds for pure convection and convection–diffusion problems. The analysis distinguishes endpoint singularities, interior singularities aligned with a mesh node, and interior singularities lying inside an element; the unfitted case has substantially lower p -convergence. For a singularity behaving as $|x - \theta|^\alpha$, attainable rates depend on α , an additional fractional regularity index, diffusion, and mesh alignment. Numerical experiments reproduce all predicted rates and correct the earlier interpretation of the diffusion example. The analysis is restricted to the one-dimensional Castillo–Cockburn–Schötzau–Schwab LDG formulation, although the framework suggests how analogous p -suboptimality may be addressed elsewhere.

Contributions

1. Introduced Caputo-derivative spaces that characterize algebraic singularities for Gauss–Radau approximation.
2. Derived sharp Legendre-coefficient projection bounds and p -optimal LDG estimates for endpoint singularities.
3. Extended the theory to mesh-fitted interior singularities and quantified their convective and diffusive rates.
4. Established lower rates for unfitted interior singularities, showing the benefit of mesh alignment.
5. Validated the endpoint, fitted, and unfitted estimates and corrected the motivating diffusion convergence rate.

2026LYVZ

ChatGPT (July 2026)

Summary

The paper enlarges the stellarator design space by combining poloidal omnigenity (PO), which suppresses orbit-averaged radial drift, with piecewise omnigenity (pwO), which relaxes the requirement that all magnetic-strength contours close poloidally. Within the OOPS framework, a restricted Landreman-Catto coordinate mapping “squeezes” the high-field region into locally closed, approximately parallelogram-shaped contours while preserving ordinary omnigenity in the low-field region. Boundary optimization produces nine PO-pwO configurations spanning two to five field periods and several aspect ratios. Their departures from exact omnigenity and effective ripple are below those of the W7-X high-mirror reference and close to a pure PO case. Reactor-scaled collisionless calculations lose less than 4% of 3.5 MeV alpha particles over 0.2 s in all nine configurations. A further design develops a vacuum magnetic well of depth about 0.023 while keeping alpha losses below 2%. MONKES calculations show low radial neoclassical transport; bootstrap-current coefficients vary because the zero-current pwO condition was not explicitly imposed, although some cases are comparable to W7-X. The method therefore demonstrates a controlled transport-versus-shaping trade-off, while leaving coil complexity and integrated reactor optimization for future work.

Contributions

1. Formulated a restricted-coordinate “squeeze” mapping that combines PO and pwO within the OOPS optimization framework.
2. Generated nine optimized PO-pwO equilibria across multiple field periods and aspect ratios rather than tuning a single configuration.
3. Demonstrated low effective ripple and reactor-scaled alpha-particle losses below 4% across the configuration family.
4. Produced a PO-pwO equilibrium with a vacuum magnetic well while retaining low neoclassical transport and below-2% alpha losses.

5. Quantified radial and bootstrap-current transport coefficients and clarified which configurations approach W7-X-level bootstrap behavior.

2026LYXB

ChatGPT (September 2026)

Summary

This paper accelerates collisionless local linear gyrokinetic eigenanalysis by using particle energy and magnetic moment as velocity-space coordinates. Passing and trapped trajectories then form independent one-dimensional orbit blocks coupled only through the field equations. The MGK solver discretizes passing orbits with Legendre–Gauss–Lobatto differentiation, trapped orbits in a periodic bounce-angle representation, and separates quadrature at the trapped–passing boundary. A Schur-complement shift-invert Arnoldi method avoids factoring the full generalized eigenproblem and reduces per-shift scaling from cubic to linear in the number of orbit blocks. Batched GPU operations extend the method to kinetic electrons, Miller geometry, and electrostatic or electromagnetic two- and three-field models. ITG, TEM, and KBM frequencies and eigenfunctions generally agree with CGYRO and other reference solvers within a few percent. Reported solves take roughly 0.01–0.1 seconds on a GPU and can be over three orders of magnitude faster than CGYRO. The demonstrated scope is local, linear, and collisionless; nonlinear and transport extensions remain future work.

Contributions

1. Decomposed collisionless gyrokinetic dynamics into orbit-invariant passing and trapped blocks.
2. Combined orbit-specific spectral discretizations with separatrix-aware velocity-space quadrature.
3. Derived a Schur-complement shift-invert eigensolver whose cost is linear in the number of orbit blocks.
4. Extended MGK to kinetic electrons, Miller geometry, and two- and three-field electromagnetic models.
5. Validated mode spectra and eigenfunctions while demonstrating sub-second GPU solutions.

2026LYZC

ChatGPT (July 2026)

Summary

This article models imperfect sliding electrical contact in railgun electromagnetic simulations without meshing the thin contact layer directly. The contact zone between armature and rail is replaced by a zero-thickness interface whose strongly discontinuous boundary conditions encode contact resistance, preload pressure, and liquid aluminum effects. An interior-penalty discontinuous Galerkin framework then modifies numerical fluxes so the discontinuous interface conditions are enforced while preserving favorable conditioning even for extreme contact conductivities. Comparisons with a classical contact-layer model show consistent current-density results, while the boundary-condition model uses far fewer elements and less computation; the reported element count is one-seventeenth of the classical model in a representative comparison. The method is then used to study velocities of 0, 100, 500, and 1000 m/s. Increasing speed shifts current concentration differently on rail and armature sides, highlighting asymmetric electromagnetic effects of preload pressure and imperfect contact.

Contributions

1. Replaced the thin armature-rail contact layer with a zero-thickness strongly discontinuous interface model.
2. Derived interface conditions that encode contact resistance, pressure, and liquid aluminum material effects.
3. Built an IPDG discretization whose numerical fluxes enforce the discontinuous interface conditions.
4. Verified agreement with the classical contact-layer model at much lower mesh and runtime cost.
5. Analyzed velocity-dependent current-density concentration on rail and armature contact surfaces.

2026LYZR

ChatGPT (July 2026)

Summary

This work develops a second-order semi-implicit integrator for the multispecies linearized Sugama collision operator used in gyrokinetic tokamak simulations. The scheme treats the collision operator as a unified update rather than splitting its test- and field-particle components. It advances stiff isothermal pitch-angle scattering and energy diffusion implicitly with a trapezoidal rule, while treating the non-isothermal model and field-particle terms explicitly. Combined with a finite-volume discretization in $(v_{\parallel}, \mu)$, this produces independent sparse or banded solves for each species instead of a fully implicit cross-species system. Implemented in the NLT code, the method permits time steps tens to thousands of times larger than explicit integration in electron and multispecies relaxation tests. It preserves particle number to machine precision and, with the complete correction terms, also preserves total momentum and energy, maintains adjointness, and satisfies the H -theorem. Neoclassical ion-energy flux and bootstrap-current profiles agree with GENE and ORB5. Positivity is not guaranteed intrinsically, and the tested global case omits kinetic electrons, equilibrium electric field, and sources.

Contributions

1. Constructed a second-order semi-implicit update without splitting the physical collision operator.
2. Isolated the stiff test-particle terms while avoiding fully implicit cross-species coupling.
3. Coupled the integrator to a conservative finite-volume velocity-space discretization.
4. Demonstrated large stable-time-step gains while retaining conservation and entropy production.
5. Validated the NLT implementation against independent neoclassical transport codes.

2026LZG

ChatGPT (July 2026)

Summary

This paper develops a quadrature method for uniformly sampled, nonperiodic integrands using local Fourier extensions. Each overlapping window is continued onto a larger periodic interval through a truncated-SVD-stabilized Fourier fit, and its integral is evaluated analytically from the Fourier coefficients. Uniform windows share the same discrete fitting matrix, so its factorization is precomputed once and reused in an offline/online implementation. The authors bound the quadrature error by the local approximation error and derive algebraic convergence for Sobolev-regular functions. Tests on smooth, oscillatory, and spatially varying-frequency functions reach near machine precision with fewer samples than composite Simpson quadrature. For continuous piecewise-smooth functions, coefficient-energy outliers identify suspect windows; a local procedure brackets the singular point, predicts one-sided values, and replaces only the affected contribution, restoring near-spectral accuracy in the reported cases. Jump discontinuities, multidimensional extensions, and a rigorous adaptive-refinement strategy remain open.

Contributions

1. Converted stabilized local Fourier extensions into an analytic quadrature rule on uniform samples.
2. Reused a precomputed truncated-SVD factorization across identical local windows.
3. Bounded integration error through local approximation error and established Sobolev convergence rates.
4. Developed coefficient-energy detection and localized correction for derivative singularities.
5. Demonstrated near-machine or near-spectral accuracy on smooth, oscillatory, and piecewise-smooth tests.

2026LZLC

ChatGPT (September 2026)

Summary

The authors combine meshfree generalized finite differences with Lagrangian particle motion and implicit time integration for weakly compressible viscous flows. They construct semi-implicit and fully implicit solvers using second-order Newmark, Crank–Nicolson, and Bathe schemes. In the semi-implicit form, particle displacement is updated separately from the implicit velocity solve; the fully implicit form couples them. Couette and Poiseuille flow comparisons with analytical solutions show second-order accuracy and global mean relative errors below 10^{-3} in the reported tests. Fully implicit coupling improves transient accuracy but costs roughly 30% more than the semi-implicit solver. Newmark with semi-implicit coupling offers the best tested balance for steady and viscous cases. Implicit stepping relaxes the explicit stability restriction, although accuracy and iteration count still limit useful time-step size. Particle shifting is needed to prevent uneven particle distributions.

Contributions

1. Formulated implicit Lagrangian generalized finite difference discretizations for weakly compressible viscous flow.
2. Compared semi-implicit displacement updates with fully coupled velocity–displacement iteration.
3. Evaluated Newmark, Crank–Nicolson, and Bathe time integrators within the particle method.
4. Verified accuracy against transient Couette and Poiseuille analytical solutions and examined spatial refinement.
5. Quantified the accuracy–cost trade-off and the role of particle shifting in stable simulations.

2026LZS

ChatGPT (September 2026)

Summary

A recursive Jacobian-free Cauchy–Kovalevskaya procedure simplifies high-order compact central schemes for hyperbolic conservation laws. The Cauchy–Kovalevskaya procedure obtains time derivatives from the governing equations; here, local Taylor polynomials supply virtual-point flux values, and repeated central differences replace analytical flux-derivative tensors. A uniform recursion accommodates arbitrary derivative orders without enlarging the underlying computational-cell stencil. The analysis establishes second-order virtual-step errors and exact equivalence to the analytical procedure for constant-coefficient linear systems, preserving their linear stability restrictions. Virtual steps balance differencing errors against roundoff. A total-variation-bounded troubled-cell detector replaces the earlier nonlinear limiter; flagged cells retain their averages but receive limited slopes and lose higher derivatives. Burgers and one- and two-dimensional Euler tests demonstrate second- through seventh-order convergence before numerical-error saturation, together with shock and small-vortex resolution. Higher order improves accuracy per computational cost in the reported smooth test, although the fourth-order recursive method is slower than its nonrecursive Jacobian-free predecessor. Demonstrated accuracy is limited to seventh order; still higher orders face increasing virtual-difference errors, and the Euler limiter parameter is chosen empirically.

Contributions

1. Constructed a single recursive formula for mixed space-time flux derivatives using locally reconstructed virtual points, eliminating order-specific symbolic derivative tensors and differencing formulas.
2. Proved the virtual-step error estimate, including propagated Taylor-reconstruction errors, and equivalence to exact Cauchy–Kovalevskaya differentiation for constant-coefficient linear fluxes.
3. Derived derivative-order-dependent virtual-step choices that balance roundoff amplification and differencing error, with an additional cap from physical mesh and time steps.
4. Adapted troubled-cell detection and characteristic slope limiting to compact central schemes, retaining conservative cell averages and enabling shock calculations through seventh order.

5. Verified convergence with smooth Burgers and Euler solutions and assessed shock capturing with shock tubes, blast waves, shock–entropy interaction, double Mach reflection, and multidimensional Riemann problems.

2026LZZ

ChatGPT (July 2026)

Summary

The authors use long-duration, instability-resolving three-dimensional particle-in-cell simulations to map anomalous cross-field electron transport in a Hall thruster. The model combines a realistic magnetic field, xenon ionization, continuum neutral evolution, dielectric charging, secondary-electron emission, and open plume outflow. Time and azimuthal averaging of $\langle n_e E_y \rangle$ over 200 samples reveals two persistent transport bands adjacent to the inner and outer walls near the downstream channel and exit, with the inner-wall pathway generally stronger. Conducting, ceramic-with-emission, and open-outflow boundaries alter pathway strength and downstream extension but not its topology. Dominant electron-drift-instability wavelengths are about 1 mm; fitted in-channel and near-exit phase speeds of 5.20 and 6.02 mm/ μ s closely match local ion-acoustic speeds. Doubling the 5 ps timestep changes the spectrum but preserves the averaged pathway. Fine-scale amplitudes and spectra are less secure than the large-scale topology because the baseline grid does not resolve the Debye length everywhere.

Contributions

1. Resolved net instability-driven transport as two wall-adjacent pathways rather than a uniform flux.
2. Combined realistic magnetics, ionization, neutral evolution, wall charging, emission, and open outflow in one 3D model.
3. Demonstrated that the near-wall topology persists across different wall and plume closures.
4. Linked the averaged pathways to millimeter-scale modes with ion-acoustic-scale phase speeds.
5. Distinguished robust transport topology from timestep- and grid-sensitive spectral details.

2026M

ChatGPT (September 2026)

Summary

This paper accelerates Steinhaus–Johnson–Trotter permutation generation through combinatorial partitioning and AVX2 byte shuffles. Two independent permutation streams occupy the 128-bit halves of a 256-bit register, with each lane generating one quarter of the factorial search space. A reflection property supplies the remaining half: permutations in the first half have element 0 before element 1, and reversing them covers the complementary half. Vectorized staircase insertion uses precomputed shuffle masks, while the lane width limits permutations to $n \leq 16$ and parallel scheduling changes their output order. The paper also describes an optimized scalar version of Knuth’s Algorithm P and separate vector kernels for benchmarking and callback-based use. On an Intel Core i7-8850H with GCC 15.2.0, the author reports approximately threefold normalized generation throughput over the optimized scalar baseline. The timed vector kernel excludes callback overhead and directly generates only half the permutations; these results therefore do not establish end-to-end application speedup or performance on other processors. Reported validation reaches $n = 11$, with a benchmark at $n = 13$.

Contributions

1. Partitions the first half of the permutation space into two streams that can advance concurrently within one AVX2 register.
2. Uses the relative order of elements 0 and 1 to justify recovering the complementary half-space by reversal.
3. Combines staircase insertion with lane-local byte-shuffle masks for vectorized permutation generation up to $n = 16$.

4. Provides distinct scalar, vector-benchmark, and callback-based C implementations, with checksum protection against dead-code elimination in the benchmark.
5. Reports hardware-specific normalized throughput gains over optimized Algorithm P while explicitly separating half-space kernel timing from callback processing.

2026MC

ChatGPT (July 2026)

Summary

The study extends symplectic split-operator integration to Hamiltonians that are nonseparable in conjugate variables and, more generally, have constant Poisson brackets. Duplicating phase space makes each Hamiltonian copy explicitly integrable; a quadratic restraint or midpoint or symmetric projection keeps the copies close, and their separation provides a cheap accuracy diagnostic. A local Hessian criterion determines restraint stability. Tests span 500 turbulent $E \times B$ guiding-center trajectories over 500 forcing periods and grid-based Kohn–Sham time-dependent density-functional theory. With an optimized fourth-order split, $E \times B$ errors follow fourth-order scaling below $\Delta t \approx 0.15$; large restraint frequency can create resonant error peaks. Midpoint and restraint reach 10^{-10} energy error in about 18 h, compared with 55 h for implicit symmetric projection, which alone preserves symplecticity in the original phase space. The density-functional variants also achieve fourth-order convergence, but stability windows and performance depend on restraint and split ordering.

Contributions

1. Extended phase-space splitting to constant Poisson brackets and time-dependent Hamiltonians.
2. Derived and validated a Hessian-based stability condition for the restraint.
3. Established copy separation as a cheap error and resonance diagnostic.
4. Quantified speed, resonance, and symplecticity trade-offs among three coupling methods.
5. Applied high-order split propagation to grid-based Kohn–Sham dynamics.

2026MCCC

ChatGPT (July 2026)

Summary

This paper reports the first coordinated non-dimensional energy-confinement experiments in closely matched negative-triangularity plasmas on DIII-D and TCV. Lower-single-null discharges varied normalized Larmor radius ρ_* or collisionality ν_* while approximately holding other dimensionless parameters and radial profiles fixed. TRANSP power-balance analysis finds near-Bohm confinement scaling on DIII-D, driven mainly by the ion channel, whereas TCV shows near-gyro-Bohm scaling for both ions and electrons. Direct cross-machine ρ_* scaling remained infeasible because TCV density peaking and rotation could not be matched to DIII-D. Both devices exhibit only a weak positive collisionality dependence, with $B_T \tau \propto \nu_*^{0.04 \pm 0.05}$ on DIII-D and $B_T \tau \propto \nu_*^{0.09 \pm 0.03}$ on TCV. A broader DIII-D engineering regression agrees with the controlled scans in its ρ_* and ν_* dependencies. The finite scan ranges and residual profile mismatches limit direct reactor extrapolation, but the measurements provide validation targets for first-principles transport models.

Contributions

1. Conducted the first self-similar, multi-device non-dimensional confinement study of strongly negative-triangularity plasmas.
2. Distinguished near-Bohm DIII-D scaling, dominated by ion transport, from near-gyro-Bohm TCV scaling.
3. Quantified the weakly favorable collisionality dependence of normalized confinement on both tokamaks.
4. Demonstrated agreement between controlled non-dimensional scans and an independently filtered DIII-D engineering regression.

5. Identified the profile-matching obstacles that currently prevent a direct cross-machine normalized-Larmor-radius scaling.

2026MEVD

ChatGPT (July 2026)

Summary

The paper tests a radio-emission mechanism in which nonpropagating Alfvénic magnetic perturbations carried into a relativistic, magnetized electron–positron shock convert into propagating superluminal ordinary modes downstream. A cold pair-plasma analysis derives frequency and wavenumber matching across the shock and shows that activation requires the shock-frame seed-wave frequency to exceed the downstream plasma frequency. One-dimensional particle-in-cell simulations reproduce the predicted downstream Alfvénic and superluminal branches, including their wavenumber jumps, dispersion, and thermally corrected cutoff. Long-wavelength perturbations remain nonpropagating, whereas sufficiently short-wavelength components generate outgoing ordinary modes. Broadband simulations therefore make the shock behave as a plasma-frequency high-pass filter, with the transmitted spectrum retaining the upstream spectral slope. Lorentz transformations connect downstream waves to observer-frame amplitudes and frequencies. The authors discuss magnetar winds and fast radio bursts, but uncertain turbulent energy at sufficiently small scales prevents a firm efficiency estimate; multidimensional, propagating-wave, and turbulence effects remain open.

Contributions

1. Derived a frequency-matching activation criterion for Alfvén-to-superluminal-mode conversion at relativistic magnetized shocks.
2. Predicted downstream wavenumber and amplitude relations for nonpropagating Alfvénic and propagating ordinary modes.
3. Validated those relations with one-dimensional relativistic pair-plasma particle-in-cell simulations.
4. Showed that a broadband shock acts as a plasma-frequency high-pass filter while preserving the transmitted spectral slope.
5. Transformed the generated waves to observer frames and identified constraints and uncertainties for magnetar fast-radio-burst applications.

2026MF

ChatGPT (July 2026)

Summary

Three-dimensional fully kinetic particle-in-cell simulations resolve how weakly magnetized counterstreaming plasmas redistribute flow energy among fields, electrons, and ions. Early electrostatic modes and the filamentary Weibel instability amplify fluctuations but transfer little energy between species. Significant thermalization begins later when drift-kink deformation and filament merging stretch magnetic-field lines, producing a second, dynamo-like amplification stage. Electrons are heated mainly by magnetic pumping through betatron acceleration in the strengthening fields of kinked filaments. This intrinsically three-dimensional mechanism persists across a broad magnetization range. Mass-ratio scans give the final scaling $T_e/T_i \propto (m_i/m_e)^{-1/2}$. Extrapolation to electron–proton plasmas yields $T_e/T_i \approx 0.03$, so electrons receive only a few percent of the initial ion kinetic energy, consistent with supernova-remnant observations.

Contributions

1. Separated early Weibel field growth from the later phase responsible for substantial species heating.
2. Identified drift-kink instability and filament merging as drivers of a second dynamo-like amplification stage.
3. Demonstrated that electron heating is dominated by betatron acceleration through magnetic pumping.
4. Established that the heating mechanism is intrinsically three-dimensional and robust across varied magnetization.
5. Derived the mass-ratio scaling $T_e/T_i \propto (m_i/m_e)^{-1/2}$ and predicted $T_e/T_i \approx 0.03$ for proton–electron flows.

2026MHET

ChatGPT (September 2026)

Summary

Measurements on Alcator C-Mod show that cross-field particle transport grows rapidly as high-density H-mode plasmas approach loss of confinement and L-mode plasmas approach their density limit. The authors infer particle flux at the separatrix, the boundary between closed and open magnetic field lines, from Lyman-alpha emission, electron profiles, and stationary particle balance. Comparison with the separatrix operational space model shows that the characteristic resistive-ballooning wavenumber organizes the flux more consistently than collisionality alone. Including the cylindrical safety factor accounts for plasma-current variation and yields an empirical operational boundary at $k_{\text{RBM}}^2 \hat{q}_{\text{cyl}} \approx 1$. Estimated perpendicular convective heat transport approaches parallel conductive transport near this boundary, consistent with a thermal-collapse model in which equilibrium solutions merge. Under additional geometric and transport assumptions, the threshold implies a density-limit dependence approximately proportional to $P_{\text{sep}}^{4/7}$. These observations connect enhanced transport to power starvation preceding density limits, while the reconstruction and interpretation depend on stationarity, poloidal symmetry, and transport closures. The measurements support resistive-ballooning-related transport without uniquely identifying the instability.

Contributions

1. Inferred separatrix particle fluxes from Lyman-alpha measurements and electron profiles for L- and H-mode discharges near high-density operational boundaries.
2. Showed that enhanced flux tracks the L-mode density limit and high-density H-L boundary, without corresponding systematic increases near the ideal-ballooning boundary.
3. Identified $k_{\text{RBM}}^2 \hat{q}_{\text{cyl}}$ as a common organizer of particle flux across confinement regimes and plasma-current variation, with an empirical limit near unity.
4. Linked the observed boundary to perpendicular convective heat transport approaching parallel conductive transport, consistent with loss of thermal equilibrium through a fold catastrophe.
5. Derived conditional pressure-gradient scale-length and power-dependent density scalings from the empirical threshold and examined their implications for ITER and SPARC.

2026MHGR

ChatGPT (July 2026)

Summary

This survey updates the description of Trilinos for heterogeneous, leadership-scale computing. It organizes more than 35 packages into five product areas: Core, Linear Solvers and Preconditioners, Nonlinear Solvers and Analysis Tools, Discretization Tools, and Framework. The modern stack replaces archived Epetra-era components with Kokkos-enabled successors centered on Tpetra, using execution and memory-space abstractions to support CPUs and accelerators from one source base. The paper maps distributed linear algebra, load balancing, Krylov and direct solvers, domain decomposition, multigrid, eigensolvers, nonlinear solution, optimization, automatic differentiation, uncertainty quantification, time integration, finite elements, and meshfree approximation onto this structure. It also documents the TriBITS-based build and test system, documentation infrastructure, PyTrilinos2 bindings, package-quality expectations, contribution pathways, and links to external scientific-software communities. The result is a current architectural and organizational reference rather than a performance benchmark.

Contributions

1. Reorganized the modern Trilinos ecosystem into five coherent product areas and mapped its packages to them.
2. Explained how Kokkos and Tpetra provide performance-portable execution, memory management, and distributed linear algebra.
3. Catalogued the current solver, analysis, and discretization capabilities and their package-level interfaces.

4. Documented build, continuous-testing, documentation, and Python-binding infrastructure for users and developers.
5. Described governance, contribution expectations, onboarding channels, and integration with the wider high-performance software ecosystem.

2026MHL

ChatGPT (July 2026)

Summary

The authors develop a finite-difference exponential-time-domain Maxwell solver for relativistic particle-in-cell simulations. High-order Taylor approximations to the Maxwell exponential propagator are combined with staggered banded finite differences, retaining the locality and domain-decomposition potential of Yee methods while approaching pseudospectral accuracy without basis transforms. Current filtering suppresses unresolved high-frequency sources, and optional twofold field-interpolation supersampling improves particle-force accuracy at lower grid resolution. Vacuum propagation tests exhibit the expected high-order convergence; roughly twelfth-order exponential expansion is needed for near-analytic precision. Laser propagation, wakefield, relativistic-particle, surface high-harmonic, and nonlinear laser-wakefield benchmarks show reduced numerical dispersion and Cherenkov radiation and more consistent convergence than the compared SMILEI and EPOCH runs. The method remains slower per update than a conventional Yee solver, finite-order propagation is not exactly norm-conserving, and high spatial frequencies require filtering. The current Cartesian implementation lacks the proposed quasi-cylindrical geometry, distributed-memory and GPU execution, ionization, and quantum-electrodynamic modules.

Contributions

1. Derived a local real-space exponential Maxwell propagator for particle-in-cell simulation.
2. Combined high-order staggered differences with filters controlling dispersion and numerical Cherenkov radiation.
3. Integrated the solver into a charge-conserving relativistic particle-in-cell loop.
4. Introduced interpolation supersampling that approaches double-resolution particle-force accuracy.
5. Validated convergence across linear and nonlinear laser-plasma benchmarks.

2026MHLH

ChatGPT (July 2026)

Summary

The study uses batch Bayesian optimization around three-dimensional particle-in-cell simulations to improve attosecond betatron X-rays from a 37 mJ laser-wakefield accelerator. It varies the position, length, and peak density of a longitudinal plasma-density spike following a down-ramp injector. The optimizer identifies a long spike, peaking near four times the background density and placed shortly after injection, that triggers a second high-charge electron injection on the spike's falling edge. This mechanism raises the optimization cost by more than 60 times, peak on-axis radiated energy by more than 25 times, and energy in the central half of the pulse by more than six times while retaining attosecond duration. Comparing batch sizes 1, 4, and 8 shows that parallel batches shorten wall time; batch size 4 best balances resources and improvement on the tested platform. Global optimality is not claimed.

Contributions

1. Coupled batch Bayesian optimization to expensive 3D particle-in-cell simulations of an attosecond betatron source.
2. Optimized density-spike position, length, and peak density instead of scanning them exhaustively.
3. Identified a high-performing long, fourfold-density spike located soon after the initial injection ramp.
4. Revealed a second down-ramp electron injection as the radiation-enhancement mechanism.
5. Quantified large radiation gains and compared sequential and parallel batch sizes.

2026MHSE

ChatGPT (July 2026)

Summary

This paper combines an Alcator C-Mod discharge database, high-resolution Thomson scattering, neutral measurements, phase-contrast imaging, and pedestal models to study the transition between ELMy and enhanced- D_α H-modes. In ELMy operation, pedestal density responds to neutral sources; in EDA operation it is comparatively insensitive, consistent with stronger particle transport. Quasi-coherent fluctuations strengthen and become more coherent across the transition before weakening at the highest densities. The Saarelma–Connor neutral-transport model reproduces ELMy pedestal densities up to about $2.0 \times 10^{20} \text{ m}^{-3}$. Adding a resistive-ballooning-mode transport term improves agreement for EDA cases up to about $3.0 \times 10^{20} \text{ m}^{-3}$. EPED scans show that the separatrix-to-pedestal density ratio shifts the peeling–ballooning branch transition and changes predicted pressure. Applied to SPARC scenarios, added edge transport weakens the near-separatrix density gradient and lowers predicted pedestal density by roughly 20%. Model coefficients and boundary inputs remain important uncertainties.

Contributions

1. Linked neutral-source sensitivity to the ELMy-to-EDA change in pedestal particle transport.
2. Mapped quasi-coherent fluctuation evolution across density and confinement regimes.
3. Tested the Saarelma–Connor pedestal-density model against high-density C-Mod data.
4. Introduced a resistive-ballooning transport term that improves EDA density predictions.
5. Quantified separatrix-density effects in EPED and propagated them to SPARC pedestal scenarios.

2026MK

ChatGPT (July 2026)

Summary

Misra and Krishan study gravitational, Rayleigh-Taylor-type instability in partially ionized magnetoplasmas using a two-fluid model. Electrons and neutrals are treated as a combined fluid with effective charge and mass, while singly charged ions form the second fluid; electron-neutral and ion-neutral collisions couple the fluids. The resulting low-frequency, long-wavelength electrostatic dispersion relation modifies the classical fully ionized result by introducing a plasma-density inhomogeneity length scale and collision-dependent changes to the Brunt-Vaisala frequency. The instability growth rate depends on the transverse wave-number ratio $\kappa = k_x/k_y$ and normalized ion-neutral collision frequency $f = \nu_{in}/\Omega_{ci}$. The paper estimates minute-to-hour instability timescales relevant to the ionospheric E region, solar chromosphere, and prominence threads. It remains a linear theory, but it provides parameter trends for future nonlinear modeling and observational comparison.

Contributions

1. Built a two-fluid partially ionized plasma model under gravity.
2. Derived modified Rayleigh-Taylor growth rates including ion-neutral collision effects.
3. Identified dependence on transverse wave-number ratio and normalized collision frequency.
4. Estimated instability timescales for ionospheric and solar partially ionized plasmas.
5. Clarified where nonlinear and more detailed models are needed to test the predictions.

2026MKC

ChatGPT (July 2026)

Summary

The paper shows that commonly used temporal Drude models can violate momentum and energy conservation when plasma carriers are removed. A hydrodynamic current equation with an explicit carrier creation/removal momentum term is used to compare no-source, rise-and-collapse-source, and collapse-only-source models. Assuming electrons are created with zero mean velocity but removed carrying the ensemble-average momentum leads to the collapse-only-source model. For a temporally triangular plasma-density pulse, this model satisfies the wavelength-averaged energy balance to numerical precision and returns both current and magnetic field consistently toward vacuum. The other models predict apparent gain without an identified energy source and leave nonphysical residual current or magnetostatic field after the carriers vanish. The analysis supplies a physically consistent basis for interpreting frequency conversion and scattering in pulsed plasmas and other time-varying polarizable media.

Contributions

1. Traced spurious postcollapse amplification to omitted momentum carried away by removed carriers.
2. Derived a hydrodynamic current equation containing an explicit carrier source/sink momentum term.
3. Formulated a collapse-only-source model consistent with carrier creation and removal physics.
4. Derived a wavelength-averaged electromagnetic energy balance for time-modulated plasma fields.
5. Showed that only the collapse-only-source model avoids unexplained gain, residual current, and residual magnetostatic field.

2026MLL

Claude (October 2026)

Summary

This preprint uses the global gyrokinetic particle-in-cell code TRIMEG to test how nonlinear thermal-plasma dynamics change energetic-particle (EP) transport by Alfvén eigenmodes in the ITPA toroidal Alfvén eigenmode (TAE) benchmark ($R_0 = 10$ m, $a = 1$ m, $B = 3$ T, $n = -6$, 400 keV deuterons) and in beta-induced (BAE) and reversed-shear (RSAE) cases. Case A pushes thermal ions, electrons, and EPs nonlinearly; Case B pushes only the EPs nonlinearly. Linear results coincide, but in Case A the field energy overshoots and decays, the dominant harmonics split and broaden radially, and the EP flux develops two peaks and is much smaller. Over EP densities from 0.2 to 4 times the reference, the saturated amplitude scales quadratically then linearly with drive in Case B but nearly linearly in Case A, so the flux scaling drops from quartic to quadratic and the flux is overestimated by up to an order of magnitude without thermal nonlinearity. Zonal flows grow at twice the TAE rate but shear far too weakly to matter. BAE and RSAE saturation is overestimated by about 5 and 2 to 4 times. Single- n runs with a reduced mass ratio limit generality.

Contributions

1. Specified the setup: gyrocenter equations (3)–(6) with $u_{\parallel} = v_{\parallel} + (q_s/m_s)\langle\delta A_{\parallel}^h\rangle$, the δf equation (7), the quasi-neutrality equation (8) without electron polarization, the iterative Ampère solver (10)–(13), and pullback (14)–(16); ad hoc equilibria with analytic ψ for $q = q_0 + q_2 r^2$ (TAE, BAE) and $q = q_0 + q_2 r^2 + q_4 r^4$ (RSAE); $n_{e0} = 2 \times 10^{19} \text{ m}^{-3}$, $T_e = T_i = 1 \text{ keV}$, $\beta_e \approx 9 \times 10^{-4}$, $\rho_{ti} = 1.52 \times 10^{-3} \text{ m}$, $m_i/m_e = 183.6$; EP profile (20) with $c_0 = 0.49123$, $c_1 = 0.298228$, $c_2 = 0.198739$, $c_3 = 0.521298$ and $n_{EP}/n_e = 0.00376$ at $r = 0.5a$; 32×10^6 , 8×10^6 , and 8 to 16×10^6 markers on a 64×128 grid; Table 1 defines Case B by dropping (5)–(6), the $\delta\dot{R}$ and $\delta\dot{u}_{\parallel}$ terms on the left of (7), and the parallel-velocity pullback (15).
2. Showed that the radial mode width grows with EP density whether or not thermal nonlinearity is included (FWHM from about 0.06 to 0.19 for $n_f/n_{f,ref}$ from 0.2 to 4, Fig. 12), that Case B exhibits a sub-exponential upward drift of the field energy after the linear stage while Case A overshoots and decays with broadened $E_P(r)$ and split $m = 10, 11$ harmonics at $t = 50R_N/v_N$ (Figs. 3, 4), and that the gyro-Bohm-normalized EP flux stays

positive with two peaks in Case A, whose thermal density perturbations exceed the EP ones but cancel in net charge, whereas in Case B they are smaller than the EP flattening (Figs. 5, 6).

3. Analyzed $|\delta f|^2$ in $(E/E_{max}, \lambda)$ space against the bounce–precession resonance $n\omega_d + p\omega_b = \omega$ (30) with $\omega \approx 4 \times 10^5$ rad/s: passing EPs resonate at $p = 3/2$, with $p = 5/2$ emerging nonlinearly, consistent with $v_{\parallel} \approx v_A/3$ and $v_A/5$ from $p\omega_b \approx \omega = k_{\parallel}v_A$, $k_{\parallel} \approx 1/(2qR)$; electrons also show $p = 3/2$ with a broader nonlinear structure in Case A, and Case B shows stronger resonance overlap (Fig. 7).
4. Found with the $n = 0$ component retained that the zonal electrostatic energy grows at $\gamma_{P,0} = 0.11744$, close to twice the TAE rate 0.058197, indicating beat-driven zonal flow, that the zonal to $n = -6$ energy ratio is about 0.6 after saturation, and that the shearing rate $-\partial_r^2 \delta \Phi / B$ is much smaller than the TAE frequency and growth rate (Figs. 8, 9), so zonal flows regulate the instability only weakly.
5. Defined transport stiffness as $S_{\Gamma_{EP}} = \partial \Gamma_{EP} / \partial n_{EP}$ (31), distinct from profile stiffness $\partial P_{EP} / \partial \text{Src}_{EP}$ (32); showed the Case B saturation level follows the pendulum-model quadratic scaling up to about $n_{f,ref}$ and then the linear radial-decoupling scaling, with Case A lower by a factor rising to about 5 (Fig. 10), and the flux overestimation exceeding an order of magnitude at $n_{f,ref}$ (Fig. 11); for the BAE the growth rate falls and the frequency rises by about 20% toward near stabilization at the highest shear, with saturation overestimated about 5 times at the lowest shear (Fig. 14), and for the RSAE the growth rate falls from about 0.054 to 0.035 and the frequency roughly halves as $\Delta q = q_{min} - 1.75$ goes from 0 to 0.05, with saturation overestimated 2 to 4 times (Fig. 16); $\delta A_{\parallel, m=12}$ has odd parity for the BAE and even parity for the RSAE.

Comments

- Section 1: “The results show that nonlinear thermal plasma dynamics can significantly modify the mode structure and lead to a stiff dependence of energetic particle transport”, whereas the title, the abstract (“the dependence of the EP flux on the EP drive becomes weaker”), and Section 7 state that thermal nonlinearity reduces the stiffness; a less stiff dependence is evidently intended.
- Section 2.2 and Fig. 1: the RSAE profiles are listed with $\bar{q}_0 = 1.765625, 1.790625, 1.790625, 1.790625, 1.815625$, but the legend of Fig. 1 gives $q_0 = 1.7656, 1.7781, 1.7906, 1.8031, 1.8156$, and Section 9 spans $\Delta q = q_{min} - 1.75$ from 0 to 0.05 with $q_{min} = \bar{q}_0 - 0.015625$ at $r_c = 0.5$; the repeated entries are evidently 1.778125 and 1.803125.
- Section 8 and Fig. 14: the BAE cases are labeled by $dq(r_c)/dr$ from 0.32 to 0.8, but the profiles $\bar{q} = \bar{q}_0 + \bar{q}_2 r^2$ of Section 2.2 with $\bar{q}_2 = 0.16, 0.24, 0.32, 0.40$ give $dq/dr = 2\bar{q}_2 r_c = \bar{q}_2$, that is 0.16 to 0.40, at $r_c = 0.5a$; the plotted abscissa equals $2\bar{q}_2$; unresolved.
- Section 10: “the case with EP-only nonlinearity exhibits an approximately quadratic scaling of the saturation level and EP flux, the fully nonlinear simulations show a scaling closer to linear”, whereas Section 7 and Fig. 11 give quartic and quadratic scalings of the flux for the EP-only and fully nonlinear cases, the quadratic and linear scalings applying to the saturation level alone.
- Section 3: “The left and middle frames show the 2D structures of the density perturbation for $n_{EP}/n_{f,ref} = 0.5$ (left) and $n_{EP}/n_{f,ref} = 2$ (right)”, but Fig. 2 has two frames, left and right.

2026MMGK

Claude (September 2026)

Summary

Accurate particle-in-cell (PIC) modeling of high-intensity laser–plasma interaction requires the laser’s field amplitude and phase, but experiments often provide only fluence images at a few transverse planes. The authors present a cylindrical implementation of the Gerchberg–Saxton algorithm with mode decomposition (GSA-MD), which iteratively fits complex Laguerre–Gauss mode coefficients to the fluence images on per-plane polar grids, and describe it together with the earlier Cartesian Hermite–Gauss version in one framework. On averaged fluence images from the DRACO laser at HZDR, which are highly rotationally symmetric, 105 Laguerre–Gauss modes reached a reconstruction error of about 3.4% in roughly 75 s, whereas 25 Hermite–Gauss modes in similar time gave an error 3.8 times larger and far larger beam-width errors. Sampling the azimuth at 36 rather than 720 points kept comparable accuracy and cut the time to 5 s. Analytical expressions map each Laguerre–Gauss mode onto azimuthal Fourier harmonics for laser injection in the PIC

code SMILEI; a vacuum-propagation run reproduces the reconstructed fluence within 3.4%. In an ionization-injection wakefield simulation, the reconstructed laser yields a broad electron spectrum with more charge than a Gaussian fit (214 versus 150 pC). The efficiency gain applies to nearly symmetric lasers; codes and data are released.

Contributions

1. Implemented a Laguerre–Gauss, cylindrical-grid version of the Gerchberg–Saxton algorithm with mode decomposition, unified with the Hermite–Gauss Cartesian version.
2. Showed by benchmark that the cylindrical version reconstructs nearly symmetric lasers more accurately at equal cost, and that Nyquist-guided coarse azimuthal sampling gives preliminary reconstructions in seconds on a laptop.
3. Derived closed-form mappings from Laguerre–Gauss coefficients to the azimuthal harmonics of the injected laser field, including rescaling to a prescribed energy and temporal profile.
4. Demonstrated reconstructed-laser PIC simulations whose wakefield results contradict the previously reported trend of lower charge and energy for realistic lasers, arguing that realistic laser models are needed to design working points.
5. Defined a rotational symmetry parameter for comparing laser fluence symmetry and a separable storage of mode fields that greatly reduces memory use.

2026MML

Claude (October 2026)

Summary

The authors extend their kinetic theory of parametric instabilities driven by a high-harmonic fast wave (HHFW) from a uniform scrape-off layer (SOL) to the inhomogeneous near-SOL pedestal with density and temperature gradients. In a slab with uniform B and a pump of frequency $\omega_0 = 30.5\omega_{ci}$, the ion Vlasov equation is solved in a frame oscillating with the ions, and the electron response, seen there as harmonics shifted by k_{0z} and ω_0 , couples the potential's Fourier components. A renormalized linearization includes nonlinear ion scattering by the resulting ion-cyclotron (IC) turbulence through $\hat{\omega} = \omega - iC_i$. The dispersion relation, truncated to the high-harmonic IC mode and two satellites, is solved with the complex Halley method for deuterium, $n_0 = 2 \times 10^{11} \text{ cm}^{-3}$, $B_0 = 1 \text{ T}$, $T_i = T_e$, and pump amplitude $\hat{E}_{0x} = 0.22$. Unstable harmonics $n = 17$ to 27 appear only for finite k_{0z} ; growth rates peak at steep density gradients, are governed by inverse ion Landau damping, rise strongly with η_i , and respond weakly to η_e . The saturated turbulence level follows from $\gamma_{Lin} = C_i$, and quasilinear moments give perpendicular ion heating exceeding parallel heating by $1/(k_z\rho_i)$, consistent with NSTX edge-ion observations.

Contributions

1. Formulated the problem in ion-oscillation-frame variables in which the pump enters only through terms of order $k_{i0}\xi_i$, neglected for HHFW, so the ion Vlasov equation (4) takes its field-free form; introduced guiding-center variables (5)–(7) and the renormalized linearization (8)–(13), which turns second-order nonlinearities into third-order ones and adds the Gaussian orbit-scattering factor $e^{-C_i(t-t_1)}$ (18) with C_i given by (31)–(32); expressed the electron density and potential in the ion frame through Bessel series in $a_{ei} = |k_i\xi_{ie}|$ with phase δ (21)–(24), yielding the coupled functional equation (25)–(26), the infinite system (33) with coefficients b_{mq} (34), and the determinant condition (35), which reduces to the current-driven instability equation as $\omega_0 \rightarrow 0$.
2. Gave the renormalized ion permittivity (29) and linear electron permittivity (30) for Maxwellian species with inhomogeneous density and temperature in terms of $\xi_\alpha = k_y v_{d\alpha}/(\sqrt{2}k_z v_{T\alpha})$, $\eta_\alpha = d \ln T_\alpha / d \ln n_\alpha$, $A_{\alpha p} = I_p(k_\perp^2 \rho_\alpha^2) e^{-k_\perp^2 \rho_\alpha^2}$, $\hat{A}_{ip}$, and $z_{ip} = (\hat{\omega} - p\omega_{ci})/(\sqrt{2}k_{iz} v_{Ti})$, noting that replacing ω by $\hat{\omega}$ in the electron term is negligible because $|z_e| \ll 1$ and $|C_i| \ll |\omega|$.
3. Found numerically that a spatially uniform pump ($k_{0z} = 0$) suppresses the inverse-electron-Landau-damping instability of the uniform SOL over $\omega_0 \sim (30\text{--}50)\omega_{ci}$, so finite k_{0z} is essential; with $(k_{0z}\rho_i)^{-1} = 16.763$, $(k_z\rho_i)^{-1} = 14.92$, $k_x\rho_i = 0.612$, $k_y\rho_i = 4.65$, $m_i/m_e = 3680$, and harmonic sums over $-50 \leq n, r \leq 50$, Figs. 1–7 ($\eta_i = 0, 0.7, 1.2, 1.5$; $\eta_e = 0, 0.7$) show growth-rate maxima at small L_n/ρ_i , largest for $n = 19\text{--}25$ and about three times smaller for $n = 17$ and 27 , instability confined to $n = 17\text{--}27$, $|z_{in}| < 3.5$ with $|z_{en}| \ll 1$,

maxima for $\eta_i = 1.5$ about twice those for $\eta_i = 0.7$, ETG-only growth rates near $10^{-3}\omega_{ci}$, and, for $\eta_i = 1.2$, peaks of $\gamma/\omega_{ci} \approx 0.17\text{--}0.19$ near $L_n/\rho_i \approx 30\text{--}60$ against about 2×10^{-3} for $\eta_i = \eta_e = 0$.

4. Developed the renormalized ion-kinetic quasimode decay theory: zeroth-order solution $\hat{\omega} = \Omega(k_i + q_0 k_{0z}) - q_0 \omega_0$ (38), the growth rate (39) valid for $\gamma < |\delta\omega|$, saturation at $\gamma_{NL} = \gamma_{Lin} - C_i = 0$ (41), the steady-state integral equation (44) whose resonance-broadening term exceeds the quasilinear term by about an order of magnitude, and the estimate $e|\bar{\varphi}_i|/T_i \sim |\delta\Omega|/(\omega_{ci}\hat{k}_{i\perp}^2\rho_i^2 F_i^{1/2})$ (47) for the turbulence level.
5. Derived from the renormalized quasilinear equation (48)–(53), with resonance function R_i (54) composed of a Lorentzian broadening part and a resonant delta part, the perpendicular heating equation (56) with rate $\nu_{i\perp} \sim \gamma(\hat{k}_i)W/(n_{0i}T_{i\perp})$ (57) and anomalous conductivity $\chi_{i\perp} \sim (\gamma/\hat{k}_{i\perp}^2)(\omega_{ci}/\delta\Omega)^2 W/(n_{0i}T_{i\perp})$ (58), the parallel heating rate $\sim k_z \rho_i \gamma W/(n_{0i}T_{iz})$ (60), hence the anisotropy ratio $k_z \rho_i \ll 1$ (61), and concluded that near-SOL growth and heating rates about an order of magnitude above uniform-SOL values make pedestal formation a likely enhancer of edge RF absorption.

Comments

- Section III: the harmonic sums are said to run “over n in Eq. (34) for ε_i ” and “over r in Eq. (35) for b_{mq} ”, but ε_i with its harmonic sum is Eq. (29), the sum over r defining b_{mq} is Eq. (34), and Eq. (35) is the determinant condition; (29) and (34) are evidently intended.
- Section III: “ $\hat{E}_{0x} = 0.22$, which corresponds to $E_{0x} = 400$ V/cm for the values of n_{0i} and T_i specified above”, but only the ratio $T_i/T_e = 1$ is specified; Eq. (36) with $n_0 = 2 \times 10^{11} \text{ cm}^{-3}$ and $E_{0x} = 400$ V/cm implies $T_i \approx 9$ eV, which is not stated.
- Sections III and IV: Section III states that for $\omega_0 = 30.5\omega_{ci}$ “unstable solutions are found only for the high-harmonic IC modes with $n = 17\text{--}27$ ”, whereas Section IV refers to “all unstable IC harmonics with $n = 21, 23, 25$, and 27 ” in Figs. 1–4, omitting $n = 17$ and 19 , which those figures show to be unstable; unresolved.
- Section II, after Eq. (15): the nonlinear phase shift $k_i \delta r_i(t)$ “is included in Eq. (15) without terms of second order”, but (15) is the definition of the phase shift itself and it enters the expansion (14); Eq. (14) is evidently intended.

2026MMWS

ChatGPT (September 2026)

Summary

Maity and coauthors model a single-stage capillary gas cell for laser wakefield acceleration. A roughly 2 mm entrance region contains helium and nitrogen to inject electrons released from nitrogen’s inner shell; a longer pure-helium region accelerates them while limiting further injection. Hydrodynamic simulations of a three-inlet cell provide gas-density profiles, which the authors pass directly to particle-in-cell simulations driven by a 100 TW-class laser. Varying inlet spacing and pressures produces different density ramps and changes beam charge, energy, and energy spread. In the reported Case 2, the ionization-injected beam exits with about 41 pC charge, 1.02 GeV mean energy, and 2.4% relative full-width-at-half-maximum energy spread. The simulations also track self-injected helium electrons: they trail the principal beam, but spectral overlap can impair the beam observed experimentally. Slowly rising density profiles suppress this secondary population. These are computational predictions for a proposed target, not measurements from an operating cell.

Contributions

1. Designed a three-inlet capillary with a localized helium–nitrogen injection region and a longer pure-helium acceleration region.
2. Generated realizable longitudinal gas-density ramps by varying the capillary geometry and inlet pressures in hydrodynamic simulations.
3. Coupled those gas profiles to particle-in-cell simulations to resolve injection and acceleration with a 100 TW-class laser.

4. Quantified the beam-quality trade-off across five gas configurations, including the 41 pC, 1.02 GeV, 2.4%-spread result for Case 2.
5. Distinguished trailing self-injected helium electrons from the nitrogen-derived beam and identified rising density ramps that suppress them.

2026MNM

ChatGPT (July 2026)

Summary

This paper constructs a fully discrete continuous Galerkin finite-element method for initial-boundary-value advection–diffusion problems using summation-by-parts operators in both space and time. Initial and boundary conditions are imposed weakly through simultaneous approximation terms. Matching the discrete integrations by parts to the continuous energy argument yields a bound controlled by initial and external boundary data, establishing energy stability for the complete space–time discretization rather than only its spatial part. The authors analyze both global temporal elements and a one-step multistage implementation. Manufactured-solution tests across diffusion- and advection-dominated regimes show superconvergence of order $\mathcal{O}(p + 2)$ in space and time for Lagrange degree $p \geq 2$. An oscillatory advection–diffusion application demonstrates that coarse temporal meshes still resolve space–time variation, while comparisons with explicit fourth-order Runge–Kutta integration report two to three orders of magnitude lower normalized computational cost in the studied cases. The present analysis is restricted to linear equations; extension to nonlinear problems is left for future work.

Contributions

1. Formulated a continuous Galerkin discretization with summation-by-parts structure in both space and time.
2. Imposed initial and boundary data weakly with simultaneous approximation terms.
3. Proved a fully discrete energy estimate controlled by prescribed data.
4. Demonstrated $\mathcal{O}(p + 2)$ spatial and temporal superconvergence for $p \geq 2$.
5. Showed coarse-mesh accuracy and large cost reductions relative to explicit RK4 in oscillatory tests.

2026MR

ChatGPT (July 2026)

Summary

This paper develops structure-preserving discretizations for three two-dimensional ideal magnetized-fluid models: reduced magnetohydrodynamics (RMHD), Hazeltine’s three-field model, and the Charney-Hasegawa-Mima (CHM) equation. Each model has a non-canonical Hamiltonian formulation with Lie-Poisson geometry, Casimir invariants, and phase flows restricted to coadjoint orbits. The authors apply Zeitlin’s matrix-hydrodynamics truncation and a compatible geometric time integrator so that finite-dimensional simulations retain this structure over long times. Randomized initial conditions reveal shared and model-specific turbulent organization. RMHD and Hazeltine dynamics form large-scale magnetic dipoles, and spectral diagnostics support inverse transfer of magnetic energy and mean-square magnetic potential. Their vorticity behavior differs sharply: RMHD develops thin filaments, growing vorticity, and a direct kinetic-energy cascade, whereas Hazeltine and CHM form large-scale vorticity structures with inverse kinetic-energy transfer and little vorticity-amplitude change. The comparison shows that structure-preserving numerics can expose distinct long-time statistics among closely related reduced plasma models.

Contributions

1. Formulated compatible matrix-hydrodynamics discretizations for RMHD, Hazeltine, and CHM dynamics.
2. Combined the spatial truncations with a geometric time integrator that preserves the underlying Lie-Poisson structure.
3. Performed long-time turbulent simulations from randomized initial data across all three models.

4. Demonstrated inverse magnetic transfer and magnetic-dipole formation in RMHD and Hazeltine dynamics.
5. Distinguished RMHD's filamentation and direct kinetic cascade from the inverse kinetic cascades of Hazeltine and CHM models.

2026MRD

ChatGPT (July 2026)

Summary

This preprint derives and implements a self-consistent electrostatic model coupling large-amplitude, slowly varying full- f drift-kinetic fields to small-amplitude, rapidly varying δf gyrokinetic fields. Critical-balance ordering reduces the electron dynamics to a drift-reduced Braginskii model, while Hermite–Laguerre expansions represent both components of the ion distribution through gyromoments. Global simulations with Large Plasma Device parameters converge with few moments because ion–ion collisions keep the drift-kinetic distribution close to a bi-Maxwellian. At the physical collisionality, gyrokinetic fields remain subdominant and do not materially change the drift-kinetic turbulence. Artificially reducing gyrokinetic collisions while strengthening gyrokinetic sources produces finer perpendicular structures. Linear analysis attributes the baseline turbulence to a drift-kinetic Kelvin–Helmholtz instability and identifies a gyrokinetic Kelvin–Helmholtz-like mode driven by short gyrokinetic density and potential gradient scales in the modified regime.

Contributions

1. Performed coupled full- f drift-kinetic and δf gyrokinetic simulations in a linear device.
2. Derived the coupled electron, ion-gyromoment, and electrostatic equations from first principles.
3. Demonstrated rapid Hermite–Laguerre spectral convergence under physical LAPD collisionality.
4. Established that gyrokinetic fields are subdominant to drift-kinetic turbulence in that regime.
5. Identified a gyrokinetic Kelvin–Helmholtz-like mechanism for enhanced small-scale structure.

2026MRG

ChatGPT (July 2026)

Summary

The paper constructs arbitrary-order discontinuous spectral-element schemes for rotating shallow-water equations with variable topography on curved manifolds. A covariant balance-law formulation, skew-symmetric tensor-divergence split, tensor-product summation-by-parts operators, and flux differencing yield methods that are well balanced, mass conservative, and either total-energy conservative or dissipative according to the interface flux. Analytic geometry and metric terms can be used directly, avoiding approximate metric identities normally required for entropy proofs. The authors prove conservation, entropy stability, and lake-at-rest well balancing on general unstructured quadrilateral grids. Cubed-sphere experiments involving unsteady solid-body rotation, isolated-mountain flow, barotropic instability, and Rossby-Haurwitz waves verify optimal order $p + 1$, machine-level conservation for entropy-conservative fluxes, controlled energy dissipation for entropy-stable fluxes, and robustness for under-resolved vortical flows and long integrations. The formulation is positioned as a foundation for a three-dimensional atmospheric dynamical core.

Contributions

1. Developed a split, flux-differencing covariant shallow-water DG spectral-element formulation on curved manifolds.
2. Derived entropy-conservative and entropy-stable two-point and interface fluxes.
3. Proved mass conservation, energy conservation or dissipation, and lake-at-rest well balancing on curved quadrilateral meshes.
4. Avoided enforced discrete metric identities through analytical geometry representation.
5. Verified optimal convergence and long-time robustness across four cubed-sphere atmospheric benchmarks.

2026MRHH

Claude (September 2026)

Summary

The authors perform integrated core–pedestal–scrape-off-layer (SOL) modeling of ARC H-modes with the ASTRA transport code to test whether high fusion power is compatible with a detached divertor. TGLF turbulent and FACIT neoclassical transport feed STRAHL, which evolves impurity charge states and radiation; a neural network trained on EPED predicts the pedestal; and the extended Lengyel model gives the SOL seeding fraction and separatrix temperature for a 2 eV divertor, all iterated to self-consistency. Pedestal and separatrix densities are prescribed, and the core seeded-impurity content follows an empirical SOL-to-core enrichment scaling imposed at the pedestal top. For an ARC V3A-like design with Ar seeding, the fusion power is about 900 MW. It varies little with pedestal density, drops to 800 MW when the separatrix-to-pedestal density ratio rises from 0.4 to 0.5 because the pedestal becomes ballooning-limited, and stays within 820–920 MW for enrichment factors of 50–200% of the scaling. Ne seeding gives lower enrichment, core accumulation, fuel dilution, lower fusion power, and less robust H-mode access. Turbulence dominates core impurity transport, giving nearly flat concentrations and weak W peaking, and modeled rotation changes little. Limitations include assumed separatrix density and enrichment, no ELM or He-ash modeling, and uncertain L–H thresholds.

Contributions

1. Coupled the X-Lengyel, EPED neural-network, and core transport models by exchanging the power crossing the separatrix, separatrix temperature, SOL seeding fraction, divertor neutral pressure, enrichment, and pedestal Z_{eff} , imposing detachment self-consistently in H-mode predictions.
2. Surveyed ARC-class designs with fields of 9.8–11.7 T and currents of 10.0–13.8 MA, finding that elongation slightly raises the mostly peeling-limited pedestal pressure while triangularity mainly shifts the density of the peeling-to-ballooning transition.
3. Traced the high predicted Ar pedestal concentrations to an enrichment of about 1.25 for ARC, against about 3 in ASDEX Upgrade, caused by the inverse dependence of the enrichment scaling on divertor neutral pressure.
4. Compared Ar, Ne, and an N-equivalent radiator, the latter giving 760–1000 MW with robust H-mode access, and showed that H-mode margins depend strongly on the chosen L–H threshold scaling.
5. Linked the lower W peaking at higher Z_{eff} to a larger W flux in standalone TGLF scans, and showed with standalone FACIT scans of density gradient, temperature gradient, and rotation that neoclassical W transport becomes large only above Mach number 0.4.

Comments

- Section 4.4, Figs. 7–9: panel (c) is labelled $P_{\text{loss}}/P_{LH, \text{Schmidtmayr}}$, whereas the text defines the plotted $f_{LH, \text{Schmidtmayr}} = P_{i, \text{loss}}/P_{LH, \text{Schmidtmayr}}$ with the ion power loss, and P_{loss} elsewhere, including panel (f), denotes the total power crossing the separatrix.
- Section 4.4, Table 2: the standard deviation of $\langle n_e \rangle$ for Ar is given as 0.26 (in 10^{19} m^{-3}) across the pedestal-density, enrichment, and separatrix-density scans, but Fig. 2(b) alone shows $\langle n_e \rangle$ from about 22.4 to 26.7 in the pedestal-density scan, which is incompatible with that value; the intended value is unresolved.
- Section 5: “Ne-seeded plasmas have shown lower fusion power (< 800 MW)” conflicts with the Abstract (“600–850 MW”), the Section 4.4 summary (“ $600 < P_{fus} < 850 \text{ MW}$ ”), and the Section 4.4 statement that Ne cases with lower accumulation reach “ $P_{fus} > 800 \text{ MW}$ ”; the likely intended reading is 600–850 MW.

2026MSCM

ChatGPT (September 2026)

Summary

In DIII-D tokamak measurements, the authors examine the edge harmonic oscillation (EHO), a phase-locked set of magnetic harmonics associated with quiescent high-confinement operation without damaging edge-localized bursts.

Bispectral analysis of internal polarimetry, magnetic coils, and edge density fluctuations detects significant nonlinear coupling among EHO harmonics during sustained quiescent operation, supporting a saturation mechanism predicted by magnetohydrodynamic simulations. After core-resonant magnetic tearing modes appear, an intermittent EHO-like edge fluctuation couples to those core modes rather than strongly to its own harmonics, while edge-localized modes persist. Internal interferometer-polarimeter measurements reveal this core–edge coupling more clearly than external coils. The observations suggest that tearing-mode interaction can interrupt coherent EHO formation; the proposed momentum-transfer mechanism and the benefit of tearing-mode control remain hypotheses for further modeling and experiments.

Contributions

1. Provided direct experimental evidence of nonlinear coupling among EHO harmonics in quiescent H-mode.
2. Detected coupling between a transient EHO-like fluctuation and core-resonant tearing modes.
3. Distinguished the coupled modes using internal magnetic, external coil, and edge-density diagnostics.
4. Associated harmonic decoupling with the return of edge-localized bursts in the measured discharge.
5. Identified core–edge mode interaction as a possible constraint on sustained quiescent confinement.

2026MTWG

Claude (September 2026)

Summary

The authors implement, in MATLAB with GPU arrays, a matrix-free Full Approximation Storage (FAS) multigrid solver, a nonlinear multigrid whose coarse grids carry full approximations rather than corrections, for finite differences on cell-centered and staggered grids. Because masked red–black updates are awkward to vectorize there, the smoother is an X-shape multicolor Gauss–Seidel (X-MCGS) sweep over four index-parity subsets in 2D, or eight in 3D, each updated as a strided sub-block, presented as an implementation convenience, not a better smoother than red–black ordering. For $p - \Delta p = f$ the solver converges nearly independently of h with second-order accuracy in 3D, and benchmarks indicate that its speed is governed mainly by memory traffic and smoothing-kernel organization. For incompressible Navier–Stokes, first- and second-order projection schemes are reorganized so that 8 instead of 12 or 15 arrays stay resident, reducing peak memory by 19–30% and run time by 18–31% and fitting 512^3 on a 24 GB RTX 4090. Lid-driven cavity flow, grain growth with up to 1189 orientations, vesicle phase separation, and two-bubble coalescence are also computed. The authors state that the transfer operators are not optimized, full V-cycles scale worse than the smoothing kernel, mixed precision gained only 5–15%, and multi-GPU execution is future work.

Contributions

1. Assigned odd–even and even–odd index pairs (i, j) to sets 1 and 2, updated from the previous sweep, and even–even and odd–odd pairs to sets 3 and 4, updated from the new values, each by $p_{i,j} = (p_{i+1,j} + p_{i-1,j} + p_{i,j+1} + p_{i,j-1} + h^2 f_{i,j}) / (4 + h^2)$ on strided index vectors, and found on a 2048^2 grid that the 1234–1234 X ordering reached $\text{res} < 10^{-9}$ in 15 V-cycles, against 21 for 1234–4321 and 31 to 43 for U- and Z-shape orderings.
2. Removed the copies $(\cdot)_{\text{old}} = (\cdot)_{\text{new}}$ of the first-order scheme by writing u^{n+1} into U_{old} and $\tilde{u}^{n+1}$ into U_{new} , which leaves the results unchanged, and in the second-order scheme executed $u^{n-1} = u^n$ right after assembling f_u , and likewise for v and w , so that the v and w momentum equations are advected with $(u^n + \tilde{u}^{n+1})/2$, and the w equation also with $(v^n + \tilde{v}^{n+1})/2$, in place of the extrapolations $(3u^n - u^{n-1})/2$ and $(3v^n - v^{n-1})/2$ of the classical implementation.
3. Timed one V-cycle at 16384^2 and 512^3 at 0.9396 s and 0.4633 s on the RTX 4090, against 36.6231 s and 21.5990 s on one i9-13900KS core, with both 8 GB GPUs out of memory, and estimated an effective bandwidth of about 217 GB/s, against a 1008 GB/s peak, for an isolated smoothing step from 4096^2 to 16384^2 under a minimum-traffic model of 96 bytes per unknown in 2D and 128 in 3D.
4. Reported errors from 1.74×10^{-3} on 32^3 to 6.75×10^{-6} on 512^3 for $p_{\text{exact}} = \prod_l \sin(\pi \sin(\pi x_l))$, temporal orders on a 4096^2 grid at $Re = 10$ for $\Delta t = 1/10$ to $1/160$ approaching 1.00 for the first-order scheme and 2.01 for the

second-order velocity, with the second-order pressure orders falling from 1.47 to 1.13, and a discrete divergence integral below 10^{-19} in magnitude in cavity runs on 4096^2 and 512^3 grids.

5. Fitted $R_t^m - R_0^m = kt$ to Fan–Chen grain growth with q up to 1189 on 512^2 and 123 on 128^3 , about 3.12×10^8 and 2.58×10^8 unknowns, obtaining m near 2 and k leveling off as q grows, showed for vesicle phase separation at $a_{ld} = 0.3$ and 0.7 the realization closest to experiment out of 10 random seeds, and ran the quasi-incompressible Navier–Stokes–Cahn–Hilliard two-bubble case on $256 \times 256 \times 1024$ with $\Delta t = 2.5 \times 10^{-5}$ at a peak of 23.46 GB, where the classical implementation exceeded memory.

Comments

- Section 2.2, Eqs. (9), (11)–(12): the model problem requires “ $p = 0$ on $\partial\Omega$ ”, but the ghost-point conditions $p_{1,N+1,j} = p_{1,N,j}$ and $p_{1,i,N+1} = p_{1,i,N}$ impose a zero normal derivative on the right and top boundaries (only $p_{1,0,j} = -p_{1,1,j}$ and $p_{1,i,0} = -p_{1,i,1}$ give $p = 0$), which the Section 3.1 test solution $\prod_l \sin(\pi \sin(\pi x_l))$, with nonzero normal derivative at $x_l = 1$, does not satisfy although Table 2 reports second-order convergence to it; the evident intended conditions are $p_{1,N+1,j} = -p_{1,N,j}$ and $p_{1,i,N+1} = -p_{1,i,N}$.
- Section 2.2: “Step 9 updates $r_0 \leftarrow L_{h_0}(p_0^{k,0}) + r_0$ without storing f_0 separately”, but in Algorithm 1 this is Step 8, $f_0 := r_0 + L_{h_0}(p_0^{k,0})$ (“Coarse source term”), whereas Step 9 opens the coarse-grid smoothing loop (Steps 9–11 in Remark 2); the evident intended reference is Step 8.
- Section 3.3.2: “Figs. 12 and 11 present the phase separation dynamics on the vesicle membrane for $a_{ld} = 0.3$ and $a_{ld} = 0.7$, respectively”, but the captions assign $a_{ld} = 0.3$ to Fig. 11 and $a_{ld} = 0.7$ to Fig. 12, whose panels show red l_d domains in the blue l_o phase and the reverse; the figure numbers in the text are swapped.
- Section 3.3.2 and Appendix E: the regularized delta function $\delta(\phi) = \epsilon^{-1} f(\phi) + \epsilon |\nabla \phi|^2 / 2 + \eta$ uses the double well $f(c) = c^2(1 - c)^2 / 4$, with minima at 0 and 1, while “ $\phi = 1$ inside, $\phi = -1$ outside” and Eq. (22) sets $\phi = \tanh((r_0 - |x|)/(2\sqrt{2}\epsilon))$, so $f(-1) = 1$ makes $\delta \approx \epsilon^{-1}$ throughout the exterior, 100 for $\epsilon = 0.01$, above the membrane value $1/(16\epsilon) = 6.25$ at $\phi = 0$, contrary to “a regularized delta function characterizing the vesicle membrane”; unresolved.
- Table 12: the errors $2.90E - 03$ for u at $\Delta t = 1/20$, and $7.43E - 02$ and $5.73E - 03$ for p at $\Delta t = 1/20$ and $1/160$, exceed the preceding entries $1.24E - 03$, $2.06E - 02$, and $1.25E - 03$, contrary to the positive orders 2.09, 1.47, and 1.13 in the same rows and to the following orders 2.04 and 1.35; $2.90E - 04$, $7.43E - 03$, and $5.73E - 04$ reproduce all printed orders to rounding and are evidently the intended values.
- Tables 8 and 10: the rows for $\tilde{v}^{n+1}$ and $\tilde{w}^{n+1}$ (Table 8, Steps 4 and 6) and for $v^{n-1} = v^n$, $\tilde{v}^{n+1}$, $w^{n-1} = w^n$, and $\tilde{w}^{n+1}$ (Table 10, Steps 5, 6, 8, and 9) place these fields in the U_{new} and U_{old} columns, which then hold $\tilde{u}^{n+1}$ and u^n , whereas Step 7 of both tables reads $\tilde{u}^{n+1}$ from U_{new} and $\tilde{v}^{n+1}$ from V_{new} , and Step 10 of Table 10 reads $\tilde{w}^{n+1}$ from W_{new} ; the entries evidently belong in the V and W columns.
- Appendix A, Listing A.1: the loop “for color = 1:2 % 1 = black, 2 = red” updates a point only “if mod(i+j,2) == color”, but mod($i + j, 2$) is 0 or 1, so for color 2 no point is updated and set 2 ($i + j$ even) of update rule (ii), stated to follow set 1 sequentially, is never smoothed; the evident intended test for color 2 selects mod($i + j, 2$) = 0.
- Appendix B, Listing B.1: sets 3 and 7 use vector_i=2:2:M_1-1, whereas the other even- i sets 2 and 6 use vector_i=2:2:M_1; for the even M_1 of all reported grids, 2:2:M_1-1 stops at $M_1 - 2$, so points with $i = M_1$ and even j are never smoothed; the evident intended range is 2:2:M_1.
- Appendix E, Eqs. (E.5)–(E.9): “Given fields at level n (and $n - 1$ for the second-order advection term), one step is: solve (E.5) for $\tilde{\mathbf{u}}^{n+1}$, solve (E.6) for $\tilde{p}^{n+1}$, update $(\mathbf{u}^{n+1}, p^{n+1})$ by (E.7), and then solve (E.8) and (E.9) to obtain (c^{n+1}, μ^{n+1}) ”, but (E.5)–(E.7) contain $\rho^{n+\frac{1}{2}}$, $\nu(c^{n+\frac{1}{2}})$, $c^{n+\frac{1}{2}}$, and $\mu^{n+\frac{1}{2}}$, which by the stated definition $(\cdot)^{n+\frac{1}{2}} = ((\cdot)^{n+1} + (\cdot)^n)/2$ require the c^{n+1} and μ^{n+1} obtained only afterwards; how these half-level values are evaluated is unresolved.
- Appendix D, Eq. (D.1): the EW restriction weights the fine edges $2i - \frac{1}{2}$ and $2i + \frac{1}{2}$ equally and the rows $2j - 1$, $2j$, $2j + 1$ by 1, 2, 1, a stencil centered at the fine cell center (x_{2i}, y_{2j}) , whereas with $x_{i+1/2} = a_1 + ih_1$ and $y_j = a_2 + (j - 1/2)h_1$ from Section 2.1 and $h_0 = 2h_1$ the coarse edge $(i + \frac{1}{2}, j)$ lies at the fine edge $2i + \frac{1}{2}$ midway between rows $2j - 1$ and $2j$, the location assumed by the prolongation (D.2); unresolved.

2026MZL

ChatGPT (July 2026)

Summary

The paper incorporates Shaing's analytic neoclassical toroidal viscosity (NTV) torque into a one-dimensional cylindrical model of nonlinear tokamak response to a single-helicity resonant magnetic perturbation. The model couples toroidal and poloidal momentum balance, magnetic-island evolution through a modified Rutherford equation, and island-phase dynamics. For representative locked and unlocked cases with uniform pressure, NTV is weak near the resonant $q = 2$ surface but distributed throughout the plasma core. It therefore modifies core toroidal, poloidal, and parallel rotation without changing whether the resonant surface is locked or rotating. With nonuniform pressure, increasing β strengthens the global NTV torque while reducing the resonant electromagnetic torque through changes in parallel flow and island phase; their competition preserves the locked state. A heat-diffusion model for island-induced temperature flattening alters local NTV structure and core flow but again leaves the locked or unlocked classification unchanged. Results are parameter-dependent predictions of a cylindrical single-fluid model rather than a full two-fluid, kinetic, or three-dimensional MHD treatment.

Contributions

1. Added an energy- and pitch-integrated analytic NTV torque to a nonlinear RMP-response model coupling momentum, island width, and phase.
2. Demonstrated that NTV can rival electromagnetic torque in the core while remaining small at the resonant surface.
3. Showed that NTV changes core rotation profiles without changing locked or unlocked mode states in uniform-pressure examples.
4. Found that increasing nonuniform-pressure β raises global NTV torque and lowers electromagnetic torque while retaining locking.
5. Incorporated island-driven temperature flattening and showed that it redistributes NTV without changing the original mode state.

2026N

ChatGPT (July 2026)

Summary

This paper studies differential inversion of the implicit Euler map $E(t, m, x)$: computing $(E')^{-1}v$ without explicitly accumulating and inverting its full Jacobian. Three approaches are compared. Naive algorithmic differentiation through Newton solves costs $\mathcal{O}(mpn^4)$; a partially symbolic implicit-function treatment reduces this to $\mathcal{O}(mn^3)$. The new fully symbolic method stores elemental Jacobians during the m Euler steps and propagates inverse actions backward through matrix-vector products, yielding $\mathcal{O}(mn^2)$ work and $\mathcal{O}(mn^2)$ storage. A C++ reference implementation using Eigen and dco/c++ tests Lotka-Volterra systems. Runtime scaling supports the theoretical advantage, while memory measurements show that the stored tape dominates the fully symbolic method. Checkpointing can trade recomputation for memory. The analysis assumes differential invertibility and sufficiently converged nonlinear solves; regularization for ill-posed inverse problems remains future work.

Contributions

1. Formulated inverse-Jacobian actions for implicit Euler integration as a differential-inversion problem.
2. Separated naive, partially symbolic, and fully symbolic algorithms with explicit complexity estimates.
3. Derived reverse elemental propagation that avoids both global Jacobian accumulation and a final dense solve.
4. Reduced asymptotic work to $\mathcal{O}(mn^2)$, an $\mathcal{O}(n^2)$ improvement over the naive route.
5. Supplied reproducible C++ implementations and runtime, memory, and numerical-accuracy experiments.

2026N2

ChatGPT (July 2026)

Summary

Third-order edge-based finite-volume schemes traditionally extrapolate physical fluxes from both nodes and average them at an edge midpoint, which complicates reuse of standard Riemann solvers. This note derives an equivalent flux-correction form. The arithmetic mean of extrapolated fluxes is replaced by a general numerical flux evaluated from left and right midpoint states, plus a correction assembled from nodal flux derivatives and reconstructed gradients. Third-order accuracy is retained when the midpoint states are exact for quadratic data; the paper obtains this property with U-MUSCL reconstruction at $\kappa = 1/2$. The reformulation lets existing edge-based solvers call unmodified numerical flux functions while adding a compact high-order correction. Manufactured-solution experiments for the three-dimensional Euler equations on irregular tetrahedral grids recover third-order convergence with both HLLC and LDFSS fluxes. The study establishes accuracy rather than application performance; shock robustness and comparative cost in production solvers remain future work.

Contributions

1. Recast a third-order edge-based discretization into a general numerical-flux plus correction form.
2. Derived the correction without sacrificing the original truncation-error cancellation.
3. Identified quadratic-exact midpoint states as the condition for third-order accuracy.
4. Realized that condition with U-MUSCL reconstruction at $\kappa = 1/2$.
5. Verified third-order convergence with HLLC and LDFSS fluxes on irregular tetrahedral grids.

2026N3

ChatGPT (September 2026)

Summary

The paper proposes a finite-volume method for hyperbolic conservation-law systems with a strictly convex mathematical entropy and finite propagation speed. At each time step, interface fluxes interpolate between componentwise flux extrema over neighboring-state intervals. Their coefficients are chosen by minimizing the total discrete entropy, translating maximal entropy dissipation into a convex optimization problem. Strict convexity gives a unique updated state when a feasible minimum exists, although the flux coefficients need not be unique; affine constraints can enforce component nonnegativity. The paper establishes a conditional weak-solution result: bounded almost-everywhere convergence of the approximations implies that their limit solves the conservation laws. In one dimension, it presents an entropy-admissibility comparison with bounded-variation solutions constructed by Glimm or wave-front tracking methods. A numerical example for an isentropic Euler system reproduces a rarefaction followed by a shock using an interior-point Newton optimizer. The formulation extends to multiple dimensions, but general convergence is not established, the admissibility comparison is restricted to one dimension, and optimization cost remains a practical limitation.

Contributions

1. Constructs conservative, consistent finite-volume fluxes by optimizing weights between componentwise flux extrema, minimizing discrete entropy at each time step.
2. Distinguishes uniqueness of the entropy-minimizing state from possible nonuniqueness of its flux weights, and incorporates affine feasibility constraints without losing state uniqueness.
3. Applies the Lax–Wendroff framework to show that a bounded, almost-everywhere convergent sequence of numerical approximations has a weak-solution limit.
4. Presents a one-dimensional comparison with Glimm and wave-front tracking solutions, relating a convergent numerical limit to maximal entropy dissipation on a nontrivial time interval under the stated regularity assumptions.
5. Demonstrates the method on an isentropic Euler Riemann problem with pressure $p(\rho) = \rho^2$, recovering the wave speeds and intermediate state of a rarefaction–shock solution.

2026N4

ChatGPT (September 2026)

Summary

A rank-23 scheme multiplies general 3×3 matrices using 52 additions or subtractions in a transformed basis. Its two input linear maps require 12 additions each and its output map 28; conversion into and out of this basis is additional work. For recursive multiplication, the paper proves that these conversions cost $O(n^2 \log n)$, leaving the leading $O(n^{\log_3 23})$ arithmetic term unchanged. The construction searches an existing database of more than 17,000 multiplication schemes over $\mathbb{F}_2$, using randomized invertible basis changes and a greedy heuristic that reuses intermediate sums. Updating only the affected part of the heuristic's potential reduces each candidate-potential evaluation from $O(mn^2)$ to $O(mn)$. A larger search finds 405 binary schemes with at most 55 additions. One 52-addition computation is lifted to signed arithmetic over general fields by solving sign constraints while preserving its computation graph. Explicit formulas and a symbolic verification script support reproducibility. The result reduces an arithmetic operation count; it does not establish optimality or measured matrix-multiplication throughput.

Contributions

1. Gives explicit rank-23 multiplication formulas with a 52-addition transformed-basis core, split into $12 + 12 + 28$ additions.
2. Proves that recursive input and output basis conversions require only $O(n^2 \log n)$ additional arithmetic operations.
3. Reduces candidate-potential evaluation from $O(mn^2)$ to $O(mn)$ by updating only affected column-pair contributions.
4. Combines randomized basis changes with an existing rank-23 database to find 405 binary schemes requiring at most 55 additions.
5. Lifts a 52-addition binary computation to signed general-field arithmetic and supplies explicit basis matrices, search instructions, and symbolic verification code.

2026NBBP

ChatGPT (July 2026)

Summary

This paper uses global gyrokinetic particle-in-cell simulations with ORB5 to determine how turbulence generates radially extended, frequency-coherent zonal structures in realistic TCV and ASDEX Upgrade magnetic configurations. Self-consistent simulations show that ion-temperature-gradient turbulence produces these global structures and identify high toroidal mode numbers as the effective drivers. The authors then extract the spatial structure of selected turbulent modes and impose it through a nonzonal electrostatic antenna, isolating the nonlinear coupling from the full turbulent system. A single $n = 80$ antenna reproduces a global zonal structure in the studied TCV case, demonstrating that generation does not require a statistical superposition of many modes. Frequency scans find a zonal response at twice the antenna frequency, consistent with three-wave coupling to an $n = -80$ sideband. Because lower- n ion-temperature-gradient frequencies approach zero, the mechanism also implies a threshold below which modes cannot drive a response in the geodesic-acoustic-mode frequency range.

Contributions

1. Produced global zonal structures self-consistently in realistic TCV and ASDEX Upgrade gyrokinetic simulations.
2. Identified the high- n ion-temperature-gradient spectrum as the source of the nonlinear excitation.
3. Implemented a realistic nonzonal antenna in ORB5 to isolate the generation mechanism.
4. Reproduced a global zonal response using a single toroidal antenna mode.
5. Connected frequency doubling and the low- n drive threshold to three-wave coupling.

2026NCE

ChatGPT (September 2026)

Summary

This paper extends the Reduced Augmentation Implicit Low-rank (RAIL) integrator from matrix-valued problems to three-dimensional convection–diffusion equations represented by Tucker tensors. Spectral spatial discretization and implicit–explicit Runge–Kutta time stepping first produce fully discrete tensor equations. At every Runge–Kutta stage, bases from earlier stages are augmented, compressed, and updated dimension by dimension; a Galerkin solve then advances the Tucker core. This construction permits rank adaptation and high-order time accuracy while treating diffusion implicitly and retaining low-rank storage. A postprocessing step truncates the tensor without sacrificing prescribed mass, momentum, and energy moments. Analysis distinguishes three-dimensional RAIL from Tucker Basis Update and Galerkin methods, whose inactive modes effectively remain frozen. Tests on linear convection–diffusion, Fokker–Planck dynamics, and viscous Burgers equations recover the designed temporal orders, track evolving ranks, and preserve the targeted moments. The richer augmented spaces improve robustness and accuracy but can cost substantially more per step than Tucker-BUG at higher Runge–Kutta order.

Contributions

1. Extended RAIL integration to third-order Tucker tensors for three-dimensional convection–diffusion equations.
2. Combined rank adaptation with high-order IMEX Runge–Kutta schemes through reduced stage-basis augmentation.
3. Developed dimension-by-dimension basis updates followed by a Galerkin core solve.
4. Preserved mass, momentum, and energy during low-rank truncation through conservative postprocessing.
5. Demonstrated designed accuracy and robustness on convection–diffusion, Fokker–Planck, and viscous Burgers tests.

2026NDLB

ChatGPT (July 2026)

Summary

This paper adapts the open-source EPOCH and Smilei particle-in-cell codes to model metals, semimetals, and nanoscale light–matter interactions as cold quantum plasmas. Four modules are presented: Fermi–Dirac carrier initialization; material-edge boundary conditions for spill-out, confinement, and scattering; bound macroparticles with damped-Lorentz forces for dispersive media; and massless Dirac carriers whose velocity is renormalized to the Fermi velocity. Benchmarks reproduce Fermi–Dirac relaxation trends, nonlocal surface-plasmon shifts, an FDTD Lorentz-medium response, silicon-nanopillar reflectance, persistent radiation from excited bound media, and analytical graphene-disk resonances. Because material components remain particles coupled self-consistently to Maxwell fields, the framework captures time-dependent media and surface dynamics that static dielectric models omit. Important limitations remain: the method is semiclassical, does not enforce Pauli exclusion during evolution, omits unresolved incoherent photon emission, and currently treats Dirac carriers at zero electronic temperature.

Contributions

1. Implemented Fermi–Dirac particle initialization and quantified its resolution-dependent relaxation.
2. Added material-boundary models for electron spill-out, work-function confinement, and surface scattering.
3. Derived a modified Boris pusher for bound macroparticles in Lorentz-dispersive media.
4. Developed a constant-Fermi-speed PIC model and validated graphene-disk resonances.
5. Demonstrated self-consistent coupling with existing PIC physics and identified semiclassical limitations.

2026NHGN

ChatGPT (July 2026)

Summary

PanoMHD is a self-supervised multimodal dynamics model that predicts both global tokamak performance and high-dimensional Mirnov-coil magnetic-fluctuation spectrograms from recent plasma states and control commands. A VQ-VAE discretizes cross-power and cross-phase spectrograms, scalar signals are quantized, and a causal decoder-only Transformer predicts the next 50 ms state from a ten-step context. Training uses 978 KSTAR discharges from 2017–2022, with 101 held out for testing. The 457-million-parameter model attains $R^2 = 0.987$ for normalized pressure and 0.956 for H_{89} , with spectrogram PSNRs of 30.1 and 23.0 dB. Predicted spectra reproduce ELM and tearing-mode signatures together with their associated performance changes. Feeding the predictions to OASIS yields 97.3% L/H classification accuracy, exceeding a dedicated 94.5% baseline without Thomson scattering or D_α input. Results are limited to one device, so multi-machine transfer remains unverified.

Contributions

1. Formulated tokamak prediction as joint generative modeling of scalar performance, controls, and magnetic-fluctuation spectrograms.
2. Developed a VQ-VAE plus causal-Transformer architecture for self-supervised next-state prediction of multimodal plasma signals.
3. Showed that widely available Mirnov-coil and standard control signals can replace costly spatial diagnostics for the evaluated tasks.
4. Improved normalized-pressure prediction and downstream L/H classification on held-out KSTAR shots.
5. Demonstrated physically coherent forecasts of ELMs and tearing modes and quantified the benefit of increasing model size.

2026NN

ChatGPT (July 2026)

Summary

This withdrawn draft attempts to derive a three-field one-fluid closure for a strongly magnetized relativistic pair plasma from a Vlasov–Boltzmann–Maxwell moment hierarchy. It introduces a scalar regulator α by projecting unresolved density, pressure-anisotropy, heat-flux, and irreversible-production moments onto an assumed isolated slow mode. The reduced variables $(\phi, A_\parallel, \alpha)$ are arranged in a GENERIC reversible–irreversible structure, yielding two electromagnetic branches and a relaxing thermodynamic branch; an extended formulation retains nonneutral charge dynamics. The draft also proposes temperature-dependent coefficients and discusses cold, relativistic, and frozen-thermodynamic limits. Its key reduction requires strong anisotropy, a finite moment truncation, a spectral gap, Markovian relaxation, and leading-order linear feedback. The authors withdrew version 2 with the statement “Mathematical Errors” and “Decided not to move forward with concept paper/draft.” Accordingly, the constructions below describe the aims and contents of recoverable version 1, not validated results, and its propositions and claimed thermodynamic consistency should not be relied upon without an independent corrected derivation.

Contributions

1. Attempted a strong-guide-field reduction of relativistic pair-plasma moment equations to $(\phi, A_\parallel, \alpha)$.
2. Defined α as a projected slow combination of four unresolved thermodynamic moments under a spectral-gap assumption.
3. Proposed a GENERIC split between reversible electromagnetic coupling and irreversible relaxation of the regulator.
4. Derived formal electromagnetic and thermodynamic branches and related limiting forms, including a nonneutral whistler–Alfvén subset.

5. Made the closure assumptions explicit enough to identify where a corrected derivation would require renewed mathematical verification.

2026NO

ChatGPT (July 2026)

Summary

The authors study mixed-precision iterative refinement for linear discrete inverse problems, where ill-conditioning and noisy data make an unregularized least-squares solution unstable. They apply refinement to standard Tikhonov regularization and express its iterates recursively through a preconditioned Landweber method with a Tikhonov-type preconditioner. This representation shows that refinement acts as a spectral filter rather than merely correcting a conventional linear solve. The algorithm permits different precisions for the regularized factorization/preconditioner, working updates, and residual evaluation, allowing costly operations to use lower precision while preserving higher precision where error sensitivity is greatest. MATLAB experiments simulate multiple precision triples on test inverse problems, including spectral reconstruction and image deblurring. Across the reported cases, mixed-precision refinement generally reaches stable relative reconstruction errors within a few decimal places of all-double-precision results and outperforms the paper's approximate iterative-refinement benchmark. Lower-precision preconditioners can increase run-to-run variability, especially with weak regularization; stronger regularization reduces that variability but may increase reconstruction error. The results support mixed precision for regularized inverse problems while showing that precision choices and regularization strength must be selected together.

Contributions

1. Formulated mixed-precision iterative refinement directly for Tikhonov-regularized inverse problems.
2. Derived a recursive filtered-solution interpretation using preconditioned Landweber iteration.
3. Separated preconditioner, working, and residual arithmetic into independently chosen precisions.
4. Demonstrated reconstruction accuracy within a few decimal places of double precision in the tested problems.
5. Quantified how low-precision preconditioning and regularization jointly affect solution variability and error.

2026NP

ChatGPT (July 2026)

Summary

This paper generalizes gyrokinetic secondary-instability theory to retain toroidal geometry, especially the radial magnetic drift and associated Stringer–Winsor force. The resulting dispersion relations unify three branches of zonal-flow-generating secondary modes: the conventional Rogers–Dorland–Kotschenreuther mode, localized ion-temperature-gradient secondary modes, and the toroidal secondary mode (TSM). The TSM is a growing zonal flow that propagates radially because zonal shearing and magnetic-drift advection create an up–down pressure asymmetry. Gyrokinetic simulations support the predicted TSM frequencies, growth, propagation, and radial-wavenumber threshold. In strongly driven tokamak ion-temperature-gradient turbulence, parallel streaming sets this threshold near $k_x \rho_i \simeq 1/(2q)$, while the TSM dominates localized secondary modes at sufficiently strong drive. The theory applies to arbitrary magnetic geometry and helps explain the short-radial-wavelength zonal structures that contribute to turbulence saturation.

Contributions

1. Generalized gyrokinetic secondary-mode theory to include toroidal radial magnetic drifts and the Stringer–Winsor force.
2. Derived dispersion relations connecting toroidal, ion-temperature-gradient, and Rogers–Dorland–Kotschenreuther secondary modes.
3. Identified how shearing and drift advection generate the pressure asymmetry that drives propagating toroidal secondary modes.

4. Quantified parameter regimes and radial-wavenumber thresholds for competing secondary instabilities.
5. Validated the toroidal-secondary-mode predictions against gyrokinetic simulations of tokamak turbulence.

2026NPBM

ChatGPT (July 2026)

Summary

Strongly driven ion-temperature-gradient turbulence in tokamaks is shown to saturate through a propagating zonal-flow mode termed the toroidal secondary mode (TSM). Nonlinear turbulence creates an up-down pressure asymmetry; together with zonal shearing and magnetic-drift advection, this drives small-scale zonal flows that propagate radially. Above a fluctuation-amplitude threshold, TSMs become unstable and shear turbulent eddies, returning the system toward marginality. Free-energy transfer diagnostics show these flows moving energy from large injection scales toward small dissipative scales. Incorporating TSM marginality into critical-balance arguments replaces perpendicular isotropy with anisotropic outer scales and produces scaling laws for heat flux, fluctuation amplitudes and spectra, correlation lengths, and zonal-flow energy. Nonlinear gyrokinetic simulations with both *stella* and *GX* validate the parameter scalings, including heat flux approximately linear in the normalized temperature gradient rather than the previously predicted cubic dependence. Rapid TSM oscillations still regulate turbulence because their temporally decorrelated advection produces diffusive cross-field displacement.

Contributions

1. Identified the toroidal secondary mode as a radially propagating zonal flow in tokamak turbulence.
2. Explained its drive through nonlinear pressure asymmetry, zonal shearing, and magnetic drift.
3. Demonstrated threshold regulation of turbulence and small-scale free-energy transfer by TSMs.
4. Derived anisotropic critical-balance scalings for transport, spectra, correlations, and zonal-flow energy.
5. Validated the scalings with independent *stella* and *GX* gyrokinetic simulations.

2026NPSB

ChatGPT (September 2026)

Summary

Stiff chemical kinetics makes repeated thermochemical time integration costly. AMORE trains one multi-output neural operator to predict temperature and species histories from initial conditions. Its DeepONet variants use either residual networks or a Kolmogorov–Arnold network in the trunk, with a partition-of-unity construction that constrains basis functions. Two gradient-free adaptive losses reweight variables and samples using previous-epoch errors. The authors also compare one-step and two-step training and several ways to enforce species mass-fraction sums, including an invertible map to $n - 1$ coordinates and a softmax output. Tests on syngas with 12 states and GRI-Mech 3.0 with 24 active states show improved prediction from adaptive weighting; the Kolmogorov–Arnold trunk yields lower error statistics in the tested cases. Two-step training improves results only marginally while taking longer. Exact unit-sum constraints need adjustment when only a subset of species is modeled.

Contributions

1. Built a single DeepONet architecture for multiple thermochemical output trajectories.
2. Introduced variable- and sample-adaptive losses requiring no loss-weight gradients.
3. Compared residual-network and Kolmogorov–Arnold trunk designs with partition-of-unity basis functions.
4. Evaluated analytical-coordinate and softmax constructions for exact mass-fraction normalization.
5. Benchmarked single- and two-step training on syngas and GRI-Mech 3.0 kinetics.

2026OAGT

ChatGPT (September 2026)

Summary

Quasi-stationary distributions describe the long-lived behavior of stochastic processes conditioned on avoiding absorption. This paper extends two numerical approaches for computing them. Its deterministic iteration solves the nonlinear quasi-stationary equation for general Markov processes with discrete or continuous state spaces and one or more absorbing states. Its Monte Carlo alternative follows a single trajectory and, after absorption, resets it using states sampled from that trajectory’s history. Tests span branching, immigration-death, random-walk, Ornstein–Uhlenbeck, circular-diffusion, SIRS, and SIS models, including cases with analytical solutions. The iterative method is usually more accurate in less time, resolves extremely small probabilities, and estimates rare-event extinction times, while the history-reset Monte Carlo method can retain bias and convergence sensitivity. Monte Carlo is nevertheless easier and sometimes preferable for multidimensional domains with complicated boundaries. Parameter studies show that mild overrelaxation and a concentrated initial distribution accelerate iteration, while spatial and temporal discretization choices impose different accuracy-cost tradeoffs in continuous problems. The comparison therefore provides both generalized algorithms and practical criteria for selecting and tuning them.

Contributions

1. Generalized nonlinear fixed-point iteration to continuous or discrete Markov processes with multiple absorbing states.
2. Introduced a single-trajectory Monte Carlo reset mechanism based on the trajectory’s empirical history.
3. Benchmarked both methods across one- and two-dimensional models with natural and artificial absorbing boundaries.
4. Demonstrated the iterative method’s advantage for high accuracy and rare-event probabilities while identifying Monte Carlo bias and convergence failures.
5. Quantified how relaxation, initialization, grid spacing, timestep, system size, and boundary complexity affect efficiency and method choice.

2026OC

ChatGPT (July 2026)

Summary

The paper develops a unifying description of Hamiltonian fluid closures for the one-dimensional Vlasov–Poisson equation. Starting from the moment hierarchy, it uses hydrodynamic Poisson brackets and their Casimir invariants to identify normal variables that simplify both the bracket and the closure relations. In these variables, every examined Hamiltonian closure—including multi-delta, waterbag, derivative-of-delta, Burby, and four-field constructions—has a polynomial parameterization: the normalized moment μ_n is a polynomial of degree $n + 1$. A recurrence shows that the entire closure is uniquely generated by the single moment μ_2 , a homogeneous cubic polynomial in $N - 2$ normal variables. Thus one generator specifies both the reduced Poisson bracket and Hamiltonian for an arbitrary number of fluid variables. The normal form also decomposes the bracket into a direct sum, suggesting split integrators that preserve individual Casimir invariants. Not every cubic generator corresponds to a valid Vlasov reduction, but the representation provides a compact search space for data-driven closure discovery.

Contributions

1. Derived normal fluid variables systematically from Casimir invariants of hydrodynamic Poisson brackets.
2. Reduced higher-moment closure relations to a recurrence generated solely by μ_2 .
3. Proved a common polynomial structure across the examined Hamiltonian Vlasov closures.
4. Connected multi-delta, waterbag, Burby, and four-field closures within one framework.
5. Exposed a direct-sum bracket useful for Casimir-preserving split integration and constrained data-driven discovery.

Summary

Electrostatic particle-in-cell simulations solve a Poisson equation each time step, where sheaths create localized steep gradients that make uniformly high-order discretizations wasteful. The authors extend the hybridizable discontinuous Galerkin spectral element (HDG-SEM) Poisson solver of PICLas from one polynomial degree to element-local degrees on curved hexahedra. In this background method, eliminating the element unknowns (static condensation) of a first-order system for the potential φ and displacement $\mathbf{D} = \varepsilon \mathbf{E}$ leaves a global system for the face potential λ . Here each face takes the larger degree of its two elements, and particle charges are projected onto the basis without shape functions. Between runs, degrees are updated by Legendre modal analysis of time-averaged potential and charge densities, with density thresholds scaled by an estimated sampling variance. Prescribed distributions match uniform-degree errors with fewer unknowns for a dielectric sphere and a plasma sheath; in an axisymmetric ion optic, indicator-driven degrees up to 5 between the grids reproduce the uniform-degree-5 axial ion-density profile to 4.6% (relative L^2) in a final run 1.276 times faster. The authors note that the threshold multipliers are empirical, that sparse particles near the axis inflate the variance estimate and block refinement there, and that conditioning under nonuniform degrees is unexamined.

Contributions

1. Following Cockburn et al., took the numerical traces $(\mathbf{D}^* \cdot \mathbf{n})|_K = (\mathbf{D} \cdot \mathbf{n})|_K + \tau_K(\varphi - \varphi^*)$ with $\tau_K = 2|K|^{-1/3}$, multiplied the weak form of $\nabla \cdot \mathbf{D} = \rho$ by -1 to obtain symmetric element matrices, and condensed them into trace blocks $\mathcal{S}_{F,F'}^K = \tilde{\mathcal{E}}_{F,K} \tilde{\mathcal{D}}_K^{-1} \tilde{\mathcal{E}}_{F',K}^T + \mathcal{C}_{F,K} \mathcal{A}_K^{-1} \mathcal{C}_{F',K}^T - \delta_{F,F'} \mathcal{F}_{F,F}^K$ with $\tilde{\mathcal{D}}_K = \mathcal{D}_K - \mathcal{B}_K \mathcal{A}_K^{-1} \mathcal{B}_K^T$, whose sizes and entries but not coupling pattern change with the local degrees, stating that for $\varepsilon > 0$, a non-degenerate mesh, and $\tau_K > 0$ the element systems stay nonsingular for any degree distribution and the trace matrix is symmetric positive definite for a nonempty Dirichlet boundary.
2. Evaluated integrals by Gauss quadrature at the Legendre–Gauss interpolation nodes, making the $\mathbf{D}$ mass matrix block diagonal with a 3×3 block $\omega_i \omega_j \omega_k \mathcal{J}_{ijk}^T \mathcal{J}_{ijk} / (\varepsilon J_{ijk})$ per node and, since $p(F) \geq p(K)$, adding no face quadrature error, and implemented variable degrees with derived element and face types stored in one-dimensional arrays, a global side index and rotation (one of five orientations) per local face, only $\tilde{\mathcal{E}}_K$ and $\tilde{\mathcal{D}}_K$ precomputed, and PETSc solves by Cholesky factorization or BoomerAMG-preconditioned conjugate gradients.
3. Nested the leapfrog PIC time loop in an outer loop of runs (Algorithm 1), each run using fixed degrees and time-averaging φ and species densities, after which the indicator is evaluated separately along each coordinate and an element’s degree may rise by one, stay, or fall to any lower value; a density mode counts as noise when $\hat{\rho}_k^2 < \varepsilon \sigma_k^2$, with $\sigma_k^2 = |K|^2 \Sigma_{kk} / N_p$ from the single-particle covariance $\Sigma = \mathbf{M}^{-1} (\int_{-1}^1 \psi \psi^T \bar{f} d\xi) \mathbf{M}^{-T} - \mu \mu^T$ of the modal coefficients evaluated for a uniform $f = 1/|K|$, and with the radial Jacobian $J_r = \iint 2\pi r J d\eta d\zeta$ in axisymmetry.
4. For the dielectric sphere ($\varepsilon_r = 10$, $R = 1$ m, analytic Dirichlet data at radius $2R$, $p_{\max} = 2$ to 5 on three meshes), prescribed degree 2 in elements inside the sphere not touching its surface, because the nonconstant Jacobian of the trilinear interior elements causes integration errors at degree 1; for the sheath ($L = 0.03$ m, particle weight $w_p = 10^6$, $\Delta t = 10^{-8}$ s), degree 4 in the two wall elements and 1 elsewhere (14 DOFs) gave a relative L^2 error of 4.520×10^{-3} , against 2.775×10^{-1} for uniform degree 1 (8 DOFs), 5.308×10^{-3} for uniform degree 4 (20 DOFs), and 6.106×10^{-3} for 32 linear elements (64 DOFs).
5. In the ion optic (10^{17} m^{-3} xenon plasma, grids at 1300 V and -300 V, $\Delta t = 10^{-11}$ s $\approx 0.2/\omega_e$, averages over 10^5 steps), set $\varepsilon_\varphi = 10$ for the potential and $200^2 \sigma_k^2$ and $60^2 \sigma_k^2$ for the ion and electron densities instead of the theoretical $(3\sigma_k)^2$, offsetting the temporal correlation of particles sampled repeatedly in one element; the relative Xe^+ density difference from uniform degree 5 along $r = 0.2$ mm was 2.254×10^{-1} at $p_{\max} = 1$, 2.734×10^{-1} at $p_{\max} = 2$, and 4.629×10^{-2} at $p_{\max} = 5$, with wall-clock times over the averaging interval of 2458 to 2870 s on 40 AMD EPYC Milan cores against 3663 s for uniform degree 5.

Comments

- Fig. 1: the colour bar is labelled “ $|E|/\text{V}$ ”, but Section 5.1 gives the field in V m^{-1} ($E_z = -1 \text{ V m}^{-1}$), and the bar, from about 0.25 (unlabelled lower end) to 2.5, spans the minimum and maximum of the analytic field magnitude

from Eq. (54) in V m^{-1} , $3|E_z|/(\epsilon_r + 2) = 0.25$ inside the sphere and $(1 + 2(\epsilon_r - 1)/(\epsilon_r + 2))|E_z| = 2.5$ just outside its poles; the evident intended unit is V m^{-1} .

2026ODDE

ChatGPT (July 2026)

Summary

STORM models three-dimensional turbulence and filament dynamics across a tokamak scrape-off layer, where turbulent transport controls plasma exhaust loads on material surfaces. It solves a drift-reduced, cold-ion, collisional fluid system in BOUT++ using a flux-surface-aligned grid, mostly finite differences, and field-aligned parallel derivatives. Parallel fluxes are placed on a staggered grid to suppress checkerboard modes, while a fully implicit, matrix-free CVODE integrator accommodates fast electron thermal conduction. The paper provides a consolidated specification of the plasma equations, Boussinesq options, sheath and radial boundary conditions, elliptic inversions, initialization procedures, sources, and synthetic probe diagnostics. It also documents BOUT++ version-4 improvements developed for STORM: geometric quantities can be represented consistently at cell faces, parallel transforms account for field location, and field-aligned derivative machinery supports wider stencils. These changes remove a prior staggered-geometry inconsistency that could restrict spatial convergence. The implementation supports turbulence and seeded-filament studies in slab and realistic tokamak geometries, but its collisional closure limits use in low-collisionality regimes, and electromagnetic operation in tokamak geometry remains unresolved.

Contributions

1. Presented a complete, reproducible specification of the three-dimensional STORM scrape-off-layer turbulence model and its BOUT++ implementation.
2. Established a staggered parallel-grid discretization that suppresses grid-scale checkerboard instabilities in the fastest plasma dynamics.
3. Upgraded BOUT++ geometry and parallel-transform infrastructure so staggered variables use consistent cell-face metrics and field-aligned derivatives.
4. Integrated practical numerical components for full simulations, including elliptic solvers, implicit CVODE time integration, sheath and radial boundaries, initialization, and source models.
5. Added simulation-facing diagnostics and workflows for both statistically steady turbulence and seeded filaments, including high-frequency synthetic Langmuir-probe output.

2026OFL

ChatGPT (July 2026)

Summary

This paper adapts B-trees to embedded devices with single-threaded processors, roughly 4–64 KB of RAM, and flash storage that may lack a file system or flash translation layer. Its VMTree uses a small memory-resident table of virtual page redirects so updated nodes can be written sequentially without propagating physical-address changes from leaf to root. VMTree-OW instead exploits NOR or DataFlash page-overwrite constraints with an append-within-page layout, eliminating the mapping table. Both designs include free-space management and recovery, and can combine a small operation log with batched insertion. Experiments on 16-bit PIC and 32-bit ARM boards compare NAND, DataFlash, and SD cards using random, environmental, and wearable-sensor data. VMTree runs efficiently on raw NAND using about 3–4 KB of memory; overwrite-aware pages deliver roughly $4\times$ speedup on DataFlash. One-page write buffers improve clustered sensor insertion by $3\text{--}5\times$ overall and by up to $9\times$ in a reported platform/data combination. Results also show when conventional B-trees remain preferable and how scarce RAM should be divided between path caching and write buffering.

Contributions

1. Introduced bounded virtual page mappings that suppress B-tree write amplification on raw flash without a full translation layer.
2. Developed VMTree for sequential writes to raw NAND and VMTree-OW for overwrite-capable flash pages.
3. Supplied low-memory free-space management and recovery mechanisms for embedded deployments.
4. Evaluated storage-specific variants on two microcontroller platforms, three storage types, and multiple data distributions.
5. Quantified how page buffers and operation write buffers should share scarce RAM for read and insertion performance.

2026OMCL

ChatGPT (July 2026)

Summary

This paper predicts ITER 15 MA baseline core transport under neon-seeded divertor constraints by coupling SOLPS-ITER power-exhaust targets with OMFIT STEP core modeling. TGYRO, using TGLF and NEO transport models, computes stationary profiles across $1.5 \leq Z_{\text{eff}} \leq 2.5$, while separatrix power P_{sep} is compared with no-ELM neon-seeded divertor solutions. The scan finds that modest impurity-content changes shift P_{sep} by more than 50% and identify a compatibility window near $Z_{\text{eff}} \approx 1.6\text{--}1.75$. Within this window, core performance can remain consistent with $Q_{\text{fus}} \gtrsim 10$ while matching roughly 100 MW divertor-compatible power exhaust. Reducing auxiliary heating to about 75% of nominal can produce similar power flow at $Z_{\text{eff}} = 1.6$. Rotation scans change P_{sep} by less than about 20%, and AURORA modeling indicates charge-exchange radiation inside the separatrix is negligible. The study provides model-constrained guidance for ITER impurity seeding and heating schedules.

Contributions

1. Coupled ITER core transport prediction to SOLPS-ITER neon-seeded divertor power constraints.
2. Scanned impurity content and auxiliary heating in OMFIT STEP/TGYRO baseline scenarios.
3. Identified a narrow Z_{eff} and heating window compatible with fusion gain and exhaust limits.
4. Quantified the modest sensitivity of separatrix power to plausible toroidal rotation changes.
5. Used AURORA to show that core charge-exchange radiation is negligible for the predicted neutral densities.

2026OPK

ChatGPT (July 2026)

Summary

The paper formalizes quadratization for spatially one-dimensional autonomous partial differential equations (PDEs): introducing auxiliary variables so every evolution equation becomes at most quadratic in the original variables, their spatial derivatives, and the auxiliaries. It proves that every polynomial PDE of spatial differential order h , under sufficient regularity, admits a monomial quadratization of differential order $3h$, while finding a minimum-order monomial quadratization is NP-hard. The authors then introduce QuPDE, a symbolic algorithm for polynomial or rational PDE formulas. It enumerates monomial decompositions, verifies candidate lifts, searches with branch and bound, and polynomializes rational expressions. Pruning and ordering heuristics make the combinatorial search practical. Tests on fourteen PDE models from fluid mechanics, space physics, chemical engineering, and biology produce low-order lifts in every case. Several lifts use fewer auxiliaries than earlier transformations, while other examples are quadratized computationally for the first time. The work concerns symbolic reformulation rather than well-posedness or numerical solution of the lifted boundary-value problems.

Contributions

1. Gave a rigorous definition of PDE quadratization, including quadratization order and differential order.
2. Proved existence of a monomial quadratization of differential order $3h$ for polynomial PDEs of order h .
3. Proved that optimal monomial PDE quadratization is NP-hard by reduction from the ordinary-differential-equation problem.
4. Developed QuPDE by combining symbolic decomposition, candidate verification, branch-and-bound search, and rational polynomialization.
5. Demonstrated low-order quadratizations on fourteen PDE benchmarks, including improved and previously unknown lifts.

2026OTWH

Claude (September 2026)

Summary

Kinetic distribution functions are expensive to store, and moment closures require recovering a distribution from finitely many moments. The authors represent a velocity distribution by nodal values on a uniform tensor grid and propose two moment-aware reconstructions. Sparse entropic quadrature (SPEQ) extends entropic quadrature by writing $f = wg$ with a prior weight w , either a discrete Maxwellian fitted to low-order moments or known data, and minimizing discrete entropy, or Kullback–Leibler divergence to a known distribution, plus $\lambda\|g\|_1$ subject to the moment constraints; the penalty acts on g because nonnegativity makes $\|f\|_1$ the conserved density. The concave dual is solved by Newton’s method, and values of g below a threshold are zeroed. Low-rank moment recovery (LRMR) minimizes the row-scaled moment residual over nonnegative tensors of Tucker rank r by projected gradient descent, without a guarantee of global minimization. On Maxwell–Boltzmann, Druyvesteyn, and bimodal Mott–Smith distributions, LRMR needs about an order of magnitude fewer degrees of freedom than SPEQ, and strong sparsity hollows out the bulk. On NuFI simulations of a Vlasov–Maxwell beam instability, divergence-based SPEQ retains filaments at high sparsity, whereas entropy-based SPEQ and LRMR lose fine detail, and the authors conclude that LRMR needs additional priors.

Contributions

1. Derived the dual of the weighted, ℓ_1 -regularized divergence problem, with primal recovery $g_i = (\phi_i/w_i) \exp(-1 - (\lambda + s_i)/(\Delta v_i w_i))$, $s = \hat{A}_m^\top y$, a Hessian $\hat{A}_m H \hat{A}_m^\top$ with diagonal H , a least-squares initial guess aimed at $g \equiv 1$, and row normalization of the moment matrix to improve conditioning.
2. Constructed the Maxwellian weight directly on the discrete grid by solving a scalar Newton equation for the temperature that reproduces the discrete energy, because the discretization breaks the continuous relation $e = \frac{3}{2}T$.
3. Showed that LRMR recovers the Maxwell–Boltzmann distribution to relative L_2 errors near 10^{-14} at every rank from 1, about 0.2% of the 40^3 nodal values, as its separable Tucker rank $(1, 1, 1)$ implies, while both methods keep the odd next-order moments of the symmetric test distributions near zero.
4. Found for the Mott–Smith case that entropy-based SPEQ errors plateau as sparsity decreases, that constraining moments up to order 6 rather than 4 lowers both the distribution and next-moment errors, and that the cutoff ε changes sparsity at small λ but hardly changes the distribution error.
5. Reported on the NuFI data that divergence-based SPEQ errors grow at later times as filamentation develops, that both SPEQ variants give sparser representations at earlier times, and that LRMR errors barely change when moments up to order 10 are constrained.

Comments

- §3.1, Eqs. (37)–(38): setting $g_i \equiv 1$ and $\lambda = 0$ in (30)–(32), as the text states, gives $s_i = \Delta v_i w_i (\log(\phi_i/w_i) - 1) = r_i$, that is, $\hat{A}_m^\top y_0 = r$, whose normal equations are $\hat{A}_m \hat{A}_m^\top y_0 = +\hat{A}_m r$, but (37) and step 6 of Algorithms 1 and 2 print $-\hat{A}_m r$; in Algorithm 2 ($w = \phi$) the printed system yields $s = -r$ and $g_i = e^{-2}$ rather than $g_i \equiv 1$, so the intended sign is $+$.

- §3.1, step 2 of the weighting procedure: the energy is printed as $e = (M_{2,0,0}/M_{0,0,0} - \bar{v}_x)^2 + (M_{0,2,0}/M_{0,0,0} - \bar{v}_y)^2 + (M_{0,0,2}/M_{0,0,0} - \bar{v}_z)^2$, which subtracts a velocity from a squared velocity, whereas Eq. (27) and step 3 ($T_0 = \frac{2}{3}e$) use e as the mean of $\|v - \bar{v}\|^2$; the intended terms are $M_{2,0,0}/M_{0,0,0} - \bar{v}_x^2$ and its y and z analogues.
- §4.1.2, Eqs. (48)–(49): $\alpha = (2\Gamma(5/4)/(3T\Gamma(3/4)))^2$ is exactly the coefficient for which $\exp(-\alpha\|v\|^4)$ has $\langle\|v\|^2\rangle = \frac{3}{2}T$, the relation that §3.1 uses with $\exp(-\|v\|^2/T)$, but (48) divides the exponent by a further T , so it scales as $T^{-3}\|v\|^4$ and gives $\langle\|v\|^2\rangle = \frac{3}{2}T^{3/2}$; the extra $1/T$ has no effect for the $T = 1$ used, and the intended exponent is $-\alpha\|v\|^4$.
- Fig. 1 and Fig. 2: the Fig. 2 panel for $\tilde{\lambda} = 0.1$ ($m_{\max} = 6$, “% of D.o.F = 9.6”) shows the four central nodes of slice $k = 20$ at no more than about 0.02 where the Maxwell–Boltzmann truth is 0.174, which alone implies a relative L_2 error of at least 0.10 on the 40^3 grid, but Fig. 1 plots about 0.018 for the $m_{\max} = 6$ point at 9.6%; what the plotted $\|f_{\text{pred}} - f_{\text{true}}\|_2 / \|f_{\text{true}}\|_2$ values measure is unresolved.
- Fig. 3 and Fig. 5: Fig. 5 shows the rank-1 LRMR reconstruction ($m_{\max} = 6$, 0.2% of D.o.F.) at about 0.17, and at least 0.15, on the four central nodes of slice $k = 20$, where the Druyvesteyn truth is 0.090, and §4.1.2 says LRMR “predicts larger values of the distribution”, which alone implies a relative L_2 error of at least 0.046, but Fig. 3 plots about 5×10^{-3} for this point; unresolved.
- Fig. 9 caption: “reconstructed distributions for various values of λ ”, but the panels are labeled “LRMR, $r = 1$ ”, “ $r = 3$ ”, and “ $r = 5$ ”, and §4.1.3 says Fig. 9 shows “the same slices of the distributions obtained with the LRMR method”; the intended caption is “for various ranks r ”, as in Fig. 5.
- §4.2: Fig. 12 is said to show the Kullback–Leibler variant “for the same values of λ and the same slice as shown on Figure 11”, but the Fig. 11 panels use $\tilde{\lambda} = 10^{-8}, 10^{-4}, 10^{-2}$ and the Fig. 12 panels use $\tilde{\lambda} = 10^{-8}, 10^{-2}, 1$; only the slice ($k = 40, t = 300$) and $\tilde{\lambda} = 10^{-8}, 10^{-2}$ are shared.
- Figs. 16–18 captions: “absolute error in next predicted moments (right)”, but the right-hand axes are labeled $E_{m_{\max}+1}^{\text{rel}}$ and §4.2 says these figures show “the relative errors in the predicted moments”; the intended word is “relative”.

2026OUZJ

ChatGPT (July 2026)

Summary

The full- f , electromagnetic, collisional gyrokinetic code GENE-X is coupled to a one-moment fluid-neutral model for edge and scrape-off-layer studies. The neutral density evolves through a diffusion–advection equation, with charge exchange represented diffusively and neutral velocity and temperature prescribed from ion moments. For electrons, ions, and atomic neutrals, the work derives Krook-like ionization and three-body-recombination operators, maps them heuristically into gyrocenter coordinates, and uses AMJUEL reaction and cooling rates. Mixing parameters enforce conservation of mass, nuclei, free-plus-bound electrons, momentum, and kinetic energy in the conservative operator; a separate electron term accounts for binding-energy and radiative losses without changing density or momentum. Velocity-space refinement shows at least fourth-order convergence and near-machine-precision conservation, limited chiefly by parallel-velocity quadrature. Relaxation tests show that atomic reactions cool electrons, alter densities, and accelerate isotropization relative to Coulomb collisions alone. In a divertor-like case, newly ionized cold, slow particles create skewed, non-Maxwellian distributions and conductive heat fluxes comparable to standard fluid estimates. The tests establish implementation consistency but stop short of nonlinear turbulence validation, which remains necessary for assessing macroscopic transport effects.

Contributions

1. Added a tightly coupled gyrokinetic-plasma/fluid-neutral framework to GENE-X.
2. Constructed gyrocenter Krook-like operators for ionization and recombination in a three-species hydrogenic system.
3. Enforced particle, momentum, and conservative-energy constraints while separately representing atomic binding and radiative losses.
4. Verified the implemented conservation laws through velocity-space convergence to near machine precision.

5. Demonstrated neutrals-driven isotropization, non-Maxwellian distortion, skewness, and heat flux in relaxation tests.

2026P

ChatGPT (July 2026)

Summary

Many hyperbolic systems implicitly preserve the curl or divergence of a vector field, but ordinary discretizations can destroy these differential constraints. This paper proves that a classical, fully discontinuous and collocated discontinuous Galerkin (DG) method can preserve them exactly when it uses vector approximation spaces compatible with a discrete de Rham complex and a suitably directed numerical-flux diffusion. Curl preservation requires diffusion parallel to face normals, whereas divergence preservation requires orthogonal diffusion. The conserved discrete quantities are defined through adjoint curl and divergence operators, which fit naturally into the weak Galerkin formulation. The framework also provides potential-based initialization of constraint-satisfying fields and, for the induction equation, a discrete transport equation for the adjoint divergence. Tests on two-dimensional Maxwell, wave, and induction systems over Cartesian, triangular, and quadrilateral meshes confirm machine-precision constraint preservation and high-order accuracy. The experiments cover linear systems; extending stable, limited schemes to nonlinear problems remains open.

Contributions

1. Proved exact DG preservation of adjoint curl and divergence constraints under explicit space-and-flux conditions.
2. Connected the required vector approximation spaces and discrete operators through a de Rham framework.
3. Identified the numerical-flux diffusion directions needed for curl and divergence preservation.
4. Derived constraint-compatible initialization and an adjoint-divergence transport equation for induction.
5. Verified sharpness, high-order accuracy, and machine-precision preservation on three linear hyperbolic systems.

2026P2

ChatGPT (September 2026)

Summary

Pavlov presents moljax, an open-source JAX method-of-lines library for stiff reaction–diffusion equations on structured grids. It combines GPU-compiled adaptive stepping, matrix-free Newton–Krylov solves with automatic-differentiation Jacobian–vector products, and Fourier-based operators and preconditioners for constant-coefficient diffusion. Its integrators include explicit Runge–Kutta, Crank–Nicolson, implicit–explicit splitting, and exponential time differencing. Controlled benchmarks on Gray–Scott, Schnakenberg, and Brusselator systems find the splitting and exponential methods 10–17 times faster than explicit RK4 on the same GPU and spatial discretization; only ETDRK4 shows monotone pointwise convergence in the reported pattern-forming tests. The FFT preconditioner reduces GMRES iterations by as much as 509-fold in diffusion-dominated cases. A tubular-reactor comparison reports an 18–40-fold speedup over DiffraX for the tested settings. FFT diagonalization assumes constant diffusion on structured grids, and initial compilation adds overhead.

Contributions

1. Implemented JIT-compatible adaptive step acceptance and rejection on GPU using JAX control flow.
2. Built matrix-free Newton–Krylov solvers with automatic-differentiation Jacobian–vector products.
3. Used FFT, sine, and cosine transforms for diffusion solves and physics-based preconditioning.
4. Compared explicit, implicit, splitting, and exponential integrators under matched benchmark conditions.
5. Released software and reproduction scripts with performance decompositions and a reactor sensitivity example.

2026PBL

ChatGPT (September 2026)

Summary

Fourier coefficients of magnetic perturbations in a tokamak depend on the chosen flux-surface coordinates, which can make estimates of coupling between external fields and resonant plasma responses inconsistent. This paper proves that the coupling matrix has coordinate-invariant singular values when the external vacuum field on the control surface uses a square-root-area Fourier weight and the resonant field on each rational surface uses a full-area weight. Under this pairing, right singular vectors reconstruct the same physical field pattern across coordinate systems. The authors define a dimensionless dominant-mode overlap using the weighted external field and toroidal-field normalization. Generalized Perturbed Equilibrium Code calculations in PEST, Boozer, and Hamada coordinates confirm the analysis. With other weights, combined mode overlaps can differ by a factor of roughly 2–3 for strongly shaped, low-aspect-ratio equilibria; the discrepancy is smaller at high aspect ratio. The result gives a precise weighting prescription for resonant-field metrics used in tokamak coil design and error-field correction.

Contributions

1. Proved that square-root-area weighting makes the external-field Fourier vector transform unitarily between flux-surface coordinates.
2. Paired that vector with full-area-weighted resonant coefficients to obtain coordinate-invariant coupling-matrix singular values and physical modes.
3. Specified a dimensionless, coordinate-invariant dominant-mode overlap normalized by the toroidal magnetic field.
4. Confirmed the invariance numerically across PEST, Boozer, and Hamada coordinates using GPEC calculations.
5. Quantified errors from alternative weights across an aspect-ratio and shaping scan, reaching roughly a factor of 2–3 in extreme cases.

2026PDD

ChatGPT (July 2026)

Summary

This paper models ion-acoustic shock waves in a magnetized, dissipative plasma containing inertial nondegenerate positive and negative ions together with inertialess relativistically degenerate electrons and positrons. Starting from normalized fluid equations, the authors apply reductive perturbation theory to derive a Korteweg-de Vries-Burgers equation for weakly nonlinear propagation. Its travelling-wave solution supports only positive-potential, compressive shocks for both acoustic modes considered. Parameter scans show that ion-density ratios, electron and positron fractions, the positive-to-negative ion temperature ratio, magnetic-field strength, and parallel and perpendicular kinematic and bulk viscosities alter shock amplitude and steepness. In the reported regime, stronger dissipation enhances the oscillatory shock structure, while larger temperature and magnetic-field parameters raise the shock potential. The evolution equation is also recast as a planar dynamical system; its equilibria include a saddle at the origin and a centre at the nonzero fixed point for the selected parameters. The model targets nonlinear electrostatic structures in dense environments such as white dwarfs and neutron stars.

Contributions

1. Formulated a four-species magnetized plasma model with nondegenerate pair ions and relativistically degenerate electron-positron populations.
2. Derived the Korteweg-de Vries-Burgers evolution equation and its shock solution by reductive perturbation theory.
3. Established positive-potential compressive ion-acoustic shocks for both acoustic modes in the studied regime.
4. Quantified how density ratios, temperature ratio, magnetic field, and viscous dissipation control shock structure.
5. Recast the evolution equation as a planar dynamical system and classified its saddle and centre equilibria.

2026PJCD

ChatGPT (July 2026)

Summary

This paper applies deflation to find physically distinct stellarator equilibria and optimization minima from a common initial guess. Known solutions are excluded either by multiplying equilibrium residuals by inverse-distance barriers or by adding nonlinear distance constraints to optimization. In DESC, repeated deflation finds 25 first-order near-axis-constrained quasi-axisymmetric equilibria sharing the same magnetic axis and on-axis transform but differing in shear, outer geometry, and magnetic-well behavior. It also recovers a helical-core branch from an axisymmetric starting point without prescribing the helical perturbation. For stage-one quasi-helical optimization, 18 of 30 runs pass transform and force-balance filters, although coordinate and toroidal-shift degeneracies expose limits of coefficient-space distance. A proposed sum-reduction operator prevents multiple deflations from unintentionally shrinking earlier exclusion regions. Stage-two optimization then finds six coil sets, five satisfying the stated engineering and normal-field constraints. The results establish deflation as a practical landscape-exploration tool while motivating more invariant distance metrics and robust least-squares formulations.

Contributions

1. Adapted residual and constraint forms of deflation to non-axisymmetric MHD equilibrium and stellarator optimization.
2. Found 25 near-axis-constrained quasi-axisymmetric equilibria with shared core constraints but diverse global geometry and profiles.
3. Recovered a helical-core equilibrium from an axisymmetric initial state without a prescient helical perturbation.
4. Produced multiple attractive stage-one quasi-helical equilibria and diagnosed coordinate-representation and toroidal-shift degeneracies.
5. Introduced sum-reduction multiple deflation and used it to obtain several distinct, mostly feasible stage-two coil sets.

2026PLBE

ChatGPT (September 2026)

Summary

This study investigates the origin and divertor energy loads of small edge-localised modes (ELMs), intermittent edge-plasma instabilities, in high-density tokamak operation. DIII-D measurements show that these events remove approximately 0.7% of stored plasma energy while weakly disturbing the edge pressure pedestal. Conditional averages, scrape-off-layer probes, and BOUT++ fluid simulations place their origin near the pedestal foot and separatrix, the boundary between closed and open magnetic field lines. Replacing pedestal-top pressure with pressure from this region in the Eich energy-fluence model brings predictions into agreement with DIII-D and ASDEX Upgrade measurements to approximately 40% on average. First-order extrapolations give peak parallel energy fluences of 0.7 MJ m^{-2} for SPARC and $0.43\text{--}2.3 \text{ MJ m}^{-2}$ for ITER, depending on the assumed origin. These estimates lie below the nominal tungsten-surface melting threshold used in the study. They do not establish divertor survival, which also depends on magnetic incidence angle, exposed leading edges, background heat loads, and extrapolation uncertainty.

Contributions

1. Quantifies average small-ELM stored-energy losses of approximately 0.7% in DIII-D, with roughly 80% of the released energy accounted for at the divertor targets.
2. Combines pedestal-profile changes, near-separatrix density fluctuations, and extrapolated probe pressures to locate small-ELM filament origins at normalized poloidal flux $\Psi_n \approx 0.99\text{--}1.0$.
3. Identifies high-toroidal-mode-number resistive ballooning activity in BOUT++ simulations, with pressure perturbations near $\Psi_n = 0.98\text{--}1.01$ and nonlinear saturation that only weakly perturbs the pedestal.

4. Modifies the Eich fluence model to use pedestal-foot pressure, correcting the factor-eight-or-greater overestimate for DIII-D small ELMs and obtaining approximately 40% average agreement across the studied measurements.
5. Provides first-order SPARC and ITER peak parallel energy-fluence projections below a nominal 12 MJ m⁻² tungsten-surface melting threshold, while identifying geometric and material limits on that comparison.

2026PMAA

Claude (September 2026)

Summary

Laser wakefield acceleration (LWFA) is limited by dephasing: trapped electrons outrun the accelerating phase of a wake moving at the driver's subluminal group velocity. As background, flying-focus pulses drive luminal wakes but bring a varying spot size, long duration, or large plasma volume. The authors build the driver in a preformed parabolic plasma waveguide from N radial modes with frequencies $\omega_q = \omega_0 + q\Omega$, so that the modal interference peak travels at c with constant spot size and ultrashort duration while the envelope moves at the group velocity. Setting the peak duration to π/ω_{p0} , with ω_{p0} the on-axis plasma frequency, fixes the matched spot size (the waveguide's lowest-mode width); matching the recurrence period $2\pi/\Omega$ to the envelope duration fixes the stage length at N/σ dephasing lengths at equal density, with σ between 0.88 and 1. Linear-regime scaling laws compare energy gain, stage length, pulse energy, and density with standard LWFA. Quasi-3D particle-in-cell simulations with ten modes at 6×10^{18} cm⁻³ give an 11-fold longer stage and energy gain, in close agreement with scaling values of 660 versus 58 MeV at normalized amplitude $a_0 = 0.5$. Stated costs are broader bandwidth, system complexity, larger matched spots, and self-focusing and depletion at $a_0 \sim 0.75$.

Contributions

1. Showed that imposing $a_{oa}(z + v_a \mathcal{T}, t + \mathcal{T}) = a_{oa}(z, t)e^{i\eta \mathcal{T}}$ on the on-axis potential cannot be met by one broadband spectrum across the dispersion relations $k_q(\omega) = c^{-1}[\omega^2 - \omega_{p0}^2 - 4(1 + 2q)c^2/w^2]^{1/2}$ but is met by one frequency per mode, $\omega_q = \omega_0 + (k_q(\omega_q) - \kappa_0)v_a$, which for $v_a = c$ is $\omega_q = \omega_0 + 4c^2q/[(\omega_0 - c\kappa_0)w^2]$ and for $(\omega_{p0}w/2c)^2 \gg 1$ has spacing $\Omega = 2\omega_0(2c/\omega_{p0}w)^2$, and derived the on-axis intensity $N^{-2}|\alpha(t - z/v_g)|^2 \sin^2[\frac{1}{2}N\Omega(t - z/c)]/\sin^2[\frac{1}{2}\Omega(t - z/c)]$, whose peaks of FWHM $\tau = (\sigma\pi/N\omega_0)(\omega_{p0}w/2c)^2$ cross an envelope of duration T in $L_D = cv_g T/(c - v_g) \approx 2(\omega_0/\omega_{p0})^2 cT$.
2. Derived from $\tau = \pi/\omega_{p0}$ the matched spot size $w = (2c/\omega_{p0})(\omega_0/\omega_{p0})^{1/2}(N/\sigma)^{1/2}$ and the FWHM spot $w_F = (2c/\omega_{p0})(\omega_0/\omega_{p0})^{1/2}(2\ln 2/\sigma)^{1/2}$, smaller than w for $N > 1$, that sets the wake width, and from $T_R \simeq T$ the stage length $L_D = (2\pi c/\omega_{p0})(\omega_0/\omega_{p0})^2 N/\sigma$, which with the wake field $e|E_z|/mc = (\pi/8)\omega_{p0}a_0^2$ give $\Delta\gamma_D = (\pi^2/4)(N/\sigma)(\omega_0/\omega_{pD})^2 a_0^2$ and, for a flattop envelope, the pulse energy $U_D/mc^2 = (\pi cN/4\sigma^2 r_e \omega_0)(\omega_0/\omega_{pD})^4 a_0^2$, both equal to the standard values for $N = \sigma = 1$ at equal density.
3. Compared DLWFA with a standard pulse of equal normalized FWHM spot, $\omega_{pD}w_{FD} = \omega_{pS}w_{FS}$: at fixed density the gain, stage-length, and pulse-energy ratios all equal N/σ with $w_D = N^{1/2}w_S$; at fixed stage length the gain ratio is $(N/\sigma)^{1/3}$ with $U_D = U_S$, $w_D = (\sigma^2 N)^{1/6}w_S$, and $n_D = (N/\sigma)^{2/3}n_S$; at fixed gain $L_D/L_S = U_D/U_S = (\sigma/N)^{1/2}$ with $w_D = \sigma^{1/2}w_S$ and $n_D = (N/\sigma)n_S$; and at equal density $P/P_{cr} = \omega_{pD}^2 w_D^2 a_0^2 / (32c^2 N)$ equals the standard value because $w_{FD} = w_{FS}$.
4. Simulated with OSIRIS, using azimuthal modes 0 and 1, the designs with $\lambda_0 = 1 \mu\text{m}$, $\tau = 23$ fs, $w_F = 20 \mu\text{m}$, $w = 17$ and $54 \mu\text{m}$, $\Lambda = (\lambda_0/\omega_0)\Omega = 13$ nm, $L_d = 2.5$ mm, and $L_D = 2.8$ cm, in a $178 \times 323 \mu\text{m}$ window of 4980×1490 cells for 109 ps with a 258-fs flattop envelope between 36-fs ramps, finding the first recurrence of the luminal peak near $\omega_{p0}t \approx 10^4$, temporal pulse splitting that settles into a steady state at $a_0 = 0.5$, and at $a_0 = 0.75$ self-focusing, depletion, redshifting, and compression that slow the wake until a burst dislocates its phase, repeatedly; within the FWHM spot both pulses and wakes are nearly identical, while outside it modal interference gives the wake multiple transverse periods.
5. Projected for $a_0 = 0.5$ a 100-GeV accelerator of five ten-mode DLWFA stages at 2×10^{17} cm⁻³ with $L_D = 4.7$ m and matched spot 690 μm , against ten 10-GeV standard stages at 3.5×10^{16} cm⁻³ with $L_d = 5.7$ m, with 2.4 times less total pulse energy, estimated the waveguide plasma volume $V \propto \pi N w^2 L = \sigma(\omega_{p0}/\omega_0)^3 L^3/\pi$

against $\pi L^3/(12f_{\#}^2)$ for the full-aperture axiparabola cone, and showed that luminal peaks resonantly driving a multi-pulse wake with $\tau = \pi\sigma/\omega_{p0}$ and $T_R = 2\pi\ell/\omega_{p0}$ need $N = 2\ell$ modes.

Comments

- Section VI: the 100-GeV example states “ $w_D = 690 \mu\text{m}$ compared with $w_S \simeq 160 \mu\text{m}$ ”, but Eqs. (18)–(19) with the equal normalized FWHM spot size $\omega_{pD}w_{FD} = \omega_{pS}w_{FS}$ assumed for Eq. (25) give $w_D/w_S = N^{1/2}\omega_{pS}/\omega_{pD}$, which reproduces the Section IV relations $w_D = N^{1/2}w_S$, $(\sigma^2 N)^{1/6}w_S$, and $\sigma^{1/2}w_S$ and for $N = 10$, $n_S = 3.5 \times 10^{16} \text{ cm}^{-3}$, and $n_D = 2 \times 10^{17} \text{ cm}^{-3}$ gives $w_D/w_S \approx 1.32$, or $w_S \approx 520 \mu\text{m}$, while Eq. (18) with $N = \sigma = 1$ at n_S gives about $760 \mu\text{m}$; neither yields $160 \mu\text{m}$, whereas the quoted $w_D = 690 \mu\text{m}$ and pulse-energy factor 2.4 follow from Eqs. (18), (24), and (25); unresolved.
- Appendix A: the mode phases “ $\Phi_q = q\Omega t_d$ ”, chosen “to allow for a desired delay t_d ”, multiply the q th mode, whose spectral amplitude is $\tilde{\alpha}(\omega - \omega_q)$ (Section II), by $e^{iq\Omega t_d}$ and so give the interference factor $\exp[-iq\Omega(t - t_d - z/c)]$ with an undelayed envelope, whereas Eq. (A1), and hence Eq. (14), keeps $\exp[-iq\Omega(t - z/c)]$ and places the delay in the envelope $\alpha(t) = \frac{1}{2\pi} \int \tilde{\alpha}(\omega) e^{-i\omega(t-t_d)} d\omega$, which requires the phase $(\omega - \omega_q)t_d$ rather than $q\Omega t_d$; unresolved.

2026PN

ChatGPT (July 2026)

Summary

This paper reconstructs a sparse or localized matrix B , especially a matrix function $f(A)$, using only black-box matrix-vector products. SpaMRAM treats each row as an unknown sparse signal and applies Gaussian sensing with normalized iterative hard thresholding, allowing recovery when the sparsity pattern is unknown and dominant entries are sufficiently distributed. BaMRAM targets banded matrices and functions with off-diagonal decay: deterministic probing vectors separate nearby diagonals, recover a prescribed band, and admit inexpensive Gaussian a posteriori error estimates. Entry-decay assumptions yield error bounds for both methods, although implicit decay constants make them chiefly explanatory rather than computable a priori. The construction extends to Kronecker sums through multidimensional probing patterns. Experiments on synthetic matrices, SuiteSparse problems, and graph communicability show accurate recovery of exponentials, square roots, logarithms, and localized network quantities. In scalability tests up to dimension 25,600, BaMRAM’s required products are essentially dimension-independent for bounded bandwidth, while SpaMRAM grows slowly and sublinearly on the tested sparse family. Krylov approximation error eventually limits accuracy.

Contributions

1. Introduced SpaMRAM for row-wise compressed-sensing recovery without a known sparsity pattern.
2. Introduced BaMRAM for deterministic recovery of banded matrices and off-diagonally decaying matrix functions.
3. Proved decay-based approximation bounds and supplied practical randomized error estimators.
4. Extended deterministic probing to matrices arising from Kronecker-sum structures.
5. Demonstrated recovery of matrix functions and network quantities with favorable scaling on synthetic and SuiteSparse examples.

2026PSHZ

ChatGPT (August 2026)

Summary

Global trans-collisional fluid simulations with GRILLIX produce three pedestal relaxation event (PRE)-like bursts over more than 5 ms of an ASDEX Upgrade I-mode discharge. Tearing parity, electromagnetic radial electron-heat transport, electron-diamagnetic propagation, and a dispersion relation consistent with a kinetic estimate identify microtearing modes as the trigger. The trajectory of density and electron-temperature gradient lengths near $\rho_{\text{pol}} = 0.95$ agrees with

a simplified slab Landau-fluid growth-rate estimate and motivates a qualitative PRE cycle. A Landau-fluid closure permits the instability and bursts, whereas a Braginskii closure with heat-flux limiters makes the microtearing mode linearly stable and produces no PREs. Simulated events reproduce magnetic activity, predominant electron-temperature relaxation, and approximately 0.3 ms duration seen experimentally. Important discrepancies remain: the modeled stored energy rises after a burst because of the boundary conditions, and the reference discharge did not exhibit PREs at the simulated time. The authors attribute premature onset partly to classical Spitzer resistivity underestimating damping at high electron temperature, exposing limits of trans-collisional fluid models at low collisionality.

Contributions

1. Produced multiple PRE-like bursts in a global, experimentally parameterized GRILLIX simulation spanning more than 5 ms.
2. Identified microtearing modes as the trigger using mode parity, transport, propagation, phase relations, and dispersion-relation fingerprints.
3. Matched the nonlinear evolution of pedestal gradients to a simplified linear Landau-fluid growth-rate model and proposed a full PRE cycle.
4. Demonstrated that the Landau-fluid closure destabilizes the relevant mode, while the Braginskii closure with heat-flux limiters suppresses it and the PREs.
5. Compared simulated and experimental event characteristics and isolated boundary conditions and low-collisionality resistivity as important model limitations.

2026PSKA

ChatGPT (September 2026)

Summary

This experimental study investigates harmonic generation in plasma sources driven at 2.45 GHz, where electron density is effectively constant over a microwave cycle. Directional couplers, spectrum analyzers, power meters, and resonance tracking distinguish harmonics generated by the plasma from those supplied by the amplifier. Both inductively coupled plasma (ICP) and capacitively coupled plasma (CCP) sources exhibit stronger nonlinear transport at low pressure. The ICP third harmonic reaches approximately 0.2% of absorbed power, whereas the CCP second harmonic reaches approximately 4%. Increasing pressure suppresses harmonic generation in both sources. The authors attribute ICP nonlinearity to a time-dependent effective electron scattering frequency associated with velocity-dependent transport, and CCP nonlinearity to rectification by the sheath, the charged boundary layer next to the wall. Their observations show that nearly constant electron density does not guarantee linear microwave conductivity. The proposed mechanisms interpret the measured harmonic signatures rather than follow from direct time-resolved measurements of scattering frequency.

Contributions

1. Measures nonlinear microwave response at pressures above 10 Pa despite electron density remaining effectively constant within the excitation cycle.
2. Extends resonance-based source diagnostics with forward and reflected harmonic measurements, separating amplifier contributions from plasma-generated signals.
3. Finds enhanced third-harmonic generation in the low-pressure ICP source, reaching approximately 0.2% of absorbed power.
4. Observes substantially stronger second-harmonic generation in the low-pressure CCP source, reaching approximately 4% of absorbed power and supporting sheath rectification as its origin.
5. Contrasts time-dependent effective scattering in ICP with sheath rectification in CCP, and relates their reduced nonlinearity at higher pressure to shorter electron mean free paths and thinner sheaths.

2026QKLM

ChatGPT (July 2026)

Summary

The Tempered Finite Element Method modifies standard finite elements so meshes containing nearly or exactly zero-measure elements can still converge. The motivating failure is not an isolated needle or flat cap but a band of cap elements that imposes an artificial locking constraint. TFEM clips the element-map Jacobian only where it appears in denominators, locally softening diffusion while preserving the usual formulation elsewhere. For two-dimensional Poisson problems with linear elements, the authors prove optimal convergence when the minimum Jacobian scales as h^3 ; numerical evidence gives the corresponding h^4 scaling in three dimensions. Exactly degenerate elements recover a nonconforming mortaring or penalty interpretation. Manufactured and randomized tests show errors close to regular-mesh finite elements, mild sensitivity to the clipping constant, and no degradation of preconditioned conjugate-gradient convergence. Numerical extensions cover elasticity, mortaring of nonconforming meshes, high-order elements, and advection. Rigorous theory is concentrated on linear Poisson elements, the optimal constant remains geometry-dependent, and general three-dimensional degeneracy criteria are unresolved.

Contributions

1. Distinguished harmless isolated degenerate elements from locking cap-band configurations.
2. Proposed a one-line Jacobian-denominator clipping rule that defines TFEM.
3. Proved optimal convergence for the two-dimensional linear Poisson setting.
4. Established the equivalence between zero-measure TFEM and nonconforming mortaring.
5. Demonstrated robust conditioning and extensions to elasticity, high order, and advection.

2026QMLC

ChatGPT (July 2026)

Summary

Qiu and coauthors optimize laser-driven proton acceleration in near-critical-density plasma using particle-in-cell simulations and analytic modeling. The study asks whether tighter focusing helps only by increasing intensity, or whether spot size matters even at fixed intensity. Simulations show that reducing the laser spot from a conventional scale to $0.8\ \mu\text{m}$ increases maximum proton energy despite lower total laser energy. The mechanism is stronger ponderomotive-force-driven electron acceleration at small spot size, producing a faster and stronger charge-separation field. The authors then design a density down-ramp that velocity-matches protons to the accelerating field, enabling phase-stable acceleration and further energy gain. The combined strategy raises proton energy while relaxing the demand for large laser facilities, suggesting that advanced focusing and tailored plasma profiles can make compact laser-plasma proton sources more efficient.

Contributions

1. Isolated the effect of laser spot size at fixed intensity in proton acceleration.
2. Showed that tighter focusing can increase proton energy even with lower laser energy.
3. Identified enhanced ponderomotive electron dynamics as the source of stronger accelerating fields.
4. Derived a density down-ramp profile for phase-stable proton acceleration.
5. Demonstrated additional energy gain from combining tight focusing with target-profile optimization.

2026QNC

ChatGPT (July 2026)

Summary

This paper extends entropy-stable reduced-order modeling for nonlinear conservation laws from finite-volume to high-order discontinuous Galerkin full-order models. A mass-matrix-aware weighted test basis supports two-stage hyper-reduction; under ideal hyper-reduction, the resulting differentiation-matrix error is orthogonal to the ROM approximation space. Dimension-by-dimension sampling reduces multidimensional volume flux evaluations, while numerical-linear-algebra-based Carathéodory pruning hyper-reduces boundary terms without a sparsity-promoting linear-programming solver and gives a predictable number of boundary nodes. Artificial-viscosity terms are projected to dissipate entropy. One- and two-dimensional advection, Burgers, and Euler tests with periodic and weakly imposed reflective boundaries keep convective entropy near machine precision and show viscous entropy dissipation, with useful speedups for small bases. The method does not provide a rigorous error analysis, relies on simple artificial viscosity and linear proper-orthogonal-decomposition spaces, and therefore does not resolve the known accuracy difficulty of transport-dominated solutions.

Contributions

1. Generalized entropy-stable reduced-order model construction from finite-volume to high-order discontinuous Galerkin full-order models.
2. Introduced a mass-matrix-aware weighted test basis and established ideal hyper-reduction error orthogonality.
3. Developed dimension-by-dimension volume hyper-reduction for multidimensional discontinuous Galerkin flux differencing.
4. Replaced linear-programming-based boundary sparsification with Carathéodory pruning having predictable sample counts.
5. Validated entropy behavior, accuracy, and runtime on one- and two-dimensional nonlinear conservation laws with periodic and reflective boundaries.

2026QSW

ChatGPT (July 2026)

Summary

This paper constructs a fast Fourier spectral algorithm for the wave kinetic equation arising as a kinetic limit of the cubic Schrödinger equation. The resonant four-wave collision operator is first rewritten as a spherical integral analogous to the classical Boltzmann operator. Fourier approximation then exposes double-convolution structures that can be evaluated with fast Fourier transforms, avoiding large precomputed operator tensors and their storage. Spherical-design quadrature handles angular integration in three dimensions. The authors test stationary distributions and time-dependent, isotropic, anisotropic, continuous, and discontinuous data in two and three dimensions. The computations exhibit spectral convergence for smooth cases, retain the expected mass and energy behavior, and resolve non-isotropic wave spectra. Timing studies show the practical gain from the FFT formulation. Relative to discrete-interaction and direct quadrature approaches, the method provides a more accurate and scalable route for simulating high-dimensional wave-turbulence kinetics.

Contributions

1. Recast the resonant wave collision operator as a Boltzmann-like spherical integral.
2. Derived double-convolution formulas suitable for fast Fourier transform evaluation.
3. Eliminated large operator precomputation and storage requirements.
4. Combined Fourier discretization with spherical-design quadrature for three-dimensional problems.
5. Verified accuracy, conservation behavior, anisotropic dynamics, and efficiency in two and three dimensions.

2026R

ChatGPT (July 2026)

Summary

The paper constructs periodic traveling waves for the one-dimensional two-species Vlasov–Poisson equation as ion and electron patches bounded by four moving velocity profiles. A Hamiltonian reformulation reduces the strip-interface dynamics to a coupled functional equation near homogeneous velocity-layer equilibria. Analytic Crandall–Rabinowitz theory then gives small-amplitude branches at simple Fourier-mode eigenvalues. Depending on the relative geometry of the ion and electron layers, the local diagram has two or four branches; the bifurcations are pitchforks, and symmetric configurations produce a hyperbolic branch geometry. A global analytic bifurcation argument continues each local family until it loops or approaches blow-up, boundary collision, or degeneracy. An affine transformation maps these kinetic patch states to traveling solutions of the two-component Euler–Poisson system, providing both small- and large-amplitude waves. The construction extends earlier electron-only work by allowing both charged species to evolve, at the cost of a higher-dimensional kernel and more intricate branch classification.

Contributions

1. Formulated traveling ion–electron layer states as coupled patch-interface equations for the two-component Vlasov–Poisson system.
2. Established local analytic bifurcation from homogeneous velocity-layer equilibria at selected Fourier modes.
3. Classified two- and four-branch diagrams, proved pitchfork structure, and identified hyperbolic geometry in symmetric cases.
4. Continued local branches globally and characterized the possible loop, blow-up, collision, and degeneracy alternatives.
5. Transferred the construction through an affine transformation to small- and large-amplitude two-component Euler–Poisson traveling waves.

2026R2

ChatGPT (July 2026)

Summary

This paper develops a kernel-based quadrature algorithm for bounded smooth surfaces known only through computational nodes, a triangulation, and identified boundary nodes; neither surface nor boundary parameterizations are required. Local coordinate maps partition the surface around triangles, while radial-basis-function interpolation augmented by polynomials produces reusable weights for integrands, derivatives, normals, and surface integrals. A special boundary-sliver construction avoids exact knowledge of the boundary curve, and a stable normal approximation removes the small-node-spacing instability of earlier work. Error estimates connect convergence to polynomial reproduction and integrand smoothness. Tests on bounded rotated Cassini surfaces recover convergence orders comparable to parameterization-based predecessors; for $m = 7$, the same orders are observed despite approximate geometry. Weight construction scales as $O(N)$. Accuracy degrades when nodes underresolve narrow surface regions, and unexpected parity-dependent superconvergence and more general scattered-data stabilization remain open questions.

Contributions

1. Removed the requirement for explicit surface and boundary parameterizations.
2. Constructed local kernel-polynomial quadrature weights from triangulated surface nodes.
3. Introduced a boundary-sliver integration treatment without exact boundary curves.
4. Stabilized surface-normal approximation as node spacing decreases.
5. Established convergence estimates and observed linear weight-construction scaling.

Summary

The author reorganizes the first nonlinear (second-order) Vlasov–Maxwell response of a homogeneous magnetized plasma with gyrotropic equilibrium, usually written as a double cyclotron-harmonic expansion with long Bessel sums and nested resonance denominators. As background, the author’s linear theory keeps the Larmor phase unexpanded and evaluates the orbit integral as the incomplete-Bessel function $G_\mu(z, \psi) = i \int_0^\infty e^{-iz \sin(\psi+\lambda)} e^{-i\mu\lambda} d\lambda = \sum_n J_n(z) e^{-in\psi} / (n + \mu)$, with z the normalized Larmor radius, ψ the gyrophase, and μ the normalized detuning. For a triad $\mathbf{k} = \mathbf{p} + \mathbf{q}$, the source is the Lorentz force of mode $\mathbf{p}$ on the linear response of mode $\mathbf{q}$; with that response in incomplete-Bessel form, the outer orbit integral yields a bivariate orbit-resolvent $G_{\mu,\nu}^{(r)}(z, \psi; w, \chi)$, whose shift $r \in \{-2, \dots, 2\}$ combines inner and outer gyrophase blocks. The second-order distribution becomes nine blocks of these kernels and their derivatives. The author proves expansion, resolvent, recurrence, and limit identities, recovers an extension of the Liu–Tripathi double-harmonic formula, derived independently, through Graf’s addition theorem, and expresses the nonlinear susceptibility through projected kernels. No numerical evaluation is given; the construction keeps the unperturbed characteristics, and its computational benefit is stated to be regime-dependent.

Contributions

1. Wrote the second-order characteristic solution in closed form (Eq. (6.3)): with inner blocks $K_{\mathbf{q},a} = e^{iz_{\mathbf{q}} \sin \psi_{\mathbf{q}} + ia\psi_{\mathbf{q}}} G_{\mu_{\mathbf{q}}-a}(z_{\mathbf{q}}, \psi_{\mathbf{q}})$, $a \in \{-1, 0, 1\}$, and outer force blocks $\mathcal{P}_{\mathbf{p},b} = e^{ib(\phi_{\mathbf{k}}-\phi_{\mathbf{p}})} [C_{\mathbf{p},b}^{(\phi)} \partial_\phi + C_{\mathbf{p},b}^{(\perp)} \partial_{v_\perp} + C_{\mathbf{p},b}^{(\parallel)} \partial_{v_\parallel}]$, the orbit integral maps $K_{\mathbf{q},a}$ to $e^{ia\psi_{\mathbf{q}}} G_{\mu_{\mathbf{k}}, \mu_{\mathbf{q}}-a}^{(a+b)}(z_{\mathbf{k}}, \psi_{\mathbf{k}}; z_{\mathbf{q}}, \psi_{\mathbf{q}})$ and its $\psi_{\mathbf{q}}$, $z_{\mathbf{q}}$, and $\mu_{\mathbf{q}}$ derivatives to the $(\partial_{\psi_{\mathbf{q}}} + ia)$, $\partial_{z_{\mathbf{q}}}$, and $\partial_{\mu_{\mathbf{q}}}$ derivatives of that kernel, and the result keeps the Larmor factor $e^{iz_{\mathbf{k}} \sin \psi_{\mathbf{k}}}$ of the linear response.
2. Proved the superposition $G_{\mu,\nu}^{(r)} = \sum_\ell T_\ell(w, \nu) e^{-i\ell\chi} G_{\mu+\ell-r}(z, \psi)$, where Newberger’s Turkin functions $T_\ell(w, \nu) = \sum_n J_n(w) J_{n-\ell}(w) / (n + \nu)$ are the Fourier coefficients of the dressed inner factor $e^{iw \sin \chi} G_\nu(w, \chi)$, the double expansion with denominators $N + \ell + \mu - r$, the resolvent identity $(i\partial_\psi + i\partial_\chi + \mu - r) G_{\mu,\nu}^{(r)} = e^{-iz \sin \psi} e^{iw \sin \chi} G_\nu(w, \chi)$, the outer Bessel equation, the recurrence $\partial_z G_{\mu,\nu}^{(r)} = \frac{1}{2} [e^{-i\psi} G_{\mu+1,\nu}^{(r)} - e^{i\psi} G_{\mu-1,\nu}^{(r)}]$, and $G_{\mu,\nu}^{(r)}(z, \psi; 0, \chi) = G_{\mu-r}(z, \psi) / \nu$ at zero inner Larmor radius.
3. Obtained from $(i\partial_\chi + \nu + w \cos \chi) [e^{iw \sin \chi} G_\nu(w, \chi)] = 1$ the Turkin recurrence $(\nu + \ell) T_\ell + \frac{w}{2} (T_{\ell-1} + T_{\ell+1}) = \delta_{\ell,0}$ and the bivariate identity $(i\partial_\chi + \nu) G_{\mu,\nu}^{(r)} + \frac{w}{2} [e^{i\chi} G_{\mu,\nu}^{(r+1)} + e^{-i\chi} G_{\mu,\nu}^{(r-1)}] = G_{\mu-r}(z, \psi)$, together with phase insertions that shift r , inner derivative rules, and a double-characteristic representation of $G_{\mu,\nu}^{(r)}$ as a two-time orbit integral.
4. Derived the double-harmonic formula directly, $\tilde{f}_{\mathbf{k},N}^{(2)} \propto [N\omega - (\Omega_{\mathbf{k}} - k_\parallel v_\parallel)]^{-1} \sum_M e^{-i[N\phi_{\mathbf{k}} - (N-M)\phi_{\mathbf{p}} - M\phi_{\mathbf{q}}]} L_{NM}^{(\mathbf{p})} \tilde{f}_{\mathbf{q},M}^{(1)}$, whose three blocks carry $J_{N-M+b}(z_{\mathbf{p}})$ and turn gyrophase derivatives into multiplication by $-iM$, and recovered it from Eq. (6.3) by expanding the kernels into inner harmonics $\sum_a T_{m+a}(z_{\mathbf{q}}, \mu_{\mathbf{q}} - a) X_{\mathbf{q},a}$, moving derivatives through $J_{M-m}(z_{\mathbf{q}})$ with $\Omega_{\mathbf{p}} \mathcal{G}_{\mathbf{p}} J_{M-m}(z_{\mathbf{q}}) = -\mu_{\mathbf{p}} q_\perp J'_{M-m}(z_{\mathbf{q}})$, and contracting the intermediate harmonic by Graf’s theorem $e^{il(\phi_{\mathbf{p}}-\phi_{\mathbf{k}})} J_l(z_{\mathbf{p}}) = \sum_h J_{l+h}(z_{\mathbf{k}}) J_h(z_{\mathbf{q}}) e^{ih(\phi_{\mathbf{k}}-\phi_{\mathbf{q}})}$, the step that produces the $z_{\mathbf{p}}$ Bessel factor.
5. Introduced projected kernels $\mathcal{K}_N^{(r,a)}(z_{\mathbf{k}}, z_{\mathbf{q}}, \Delta; \mu_{\mathbf{k}}, \nu)$, angular Fourier projections at relative angle $\Delta = \phi_{\mathbf{k}} - \phi_{\mathbf{q}}$ with the single-sum form $\sum_n e^{in\Delta} T_{n+a}(z_{\mathbf{q}}, \nu) T_{N-n}(z_{\mathbf{k}}, \mu_{\mathbf{k}} + n + a - r)$, showed that every angular contraction of the nonlinear current reduces to $\mathcal{K}_N^{(r,a)}$ and its ∂_Δ , $\partial_{z_{\mathbf{q}}}$, and ∂_{ν} derivatives, wrote the second-order susceptibility $\chi_{\tau\sigma\rho}^{(2)}$ in the circular basis as nine (a, b) blocks with polarization-resolved coefficients followed by a two-dimensional velocity integral, and argued that the n th order is governed by n -variate orbit kernels.

Comments

- Section 10: the inner amplitudes are given as $X_{\mathbf{q},1;1} = \frac{e^{i\phi_{\mathbf{q}}}}{\sqrt{2}} \mathcal{G}_{\mathbf{q},\Omega_{\mathbf{q}}} f_0$, $X_{\mathbf{q},-1;-1} = \frac{e^{-i\phi_{\mathbf{q}}}}{\sqrt{2}} \mathcal{G}_{\mathbf{q},\Omega_{\mathbf{q}}} f_0$, and $X_{\mathbf{q},\pm 1;0} = \frac{q_\perp}{2\Omega_{\mathbf{q}}} \mathcal{H} f_0$, but neither $\mathcal{G}_{\mathbf{q},\Omega_{\mathbf{q}}}$ nor $\mathcal{H}$ is defined anywhere in the paper; the definitions of $X_{\mathbf{q},\pm 1}$ in Section 3 use $\mathcal{G}_{\mathbf{q}}$ and $\mathcal{G} = v_\parallel \partial_{v_\perp} - v_\perp \partial_{v_\parallel}$, so the evident intended operators are $\mathcal{G}_{\mathbf{q}}$ and $\mathcal{G}$.

2026RARB

ChatGPT (July 2026)

Summary

This paper documents major SUNDIALS capabilities introduced from versions 6.2.0 through 7.4.0. ARKODE gains low-storage Runge–Kutta methods, including strong-stability-preserving and super-time-stepping families; symplectic partitioned Runge–Kutta methods up to order ten; and flexible standard and forcing-based operator splitting through a common SUNStepper interface. Multirate methods now adapt slow and fast time scales, while explicit Runge–Kutta methods support discrete adjoint sensitivity analysis with checkpointing. KINSOL adds runtime control of Anderson-acceleration damping and depth, and library-wide logging and error handling become more uniform. The implementations reuse SUNDIALS abstractions for shared-memory, distributed-memory, and GPU data. Gray–Scott experiments recover splitting orders through six and show more than an order-of-magnitude RKC speedup over a conventional second-order explicit method at loose tolerances. Lotka–Volterra tests recover the expected third- through fifth-order convergence for forward states and adjoint gradients.

Contributions

1. Added low-storage SSP and super-time-stepping Runge–Kutta integrators for memory-constrained and diffusion-dominated simulations.
2. Added high-order symplectic partitioned Runge–Kutta methods for separable Hamiltonian systems.
3. Introduced flexible operator-splitting modules and a common interface for composing SUNDIALS, custom, or exact subintegrators.
4. Extended multirate temporal adaptivity and explicit Runge–Kutta discrete adjoint sensitivity analysis.
5. Improved Anderson acceleration, error handling, and logging and validated representative convergence and performance behavior.

2026RBCH

ChatGPT (July 2026)

Summary

The paper presents a pipeline for producing curved, block-structured hexahedral meshes around atmospheric-reentry vehicles. Starting from a tetrahedral background mesh and a quadrilateral blocking of the vehicle surface, distance and direction fields guide an advancing front that extrudes successive hexahedral block layers toward the far-field boundary. Local topological patterns and smoothing improve the linear block structure while retaining user control over cell size and orientation. The blocks are then curved with Bézier representations so the wall is captured before the final mesh is subdivided. A constrained interval-assignment procedure makes edge subdivision counts compatible across neighboring blocks and creates a conforming linear mesh. Tests on several vehicle shapes assess scaled Jacobian and skewness distributions and show how pattern placement, smoothing, and curving affect quality. Supersonic simulations around the RAM-C II geometry contain no inverted cells and recover the expected bow shock, subsonic nose region, and wake topology. The current method still assumes a pre-blocked, quasi-convex single vehicle.

Contributions

1. Extended a field-guided advancing-front block-generation method from two to three dimensions.
2. Added local topological patterns and smoothing to control block quality, size, and orientation.
3. Developed an a posteriori Bézier-curving procedure for body-fitted block structures.
4. Enforced conforming hexahedral subdivision through constrained interval assignment.
5. Validated generated meshes using geometric quality metrics and supersonic CFD simulations.

2026RCC

ChatGPT (July 2026)

Summary

The paper develops an efficient modal discontinuous Galerkin method for multicomponent compressible Euler flows. Conservative variables are advanced in time, while the spatial residual is evaluated from the L^2 projection of entropy variables onto the discontinuous Galerkin space. This entropy-projection formulation avoids the elementwise matrix inversions required when entropy variables are evolved directly. A Direct Enforcement of Entropy Balance correction supplies semidiscrete entropy conservation or stability without costly over-integration, and directional artificial diffusion controls oscillations near shocks. Manufactured-solution tests verify high-order accuracy in one and two dimensions. Compared with an over-integrated entropy-variable solver, the proposed method reduces CPU time by 30–78% while retaining the designed entropy behavior. Multicomponent Riemann problems confirm robust shock treatment, and two-dimensional implosion and shock–inhomogeneity tests show sharp resolution of interacting waves, material-interface deformation, vortex formation, and small-scale interfacial instabilities.

Contributions

1. Evolved conservative variables while constructing residuals from projected entropy variables.
2. Enforced entropy conservation or stability without numerical over-integration.
3. Added directional shock capturing for discontinuity-induced oscillations.
4. Verified high-order convergence and 30–78% lower cost than the comparison method.
5. Resolved multicomponent waves and interfaces in demanding one- and two-dimensional tests.

2026RECB

ChatGPT (July 2026)

Summary

Physics-based plasma-control models must meet deterministic cycle deadlines, which general-purpose threading runtimes do not target. The authors develop a real-time-safe C11 multithreading library for the DIII-D plasma control system. Worker threads are initialized before a shot, run on isolated cores, and synchronize through acquire–release atomics and spin waiting, avoiding kernel-mediated synchronization during real-time execution. The library parallelizes two production physics codes. Real-time TORBEAM traces independent electron-cyclotron beams for four gyrotrons consistently within a roughly 20 ms cycle. Real-time STRIDE partitions state-transition-matrix integrations around rational surfaces and computes ideal-stability quantities in about 100 ms on 72 cores. STRIDE’s remaining bottleneck is serial equilibrium-coordinate preprocessing, which consumes most of the runtime. The work demonstrates deployable deterministic parallelism on DIII-D, while the reported timing depends on isolated-core hardware and does not establish portability to every control environment.

Contributions

1. Designed a C11 multithreading library around deterministic real-time plasma-control constraints.
2. Used precreated workers and acquire–release atomics to avoid nondeterministic kernel synchronization during shots.
3. Deployed real-time TORBEAM for parallel gyrotron ray tracing with approximately 20 ms execution cycles.
4. Parallelized STRIDE’s stability calculation to execute in approximately 100 ms on DIII-D hardware.
5. Identified serial coordinate preprocessing as the dominant remaining STRIDE performance bottleneck.

2026RF

ChatGPT (July 2026)

Summary

This paper gives a mathematical foundation for Boltzmann generators, which use invertible normalizing flows to map an easy reference distribution to a molecular Boltzmann distribution. Physical Coulomb and Lennard–Jones singularities make the target density vanish at atomic collisions, outside standard smooth-transport assumptions. The authors regularize these densities, extend the Moser transport construction to positive low-regularity endpoint densities, and show that its map is invertible and bilipschitz. Sobolev approximation results then provide invertible RealNVP networks whose push-forward measures converge narrowly and, on bounded configuration domains, approach the target arbitrarily closely in Wasserstein-2 distance. A two-dimensional double-well experiment reports joint $W_2 = 0.0993$ and reproduces the observed metastable switching count. An alanine-dipeptide experiment learns the principal Φ – Ψ conformational basins from 60 ns of molecular-dynamics data and reports a 9.21° Wasserstein distance. The analysis proves existence and expressivity rather than training convergence; controlling inverse-map conditioning through a bias toward Moser-like transports remains open.

Contributions

1. Extended Moser transport maps to low-regularity densities relevant to singular molecular interactions.
2. Proved existence of invertible bilipschitz transports between regularized reference and Boltzmann measures.
3. Established Sobolev approximation of these maps by invertible RealNVP networks.
4. Derived arbitrarily accurate Wasserstein-2 approximation for molecular Boltzmann distributions on bounded domains.
5. Validated equilibrium and metastable behavior on a double well and alanine dipeptide.

2026RFR

ChatGPT (July 2026)

Summary

This study uses one-dimensional azimuthal and two-dimensional radial-azimuthal particle-in-cell simulations to determine how an oscillatory axial electric field changes instability-driven electron transport in idealized $E \times B$ plasmas. The response depends strongly on forcing frequency and amplitude. Modulation below 150 MHz, especially near 40 MHz, weakens the electron cyclotron drift instability, shifts spectral energy toward lower frequencies and longer wavelengths, and reduces axial electron transport by up to 30%. Near the electron cyclotron frequency, forcing broadens the spectrum and can greatly enhance transport; at 560 MHz the response is non-monotonic in amplitude, with high-amplitude cases increasing axial current by more than two orders of magnitude. Fourier and bicoherence analyses identify modulation sidebands, harmonics, and coherent three-wave coupling with a low-frequency mode near 2 MHz. Transport changes are further traced to the phase correlation between electron-density and azimuthal-electric-field fluctuations. Within the local slab model, forcing near 40 MHz and amplitudes of approximately 0.1 – $0.3E_y$ give the strongest suppression.

Contributions

1. Mapped instability spectra and electron mobility over wide modulation-frequency and amplitude ranges.
2. Identified a forcing regime near 40 MHz that reduces cross-field transport by up to 30%.
3. Demonstrated amplitude-sensitive transport enhancement near the electron cyclotron frequency.
4. Used bicoherence to resolve sideband, harmonic, and three-wave coupling pathways created by modulation.
5. Connected transport suppression and enhancement to changes in density-electric-field phase alignment.

Summary

The authors introduce TokaMark, an open benchmark for machine-learning models on real experimental data from the Mega Ampere Spherical Tokamak (MAST), addressing the scarcity of curated, standardized fusion datasets. From the 11,573 shots of the open FAIR-MAST dataset they select 39 signals (magnetics, kinetics, radiation, currents, coil voltages, fueling and control references, and equilibrium reconstructions) sampled at 0.2–500 kHz and kept at native rates without resampling or imputation. Fourteen window-based tasks in four groups cover instantaneous equilibrium reconstruction, short-term magnetic dynamics, kinetic-profile evolution, and long-horizon forecasting of magnetohydrodynamic (MHD) activity and disruption precursors, posed as reconstruction, autoregressive forecasting of the input diagnostics, or forecasting of other signals. Scores are hierarchical normalized root-mean-square errors (NRMSE) masked to available ground truth. Baselines are mean and persistence predictors and a multi-branch convolutional encoder–decoder with a multilayer-perceptron (CNN) or long short-term memory (CNN+LSTM) fusion backbone. Under a random shot-level 80-10-10 split the networks beat the naive predictors, most strongly on equilibrium and magnetic tasks; holding out the last two campaigns degrades them, sometimes below the naive predictors, which the authors suggest may reflect distribution shift such as sensor recalibration. They note that the baselines are generic, without physics priors or task-specific tuning.

Contributions

1. Organized the signals by category, origin (diagnostic, actuator, or derived from EFIT reconstruction), sampling rate, and modality (time series, profiles, or video as 1D–3D tensors), and separated Markovian tasks with 5 ms input windows from non-Markovian tasks that use long histories with 5–100 ms horizons, including a Fourier-domain Mirnov-coil target in Task 4-5.
2. Quantified structured missingness in the frozen data, 38.2–55.2% for most equilibrium quantities (concentrated at shot start and end), 46.1% for Thomson scattering, and 0–5.4% for coil currents and voltages, plasma current, references, gas puffing, and neutral-beam power, and published per-task shot counts for both splits.
3. Defined the metric so that window RMSEs are combined by a quadratic mean within each shot, normalized by the signal’s standard deviation over the evaluation shots, averaged with equal weight over a task’s signals, and only then averaged over test shots, and also reported normalized mean absolute errors for all 57 task–signal pairs.
4. Built the baseline from independent 1D, 2D, or 3D convolutional encoders for each input and actuator on non-overlapping 5 ms chunks, with z-scored, zero-filled inputs, a shared backbone projecting to $D = 16$, and mirrored transposed-convolution decoders with cropping ($N = 3$, $K = 3$, $s = 3$), giving 0.27–1.26 million parameters per task.
5. Found that under the temporal split the CNN+LSTM kept the Group 2 score at 0.2928 against 0.7068 for the CNN, persistence beat both networks on soft-X-ray forecasting (Task 4-1: 0.1597 against 1.0970 and 1.1788), and on the random split both networks only matched the mean predictor on Mirnov spectral forecasting (0.6507 and 0.6488 against 0.6533).

Comments

- Section 4.2, Table 3: the temporal-split Group 4 score of CNN+LSTM is given as 0.8062, but Section 3.3 defines the group score as the average of its task scores, and the five Task 4-1 to 4-5 entries (1.1788, 1.1445, 0.8198, 0.4954, 0.7309) average 0.8739; 0.8062 is the average of the four tasks other than Task 4-2, whereas every other group entry in Table 3 agrees with its task average to within 0.0002.
- Section 4.2: “soft x-rays forecasting (Task 4-1 and Task 4-2) is by far the worst and exceeds unity” conflicts with Table 3 under temporal splitting, where the CNN+LSTM scores worse on Task 3-3 (1.4718) than on Tasks 4-1 (1.1788) and 4-2 (1.1445), and the CNN scores worse on Tasks 2-1 (1.7570) and 4-3 (1.3249) than on Task 4-1 (1.0970); only the CNN on Task 4-2 (3.0603) is the worst entry.
- Table D.3: the Mean column for Task 3-3 contains only “—”, the mark used for inapplicable models, although Section 4.1.1 states that the mean model predicts all tasks, Table 3 reports a temporal-split Task 3-3 Mean score of 0.9859, and Table D.4 lists its NMAE values for the same five signals.

- Appendix C.1, Eqs. (6)–(7): the decoder output has size $L_d^{(N,v_o)}$ and Eq. (7) uses $L_d^{(0,v_o)}$ as the compressed map that transposed convolutions upsample, but Eq. (6), $L_d^{(\ell,v_o)} = \lceil s(L_d^{(\ell+1,v_o)} - 1) - 2p + K + o_p \rceil$, gives $L_d^{(\ell,v_o)} = 3L_d^{(\ell+1,v_o)} - 1$ for the Appendix C.2 values ($K = 3, s = 3, p = 1, o_p = 1$), so $L_d^{(0,v_o)}$ would exceed the target size; the standard transposed-convolution relation has ℓ and $\ell + 1$ interchanged, and whether that or an inverse relation was intended is unresolved.

2026RLLW

ChatGPT (July 2026)

Summary

This work ports RMC criticality calculations to GPUs through a unified CPU–GPU data-transfer framework and four GPU-oriented techniques. Refined Event Parallelism groups particle work by spatial cell and reaction type to reduce thread divergence. Parallel Atomic Tally distributes scoring across small atomic arrays, while Parallel Atomic Bank stores fission neutrons with near-optimal memory use. A pool-managed hierarchical cache retains geometry levels and cross sections. Tests on a fuel assembly, KRUSTY, VENUS-II, and a full pressurized-water-reactor core agree with CPU RMC within 2 pcm in k_{eff} ; full-core mesh-flux errors remain below 1%, averaging about 0.1%. At 10^6 particles per generation, one RTX 4090 delivers performance comparable to roughly 140–150 CPU cores, and Refined Event Parallelism is up to 5.35 times faster than GPU history-based tracking. Benefits diminish for small particle populations and simple geometries because kernel-launch and scheduling costs dominate.

Contributions

1. Designed an automatic CPU–GPU class-data mapping framework that reduces porting boilerplate by more than 90%.
2. Developed Refined Event Parallelism using spatial-cell and reaction-type hashing to suppress branch divergence.
3. Introduced Parallel Atomic Tally, improving transport efficiency by 20–60% over a single atomic tally.
4. Introduced Parallel Atomic Bank, reaching at least 96% relative storage efficiency at 10^6 particles.
5. Validated RMC-GPU on four reactor models and quantified its accuracy, scaling, and workload-dependent limits.

2026RPRT

ChatGPT (July 2026)

Summary

This study models runaway-electron heating of a realistic, though not final, tungsten-heavy-alloy tile from SPARC’s outboard off-midplane limiter. A one-way workflow combines parametric and DREAM energy–pitch distributions, Geant4 volumetric energy deposition, and MEMENTO heat transfer for 1 and 10 ms loads. Curvature, magnetic-field inclination, finite Larmor radii, and redeposition of backscattered electrons produce three-dimensional energy partitioning that cannot be reproduced by a simple $\sin \alpha$ optical model or a single representative electron energy and pitch. At 20 kJ, electrons of 10 MeV or more do not melt the limiter, whereas approximately 1 MeV electrons can melt about $100 \mu\text{m}$. At 100 kJ, high-energy populations produce melt depths from several hundred micrometres to roughly 1 mm; low-energy loading over 10 ms can vaporize about $500 \mu\text{m}$. DREAM distributions predict roughly 1 mm melting on the curved tile side. The model omits the complete viscoplastic, hydrodynamic, and fracture response required to predict explosive damage.

Contributions

1. Built a SPARC-specific DREAM–Geant4–MEMENTO workflow with realistic limiter geometry and temperature-dependent alloy properties.
2. Compared two DREAM avalanche models and demonstrated why full energy–pitch distributions matter.

3. Quantified curvature, inclination, Larmor-radius, and backscatter-redeposition effects absent from simple optical models.
4. Mapped melting and vaporization across incident energies, pitches, total loads, and deposition times.
5. Resolved particle-loss channels and showed that high-energy photon losses require assessment of surrounding structures and magnets.

2026RS

ChatGPT (July 2026)

Summary

This paper develops an asymptotic-preserving finite-volume solver for Boltzmann kinetic equations using a micro–macro decomposition. The distribution is separated into an equilibrium component represented by macroscopic moments and a nonequilibrium component retained on a velocity grid, reducing velocity-domain requirements near equilibrium. The work extends earlier one-dimensional BGK schemes to two spatial and two velocity dimensions, the transport-correct ES-BGK collision operator, and diffusely reflecting walls. The macro equations use kinetic flux-vector splitting and receive heat flux from the micro distribution; the projected micro equation uses upwind transport. L-stable implicit collision updates and operator splitting preserve stability as the Knudsen number tends to zero. Manufactured solutions establish convergence in one and two dimensions, while shock tubes, cylindrical shocks, wall-driven flow, and a lid-driven cavity exercise continuum through rarefied-flow regimes. An open-source C++/MPI implementation achieves about 74% weak-scaling efficiency at the largest reported processor count and favorable strong scaling for the fixed test problem.

Contributions

1. Extended micro–macro kinetic discretization to multidimensional ES-BGK dynamics.
2. Coupled kinetic flux-vector-split macro updates to projected upwind micro updates through nonequilibrium heat flux.
3. Derived compatible diffusely reflecting boundary conditions for both components.
4. Verified asymptotic-preserving behavior and convergence on one- and two-dimensional kinetic benchmarks.
5. Implemented and evaluated a parallel C++/MPI solver with weak- and strong-scaling studies.

2026RS2

ChatGPT (September 2026)

Summary

This review and perspective compares thermal, inertial, beam-driven, plasmon-assisted, and muon-catalyzed fusion through nuclear reaction networks, fuel supply, stopping, radiation losses, and neutron production. The authors argue that deuterium–tritium systems must be assessed together with tritium breeding, activation, shielding, and the energy carried by 14 MeV neutrons. They show that deuterium–deuterium burn produces tritium and helium-3 whose secondary reactions materially change heating and neutron balances, and they quantify how helium-3 additions alter thermal power balance. For nonthermal fusion, the paper estimates beam interaction lengths, reviews nonlinear electron screening near heavy nuclei, and examines proton–boron, boron nitride, and proposed beryllium–helium-3 reaction cycles. It also evaluates reported nanostructured-target signatures and routes to improve muon-catalyzed fusion. Several proposals remain conceptual: sustaining the necessary beams, separating fusion products from other signals, and achieving sufficient fusions per muon require experimental and system-level validation. One author discloses a related patent interest.

Contributions

1. Integrated reaction-network, neutron, tritium, and fuel-cycle constraints into comparisons of fusion concepts.
2. Quantified secondary burn channels in deuterium fuel and the thermal effect of helium-3 admixture.

3. Analyzed interaction lengths and electron screening for beam-driven fusion in dense targets.
4. Assessed proton–boron and nanostructured-target evidence and proposed coupled boron-nitride and beryllium–helium-3 cycles.
5. Revisited muon-catalyzed fusion thresholds and possible muon-production and reactivation routes.

2026RVKK

ChatGPT (July 2026)

Summary

The paper proposes an equation-independent adaptive hyperviscosity stabilization for radial-basis-function-generated finite differences on periodic, boundaryless domains. It assembles the RBF-FD evolution matrix, uses its largest-magnitude eigenvalue to determine whether the selected time integrator is stable, and applies bisection to find the smallest hyperviscosity constant that places the spectrum inside the stability region. An error estimate shows that the high-order hyperviscosity operator can use lower-degree monomial augmentation consistently, reducing stencil size and sparsity costs. The authors also recommend different polyharmonic-spline orders for advection and hyperviscosity, using order three for advection and $2\alpha + 1$ for a 2α -order stabilizer. Linear-advection and nonlinear Burgers tests identify stable, low-dissipation parameter ranges and show why the optimal constant must be recomputed as nonlinear dynamics evolve. Limitations include costly repeated eigenvalue and root calculations, quasi-uniform nodes, global rather than local parameters, and no bounded-domain treatment.

Contributions

1. Gave a spectral-radius-based algorithm that adaptively selects a stable hyperviscosity constant without equation-specific von Neumann tuning.
2. Proved consistency for hyperviscosity RBF-FD approximations using monomial degree below the derivative order, enabling smaller stencils.
3. Introduced a hybrid spline-order prescription using order three for advection and $2\alpha + 1$ for hyperviscosity.
4. Mapped interactions among hyperviscosity, spline order, stencil size, time step, stability, and dissipation in linear advection.
5. Demonstrated time-dependent stabilizer recomputation for nonlinear Burgers flow and quantified sensitivity to recomputation frequency.

2026RXPS

ChatGPT (July 2026)

Summary

This work develops physics-based objectives for reducing ion-temperature-gradient turbulence in quasi-isodynamic stellarators. An updated model estimates the critical gradient of localized toroidal modes and motivates “mode splitting”: separating unfavorable-curvature wells so that unstable modes localize independently and require a steeper temperature gradient to grow. Optimization produces the six-field-period QICG configuration with $a/L_{T,\text{crit}} \approx 2.14$, low neoclassical transport, and a vacuum magnetic well, although its above-threshold nonlinear heat flux is excessive. A second, inverse-mirror-like configuration adds a target combining field-line variation of $|\nabla\alpha|$ with kinetic-electron mode inertia. It retains a critical gradient near 2 while shifting unfavorable curvature away from the magnetic-field minimum. Nonlinear gyrokinetic calculations with kinetic electrons show heat flux below or comparable to W7-X high mirror for density gradients through $a/L_n = 2$. Fast-particle confinement and tighter quasi-isodynamic quality remain open design constraints.

Contributions

1. Updated the localized toroidal-mode critical-gradient model for quasi-isodynamic geometry.
2. Introduced mode splitting as a strategy for raising the ion-temperature-gradient threshold.

3. Produced a six-field-period configuration with $a/L_{T,\text{crit}} \approx 2.14$ and a vacuum magnetic well.
4. Designed an inverse-mirror target that accounts for kinetic-electron mode inertia above threshold.
5. Demonstrated kinetic-electron heat flux competitive with W7-X high mirror over finite density gradients.

2026S

ChatGPT (July 2026)

Summary

Strang revisits high-order finite-difference time stepping for the constant-coefficient wave equation through the lens of the Lax equivalence theorem, the CFL condition, and von Neumann stability. The note contrasts the domain-of-dependence requirement with the sharper amplification-factor test needed for convergence. Two polynomial-interpolation families are discussed: one using $2N$ points with order $2N - 1$, and a Lax-Wendroff generalization using $2N + 1$ points with order $2N$. For the higher-order Lax-Wendroff family, the paper sketches the stability analysis of the complex growth factor and illustrates the fourth-order case. Although the CFL condition can allow larger Courant numbers up to the interpolation radius, the amplification factor exceeds one when $r > 1$ in the shown example. The central message is that high-order Lax-Wendroff accuracy is compatible with stability for $|r| \leq 1$, but CFL admissibility alone is not a sufficient convergence test.

Contributions

1. Reframed high-order Lax-Wendroff methods using the Lax equivalence and von Neumann stability tests.
2. Distinguished CFL domain-of-dependence necessity from the stronger amplification-factor stability condition.
3. Summarized two interpolation-based high-order finite-difference families for hyperbolic equations.
4. Illustrated fourth-order stability and instability through the growth factor for different Courant numbers.
5. Clarified that maximal formal accuracy remains stable only under the sharper $|r| \leq 1$ restriction.

2026S2

ChatGPT (July 2026)

Summary

Sladkov presents AKAM, a three-dimensional hybrid PIC–fluid framework for laser-plasma interaction in high-energy-density regimes. Ions are treated kinetically, while electrons are modeled as a ten-moment fluid, allowing pressure anisotropy, non-Maxwellian ion distributions, and ion-scale dynamics without the cost of a fully kinetic electron treatment. The laser is represented by a slowly varying envelope, so energy deposition and ponderomotive heating are modeled without resolving optical carrier oscillations. Collisions, ionization, inverse bremsstrahlung, and charge-state evolution provide self-consistent plasma-energy coupling. Demonstration simulations show localized magnetic structures, rapid electron heating near the laser focus, delayed ion energization, high-energy ion tails, and spectra consistent with multi-scale magnetic evolution. The paper positions AKAM as a scalable middle ground for XUV sources, laser-driven plasma jets, ablation, and laboratory high-energy-density experiments.

Contributions

1. Introduced a 3D hybrid PIC–fluid model for laser-plasma interaction.
2. Combined kinetic ions with ten-moment fluid electrons to retain ion-scale physics.
3. Used a laser-envelope model to avoid optical-cycle resolution while retaining energy deposition.
4. Coupled ionization, collisions, and inverse bremsstrahlung to charge-state and energy evolution.
5. Demonstrated magnetic-structure formation, electron heating, and nonthermal ion tails in simulations.

2026S3

ChatGPT (July 2026)

Summary

Relativistic particle-in-cell simulations require particle pushers that accurately integrate charged-particle trajectories in electromagnetic fields. This paper compares explicit relativistic pushers—including Boris, Vay, Higuera–Cary, exact-rotation, and proper-time schemes—against an implicit midpoint benchmark across homogeneous, spatially varying, time-dependent, and plane-wave fields. No method dominates every test. Higuera–Cary is consistently competitive among explicit schemes and substantially improves the scenario for which it was designed. Proper-time methods can be exact in constant fields but may lose order in inhomogeneous or time-dependent fields and are sensitive to root-solver tolerances. The implicit midpoint method often preserves invariants to machine precision but costs more and is not uniformly best. The paper identifies volume-preserving Boris-like maps and applies symmetric Yoshida compositions to construct arbitrary even-order variants. Fourth-order versions converge faster and reduce errors when physical timescales are resolved, but they do not rescue under-resolved steps. The study excludes field-grid and interpolation errors and does not benchmark computational cost.

Contributions

1. Compared a broad set of relativistic explicit particle pushers across six representative electromagnetic-field tests.
2. Identified Higuera–Cary as a consistently competitive general-purpose explicit alternative to Boris.
3. Showed that proper-time schemes can lose second-order accuracy and depend strongly on nonlinear-solver tolerances.
4. Derived arbitrary even-order, volume-preserving extensions of Boris-like pushers using symmetric Yoshida compositions.
5. Demonstrated fourth-order convergence while showing that higher order cannot compensate for unresolved physical timescales.

2026S4

ChatGPT (July 2026)

Summary

The state-dependent Riccati equation (SDRE) method approximates nonlinear optimal feedback control by expressing nonlinear dynamics in a state-dependent linear form and repeatedly solving algebraic Riccati equations. This paper relates the resulting quadratic value-function approximation to the Hamilton–Jacobi–Bellman equation. It derives a computable residual from Lyapunov equations for derivatives of the Riccati solution and bounds the value-function error by residual integrals along optimal and SDRE-controlled trajectories; an explicit local bound follows under exponential stability. Because semilinear representations are nonunique, the paper characterizes residual-minimizing representations and proves the existence of a zero-residual choice when two admissible representations produce opposite residual signs. Computationally, it compares a first-order offline–online approximation with a cascade Newton–Kleinman method warm-started from the previous time step. Tests on a controlled nonlinear Zeldovich reaction–diffusion equation show that cascade Newton–Kleinman matches direct Riccati solves at much lower cost and remains stable in cases where the offline–online method diverges.

Contributions

1. Expressed SDRE suboptimality through the Hamilton–Jacobi–Bellman residual of its approximate value function.
2. Derived residual-integral error bounds and a local norm bound under exponential stability.
3. Established conditions for a semilinear representation with vanishing residual and formulated residual minimization otherwise.
4. Gave a computable sufficient condition for stability of the offline–online closed-loop approximation.

5. Showed that cascade Newton–Kleinman is faster than direct Riccati solves and more robust than the offline–online approximation on a nonlinear PDE control problem.

2026S5

ChatGPT (July 2026)

Summary

FastLSQ represents solutions with frozen sinusoidal random Fourier features whose derivatives of any order have closed forms. This permits graph-free assembly of differential operators: linear PDEs require one least-squares solve, while nonlinear PDEs use Newton–Raphson iterations that reuse analytical assembly. Across 17 PDE benchmarks spanning one to six dimensions, the method attains relative L^2 errors near 10^{-7} within 0.1 s for linear problems and 10^{-8} – 10^{-9} within 9 s for five nonlinear solver problems on a single NVIDIA T4. Controlled comparisons attribute substantial accuracy gains over a one-shot tanh basis to the sinusoidal features. Exact derivatives also support noisy PDE discovery, inverse heat-source localization, and sparse-sensor coil recovery. Current limitations include problem-specific bandwidth selection, conditioning growth for high derivatives, penalty tuning, and a focus on box domains.

Contributions

1. Derived an operator-agnostic closed-form formula for arbitrary-order derivatives of sinusoidal random features.
2. Reduced linear PDE solution to one analytical least-squares assembly and solve without automatic differentiation.
3. Extended the framework to nonlinear PDEs through regularized, warm-started Newton iterations with line search and continuation.
4. Benchmarked value, gradient, residual, runtime, and ablation behavior on 17 linear and nonlinear PDEs.
5. Demonstrated differentiable inverse problems and substantially cleaner derivative estimation for sparse-regression PDE discovery.

2026S6

ChatGPT (July 2026)

Summary

This paper constructs a transparent boundary condition for two-dimensional exterior Helmholtz problems with compactly supported forcing. The exterior solution is represented by the method of fundamental solutions on concentric source and collocation circles. Equally spaced points make the discrete Dirichlet-to-Neumann map circulant, so it is diagonalized and applied efficiently with the discrete Fourier transform. Coupling this map to a finite-element formulation truncates the unbounded domain without replacing the inhomogeneous equation by a boundary-integral volume treatment. Numerical tests separately examine high-wavenumber map accuracy and the complete finite-element solution for circular and irregular interior boundaries. They show high accuracy when the source circle is chosen appropriately and confirm that the resulting transparent condition reproduces manufactured exterior solutions. The analysis also relates the circulant discretization to the Fourier eigenfunctions of the continuous map. Extension to three dimensions is plausible through block-circulant structure, while convergence of the discrete operator in a suitable topology remains open.

Contributions

1. Constructed a method-of-fundamental-solutions discretization of the exterior Helmholtz Dirichlet-to-Neumann map.
2. Exploited concentric circular point sets to obtain a circulant map with FFT-based application.
3. Embedded the discrete map as a transparent boundary condition in a finite-element formulation.
4. Treated compactly supported inhomogeneous forcing on unbounded exterior domains without domain meshing at infinity.
5. Verified high-wavenumber map accuracy and complete solutions on circular and irregular geometries.

2026S7

ChatGPT (September 2026)

Summary

This paper develops a full-approximation-scheme multigrid solver for elliptic problems discretized by high-order discontinuous Galerkin spectral elements on hexahedral meshes. Local h - and p -refinement uses level-specific active, twig, leaf, and frozen elements, coupling refinement zones through transfer operators without hanging nodes. The smoother employs weighted overlapping, element-centered Schwarz subdomains that include compatible face, edge, and vertex neighbors. Replacing each curved subproblem by a volume-preserving Cartesian tensor-product surrogate enables fast diagonalization with $O(p)$ work per subdomain node; an inexact Schwarz-preconditioned conjugate-gradient iteration improves robustness. Tests cover aspect ratios through 48, distorted meshes, irregular topology, and a compressor-blade geometry. Local blade refinement reduces memory by about 93% and runtime by 90% relative to global refinement, while a dynamically adapted spherical-front benchmark saves 93–98% of the degrees of freedom. Full-multigrid initialization and Hilbert-curve partitioning support parallel adaptation, and a cylinder-flow example provides a preliminary extension to incompressible Navier–Stokes problems.

Contributions

1. Formulated a conservative DG full-approximation multigrid hierarchy for local hp -refinement without hanging nodes.
2. Constructed an overlapping Schwarz smoother that incorporates compatible face, edge, and vertex neighbors on irregular curvilinear meshes.
3. Combined tensor-product surrogate subproblems, fast diagonalization, and inexact Krylov acceleration for $O(p)$ work per subdomain node.
4. Demonstrated robustness to high aspect ratios, element deformation, and irregular mesh topology.
5. Quantified large memory and runtime savings from parallel dynamic adaptation and demonstrated an initial incompressible-flow application.

2026S8

ChatGPT (September 2026)

Summary

This paper compares implicit data mapping, explicit device allocation, and unified-address-space operation for OpenMP Target GPU offloading. Default mapping is convenient for simple arrays but can repeatedly transfer buffers, hide slowdowns, allow host and device copies to diverge, and fail to translate nontrivial pointer structures safely. Long-lived mappings reduce transfers but retain synchronization and duplicated-memory risks; on shared-memory accelerators such as AMD MI300A systems, redundant host and device buffers can substantially reduce usable capacity. Allocating GPU-only storage with `omp_target_alloc` removes unused host copies and makes ownership and locality explicit. Kernels identify resident pointers with `is_device_ptr`, preserving abstraction for nested C++ objects because callers need not describe their internal buffers. The costs are nonstandard allocation, manual lifetime management, and especially intrusive Fortran pointer handling. Unified addressing is simpler when supported, but hardware, driver, and GPU-library assumptions limit portability. The paper provides conceptual guidance and illustrative C++ and Fortran code rather than quantitative benchmarks.

Contributions

1. Cataloged hidden transfer costs and synchronization hazards in default OpenMP Target buffer mapping.
2. Explained correctness and abstraction failures involving stale replicas and nested C++ pointers.
3. Identified the capacity cost of duplicate host and device buffers, including on shared-memory accelerators.
4. Demonstrated explicit GPU-resident allocation and kernel access using `omp_target_alloc` and `is_device_ptr`.

5. Compared explicit ownership with unified addressing and documented their portability, usability, and library-integration tradeoffs.

2026SBBM

ChatGPT (July 2026)

Summary

This study couples radially global GENE gyrokinetic simulations to the Tango transport solver to predict steady electron-temperature, ion-temperature, and density profiles for the 15 MA ITER baseline scenario from near the magnetic axis to $\rho_{\text{tor}} = 0.6$. Electromagnetic simulations moderate the strong density peaking found electrostatically and yield a fusion gain of $Q = 12.2$. Spectral analysis identifies microtearing, kinetic-ballooning, and Alfvénic ion-temperature-gradient modes at low binormal wavenumber as important transport channels that require high numerical resolution. Local flux-tube calculations recover the dominant modes but respond more stiffly to profile gradients than the global model. Realistic imposed toroidal rotation has little effect, whereas the safety-factor profile is decisive: flat $q \approx 1$ regions with weak magnetic shear destabilize core kinetic-ballooning modes and worsen confinement. Electron-temperature-gradient transport is eventually suppressed by slowly evolving zonal flows at ITER-like low collisionality. Fusion-born alpha-particle effects and broader reduced-model validation remain future work.

Contributions

1. Produced the first radially global gyrokinetic profile prediction for the 15 MA ITER baseline scenario using coupled GENE–Tango simulations.
2. Predicted pronounced but electromagnetically moderated density peaking and a fusion gain of $Q = 12.2$.
3. Identified the electromagnetic modes and numerical resolution needed to represent core turbulent transport.
4. Quantified differences in profile stiffness between global and local flux-tube simulations.
5. Showed that safety-factor shaping and long-timescale zonal-flow dynamics strongly affect confinement predictions.

2026SC

ChatGPT (September 2026)

Summary

The paper extends a spatio-spectral graph neural operator to learn solution maps for steady and time-dependent partial differential equations on irregular and varying geometries. The proposed π G-Sp²GNO combines local graph interactions with spectral features for multiscale behavior. Geometry enters through either a projection-based construction or a trainable encoder, allowing evaluation on geometries absent from training. Governing-equation residuals provide training signals without labeled solution data; spatial derivatives are computed by stochastic projection instead of differentiating the network with respect to coordinates. For time-dependent problems, a hybrid physics loss couples higher-order time marching to that projection. Benchmarks include changing geometries, chaotic dynamics, and long-time prediction, with favorable comparisons to other physics-informed neural operators. The authors note sensitivity to graph-neighbor and loss-weight choices, elevated errors near holes in a plate example, and limited handling of topology changes.

Contributions

1. Added two geometry-aware mechanisms to the spatio-spectral graph neural operator.
2. Trained solution operators from PDE residuals without supervised solution pairs.
3. Used stochastic projection to evaluate spatial derivatives on irregular point sets.
4. Developed a hybrid time-dependent physics loss using higher-order marching.
5. Tested generalization across geometries and documented errors near geometric singularities and topology limits.

2026SCL

ChatGPT (July 2026)

Summary

This paper develops a mass-lumping technique that enables fully explicit time integration for the shifted-basis extended finite element method (SBXFEM) in three-dimensional dynamic fracture simulations. SBXFEM avoids blending elements, but its enriched degrees of freedom create coupled mass matrices and conditioning issues that complicate explicit dynamics. The authors introduce a mass-scaling factor for the enriched mass-matrix contribution, derive an approximate critical time step, and analyze the resulting kinetic-energy error. At the single cracked-element level the scaling factor matters, but its influence becomes negligible when cracked elements are a small fraction of a larger structure. The method is implemented as an Abaqus Explicit VUEL and tested on 1D and 3D benchmarks, including Mode I and Mode II ductile crack propagation under large deformation. Comparisons with implicit and native Abaqus references show credible fracture behavior with lower computation time, while future work targets self-heating, micro-inertia, cohesive-phase refinements, and contact.

Contributions

1. Proposed enriched-mass scaling for explicit SBXFEM dynamics in cracked 3D elements.
2. Derived kinetic-energy error and critical-time-step behavior for the lumped enriched mass matrix.
3. Implemented the approach as an Abaqus Explicit user element.
4. Validated the method on increasingly complex dynamic fracture benchmarks.
5. Showed that enriched-mass errors diminish for large structures with few crack-containing elements.

2026SCSS

ChatGPT (July 2026)

Summary

This preprint organizes cold-fluid dispersion theory for the purely transverse Weibel instability across nonrelativistic and relativistic single- and multispecies counterstreaming plasmas. Known quartic relations are reduced to quadratics in ω^2 and used to map growth rates, characteristic scales, relativistic suppression, and mass-ratio effects. The resulting regime charts quantify when simpler formulas cease to be reliable and where oblique or two-stream modes may compete. Applied to a tabletop laser experiment, the model predicts an ion skin depth of about $31.7 \mu\text{m}$, close to the reported $31 \mu\text{m}$ filament spacing; its growth-rate and saturation-field estimates are explicitly treated as cold-fluid upper bounds. Analyses of two MMS bow-shock crossings place perpendicular magnetic spectral breaks near $kd_i = 1$. The paper emphasizes that warm-beam kinetics, nonlinear saturation, and background magnetic fields are required for quantitative growth predictions beyond these scale comparisons.

Contributions

1. Unified cold-fluid Weibel derivations for four velocity and species regimes.
2. Quantified regime-transition errors from velocity and particle-mass ratios.
3. Produced diagnostic maps for relativistic suppression and competing instability domains.
4. Compared the ion-inertial scale prediction with a tabletop laser experiment.
5. Tested $kd_i \approx 1$ against two MMS bow-shock magnetic spectra.

2026SDL

ChatGPT (July 2026)

Summary

The authors develop higher-order residual-based variational multiscale (VMS) schemes for incompressible turbulent flow using half-explicit Runge–Kutta (HERK) integration. Applying time discretization before space discretization respects the Navier–Stokes equations’ index-2 differential–algebraic structure and avoids the order reduction of conventional multistage treatments. Each Runge–Kutta stage produces a Darcy-like subproblem, allowing the subgrid stabilization parameter to be derived analytically by bubble enrichment without additional linearization assumptions. Fourier dispersion–dissipation analysis shows favorable spectral behavior. Algebraic scaling isolates all stage and time-step dependence, so a single symmetric block matrix can be reused across stages and steps, reducing memory and solve cost. Taylor–Green vortex tests improve energy-spectrum and dissipation predictions over generalized- α VMS, while open-cavity simulations recover the theoretical Hopf threshold and limit-cycle amplitude. Explicit convection and viscosity impose a stricter stability limit, but the method supplies accurate high-order dynamics for transient and turbulent flows.

Contributions

1. Formulated HERK-based VMS discretizations consistent with the index-2 incompressible-flow system.
2. Derived a subgrid stabilization parameter from bubble enrichment of the stagewise Darcy problem.
3. Established improved dispersion–dissipation behavior for the resulting space–time discretization order.
4. Reused one symmetric block matrix across Runge–Kutta stages and time steps.
5. Improved turbulent-spectrum, dissipation, Hopf-threshold, and limit-cycle predictions in benchmark flows.

2026SDLM

ChatGPT (July 2026)

Summary

Variable-density shallow-water flow coupled to solute transport must preserve hydrostatic equilibria while keeping water depth and concentration nonnegative. This paper reformulates the coupled system with auxiliary variables that expose a still-water equilibrium even when bottom topography and mixture density vary. A discontinuous Galerkin discretization with Lax–Friedrichs fluxes is designed to preserve that equilibrium exactly for the specified auxiliary-variable choices. The authors derive sufficient positivity conditions for cell averages and add a scaling limiter that enforces nonnegative depth and concentration without destroying high-order accuracy or the well-balanced state. One- and two-dimensional tests verify the designed convergence order, preservation of stationary solutions, wet/dry robustness, and consistent transport of constant concentration through rapidly changing flows. The present two-dimensional implementation uses structured grids and therefore targets simple domains; extension to unstructured meshes and complex geometries is deferred.

Contributions

1. Formulated coupled variable-density shallow-water and solute equations with equilibrium-compatible auxiliary variables.
2. Constructed a discontinuous Galerkin scheme that exactly preserves the targeted still-water state.
3. Proved sufficient conditions for nonnegative water depth and solute concentration.
4. Added a positivity limiter compatible with high-order accuracy and well-balancedness.
5. Verified accuracy, equilibrium preservation, wet/dry behavior, and transport consistency numerically.

2026SE

ChatGPT (July 2026)

Summary

This paper analyzes global, non-axisymmetric instabilities of differentially rotating cylindrical plasmas in incompressible Hall magnetohydrodynamics. The Hall term splits the ideal-MHD wave degeneracy into whistler and ion-cyclotron branches, both of which can extract energy from shear. Eigenvalue calculations with CYL_SPEC and initial-value simulations with NIMROD agree on growth rates and radial structures. At modest normalized ion skin depth, whistler modes can grow much faster than corresponding MHD modes; at stronger fields, long-wavelength ion-cyclotron modes remain unstable after ideal-MHD modes are stabilized because Hall physics weakens ion field-line bending. In the electron-MHD limit, the authors derive local and global stability conditions. For an azimuthal field, a zero-axial-wavenumber subset is driven by a global corotation-amplifier mechanism that local WKB dispersion relations miss. The identified parameter ranges may be relevant to weakly ionized accretion disks, although equilibrium stratification and nonlinear momentum transport are not included.

Contributions

1. Identified distinct global whistler and ion-cyclotron instability branches in rotating Hall-MHD plasma.
2. Showed that non-axisymmetric Hall modes can outgrow and outlast ideal-MHD instabilities.
3. Verified eigenvalue results with independent NIMROD initial-value calculations.
4. Derived electron-MHD local and global stability criteria including cylindrical curvature.
5. Discovered a global corotation-amplified whistler mode absent from local dispersion analysis.

2026SG

ChatGPT (July 2026)

Summary

This paper formulates a direct sampling method for locating time-dependent electromagnetic sources from sparse, multi-frequency far-field measurements. A temporal Fourier transform converts Maxwell's equations into frequency-domain systems with frequency-dependent sources, while suitable polarization vectors reduce the vector data to scalar indicators. For an impulsive source, combining measurements from two opposite observation directions identifies the unknown excitation time. Once that time is known, data from one direction recover the smallest slab between parallel hyperplanes containing the source support; intersecting slabs from several sparse directions approximates its Θ -convex hull. The construction also treats sources radiating over a finite interval when either the starting or ending time is known. The indicators require neither iterative forward solves nor a divergence-free source assumption. Three-dimensional synthetic experiments recover excitation times, shifted slabs, and multiple source geometries. Frequency integration provides substantial noise tolerance: the principal slab remains visible in the reported tests even with 80% multiplicative noise, although its boundary becomes more oscillatory.

Contributions

1. Extended direct sampling to frequency-dependent electromagnetic inverse sources governed by Maxwell's equations.
2. Constructed opposite-direction indicators that recover an impulsive source's excitation time.
3. Recovered directional support slabs and their multi-direction Θ -convex-hull intersection.
4. Extended the reconstruction to finite radiation intervals with one known temporal endpoint.
5. Demonstrated three-dimensional reconstruction without divergence-free assumptions and under high synthetic noise.

2026SGSH

ChatGPT (July 2026)

Summary

Using consistently analyzed H-mode measurements from 111 Alcator C-Mod, 43 ASDEX Upgrade, and 45 JET discharges, the study models the electron separatrix density $n_{e,\text{sep}}$ from engineering parameters. A log-link regression finds positive dependence on divertor neutral pressure and $P_{\text{SOL}}/R_{\text{geo}}$, negative dependence on toroidal field and machine size, and negligible plasma-current dependence, achieving $R^2 = 0.91$ and 19% normalized RMSE. An independently derived two-point-model scaling coupled to divertor geometry gives similar device constants and parameter trends. The resulting hybrid formula retains physics-based normalization and regression exponents, predicts the three-machine database within a factor of 1.5, and gives projections for ITER, SPARC, DTT, JT-60SA, and COMPASS-U consistent with available SOLPS results. Applicability is presently limited mainly to similar, mostly unseeded closed-divertor H-mode conditions.

Contributions

1. Built a cross-machine H-mode separatrix-density database using a consistent analysis procedure.
2. Quantified engineering-parameter dependencies and device-specific constants with generalized linear regression.
3. Derived an engineering-input two-point-model expression including divertor and magnetic geometry.
4. Combined physics-based normalization with empirical exponents into a formula accurate within a factor of 1.5.
5. Projected separatrix densities for five next-step devices and checked them against available SOLPS simulations.

2026SH

ChatGPT (July 2026)

Summary

The study evaluates a single-source fluid solver that combines nodal discontinuous Galerkin spatial discretization, explicit Runge–Kutta time integration, MPI domain decomposition, and Kokkos portability across CPUs and GPUs. The code solves linear advection and isothermal Euler equations in one, two, and three dimensions with spatial orders through eight and temporal orders through six. Fixed-accuracy experiments show that high-order discretizations require far fewer degrees of freedom and less runtime than third- or fourth-order alternatives despite greater per-step cost; classical fourth-order Runge–Kutta generally offers a practical temporal balance. Local element operations map effectively to many-core hardware, and the same implementation exhibits near-perfect weak scaling beyond 6,000 CPU cores and across 64 GPUs without architecture-specific rewrites. Hardware comparisons reveal a workload-size crossover: CPUs are faster and can use less energy for small grids, while GPUs dominate large cases above roughly 10^7 degrees of freedom. Energy estimates also expose a rebound effect, because newer GPUs require increasingly large problems to realize their efficiency advantage. The benchmarks support portable high-order simulation, but cover a limited equation set and defer direct comparison with pseudo-spectral solvers and architecture-specific tuning.

Contributions

1. Implemented a portable high-order fluid solver combining nodal discontinuous Galerkin, Runge–Kutta, MPI, and Kokkos.
2. Quantified the memory and runtime advantage of eighth-order spatial accuracy at fixed global error.
3. Demonstrated one code base across multiple CPU and GPU vendors without substantial architecture-specific tuning.
4. Measured near-perfect weak scaling beyond 6,000 CPU cores and on 64 GPUs.
5. Established problem-size-dependent CPU/GPU performance and energy crossovers, including a hardware-generation rebound effect.

Summary

The authors present grid generation and geometry computation for a standard field-aligned coordinate system (ψ, α, θ) that enables two-dimensional axisymmetric gyrokinetic simulations spanning the core, scrape-off layer, and private-flux regions of diverted tokamaks. Such coordinates are singular at X-points, where the poloidal field vanishes and the Jacobian $J_c = s(\psi)R/|\nabla\psi|$ diverges. Here θ is poloidal arc length along a flux surface normalized to 2π , and α is a field-line label giving $B = \nabla\psi \times \nabla\alpha$. Each topologically distinct region is split at the X-point into blocks, 12 for double null and 6 for single null, whose cell corners meet exactly at the X-point, leaving no gap. Because the discontinuous-Galerkin scheme of Gkeyll evaluates characteristics and fluxes only at interior and surface Gauss–Legendre points, no geometric quantity is evaluated at the X-point. Grid points come from root finding on a piecewise-polynomial $\psi(R, Z)$, and metric coefficients follow from analytic θ -derivatives and finite differences in ψ . Enclosed-volume, manufactured-solution Poisson, and Gaussian-advection tests converge at average orders above one, though more slowly near the X-point. A double-null STEP simulation remains well behaved at the X-points and conserves particles to machine precision. The authors note coarse grid spacing near the X-point and leave 3-D turbulence simulations to future work.

Contributions

1. Wrote the drift-kinetic gyrokinetic equation and Poisson equation in general computational coordinates and their axisymmetric limit with α ignorable, evolving $JJ_c f$ in conservative form.
2. Chose the parallel coordinate as normalized poloidal arc length, which, unlike the poloidal angle or Z , works on both open and closed field lines and allows arbitrarily shaped divertor plates.
3. Kept block boundaries consistent by sharing surface quadrature nodes and normal vectors and by rescaling fluxes with the block Jacobian normalizations at radial block boundaries, the only place they change.
4. Quantified the X-point penalty: the average enclosed-volume order drops from 3.31 to 1.42, and the high-resolution Poisson order from 1.44 away from the X-point to 1.2 on it, while advection past the X-point converges at 1.55.
5. Generated multi-block grids for double-null STEP and single-null ASDEX Upgrade equilibria and ran a 100 MW deuterium STEP simulation with ad hoc diffusivities chosen for a 2 mm heat-flux width, releasing the input files.

Comments

- Section 2: “At the X- and O-points of a tokamak configurations, however, we have $J_c = 0$ for field-line-following coordinates” conflicts with Eq. (4.4), $J_c = s(\psi)R/|\nabla\psi|$, and with Sections 4.1, 5, and 6.1, which state that the vanishing $\nabla\psi$ makes the Jacobian diverge; the evident intended reading is that J_c diverges ($J_c^{-1} \rightarrow 0$).
- Section 3.3, Eq. (3.26): the last term of z^3 is printed as $+\frac{b_2}{qJ_c B_{\parallel}} \frac{\partial H}{\partial z^1}$, but taking $i = 3$ in Eq. (3.20) gives $\epsilon^{321} b_2 \partial H / \partial z^1$ with $\epsilon^{321} = -1$; with the printed sign, Eqs. (3.25)–(3.27) do not conserve H , so the evident intended sign is minus.
- Section 5, Eq. (5.1): the equation for $F = JJ_c f$ is written in advective form, $\partial_t F + z^i \partial F / \partial z^i + v_{\parallel} \partial F / \partial v_{\parallel} = 0$, whereas Eq. (3.16) is the conservative form, and the weak form Eq. (5.3), said to follow from Eq. (5.1) by integration by parts, is that of the conservative form; the conservative form is evidently intended.
- Fig. 1(a): the sub-caption says tracing “starts in each region at $Z_{\text{lower}}(\psi)$ (marked in blue) and stops at $Z_{\text{upper}}(\psi)$ (marked in red)”, but the panel legend marks the upper boundary in blue and the lower boundary in red, as does the Fig. 1(b) sub-caption.
- Section 7.1, Table 1: the caption gives the average order of convergence without the X-point as 3.37, whereas the text states 3.31, which is the mean of the tabulated orders (0.58, 5.02, 2.16, 5.47).
- Section 7.3: the simulation is said to run for “ 2.0×10^6 s”, but Fig. 7(b) is labelled $t = 2.0\text{e}-06$ s, and only 2×10^{-6} s moves the bump the stated 0.2 m at $v_0 = 10^5$ m s $^{-1}$.
- Section 7.3: the initial and final bumps are centred by $(Z - 2.1)$ and $(Z - 1.9)$, i.e. at $Z = +2.1$ and $+1.9$, which would move them in $-\hat{Z}$ against $v = v_0 \hat{Z}$, whereas Fig. 7 shows them at $Z \approx -2.1$ and -1.9 beside the X-point at $(2, -2)$; the evident intended reading is $(Z + 2.1)$ and $(Z + 1.9)$.
- Section 7.3: both “Gaussian” profiles are written $n = e^{[(R-2.2)^2 + (Z \mp \dots)^2]/0.1^2}$ without the minus sign in the exponent, which makes them grow away from the centre, contrary to the bumps shown in Fig. 7.

2026SJKY

ChatGPT (July 2026)

Summary

This paper develops a multi-species nonlinear Rosenbluth-Fokker-Planck collision operator for discontinuous Galerkin kinetic plasma simulations. Distribution functions are represented with DG basis functions, while Rosenbluth potentials are computed with B-spline finite elements in a reduced two-dimensional velocity space aligned with a strong background magnetic field. To handle mass-disparate species, the potential is split into analytic and correction parts, and expensive Gaussian quadrature contributions are stored as precomputed integration matrices. Because standard upwinding is incompatible with those matrices, the method uses an HLL flux formulation for advective boundary terms. Mass conservation follows from the divergence structure, while added correction operators enforce momentum, energy, and stable collisional equilibrium. Benchmarks against hydrogen, electrons, tungsten, five-species relaxation, implicit large-time-step tests, and a circular-tokamak bootstrap-current case show good agreement with theory and near-ideal scaling with species count. The paper leaves adaptive velocity grids for strongly varying temperatures and flows as future work.

Contributions

1. Formulated a DG-compatible nonlinear Rosenbluth-Fokker-Planck operator for multi-species kinetic plasma simulations.
2. Combined DG distribution functions with finite-element Rosenbluth potentials and precomputed quadrature matrices.
3. Introduced HLL fluxes compatible with matrix precomputation for advective terms.
4. Added correction operators for momentum, energy, and stable interspecies equilibrium under large mass ratios.
5. Validated accuracy, conservation, efficiency, implicit stepping, and bootstrap-current behavior through numerical benchmarks.

2026SJNB

Claude (September 2026)

Summary

Sharp ridges, creases of large principal curvature along which field lines collimate, appear mostly on the inboard side in compact quasisymmetric (QA) stellarators. The authors extend their vacuum ridge theory to ideal magnetohydrostatic equilibria with pressure and current by expanding the strong-quasisymmetry generalized Grad-Shafranov system in the deviation from axisymmetry, using a basis that stays nondegenerate in that limit. The first-order system is linear but overdetermined, and a volumetric solution is deemed unlikely; defining ridges as minima of $|\nabla\psi|$ and imposing $\nabla\psi_0 \cdot \nabla\psi_1 \approx 0$ closes it on a flux surface. For many field periods N , local periodic solutions exist only where a discriminant is nonnegative, which confines $\mathcal{A}_0 = \nabla\psi_0 \cdot \nabla|\nabla\psi_0|^2/(2|\nabla\psi_0|^2)$ to a band and hence the Gaussian curvature $\mathcal{K}$ of the axisymmetric background to a mostly negative range whose positive part is smaller by $O(\varepsilon_p^2)$, with $\varepsilon_p = |\nabla\psi_0|/F$; the perturbation therefore vanishes on the outboard side, where $\mathcal{K}$ is largest. A ballooning-type analysis for arbitrary N gives the same condition. Numerical solutions on a Solov'ev tokamak surface are localized inboard, and in SIMSOPT-optimized QA configurations with $n_{fp} = 2, 3, 4$, and 6, ridges lie where $\mathcal{K}$ is strongly negative and $\nabla\psi_0 \cdot \nabla\psi_1$ is small. The analysis is local, linear, and single-surface.

Contributions

1. Recast the strong-quasisymmetry Grad-Shafranov system in the $(u, \nabla\psi, B)$ basis and grouped its first-order expansion so that u_1 follows from ψ_1 and Λ_1 by on-surface relations, the generalized Grad-Shafranov equation fixes the radial gradient of Λ_1 , and a surface-averaged compatibility condition fixes the magnetic shear.
2. Derived the large- N ridge ordering $\Lambda_1^{(0)} = 0$ and $\nabla\psi_0 \cdot \nabla\psi_1/|\nabla\psi_0|^2 = O(N^{-1})$, with radial derivatives of other perturbations of order N^2 for irrational and order one for rational rotational transform, the latter requiring

- $\mathcal{C}_\psi = 0$ and giving X-point ridges along closed field lines on both the inboard and outboard sides.
3. Reduced the lowest-order on-surface problem to a hyperbolic Laplace equation from conjugate-net theory, whose local solutions $\exp iN(k_\theta \theta \pm \phi)$ obey a quadratic dispersion relation for the ridge pitch k_θ , and showed that in the ballooning formulation zeros of $\mathcal{C}_1$ are regular singular points with indices 0 and $1 + 2ix_0$ that act as infinite barriers localizing the solutions.
 4. Expressed the Gaussian curvature of an axisymmetric flux surface through ψ_0 as $\mathcal{K} = (\partial_R \psi_0 / R)(\partial_R^2 \psi_0 + \partial_z^2 \psi_0 - \mathcal{A}_0) / |\nabla \psi_0|^2$, linking it to the Grad–Shafranov operator and identifying $\mathcal{A}_0 = 0$ with the Aronsson equation.
 5. Posed the overdetermined ridge equations as minimizing $|\mathbf{A}\mathbf{x}|$ at fixed $|\mathbf{x}|$, solved as the eigenproblem $\mathbf{A}^\top \mathbf{A}\mathbf{x} + \lambda \mathbf{x} = 0$ with a Fourier–Galerkin discretization for $N = 8$ and linear plus mapped Zienkiewicz infinite elements for $k_\alpha = 4$, with normalized residuals from 6×10^{-4} to 3.8×10^{-3} .

Comments

- §4.3: the text states that “ $\delta_J = -(R^2 p' + FF')$ from lowest-order force-balance (3.2)”, but (4.18) defines $2\delta_J = \Delta^* \psi_0$ and (3.2) gives $\Delta^* \psi_0 = -(FF' + R^2 p')$, so $\delta_J = -(R^2 p' + FF')/2$; the vacuum limit $\delta_J = 0$ used afterwards is unaffected.
- §4.3, Eq. (4.22): in the vacuum limit (4.21) gives the bounds $\mathcal{A}_{00} \pm |\mathcal{A}_{01}| = \delta_R(1 + \varepsilon_p^2) \pm |\delta_R| \sqrt{1 + \varepsilon_p^2}$, whose upper value for $\delta_R > 0$ expands as $\varepsilon_p^2 \delta_R(2/\varepsilon_p^2 + 3/2 + O(\varepsilon_p^2))$, not the printed $\varepsilon_p^2 \delta_R(1/\varepsilon_p^2 + 3/2 + O(\varepsilon_p^2))$; through (4.23c) the lower curvature bounds in (4.24) and (4.25) should then read $-(\varepsilon_p \delta_R / F)^2(2/\varepsilon_p^2 + 1/2 + O(\varepsilon_p^2))$ and $-2(R_0 \delta_R / F)^2$.
- §4.5, Eq. (4.35): the characteristic variable is printed as $\varphi \equiv \phi - \int (dR/R)(F/|\nabla \psi_0|) \partial_z \psi_0$, but the operator of (4.31), $\partial_R + (F/R)(\partial_z \psi_0 / |\nabla \psi_0|^2) \partial_\phi$, has characteristics $d\phi/dR = F \partial_z \psi_0 / (R |\nabla \psi_0|^2)$, and only $|\nabla \psi_0|^2$ makes φ dimensionless; the intended denominator is $|\nabla \psi_0|^2$.
- §5, Eq. (5.5): eliminating $\psi_1^{(0)}$ between (5.4a) and (5.4b) gives the first-derivative term $2i(\mathcal{C}_2 - \mathcal{C}_1)k_\alpha^{-1} \partial_\parallel \Lambda_1^{(1)}$, not the printed $2i(\mathcal{C}_1 - \mathcal{C}_0)k_\alpha^{-1} \partial_\parallel \Lambda_1^{(1)}$; only $2i(\mathcal{C}_2 - \mathcal{C}_1)$ yields the phase S_c with integrand proportional to $(\mathcal{C}_2 - \mathcal{C}_1)/\mathcal{C}_1$ in (5.6), the potential $\mathcal{D}_B = \mathcal{C}_2^2 - \mathcal{C}_1 \mathcal{C}_0$ in (5.7), and the phase derivative $(\mathcal{C}_1 - \mathcal{C}_2 \pm \sqrt{\mathcal{D}_B})/\mathcal{C}_1$ in (5.9), so $2i(\mathcal{C}_2 - \mathcal{C}_1)$ is intended.
- §5, Eq. (5.9): (5.4a) with the printed phase S gives $\psi_1^{(0)} = -i(\mathcal{C}_2 \mp \sqrt{\mathcal{D}_B})\Lambda_1^{(1)} / (F\mathcal{C}_0\mathcal{C}_1)$, whereas (5.9) prints $\psi_1 \approx -i\Lambda_1/\mathcal{C}_0$ for the same amplitude $\chi_{1a} e^{ik_\alpha S} \sqrt{\mathcal{C}_1/\sqrt{\mathcal{D}_B}}$, without the factor $(\mathcal{C}_2 \mp \sqrt{\mathcal{D}_B})/(F\mathcal{C}_1)$; unresolved.
- §5, Eq. (5.11): for $\mathcal{C}_1 = 0$, (5.4) gives $\partial_\parallel S_{\mathcal{C}_1} = 1 - \mathcal{C}_0/(2\mathcal{C}_2)$ and $\psi_1^{(0)} = -i\Lambda_1^{(1)}/(2F\mathcal{C}_2)$, and (5.5) as printed gives $\partial_\parallel S_{\mathcal{C}_1} = 1/2 - \mathcal{C}_2/\mathcal{C}_0$, but the printed $S_{\mathcal{C}_1}$ has $\partial_\parallel S_{\mathcal{C}_1} = \mathcal{C}_2/\mathcal{C}_0 - 1/2$, and even for this phase (5.4a) gives $\psi_1^{(0)} = +i\Lambda_1^{(1)}(\mathcal{C}_2/\mathcal{C}_0 - 3/2)/(F\mathcal{C}_0)$, opposite in sign to the printed expression; unresolved.
- §6.1: $R_0 = 1$ m is called “the radius of the magnetic axis”, and $\theta = \arctan[z/(R - R_0)]$ is referred to it, but (6.1) with the printed C, d_1, d_2, d_3 has $\partial_R \psi_0 = 0$ on $z = 0$ at $R = \sqrt{-2d_2/(1/2 + 4d_3)} = 1.050$ m; 1 m is the midpoint of the midplane boundary intercepts $R = 0.680$ and 1.320 m, and which value defines θ is unresolved.
- §6.2: “the results or our linear perturbation results shown in Figure 4 agree with the fully nonlinear ridges shown in Figure 3” reverses the figures, since Fig. 3 shows contours of $\psi_0 + 0.1\psi_1$ from the linear ballooning solutions of Fig. 2 and Fig. 4 shows the optimized configurations; the linear results are in Fig. 3 and the nonlinear ridges in Fig. 4.
- Fig. 4 caption: the sections are said to be at “ $(2/n_{fp})\phi = 0, \pi/2, \pi$ and $3\pi/2$ ”, which for $n_{fp} = 4$ gives $\phi = 0, \pi, 2\pi, 3\pi$, all multiples of the field period $\pi/2$, and for $n_{fp} = 6$ only two distinct phases within a period, yet each panel shows four distinct boundaries; the actual angles are unresolved.
- Figs. 5 and 7 captions: the quantities are said to be shown “on half a field period”, but the horizontal axes span $\phi/\pi \in [0, 1], [0, 2/3], [0, 1/2]$, and $[0, 1/3]$ for $n_{fp} = 2, 3, 4$, and 6, that is, $2\pi/n_{fp}$, one full field period, as the §6.2 text says (“on a field period”); the captions also name k_2 as the bottom row, whereas the bottom row is $1/|\nabla \psi|$.
- Fig. 6 caption: the plotted combination is defined as “ $\Lambda_1 = R\mathbf{u} \cdot \nabla \psi_0 / Z - Z\mathbf{u} \cdot \nabla \psi_0 / R$ ”, which is $(R/Z - Z/R)\mathbf{u} \cdot \nabla \psi_0$, a multiple of $\mathbf{u} \cdot \nabla \psi_0 = -\partial_\phi \psi_1$ at first order by (C 10a), not the $\Lambda_1 \equiv \partial_z \psi_0 \Lambda_{1R} - \partial_R \psi_0 \Lambda_{1z}$ of (3.5); the quantity plotted as Λ_{1T} is unresolved.
- Appendix E, Eq. (E 6): $\mathcal{K} = \kappa_1 \kappa_2$ with $\kappa_2 = \partial_\theta Z / (R \sqrt{(\partial_\theta R)^2 + (\partial_\theta Z)^2})$ from (E 5) carries a factor $1/R$ that

the printed $\mathcal{K}$ lacks; (E 8) contains it as $\partial_R \psi_0 / R$, so the intended (E 6) has R in the denominator.

2026SKIS

ChatGPT (July 2026)

Summary

The paper addresses conservation errors caused by child-to-parent transfer during adaptive coarsening on balanced octree meshes with continuous Galerkin finite elements. Whereas parent-to-child interpolation preserves integral quantities, standard injection discards fine-grid degrees of freedom and can accumulate mass drift over long simulations. The proposed operator first performs a local tensor-product L^2 restriction of fine-element quadrature values to each coarse parent, then recovers coarse nodal degrees of freedom through a global L^2 projection. Manufactured diffusion tests retain optimal second- and third-order convergence for linear and quadratic elements while conserving mass to numerical precision. Two- and three-dimensional Cahn–Hilliard tests, with polynomial and Flory–Huggins free energies, and a Cahn–Hilliard–Navier–Stokes rising-bubble test likewise eliminate injection-induced mass drift without disrupting energy decay. Coarsening energy mismatch falls by nearly two orders of magnitude. The implementation is communication-local before the mass solve and adds about 10% cost for Cahn–Hilliard and 7% for the coupled flow case.

Contributions

1. Identified injection-based coarsening, rather than refinement, as the systematic conservation failure in continuous-Galerkin octree AMR.
2. Derived a dimension-independent tensor-product L^2 restriction followed by a global mass-matrix projection to coarse nodes.
3. Retained optimal convergence for linear and quadratic elements while reducing manufactured-solution mass drift to numerical precision.
4. Validated robust conservation and physical energy decay across Cahn–Hilliard and coupled flow problems.
5. Provided a distributed-octree implementation with communication-local restriction and measured overheads of about 10% and 7%.

2026SLMW

ChatGPT (September 2026)

Summary

ORB5X is a C++17/Kokkos translation of the global electromagnetic gyrokinetic particle-in-cell code ORB5. It retains ORB5’s variational gyrokinetic model, δf marker particles, B-spline field representation, Fourier filtering, MPI toroidal decomposition and cloning, and HDF5 diagnostics while replacing Fortran, OpenMP, and OpenACC data and kernels with typed C++ objects and Kokkos Views. The authors used coding agents for bounded translation tasks, followed by human review and staged CPU, GPU, and multi-GPU validation. ITG, ITPA, and chirping benchmarks reproduce ORB5 fields and diagnostics to near machine precision. Strong-scaling tests extend to 2,048 AMD GPUs on LUMI-G and Alps, with near-ideal scaling in selected regimes and execution across laptops and multiple clusters. The work establishes a portable, maintainable baseline rather than a new physical model: the original Fortran code remains faster where available, and some FFTW, HDF5, LAPACK, and MPI paths still stage data on the host.

Contributions

1. Reimplemented ORB5’s electromagnetic gyrokinetic PIC algorithm in C++17 with Kokkos portability.
2. Preserved the established MPI decomposition, particle cloning, and HDF5 diagnostics across the translation.
3. Defined a human-supervised agentic workflow with bounded patches and progressively broader validation.
4. Demonstrated numerical agreement with ORB5 on ITG, ITPA, and chirping benchmarks.
5. Evaluated portability and strong scaling through 2,048 GPUs while identifying remaining host-staged operations.

2026SNMD

ChatGPT (July 2026)

Summary

Slendebroek and coauthors compare positive-triangularity and negative-triangularity tokamak power-plant designs using constrained multi-objective optimization in FUSE. More than 200,000 integrated design evaluations explore capital-cost minimization versus operational reliability while enforcing net electric power, tritium breeding, power exhaust, and flat-top constraints. Both shapes can reach similar cost-performance Pareto fronts, but through different risks. Positive-triangularity designs tolerate substantial pedestal degradation by adjusting current and profiles, yet their viable space is squeezed between L-H threshold requirements and divertor power-exhaust limits, favoring larger, lower-field machines. Negative-triangularity designs avoid those constraints, enabling compact high-field solutions that align with HTS magnet development but carry higher baseline capital cost. Sensitivity studies show that uncertainty in L-H threshold scaling or divertor power handling can shift which configuration is favored, making configuration choice a risk-informed design decision.

Contributions

1. Performed a large constrained optimization study of tokamak power-plant design space.
2. Compared positive- and negative-triangularity concepts under common engineering constraints.
3. Showed that positive triangularity is limited mainly by L-H threshold and exhaust margins.
4. Identified compact high-field negative-triangularity designs enabled by relaxed constraints.
5. Quantified how physics and exhaust uncertainties can change the preferred configuration.

2026SNSK

ChatGPT (July 2026)

Summary

This paper constructs an L^2 -conforming discontinuous Galerkin discretization of Liu's consistent splitting scheme for incompressible Navier–Stokes flow. Each time step decouples pressure and velocity into one pressure Poisson solve followed by one vector convection–diffusion–reaction solve. Consistent pressure boundary conditions cover Dirichlet, open, and traction boundaries, avoiding the artificial pressure boundary layers and temporal splitting errors of standard projection methods. Symmetric interior penalty handles second derivatives, while linearly implicit convection relaxes explicit CFL restrictions without requiring nonlinear solves. A Leray projection together with divergence and normal-continuity penalties improves mass conservation. Manufactured-solution tests recover optimal spatial and temporal rates, including higher-order BDF formulas, and cylinder-flow and three-dimensional Taylor–Green-vortex benchmarks agree with established solvers. The analysis also identifies a lowest-order accuracy limitation when the viscous curl–curl term is under-resolved.

Contributions

1. Formulated an L^2 discontinuous Galerkin consistent-splitting method for incompressible flow.
2. Derived consistent pressure boundary treatments for closed, open, and traction boundaries.
3. Combined linearly implicit convection with decoupled pressure and momentum solves.
4. Improved discrete mass conservation using Leray, divergence, and normal-continuity stabilization.
5. Verified high-order convergence and benchmark accuracy for two- and three-dimensional flows.

2026SP

Claude (September 2026)

Summary

The authors analyse two-dimensional incompressible flow past two side-by-side square cylinders at Reynolds number 200 with a gap equal to the cylinder side, where a flapping gap jet disrupts vortex shedding. Finite-volume simulations and spectral proper orthogonal decomposition (SPOD) identify three frequency bands: jet flapping ($St \approx 0.063$), vortex shedding ($St \approx 0.168$), and a subharmonic linked to downstream vortex merging. Linearising the Navier–Stokes equations about the chaotic trajectory, restricted to two-dimensional perturbations, they compute six Lyapunov exponents and the corresponding covariant Lyapunov vectors (CLVs), the generally non-orthogonal directions of asymptotic growth and decay. The flow has two positive exponents, the largest about 0.1. The leading CLV is concentrated in the near wake and, through SPOD, carries the flapping and shedding frequencies; the second peaks farther downstream and captures the subharmonic merging instability. Angles between CLVs show that hyperbolicity is violated mainly by near-tangency of neutral and stable directions, in up to 2.4% of samples. Global linear stability analysis of the time-averaged flow yields a neutral mode matching the leading CLV’s shedding structure and frequency but misses the growth of that direction, jet flapping, and the subharmonic. A Reynolds-number sweep shows the Lyapunov spectrum marking the transitions to periodic, chaotic, and flip-flop regimes.

Contributions

1. Computed Lyapunov exponents and covariant Lyapunov vectors for a chaotic two-cylinder wake, finding two unstable directions with footprints in the near wake and farther downstream and the most stable computed direction active near the outlet.
2. Applied SPOD to covariant Lyapunov vectors, which the authors describe as the first such use, linking the leading vector to jet flapping and shedding and the second to subharmonic vortex pairing.
3. Assessed hyperbolicity from the statistics of angles between vectors, finding violations concentrated in the neutral and stable subspaces and near-orthogonality for vectors with non-overlapping footprints.
4. Compared Lyapunov and mean-flow global stability analyses explicitly for a chaotic flow, showing that the mean-flow analysis labels as neutral a direction that is unstable.
5. Used the three leading exponents to identify transitions at $Re \approx 67$ (periodic), 85 (chaotic), and 125 (flip-flop, second positive exponent), and released open-source code for the exponent and vector computations.

2026SQHZ

ChatGPT (July 2026)

Summary

This paper proposes an adaptive-rank solver for multiscale BGK kinetic equations based on greedy sampling and adaptive cross approximation. Instead of forming low-rank decompositions of nonlinear local Maxwellians, the method identifies important rows and columns of the solution matrix, advances only those samples with a high-order semi-Lagrangian local solver, and reconstructs the full solution through cross approximation. A locally macroscopic conservative correction enforces mass, momentum, and energy conservation by coupling the low-rank kinetic update to a high-order macroscopic system. The macroscopic equations use stiffly accurate DIRK time stepping and matrix-free Jacobian-free Newton-Krylov solves, with the provisional low-rank solution serving as a closure and initial guess. The paper proves a conditional asymptotic-preserving property and tests smooth, discontinuous, and mixed-regime problems spanning several Knudsen-number orders of magnitude. Results show strict conservation, high-order behavior, large-time-step capability, and linear scaling in one benchmark, with high-dimensional extensions and preconditioning left for future work.

Contributions

1. Introduced a greedy adaptive-rank BGK solver that avoids explicit low-rank Maxwellian decompositions.

2. Used adaptive cross approximation to update selected rows and columns and reconstruct the kinetic solution.
3. Enforced mass, momentum, and energy through a locally macroscopic conservative correction.
4. Coupled semi-Lagrangian kinetic updates with DIRK and matrix-free Newton-Krylov macroscopic solves.
5. Demonstrated conservation, conditional asymptotic preservation, and performance across kinetic-fluid regimes.

2026SRHK

Claude (September 2026)

Summary

The authors develop a time integrator for a zonal quasineutral Vlasov model of strongly magnetized plasmas: kinetic ions in a constant magnetic field with adiabatic electrons, supplemented by the condition that no net charge is transported between flux surfaces (planes normal to x in slab geometry), $\partial_t \langle n \rangle_F = 0$. The electrostatic potential is split into a part $\tilde{\phi}$ with zero flux-surface average, given by the adiabatic-electron relation, and a flux-surface average $\langle \phi \rangle_F(x)$ chosen so that one semi-Lagrangian Strang splitting step (half velocity advection, spatial advection, half velocity advection) preserves $\langle n \rangle_F$. Tracing the density change back through the splitting with a Taylor expansion in the characteristic shifts gives an equation for $\langle \phi \rangle_F$ accurate to $O(h^3)$ per step, derived explicitly for a velocity grid rotating at the ion cyclotron frequency and outlined for a fixed grid. The BSL6D implementation uses high-order Lagrange interpolation and Fourier derivatives. In a 1D–2V equilibrium test, $\langle n \rangle_F$ stays constant after an initial jump and the cross-surface particle flux converges at second order in the time step. In 2D–2V runs, simulated ion Bernstein wave spectra agree qualitatively with analytic dispersion relations. The authors observe late-time noise growth and leave stability and well-posedness analysis open.

Contributions

1. Extended the adiabatic-electron quasineutral model of R  th and Hallatschek with a net-zero charge transport constraint enforced through a flux-surface-averaged potential, which also produces zonal radial electric fields.
2. Derived an explicit equation for $\langle \phi \rangle_F$ in terms of the density, current, stress tensor, and a third-order moment of the distribution by combining a multi-index Taylor expansion of the spatial shift with an integral transformation for the velocity shift.
3. Showed that the final velocity half-step leaves the density unchanged, so only the first half-step and the spatial advection enter the constraint, while keeping the full Strang step to retain second-order convergence.
4. Derived the linear dispersion relations of the constrained model for neutralized ion Bernstein waves along the flux surfaces and pure ion Bernstein waves across them.
5. Reproduced pure ion Bernstein waves in simulation, which the authors state had not been done before, and maintained the continuity condition for more than 20000 time steps.

Comments

- Section 2, text after Eq. (1): the density is defined by “ $n_0 - n(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{v}, t) d^3v$ ”, which conflicts with the same sentence’s split of f into a background n_0 and a perturbation n , with Algorithm 1 ($\phi = \int f d^3v - \langle \int f d^3v \rangle_F + \langle \phi \rangle_F$), and with Section 4.1 ($n(x, t) = \int f d^2v$); the evident intended reading is $n_0 + n = \int f d^3v$.
- Section 3.1, first paragraph: “In section 4 we observe, that the rotating grid is superior with respect to accuracy and stability against the discretization on the physical domain”, but Section 4 presents only rotating-grid results (Figs. 1 and 2), and the end of Section 3.1 attributes this superiority to Ref. [25].
- Section 4.1: the potential said to maintain the equilibrium, $\phi(x) = n(x) - n_0 = \int f d^2v - n_0$, has the opposite sign to that required by Eq. (24) for the initial state $(1 + 0.01 \sin(k_j x))f_M$ to remain stationary, since the pressure and electric forces cancel only for $\partial_x \phi = -(T/e) \partial_x \ln n$; unresolved.
- Section 4.1: “ $k_j = jL_x/2\pi$ ” has the dimension of length and does not give a mode periodic on $[0, L_x]$; the evident intended reading is $k_j = 2\pi j/L_x$.
- Appendix A.2, Eqs. (A.17)–(A.18): the stated intermediate factors do not produce the normalization of Eq. (A.20): $\int_0^{2\pi} e^{i(p'-p)\varphi} d\varphi$ equals $2\pi \delta_{p,p'}$, not $\delta_{p,p'}$; the prefactor $n_0/(\pi^{3/2} v_{th}^3)$ does not normalize $e^{-v^2/(2v_{th}^2)}$; and

$\int_0^\infty e^{-v_\parallel^2/(2v_\text{th}^2)} dv_\parallel = \sqrt{\pi/2} v_\text{th}$, not $\sqrt{\pi} v_\text{th}$, with the $v_\parallel$ integral needing the full real line; together they give n_0/π in place of n_0 .

- Appendix A.3, Eq. (A.21): the step from $\hat{\phi}_1 = \hat{\phi}_1 e^{-k_y^2} \sum_p \frac{p}{\omega-p} I_p(k_y^2)$ uses the identity “ $\sum_{p=-\infty}^\infty I_p(x) = 1$ ”, whereas $\sum_p I_p(x) = e^x$; with the correct identity the preceding line gives $e^{-k_y^2} \sum_p \frac{\omega}{\omega-p} I_p(k_y^2) = 2$, not the printed 0, which would instead follow from $\hat{\phi}_1 = -\hat{n}_1$.

2026SS

Claude (September 2026)

Summary

The authors propose a multi-resolution (MR) limiter for Runge–Kutta discontinuous Galerkin schemes for hyperbolic conservation laws that assigns each cell a polynomial degree matching local smoothness, rather than classifying cells as good or troubled (flagged for limiting). A baseline IS^0 is the minimum, over substencils containing the target cell, of their largest cell-average difference from it. From the DG degree $m = k$ downward, IS^m sums the absolute differences between the averages of the degree- m truncated polynomial over adjacent cells and their cell averages; while $IS^m > C_m IS^0$, the degree- m modes are removed, preserving the cell average, with a minmod slope reconstruction if no degree $m \geq 1$ passes. Neighbours enter only through cell averages, so limiting them leaves the indicators unchanged, as do affine coordinate maps, and affine rescaling of the solution leaves the comparisons unchanged. With $C_m \geq 2.86$ one-dimensional advection converges optimally for $k = 1$ to 6, and with $C_m = 4$ two-dimensional L^1 errors do for $k \leq 3$, and Euler problems with shocks and strong rarefactions, on two-dimensional quadrilateral and triangular meshes, show essentially no oscillations, often at intermediate degrees. The authors report suboptimal two-dimensional L^∞ orders for $k = 1$ and round-off-induced polynomial-degree reductions in constant regions, and suggest degree-dependent C_m , less dissipative slope limiting, and less diffusive fluxes.

Contributions

1. Built the triangular-mesh baseline from the substencils $\{K_0, K_l, K_{l1}, K_{l2}\}$, $l = 1, 2, 3$, of the ten-cell stencil of edge neighbours K_l and their neighbours, omitting substencils outside the domain, and the fallback linear polynomial from the componentwise minmod of three slopes, each matching the averages of K_0 and a pair of edge neighbours, so that only the baseline stencil is non-compact; polygonal and polyhedral meshes are stated to need only a baseline adapted to the number of neighbours.
2. Contrasted the reconstruction, affine-coordinate, and scale invariance of the MR indicators with the KXRCF indicator of Krivodonova et al., an inflow-edge jump divided by $h^{(k+1)/2} |\partial K_0^-| \|p_0\|$, which has none of them, and the Fu–Shu indicator, summed deviations of the neighbour polynomials’ K_0 -averages from $\bar{p}_0$ divided by $\max_j |\bar{p}_j|$, which is only affine-coordinate invariant, and kept the accuracy and robustness of combined-wave advection with the solution scaled by one factor per k between 10^{-10} and 10^{10} .
3. Used $\Delta t = 0.25h^{4/3}$, $0.3h^{5/3}$, $0.5h^2$, and $0.4h^{7/3}$ for $k = 3$ to 6 so that SSP-RK3 does not limit the spatial order, and quadruple precision for $k \geq 4$, in one-dimensional advection of $\sin(\pi x)$, where no cell lowers its degree on meshes with $h \leq 0.1$ once $C_m \geq 2.86$, the thresholds for $k = 2$ to 5 being 0.08, 0.14, 0.012, and 3.7×10^{-4} .
4. Reported two-dimensional advection L^∞ orders as low as 1.42 on uniform quadrilaterals and 0.56 on quasi-uniform triangles for $k = 1$, attributed to IS^1 , a second-order quantity, possibly exceeding the baseline at extrema and to $C_m = 4$ being suboptimal in two dimensions, and orders 3.33, 5.03, and 3.34 for $k = 3$ on triangles, where no cell lowered its degree and the unlimited scheme gave identical errors, tentatively attributed to non-uniform refinement of the triangular meshes.
5. Took the density as indicator variable for the Euler equations, applied the minmod fallback to characteristic variables with a single rotated characteristic decomposition along the density gradient, and applied the Zhang–Shu positivity-preserving limiter after the MR limiter; with 250 cells the $k \geq 2$ blast-wave solutions almost recover the density peak, and in double Mach reflection small Kelvin–Helmholtz vortices appear for $k \geq 2$ on 960×240 quadrilaterals and already for $k = 1$ on triangles with boundary edge length $1/240$.

Comments

- Section 3.1, Eq. (8), and Section 3.2, Eq. (13): the indicators are printed as $IS^m = \frac{1}{h_{j-1}} \left| \int_{K_{j-1}} p_j^m(x) dx - \bar{p}_{j-1} \right| + \frac{1}{h_{j+1}} \left| \int_{K_{j+1}} p_j^m(x) dx - \bar{p}_{j+1} \right|$ and $IS^m = \sum_{l=1}^3 \frac{1}{|K_l|} \left| \int_{K_l} p_0^m(x, y) dx dy - \bar{p}_l \right|$, with the prefactor also dividing the cell averages, so (8) is not equivalent to the form stated after Algorithm 1, $\frac{1}{h_{j-1}} \left| \int_{K_{j-1}} (p_j^m(x) - p_{j-1}^m(x)) dx \right| + \frac{1}{h_{j+1}} \left| \int_{K_{j+1}} (p_j^m(x) - p_{j+1}^m(x)) dx \right|$, and (13) satisfies neither (18) for $\sigma \neq 0$ nor the invariance under affine coordinate maps claimed in Section 3.3; the evident intended terms are $\left| \frac{1}{h_{j\pm 1}} \int_{K_{j\pm 1}} p_j^m(x) dx - \bar{p}_{j\pm 1} \right|$ and $\left| \frac{1}{|K_l|} \int_{K_l} p_0^m(x, y) dx dy - \bar{p}_l \right|$, the form used in the FS indicator (17).
- Section 3.1, first item of the analysis after Algorithm 1: “when the solution has at most one discontinuity on the global stencil S , at least one of S_L and S_R is smooth, and we always have $IS^0 = \min(IS_L, IS_R) = \mathcal{O}(h_j)$ ”, but by (5)–(6) both substencils contain K_j , so a single discontinuity inside K_j lies in both and makes IS_L, IS_R , and IS^0 generally of the order of the jump; for that cell the second conclusion, “ IS^0 is a higher-order quantity than IS^m ($m \geq 1$)”, does not follow, and the same applies to the statement for S^Δ in Section 3.2, whose substencils all contain K_0 ; unresolved.
- Section 3.3, after Definition 3.3: the change of variables is written $|K'_j| = \int_{K'_j} dx' dy' = \int_{K_j} \det(A) dx dy = \det(A) |K_j|$ for a general nonsingular A , which gives a negative area when $\det(A) < 0$; the Jacobian factor is $|\det(A)|$, and the conclusion $\bar{p}'_j = \bar{p}_j$ is unaffected.
- Section 4.2, Example 4.5: “For the cases of $k > 1$, although some troubled cells are detected, the polynomial order does not reduce to 0”, with only “the troubled cells detected outside the expansion region” attributed to truncation errors, but the time histories in Fig. 8(h) and (j)–(l) show cells in the order-0 legend colour at the centre of the expansion region, $x \approx 0$, for t between about 0.14 and 0.27, a streak at least one cell wide for $k = 4, 5$, and 6, whereas the $t = 0.6$ profiles in Fig. 8(b)–(f) have no order-0 cells inside the fans; unresolved.
- Section 4.2, Example 4.7: the initial density for $x \geq -4$ is “ $1 + 0.2 \sin(5\pi x)$ ”, but in Fig. 10 the undisturbed entropy wave ahead of the shock has maxima near $x \approx 2.8$ and 4.1 and minima near 3.45 and 4.7, a period of about 1.26 that matches $\sin(5x)$ (period $2\pi/5$) rather than $\sin(5\pi x)$ (period 0.4); the evident intended density is $1 + 0.2 \sin(5x)$.
- Section 4.2, Example 4.8: the blast waves are computed “in the computational domain $[-1, 1]$ ”, but the initial data are defined only for $0 \leq x \leq 1$ and all panels of Fig. 11 span $0 \leq x \leq 1$; the evident intended domain is $[0, 1]$.
- Section 4.2, Example 4.8: “Fig. 11 shows the distributions of the density and the polynomial order at $t = 1.8$ ”, but the Fig. 11 caption states $t = 0.038$ and its time-history panels end at about $t = 0.038$; the evident intended time is $t = 0.038$.

2026SSEH

Claude (September 2026)

Summary

The authors study adaptive sparse-grid discontinuous Galerkin (DG) discretizations of the Bhatnagar–Gross–Krook (BGK) kinetic model in four- and six-dimensional phase space, where full-grid DG is impractical. The distribution is expanded in Alpert multiwavelets, and coefficient decay across levels drives simultaneous refinement and coarsening of an ancestor-complete grid after each step of an implicit-explicit Runge–Kutta scheme with explicit upwind advection and implicit relaxation. Interpolating the nonseparable Maxwellian directly on the sparse grid violates the collision invariants when velocity resolution is coarse, so they introduce a hybrid interpolation: interpolation in position only, with exact error-function velocity projections that exploit velocity separability. They prove that it reproduces density, momentum, and energy for polynomial degree $k \geq 2$. Tests comprise a 3x3v relaxation problem, a radial 2x2v Sod problem at Knudsen numbers 0.001, 0.1, and 1, a shear flow, and expansion problems from dynamical low-rank studies. In the relaxation test with $k \geq 2$, the hybrid Maxwellian conserves moments to roundoff, whereas phase-space interpolation errs by 0.1–9%; fluid-regime results match Euler references; rarefied velocity distributions depart from the Chapman–Enskog perturbation; and active degrees of freedom shrink by factors from about three to over five thousand

relative to full grids. All computations use the ASGarD library.

Contributions

1. Introduced a hybrid Maxwellian interpolation and proved that it preserves the discrete BGK collision invariants on ancestor-complete adaptive sparse grids for $k \geq 2$.
2. Assembled an adaptive sparse-grid DG–IMEX algorithm for BGK that refines and coarsens in one pass, keeps grids ancestor complete, and adds nonlinear refinement driven by the interpolated Maxwellian.
3. Demonstrated that a phase-space-interpolated Maxwellian causes conservation errors and oscillations that densify the grid, whereas the hybrid version needs orders of magnitude fewer degrees of freedom in the shear-flow test.
4. Resolved sharp gradients first in position and then in velocity for a radial Sod problem from fluid to transitional regimes, showing kinetic perturbations differing from the Navier–Stokes correction by nearly a factor of four at $\nu = 1$, where that correction also turns negative.
5. Showed on low-rank benchmarks that sparse grids compress distributions whose rank keeps growing, with phase-space degrees of freedom decreasing after an initial rise, and ran $3 \times 3 \nu$ cases with about 0.12% of the full-grid degrees of freedom.

2026ST

ChatGPT (July 2026)

Summary

This paper compares explicit integrators for obstacle-potential phase-field models coupled to concentration diffusion. It treats forward Euler, strong-stability-preserving Runge–Kutta schemes, and first- and second-order Runge–Kutta–Legendre super-time-stepping methods. The implementation projects every phase-field stage onto the Gibbs simplex, adapts error estimates to compact-support phase storage, and uses a PID-type step controller. Rather than comparing only with refined numerical references, the authors construct equilibrium and kinetic benchmarks with sharp-interface solutions, allowing temporal, spatial, and diffuse-interface errors to be separated. Across practical resolutions, spatial and interface-width errors usually dominate time-integration error; strict discrete energy stability has little effect on the sharp-interface error in the tested cases. Super-time-stepping is therefore substantially more efficient than forward Euler without appreciable accuracy loss. Three-dimensional grain-growth examples reproduce unpinned growth, Zener-like limiting behavior from immobile particles, and slowed continued growth from mobile pores that move and merge. The largest gain occurs when diffusion creates strongly separated time scales.

Contributions

1. Defined reproducible equilibrium and kinetic phase-field benchmarks with analytical sharp-interface solutions.
2. Adapted strong-stability-preserving and Runge–Kutta–Legendre schemes to simplex-constrained, compactly stored phase fields.
3. Developed field-aware error estimation and PID-controlled adaptive stepping for coupled phase and concentration variables.
4. Separated temporal, spatial, and interface-width errors and evaluated the practical role of energy stability.
5. Demonstrated the methods on three-dimensional grain growth with no particles, immobile pinning particles, and mobile coalescing pores.

2026STCR

ChatGPT (July 2026)

Summary

This paper redesigns PLUTO’s Lagrangian Particle module for GPU-enabled gPLUTO simulations of nonthermal particles in relativistic magnetized astrophysical flows. Each computational particle represents an ensemble whose energy spectrum evolves through adiabatic changes, synchrotron and inverse-Compton losses, and first-order Fermi

acceleration at shocks. The C++ implementation uses OpenACC for NVIDIA GPU offloading and MPI across nodes, restructures particle data for coalesced access, compacts deleted particles, and overlaps particle exchange with container resizing. Verification recovers the expected second- and third-order particle-advection convergence and matches the analytical post-shock spectral slope within 0.5%. Strong- and weak-scaling tests on MareNostrum 5 and Leonardo cover up to 28,672 CPU cores and 1,024 GPUs. Weak-scaling efficiency remains above 90% on CPUs and approximately 80–90% on GPUs. Comparing complete nodes, four GPUs outperform 112 CPU cores by about $4\times$ for the shock case and $6\times$ for advection. OpenACC currently limits accelerator portability to NVIDIA hardware; an OpenMP port is planned.

Contributions

1. Re-engineered PLUTO’s nonthermal Lagrangian-particle module as GPU-compatible C++ within gPLUTO.
2. Implemented OpenACC/MPI particle storage, compaction, boundary exchange, and spectral-update workflows.
3. Verified particle-advection order and shock-accelerated spectral evolution against analytical solutions.
4. Demonstrated strong and weak scaling through 28,672 CPU cores and 1,024 GPUs on two EuroHPC systems.
5. Measured full-node GPU speedups of roughly $4\times$ for shock evolution and $6\times$ for advection.

2026SV

ChatGPT (July 2026)

Summary

This paper analyzes an embedded Trefftz discontinuous Galerkin method for the Helmholtz equation. It begins with a polynomial DG space and imposes local PDE constraints through an embedding, avoiding explicit construction of specialized Trefftz basis functions. Global coupling uses a simple symmetric interior-penalty form rather than stronger complex or derivative stabilization. Because this formulation is noncoercive, the analysis combines T -coercivity with a Schatz duality argument. Under the explicit resolution condition $(1 + \omega^2)h \leq C_\Omega$, it establishes wavenumber-explicit stability, quasi-optimality, optimal DG-norm convergence, and an L^2 estimate. Numerical experiments confirm $O(h^p)$ DG error and $O(h^{p+1})$ L^2 error, exponential pre-asymptotic convergence under p -refinement, and comparable accuracy to standard polynomial DG with fewer unknowns. High-wavenumber and spatially varying-wavenumber tests suggest useful behavior beyond the proved setting.

Contributions

1. Formulated a polynomial embedded Trefftz DG discretization with local Helmholtz constraints and symmetric interior-penalty coupling.
2. Transferred stability to the constrained space through a T -coercive Schatz-type framework.
3. Proved wavenumber-explicit stability and quasi-optimality under an explicit mesh-resolution condition.
4. Derived optimal-order DG- and L^2 -error estimates using polynomial best approximation and the Trefftz embedding.
5. Verified h - and p -convergence and explored high-frequency and variable-wavenumber regimes numerically.

2026SWJ

ChatGPT (July 2026)

Summary

This work introduces the projection enhanced triangular finite element method (PET-FEM) for polygonal meshes. Each polygon is subdivided into triangles, while internal triangle degrees of freedom are recovered from a least-squares polynomial projection of the polygonal degrees of freedom. The construction retains familiar FEM interpolation and avoids problem-dependent stabilization parameters. Higher-order elements use a serendipity-like reduction of internal unknowns and remain less sensitive to distortion than classical serendipity FEM. For first-order analysis, the authors also combine PET-FEM with virtual-element stabilization to produce the VPF-12 element and examine under-integrated

variants. Polynomial reproduction and Poisson tests attain the expected optimal convergence rates. Linear and nonlinear structural benchmarks—including patch tests, thin-beam bending, Cook’s membrane, and punch loading—show robust behavior on regular and Voronoi meshes. Under-integrated PET-FEM reduces shear locking without hourglass modes, while VPF-12 gives accurate nearly incompressible results on coarse meshes and converges in cases where conventional triangular FEM or VEM variants fail.

Contributions

1. Defined polygonal PET-FEM through triangular submeshes and least-squares recovery.
2. Extended the construction to higher order with serendipity-style internal moments.
3. Improved robustness to polygon distortion compared with classical serendipity elements.
4. Introduced the locking-resistant VPF-12 hybrid for nearly incompressible materials.
5. Established optimal Poisson convergence and robust nonlinear structural performance.

2026SWLA

Claude (September 2026)

Summary

Kilohertz laser-wakefield accelerators (LWFA) driven by few-millijoule pulses reach only about 10 MeV; higher energies need much more laser average power. The authors drive an LWFA with the SYLOS 3 optical parametric chirped-pulse amplification (OPCPA) laser at ELI-ALPS: 70 mJ, 8.1 fs pulses at 850 nm on target, focused to 10 μm FWHM ($a_0 = 1.5$) into a continuous-flow He(99%)–Ar(1%) gas jet with electron density about $3 \times 10^{19} \text{ cm}^{-3}$, where argon enables ionization injection. Operated at 100 Hz because of radioprotection and beam-dump limits, the accelerator produced quasi-monoenergetic 47 MeV beams with 15 pC core charge. Over 10,000 consecutive shots, peak-energy and charge fluctuations were 1.4% and 12% rms, and pointing jitter was 1.2–1.3 mrad rms. Deconvolving the 4.8 MeV spectrometer resolution bounds the intrinsic energy spread below 10% FWHM at 95% confidence. Principal-component analysis of beam profiles separates charge, pointing, and width fluctuations plus one unexplained core–halo mode, and shows variability dominated by drifts below about 1 Hz; recorded laser energy and carrier-envelope phase do not significantly explain it. Particle-in-cell simulations with the measured wavefront reproduce spectrum, charge, and divergence, attributing the narrow spectrum to injection localized by rapid self-focusing, but only with 0.1% instead of 1% argon, an unexplained discrepancy.

Contributions

1. Demonstrated OPCPA-driven LWFA producing 47 MeV, 15 pC quasi-monoenergetic beams at 100 Hz with a kHz-compatible laser and gas target, which the authors report as a significant energy increase and a 10-fold charge increase over kHz LWFA driven by post-compressed Ti:sapphire lasers.
2. Measured shot-to-shot stability over 10,000 consecutive shots, with pointing jitter about one order of magnitude below the 15.6 and 18.8 mrad FWHM beam divergences.
3. Bounded the intrinsic energy spread by fitting an instrument-response-convolved model to a spectral peak comparable in width to the resolution, obtaining a 3.8 MeV best fit and a 4.6 MeV upper limit from a profile- χ^2 analysis.
4. Validated the principal components against core charge, position, and width (Pearson $r = 0.48$ – 0.98) using a circular time-shift test that accounts for slow drifts, and argued that feedback well below the repetition rate could correct the dominant fluctuations.
5. Identified with FBPIC simulations self-truncated ionization injection, mainly from Ar^{9+} to Ar^{11+} and most of the charge within about 100 μm , and explained the small Ar^{8+} contribution through the trapping condition at its early ionization phase.

2026SYTN

ChatGPT (September 2026)

Summary

Shirakawa and coauthors address anisotropic electron diffusion in magnetized-plasma fluid models, where field-aligned transport and cross-derivative terms can make standard discretizations inaccurate or poorly conditioned. Their predictor–corrector method alternates a flux prediction with an isotropic potential-correction solve. Direction-dependent relaxation removes cross-diffusion from that solve and allows a five-point finite-difference stencil with a standard iterative solver. Analysis estimates a stable relaxation range under stated assumptions. In analytic-solution tests, the method maintains approximately second-order accuracy across strong anisotropy, including a case with a Hall parameter of 10^6 . Further Cartesian and axisymmetric tests with nonuniform, curved magnetic fields show improved convergence over the transverse-flux method. The paper demonstrates two-dimensional cases; extension to three dimensions is proposed but untested.

Contributions

1. Converted anisotropic electron transport into repeated isotropic potential-correction solves.
2. Avoided explicit cross-diffusion terms with a staggered, second-order finite-difference discretization.
3. Estimated a stability range and near-optimal relaxation parameter for the predictor–corrector iteration.
4. Verified approximately second-order accuracy against an analytic solution under extreme anisotropy.
5. Extended the method to axisymmetric coordinates and tested curved magnetic-field configurations.

2026TBA

ChatGPT (September 2026)

Summary

The authors reformulate the electrostatic Vlasov–Ampère system by evolving a conditional distribution in a locally centered and thermal-speed-scaled velocity coordinate. Its density, momentum, and energy moments are fixed, while separate moment equations evolve those physical quantities and use stress and heat flux from the conditional distribution as closures. This division lets positivity controls act on the kinetic equation without corrupting conservation, and concentrates Gauss’s law and the quasi-neutral limit in a staggered moment–field subsystem. A slow-manifold reduction removes fast electron time scales from the kinetic equation, and a coupled fixed-point/quasi-Newton solver advances the resulting 1D1V discretization. Tests recover Landau damping and ion-acoustic-shock dynamics while preserving mass, momentum, energy, Gauss’s law, and distribution positivity to machine precision. At $\epsilon = 10^{-7}$, charge and current scale as $O(\epsilon^2)$ and the electric field agrees with the reduced Ohm’s law. The present method is only first-order accurate in space and time and remains explicitly CFL-limited in the kinetic subsystem.

Contributions

1. Derived an exactly equivalent conditional Vlasov formulation whose first three velocity moments are invariant.
2. Localized conservation laws, Gauss’s-law preservation, and quasi-neutral asymptotics in a separate moment–field subsystem.
3. Constructed a staggered finite-difference scheme that simultaneously maintains conservation, positivity, and asymptotic consistency.
4. Demonstrated machine-precision invariants and the expected $O(\epsilon^2)$ quasi-neutral scaling on an ion-acoustic shock.
5. Identified first-order numerical dissipation and the explicit kinetic CFL constraint as the principal limits of the current implementation.

Summary

The authors couple a plasma fluid model of capacitively coupled plasma (CCP) discharges with an external circuit model (an L-matching network) in a finite-volume solver on ultraMPP. The CCP voltage is not an explicit function of charge or current, and explicit coupling can diverge even for a plasma-free capacitor, so the potential is split into a Laplace part scaled by the electrode voltage and a charge-driven part, giving a vacuum capacitance C_{CCP} and a plasma-induced charge Q_{pm} , which alone is extrapolated while the circuit equations are integrated implicitly at second order. Runs stop when relative two-norm differences of plasma fields and electrode voltage between successive radio-frequency (RF) cycles fall below a threshold, linked to the slowest exponential time constant. A time-step size refinement (TSSR) algorithm halves Δt from $T/100$ (T the RF period), restarting from the interpolated periodic state, until the discharge power changes by less than 1%. An unignited test matches the analytic circuit solution with errors generally below 10^{-3} . For a quasi-one-dimensional 20-W, 0.2-torr argon discharge, TSSR reached $T/3200$ after six iterations and 709,700 steps, against over 1.28 million at fixed $T/3200$. The authors note that savings are problem-dependent and small two-norm differences do not guarantee convergence.

Contributions

1. Split the potential as $\phi = V_{CCP}\phi_{base} + \phi_p$, with ϕ_{base} the Laplace solution for unit voltage on the powered electrode and ϕ_p the Poisson solution with both electrodes grounded, so that the powered-electrode charge is $Q_p = C_{CCP}V_{CCP} + Q_{plasma,p}$ with $C_{CCP} = \int \epsilon \vec{E}_{base} dA$ and $Q_{plasma,p}$ the sum of the $\vec{E}_p$ displacement charge and the conduction charge $\int \sum_j q_j \vec{\Gamma}_j dA$, and kept the Kirchhoff system for (Q_{CCP}, I_{CCP}, Q_m) in descriptor form with the matrix A containing L_1 and R_s rather than premultiplying by A^{-1} , which would introduce $1/L$ terms, problematic for circuits without inductors.
2. Blended linear and quadratic extrapolation as $Q_{pm}^{n+1} \approx (1 - \omega)(2Q_{pm}^n - Q_{pm}^{n-1}) + \omega(3Q_{pm}^n - 3Q_{pm}^{n-1} + Q_{pm}^{n-2})$ with $\omega = \min(N_t/500, 1)$, since the linear form underestimated and the quadratic form overestimated the discharge power and the quadratic form was unstable for $\Delta t \gtrsim T/500$, after testing backward Euler, midpoint, trapezoidal, Lobatto IIIC, Radau IIA, modified extended BDF, BDF2, and BDF3 circuit integrators and finding BDF2 accuracy sufficient, and matched the target power by updating the supply amplitude once per cycle as $V_{pp}^{(new)} = V_{pp}^{(old)} + 0.2(P_{p,tar}V_{pp}^{(old)}/P_p^n - V_{pp}^{(old)})$ whenever the relative power error exceeded 0.1%.
3. Showed for a plasma-free CCP driven through series R and C by explicit coupling with $Q_{CCP}^n = C_{CCP}V_{CCP}^n$ that the amplification polynomial $r^2 + (\xi + \zeta)r - \xi = 0$, with $\xi = C_{CCP}R/\Delta t$ and $\zeta = C_{CCP}/C$, is stable only if $2\xi < 1 - \zeta$, and that adding a series L yields the cubic $r^3 + (\eta + \xi + \zeta)r^2 - (2\eta + \xi)r + \eta = 0$ with $\eta = C_{CCP}L/\Delta t^2$, whose root product $-\eta$ makes $\Delta t > \sqrt{C_{CCP}L}$ necessary, a step at the intrinsic LC time scale.
4. Modeled a transient variable as $X = X_{ave,qs}(1 - e^{-t/\tau})H$ with H periodic in the RF cycle and bounded the cycle-to-cycle difference by $D_X \lesssim (T/\tau_{max})e^{-t/\tau_{max}}$, giving the 1% threshold $D_X < (T/\tau_{max}) \cdot 1\%$; in a 2000-cycle run at $\Delta t = T/400$ with $\omega = 0$, slopes gave $\tau_1 \approx 48.64T$ for the electron and Ar^+ densities and V_{CCP} (1% after 223.7T) and $\tau_2 \approx 385.5T$ for metastable argon (1773T), and the authors recommended stopping at about 474 cycles ($D_{V_{CCP}} \leq 10^{-5}$) for macroscopic argon quantities and at $D_{V_{CCP}} \leq 10^{-6}$ for other chemistries or boundary conditions without prior convergence benchmarks.
5. With $R_s = 50 \Omega$, $C_m = 76.10 \text{ pF}$, $C_1 = 200 \text{ pF}$, $L_1 = 4.181 \mu\text{H}$, $R_1 = 0.5 \Omega$, a nine-reaction argon set with five BOLSIG+ rate coefficients, and 60 nonuniform cells of minimum width 0.2 mm, obtained $Z_{CCP} = 17.96 - 239.4j \Omega$ from $V_{CCP,max} = 220.2 \text{ V}$, $I_{CCP,max} = 0.9173 \text{ A}$, and a phase of -85.71° , against $5.326 - 253.2j \Omega$ from two 3-mm vacuum sheaths and a resistive bulk with $\sigma_p = 0.08366$, and found in the TSSR run that the peak electron density rose from about 2.4×10^{15} to $3.2 \times 10^{15} \text{ m}^{-3}$, the sheath electron temperature from 3.9 to 4.4 eV, and the plasma potential from 110 to over 120 V.

Comments

- Section II-B, Eqs. (36)–(40): with $\vec{y} = (Q_{CCP}, I_{CCP}, Q_m)^T$, the first row of $A d\vec{y}/dt = B\vec{y} + c$ gives $dQ_{CCP}/dt = -I_{CCP}$, contradicting the definition of Q_{CCP} as the time integral of I_{CCP} , and the other rows give $L_1 \dot{I}_{CCP} + R_s \dot{Q}_m = (1/C_1 + 1/C_{CCP})Q_{CCP} + (R_1 + R_s)I_{CCP} + V_p + Q_{pm}/C_{CCP}$ and $R_s \dot{Q}_m = R_s I_{CCP} + Q_m/C_m + V_p$, whereas (21), (22), and (35) with $Q_p = Q_{CCP}$ give every $B\vec{y}$ term with the opposite sign and the same c ; the evident intended system is $A d\vec{y}/dt = -B\vec{y} + c$, so that $\tilde{A} = -A^{-1}B$.
- Section II-C: with $f_p^{-1} \approx T/36.3$, the resolution “ $\Delta t \lesssim T/100$ ” is said to correspond “to approximately $3f_p^{-1}$ ”, but $T/100 \approx 0.36f_p^{-1}$; the evident intended statement is about three time steps per f_p^{-1} , that is, $\Delta t \approx f_p^{-1}/3$.
- Section II-D, Eq. (52): $X_{ave,qs}(\vec{r})$ is introduced as “The spatially varying quasi-steady-state average” and then as “The spatially averaged quasi-steady-state value”, but (52) defines it at fixed $\vec{r}$ as the limit of the one-period average $\frac{1}{T} \int_t^{t+T} X(\vec{r}, \tilde{t}) d\tilde{t}$, a cycle average, consistent with $\lim_{t \rightarrow \infty} a_0 = 1$ in (54).
- Section II-D, Eq. (61): with $H(t-T) \approx H(t)$, the denominator $X(\vec{r}, t-T)$ on the left becomes proportional to $1 - e^{-(t-T)/\tau_{max}}$, but the right side has $1 - e^{-t/\tau_{max}}$; the difference disappears once the denominator is approximated by 1 for $t \gg \tau_{max}$.
- Section II-D, Eq. (62): the first two lines, $|(1 - e^{\frac{t-T}{\tau_{max}}}) - (1 - e^{\frac{t}{\tau_{max}}})|$ and $T \frac{d}{dt}(1 - e^{\frac{t}{\tau_{max}}})$, drop the minus signs of the exponents in (61); with positive exponents the derivative is $-\frac{T}{\tau_{max}} e^{t/\tau_{max}}$, whereas the stated result $\frac{T}{\tau_{max}} e^{-t/\tau_{max}}$ follows from the exponents $-(t-T)/\tau_{max}$ and $-t/\tau_{max}$ of (61).
- Fig. 7 caption: “(c) Electrostatic potential. (d) Electron temperature.”, but panel (c) is labeled “Ave. T_e [eV]” and panel (d) “Ave. ϕ [V]”, as in the text of Section III-A (“Fig. 7(c) demonstrates the transient evolution of the electron temperature”, “In Fig. 7(d), the electrostatic potential”); the labels in the caption are swapped.
- Section III-A, discussion of Fig. 8(b): the Ar^+ density is stationary because “the ion plasma frequency is typically in the kilohertz range”, but Section II-C gives $\omega_p = 3.089 \times 10^9$ rad/s for $n_e = 3 \times 10^{15} \text{ m}^{-3}$, and scaling by $(m_e/m_{\text{Ar}})^{1/2} \approx 3.7 \times 10^{-3}$ gives an Ar^+ plasma frequency of about 1.1×10^7 rad/s, or 1.8 MHz, still below but not far below the 13.56-MHz drive.
- Section III-B: “The converged power approaches 9.5 W, while simulations using larger time-steps report values below 8 W”, but in Fig. 13 the discharge power at the largest step, 100 time steps per cycle, is about 8.6 W, no plotted value is below 8 W, and the last two points, at 1600 and 3200 steps per cycle, are about 9.8 and 9.7 W; unresolved.
- Appendix A, after Eq. (A7): “when $C_{CCP} > C$, the system is always stable as the stability condition becomes negative”, but for $\zeta = C_{CCP}/C > 1$ the condition (A6), $2\xi < 1 - \zeta$ with $\xi = C_{CCP}R/\Delta t > 0$, holds for no Δt , and then $|r_-| > \xi + \zeta > 1$, so the explicit coupling is unstable for every time step; (A7) follows from (A6) only for $C_{CCP} < C$, since dividing by the negative $1/C_{CCP} - 1/C$ reverses the inequality.

2026TDAZ

ChatGPT (July 2026)

Summary

This work extends the hermes-3 drift-reduced fluid model to rapidly rotating, low- β plasmas by adding centrifugal inertia and relaxing the Boussinesq approximation, then studies Kelvin–Helmholtz (KH), curvature-driven interchange (CDI), and rotation-driven interchange (RDI) modes in two-dimensional simulations. The implementation reproduces theoretical KH and CDI growth rates and first-order finite-Larmor-radius stabilization. For RDI, the authors distinguish weak-shear, strong-shear, and fully shear-stabilized regimes and propose the local susceptibility condition $|L_v|/L_n > -1$. Profiles satisfying it everywhere avoid full RDI turbulence in the tested runs, but stability is highly profile-sensitive. Finite-Larmor-radius effects are appreciable for $\rho_i/a = 0.16$ and small for 0.01. A KH-unstable quartic velocity profile can trigger interchange crashes even when the initial criterion predicts stabilization. The analysis is limited to $k_{\parallel} = 0$ azimuthal dynamics; three-dimensional mirror geometry may change the conclusions.

Contributions

1. Added centrifugal-force physics and a non-Boussinesq vorticity formulation to hermes-3.

2. Verified simulated KH and CDI growth rates against analytic or semi-analytic predictions.
3. Demonstrated gyroviscous finite-Larmor-radius stabilization and its dependence on ρ_i/a .
4. Identified three RDI regimes and proposed the profile criterion $|L_v|/L_n > -1$.
5. Showed that global KH modes can defeat nominal RDI shear stabilization.

2026TG

ChatGPT (July 2026)

Summary

The paper presents a Python object-oriented framework that symbolically derives equations for density and correlation matrices at a requested order of the quantum BBGKY hierarchy. Classes represent density matrices, correlations, Hamiltonian terms, commutators, partial traces, index systems, and statistically infinite subsystem ensembles; their methods encode the required operator algebra and cluster expansions. Users specify subsystem families and allowed pairwise interactions, and the program emits the resulting kinetic equations in LaTeX form. Tests on two quantum-optical systems involving emitter, medium, and field-mode ensembles reproduce previously derived equations through third order, including terms omitted by earlier physical approximations. The software then generates new fourth-order correlation equations in roughly 2–48 seconds on the reported laptop, with modest memory use. The framework reduces algebraic effort and transcription errors but does not close or numerically solve the hierarchy: physical truncation and approximation remain the user's responsibility. Empirical validation is limited to pairwise-interacting light-matter models, and scalability beyond fourth order is not established.

Contributions

1. Encoded quantum hierarchy objects and their algebra as reusable Python classes.
2. Represented finite and statistically infinite subsystem families with configurable pair interactions.
3. Automated cluster expansion, partial tracing, and LaTeX generation for BBGKY correlation equations.
4. Reproduced published first- through third-order light-matter equations without manual derivation.
5. Generated previously unavailable fourth-order equations in under one minute.

2026TGRV

ChatGPT (July 2026)

Summary

Coupled three-dimensional–one-dimensional transport models represent solute exchange between a bulk domain and a thin embedded structure without resolving the structure as a full three-dimensional region. The paper defines a fully discrete approximation that combines a conforming finite-element method for the three-dimensional concentration with an interior-penalty discontinuous Galerkin method for the one-dimensional concentration. The coupling transfers solute between the two discrete domains while allowing their mesh sizes to be chosen independently. Assuming sufficient regularity of the weak solution, the authors derive error estimates for both concentrations. The bounds are optimal with respect to the time-step size and the respective three- and one-dimensional mesh sizes under the stated assumptions. Numerical convergence experiments support the theoretical rates. The results provide a rigorous accuracy foundation for mixed-dimensional transport calculations; their guarantees remain conditional on the solution regularity and model assumptions used in the analysis.

Contributions

1. Defined a coupled discretization for time-dependent three-dimensional–one-dimensional solute transport.
2. Combined conforming finite elements in the bulk domain with interior-penalty discontinuous Galerkin approximation on the embedded line.
3. Derived simultaneous error bounds for the three-dimensional and one-dimensional concentrations.
4. Proved optimal dependence of the estimates on the time step and both spatial mesh sizes under sufficient regularity.

5. Verified the theoretical convergence behavior through numerical experiments.

2026TK

ChatGPT (July 2026)

Summary

This paper develops a GPU-native clustered neighbor list for short-range particle interactions using the Hilbert space-filling-curve ordering of a Cornerstone octree. Consecutive particles form fixed-size clusters without an extra reorder, while GPU octree traversal constructs sorted cluster neighborhoods and supports particle-dependent cutoff radii. A new encoding stores interaction masks and compressed deltas between neighbor-cluster indices, reducing storage for roughly 200 neighbors to under four bytes per particle. Benchmarks on NVIDIA GH200 and AMD MI300A GPUs show neighborhood-pass performance comparable to the specialized GROMACS implementation, with hardware-dependent trade-offs between full and clustered lists. The method handles strongly nonuniform distributions and integrates naturally with octree domain decomposition and multipole long-range solvers. In a coupled smoothed-particle-hydrodynamics and gravity Evrard-collapse test, it weak-scales to 1024 GH200 GPUs and agrees with the reference solution. Cluster overhead remains significant for small neighbor counts, and pathological Hilbert-curve clusters can increase traversal and masked-interaction work.

Contributions

1. Constructed clustered neighbor lists entirely on GPUs from an existing Hilbert-ordered octree without an additional particle reorder.
2. Designed a delta-compressed cluster-index representation requiring under four bytes per particle for approximately 200 neighbors.
3. Supported nonuniform particle distributions and particle-specific interaction radii within the same neighbor-search structure.
4. Compared construction cost, pair-kernel performance, cluster overhead, and memory use against GROMACS- and LAMMPS-style approaches on GH200 and MI300A GPUs.
5. Demonstrated coupled hydrodynamic and gravitational Evrard-collapse simulations with weak scaling to 1024 GPUs and agreement with the reference solution.

2026TKCD

ChatGPT (July 2026)

Summary

This paper directly programs NVIDIA FP64 tensor cores to accelerate small, irregular matrix products in matrix-free high-order finite-element kernels. Tensor contractions in MFEM are decomposed into small dense products and mapped to DMMA PTX instructions for Grace Hopper GH200 and Grace Blackwell GB200 GPUs; custom layouts handle unsupported shapes, and kernel fusion further reduces data motion. The tensor-core kernels cut shared-memory reads by $4.6\times$, improve individual kernel throughput by up to 59%, and raise energy efficiency by up to 27%. Combined with fusion, they deliver up to $2\times$ speedup and up to 83% energy savings. In a production tsunami digital-twin application, all kernel variants show near-ideal weak scaling, while the optimized code reaches 90% strong-scaling efficiency across 9,216 GH200 GPUs on Alps. The optimizations are under review for inclusion in the public MFEM repository.

Contributions

1. Designed direct FP64 tensor-core mappings for the small, irregular matrix products in high-order finite-element operators.
2. Identified shared-memory traffic as the main bottleneck and reduced reads by $4.6\times$.
3. Combined DMMA kernels with operator fusion to obtain up to $2\times$ speedup and 83% energy savings.
4. Compared GH200 and GB200 performance and energy behavior for small FP64 tensor-core operations.

5. Demonstrated near-ideal weak scaling and 90% strong-scaling efficiency on up to 9,216 GPUs in a production tsunami application.

2026TLF

ChatGPT (July 2026)

Summary

Fully kinetic particle-in-cell (PIC) simulations often reduce the speed of light or ion-to-electron mass ratio to make multiscale calculations affordable, but these substitutions distort Coulomb collision rates. This paper derives separate scaling factors for electron-electron, ion-ion, electron-ion, and ion-electron collisions instead of applying one global correction. The construction matches total electron and ion relaxation rates to their respective electromagnetic timescales while preserving electron-ion energy equilibration and important transport properties within weakly collisional plasma theory. Binary Monte Carlo PIC benchmarks reproduce theoretical relaxation rates, and comparisons with fluid theory confirm the retained transport behavior. The scaling applies to nonrelativistic systems satisfying $m_i T_e / (m_e T_i) \gg 1$; relativistic effects remain future work. Because the method changes only collision-rate factors, it can be incorporated into existing collisional PIC and other kinetic codes while retaining the computational savings of artificial physical parameters.

Contributions

1. Quantified how reduced light speed and mass ratio alter distinct inter- and intra-species collision processes.
2. Derived species-dependent collision-rate factors instead of a single global rescaling.
3. Preserved electron and ion relaxation times, energy equilibration, and key transport properties simultaneously.
4. Validated the scaling against theoretical relaxation rates using binary-collision PIC benchmarks.
5. Connected the kinetic correction to fluid transport theory and identified its nonrelativistic validity regime.

2026TMS

ChatGPT (July 2026)

Summary

The paper develops prescribed-tolerance error control for rational approximations to e^{Ab} when finite-element semidiscretization produces $A = \tau M^{-1}K$ with a well-conditioned symmetric positive-definite mass matrix M . Directly enclosing the numerical range $W(A)$ can be difficult and overly conservative. The authors instead use the similar matrix $\hat{A} = M^{1/2}AM^{-1/2}$ and prove an error bound involving $W(\hat{A})$ and $\kappa(M)^{1/2}$. An enclosing rectangle is obtained from generalized eigenvalue problems for the symmetric and skew-symmetric parts of K ; moreover, $W(\hat{A})$ remains in the left half-plane when $W(K)$ does. Tests use P_1 and P_2 finite elements for advection-diffusion problems on square and star-shaped domains, together with subdiagonal Pad'e and AAA-based rational approximations. Errors meet requested tolerances in the baseline and roughly 10^4 -unknown sparse tests. Three stringent large-step AAA cases fail during scalar approximation construction, exposing an implementation limit rather than invalidating the derived bound.

Contributions

1. Derived a numerical-range error bound for rational approximations to $e^{\tau M^{-1}K}b$ through a similarity transform.
2. Reduced numerical-range enclosure to generalized eigenvalue problems involving the finite-element matrices M and K .
3. Proved inheritance of the left-half-plane property from $W(K)$ to $W(\hat{A})$.
4. Turned the bounds into a prescribed-tolerance algorithm applicable to multiple rational approximation schemes.
5. Validated the framework on two geometries, two finite-element orders, diffusion regimes, time-step scales, and sparse matrices.

2026TOMP

ChatGPT (July 2026)

Summary

This paper verifies the ssV code, a semi-Lagrangian Vlasov solver for a hybrid kinetic-gyrokinetic plasma model with fully kinetic ions and drift- or gyrokinetic electrons. The model is intended for weakly magnetized and multiscale regimes where fluid, standard hybrid, or fully gyrokinetic approximations miss important ion-scale and electron-response physics. The authors derive the coupled field formulation, discuss the Ampere cancellation issue, and compare several advection schemes, including PFC, FCV, FCV-Umeda, and SLMP5. Benchmarks cover linear and nonlinear Landau damping, ion-acoustic waves, ion Bernstein waves, and kinetic Alfvén waves. The tests show agreement with analytic predictions and previous results, with SLMP5 giving the best balance of accuracy, monotonicity, and numerical stability. The implementation provides a practical reduced kinetic route for space and laboratory plasma studies, while future work targets 3D3V/3D1V extensions and GPU acceleration.

Contributions

1. Presented a hybrid kinetic-gyrokinetic formulation coupling fully kinetic ions to reduced electron dynamics.
2. Implemented the model in ssV with high-order semi-Lagrangian transport options.
3. Compared PFC, FCV, FCV-Umeda, and SLMP5 on smooth and nonsmooth Vlasov benchmarks.
4. Verified Landau damping, ion-acoustic, ion Bernstein, and kinetic Alfvén wave behavior against theory.
5. Identified SLMP5 as the most robust tested scheme for accurate, stable reduced kinetic plasma simulations.

2026TP

ChatGPT (July 2026)

Summary

This paper analyzes neoclassical transport in large-aspect-ratio tokamak barriers where density, temperature, and potential vary over the ion poloidal gyroradius ρ_p , yet the banana-orbit width remains smaller by $\sqrt{\epsilon}$. This intermediate ordering retains an analytic drift-kinetic treatment while excluding finite-orbit-width loss. Strong radial electric fields shift the trapped-particle region, and small poloidal variations of density, potential, parallel flow, and ion temperature alter transport at leading order. The ion parallel-momentum balance couples momentum and particle flux: without a parallel-momentum source, neoclassical particle transport is small, whereas external or turbulence-mediated momentum input can drive a much larger ion particle flux while preserving ambipolarity. Coupling quasineutrality to the transport equations makes profile prediction nonlinear. Examples yield up to three density-gradient solutions for a prescribed particle flux and four radial-electric-field solutions for fixed local profiles; the latter have energy fluxes differing by 56%. The authors suggest that transitions among coexisting roots could relate to low- and high-transport states or H–L back-transitions, but this interpretation remains conjectural and the theory is restricted to the stated ordering.

Contributions

1. Clarified the strong-gradient neoclassical ordering between local and finite-orbit-width regimes.
2. Showed that poloidal profile variation modifies trapping and neoclassical fluxes at leading order.
3. Derived the coupling between parallel-momentum sources and ion particle transport in barriers.
4. Formulated nonlinear profile prediction through quasineutrality-dependent transport coefficients.
5. Demonstrated multiple density-gradient and electric-field solutions with distinct energy transport.

2026TSRC

ChatGPT (July 2026)

Summary

The authors construct a fully analytical manufactured solution for the collisionless electrostatic gyrokinetic Vlasov–Poisson model with adiabatic electrons in cylindrical geometry. Its spatially varying magnetic field, density, electrostatic potential, gyroaveraging, and ∇B drift exercise central gyrokinetic particle-in-cell operations while parameters suppress boundary contamination. Extending earlier Vlasov–Poisson analysis, they prove that convergence of the electric field—and, through elliptic regularity, the mesh charge density—is sufficient to establish convergence of the particle-represented distribution, avoiding direct noisy phase-space reconstruction. They implement the forced problem in XGC and perform simultaneous refinements of mesh spacing, timestep, and particles per vertex. XGC exhibits the expected second-order temporal and Monte Carlo $N_p^{-1/2}$ behavior. Spatial convergence lies between first and second order: lumping the finite-element mass matrix makes the particle-to-mesh density gather formally first order locally, while mesh symmetry improves the global L^2 behavior to about $O(\Delta r^{3/2})$.

Contributions

1. Derived a manufactured solution exercising gyroaveraging, ∇B drift, and the gyrokinetic Poisson solve.
2. Extended deterministic PIC verification theory to forced multidimensional gyrokinetic systems.
3. Proved that field or mesh-density convergence verifies the represented distribution.
4. Integrated the forcing into XGC and verified timestep and particle-number scaling.
5. Diagnosed fractional global spatial convergence from lumped-mass gathering and mesh geometry.

2026VANV

ChatGPT (July 2026)

Summary

Using the one-dimensional DREAM disruption framework, the study reassesses runaway-electron production during ITER shattered-pellet-injection mitigation. It adds four reactor-relevant effects: runaway scrape-off during vertical plasma motion, semi-analytic plasmoid drift and deposition, adaptive hyper-resistive current-profile relaxation, and an updated Compton seed for ITER’s tungsten first wall. Parameter scans cover 15 MA non-nuclear L-mode, 15 MA nuclear deuterium–tritium H-mode, and 7.5 MA non-nuclear H-mode scenarios. Avoiding multi-megaampere beams requires high deuterium assimilation, limited neon, a sufficiently long pre-thermal-quench interval to suppress hot-tail seeds, and losses faster than avalanche growth. Plasmoid drift penalizes hot H-mode plasmas by shifting material outward; nuclear seeds make full-current mitigation especially difficult. Hyper-resistive flattening and vertical scrape-off reduce avalanche multiplication, while current-quench magnetic perturbations above a threshold suppress beams below ITER’s nominal limit. A proposed two-stage injection yields a 0.24 MA representative beam, near but above the cited 0.15 MA target, making it promising rather than robustly acceptable.

Contributions

1. Extended DREAM with models for vertical scrape-off, plasmoid drift, hyper-resistive relaxation, and tungsten-wall Compton seeding.
2. Performed consistent scans of 15 MA L-mode, 15 MA deuterium–tritium H-mode, and 7.5 MA H-mode injection scenarios.
3. Identified the coupled conditions for avoiding runaway beams: long pre-quench evolution, high hydrogenic assimilation, low neon, small seeds, and rapid losses.
4. Found that $\delta B/B \sim 4 \times 10^{-4}$ suppresses all studied beams below 150 kA and that hyper-resistivity can reduce favorable-case currents by about an order of magnitude.
5. Proposed a two-stage H-mode injection sequence that gives a 15 mA seed and a 0.24 MA representative runaway current.

2026VBFP

ChatGPT (July 2026)

Summary

This paper automates the construction and shape optimization of stellarator island divertors within a fixed magnetic equilibrium. Two starting coordinates on the island separatrix define curved V-shaped plates subject to a maximum field-line incidence angle. The FLARE field-line diffusion model estimates surface heat loads, and the objective penalizes both peak heat flux and trajectories striking end plates. A parameter grid finds a best design with $q_{\text{peak}} = 2.915 \text{ MW/m}^2$ and 3.487% end-plate hits, below the 10 MW/m^2 engineering limit used in the study. Gaussian-process Bayesian optimization recovers a nearly identical design, 3 MW/m^2 and 3.53%, after simulating 95% fewer divertors. Repeated optimizations expose the trade-off between the two objectives, and cross-field-diffusivity scans test heat-flux-width robustness until increasing particle losses to the vessel invalidate the expected scaling. This is a proof of principle: support, cooling, manufacturing tolerances, neutral and impurity transport, equilibrium co-optimization, and validation with higher-fidelity EMC3-EIRENE modeling remain outside its reduced model.

Contributions

1. Created a two-parameter algorithm that automatically constructs curved stellarator island-divertor plates.
2. Coupled divertor geometry generation to FLARE heat-load evaluation and engineering-oriented objectives.
3. Found shapes with peak loads near 3 MW/m^2 and low end-plate interception.
4. Reduced the number of simulated candidates by 95% using Bayesian optimization.
5. Quantified objective trade-offs and sensitivity to cross-field heat diffusivity while stating reduced-model limits.

2026VBGM

ChatGPT (July 2026)

Summary

This work develops backward semi-Lagrangian advection for Vlasov-type equations on multi-patch meshes using local cubic B-splines. Merely enforcing C^0 continuity at patch interfaces can be unstable, so the method imposes Hermite conditions with C^1 continuity and reconstructs interface derivatives. A generalized derivative formula handles uniform or nonuniform cells, conforming interfaces, selected nonconforming interfaces, and T-joints. In the conforming case, solving a small global system makes the local splines coincide with the corresponding global spline and can recover C^2 continuity. A cheaper localized approximation is also derived: distant data have decreasing influence, and about 30 cells per patch reach machine precision in the tested uniform configurations. Implemented in Gyselalib++, the method reproduces global-spline interpolation, fourth-order convergence when sufficiently many cells enter the derivative approximation, and two-dimensional guiding-center advection through patch interfaces. Nonconforming refinement achieves accuracy comparable to a refined global mesh with fewer cells. Tests include O-point and T-joint geometries; treating the X-point needed for a complete tokamak cross-section remains future work.

Contributions

1. Established that C^0 patch coupling can be unstable and adopted C^1 Hermite coupling.
2. Generalized interface-derivative reconstruction to nonuniform, conforming, and selected nonconforming meshes.
3. Recovered the equivalent global cubic spline through a small interface system in conforming cases.
4. Derived controllably local approximations whose distant-point influence decays with patch size.
5. Validated the method in Gyselalib++ on O-point, nonconforming, and T-joint guiding-center problems.

2026VC

ChatGPT (July 2026)

Summary

Entropy-correction artificial viscosity (ECAV) enforces a semi-discrete cell entropy inequality by adding only the dissipation needed to offset a local entropy residual. This paper analyzes ECAV when its viscous term is discretized by the local discontinuous Galerkin (LDG) method. Unlike the earlier BR-1 choice, whose discrete gradient has spurious nonconstant null modes, the LDG gradient admits a lower bound that yields an $O(h)$ upper bound on the elementwise viscosity coefficient. Thus ECAV does not impose an asymptotically stricter time-step condition than the hyperbolic CFL restriction. The LDG formulation retains the global entropy-dissipation estimate and, with a contact-preserving interface flux, exactly preserves representable stationary contact waves. Tests on Burgers and compressible Euler problems verify entropy stability, high-order accuracy, contact preservation, shock robustness, and minimal dissipation. In one Burgers test, LDG required 474 adaptive steps versus 20,932 for BR-1; a nodal BR-1 run stalled beyond 100,000 steps. Smooth density-wave tests also show smaller viscosity and error than a tuned modal shock-capturing method.

Contributions

1. Recast ECAV with an LDG discretization of the entropy-dissipative viscous term.
2. Proved a lower bound for the LDG gradient and an $O(h)$ upper bound for ECAV viscosity.
3. Established that LDG ECAV satisfies the required global semi-discrete entropy inequality.
4. Proved and numerically verified preservation of stationary contact waves.
5. Demonstrated high-order accuracy, low dissipation, and substantially less time-step restriction than pathological BR-1 cases.

2026VCG

ChatGPT (July 2026)

Summary

The paper proposes controlling stellarator bootstrap current by adding piecewise-omnigenous perturbations to an approximately quasi-axisymmetric magnetic field. Quasi-axisymmetry offers tokamak-like neoclassical confinement and comparatively simple coils, but its large bootstrap current conflicts with island divertors, which require little edge toroidal current. The authors derive a low-collisionality condition under which regions with differently directed constant- B contours cancel their contributions to the bootstrap current. They construct a smooth Fourier-represented quasi-axisymmetric, piecewise-omnigenous field and use MONKES calculations to show adjustable D_{31} , including a vanishing bootstrap current at reactor-relevant collisionality, while D_{11} retains favorable transport without a $1/\nu$ regime. A reactor-profile estimate finds roughly order-of-magnitude current reduction relative to a quasi-axisymmetric reference and shows that opposite-sign core and edge contributions can keep the rotational-transform change $\Delta\iota$ small. Further optimization is needed, and effects on flow damping, impurity screening, full-volume equilibria, and practical coils remain unresolved.

Contributions

1. Derived a condition for zero bootstrap current in a prototypical quasi-axisymmetric, piecewise-omnigenous field.
2. Constructed a smooth Fourier magnetic-field model that realizes the proposed cancellation mechanism.
3. Demonstrated continuous control over the sign and magnitude of the bootstrap-current coefficient D_{31} .
4. Preserved low radial neoclassical transport without introducing a $1/\nu$ regime in the model fields.
5. Estimated reactor-scale rotational-transform changes and identified core–edge current cancellation as an island-divertor pathway.

2026VLN

ChatGPT (July 2026)

Summary

The paper addresses recovery of the initial velocity in a compressible anisotropic Navier–Stokes system from noisy lateral Cauchy data, assuming density, pressure, body force, and the fourth-order viscosity tensor are known. It projects the velocity onto an exponentially weighted Legendre basis in time, converting the evolutionary inverse problem into a coupled nonlinear elliptic system for spatial coefficients. An asymptotic commutation result justifies projecting the time derivative and convection nonlinearity for sufficiently regular solutions. The overdetermined elliptic system is regularized by quasi-reversibility and solved with a damped Picard iteration. Two-dimensional manufactured tests with 10% multiplicative boundary noise recover diagonal, disconnected, multiscale, and sign-changing initial structures, albeit with smoothing. Truncation acts as a temporal low-pass filter, and weighted Legendre modes markedly outperform unweighted modes. Global convergence of the nonlinear iteration in the fully anisotropic setting remains open.

Contributions

1. Formulated an inverse initial-velocity problem for compressible anisotropic Navier–Stokes flow from lateral boundary observations.
2. Built an exponentially weighted Legendre basis suited to temporal differentiation and coefficient decay.
3. Proved asymptotic commutation of temporal projection with both the time derivative and convection nonlinearity.
4. Reduced the evolutionary problem to coupled elliptic equations solved by quasi-reversibility and damped Picard iteration.
5. Demonstrated stable reconstructions under 10% noise and quantified the benefit of exponential weighting.

2026VPBA

Claude (September 2026)

Summary

The authors compare the optimal convergence rates of two classes of distributed solvers for $Ax = b$, $A \in \mathbb{R}^{N \times n}$, whose rows are partitioned among m machines coordinated by a taskmaster: projection-based methods, chiefly Accelerated Projection-Based Consensus (APC), whose known rate depends on $\kappa_+(S)$ for $S = \sum_i A_i^\top A_i$, and gradient-based methods, chiefly the Distributed Heavy-Ball Method (D-HBM), whose rate depends on $\kappa(A^\top A)$. They introduce cross-machine angular heterogeneity θ_H , the smallest principal angle between the row spaces of two machines, and local angular heterogeneity $\phi_{\min}$, the smallest angle between two rows on one machine. Mutually orthogonal row spaces give $\kappa_+(S) = 1$, so APC converges in one step; in general $\kappa_+(S) \leq (1 + (m-1) \cos \theta_H) / (1 - (m-1) \cos \theta_H)$ when $(m-1) \cos \theta_H < 1$, and $\kappa(S) \geq 1 / \sin^2 \theta_H$ for square invertible A . Since $\kappa(A^\top A) \geq 1 / \sin^2 \phi_{\min}$, APC is favored over D-HBM when cross-machine heterogeneity is high and local heterogeneity low; the authors leave other regimes partly inconclusive. Monte Carlo experiments with square Gaussian matrices show APC faster than D-HBM, approaching it as m grows, and nearly insensitive to n for large n . Numerical evaluation suggests $\theta_H \rightarrow \pi/2$ as $n \rightarrow \infty$ for Gaussian data with fixed local size p , but the authors state this is too slow by itself to guarantee $(m-1) \cos \theta_H < 1$, and their lower bounds for projection methods require square A .

Contributions

1. Gave a distributed procedure (Algorithm 3.2) that computes θ_H from local QR factorizations of $A_i^\top$ and SVDs of $Q_i^\top Q_j$, costing $O(Nnp)$ in parallel against $O(N^2n)$ for the centralized loop when each machine holds $O(p)$ rows.
2. Bounded $\kappa_+(S)$ independently of m by $(2+D)/(2-D)$ with $D = (N-p) \cos \theta_H + \sqrt{(N-p)^2 \cos^2 \theta_H + 4(N-p)p \cos \theta_H}$ for any machine holding p equations when $D < 2$, using a QR factorization of the stacked orthonormal bases, the generalized matrix determinant lemma, and Gershgorin's theorem.

3. Proved for any full-column-rank A that $\kappa(A) \geq \max_k \|\tilde{a}_k\| / \min_{\ell} \{\|\tilde{a}_\ell\| \sin \gamma_{\min}^{(\ell)}\}$ in terms of column norms and inter-column angles, with a row version for invertible square submatrices, attained with equality by diagonal matrices and implying that one row of atypical norm makes $\kappa(A^\top A)$ large while leaving APC unaffected.
4. Converted the bounds into explicit rate bounds for the Mou–Liu–Morse, block Cimmino, APC, DGD, D-NAG, and D-HBM methods, e.g. $2/(1 + \sin \theta_H) - 1 \leq \rho_{\text{APC}} \leq (m-1) \cos \theta_H / (1 + \sqrt{1 - (m-1)^2 \cos^2 \theta_H})$, and derived the sufficient condition $2/(1 + 1/((m-1) \cos \theta_H)) < \cos^2 \phi_{\min}$ for $\rho_{\text{APC}} < \rho_{\text{HBM}}$.
5. Constructed a block-diagonal system in which D-HBM is strictly slower than one-step APC unless every $A_i^\top A_i = cI$ with a common c , and the system $A = J_n - \varepsilon I_n$, in which θ_H and $\phi_{\min}$ both tend to zero as $\varepsilon \rightarrow 0$ and both methods become arbitrarily slow.

Comments

- Section 6, Experiment 1, Key Inference 3: “APC converges faster when the coefficient mean μ is increased”, but in Figure 3 ρ_{APC} at $\mu = 1$ is below its $\mu = 0$ value only for $m \leq 10$ (about 0.9918 versus 0.9955 at $m = 2$), about equal at $m = 12$, and above it for $m \geq 15$ (about 0.9987 versus 0.9976 at $m = 60$ and 0.9989 versus 0.9976 at $m = 120$); what holds for every m is a larger gap between APC and D-HBM at $\mu = 1$, which the experiment’s setup paragraph anticipates, so whether the absolute or the relative statement was intended is unresolved.

2026VRDK

ChatGPT (July 2026)

Summary

The authors parallelize an advancing-front Poisson-disc node generator for meshless discretizations by coupling a spatial-index hypertree with a density-aware work-distribution hypertree. Work-tree leaves approximate balanced units; threads claim non-neighboring leaves, advance independent fronts, and restart from queued cells. Point-placement constraints permit insertion into the partially prebuilt spatial tree without locking it, while collision avoidance is synchronized only at work-leaf granularity. On a dual-socket AMD EPYC 7702 system, constant-spacing disc tests producing roughly 2×10^6 and 4×10^7 points show about a twofold advantage over Pfill where scaling remains favorable. Throughput rises through 64 threads but per-thread efficiency falls to about 20%; performance drops at 128 threads and exhibits unexplained run-to-run bifurcation. The benchmarks exclude sequential tree prebuilding, use a uniform disc and constant density, and lack adaptive spatial-leaf splitting, so variable-density and distributed-memory claims remain future directions.

Contributions

1. Designed a density-informed work hypertree alongside a separate spatial-index hypertree.
2. Developed non-neighboring leaf claims that reduce synchronization during point insertion.
3. Made spatial-tree insertion lock-free and retained queued restarts for unfinished cells.
4. Demonstrated roughly twofold improvement over Pfill in the favorable scaling range.
5. Diagnosed cross-socket, efficiency, front-exhaustion, and run-to-run scaling limits.

2026VS

ChatGPT (July 2026)

Summary

The paper proposes $\tau_{LR} = L_i/R$, evaluated at stationary Ohmic flat-top conditions, as a cross-machine reference time for reducing tokamak plasma current to 20% of its initial value. A Poynting-theorem energy balance motivates the metric, and coupled current- and heat-diffusion simulations with RAPTOR test it on TCV, JET, ITER, and DEMO. Ramp-downs lasting τ_{LR} produce approximately self-similar current-density peaking, avoid substantial reverse edge current, and keep normalized internal inductance near or below 2; the corresponding times are 0.033, 2.87, 63.2, and 166.9 seconds. At

$0.6\tau_{LR}$, constant-shape cases develop reverse current carrying roughly 20–40% of total current and internal inductance near 3. Reducing plasma volume and elongation suppresses this reversal in ITER and DEMO and improves vertical-control margin, but may narrow magnetohydrodynamic stability margins. A zero-dimensional engineering model reproduces simulated τ_{LR} values within about 10% under baseline assumptions and exposes sensitivity to effective charge, density, and confinement. Burning-plasma exit, auxiliary heating, radiation, and full magnetic control remain outside the model.

Contributions

1. Derived $\tau_{LR} = L_i/R$ as an operationally accessible termination timescale from magnetic-energy balance.
2. Demonstrated approximately self-similar τ_{LR} ramp-down behavior across four tokamaks with RAPTOR.
3. Quantified reverse current and current peaking produced by faster $0.6\tau_{LR}$ terminations.
4. Showed how coordinated volume and elongation reduction can mitigate fast-ramp current reversal and vertical-control risk.
5. Developed a zero-dimensional model for estimating τ_{LR} and its dependence on engineering and plasma parameters.

2026W

ChatGPT (July 2026)

Summary

This paper develops a gyrokinetic theory for radial electric fields, or zonal flows, generated by radial redistribution of energetic-ion pressure in tokamaks. Starting from the ensemble-averaged nonlinear gyrokinetic equation and quasineutrality, it treats redistribution of density, energy, and toroidal momentum and includes neoclassical polarization and Reynolds-stress terms. For isotropic Maxwellian or slowing-down energetic ions, the derived field depends primarily on the energetic-ion pressure-gradient perturbation because their mass density is small relative to that of the thermal ions. A generalized Rosenbluth–Hinton test with the NLT gyrokinetic code agrees with this relation. Extending the calculation to a trapped energetic-ion distribution yields an enhancement proportional to the inverse square root of inverse aspect ratio, making trapped ions more effective zonal-flow drivers than isotropic ions. Applied to ITER-like deuterium–tritium fusion, the theory estimates that fusion-born alpha-particle pressure redistribution could increase the mean radial field by roughly 30 kV/m and improve core confinement. The prediction is transport-based and relatively insensitive to the instability causing redistribution, but it does not apply directly to steady stochastic magnetic-field-line transport.

Contributions

1. Derived a nonlinear gyrokinetic relation between energetic-ion pressure redistribution and the radial electric field.
2. Combined density, pressure, toroidal-momentum, neoclassical-polarization, and Reynolds-stress effects.
3. Confirmed the isotropic energetic-ion relation with a generalized Rosenbluth–Hinton test using NLT.
4. Showed that trapped energetic ions enhance the drive by a factor scaling as $\epsilon^{-1/2}$.
5. Estimated an ITER-scale alpha-particle contribution of about 30 kV/m and stated the stochastic-field limitation.

2026W2

ChatGPT (July 2026)

Summary

The paper develops an exact functional expansion built from recursive differential identities rather than local Taylor polynomials. In multiple dimensions, it writes a smooth function as divergence terms whose polynomial weights and derivatives are systematically related. Applying the divergence theorem converts volume integrals over flat-faced polytopes into facet integrals and recursively into lower-dimensional boundary data, avoiding interior tessellation. A complex displacement of the expansion center, chosen from roots of unity, cancels selected low-order remainder terms; conjugate pairing then produces real quadrature formulas with higher polynomial exactness and few function evaluations.

The paper also derives the affine geometry needed for implementation, including transformations of normals and surface measures between parameter and physical facets. The framework provides a systematic route to high-order integration on arbitrary polytopes, although broad numerical benchmarking is not presented.

Contributions

1. Derived a one-dimensional exact functional expansion from repeated product-rule identities.
2. Extended the construction to $\mathbb{R}^n$ as a recursive divergence expansion.
3. Used complex centers and roots of unity to cancel lower-order quadrature errors while retaining real-valued rules.
4. Formulated recursive polytope integration without simplex tessellation.
5. Proved affine transformation laws for facet normals and surface measures.

2026WBBC

ChatGPT (July 2026)

Summary

This tutorial organizes inversion-based shape control (IBSC), the common tokamak-control strategy that linearizes an equilibrium to map actuator currents or voltages to shape changes and then inverts that map. It unifies static current-response and dynamic voltage-response formulations, pseudoinverse and quadratic-program inversions, actuator and shape weighting, regularization, filtering, and saturation constraints. The authors emphasize failure modes caused by singular directions, poor units or scaling, actuator limits, model mismatch, and interactions between shape and vertical-position loops. They provide a systematic design and validation workflow, supported by extensive appendices, and apply it to NSTX-U. The original controller allowed shape commands to interfere with rapid vertical stabilization, producing a vertical-position “bobble.” A decoupling construction removes this interaction in simulation while retaining upper- and lower-null control. The actuator study also shows that adding PF1A and PF2 to the vertical-control set improves simulated phase margin by 6°. The results are control-design predictions based on linearized models and simulations; commissioning performance will still depend on model fidelity, nonlinear limits, diagnostics, and the actual power-supply dynamics.

Contributions

1. Unified major static, dynamic, unconstrained, and constrained variants of IBSC in one framework.
2. Catalogued conditioning, scaling, saturation, modeling, and loop-interaction failure modes.
3. Supplied a systematic controller-design procedure and implementation-oriented tutorials.
4. Diagnosed NSTX-U’s vertical bobble as coupling between shape and vertical-control commands.
5. Removed the simulated bobble through decoupling and identified PF1A/PF2 as phase-margin-improving actuators.

2026WCMT

ChatGPT (July 2026)

Summary

Exascale particle-in-cell Monte Carlo plasma simulations can generate data faster than file-based workflows can store and analyze it. This work modernizes the electrostatic BIT1 code with OpenMP task parallelism in the particle mover and an openPMD data layer backed by ADIOS2. Besides conventional BP4 files, the Sustainable Staging Transport engine streams checkpoints directly from simulation processes to analysis and visualization consumers in memory. Time-dependent checkpointing lets downstream tools inspect evolving particle and field data without stopping the simulation or routing every step through persistent storage. Profiling with gprof, perf, IPM, and Darshan separates computational, communication, and I/O costs and compares the legacy path with BP4 and SST configurations. The resulting hybrid MPI+OpenMP workflow improves runtime, data availability, and interactive analysis while reducing pressure on the storage system. The paper demonstrates an integrated route to in-situ plasma diagnostics rather than treating simulation, output, and visualization as separate stages.

Contributions

1. Added OpenMP task-based parallelism to the BIT1 particle mover.
2. Integrated openPMD as a portable interface for particle and field checkpoints.
3. Enabled in-memory ADIOS2 SST streaming from simulation to analysis consumers.
4. Supported time-dependent in-situ visualization without interrupting the running calculation.
5. Profiled and compared legacy file I/O, ADIOS2 BP4, and SST workflows.

2026WEK

ChatGPT (July 2026)

Summary

The paper combines goal-oriented adaptive $h(p)$ refinement with a discontinuous Bubnov–Galerkin, NURBS-based isogeometric discretization of the first-order discrete-ordinates neutron transport equation. Dual-weighted residual estimates refine the mesh to reduce error in a prescribed quantity of interest. To accelerate scattering-dominated transport solves on the resulting irregular rational geometries, the authors derive a partially consistent diffusion synthetic acceleration scheme using a modified symmetric interior-penalty discretization. Its penalty is computed for general elements and bounded below in optically thick cells. The resulting diffusion systems are symmetric positive definite and amenable to preconditioned conjugate gradients; the same operator also preconditions GMRES. Manufactured-solution and reactor-physics benchmarks show stable compatibility with adaptive meshes, with DSA-preconditioned GMRES consistently the strongest solver considered. Scope is restricted to one-speed, isotropically scattering, fixed-source problems. High NURBS degrees can cause ill-conditioning, and diffusion solves can approach half the total runtime when standard algebraic multigrid loses effectiveness.

Contributions

1. Combined dual-weighted-residual-driven $h(p)$ adaptivity with discontinuous Bubnov–Galerkin NURBS isogeometric neutron transport.
2. Derived a modified symmetric interior-penalty diffusion acceleration discretization.
3. Chose nonarbitrary general-element penalties with a lower bound for optically thick cells.
4. Produced symmetric-positive-definite diffusion systems usable with conjugate gradients and as GMRES preconditioners.
5. Demonstrated stable acceleration on adaptive rational-geometry reactor benchmarks.

2026WFZ

ChatGPT (July 2026)

Summary

The paper proves convergence of entropy-stable split-form continuous summation-by-parts (C-SBP) semi-discretizations for scalar and symmetric hyperbolic conservation laws with homogeneous fluxes. Assuming periodic domains, a smooth exact solution and flux, and globally bounded second flux derivatives, the analysis works directly with the semi-discrete error equation instead of linearization or a projection-based decomposition. Taylor remainders, consistency estimates, and discrete inverse inequalities yield a nonlinear error inequality bounded by a constant-coefficient Riccati initial-value problem. For sufficiently small mesh spacing and degree- p C-SBP operators in d dimensions satisfying $p > 1 + d/2$, the Riccati envelope remains finite on any fixed finite time interval and vanishes under refinement. This establishes convergence even when the discretization has local linear instabilities. The bound also separates contributions from consistency, flux-Jacobian variation, and discrete gradients, suggesting quantitative criteria for comparing schemes. Extensions to discontinuous SBP methods, state-dependent symmetrizers, nonhomogeneous splittings, and long-time behavior remain open.

Contributions

1. Proved convergence for entropy-stable split-form C-SBP discretizations of scalar and symmetric hyperbolic systems.
2. Derived the estimate directly from the nonlinear semi-discrete error equation without linearization or projection splitting.
3. Bounded nonlinear error growth by an explicit constant-coefficient Riccati problem.
4. Established the sufficient condition $p > 1 + d/2$ for finite-time convergence under mesh refinement.
5. Identified how consistency, flux variation, and discrete gradients enter the error-growth coefficients used to compare schemes.

2026WGL

ChatGPT (July 2026)

Summary

Using a validated EAST passive-conductor representation and the Tokamak Simulation Code, the study constructs diverted equilibria and extracts vertical growth rates from free-drift trajectories of the plasma center. Parameter scans show that outward radial displacement improves passive stabilization; increasing minor radius strongly lowers growth by simultaneously reducing elongation and moving the plasma nearer the passive conductor; elongation itself increases growth approximately linearly. Growth rises with plasma current, with a much steeper slope above roughly 800 kA, and increases with internal inductance as the current profile becomes more peaked. A database of 160 equilibria supports a compact backpropagation neural network mapping seven global parameters to growth rate. Training on 120 cases gives mean absolute error 8.11, while 40 test cases give 10.45; randomized-input tests indicate acceptable robustness through about 3% relative error. The model is intended for predesign stability screening, not replacement of detailed equilibrium calculations.

Contributions

1. Built a TSC free-drift workflow using an experimentally and TokSys-validated EAST passive-structure model.
2. Established how radial position and minor radius alter passive vertical stabilization.
3. Found approximately linear elongation dependence and a steeper current dependence above about 800 kA.
4. Showed that more peaked current profiles, represented by higher internal inductance, increase growth rate.
5. Trained a seven-input neural network on 160 equilibria and quantified its prediction error and input robustness.

2026WH

ChatGPT (July 2026)

Summary

Wang and Hu improve the local-relabeling bidirectional buffer used for open boundaries in SPHinXsys, targeting particle flows with strong backflow or several closely arranged inlets and outlets. A hysteresis-like offset separates particle relabeling from particle generation, preventing repeated creation and overlap during backflow. Unique buffer identifiers and geometric containment checks stop one buffer from deleting or processing another buffer's particles. A reproducing-kernel gradient correction with a mirror treatment raises the pressure boundary from zeroth- to first-order consistency. The modifications apply to laminar weakly compressible smoothed particle hydrodynamics and turbulent Reynolds-averaged calculations and require limited changes to the existing framework. Straight and U-shaped channels, laminar and turbulent plane jets, and a three-dimensional self-rotational micromixer demonstrate improved stability and agreement with finite-volume or experimental references. Residual equation-of-state pressure fluctuations, resolution dependence, and an empirically selected buffer offset remain limitations.

Contributions

1. Introduced an offset generation boundary that prevents runaway particle creation under strong backflow.
2. Added buffer ownership identifiers that eliminate erroneous cross-buffer particle deletion.
3. Constrained effective buffer domains geometrically so arbitrarily arranged inlets and outlets remain independent.
4. Derived a mirror-corrected reproducing-kernel pressure treatment with first-order boundary consistency.
5. Validated laminar and turbulent flows in channels, jets, and a 1.35-million-particle three-dimensional micromixer.

2026WH2

ChatGPT (July 2026)

Summary

This paper presents a dynamical low-rank solver that compresses the three-dimensional velocity dependence of kinetic equations in tensor-train (TT) form while treating physical space parametrically. A projector-splitting integrator updates the TT cores through directional sweeps, avoiding the full high-dimensional step followed by truncation used in step-and-truncate methods. The local representation limits rank growth caused by coupling space and velocity, and it exploits the rank-one structure of Maxwellian equilibria. The formulation is combined with problem-specific spatial and temporal discretizations and tested on homogeneous and inhomogeneous kinetic models, including Vlasov–Ampère–Fokker–Planck dynamics. Experiments reproduce relaxation, Landau damping, and two-stream behavior with small prescribed ranks, and diagnostics show that effective ranks remain low in collisional or near-equilibrium regimes. Collisionless phase mixing can demand higher ranks, which the authors identify as a limitation and motivation for local, anisotropic, and adaptive rank strategies.

Contributions

1. Formulated a velocity-local tensor-train representation for multidimensional kinetic equations.
2. Adapted projector splitting to evolve TT cores without forming or truncating a full velocity tensor.
3. Exploited the rank-one Maxwellian structure to target near-equilibrium and collisional regimes efficiently.
4. Verified the method on homogeneous relaxation and spatially inhomogeneous plasma benchmarks.
5. Quantified effective-rank behavior and identified local and anisotropic rank adaptation as natural extensions.

2026WJPA

ChatGPT (July 2026)

Summary

EDAM, the Equilibrium Derivative in Arbitrary Mesh code, computes first-, second-, and third-order derivatives of tokamak equilibrium poloidal flux directly at nodes of an unstructured triangular mesh. After transforming the Grad–Shafranov operator to a Cartesian Laplacian form, the method derives a second-kind Fredholm boundary-integral equation for the normal flux derivative. Spectral tangential differentiation and the Grad–Shafranov equation supply higher boundary derivatives, while independently solved linear derivative equations propagate them through the interior by finite elements. Validation against Cicogna’s nonlinear D-shaped analytic equilibrium shows approximately second-order L^2 convergence for all three derivative orders, avoiding the progressive accuracy loss of repeated direct differentiation. Applied to KSTAR discharge 18602 on a mesh of 19,838 triangles, EDAM yields smooth derivatives through third order at arbitrary nodes. The current boundary-integral formulation assumes a constant-flux boundary and is therefore unsuitable when a computational boundary outside the plasma crosses a diverted X-point. Third derivatives can also inherit spectral boundary error, and broader device validation remains needed.

Contributions

1. Derived a boundary-integral equation for the normal derivative on constant-flux boundaries.
2. Constructed boundary formulas for second- and third-order poloidal-flux derivatives.

3. Formulated linear derivative Grad–Shafranov equations on arbitrary triangular meshes.
4. Implemented the modular EDAM workflow for equilibrium data, meshing, and finite-element solves.
5. Demonstrated uniform second-order convergence and a practical KSTAR reconstruction.

2026WKN

ChatGPT (July 2026)

Summary

The paper constructs nonlinear numerical boundary fluxes for open-boundary hyperbolic problems whose entropy or energy growth is provably bounded only by prescribed exterior data. It rewrites the entropy flux as a quadratic form in nonlinear characteristic variables and derives an algebraic compatibility condition between a characteristic penalty and the numerical flux. The framework applies directly to split-form discontinuous Galerkin spectral element and related summation-by-parts discretizations. Explicit fluxes are obtained for Burgers’ equation and for every subcritical or supercritical inflow/outflow regime of the two-dimensional shallow-water equations; because physical energy is an entropy for these systems, the bounds also establish energy stability. Manufactured-flow tests verify accuracy, while a geostrophic-adjustment experiment shows that the new outflow flux damps reflections and runs to completion. Standard linearly derived LLF and HLL boundary treatments crash near $t = 19.5$ unless extra interior dissipation is introduced.

Contributions

1. Proved a general algebraic condition ensuring that a nonlinear boundary flux bounds the discrete entropy rate solely by external data.
2. Connected nonlinear characteristic SAT penalties to numerical boundary-flux construction for split-form SBP/DGSEM schemes.
3. Derived an energy-stable Burgers inflow flux and showed why common entropy-conservative and LLF choices lack the required data-only bound.
4. Constructed explicit shallow-water fluxes for all four subcritical and supercritical inflow and outflow regimes.
5. Demonstrated accurate, robust open-boundary behavior where conventional linearly derived LLF and HLL boundary treatments fail.

2026WKRS

ChatGPT (September 2026)

Summary

Willems and coauthors develop a meshless scheme for rarefied-gas flow around moving boundaries and rigid bodies. They write the Bhatnagar–Gross–Krook (BGK) kinetic equation in arbitrary Lagrangian–Eulerian form and move spatial points with the local gas velocity. Moving least squares supplies MUSCL-type spatial reconstruction on the resulting irregular point cloud; an implicit–explicit Runge–Kutta method handles transport and stiff relaxation. A multidimensional optimal-order-detection (MOOD) fallback limits oscillations near discontinuities, while a new diffuse-reflection boundary treatment avoids iterative extrapolation. Convergence studies report fourth order in one dimension and second order in two dimensions. Shock-tube, moving-plate, driven-cavity, and shear-layer tests assess discontinuities and moving geometry. Because the least-squares discretization is not strictly conservative, the paper also measures conservation errors. The demonstrations cover one- and two-dimensional, single-species gas models.

Contributions

1. Coupled a moving-point BGK formulation to meshless MUSCL reconstruction for irregular, moving domains.
2. Combined explicit transport and implicit BGK relaxation in higher-order Runge–Kutta time integration.
3. Adapted MOOD detection to retain accuracy at smooth extrema while controlling oscillations at shocks.
4. Introduced a diffuse-reflection boundary update without iterative solution or extrapolation.
5. Quantified convergence and conservation behavior and demonstrated moving-wall and rigid-body simulations.

2026WL

ChatGPT (July 2026)

Summary

High-resolution ordinary differential equations use backward error analysis to expose behavior hidden by continuous limits of optimization algorithms. Existing $O(s^r)$ -resolution theory assumes the update map satisfies $g(z, 0) = z$, excluding many momentum methods. This paper introduces an $O((\sqrt{s})^r)$ -resolution framework that allows momentum, lifted states, and variable parameters, covering heavy-ball, Nesterov acceleration, and accelerated mirror methods. It proves local and finite-time global approximation bounds and derives correction terms systematically. At low resolution, heavy-ball and Nesterov methods share the same ODE; at $O(\sqrt{s})$, Nesterov contains Hessian-driven damping while heavy-ball contains only a velocity correction, explaining their different stability. The authors then design high-resolution corrections for primal–dual hybrid gradient and heavy-ball methods. Lyapunov analysis proves globally optimal convergence rates for the corrected schemes. Numerical tests show that corrected PDHG removes limit cycles on bilinear saddle problems and that corrected heavy-ball converges more stably and rapidly on a standard piecewise-quadratic counterexample.

Contributions

1. Extended high-resolution backward-error ODE analysis to momentum methods that violate $g(z, 0) = z$.
2. Established systematic $O((\sqrt{s})^r)$ -resolution models with local and finite-horizon approximation bounds.
3. Distinguished Nesterov and heavy-ball dynamics through Hessian-driven damping versus velocity correction.
4. Derived corrected primal–dual hybrid gradient and heavy-ball algorithms from their high-resolution ODEs.
5. Proved optimal convergence rates and validated the predicted stability improvements numerically.

2026WLST

ChatGPT (September 2026)

Summary

This paper develops an efficient compatible finite-element solver for the stiff magnetic-wave subsystem of low- β resistive magnetohydrodynamics in realistic three-dimensional tokamak geometry. An implicit–implicit time split separates fast Alfvén waves and anisotropic heat transport from slower acoustic dynamics while retaining the required coupling. A curl-conforming magnetic-field discretization preserves a discrete divergence constraint. The authors design a matching velocity space so that the Lorentz-force coupling retains an anisotropic curl–curl structure after block elimination. A physics-based block preconditioner then forms a magnetic-field Schur complement and applies auxiliary-space Maxwell multigrid to its dominant operator. Spatial-convergence tests confirm the intended accuracy and discrete energy behavior. Refinement studies on an ITER-like configuration show moderate robustness on anisotropic meshes and demonstrate that the tailored velocity space is necessary to keep iteration counts low at larger timesteps and finer resolutions. Longer nonlinear runs for near-equilibrium, resistive-decay, and kink-instability scenarios produce the expected field evolution with practical iteration counts and runtimes, establishing a scalable route beyond geometry-specific direct solvers.

Contributions

1. Combined constrained-transport finite elements with an implicit–implicit split tailored to stiff tokamak MHD dynamics.
2. Designed a velocity space that preserves the Lorentz-induced curl–curl structure in the magnetic Schur complement.
3. Constructed a physics-based block preconditioner using auxiliary-space Maxwell multigrid.
4. Verified spatial accuracy, discrete magnetic divergence preservation, and consistent energy behavior.
5. Demonstrated robust nonlinear performance on anisotropic three-dimensional tokamak tests, including a kink instability.

2026WLWC

ChatGPT (July 2026)

Summary

Wang, Li, Wu, and Cui reassess proton-boron fusion using newer p - ^{11}B cross-section measurements. They build a continuous analytical parameterization of the astrophysical S -factor and cross-section over 0–10 MeV, including the dominant resonance near 0.6 MeV and newer information around 4.7 MeV. The resulting fit is used to compute thermonuclear reactivity and fusion power density. The energy-balance analysis compares fusion power with electron bremsstrahlung losses under two thermal assumptions. If electrons and ions are forced to share a temperature, the conventional bremsstrahlung obstacle remains. When the electron temperature is instead determined self-consistently from the electron energy balance, the model admits a finite ion-temperature interval where fusion power exceeds bremsstrahlung. The result does not make p - ^{11}B fusion easy, but it argues that updated nuclear data and thermal modeling no longer fundamentally exclude it.

Contributions

1. Produced a 0–10 MeV analytical parameterization of the p - ^{11}B cross-section.
2. Incorporated recent experimental data, including the high-energy resonance region.
3. Recomputed thermonuclear reactivity from the updated cross-section model.
4. Compared fusion power density with bremsstrahlung under different temperature assumptions.
5. Identified a self-consistent thermal window where bremsstrahlung need not dominate.

2026WRS

ChatGPT (July 2026)

Summary

The authors use an ensemble of three-dimensional incompressible magnetohydrodynamic direct numerical simulations to isolate how unstable reconnection creates turbulence without external forcing. An initially weak uniform magnetic field is stretched by an unstable jet until current sheets form. A three-dimensional current-sheet instability then drives a transition from laminar Sweet–Parker-like reconnection through stochastic reconnection to a decaying turbulent state. Reynolds-decomposed kinetic and magnetic energy budgets show that turbulent electromotive-force interaction with the mean magnetic shear is the dominant source of fluctuating magnetic energy. Most energy extracted from the mean magnetic field first enters Alfvénic fluctuations and is then exchanged with kinetic fluctuations through a nonlinear, nearly conservative inter-domain cascade. Turbulent production remains localized near reconnection sites, while local negative electromotive work in outflows returns some energy to the mean field. The study supplies a mechanistic energy pathway for reconnection-driven turbulence and distinguishes it from turbulence-triggered reconnection.

Contributions

1. Demonstrated a self-sustained laminar-to-turbulent reconnection transition in unforced three-dimensional magnetohydrodynamic simulations.
2. Used a five-realization ensemble and Reynolds and phase averaging to separate coherent current sheets from stochastic fluctuations.
3. Derived turbulent kinetic- and magnetic-energy budgets covering production, dissipation, transport, and cross-domain transfer.
4. Identified electromotive-force–mean-magnetic-shear work as the dominant production route from mean magnetic energy to Alfvénic fluctuations.
5. Showed linear magnetic-energy-annihilation-rate growth during the current-sheet instability and localized turbulence production to reconnection regions.

2026WTCJ

ChatGPT (September 2026)

Summary

BIT1, an electrostatic particle-in-cell Monte Carlo code with one spatial and three velocity dimensions, is modernized for heterogeneous supercomputers. The hybrid MPI+OpenMP implementation combines contiguous particle arrays, persistent device memory, pinned host memory, direct device-pointer access, and asynchronous target tasks with explicit dependencies. These changes reduce transfer and synchronization overhead while supporting NVIDIA and AMD accelerators. An ionization benchmark on four NVIDIA H100 GPUs falls from approximately 1919 to 113 seconds, a seventeenfold improvement over the initial GPU implementation. Divertor-sheath benchmarks retain 97–98% weak-scaling efficiency with minimal diagnostics through 100 nodes. With heavier diagnostics on 2000 Frontier nodes, openPMD and ADIOS2 streaming achieve 73.6% weak-scaling efficiency and a 5.25-fold strong-scaling speedup relative to 50 nodes. The study demonstrates how memory management, asynchronous execution, and scalable output jointly improve an established plasma code. Results concern the selected ionization and sheath workloads; Intel GPU support remains future work.

Contributions

1. Implemented portable asynchronous MPI+OpenMP execution on NVIDIA and AMD GPUs using explicit task dependencies and direct device-pointer access.
2. Combined contiguous particle arrays, persistent device allocations, and pinned host memory to reduce transfer and allocation overhead.
3. Demonstrated approximately seventeenfold total-runtime improvement on four H100 GPUs through an incremental ionization-benchmark optimization study.
4. Demonstrated 97–98% weak-scaling efficiency for sheath simulations with minimal diagnostics through 100 nodes.
5. Integrated openPMD with ADIOS2 file and streaming backends, achieving 73.6% weak-scaling efficiency for simulations with heavy diagnostic output on 2000 Frontier nodes.

2026WW

ChatGPT (July 2026)

Summary

Global gyrokinetic simulations investigate ion-temperature-gradient instability and internal-transport-barrier formation close to a tokamak's magnetic axis, where conventional ballooning representations fail. A mean radial-electric-field well raises the critical temperature gradient for linear instability when central magnetic shear is weak, but not when it is strong. The distinction follows from geometric curvature: near the axis it nearly cancels the $E \times B$ shear in the inner half of the well and roughly doubles it in the outer half. Consequently, the electric-field well stabilizes high-toroidal-mode-number turbulence outside, while weak magnetic shear must suppress the low-mode-number instability inside. Nonlinear simulations including ion heating and ion-ion collisions produce an internal transport barrier near $r = 0.13a$ only for weak shear and support approximate radial force balance for the mean electric field. The authors therefore propose a complementary mechanism in which weak shear and curvature-modified $E \times B$ stabilization act in different radial regions. Electron neoclassical fluxes and the electric-field generation mechanism are outside the model's scope.

Contributions

1. Simulated near-axis ion-temperature-gradient modes without relying on the usual ballooning approximation.
2. Identified a weak-shear-specific linear upshift of the critical ion-temperature gradient.
3. Derived how geometric curvature suppresses or amplifies $E \times B$ shear across the radial-electric-field well.
4. Separated the radial roles of weak magnetic shear and electric-field shear in turbulence suppression.
5. Confirmed the proposed internal-transport-barrier scenario with nonlinear gyrokinetic simulations.

2026WWQW

ChatGPT (July 2026)

Summary

2DESR directly solves the two-dimensional gyrokinetic eigenvalue problem for electrostatic ion-temperature-gradient modes in tokamaks. Starting from the linear Vlasov–Poisson system with adiabatic electrons, the formulation retains passing and trapped ions, finite-Larmor-radius effects, and coupled poloidal harmonics in Fourier coordinates (z, m) with $z = nq(r) - m$. Finite differences in z and parallel velocity, Gauss–Laguerre quadrature in magnetic moment, multipoint gyro-averaging, PARDISO linear solves, and a Newton eigenvalue iteration produce eigenfrequencies and global mode structures. Cyclone-case scans over toroidal mode number agree with GENE and NLT eigenvalues and with NLT eigenfunctions at both low and high n . Unlike initial-value solvers, 2DESR resolves two simultaneously unstable ITG branches, with different radial localization and growth-rate peaks near $n = 20$ and $n = 30$. The branch competition explains a prior GENE–GKW frequency mismatch and the unusually long time evolution required near $n = 35$.

Contributions

1. Derived a two-dimensional poloidal-Fourier gyrokinetic eigenproblem retaining full ion kinetics and coupled radial and poloidal structure.
2. Implemented finite-difference, Gauss–Laguerre, gyro-average, sparse-linear-solve, and Newton components in the 2DESR code.
3. Used (z, m) position coordinates and $(v_{\parallel}, \mu)$ velocity coordinates to treat passing and trapped ions uniformly with manageable cost.
4. Benchmarked eigenvalues and global eigenmode structures against the GENE and NLT initial-value codes for low- and high- n Cyclone cases.
5. Identified two coexisting ITG branches with distinct radial localization and explained an earlier inter-code real-frequency discrepancy.

2026WWZ

ChatGPT (July 2026)

Summary

The paper constructs variable-step, second-order exponential time-differencing schemes for periodically forced incompressible Navier–Stokes equations. Its main ETD-mr-ccSAV method combines a mean-reverting scalar auxiliary variable, a concurrent second-order auxiliary-variable correction, and multistep ETD treatment of nonlinear advection. For arbitrary positive step sequences and bounded forcing, the discrete velocity or vorticity has a uniform-in-time L^2 bound independent of both step size and viscosity, while nonlinear advection remains explicit. Each step requires two constant-coefficient heat or Stokes solves and one scalar cubic equation. A first-order embedded companion supplies adaptive error estimates and step selection. Two-dimensional Fourier pseudospectral experiments verify second-order temporal accuracy and stable long integrations where classical ETD-MS2 and a correction-only variant can grow unstable. In Kolmogorov-flow tests, adaptive integration reproduces fixed-step mean enstrophy and flow scales while reducing computational cost; rare-event tail statistics are less accurate. Rigorous convergence and invariant-measure convergence are left open.

Contributions

1. Combined mean-reverting and concurrent-correction SAV mechanisms with multistep ETD to obtain variable-step second-order schemes.
2. Proved uniform-in-time L^2 bounds under general bounded forcing for arbitrary positive step sizes and all viscosity values.

3. Reduced each update to two constant-coefficient heat or Stokes solves plus one scalar cubic equation while treating advection explicitly.
4. Built an embedded first-order companion scheme for adaptive error estimation and automatic step-size control.
5. Verified second-order accuracy, long-time robustness, statistical fidelity, and computational savings in periodic two-dimensional flow experiments.

2026WXXM

ChatGPT (July 2026)

Summary

Quasi-three-dimensional particle-in-cell simulations propose an all-optical route to nanometer-scale pre-bunched electron beams for compact X-ray free-electron lasers. Two low-intensity counter-propagating lasers establish a plasma-density grating before an intense drive laser arrives. The grating periodically modulates the phase velocity of the expanding laser wake, switching self-injection on and off; longitudinal mapping from injection time to wake phase converts this gating into beam microbunching. Simulations produce multi-10 kA beams whose main body has approximately 30 nm slice emittance and 0.5 MeV slice energy spread. Polarization matters: equal-frequency parallel drive and grating fields disrupt the wake modulation, while perpendicular polarization or a sufficiently different colliding-laser wavelength restores bunching. A 1600 nm case yields a 16 nm period and bunching factor 0.025. Adjusting laser chirp, duration, and delay generates chirped-period or partially bunched structures. These are simulation predictions; beam transport, preservation of nanometer structure, and X-ray production remain to be demonstrated experimentally.

Contributions

1. Proposed density-grating modulation of wake phase velocity as an all-optical periodic injection gate.
2. Demonstrated nanometer-scale microbunching in fully self-consistent quasi-three-dimensional simulations.
3. Quantified simulated current, slice emittance, energy spread, bunching period, and bunching factor.
4. Identified laser-polarization interference and showed how wavelength separation can suppress it.
5. Tuned laser chirp, duration, and timing to create chirped-period and partially pre-bunched beam structures.

2026WZJL

ChatGPT (July 2026)

Summary

The authors build a two-dimensional finite-element model of electromagnetic transmission through the magnetized plasma sheath surrounding a reentry vehicle. Plasma flow fields at several altitudes are generated with USIM and transferred to COMSOL, where spatially varying plasma properties and an imposed magnet field determine full-wave propagation. Field maps and dispersion behavior verify the implementation. The simulations quantify the magnetic-window mechanism: transmission depends jointly on magnetic-field strength, the background-field excitation position, and wave frequency, with lower-frequency whistler waves improving penetration under realistic field limits. For a maximum field of 2 T, frequency tuning can keep the wave-loss coefficient near or above -40 dB across the studied altitudes. Under a stricter 1 T limit, locating the magnetic window about 1.1 m from the reference position supports transmission over 0–0.5 GHz. The coupled workflow also reduces per-frequency solution time relative to performing the electromagnetic calculation inside the flow solver.

Contributions

1. Coupled reentry plasma-flow data to a two-dimensional full-wave finite-element model.
2. Verified magnetized-sheath propagation through dispersion trends and field-distribution diagnostics.
3. Quantified how magnetic strength and signal frequency control whistler-wave transmission loss.
4. Identified an effective magnetic-window position under a realistic 1 T field constraint.
5. Demonstrated a faster multiphysics workflow for evaluating plasma-blackout mitigation strategies.

2026X

ChatGPT (July 2026)

Summary

The Landau–Lifshitz–Gilbert equation constrains the magnetization to unit length, while a fully implicit Crank–Nicolson discretization is nonlinear. This paper proposes a linear fractional-step alternative based on componentwise Gauss–Seidel updates of a Crank–Nicolson-type system. Each magnetization component is solved successively through linear elliptic subproblems, avoiding an external projection step. The construction is designed to preserve the discrete length constraint and retain the favorable stability behavior of the implicit formulation. The paper presents consistency and stability arguments and reports first-order temporal and second-order spatial accuracy. One-dimensional convergence tests and three-dimensional magnetization examples assess accuracy, norm preservation, and stability for several initial configurations. The method offers a simple structure-preserving update, but the exposition leaves a fuller convergence and stability theory, finite-element spatial discretization, and realistic micromagnetic applications as future work.

Contributions

1. Proposed a linear Crank–Nicolson-type fractional-step method for the Landau–Lifshitz–Gilbert equation.
2. Used implicit Gauss–Seidel component updates to avoid nonlinear coupled solves.
3. Preserved the magnetization-length constraint without a separate projection step.
4. Presented first-order-in-time and second-order-in-space accuracy and stability analysis.
5. Tested accuracy, norm preservation, and stability in one- and three-dimensional examples.

2026XDZW

ChatGPT (July 2026)

Summary

Experiments on the full-metal-wall EAST tokamak demonstrate an edge-localized-mode-free high-confinement regime sustained for 50–70 s while the divertor remains partially detached. Feedback-controlled nitrogen seeding lowers target temperature and suppresses tungsten sources without the usual pedestal degradation; a lower- q_{95} discharge raises H_{98y2} from about 0.85 to 1.2. The proposed detached, turbulence-dominated-pedestal mechanism begins when power starvation reduces recycling-neutral ionization in the closed divertor. Higher neutral pressure enhances pumping, weakens pedestal cooling, and steepens the electron temperature gradient. High-frequency broadband turbulence then supplies outward particle and heat transport that arrests pedestal growth and prevents ELMs. GENE calculations identify this turbulence as an η_e -driven trapped-electron mode, consistent with its frequency, wavenumber, gradient dependence, collisional stabilization, and nonlinear fluxes. Independent GENE and CGYRO calculations predict the same mode can overcome $E \times B$ shear in an ITER baseline pedestal. ITER would require a nonreactive seed impurity such as neon rather than nitrogen.

Contributions

1. Demonstrated minute-scale ELM-free H-mode operation with divertor partial detachment and strong confinement.
2. Showed that feedback-controlled seeding reduces divertor heat and metal-impurity sources without cooling the pedestal.
3. Identified high-frequency broadband pedestal turbulence correlated with complete ELM suppression.
4. Matched the measured turbulence to an η_e -driven trapped-electron mode using linear and nonlinear gyrokinetics.
5. Predicted with two gyrokinetic codes that the mechanism is accessible in the ITER baseline pedestal.

2026XL

ChatGPT (July 2026)

Summary

This paper analyzes the spectral sign conditions underlying Van Leer and original AUSM flux-vector splittings for the one-dimensional Euler equations. By transforming between conservative variables and (ρ, a, M) primitive variables, it factors the Van Leer split-flux Jacobian into one zero eigenvalue and a quadratic for the remaining pair. Explicit coefficient signs and a Sturm-sequence proof of discriminant nonnegativity establish that, for $1 \leq \gamma \leq 3$, $|M| < 1$, and $a > 0$, the two nonzero eigenvalues of $\partial F^+ / \partial U$ are real and positive. This supplies the detailed proof omitted from Van Leer's original presentation; symmetry gives the corresponding negative-split result. Applying the same framework to Liou–Steffen AUSM variants reveals that linear pressure splitting violates the common-sign condition. For second-order pressure splitting, the characteristic cubic has positive coefficients, and numerical scans support a nonnegative discriminant and hence three positive real eigenvalues. The latter reality claim remains numerical rather than a complete analytic proof.

Contributions

1. Derived explicit primitive-variable Jacobians and characteristic polynomials for Van Leer and two original AUSM pressure splittings.
2. Identified the rank deficiency that gives the Van Leer positive split one zero eigenvalue.
3. Proved discriminant nonnegativity with a Sturm sequence and established the Van Leer eigenvalue sign condition over the stated parameter domain.
4. Demonstrated analytically that AUSM with linear pressure splitting fails to keep all split-Jacobian eigenvalues of one sign.
5. Proved positive cubic coefficients for second-order AUSM pressure splitting and supplied numerical evidence that its three eigenvalues are real and positive.

2026XLDM

ChatGPT (July 2026)

Summary

Proton–boron spherical tokamaks combine large ion-mass disparity with neutral-beam-driven toroidal rotation, so centrifugal separation and electrostatic polarization can invalidate single-fluid equilibria. The paper derives a reduced axisymmetric multi-fluid model that retains species-dependent toroidal inertia and a self-consistent electrostatic potential while neglecting poloidal-flow inertia, pressure anisotropy, and finite-orbit-width corrections. Species Bernoulli relations and pointwise quasineutrality close a generalized Grad–Shafranov equation. The accompanying NFEQ solver uses nested Picard and Newton iterations and adjusts the toroidal-field profile to enforce a prescribed plasma current. Fixed-boundary calculations for experimental EHL-2 and reactor-scale EHL-3B recover single-fluid behavior for boron Mach number below about 0.5. Above unity, boron shifts toward the low-field side; at reactor-relevant high rotation, the density becomes strongly crescent-shaped and the self-consistent potential reaches roughly 10–15 kV. The resulting fuel mismatch, pressure redistribution, and altered safety-factor profile show why multi-fluid effects matter for equilibrium targets, although stability, transport, anisotropy, and free-boundary dynamics remain future work.

Contributions

1. Derived a reduced multi-fluid equilibrium system coupling species-specific Bernoulli distributions, quasineutrality, and a generalized Grad–Shafranov equation.
2. Preserved an elliptic field problem while retaining the dominant centrifugal and electrostatic physics of strongly rotating $p\text{--}^{11}\text{B}$ plasmas.
3. Developed the NFEQ fixed-boundary algorithm with a pointwise analytic-Jacobian Newton solve, outer Picard iteration, and feedback control of total plasma current.

4. Recovered the static single-fluid limit and identified weak changes for boron Mach number below 0.5 but substantial separation above unity.
5. Showed that high rotation in EHL-2 and EHL-3B produces low-field-side boron accumulation, multikilovolt potentials, outboard-peaked current, and modified safety-factor profiles.

2026XM

ChatGPT (September 2026)

Summary

Discontinuous Galerkin spectral element implementations need consistent interface fluxes across element shapes, basis choices, and differing polynomial orders. The authors analyze two ways to evaluate nonconforming interfaces in a general Nektar++ framework. A shared trace space gives both sides the same quadrature and flux evaluation, preserving the symmetry needed by conjugate-gradient solves of symmetric interior-penalty Helmholtz systems. Point-to-point interpolation can be cheaper, but inconsistent interface evaluation can make the matrix nonsymmetric. Their matrix-free workflow keeps most operations in physical space and limits expensive transforms to one backward transform and one inner product. CPU implementations reuse cached data and vectorize batches across hexahedra, prisms, and tetrahedra. Benchmarks compare interface strategies, polynomial orders, and element types; interpolation is faster than trace inner products but carries the stated numerical caveat. The study treats polynomial nonconformity, with geometric nonconformity left for future work.

Contributions

1. Identified how inconsistent nonconforming flux evaluation breaks interior-penalty matrix symmetry.
2. Developed shared-trace and point-to-point interface strategies for general spectral elements.
3. Reduced coefficient-physical transforms in a matrix-free discontinuous Galerkin workflow.
4. Implemented cache reuse and SIMD batching for several unstructured element shapes in Nektar++.
5. Measured CPU throughput and exposed the speed-symmetry trade-off between interface strategies.

2026XMY

ChatGPT (July 2026)

Summary

The paper proposes a projection-free finite-difference integrator for the Landau–Lifshitz equation, whose magnetization must satisfy the pointwise unit-length constraint. It combines an implicit Gauss–Seidel predictor, a damping update, a second diffusion solve, and a Crank–Nicolson-type rotation step. The final step is based on a Cayley transform of a skew-symmetric cross-product operator, making that update orthogonal and therefore exactly norm preserving without spherical projection. The resulting method is reported as first order in time and second order in space. Manufactured one- and three-dimensional tests examine temporal and spatial errors, compare with the first-order Gauss–Seidel projection method, and test several initial magnetization fields; the authors report consistent, stable, and robust behavior. A significant qualification is required: despite “energy stable” in the title, the manuscript does not present a theorem proving energy stability of the full composite method, and its conclusion says stability analysis is future work. The evidence therefore establishes exact norm preservation and numerical stability, not a rigorous full energy-stability proof.

Contributions

1. Constructed a projection-free composite integrator from Gauss–Seidel, damping, double-diffusion, and Crank–Nicolson-type stages.
2. Used the orthogonality of a Cayley-transform update to establish exact norm preservation for the final rotational step.
3. Avoided the nonlinear spherical projection used by the conventional Gauss–Seidel projection method.

4. Reported first-order temporal and second-order spatial behavior in manufactured one- and three-dimensional accuracy tests.
5. Provided numerical comparisons and varied-initial-condition experiments supporting consistency and practical robustness while leaving rigorous full-method stability analysis open.

2026YCW

ChatGPT (July 2026)

Summary

The paper develops a Fourier spectral algorithm for the Boltzmann collision operator linearized around a local Maxwellian for variable hard-sphere gases. By reformulating the operator in terms of Fourier coefficients of the relative distribution, it removes the angular-quadrature factor M from the leading cost, reducing $O(MN^4 \log N)$ work to $O(N^4 \log N)$. A cutoff prevents catastrophic cancellation when dividing small distribution values by the Maxwellian. Homogeneous tests show spectral accuracy, close agreement with the quadratic collision operator, and substantial speedups. The algorithm is then embedded in a modified Newton method for steady Boltzmann problems, replacing the exact Jacobian by a local-Maxwellian linearization. Couette- and Fourier-flow experiments show convergence in only a few outer iterations while limiting expensive quadratic-operator evaluations.

Contributions

1. Derived a fast Fourier representation of the local-Maxwellian linearized collision operator without the angular-quadrature prefactor M .
2. Identified catastrophic cancellation in transforming $f/\mathcal{M}$ and introduced a gain-term cutoff that restores double-precision accuracy.
3. Verified spectral convergence and quantified differences between linearized and quadratic collision dynamics.
4. Demonstrated substantial timing gains over quadratic collision solvers in homogeneous tests.
5. Built a modified Newton steady Boltzmann solver that converges in two to five outer iterations for Couette and Fourier flows.

2026YF

ChatGPT (July 2026)

Summary

The authors combine an entropy-stable nodal discontinuous Galerkin spectral-element method for the two-dimensional compressible Euler equations with a localized oscillation-eliminating procedure on general curvilinear quadrilateral meshes. Summation-by-parts operators, discrete metric identities, and entropy-stable interface fluxes yield a semidiscrete global entropy inequality. Projection operators replace the original method's dependence on local orthogonal modal bases, while the zero-order damping coefficient becomes a troubled-cell sensor and a scale factor controls dissipation. Smooth-vortex tests on deformed meshes recover $O(h^{N+1})$ convergence and non-increasing total entropy. Shock benchmarks capture Riemann interactions, Mach-10 double reflection, Mach-3 cylinder flow, and an explosion among eight cylinders. In double-Mach reflection, localization reduces CPU time from 3123.8 s with all cells damped to 2570.3 s at threshold 0.02 and 2128.3 s at 0.1. Accuracy and dissipation depend materially on the chosen threshold and scale.

Contributions

1. Proved semidiscrete entropy stability for DGSEM on curvilinear meshes.
2. Extended oscillation elimination to curved quadrilateral elements using projections.
3. Reused the damping coefficient as a troubled-cell indicator.
4. Demonstrated high-order convergence and entropy control on deformed meshes.
5. Quantified robustness and cost reductions on strong-shock, complex-geometry tests.

2026YG

ChatGPT (July 2026)

Summary

This study asks whether SPARC, whose vessel, divertor, and poloidal-field coils were designed for positive triangularity, can support negative-triangularity plasmas. More than 600 FreeGS free-boundary Grad–Shafranov calculations scan triangularity $-0.7 \leq \delta \leq 0.7$ and elongation $1.0 \leq \kappa \leq 2.1$ under coil-current and wall-clearance constraints. At a reduced toroidal field of 8 T, a feasible double-null case has $\delta = -0.35$, $\kappa = 1.68$, plasma current 2.1 MA, and major radius 1.75 m. Relative to the primary-reference design, its plasma volume is about 42–43% smaller and its connection length is roughly 40% shorter, limiting fusion performance and divertor compatibility. Central-solenoid demand falls by 54% relative to an 8 T positive-triangularity comparison, but the PF3 shaping-coil current grows by about a factor of 5.5. The computed equilibria satisfy Troyon-beta and kink-safety-factor criteria, although vertical stability, transport, and integrated discharge evolution remain for future analysis.

Contributions

1. Mapped the negative-triangularity equilibrium space accessible within SPARC’s engineering limits.
2. Constructed a feasible 8 T, 2.1 MA negative-triangularity double-null equilibrium.
3. Quantified volume and connection-length penalties imposed by positive-triangularity hardware geometry.
4. Exposed a trade-off between reduced central-solenoid demand and increased PF3 shaping current.
5. Verified basic beta and kink margins while identifying stability and scenario-modeling work still required.

2026YLFP

ChatGPT (July 2026)

Summary

This paper uses two-dimensional direct numerical simulations to quantify how insoluble surfactants alter nonlinear gravity-capillary waves. A coupled volume-of-fluid and phase-field method resolves surfactant transport, a nonlinear surface-tension equation of state, and Marangoni stresses across wave steepness ak , Bond number Bo , Reynolds number Re , and surfactant strength β . Surfactants act through two competing mechanisms: reduced local surface tension shifts transitions among parasitic-capillary, gravity-wave, and breaking regimes, while concentration gradients generate Marangoni stresses that smooth the surface. Consequently, roughness, vorticity, and dissipation vary non-monotonically with β ; Marangoni forcing peaks near intermediate concentration, around $\beta \approx 0.3$, suppressing capillary ripples but increasing near-surface vorticity and energy loss. A surfactant-dependent effective Bond number predicts regime changes, while the rescaled parameter $BoRe^{-1/2}(ak)^{-1}$ collapses boundaries of the Marangoni-dominated regime. At sufficiently strong surfactant loading, dissipation collapses against a local Marangoni number. Grid and parameter studies support the numerical trends.

Contributions

1. Simulated surfactant-laden gravity-capillary waves across a broad four-parameter regime space.
2. Separated wave-regime changes caused by surface-tension reduction from smoothing caused by Marangoni stress.
3. Established the non-monotonic surfactant dependence of roughness, vorticity, and energy dissipation.
4. Introduced effective-Bond-number and force-balance scalings that collapse transition boundaries.
5. Linked enhanced intermediate-concentration dissipation to Marangoni-driven near-surface vorticity.

2026YM

ChatGPT (July 2026)

Summary

This paper analyzes an explicit Adams-Bashforth-type integrator (ABTI) built from complex-time sampling and a Cauchy-integral discretization. Although numerical evidence had suggested that its absolute-stability regions approach a nonzero limiting semicircle as the order grows, harmonic-analysis arguments disprove that conjecture: the parabolic stability radius ultimately tends to zero, albeit much more slowly than for classical Adams-Bashforth formulas. The authors derive a characteristic polynomial for the amplification matrix and a computable condition relating a prescribed parabolic radius to the largest admissible approximation order. They also show that the original reconstruction loses one order of accuracy and give a correction that restores the intended order. For parabolic partial differential equations, tensor-product spectral analysis yields a sharp CFL restriction, L^2 stability, and error estimates. Numerical tests on ordinary and parabolic differential equations confirm the predicted orders and show instability when the time step crosses the derived CFL boundary.

Contributions

1. Disproved the conjectured nonzero limiting stability region for arbitrary-order ABTI schemes.
2. Derived a characteristic polynomial and an order-versus-parabolic-radius stability criterion.
3. Identified the source of the original method's one-order accuracy loss and supplied a correction.
4. Established CFL-based L^2 stability and error bounds for parabolic PDE discretizations.
5. Verified the corrected convergence orders and sharp stability threshold numerically.

2026YN

ChatGPT (July 2026)

Summary

The authors introduce a basis-independent diagnostic for velocity-space complexity in gyrokinetic turbulence by applying singular value decomposition to the time-dependent perturbed ion distribution at each perpendicular Fourier mode and interpreting normalized singular-value weights through a von Neumann entropy. Nonlinear flux-tube gyrokinetic-Poisson simulations show low entropy at low wavenumber, where a few singular modes reconstruct the distribution, followed by a rapid increase around $k_{\perp}\rho_{ti} \sim 1$, where many modes become dynamically relevant. Linear growth-rate comparisons place the dominant low-wavenumber electrostatic power near the ion-temperature-gradient unstable range. Hermite and Laguerre decompositions separate parallel Landau phase mixing from perpendicular finite-Larmor-radius mixing: both are active, but broadening of the Hermite spectrum correlates more strongly with the high-wavenumber entropy rise. The diagnostic compactly maps scale-dependent kinetic complexity without prescribing a velocity-space basis.

Contributions

1. Defined an SVD-based von Neumann entropy for time-evolving gyrokinetic distribution functions.
2. Mapped velocity-space complexity across the full simulated perpendicular-wavenumber spectrum.
3. Identified a sharp complexity increase near $k_{\perp}\rho_{ti} \sim 1$.
4. Connected low-wavenumber turbulent power to the linear ion-temperature-gradient unstable range.
5. Used Hermite-Laguerre spectra to attribute the entropy rise mainly to enhanced parallel phase mixing.

2026YT

Claude (September 2026)

Summary

Alfvén eigenmodes destabilised by energetic ions can carry the ions' energy from the region where the mode is driven to the region where it is damped, a process called spatial channelling; because the mode has toroidal mode number n , it also carries toroidal angular momentum, $\Gamma_{J_\phi} = (n/\omega)\Gamma_E$. From resonant motion in action–angle variables, the authors relate the radial current of resonant ions to the power they transfer to the mode, separately for trapped and passing ions. Treating the energetic ions kinetically and the thermal species as fluids, and assuming $v_A/c \ll 1$ so that the thermal polarisation current nearly cancels the fast-ion current, they obtain the torque on the thermal plasma. For trapped ions, the thermal plasma receives essentially all the momentum given to the mode; for passing ions, a fraction, small for low-frequency modes with $n \gg 1$, stays with the energetic ions. Separated drive and damping regions therefore exert opposite torques that tend to generate sheared toroidal flow. An estimate for illustrative DIII-D-like parameters gives rotation of about 85 km/s and a shearing rate of $2.5 \times 10^4 \text{ s}^{-1}$, below an estimated turbulence growth rate of $5.2 \times 10^4 \text{ s}^{-1}$. The authors stress that the estimate depends on uncertain parameters and neglects viscosity and mode-induced momentum redistribution.

Contributions

1. Extended the theory of momentum channelling from well-passing to trapped energetic ions, deriving relations between the radial resonant-ion current and the power transferred to the mode for both orbit classes.
2. Derived toroidal and poloidal acceleration equations with a non-hydrodynamic energetic-ion population, recovering the near cancellation of the fast-ion current by the thermal polarisation current.
3. Showed that the momentum fraction retained by passing energetic ions is bounded by about $2/(nq)$ for low-frequency modes, becoming significant only for $n \sim 1$ or high-frequency, cyclotron-resonant modes.
4. Identified momentum channelling as a mechanism for sheared-flow generation and gave a steady-state scaling of the induced rotation with mode number, drive rate, perturbation amplitude, mode width, and confinement time.
5. Contrasted momentum channelling with nonlinear zonal-flow excitation: it needs spatially separated drive and damping and produces flow acceleration proportional to the squared mode amplitude rather than an exponential growth rate.

2026YZLY

Claude (September 2026)

Summary

TokaGLINT is a GPU linear solver for the implicit electromagnetic field update in SymPIC, a symplectic, charge-conserving particle-in-cell code for tokamaks. Replacing the explicit field step with a Crank–Nicolson finite-difference time-domain discretization removes the electromagnetic Courant limit while keeping the symplectic structure, but requires solving a metric-weighted curl–curl system on a cylindrical mesh at every step. The solver uses pipelined BiCGStab, which hides global reductions behind local work, preconditioned by a hierarchical additive Schwarz method: overlapping subdomains across GPUs, each split into several overlapping subdomains on one GPU. Each small subdomain is solved exactly by singular-value-based discrete transforms of the difference operators, which decouple unknowns in two directions despite the radial metric and leave diagonal plus $(2n_x) \times (2n_x)$ block systems with precomputed inverses. Equal-radius subdomains share operators and are binned so transforms run as batched GEMMs and block solves fuse into GEMMs. The preconditioner keeps BiCGStab near 7–8 iterations in weak scaling and far below Jacobi, SOR, ISAI, ILUT, and algebraic multigrid counts at large time steps. Results include 2.67–3.03× speedups over unpreconditioned HYPRE BiCGStab, 90.1% weak-scaling efficiency from 16 to 10,000 GPUs, 53.9% strong-scaling efficiency from 672 to 10,752 GPUs, and energy held within $\pm 0.1\%$ over 10^6 steps.

Contributions

1. Recast SymPIC's electromagnetic update as a field-implicit Crank–Nicolson scheme on cylindrical meshes, expressed as a metric-weighted curl–curl system for the discrete 1-form unknowns.
2. Extended discrete-transform fast exact solvers from Cartesian to cylindrical curl–curl operators, reducing each subdomain solve to diagonal and radial-line block systems with $O(N^{4/3})$ storage.
3. Designed a two-level additive Schwarz preconditioner that places several overlapping subdomains on each GPU, exchanging overlaps through high-bandwidth memory, and paired it with communication-hiding BiCGStab.
4. Introduced geometry-driven binning of equal-radius subdomains with custom data layouts, turning transforms into batched GEMMs and fusing block-diagonal solves, with one MPI process managing all subdomains on a GPU.
5. Integrated the solver into SymPIC and validated it with toroidal wave-propagation and long-time magnetized-plasma energy-conservation tests.

2026Z

ChatGPT (September 2026)

Summary

A flux-driven global simulation with the electromagnetic, drift-reduced Braginskii code GDB develops a centrally peaked density profile from an initially flat profile with particle fueling localized near the last closed flux surface. The 8.4 ms calculation uses a collisional, C-Mod-like, inner-wall-limited tokamak edge with freely evolving density boundaries. Two inward-transport mechanisms emerge. Initially, opposing density and temperature gradients support drift-wave turbulence whose electron thermal-diffusion response drives inward particle transport despite outward heat transport. Linear phase analysis and a control run suppressing electron-temperature evolution support this interpretation. Later, poloidally asymmetric equilibrium density and potential produce an inward $\mathbf{E} \times \mathbf{B}$ flux. The measured poloidal phase offset, approximately -0.08π , is attributed to nonideal corrections to electron force balance; the leading-order Boltzmann response would cancel this flux. Removing transverse heat fluxes erases the up-down asymmetry and relaxes density peaking. Magnetic flutter also carries inward particles. The late-time particle budget does not close, so its numerical channel fractions are indicative rather than definitive. The demonstrated mechanism concerns a collisional edge model; extension to hotter, less collisional plasmas requires improved closures.

Contributions

1. Demonstrates transport-timescale density peaking with edge-localized fueling, without prescribed core density or an imposed pinch velocity.
2. Uses linear cross-phase analysis and an electron-temperature control simulation to identify thermal diffusion as an early inward turbulent-transport mechanism.
3. Identifies a late inward equilibrium $\mathbf{E} \times \mathbf{B}$ channel associated with poloidal density–potential phase shifts beyond the leading-order Boltzmann response.
4. Shows that suppressing transverse heat fluxes removes the up-down asymmetry and relaxes the centrally peaked density profile.
5. Decomposes particle transport into turbulent, equilibrium, and magnetic-flutter contributions while exposing limitations of the late-time accumulation budget.

2026ZCLB

ChatGPT (July 2026)

Summary

Flux-tube gyrokinetic calculations treat one stellarator field line at a time, but nonaxisymmetry couples field lines across a flux surface. This paper uses the global gyrokinetic particle-in-cell code GTC to study linear electrostatic ion-temperature-gradient modes in quasi-axisymmetric, quasi-helically symmetric, and Wendelstein 7-X configurations.

The eigenmodes are nonuniform across field lines and localize downstream of the ion diamagnetic drift, with stronger localization as the normalized gyroradius decreases. A WKB analysis of a reduced surface-global model attributes this behavior to a nonzero imaginary component of the binormal wavenumber: a single global eigenfrequency lies below the growth rate of the most unstable flux tube, so satisfying the dispersion relation away from that line requires analytic continuation to complex wavenumber. The authors then reconstruct localized surface-global modes from local stella and GX calculations by first solving each real-wavenumber dispersion relation and continuing it into the complex plane. Agreement with GTC shows that real-wavenumber local spectra alone cannot recover the cross-field-line envelope. The analysis is limited to linear electrostatic modes with adiabatic electrons, but it provides a mechanism for interpreting symmetry-breaking localization seen in nonlinear stellarator turbulence.

Contributions

1. Demonstrated surface-global localization of linear ion-temperature-gradient eigenmodes across three stellarator configurations.
2. Identified the downstream ion-diamagnetic-drift direction as the consistent location of peak mode amplitude.
3. Explained localization through the imaginary binormal wavenumber obtained from a WKB surface-global model.
4. Showed that the global growth rate lies below that of the most unstable local flux tube.
5. Reconstructed localized global eigenmodes from stella and GX results by analytically continuing local dispersion relations to complex wavenumbers.

2026ZCXY

ChatGPT (July 2026)

Summary

Experiments in ohmically heated HL-2A tokamak plasmas compare hydrogen- and deuterium-dominated discharges to determine how isotope mass affects scrape-off-layer transport. In deuterium plasmas, the $E \times B$ shear near the last closed flux surface weakens while low-frequency boundary turbulence and outward turbulence spreading increase. Phase, correlation, and linear-stability analyses identify the principal edge fluctuation as an interchange mode driven by the electron-temperature gradient inside the confined region. The turbulence crosses the last closed flux surface into the scrape-off layer despite reduced local growth there, distinguishing spreading from locally generated turbulence. Across multiple discharges, stronger scrape-off-layer fluctuations correlate with a wider heat-flux profile and lower parallel heat flux. The results support turbulence spreading as a mechanism for isotope-dependent exhaust broadening and suggest isotope-mass control as a possible means of reducing heat loads without sacrificing core confinement.

Contributions

1. Showed wider scrape-off-layer profiles and reduced parallel heat flux in matched deuterium-dominated HL-2A plasmas.
2. Measured at least a 45% reduction of the $E \times B$ shear peak near the last closed flux surface in deuterium.
3. Identified the source fluctuation as interchange turbulence through phase, correlation, and linear-stability analyses.
4. Distinguished outward turbulence spreading from local generation across the last closed flux surface.
5. Quantified the correlation between stronger normalized turbulence, increased scrape-off-layer width, and reduced parallel heat flux.

2026ZCXZ

ChatGPT (July 2026)

Summary

The authors develop THEMIS, a three-dimensional electromagnetic model for helicon-wave propagation, damping, and antenna coupling in the toroidal Helimak boundary plasma. A finite-temperature dielectric tensor, measured magnetic-field and density profiles, and several damping channels show that fast waves are inaccessible, whereas slow

Trivelpiece–Gould modes propagate and deposit power locally. Electron Landau and collisional damping provide more than 95% of absorption under Helimak conditions. Simulations of four experimental antennas explain the initial campaign’s null density response: the protruding dielectric window traps fields in the waveguide, and the low-density scrape-off-layer gradient causes a slow-wave cutoff. Recessing the window reduces the evanescent layer and improves plasma access, with absorption depending non-monotonically on recess depth. Geometry scans favor open-circuit termination, longer and wider current straps, suitable inter-turn spacing, and clearance from metallic walls. An optimized racetrack spiral raises simulated absorption by over an order of magnitude relative to the short-circuited rectangular spiral. The results are qualitative design guidance: Helimak differs substantially from tokamak edges, and the linear model omits nonlinear high-power response and engineering constraints.

Contributions

1. Built a full three-dimensional thermal-dielectric model of helicon coupling in realistic Helimak geometry.
2. Identified slow-wave accessibility and electron Landau plus collisional damping as the dominant absorption physics.
3. Explained poor core penetration in the protruding-window experiment through cutoff and geometric trapping.
4. Quantified how window recess, termination, strap dimensions, turn spacing, and wall clearance affect coupling.
5. Designed a racetrack spiral antenna with more than tenfold simulated absorption improvement.

2026ZD

ChatGPT (July 2026)

Summary

This paper analyzes a steady, one-dimensional, collisionless plasma sheath containing cold positive ions, wall-emitted secondary electrons, and nonthermal primary electrons described by a Cairns distribution. Substitution of the distribution into the sheath fluid and Poisson equations yields a generalized Bohm criterion and corresponding ion entry speed. Current balance at a floating wall gives a modified wall potential, and the transition between classical and inverse-sheath behavior gives a critical secondary-electron-emission coefficient. All three relations recover their Maxwellian forms when the Cairns nonthermality parameter α vanishes. Parameter sweeps show that increasing α raises the generalized Bohm speed and lowers the floating potential in the adopted normalization. The critical emission threshold also becomes more sensitive to wall potential and ion Mach number as nonthermality increases. These results quantify how a suprathermal primary-electron population changes sheath entry and wall-current balance within the paper’s cold-ion, planar, steady-state assumptions.

Contributions

1. Incorporated Cairns-distributed primary electrons into a secondary-emitting sheath model.
2. Derived a generalized Bohm criterion and nonthermal Bohm speed.
3. Obtained the floating-wall potential from the modified current balance.
4. Derived a nonthermality-dependent critical secondary-electron-emission coefficient.
5. Mapped how α , wall potential, emission, and ion Mach number affect the sheath properties.

2026ZGZ

ChatGPT (July 2026)

Summary

The authors develop a predictor–corrector discrete unified gas kinetic scheme (PRC-DUGKS) for Boltzmann transport equations containing both conservative and non-conservative collisions. Unlike standard DUGKS, the method directly advances the original distribution function, so macroscopic moments and boundary conditions remain well defined when absorption, fission, or other non-conservative processes are present. A unified trapezoidal predictor–corrector treatment removes the implicit coupling without introducing the splitting errors of Strang-split DUGKS. Multigroup

neutron-transport tests range from one-group problems to the TWIGL benchmark and pin-resolved mixed-oxide reactor cores. PRC-DUGKS captures steep flux and power profiles with large time steps and consistently improves on the split formulation near strong absorbers. In the reported reactor tests it attains comparable or better accuracy while providing speedups of about $2.6\times$ for TWIGL and roughly $11\times$ for the full-core cases.

Contributions

1. Extended DUGKS to transport models combining conservative and non-conservative collision operators.
2. Directly evolved the physical distribution function, enabling consistent moment recovery and boundary treatment.
3. Coupled transport and collision processes with a second-order predictor–corrector update without operator splitting.
4. Validated the scheme on multigroup neutron benchmarks from simple media to pin-resolved reactor cores.
5. Demonstrated improved large-step accuracy and substantial speedups over Strang-split DUGKS.

2026ZHKB

ChatGPT (July 2026)

Summary

Two-temperature, single-fluid full-magnetohydrodynamic JOREK simulations model flux pumping in ASDEX Upgrade discharge 36663 and scan resistivity, viscosity, and plasma beta. A pressure-gradient-driven $m/n = 1/1$ quasi-interchange mode generates a core dynamo of order mV/m that flattens current density near 2.5 MA/m^2 , holds $q_0 \approx 1$, and quantitatively reproduces the measured flux-pumping state. Decreasing Hartmann number produces a sequence from flux pumping through sawteeth and a single crash to a quasi-stationary island with failed dynamo; representative transitions span $H \sim 10^8$ to 10^3 . Flux pumping is favored when normalized $1/1$ -mode growth exceeds roughly 10^{-1} and the current-peaking time exceeds 10^4 . A rough operating estimate uses $H_c \approx 3 \times 10^7$ and $\beta_{p,c} \approx 2$. The model nevertheless retains flux pumping near $\beta_N \approx 2$, where the experiment saw sawteeth, indicating that two-fluid and energetic-particle physics are needed for a predictive boundary.

Contributions

1. Quantitatively reproduced core-current flattening and q_0 clamping through an $m/n = 1/1$ dynamo.
2. Mapped four helical-core relaxation regimes across broad dissipation scans.
3. Showed that higher- n harmonics can weaken the dynamo and trigger delayed sawteeth.
4. Related flux-pumping access to mode growth and the current-peaking timescale.
5. Estimated a density–temperature operating window while documenting model limitations.

2026ZHWD

ChatGPT (July 2026)

Summary

This paper identifies a kinetic mechanism for rapid ion-electron energy exchange in moderate-Alfvénic-Mach-number, quasiparallel magnetized collisionless shocks. Two-dimensional OSIRIS particle-in-cell simulations model counter-streaming ion populations in an electron background, while a linear multifluid dispersion solver identifies unstable modes. Magnetized runs transfer ion drift energy to electrons much faster than Coulomb collisions; in a representative shock-transition case electrons reach about 450 eV and $T_e/T_i > 0.2$, whereas unmagnetized runs exchange much less energy. The mechanism has two stages: resonances between electron-whistler and drifting-ion Alfvén or magnetosonic waves extract ion drift energy, then Landau resonance transfers wave-mediated energy to ions and electrons. Oblique-field and physical-mass-ratio simulations retain the predicted modal structure. The authors propose that the resulting coupling could inform two-temperature magnetohydrodynamic or hybrid models of astrophysical and fusion plasmas.

Contributions

1. Reproduced faster-than-Coulomb ion-electron energy exchange in a simplified kinetic shock-transition model.
2. Demonstrated that an imposed magnetic field is essential for strong coupling in the studied regime.
3. Identified electron-whistler resonances with ion magnetosonic and Alfvén waves using multifluid theory.
4. Showed that Landau resonance mediates subsequent ion and electron thermalization.
5. Verified the mode picture for oblique fields and the physical ion-electron mass ratio.

2026ZJZU

ChatGPT (July 2026)

Summary

A 50 ms flux-driven, global full- f GRILLIX simulation follows an ASDEX Upgrade edge and scrape-off layer through a heating-power ramp from L-mode-like conditions. Profiles evolve slowly for about 32 ms before radial transport transiently collapses and the edge pedestal steepens within roughly $100\ \mu\text{s}$. The transition is not caused by smaller fluctuations: density, temperature, and potential fluctuations increase while their cross-phases abruptly change, switching the dominant turbulence from drift waves to kinetic ballooning modes and briefly producing inward flux. Self-consistent $E \times B$ shear and Reynolds stress grow beforehand; geodesic-acoustic and Alfvénic oscillations intensify and then disappear. Subsequent low-frequency transport and profile oscillations resemble a dithering I-phase or ELM-like cycle. The calculation supplies a realistic global three-dimensional turbulence demonstration of this rapid pedestal-forming transition, while noting limitations of the drift-fluid model for quantitative L-H threshold prediction.

Contributions

1. Performed a confinement-time-scale global full- f electromagnetic edge-SOL power-ramp simulation.
2. Reproduced abrupt pedestal buildup in about $100\ \mu\text{s}$ after 32 ms of slow profile evolution.
3. Identified a cross-phase switch from drift-wave to kinetic-ballooning turbulence as the microscopic trigger.
4. Separated gradual Reynolds-stress and $E \times B$ shear buildup from the fast turbulence transition.
5. Resolved geodesic-acoustic, Alfvénic precursor, and later low-frequency limit-cycle oscillations.

2026ZL

ChatGPT (July 2026)

Summary

Seven solvers for the generalized probability density evolution equation are compared within a common probability-density-evolution framework. The study evaluates finite-difference, finite-volume, and discontinuous-Galerkin grid methods; radial-basis-function and reproducing-kernel-particle meshfree collocation; and physics-informed neural networks with multilayer-perceptron and Kolmogorov–Arnold architectures. Analytical Euler–Bernoulli beam and nonlinear Kirchhoff-plate examples expose how independent probability sampling and response-space discretization cause grid sensitivity. Meshfree methods reduce that sensitivity and numerical dispersion for pure density evolution, whereas grid methods naturally support upwind absorbing boundaries for time-dependent reliability; finite-volume WENO gives the smallest reported reliability error. The neural solvers are flexible but require separate training per sample and run one to two orders of magnitude slower than classical methods. The work extends the framework to residual probability under absorbing failure boundaries, but evaluates only two systems and one failure threshold. Meshfree and neural methods still lack an intrinsic treatment of asymmetric absorbing boundaries, optimal grid density is not known in advance, and the two principal error sources are not separated a posteriori.

Contributions

1. Benchmarked seven grid, meshfree, and neural GDEE solvers against analytical solutions.
2. Diagnosed probability-sampling and response-grid imbalance as the source of grid sensitivity.

3. Established meshfree advantages for low-dispersion probability-density evolution.
4. Extended the framework to time-dependent reliability with absorbing boundaries.
5. Quantified finite-volume reliability accuracy and the large neural-network runtime penalty.

2026ZLP

ChatGPT (July 2026)

Summary

This paper links loss of pressure and internal-energy positivity in low- β magnetohydrodynamic calculations to inconsistency between reconstructed magnetic fields and energies inside HLL-family Riemann fans. It imposes an intermediate-state condition requiring total energy to equal the sum computed from the same density, momentum, and magnetic field. The resulting HLLC-ec solver adopts a revised two-state approximation that improves magnetic-field accuracy at the cost of contact resolution; HLLD-ec replaces constant total pressure with three intermediate thermal-energy states. Analytical examples and one- and two-dimensional tests show similar ordinary shock resolution but substantially smaller false waves, density dips, internal-energy errors, and divergence errors in difficult low- β cases. HLLD-ec survives some extreme tests in which the classical solver fails, although both can still break down when uncontrolled multidimensional divergence error grows. The formulation addresses conservative ideal MHD only, does not replace a divergence-control method, and does not isolate every modification's individual effect. HLLC-ec is also more diffusive across fluid contacts.

Contributions

1. Identified inconsistent intermediate magnetic fields and energies as a route to negative internal energy.
2. Formulated a general intermediate-state energy-consistency condition for HLL-family solvers.
3. Derived an energy-consistent HLLC variant prioritizing magnetic accuracy.
4. Derived an HLLD variant with three consistent intermediate thermal-energy states.
5. Connected state inconsistency to time-integrated errors and reduced low- β numerical failures.

2026ZMFA

Claude (September 2026)

Summary

The authors study the modulational instability of ion-acoustic waves in an unmagnetized, collisionless plasma of cold fluid ions and electrons following a Tsallis–Gurevich distribution, which combines nonextensive statistics with entropic index q for free electrons and a flat-topped population adiabatically trapped in the potential well. They derive closed-form electron densities in terms of hypergeometric functions for $1/3 < q < 1$ and $q > 1$; the small-potential expansions contain a $\Phi^{3/2}$ trapping term and reduce to the Tsallis and Gurevich limits. The reductive perturbation method gives a modified nonlinear Schrödinger equation with a $|\Phi|^{1/2}\Phi$ nonlinearity, dispersion coefficient P , and nonlinear coefficient Q . For $PQ > 0$ it admits bright sech^4 envelope solitons; for $PQ < 0$ only unbounded sec^4 solutions exist, so dark solitons are absent. Linear stability analysis predicts instability for envelope wavenumbers below $K_{\max} \propto \Phi_0^{1/4}$, with maximum growth rate $Q\Phi_0^{1/2}/4$. In parametric plots, smaller q lowers the wave frequency and long-wavelength group velocity, trapping barely affects linear dispersion, PQ stays positive for all examined parameters, and both $K_{\max}$ and the peak growth rate increase with q , especially for $q > 1$, with the growth rate peaking at a critical carrier wavenumber. The model assumes one-dimensional, small-amplitude propagation.

Contributions

1. Derived exact Tsallis–Gurevich electron densities for both ranges of q and verified their Tsallis and Gurevich limits.

2. Obtained, via reductive perturbation, an envelope equation with a fractional $3/2$ -order nonlinearity and explicit q -dependent coefficients for ion-acoustic wave packets.
3. Constructed traveling-wave solutions of this equation, showing that it supports bright solitons but, unlike the cubic nonlinear Schrödinger equation, no bounded dark solitons.
4. Derived the instability band $K < K_{\max} \propto \Phi_0^{1/4}$, rather than $\Phi_0^{1/2}$ as for the cubic equation, and the maximum growth rate, with the strongest instability at carrier wavenumber $K^2 = (q + 1)/4$.
5. Mapped how electron nonextensivity controls dispersion, group velocity, the instability window, and growth rates, finding modulational instability throughout the examined subextensive and superextensive regimes.

Comments

- §III-B, last paragraph: the text states that “for the current problem $P/Q < 0$, and the resulting soliton is of the bright type.” This contradicts the same section, where bright sech^4 solitons, Eqs. (37)–(38), require $P/Q > 0$ and $P/Q < 0$ gives only the unbounded solution of Eqs. (39)–(40), as well as Fig. 4, §III-D, the abstract, and the conclusion, which all report $PQ > 0$. The intended statement is most likely $P/Q > 0$.

2026ZPSR

ChatGPT (July 2026)

Summary

Zimmermann and coauthors compare benign and non-benign termination of high-current runaway-electron beams in JET and DIII-D after hydrogenic injections. The study combines fast magnetic measurements, EFIT reconstructions, and CASTOR3D linear resistive-MHD modeling for about 40 JET and 20 DIII-D discharges. In JET, unsuccessful high-current cases occur near low edge safety factor, $q_{\text{edge}} \approx 2$, and are preceded by intermittent non-terminating MHD activity. Larger internal inductance indicates more peaked runaway-current profiles, which may maintain ionization and prevent effective recombination. Benign JET cases typically terminate at $q_{\text{edge}} \geq 3$ with less peaked profiles. DIII-D spans a wider range of edge safety factors but shows the same central role for current-profile peaking. Growth rates alone do not distinguish outcomes; perturbation amplitude and the interplay of ideal and resistive dynamics matter for deconfinement.

Contributions

1. Compared runaway-electron beam termination behavior across JET and DIII-D datasets.
2. Linked benign termination likelihood to edge safety factor and internal inductance trends.
3. Identified peaked runaway-current profiles as a key factor in non-benign outcomes.
4. Used CASTOR3D modeling to relate observed boundaries to resistive kink modes.
5. Showed that MHD growth rate alone is insufficient to predict efficient runaway deconfinement.

2026ZRCE

ChatGPT (July 2026)

Summary

This paper documents the horizontal, vertical, and tangential infrared (HIR, VIR, and TIR) diagnostics on the Tokamak à Configuration Variable. Their Equus 81k M cameras measure graphite-tile temperatures at standard rates near 200 Hz and, with reduced frames, above 10 kHz. A two-stage calibration combines heated tiles and thermocouples with pixelwise non-uniformity corrections. The Python THEODOR code then solves a heat-diffusion inverse problem to infer perpendicular heat flux, parallel heat flux, and deposited power, while modeling a surface layer whose heat-transmission factor remains uncertain. New measurements of graphite properties replace older estimates: conductivity is 25-40% higher below 500 °C, although resulting peak heat fluxes increase by less than 15%. A heated, inclined VIR valley tile and TIR rooftop tiles increase signal for fast transients. Long-pass filters suppress the dominant 4051 nm deuterium

line, substantially reducing parasitic plasma light. VIR and TIR heat-flux profiles agree at low density after geometry correction, but VIR overestimates background power as density rises. Residual plasma emission and the surface-layer transmission factor therefore remain the leading diagnostic uncertainties.

Contributions

1. Consolidated the geometry, operating modes, calibration, and performance limits of TCV's three infrared diagnostic systems.
2. Detailed the temperature-to-heat-flux workflow implemented in Python THEODOR, including surface-layer and magnetic-incidence corrections.
3. Measured temperature-dependent graphite density, heat capacity, diffusivity, and conductivity and quantified their effect on inferred heat flux.
4. Designed heated and inclined VIR and TIR tiles that increase infrared signal for high-frequency transient measurements.
5. Demonstrated the benefit and remaining limitations of 4095 nm long-pass filtering through cross-comparison of VIR and TIR heat fluxes.

2026ZRZS

ChatGPT (July 2026)

Summary

Steady-state and time-dependent UEDGE simulations identify the trigger of a detachment bifurcation associated with the abrupt outer-target electron-temperature cliff observed in DIII-D. In the forward toroidal-field configuration, a high-field-side radiation front crosses the last closed flux surface while the outer divertor remains attached and settles above the X-point. The local electron temperature then collapses from about 70 eV to 10 eV, changing the electrostatic potential and reversing the $E \times B$ flow in a thin private-flux-region layer below the X-point in less than 0.5 ms. This reversal transports plasma toward the outer leg and initiates feedback that drops the outer-target temperature within 1–2 ms, followed by deep detachment. The bifurcation can occur without a visually sharp cliff when the target is already cold. The simulations further connect the transition to in–out divertor asymmetry: the corresponding radiation-front behavior and deep detachment are absent for reverse field. Direct experimental confirmation of the predicted flow reversal remains needed.

Contributions

1. Identified private-flux-region $E \times B$ flow reversal as the immediate trigger of the outer-target temperature cliff.
2. Resolved the detachment transition into two successive sub-millisecond and millisecond phases.
3. Connected radiation-front penetration across the separatrix to the X-point temperature and potential collapse.
4. Distinguished the general bifurcation from the special case in which it appears as a sharp temperature cliff.
5. Related bifurcation and deep-detachment access to toroidal-field direction and in–out divertor asymmetry.

2026ZUM

ChatGPT (July 2026)

Summary

This paper gives implementable Monte Carlo procedures for loading nine non-Maxwellian velocity distributions into kinetic particle simulations: (r, q) , regularized Kappa, subtracted Kappa, finite-width ring and shell, ring and shell Maxwellian, super-Gaussian, and filled-shell distributions. The algorithms reduce sampling to uniform, normal, and gamma variates, supplemented by tailored beta-prime or rejection schemes. Parameter-space tests identify where each method is valid and efficient, and numerical experiments verify sampled densities and moments. A loss-cone filling parameter is incorporated into the subtracted-Kappa recipe. Newly introduced ring and shell Maxwellians have simpler samplers and analytically accessible moments, remain well behaved near zero velocity, and receive a physical

interpretation through radial or spherical scattering of a seed population. The paper also identifies GPU limitations caused by divergent rejection loops and missing gamma generators in common vendor libraries.

Contributions

1. Provided numerical recipes for nine non-Maxwellian plasma velocity distributions.
2. Mapped validity and efficiency regions for beta-prime, post-rejection, and piecewise-rejection samplers.
3. Developed a sampler for the subtracted-Kappa loss-cone model, including partial loss-cone filling.
4. Introduced ring and shell Maxwellians as physically motivated, numerically simpler alternatives to conventional finite-width models.
5. Validated distributions and moments numerically while documenting practical GPU implementation risks.

2026ZW

ChatGPT (September 2026)

Summary

High-order implicit finite volume methods can produce negative density or pressure while iteratively solving the compressible-flow equations. This paper develops a positivity-preserving procedure for dual-time stepping on unstructured grids. A physical-time-step limit uses the current residual to estimate the next cell average and constrains that estimate with a lower bound that accounts for estimation error. During the inner nonlinear solve, pseudo-time-step limiting and a correction to the computed increment keep intermediate cell averages admissible. A scaling limiter then makes reconstruction polynomials admissible without changing their cell averages. The authors analyze why the procedure retains high-order accuracy and test it with fourth-order spatial reconstruction and implicit time integration on inviscid and viscous benchmarks. The reported tests show robust, well-resolved solutions; admissibility is enforced during the iterations rather than checked only after convergence.

Contributions

1. Introduced physical-time-step limiting to ensure an admissible next-time-level cell average.
2. Combined pseudo-time-step limiting with increment correction for positivity during inner iterations.
3. Coupled cell-average protection to polynomial scaling for positive reconstructed states.
4. Analyzed the method's accuracy preservation and dependence on limiting parameters.
5. Demonstrated the approach on high-order unstructured-grid compressible-flow benchmarks.

2026ZXML

Claude (September 2026)

Summary

The authors present VEQ (Veloce Equilibrium), a parametric solver of axisymmetric fixed-boundary tokamak equilibria for repeated geometry queries, and benchmark its open-source realization VEQPy. With the last closed flux surface prescribed, the unknowns are shifted-Chebyshev coefficients of Miller extended harmonic (MXH) shaping profiles and, per input route, of normalized poloidal-flux or toroidal-field profiles. The main solve enforces a Grad-Shafranov residual projected onto test functions induced by constrained flux-surface motion, with moment conditions closing the profile coefficients. Six input routes pair a pressure gradient with toroidal-field, flux-gradient, current, current-density or safety-factor data and map them to one residual. Routes agree for smooth, mutually compatible inputs, deviating most in the near-edge safety factor q_{95} . On Solov'ev, CHEASE H-mode and EFIT X-point G-EQDSK files, the last a smoothed diverted boundary, the most accurate Pareto-selected reductions, with 9, 65 and 94 parameters, give RMS flux-surface errors of $1.1\text{--}1.9 \times 10^{-3}$ of the minor radius in 1.6–19 ms solve-only time. Sampled pointwise residuals show enrichment mainly improves interior force balance, while H-mode and X-point residuals remain concentrated near the boundary. The authors note that the projected solve does not certify pointwise force balance, timings exclude setup, and only the PF route was applied to G-EQDSK data.

Contributions

1. Derived the transformed residual $\mathcal{G} = (J/R)(FF_\psi + \mu_0 R^2 P_\psi) + (g_{\theta\theta}\psi_\rho/JR)_\rho - (g_{\rho\theta}\psi_\rho/JR)_\theta$ from the Euler–Lagrange operator of the axisymmetric energy functional and projected it onto shape test functions $\chi_k = \nabla\psi \cdot \partial\mathbf{r}/\partial p_k$ obtained from the convective-variation identity $\delta_r\psi + \nabla\psi \cdot \delta\mathbf{r} = 0$.
2. Separated physical flux and source scales, $\psi_\rho = \alpha_2 \hat{\psi}_\rho$ and $FF_\psi = \alpha_1 \hat{F}\hat{F}_\psi$, fixed afterward by I_p or β_t constraints, and derived route closures from the flux-surface-averaged balance, including a square-root quadrature for $\hat{\psi}_\rho$ in the PF route and first-order ODEs for F or F^2 in the safety-factor (PQ) route.
3. Swept 8022, 9483 and 9478 reduced truncations of the D-shaped, H-mode and X-point cases on a fixed 32×32 grid with cold-start Powell-hybrid solves; all returned finite values, all but 299 H-mode candidates met the 10^{-6} projected-residual tolerance, and the selected D-shaped High row reached $E_{\text{ref}}/a = 6.24 \times 10^{-5}$ with 9 parameters.
4. Showed that an unanchored point-collocation least-squares polish on the same manifold lowers the global sampled residual of the H-mode and X-point rows (ratios 0.05–0.68) but raises their G-EQDSK shape error 3.3–8.5-fold and the total time 2.8–16.7-fold, which the authors cite for keeping the variational solve.
5. Propagated the geometry factors V' and $V'\langle|\nabla\psi|^2\rangle$ through a fixed-source 1-D flux-coordinate heat-diffusion problem, finding temperature RMS errors of at most 3.1×10^{-3} , stored-energy errors of at most 6.6×10^{-3} and beta-proxy errors of at most 1.3×10^{-2} despite V' RMS errors up to 4.7×10^{-2} .

Comments

- Section 4.5, Table 2 and Fig. 5: Table 2 gives PJ2 errors $E_{\text{coeff}} = 8.037 \times 10^{-6}$, $\Delta_{I_p} = 7.581 \times 10^{-6}$ and $\Delta_{\beta_t} = 3.648 \times 10^{-6}$ at $N_\rho = N_\theta = 32$, but the PJ2 curves in Fig. 5(a)–(c), plotted against N_ρ at $N_\theta = 32$, read about 6×10^{-4} , 8×10^{-4} and 3×10^{-5} at $N_\rho = 32$ and are the largest of the six routes in panels (a) and (b); the other five rows of Table 2 agree with the figure, so which PJ2 values are correct is unresolved.
- Appendix B, Table B.8: the text states that “The polish reduces the global sampled RMS and maximum residual in every row”, but Table B.8 lists $r_{|\mathcal{G}_{\text{std}}|_{\text{max}}} = 1.007$ for D-shaped (75), which by the caption’s convention (values below one indicate reduction) is a slight increase of the maximum; the statement holds for every other row.

2026ZYH

ChatGPT (July 2026)

Summary

Wall-modeled large-eddy simulation reduces the cost of high-Reynolds-number turbulence calculations, but conventional and offline-learned wall models often fail for nonequilibrium flows or when deployed with different grids and discretizations. This paper develops a differentiable WMLES framework in which forward-mode automatic differentiation propagates exact gradients through the LES solver and enables in-situ optimization of a physics-embedded wall model. The model augments a thin-boundary-layer ordinary differential equation with neural-network-controlled forcing terms. Gradient accuracy is verified on lid-driven cavity flow. Models trained in channel and periodic-hill simulations account for the deployed numerical environment and generalize across Reynolds numbers, grid skewness, hill slopes, and unseen bump, hump, and ramp geometries. A channel-trained model extrapolates six orders of magnitude in Reynolds number, while the periodic-hill-trained model predicts separation and reattachment on new geometries; ramp-flow mean-velocity errors remain below 2% at tested stations.

Contributions

1. Introduced an end-to-end differentiable wall-modeled large-eddy simulation solver using forward-mode automatic differentiation.
2. Embedded a neural forcing term in a thin-boundary-layer wall-model equation.
3. Verified automatic gradients against finite differences using lid-driven cavity flow.
4. Demonstrated Reynolds-number and grid-discretization transfer from in-situ channel-flow training.
5. Transferred a single periodic-hill-trained model to varied hills and unseen bump, hump, and ramp flows.

2026ZZHD

ChatGPT (September 2026)

Summary

This paper introduces a subcell entropy-constrained (SEC) limiter for entropy-stable discontinuous Galerkin spectral element methods. Assigning different high/low-order flux blending weights to subcell interfaces improves local control of shock dissipation but can break an element's semi-discrete entropy inequality. The authors restore it by optimizing pairwise corrections to the high-order entropy-conservative flux, minimizing the corrections' deviation from unity. A two-level detector combines element modal energy with subcell oscillation measurements to choose the weights. They derive an entropy-only variant, SEC(EC), and a variant with an additional flux-consistency constraint, SEC(EC+CC). Both preserve constant states and show convergence in smooth-flow tests. In advection, shock tubes, double Mach reflection, Kelvin–Helmholtz flow, and a nonconvex scalar problem, SEC(EC) generally resolves small features more sharply than the comparison limiters while suppressing spurious oscillations. The additional consistency constraint can distort some solutions and increases runtime; the entropy-only variant is cheaper. The method addresses semi-discrete entropy stability, but the paper does not claim a general positivity guarantee.

Contributions

1. Identified why nonuniform subcell blending weights can destroy the entropy cancellation of an entropy-stable DG scheme.
2. Derived pairwise high-order flux corrections through a constrained quadratic minimization with an explicit solution.
3. Developed entropy-only and entropy-plus-consistency limiter variants and analyzed consistency and convergence.
4. Combined element modal energy and subcell oscillation sensing to set interface-specific blending weights.
5. Compared accuracy, shock resolution, entropy behavior, and runtime against HG and Lin limiters on one- and two-dimensional problems.

2026ZZLW

ChatGPT (September 2026)

Summary

AlgoPlasma v1.0 is an Apache-2.0 library that exposes plasma-modeling algorithms as independently selectable components instead of prescribing a complete simulation application. Its primarily Fortran implementation, supported by C/C++, Python, MPI, OpenMP, HYPRE, and HDF5, groups functionality into ten categories covering particle initialization and advancement, particle-grid coupling, Poisson and Maxwell solvers, Monte Carlo collisions, fluid updates, distributed-memory exchange, and input/output. Components are paired where possible with focused verification or validation cases and documentation connecting mathematical formulations to interfaces, implementations, and reference results. The paper demonstrates assembly through a two-dimensional, three-velocity electrostatic particle-in-cell simulation of an oblique two-stream instability. A 26-line top-level driver combines initialization, deposition, Poisson solution, field reconstruction, interpolation, Boris pushing, and output while leaving boundary conditions and execution order to the application. The measured growth rate, $0.2603\omega_{pe}$, agrees within 0.5% with the kinetic-theory value $0.2616\omega_{pe}$, and the maximum relative total-energy error is 0.0183%.

Contributions

1. Defined an algorithm-centered library architecture whose caller-managed components can be inspected, tested, replaced, and composed independently.
2. Supplied ten component families spanning particle, field, collision, fluid, communication, and input/output operations.
3. Connected production routines to focused verification and validation cases and source-linked documentation.

4. Demonstrated composability with a compact application driver for an oblique two-stream particle-in-cell simulation.
5. Validated the assembled workflow through agreement with kinetic-theory growth and tight total-energy conservation.

2026ZZY

Claude (September 2026)

Summary

Electron drift instabilities (EDIs), azimuthal kinetic instabilities driven by the $\mathbf{E} \times \mathbf{B}$ electron drift, are linked to anomalous cross-field electron transport in Hall thrusters. The authors study how the magnetic field shapes this transport with Cartesian 3-D particle-in-cell simulations of channel and plume: 0.1 mm cells, $\Delta t = 1.5 \times 10^{-11}$ s, a periodic 0.64 cm azimuthal extent, ionisation-only Monte Carlo collisions, and an axial neutral continuity solver. They compare an analytic radial-only field with FEMM-computed fields having radial and axial components and centreline peaks near 200 G (strong) and 100 G (weak). Earlier runs are restarted at 15 μ s with up to tenfold more macro-particles. Averaging $\langle n_e E_y \rangle$ over the azimuth and time gives a correlation-based turbulent mobility $-\langle n_e E_y \rangle / (B_x n_e E_z)$ and a field-perpendicular variant. In the strong field, transport concentrates near the exit in the upper radial region, where the field is weaker. The analytic field places it at the exit and along the channel walls, and the weak field gives collapsed ion density and abnormal discharge. The turbulent mobility follows the spatial trends of the direct average $\langle u_{ez} / E_z \rangle$. Stated limitations include cells possibly exceeding the Debye length, absorbing walls without secondary emission or recycling, and a prescribed cathode current.

Contributions

1. Compared particle-number scaling factors 1, 3, and 10 in the strong-field case, finding mean downstream ion currents of 0.020–0.024 A across runs and diagnostics but axial electron density and temperature profiles for the unscaled run that differ from the two scaled runs.
2. Initialized the neutral density from the steady axial continuity equation with a cosine ionisation source, giving $n_a = \frac{n_{a0}}{2} [1 - \sin(\pi(z - z_m)/(z_2 - z_1))]$ and $n_{a0} = 2S_0(z_2 - z_1)/(\pi u_z)$, and tripled the implied 304 m/s neutral speed to 912 m/s to accelerate convergence.
3. Showed that the strong-field correlation $\langle n_e E_y \rangle$ changes in magnitude and path over the breathing cycle, falling from about 5×10^{19} to 2×10^{19} V m $^{-4}$ near the ion-density minimum and reaching the 10^{20} range near the peak, with the axial electron current density rising with it.
4. Replaced B_x and E_z by the field magnitude and the in-plane electric field perpendicular to the local field, which suppresses the sign-changing line that the axial mobility estimate shows in the upper radial region of the strong-field case.
5. Proposed the fluctuation force $\mathbf{R}_e^{\text{urb}} = q \langle \delta n_e \delta \mathbf{E} \rangle$, distinct from Coulomb drag, and the rate $\nu_{\text{urb}}^{\text{eff}} = |R_{e,y}^{\text{urb}}| / (m n_e |v_{de,y}|)$ as an MCC-like subgrid closure for reduced radial–axial PIC models, and reported inverse Hall parameters whose axial profiles differ between the centreline and the upper wall.

Comments

- §2 and §2.4: §2 states that at the restart each macro-particle “is split into f_{pn} macro-particles with the same position, while their velocities are resampled during the restart procedure”, whereas §2.4 states that each loaded macro-particle “will be copied f_{pn} times with the same position and velocity”; which procedure was used is unresolved.
- Table 1: the macro-particle weight is listed as $w_0 = 1955.5$, whereas §2 defines $w_0 \approx 19\,555$ as the weight of the earlier runs and the reduced weight after the restart as $w_p = w_0 / f_{pn}$ in Eq. (2.1); since the note after Table 1 says its macro-particle parameters are for $f_{pn} = 10$, the tabulated value is w_p .
- Table 1: the final time is $T_{\text{final}} = 35 \times 10^{-6}$ s, but the strong-B curves in Figs. 5, 6, and 9, including the $f_{pn} = 10$ run to which Table 1 refers, continue to about 37 μ s, while the weak-B and analytic-B curves in Fig. 9 end near 33 and 27 μ s; unresolved.

- §3.2, Eqs. (3.3)–(3.4): for a purely radial field, Eq. (3.4) gives $E_{\perp} = -B_x E_z / |B_x|$, so Eq. (3.3) reduces to $\mu_{\perp}^{turb} = \langle n_e E_y \rangle / (B_x n_e E_z) = -\mu_z^{turb}$ instead of to Eq. (3.2), which it is introduced to generalize; with $B_x > 0$ on the centreline (Figs. 3(a) and 4), Eq. (3.4) also gives $\langle E_{\perp} \rangle \approx -\langle E_z \rangle$ just downstream of the exit, where the field lines are radial, but Fig. 22 shows $\langle E_{\perp} \rangle$ positive there, like $\langle E_z \rangle$ in Fig. 19, and $\mu_{\perp}^{turb}$ negative in the plume, like μ_z^{turb} ; the plotted quantities correspond to $E_{\perp} = (B_x E_z - B_z E_x) / B$, the component whose $\mathbf{E} \times \mathbf{B}$ drift is along $+y$.
- §3.4: the velocities in Fig. 26 are said to be averaged over a “time period of 12.3 μs as the same as that in figure 24 of the strong-B case”, but in Fig. 24 the strong-B panel is labelled $T_{mean} = 18.0 \mu\text{s}$ and 12.3 μs is the averaging period of the analytic-B panel; the period used for Fig. 26 is unresolved.

2025A

ChatGPT (July 2026)

Summary

This paper presents a machine-checked proof of a C routine that converts a coordinate-form sparse matrix into compressed sparse row form. The algorithm sorts coordinate triples by row and column, combines duplicate entries, and fills the CSR value, column-index, and row-pointer arrays from left to right. Its implementation is short, but correctness depends on detailed relationships among partially consumed COO entries, partially constructed CSR arrays, duplicate sums, row boundaries, allocation sizes, and machine-integer bounds. The author develops the invariants bottom-up from the obligations of the Verified Software Toolchain’s separation-logic proof, while using Coq definitions to specify when COO and CSR matrices represent the same mathematical matrix. The specification and supporting lemmas required 1,571 lines of Coq, followed by 412 lines of VST proof. Formalization uncovered five implementation bugs: four off-by-one errors and one unsigned-integer initialization defect. The resulting verified module is intended as a composable component for larger verified numerical programs, including finite-element software that constructs sparse matrices.

Contributions

1. Produced a machine-checked functional-correctness proof for a C COO-to-CSR conversion routine.
2. Specified duplicate aggregation and mathematical equivalence between COO and CSR representations.
3. Demonstrated a bottom-up method for deriving intricate loop invariants from proof obligations.
4. Found and corrected five concrete implementation defects during formal verification.
5. Delivered a verified sparse-matrix construction component suitable for composition in larger numerical programs.

2025A2

ChatGPT (July 2026)

Summary

BiCGSTAB is a widely used Krylov method for large sparse nonsymmetric linear systems, but residual stagnation or algorithmic breakdown can prevent convergence. The paper embeds BiCGSTAB as the inner iteration of the generalized residual cutting (GRC) method. Each inner solve reduces the current residual to a prescribed fraction; the outer GRC iteration then forms a new approximation from the current BiCGSTAB output and stored correction vectors, stabilized by modified Gram–Schmidt orthogonalization. This construction, called GRC-BiCGSTAB, targets reliability rather than iteration speed. Tests in GNU Octave include a nonsymmetric indefinite discretized partial-differential-equation matrix and matrices from a standard sparse collection. GRC-BiCGSTAB converges on several cases where ordinary BiCGSTAB breaks down, while ordinary BiCGSTAB remains faster on cases where it succeeds. The results therefore support a robustness–cost trade-off rather than uniform superiority. The experiments are limited in number and do not provide a broad timing, memory, or preconditioning study.

Contributions

1. Proposed GRC-BiCGSTAB, using BiCGSTAB as the inner solver of a generalized residual-cutting iteration.
2. Combined residual cutting with modified Gram–Schmidt stabilization of stored correction vectors.
3. Specified an outer–inner stopping strategy that periodically stabilizes BiCGSTAB residual progress.
4. Demonstrated convergence on multiple sparse systems where standard BiCGSTAB encounters breakdown.
5. Quantified the central trade-off: improved robustness at the cost of slower convergence when BiCGSTAB already succeeds.

2025AA

ChatGPT (July 2026)

Summary

This paper computes thermalization times for energetic light ions in tokamak-relevant plasmas, focusing on deuterium-beam heating and 3.5 MeV alpha particles from D-T fusion. The authors use a quantum dielectric plasma model and derive accurate analytical approximations that separate electronic and ionic stopping contributions. A key improvement over common slowing-down formulas is the use of different logarithmic factors for projectile interactions with plasma electrons and ions, including velocity dependence in the ion term. Calculations span temperatures from 10^7 to 10^9 K and electron densities from 10^{13} to 10^{15} cm⁻³. The results show that plasma ions can significantly reduce thermalization times, while the tritium isotope effect and modest alpha impurities have small effects in the main velocity range because electron stopping dominates. The paper provides scaling laws, fitting formulas, and reference tables for fusion modeling.

Contributions

1. Computed stopping powers, ranges, and thermalization times for light ions in D and D-T plasmas.
2. Combined quantum electronic stopping with classical ion stopping in an analytical approximation.
3. Improved on single-Coulomb-log treatments by separating electron and ion logarithmic factors.
4. Derived scaling laws for projectile charge, mass, temperature, and plasma density.
5. Supplied fitting formulas and reference tables for deuteron heating beams and fusion alpha particles.

2025AB

ChatGPT (July 2026)

Summary

Modern GPU nodes leave many CPU cores idle because applications commonly assign one MPI process to each GPU. This paper exploits those cores to accelerate large GPU-aware all-reduce operations. It partitions each GPU buffer among multiple MPI processes, shares device memory through CUDA or HIP interprocess communication where supported, and lets processes progress disjoint reductions asynchronously. The approach is combined with a node-aware multi-lane algorithm whose reduce-scatter, inter-node all-reduce, and all-gather stages reduce large off-node messages. Benchmarks cover NVIDIA A100 nodes on Delta using OpenMPI host-copy communication and AMD MI300A nodes on Tuolumne using Cray MPICH GPUDirect RDMA, including SPX, TPX, and CPX GPU partitioning modes. Large-buffer speedups reach $2.45\times$ on Delta, $1.17\times$ for the standard multi-process method on Tuolumne, and $3\times$ for multi-process multi-lane all-reduce in CPX mode. Benefits depend on topology, message size, and process count, so the method complements rather than universally replaces system collectives.

Contributions

1. Extended node-aware multi-lane all-reduce to GPU-resident buffers with host-copy and GPUDirect RDMA communication.
2. Introduced a multiple-processes-per-GPU design using interprocess-shared buffers and asynchronous, disjoint buffer partitions.

3. Benchmarked the method on NVIDIA A100/Delta and AMD MI300A/Tuolumne against OpenMPI and Cray MPICH.
4. Showed up to $2.45\times$ large-buffer speedup on Delta and $1.17\times$ for the standard GPUDirect RDMA variant on Tuolumne.
5. Evaluated MI300A SPX, TPX, and CPX modes and obtained up to $3\times$ speedup for the multi-process multi-lane method in CPX mode.

2025ABGP

ChatGPT (July 2026)

Summary

High-order discontinuous Galerkin approximations on polytopic meshes can overshoot physical solution bounds. The paper constructs an arbitrary-order composite method for second-order elliptic problems that enforces prescribed bounds at user-selected points inside each polytope. Starting from an interior-penalty DG discretization, it introduces a local simplicial submesh, projects recovered values onto a convex nodal constraint set, and stabilizes the projection's kernel through a nonlinear term. Submesh nodes may be quadrature points, yet the global unknown count remains unchanged. The DG penalty depends only on the coarse polytopic mesh, not submesh granularity, and the formulation reduces to the baseline DG method where no bound is violated. Strong monotonicity and the Browder–Minty theorem establish existence and uniqueness. Because standard interpolants do not meet these constraints, the authors construct a specialized bound-preserving interpolant and derive optimal a priori estimates for sufficiently smooth solutions. A matrix formulation permits reuse of existing DG operators, while a semismooth Newton iteration solves the nonlinear system. Tests with degree-one through degree-three elements recover optimal rates and remove overshoots and undershoots around boundary and interior layers over strongly varying diffusion parameters. The guarantee is nodal at chosen points, not a pointwise bound everywhere.

Contributions

1. Developed an arbitrary-order nodally bound-preserving DG method for second-order elliptic problems on general polytopic meshes.
2. Allowed bounds to be enforced at freely selected submesh nodes without adding global degrees of freedom or changing the coarse-mesh penalty parameter.
3. Designed a projection-kernel stabilization with the monotonicity needed to prove existence and uniqueness.
4. Constructed a nonstandard bound-preserving interpolant and obtained optimal a priori error estimates for smooth solutions.
5. Demonstrated optimal convergence and suppression of nonphysical oscillations for smooth, boundary-layer, and interior-layer problems.

2025ACLB

ChatGPT (July 2026)

Summary

This paper proposes an in situ axion generation and detection scheme based on laser-plasma wakefield interactions. In the proposed setup, the strong electromagnetic fields in a wakefield drive axion production through the Primakoff process, replacing the conventional axion-generation stage in light-shining-through-a-wall concepts. The wakefield fields can exceed 10^{11} V/m, which the authors estimate can enhance axion production rates by about two orders of magnitude relative to conventional configurations. For axion masses below 0.1 meV, the scheme could probe axion–photon coupling near $g_{a\gamma\gamma} \sim 10^{-10} \text{ GeV}^{-1}$. The same plasma environment can also convert generated axions back into photons, producing axion-regenerated electromagnetic fields with distinctive polarization, frequency, and transverse-mode signatures. By filtering these regenerated fields from the background laser and plasma signals, the

method offers a compact plasma-based route for axion searches that couples generation and detection in one interaction region.

Contributions

1. Proposed laser-plasma wakefield interaction as an in situ source and detector for axions.
2. Used the Primakoff process in wakefield electromagnetic fields to enhance axion production.
3. Estimated roughly two orders of magnitude higher production rates than conventional light-shining-through-a-wall generation stages.
4. Identified sensitivity to $g_{a\gamma\gamma} \sim 10^{-10} \text{ GeV}^{-1}$ for sub-0.1 meV axion masses.
5. Described axion-regenerated electromagnetic-field signatures in polarization, frequency, and transverse mode structure.

2025ADFM

ChatGPT (July 2026)

Summary

The paper analyzes an Ozaki-style scheme that emulates binary64 matrix multiplication with fast integer matrix-multiply-accumulate units. Rows of one factor and columns of the other are scaled, decomposed into short integer slices, multiplied exactly, and accumulated in floating-point arithmetic. New forward-error bounds expose separate row- and column-scaling factors and permit different slice counts for the two operands. Because those factors are available during slicing, the authors propose a cheap search for the smallest pair of slice counts meeting a target bound. Experiments confirm that asymmetric slicing can retain accuracy with fewer products, but badly scaled operands may require impractically many slices. On random matrices, a prototype NVIDIA integer implementation reaches binary64-like accuracy with seven slices; some higher-level MAGMA factorizations require eight. Performance gains at acceptable accuracy appear on a Blackwell B200, whose INT8-to-binary64 throughput ratio is large, but not on the tested Grace-Hopper system. The scheme is therefore not a general binary64 replacement, especially for extreme scaling or IEEE special values.

Contributions

1. Derived error bounds for integer-sliced floating-point matrix multiplication.
2. Quantified the separate effects of badly scaled rows and columns on accuracy.
3. Proposed an inexpensive input-dependent search for asymmetric slice counts.
4. Tested accuracy inside block LU, QR, eigensolver, and singular-value algorithms.
5. Measured the accuracy-performance tradeoff on NVIDIA Grace-Hopper and Blackwell GPUs.

2025AF

ChatGPT (July 2026)

Summary

Numerical time integrators often preserve one invariant but cannot simultaneously enforce several conservation laws and dissipation inequalities. This paper presents a general auxiliary-variable framework for semidiscrete initial-value problems. Each quantity of interest is represented by an associated test function; projecting these functions onto the discrete test space defines auxiliary variables that modify the weak equations. The resulting arbitrary-order schemes reproduce multiple invariants and dissipation structures without reparametrizing the state to make them linear. For incompressible Navier–Stokes equations, the construction generalizes an energy-dissipating, helicity-tracking method to arbitrary temporal order. For compressible Navier–Stokes equations, it yields a time discretization that conserves mass, momentum, and energy while provably dissipating entropy. The framework also covers multiconservative Hamiltonian ordinary and partial differential equations. Reproducible implementations use Firedrake, PETSc, and MUMPS. The

main practical limitation is the large coupled nonlinear system created by multiple auxiliary variables and high-order implicit stages, making scalable solvers an important next step.

Contributions

1. Formulated a general test-function representation of conservation laws and dissipation inequalities.
2. Introduced projected auxiliary variables that enforce several structures simultaneously.
3. Generalized energy-dissipating and helicity-tracking incompressible-flow schemes to arbitrary order.
4. Derived a compressible-flow integrator conserving mass, momentum, and energy while dissipating entropy.
5. Supplied reproducible software and identified coupled nonlinear solution as the central performance challenge.

2025AKHW

ChatGPT (July 2026)

Summary

GPU choice and tuning strongly affect molecular-dynamics throughput, but the response depends on workload size and bottlenecks. This study benchmarks GROMACS 2024.4 on NVIDIA A40, A100, L4, and L40 accelerators using six biomolecular systems spanning about 20,000 to 1.07 million atoms. Exhaustive graphics-clock sweeps and power caps are interpreted against a compute-bound Pi Solver and the memory-bound BabelStream Triad kernel. Small GROMACS systems gain steeply from higher core frequencies; medium systems saturate earlier; and the largest systems become comparatively insensitive as memory movement and interconnect costs dominate. At maximum settings, the L40 delivers the highest throughput across the tested systems, while the A100 retains strong large-system performance with little clock sensitivity. Moderate power caps generally preserve throughput until device- and workload-specific thresholds are crossed. The A100 and L40 tolerate reductions well, whereas the lower-power L4 loses performance more sharply under aggressive caps. The synthetic kernels clarify frequency limits but draw too little power to serve as useful power-intensive corner cases. Results therefore support workload-specific hardware and power policies rather than a single optimal GPU setting, within the tested single-GPU configurations and software stack.

Contributions

1. Characterized GROMACS frequency and power-cap behavior across four NVIDIA data-center GPU architectures.
2. Evaluated six realistic systems over a wide size range using a controlled, reproducible GROMACS configuration.
3. Identified a progression from frequency-sensitive small systems to bandwidth- and interconnect-limited large systems.
4. Used compute- and memory-bound synthetic kernels to bound and interpret the observed frequency-scaling behavior.
5. Derived architecture-specific tuning guidance, including the broad power-cap tolerance of the A100 and L40 and the greater sensitivity of the L4.

2025ALSE

ChatGPT (July 2026)

Summary

Beam-driven plasma wakefield accelerators must extract most of a high-energy drive bunch while transferring tens to hundreds of joules per collider stage. This experiment uses a laser-ionized, all-optical hydrogen plasma source to push both depletion and transferred energy toward that regime. A 1.52 nC, 10 GeV electron bunch transfers at least 5.6 J to a plasma with density $4.5 \times 10^{16} \text{ cm}^{-3}$, corresponding to a measured drive-to-wake efficiency of at least $37 \pm 3\%$. Some driver electrons decelerate below 0.93 GeV, losing more than 90% of their initial energy, while as much as 90% of the bunch charge participates in the interaction. The results demonstrate multi-joule wake excitation with deep depletion in one stage, although the efficiency remains below the greater-than-70% target envisioned for collider operation.

Contributions

1. Demonstrated at least 5.6 J of drive-beam energy transfer to a plasma wake.
2. Measured a drive-to-wake energy-transfer efficiency of at least $37 \pm 3\%$.
3. Decelerated some 10 GeV electrons to below 0.93 GeV.
4. Achieved wakefield participation by up to 90% of the drive-bunch charge.
5. Established an all-optical plasma source as a route toward collider-scale depletion experiments.

2025ART

ChatGPT (July 2026)

Summary

This paper computationally compares best L^2 and L^∞ rational approximations of analytic functions on the unit disk. Although existence and characterization theory for these approximation problems is classical, practical computation has been much less developed. For the L^∞ problem, the authors use the AAA-Lawson method; for the L^2 problem, they formulate computations through TF-IRKA and a barycentric representation. The paper derives Hermite interpolation conditions for exact barycentric rational functions and shows how Loewner or barycentric linear systems can enforce the required interpolation and derivative constraints. Numerical examples compare the two norms, pole locations, approximation errors, and cases with nonunique global best approximants. The authors find no dramatic difference in approximation power between L^2 and L^∞ rational approximants on bounded domains, while emphasizing that rational functions remain especially valuable for singularities near boundaries, nonconvex domains, and model-reduction-style problems.

Contributions

1. Compared best L^2 and L^∞ rational approximations on the unit disk computationally.
2. Applied AAA-Lawson iteration to the minimax rational approximation problem.
3. Formulated the L^2 problem through TF-IRKA and barycentric rational forms.
4. Derived barycentric Hermite interpolation conditions using rectangular Loewner matrices.
5. Documented numerical behavior, pole patterns, and nonuniqueness phenomena in rational approximants.

2025B

ChatGPT (July 2026)

Summary

This paper revisits magnetic reconnection when electron inertia is the only non-ideal effect. Boozer gives a direct three-dimensional derivation showing that ordinary magnetic-field lines can reconnect, but the lines of the modified field $\vec{\mathcal{B}} = \vec{B} + \nabla \times ((c/\omega_{pe})^2 \mu_0 \vec{j})$ do not. The paper then examines what this invariant field means physically. Guiding-center motion receives electron-inertia corrections proportional to $(c/\omega_{pe})^2$, which are usually smaller than gyroradius-order drift effects unless current density is extremely large. The analysis also emphasizes that two-coordinate reconnection models can be misleading because fully three-dimensional magnetic-line flows are characteristically chaotic. In three dimensions, even weak diffusive non-ideal effects can make $\vec{\mathcal{B}}$ undergo large-scale reconnection within only one or two orders of magnitude of the chaotic-advection timescale. Thus the non-reconnection theorem is mathematically sharp but often practically fragile in realistic three-dimensional plasmas.

Contributions

1. Gave a direct three-dimensional proof for the non-reconnection of the electron-inertia modified field.
2. Distinguished reconnection of $\vec{B}$ from topology preservation of $\vec{\mathcal{B}}$.
3. Derived guiding-center corrections associated with electron inertia.
4. Assessed when electron-inertia effects are small relative to gyroradius-scale drifts.

5. Explained why chaotic three-dimensional field-line flow limits the practical protection from the invariant field.

2025B2

ChatGPT (July 2026)

Summary

This review develops physical intuition for magnetic-field-line chaos, cantori, and turnstiles and argues that they organize several otherwise disconnected problems in toroidal plasmas. Magnetic field-line trajectories form a Hamiltonian system; when irrational magnetic surfaces break, their remnants are cantori. Small holes in a cantorus admit paired inward and outward, highly collimated flux tubes called turnstiles. Fourier analysis of long field-line trajectories is proposed as a more informative alternative to conventional Poincaré plots for reconstructing nearly invariant surfaces and locating leakage. The framework explains how chaotic ideal evolution exponentially reduces separations between flux tubes, so weak non-ideal effects can trigger large-scale reconnection on a timescale that depends only logarithmically on resistivity. It also connects turnstile formation to localized runaway-electron wall strikes during tokamak disruptions and to the long connection lengths and resilient strike patterns of non-resonant stellarator divertors. The paper further discusses electromagnetic consequences of nominally electrostatic microinstabilities and Bohm-like transport when pressure varies along chaotic field lines. These are synthesis and theory arguments rather than a new experimental validation, and the review identifies unresolved needs in simulation fidelity, Alfvén-wave dissipation, and controllable non-resonant divertors.

Contributions

1. Unified magnetic-field-line chaos, cantori, and turnstiles in a Hamiltonian description tailored to toroidal plasma applications.
2. Explained how paired turnstiles transport magnetic flux through holes in broken irrational surfaces while remaining strongly collimated.
3. Proposed Fourier reconstruction of long field-line trajectories to identify near-surfaces, quantify chaos, and locate turnstile leakage.
4. Connected chaotic field evolution to logarithmically fast reconnection, disruption dynamics, and concentrated runaway-electron damage.
5. Applied the same concepts to resilient non-resonant stellarator divertors and identified open transport and simulation questions.

2025BBEG

ChatGPT (July 2026)

Summary

This paper analyzes photon acceleration in magnetized electron–ion plasmas. Strong magnetic fields change the electromagnetic dispersion relation and therefore change how radiation interacts with relativistic plasma-wave refractive-index modulations. The authors derive the magnetized wave dynamics, compare ordinary and extraordinary electromagnetic modes, and formulate Hamiltonian descriptions for wave-packet evolution through moving plasma-wave structures. In the photon-acceleration regime, a wave packet can gain frequency and amplitude by interacting with a relativistically moving modulation; Lorentz invariance makes the field amplitude scale with the same frequency-upshift factor. The magnetic field produces both quantitative and qualitative changes, with the strongest enhancement for extraordinary right-circularly polarized waves propagating along the static field. Ordinary and left-polarized modes show weaker changes in the cases considered. The paper connects the mechanism to laboratory high-field laser-plasma experiments and to astrophysical environments where strong magnetization and relativistic plasma waves coexist.

Contributions

1. Derived magnetized electromagnetic-wave dispersion relevant to photon acceleration.

2. Built Hamiltonian wave-packet descriptions for interaction with relativistic plasma waves.
3. Compared ordinary and extraordinary modes in magnetized electron–ion plasma.
4. Identified extraordinary right-polarized waves as the case with strongest magnetic enhancement.
5. Connected frequency upshift and amplitude gain to laboratory and astrophysical magnetized plasmas.

2025BG

ChatGPT (July 2026)

Summary

Distributed matrix-multiplication libraries usually dispatch among separate algorithms whose applicability depends on operand partitioning and replication. Unsupported layouts require costly redistribution. This paper instead presents one one-sided algorithm for dense $C = AB$ that accepts arbitrary one- or two-dimensional block partitionings, misaligned tiles, and replication factors. Each process uses index slicing to identify overlapping local and remote tiles, generates the required local multiplications and communications, and chooses stationary- A , stationary- B , or stationary- C data movement. Operations may execute directly or be lowered to an intermediate representation whose greedy scheduler overlaps communication, multiplication, and accumulation. The C++ PGAS implementation performs direct GPU-to-GPU communication with Intel SHMEM or NVSHMEM while using vendor GEMM kernels. Tests on 12 Intel Data Center GPU Max tiles and eight NVIDIA H100 GPUs cover transformer-like matrix shapes and varied layouts. Good partitioning choices generally match or exceed PyTorch DTensor, although some H100 cases remain within about 5% of DTensor because accumulation interferes with local GEMM. Performance still depends strongly on selecting a suitable layout and stationary operand; automatic partition selection and production integration remain future work.

Contributions

1. Introduced a single distributed dense-matrix-multiplication algorithm covering arbitrary block partitionings and replication factors.
2. Used slice-overlap arithmetic to generate local products without requiring aligned tiles or operand redistribution.
3. Unified stationary- A , stationary- B , and stationary- C one-sided communication within the same implementation.
4. Added an intermediate representation and greedy schedule for overlapping GPU communication, multiplication, and accumulation.
5. Benchmarked the implementation against PyTorch DTensor on Intel and NVIDIA multi-GPU systems across transformer-derived shapes and layouts.

2025BKKS

ChatGPT (July 2026)

Summary

Grid-based low-rank approximation of multivariate functions can require constructing a large tensor before compression. This paper introduces a mesh-free, two-level Chebyshev–Tucker representation. It samples the target only at a moderate tensor product of Chebyshev nodes, forms the coefficient tensor, and compresses that tensor with Tucker alternating least squares. Discretized Chebyshev basis functions then transfer the result to an arbitrarily fine grid without repeating the rank optimization. The Tucker ranks can be much smaller than the interpolation degree and approach those obtained by applying Tucker decomposition directly to the full grid. Error and complexity estimates relate interpolation accuracy, coefficient compression, and the final approximation. For functions with singularities, the method targets the smooth long-range component supplied by a range-separated tensor representation. Tests on biomolecular electrostatic potentials and monopole/dipole lattice systems show controllable accuracy, weak dependence on particle count, and quasi-optimal ranks. Extensions to higher dimensions require other hierarchical tensor formats to avoid dimensional growth.

Contributions

1. Introduced a mesh-free hybrid Chebyshev–Tucker format for multivariate function approximation.

2. Separated function sampling and rank optimization from final-grid resolution.
3. Derived error and complexity estimates for the two-level construction.
4. Extended the method to long-range components of singular many-particle potentials in range-separated form.
5. Demonstrated quasi-optimal ranks and controllable accuracy for biomolecular and lattice electrostatics.

2025BOMG

ChatGPT (July 2026)

Summary

π -PIC is a Python-controlled framework for developing, combining, and comparing particle-in-cell (PIC) algorithms without embedding every experiment in a monolithic code. A precompiled C++ core, exposed through pybind11, stores particles cell by cell and supports multithreading, while Python callbacks and modular extension and solver interfaces permit rapid experimentation without moving performance-critical work into Python. The paper demonstrates extensions for masked absorbing boundaries, moving windows, and mapping tightly focused laser pulses into smaller periodic domains. It also implements several solvers, including a simple electrostatic example and a second-order explicit energy-conserving method modified to reduce momentum error. In laser-wakefield benchmarks, the energy-conserving and Fourier-Boris solvers give similar plasma responses. Compared with Smilei, π -PIC retains greater qualitative accuracy at coarse resolution, but converges more weakly as resolution increases; the authors attribute the remaining discrepancy primarily to particle shape functions. The framework currently lacks multi-node and GPU execution, so its contribution is a modular research platform rather than a replacement for mature production-scale PIC codes.

Contributions

1. Defined a unified Python/C++ interface that separates simulation control, reusable extensions, and interchangeable PIC and field solvers.
2. Designed a thread-aware execution model that lets independently developed extensions access particles and fields while remaining compatible with multiple solvers.
3. Implemented and tested reusable extensions for absorbing boundaries, moving windows, and tightly focused laser-pulse mapping.
4. Extended an explicit energy-conserving PIC solver with corrections that eliminate two identified sources of momentum error.
5. Benchmarked π -PIC against Smilei in laser-wakefield acceleration, exposing complementary low-resolution accuracy and high-resolution convergence behavior.

2025BPQM

ChatGPT (July 2026)

Summary

This paper analyzes photon dynamics in a beam-driven plasma wakefield accelerator as a mechanism for frequency up-shifting laser light. A tailored plasma density can keep the wake's accelerating phase matched to the group velocity of a witness laser pulse, allowing large frequency shifts limited primarily by the drive beam energy. The authors study photons near this phase-matching condition using a photon kinetic model and validate the calculations against full electromagnetic particle-in-cell simulations. Those computations motivate a linear theory for photon phase-space evolution. The theory predicts scalings for the witness pulse and a self-similar phase-space solution. A notable conclusion is that, within the model, the pulse can be compressed without a lower bound on duration, suggesting a possible route to sub-femtosecond short-wavelength radiation. The work is primarily theoretical and computational, but it identifies design rules for photon acceleration experiments in plasma wakefields.

Contributions

1. Modeled photon phase-space dynamics near wake-phase matching.

2. Validated photon kinetic calculations against electromagnetic particle-in-cell simulations.
3. Derived linear-theory scalings for the witness pulse.
4. Found a self-similar solution for photon phase-space evolution.
5. Identified photon acceleration as a route to ultrashort short-wavelength radiation.

2025BSBB

ChatGPT (July 2026)

Summary

This paper uses global nonlinear gyrokinetic simulations to study how energetic ions modify ion-scale turbulence near safety-factor rational surfaces. The authors separate several competing mechanisms: dilution of thermal profiles, quasi-resonant fast-ion interaction, and turbulence self-interaction that generates zonal shearing layers. They find that fast ions reduce the destabilization threshold for zonal modes, strengthening shearing structures near the $q = 1$ surface and producing regions where turbulent transport can fall by about 90%. Tests with fast ions treated as a dilution species confirm that dilution is central to this threshold reduction, while quasi-resonant effects compete with it. At higher fast-ion temperature, the simulations excite an $n = m = 1$ fishbone at beta-induced Alfvén eigenmode frequency. The fishbone can further reduce turbulence through beat-driven zonal structures when it does not strongly flatten thermal profiles; otherwise, redistribution increases thermal fluxes. The results show that rational surfaces and energetic particles can jointly support turbulence suppression in high-confinement tokamak scenarios.

Contributions

1. Simulated fast-ion effects on core turbulence with global nonlinear gyrokinetics.
2. Identified fast-ion dilution as a mechanism that lowers the zonal-mode destabilization threshold.
3. Quantified strong local transport suppression near rational surfaces from enhanced shearing layers.
4. Separated dilution, quasi-resonant interaction, and turbulence self-interaction effects.
5. Analyzed how an energetic-particle-driven fishbone can either suppress or enhance thermal transport.

2025BSG

ChatGPT (July 2026)

Summary

This paper reformulates laser–plasma wakefield acceleration in capillary discharges as coupled operators acting on modal representations of the laser field and plasma-density response. A transverse operator describes guiding and linear mode coupling; nonlinear, plasma-oscillation, and ponderomotive-source operators encode feedback and wake excitation. Linear evolution preserves modal invariant subspaces, whereas nonlinear interactions break them and mix modes. Expanding transverse and longitudinal fields in orthonormal bases makes energy transfer and the effects of complex geometries more explicit than in the original coupled partial differential equations. Bloch–Floquet analysis extends the framework to periodic plasma modulations. The authors also propose learning the expensive nonlinear and nonlocal operators from high-fidelity simulations with neural operators, yielding a differentiable reduced model for optimization and control. The work is primarily a mathematical and modeling framework; its claims about acceleration and computational savings require systematic validation against full simulations and experiments.

Contributions

1. Expressed coupled laser and plasma dynamics through four physically interpretable operators.
2. Connected linear modal stability to invariant subspaces and nonlinear evolution to mode mixing.
3. Represented transverse and longitudinal wakefield dynamics in a common modal framework.
4. Incorporated periodic plasma structures through Bloch–Floquet analysis.
5. Proposed neural-operator surrogates for nonlinear response and ponderomotive-source evaluations.

2025BSS

ChatGPT (July 2026)

Summary

In dense-plasma wakefield accelerators, energetic electrons radiate during transverse betatron oscillations. This work shows that the resulting radiation reaction does more than damp emittance: because radiative loss grows nonlinearly with oscillation amplitude, different phase-space regions cool at different rates and the beam develops an amplitude population inversion. Starting from the Landau–Lifshitz force in an ion channel, the authors derive a nonlinear damped-oscillator model and characteristic timescales for the formation and contraction of ring-like structures in transverse position-momentum space. Multidimensional OSIRIS particle-in-cell simulations of a driver and witness beam reproduce the predicted rings only when radiation reaction is enabled, supporting the analytic mechanism. For representative parameters near 10 GeV and $5 \times 10^{19} \text{ cm}^{-3}$, the analysis estimates submillimeter phase-space structuring and a few-millimeter ion-channel-laser gain length. As cooling narrows the spread in the betatron strength parameter, the structured beam may satisfy conditions for coherent betatron emission, although the paper presents this as a proposed consequence rather than a completed demonstration. The mechanism applies to both beam- and laser-driven wakefields and becomes important for high-energy beams in dense plasmas.

Contributions

1. Identified betatron-cooling-induced population inversion as a kinetic phase-space-shaping mechanism.
2. Derived an amplitude-dependent radiation-reaction model and analytic structuring timescales.
3. Predicted contracting, ring-like transverse phase-space distributions with positive radial gradients.
4. Confirmed the structures in multidimensional particle-in-cell simulations by comparing runs with and without radiation reaction.
5. Established experimentally relevant parameter estimates and conditions under which the cooled beams may drive coherent betatron emission.

2025C

ChatGPT (July 2026)

Summary

HYLU is a sparse direct solver for shared-memory multicore systems that adapts its numerical factorization to the matrix's sparsity pattern. Rather than relying uniformly on supernodes and Level-3 BLAS, it combines row-row, supernode-row, and supernode-supernode update kernels, then selects among them using symbolic information such as predicted work and supernode structure. Preprocessing applies matching and scaling, fill-reducing ordering, and symbolic factorization. Numerical factorization uses a dependency graph with bulk parallelism in wide levels and pipelined execution along dependency chains. On 37 SuiteSparse matrices, HYLUs factorization was geometrically 2.36 times faster than Intel MKL PARDISO for one-time solves and 2.90 times faster for repeated solves. For repeated solves, factorization plus substitution was faster on every tested matrix, with a 2.53-fold geometric-mean speedup. The reported residuals were also smaller on average, although both solvers struggled with an extremely ill-conditioned case. The results support hybrid kernel selection as an alternative to applying dense-block kernels indiscriminately across sparse problems.

Contributions

1. Developed a general-purpose sparse LU solver that adapts to matrices with widely differing sparsity patterns.
2. Unified row-row, supernode-row, and supernode-supernode updates in one factorization data structure.
3. Selected numerical kernels from symbolic sparsity and work estimates instead of using a fixed supernodal strategy.
4. Combined level-based bulk parallelism with pipelined execution along strong dependency chains.
5. Demonstrated substantial factorization and repeated-solve speedups over Intel MKL PARDISO on 37 SuiteSparse matrices.

2025CGWZ

ChatGPT (July 2026)

Summary

Strong nonlinearities make direct finite-element solves of variational PDEs expensive and can restrict Newton methods to local convergence. The DualTPD framework first applies Fenchel–Rockafellar duality to separate nonlinear constitutive terms from differential operators. Mixed finite elements place the nonlinear variable in a discontinuous space, producing element-local block-diagonal mass matrices with closed-form Jacobians and inverses. A transformed primal–dual iteration then improves the monotonicity of the saddle-point dynamics, while elliptic Schur complements are handled by problem-specific multigrid preconditioners. Unlike a nested Newton solve, linearization and solution proceed in one globally convergent iteration. Experiments on elliptic p -Laplacian and nonlinear $H(\text{curl})$ problems report global, mesh-independent convergence and strong gains over primal formulations in highly nonlinear cases. The paper establishes the algorithmic framework and numerical evidence; a complete convergence analysis is deferred.

Contributions

1. Introduced DualTPD, a duality-based preconditioning framework for nonlinear variational PDEs.
2. Decoupled nonlinear and differential operators through Fenchel–Rockafellar duality.
3. Used discontinuous mixed finite elements to obtain element-local nonlinear updates and inverses.
4. Combined transformed primal–dual dynamics with multigrid-preconditioned elliptic Schur complements.
5. Demonstrated global, mesh-independent convergence for p -Laplacian and nonlinear $H(\text{curl})$ problems.

2025CHHM

ChatGPT (July 2026)

Summary

This experiment measures energy changes of an initially monoenergetic ~ 450 MeV chromium-ion beam crossing a magnetized plasma formed by colliding laser-ablated jets. Interferometry, ion deflectometry, time-of-flight detection, and radiation-MHD simulations constrain the plasma and beam interaction. Single-jet shots show little energy shift or broadening, whereas double-jet shots exhibit larger, stochastic acceleration and diffusion, including separate regimes dominated by net energization or spectral broadening with mean-energy loss. Interferometry bounds fluid-scale velocity fluctuations, while deflectometry estimates magnetic fields of roughly 40–230 kG. Coulomb stopping and diffusion cannot explain the measurements, and the predicted second-order Fermi energy change is below the beam spread and detector resolution. Short-scale electrostatic turbulence driven by the lower-hybrid drift instability is quantitatively compatible with the observed scale of energy changes, and beam-driven or slowly growing electromagnetic instabilities are disfavored. The experiment does not uniquely identify the wave mode, so other wave–particle interactions remain possible.

Contributions

1. Measured acceleration and energy diffusion of heavy ions in a laser-driven magnetized plasma.
2. Distinguished weak single-jet responses from stronger stochastic double-jet interactions.
3. Constrained density, flow fluctuations, and magnetic-field strength with complementary diagnostics and simulations.
4. Ruled out Coulomb transport and fluid-scale second-order Fermi acceleration at the measured magnitude.
5. Showed that lower-hybrid-scale electrostatic turbulence is consistent with, but not uniquely proven by, the observations.

2025CHLW

ChatGPT (July 2026)

Summary

Stiff collisions and multiscale transport make high-order Boltzmann simulations difficult without resolving the Knudsen number. This work combines a conservative semi-Lagrangian discontinuous Galerkin discretization with BGK penalization and implicit-explicit Runge–Kutta time integration. Its central device is a moments update derived from a Shu–Osher representation of the IMEX scheme, allowing the discrete method to recover the correct hydrodynamic limit. The authors derive temporal order and asymptotic-accuracy conditions that account for characteristic tracing, give sufficient conditions for positivity, and use hypocoercivity to prove stability for a linearized model. Numerical tests across kinetic and fluid regimes confirm the designed order, asymptotic-preserving behavior, and robustness. Stability for the full nonlinear Boltzmann equation remains outside the analysis.

Contributions

1. Constructed an asymptotic-preserving semi-Lagrangian DG scheme for the Boltzmann equation with BGK penalization.
2. Derived a Shu–Osher-based moments update that captures the correct limiting macroscopic system.
3. Established order conditions for the IMEX-RK stages and the moments update in the characteristic-tracing setting.
4. Proved sufficient positivity conditions and hypocoercive stability for the linearized semi-discrete model.
5. Verified accuracy, asymptotic preservation, and robustness numerically over multiple regimes and test problems.

2025CJBL

ChatGPT (July 2026)

Summary

Differentiable multiphysics workflows are difficult when an automatic-differentiation-native model must call a production C or CUDA solver. This case study models a pulsed-power circuit and a compressing Z pinch with a JAX ODE solver. At every timestep, Newton iteration determines the plasma voltage needed to produce the circuit current, using a steady Vlasov–Poisson–Fokker–Planck calculation as the closure. The Tesseract abstraction and its JAX adapter expose this closure through a common differentiable interface. The same outer solver can therefore interchange a high-fidelity Gkeyll implementation, a symbolic-regression surrogate, and a closed-form approximation without rewriting the optimization loop. Finite differences supply derivatives for non-differentiable Gkeyll, while native differentiation is used where available. Demonstrations compare density–temperature trajectories across the three fidelities and optimize initial temperature and capacitance with L-BFGS. The architecture enables progressive prototyping, but the study is a small case study: one Gkeyll trajectory requires 18 A100 GPU-hours, and quantitative fidelity validation is limited.

Contributions

1. Built an end-to-end differentiable circuit–plasma model for a compressing Z pinch.
2. Embedded a nonlinear plasma-impedance closure inside a JAX ODE solve through Newton iteration.
3. Wrapped the non-differentiable Gkeyll C/CUDA solver in a JAX-compatible Tesseract interface.
4. Made high-fidelity, surrogate, and analytical closures interchangeable without changing the outer workflow.
5. Demonstrated gradient-based design exploration across model fidelities with explicit accuracy–cost trade-offs.

2025CJV

ChatGPT (July 2026)

Summary

This paper analyzes Anderson acceleration by rewriting the discrete method as an adaptive multistep momentum scheme and taking a high-resolution continuous-time limit. The resulting second-order ODE has variable effective mass, which gives a physical explanation for a known failure mode of standard Anderson acceleration: unchecked mass growth acts like negative damping and injects energy into stiff nonlinear iterations. The same high-resolution expansion also exposes an implicit Hessian-driven damping term that can stabilize high-curvature directions. Building on this interpretation, the authors propose Energy-Guarded Anderson Acceleration, which bounds mass growth and enforces energy dissipation. Their convergence analysis uses an acceleration-gain factor to show that the guarded method preserves useful geometric contraction while suppressing nonlinear approximation error. Numerical tests on ill-conditioned convex composite problems show better stability and convergence than standard Anderson mixing and Nesterov acceleration. The paper frames acceleration as a balance between inertial gain and thermodynamic consistency.

Contributions

1. Recast Anderson acceleration as an adaptive momentum method.
2. Derived a high-resolution variable-mass ODE limit for Anderson acceleration.
3. Identified effective-mass growth as a negative-damping instability mechanism.
4. Proposed Energy-Guarded Anderson Acceleration to enforce dissipative dynamics.
5. Supported the theory with convergence analysis and ill-conditioned optimization experiments.

2025CKKS

ChatGPT (July 2026)

Summary

The paper develops asymptotic-preserving conservative semi-Lagrangian discontinuous Galerkin (AP-CSLDG) schemes for the one-dimensional Vlasov–Poisson system as the normalized Debye length λ approaches zero. Direct discretization becomes stiff and the Poisson equation degenerates in this quasi-neutral limit. The authors instead combine a conservative semi-Lagrangian Vlasov solver with a reformulated Poisson equation obtained from the kinetic equation’s moments. Semi-Lagrangian transport removes the CFL restriction and conserves particle number, while high-order DG approximation resolves fine phase-space structure. The paper proves asymptotic consistency and stability, showing that the schemes remain well behaved without resolving the Debye length in space or time. Landau damping, two-stream instability, and bump-on-tail tests assess accuracy and behavior in both quasi-neutral and non-quasi-neutral regimes; the AP variants retain reliable solutions on under-resolved meshes at per-step costs comparable to explicit schemes. The analysis and experiments are restricted to a one-dimensional electrostatic model with well-prepared initial data; electromagnetic, collisional, multidimensional, and unsplit extensions are left open.

Contributions

1. Combined conservative semi-Lagrangian DG transport with a moment-based reformulated Poisson equation.
2. Proved asymptotic consistency and stability as the Debye length tends to zero.
3. Removed both CFL and Debye-length resolution restrictions while exactly conserving total particle number.
4. Constructed splitting variants with per-step costs comparable to explicit Vlasov–Poisson solvers.
5. Validated the schemes on Landau damping, two-stream, and bump-on-tail problems across quasi-neutral regimes.

2025CLLZ

ChatGPT (July 2026)

Summary

The authors propose a high-order energy-conserving semi-Lagrangian discontinuous Galerkin (ECSLDG) method for the Vlasov–Ampère system. The Vlasov equation is reduced by operator splitting to one-dimensional advection problems solved with SLDG, providing high-order spatial accuracy, local mass conservation, and freedom from a CFL stability restriction. A semi-implicit coupling between Ampère’s law and velocity moments removes the requirement to resolve the electron plasma period without introducing nonlinear iterations. The split subsystems preserve energy, and a fourth-order composition raises temporal accuracy while retaining exact fully discrete mass and total-energy conservation. Analysis establishes unconditional stability. One-dimensional, one-velocity-dimensional tests verify convergence, conservation, and long-time behavior; compared with second-order splitting, the high-order scheme enforces Gauss’s law more accurately and maintains better fidelity at large CFL numbers. The current results do not establish exact Gauss-law preservation, and multidimensional or full Vlasov–Maxwell extensions are not demonstrated.

Contributions

1. Built a high-order SLDG discretization for the Vlasov–Ampère system with no CFL stability restriction.
2. Semi-implicitly coupled fields and distribution moments without requiring nonlinear solves or plasma-period resolution.
3. Preserved mass and total energy exactly at the fully discrete level.
4. Used fourth-order operator composition to obtain high-order accuracy in both space and time.
5. Demonstrated improved Gauss-law enforcement and numerical fidelity over second-order schemes at large CFL numbers.

2025CMPV

ChatGPT (July 2026)

Summary

Hypoxia-activated prodrugs target oxygen-poor tumor regions, but their activation, transport, metabolism, and cytotoxicity are strongly coupled. This work builds a mixed-dimensional model for tirapazamine that combines three-dimensional tissue transport, one-dimensional blood flow in a synthetic capillary network, and a zero-dimensional pharmacokinetic/pharmacodynamic surrogate. The framework represents vascular delivery, oxygen diffusion and consumption, drug exchange and metabolism, and cell survival. One-way coupling from the reduced nonlinear surrogate to the spatial finite-element model avoids solving cellular-scale kinetics throughout the tissue. A two-stage Morris global sensitivity analysis first screens parameters with the inexpensive surrogate, then studies seven influential parameters in the 3D–1D–0D model. Vascular drug concentration dominates tissue exposure, while drug metabolism and oxygen-related parameters strongly influence survival. The spatial model preserves similar importance rankings but reveals larger interaction effects and heterogeneous drug and survival distributions tied to vascular density. Signed sensitivities also show that oxygen transport can affect survival non-monotonically through competing activation and accumulation mechanisms. The results are exploratory: they use one synthetic network, quasi-steady oxygen transport, and no calibration to experimental or patient data, so the model generates hypotheses rather than clinical predictions.

Contributions

1. Formulated a mixed-dimensional 3D–1D model coupling tissue transport to blood flow in an embedded microvascular network.
2. Integrated oxygen-dependent tirapazamine metabolism and cell survival through a reduced pharmacokinetic/pharmacodynamic model.
3. Reduced the global nonlinear cost through one-way coupling of the 0D surrogate to the spatial finite-element model.

4. Designed a two-stage Morris analysis that screens parameters cheaply before evaluating a reduced set in the multiscale model.
5. Identified vascular drug concentration, oxygen availability, and metabolic rates as dominant drivers while exposing spatial heterogeneity and nonlinear interactions.

2025COWC

ChatGPT (July 2026)

Summary

Replacing part of an electron population with negative muons creates a hybrid plasma mode that can mediate laser amplification. The paper derives this μ -wave, which behaves like an ion-acoustic wave at long wavelengths and a Langmuir wave at short wavelengths, and analyzes its kinetic damping and three-wave instability. One- and two-dimensional particle-in-cell simulations reproduce the predicted resonance and growth trends. Reducing the electron fraction enhances Landau damping of ordinary Langmuir waves, suppressing spontaneous Raman scattering that would otherwise deplete the pump. In a representative case with a 20% electron fraction, μ -wave amplification raises the seed intensity above the simulated Raman and strong-coupling Brillouin alternatives while retaining a Gaussian envelope and narrow spectrum. Two-dimensional tests also show weaker filamentation than Brillouin amplification because the reduced electron plasma frequency lowers the instability growth rate. The mechanism therefore combines efficient pump-to-seed transfer with suppression of parasitic modes and pulse splitting. Practical realization remains difficult: the strongest simulated effect requires an 80% muon fraction and depends on future progress in producing and cooling sufficiently dense muon populations, although smaller fractions still provide measurable damping benefits.

Contributions

1. Derived the dispersion and damping of a hybrid μ -wave in electron–muon–ion plasmas.
2. Formulated a three-wave laser-amplification mechanism mediated by the new plasma mode.
3. Verified theoretical resonance and growth-rate predictions with one-dimensional particle-in-cell simulations.
4. Demonstrated suppression of spontaneous Raman scattering, pulse splitting, and spectral distortion through enhanced Langmuir-wave damping.
5. Showed in two dimensions that μ -wave amplification preserves the seed profile and reduces filamentation relative to strong-coupling Brillouin amplification.

2025CR

ChatGPT (July 2026)

Summary

This paper reinterprets the Discrete de Rham method as a conforming finite element system rather than as a nonconforming collection of cellwise unknowns. The authors define local spaces through PDE-based extensions on mesh cells, obtaining subcomplexes of Sobolev de Rham complexes with stronger conformity than in earlier DDR analyses. Although the local PDE solutions are not directly computable, the degrees of freedom still determine enough polynomial moments to assemble a fully discrete method. This viewpoint yields a computable and consistent discrete L^2 product, giving a Strang-lemma route to analysis without changing the numerical method itself. The paper also relates DDR to virtual element methods and earlier finite element systems based on harmonic extensions, clarifying how these polytopal methods share degrees of freedom and reconstruction principles. The result is mainly conceptual and analytical: DDR can inherit finite element system compactness, consistency, and convergence arguments once the appropriate conforming spaces and products are identified.

Contributions

1. Interpreted DDR spaces as conforming finite element systems defined by local PDEs.
2. Established stronger conformity and consistency properties for the resulting de Rham complex.

3. Constructed a computable discrete L^2 product compatible with the finite element system view.
4. Connected DDR analysis with Strang-type variational-crisis arguments.
5. Clarified relationships among DDR, virtual elements, and harmonic-extension finite element systems.

2025DBML

ChatGPT (July 2026)

Summary

This paper uses Bayesian optimization and particle-in-cell simulations to improve channel-guided laser wakefield acceleration through spectral pulse shaping. The accelerator model uses a realistic plasma profile based on optical-field-ionized channel formation, ionization injection, and low on-axis density to maximize energy gain in a trailing electron bunch. Spectral shaping changes the laser temporal profile while holding total laser energy fixed, so it can alter injection and acceleration without simply increasing driver energy. Because the combined laser-shape and plasma-channel design space is broad, the authors use numerical optimization to find high-performing configurations. The optimized pulses include additional spectral content and, together with tuned channel parameters, produce electron bunches with roughly an order-of-magnitude higher charge than a baseline Gaussian case while also increasing mean energy. The study demonstrates how machine-learning-guided optimization can expose useful driver profiles in laser-plasma accelerator designs where intuition alone may not identify the best coupled laser and plasma settings.

Contributions

1. Applied Bayesian optimization to spectral pulse shaping for channel-guided laser wakefield acceleration.
2. Coupled optimization to particle-in-cell simulations with realistic optical-field-ionized plasma-channel profiles.
3. Varied laser temporal structure at fixed laser energy to isolate spectral-shaping effects.
4. Found optimized profiles that increase electron-bunch charge by about an order of magnitude over a Gaussian baseline.
5. Demonstrated simultaneous improvements in bunch charge and mean energy through coupled laser and plasma-channel tuning.

2025DDGG

ChatGPT (July 2026)

Summary

Weakly collisional magnetic-mirror transport spans fast ion transit and slow collision times, while strong mirror forces impose prohibitive explicit time steps. This work adapts COGENT's implicit-explicit continuum kinetic framework to 1D–2V simulations with kinetic ions, nonlinear Fokker–Planck collisions, beam fueling, and Boltzmann electrons under WHAM-relevant parameters. High-order additive Runge–Kutta integration treats the Vlasov operator implicitly through Jacobian-free Newton–Krylov iterations, with an approximate-ideal-restriction algebraic multigrid preconditioner built from a lower-order discretization. Stable steps reach 2.5×10^4 times the explicit CFL limit and yield about a 2500-fold speedup, allowing a 20 ms low-resolution run in roughly 25 minutes on 512 cores. A fully implicit 0D–2V bounce-averaged square-mirror model reproduces the analytic Pastukhov confinement problem and helps assess phase-space resolution. It reveals first-order convergence when the loss-cone transition is under-resolved and substantial differences between nonlinear Fokker–Planck and simplified Lenard–Bernstein–Dougherty collisions. Fifth-order upwinding is necessary to control long-time numerical diffusion. The present Boltzmann-electron model, fixed-field preconditioner, and baseline resolution limit accuracy outside the trap and omit kinetic-electron and unlike-species effects.

Contributions

1. Developed a semi-implicit COGENT formulation for full- F continuum kinetic transport in strongly magnetized mirror geometry.

2. Combined Jacobian-free Newton–Krylov iteration with approximate-ideal-restriction multigrid preconditioning for the stiff Vlasov solve.
3. Achieved time steps 2.5×10^4 above the explicit limit and approximately 2500-fold end-of-run speedup.
4. Implemented and verified a bounce-averaged square-mirror model against the analytic Pastukhov problem.
5. Exposed resolution-sensitive loss-cone convergence, collision-operator differences, and the need for fifth-order upwinding over collisional timescales.

2025DFHL

ChatGPT (July 2026)

Summary

Solving $AX + XB = C$ with a low-precision Schur decomposition can save work, but the resulting quasi-triangular factors are perturbed and their Schur vectors are not unitary to working precision. This paper derives an iterative-refinement scheme for perturbed quasi-triangular Sylvester equations, with sufficient convergence conditions and a bound on the attainable relative residual. It then builds two mixed-precision solvers: both compute Schur data and an approximate solution in low precision, refine the triangular solve in working precision, and recover the original solution either by re-orthonormalizing the Schur vectors or explicitly inverting the nearly unitary factors. The authors also analyze Schur-preconditioned GMRES refinement and find it generally less accurate and more expensive in the two-precision setting. Tests on Sylvester and Lyapunov problems match the accuracy of high-precision Bartels–Stewart methods. Operation-count models predict speedups on hardware with sufficiently fast native low precision, rather than reporting measured wall-clock gains.

Contributions

1. Developed iterative refinement for perturbed quasi-triangular Sylvester equations.
2. Proved sufficient convergence conditions and bounded the attainable relative residual.
3. Constructed two mixed-precision solvers using re-orthonormalization or explicit inversion of Schur factors.
4. Explained why Schur-preconditioned GMRES refinement is unattractive in the targeted two-precision regime.
5. Demonstrated high-precision accuracy and quantified conditions under which low-precision work should reduce cost.

2025DHYD

ChatGPT (July 2026)

Summary

This Physical Review Letters paper reports DIII-D observations in which toroidicity-induced Alfvén eigenmodes suppress low- k ion-temperature-gradient turbulence. Beam emission spectroscopy shows that turbulence mitigation starts as the dominant TAE departs from ordinary shear-Alfvén polarization, followed by a transition to more discrete, higher-amplitude, radially localized modes. During full suppression, a narrow shear-flow layer forms with a shearing rate exceeding the turbulence decorrelation rate. The authors connect this sequence to an imbalance between Reynolds and Maxwell stress forces during nonlinear TAE evolution. The turbulence suppression is radially localized but correlates with improved thermal confinement, including electron-temperature and stored-energy changes. Alternative explanations such as fast-ion dilution and profile changes are examined and disfavored. The result provides direct experimental evidence for energetic-particle-mode control of microturbulence and suggests that externally driven edge Alfvén spectra might be used to form transport barriers in future devices.

Contributions

1. Observed localized low- k turbulence mitigation and suppression during nonlinear TAE evolution in DIII-D.
2. Measured a shear-flow layer whose shearing rate exceeds the turbulence decorrelation rate.
3. Linked turbulence reduction to Reynolds-Maxwell stress imbalance and altered TAE polarization.

4. Connected the localized suppression to improved thermal-plasma confinement indicators.
5. Excluded fast-ion dilution and profile-change explanations for the observed suppression phase.

2025DS

ChatGPT (July 2026)

Summary

Jacobi diagonalization, Gaussian elimination, and Gram–Schmidt appear to solve different matrix factorizations, yet each repeatedly transforms selected columns. The paper places one- and two-sided variants of these algorithms in a common procedure that can produce eigendecompositions, singular value decompositions, Cholesky factorizations, or QR factorizations by constraining each local transformation. It then chooses a subset of k indices uniformly at random and exactly factors the corresponding principal block. A potential function measuring normalized nonorthogonality or nondiagonality yields the same expected linear contraction factor, $1 - k(k-1)/(n(n-1))$, for every supported factorization. Consequently, sufficiently well-conditioned inputs reach an approximate factorization in $O(n^2 \log(n\hat{\kappa}/\delta))$ pairwise-size iterations, and the analysis also confirms a conjectured convergence rate for the Kaczmarz–Kac orthogonalization walk. For finite arithmetic, martingale arguments control accumulated local errors and show stability for a broad class of randomized orthogonalization methods. Applied to randomized Jacobi iteration, the analysis bounds the diagonal-normalized condition numbers of all iterates polynomially with high probability. This supplies the missing theoretical condition behind Demmel and Veselić’s relative-accuracy analysis without preconditioning. The precision exponents are explicit but not optimized, so the results establish polynomial stability rather than practical superiority at standard precision.

Contributions

1. Unified Jacobi eigendecomposition and SVD algorithms with elimination and Gram–Schmidt methods for Cholesky and QR factorizations.
2. Defined a uniformly random size- k pivoting rule applicable across the entire factorization class.
3. Proved a factorization-independent expected linear convergence rate using a common potential function.
4. Derived finite-precision stability guarantees for a broad family of randomized orthogonalization procedures through martingale error control.
5. Proved polynomial high-probability control of Jacobi iterates’ normalized condition numbers, resolving the unproved assumption in the Demmel–Veselić analysis.

2025DUA

ChatGPT (July 2026)

Summary

The viscous Burgers equation couples nonlinear advection with diffusion and develops increasingly sharp fronts as viscosity decreases. This paper constructs a collocation method that uses B-splines in both independent variables: quadratic B-splines first integrate the equation in time, after which cubic B-splines discretize the resulting spatial problem. Collocation and the boundary conditions reduce each update to a recursive algebraic system for the spline coefficients. Two one-dimensional benchmarks with analytical solutions test shock-like and traveling-front propagation. The reported L_∞ and L_2 errors generally decrease as equal time and space increments are refined, and the computed profiles become smoother. Accuracy deteriorates for lower viscosity and near the steep front at the right boundary; the second benchmark retains substantial maximum errors even on the finest tested grid. The study therefore demonstrates the feasibility of a time–space spline construction, but it does not provide a formal stability or convergence analysis. The authors identify higher-order splines in time and space as the next route for improving accuracy.

Contributions

1. Formulated a Burgers-equation integrator that uses B-spline collocation in both time and space.

2. Combined quadratic temporal splines with cubic spatial splines in a sequential discretization.
3. Derived the recursive coefficient system, including initialization and boundary treatment.
4. Tested the method on analytical shock-wave and traveling-front solutions across multiple viscosities and grid increments.
5. Quantified the loss of accuracy for sharper low-viscosity fronts and identified higher-order splines as a concrete extension.

2025E

ChatGPT (July 2026)

Summary

The Dirichlet–Neumann alternating method is well understood for linear elliptic equations, but general convergence results for nonlinear, nonoverlapping decompositions were largely restricted to one dimension. This paper proves local linear convergence for a class of semilinear elliptic equations on Lipschitz domains in two and three dimensions without cross-points. It first establishes an abstract convergence theorem for nonlinear preconditioned iterations in Hilbert spaces. The domain-decomposition iteration is then written through nonlinear Steklov–Poincaré interface operators, whose differentiability, Lipschitz continuity, symmetry of derivatives, and uniform monotonicity provide the theorem’s hypotheses. For a sufficiently small relaxation parameter and an initial interface guess sufficiently near the solution, both interface and subdomain iterates converge linearly. Finite-element experiments confirm mesh-independent rates on two nonlinear examples; they also show faster convergence than Robin–Robin iteration, while Neumann–Neumann fails on one example. Convergence degrades as the number of subdomains increases, motivating a future nonlinear coarse correction.

Contributions

1. Proved convergence of the Dirichlet–Neumann method for semilinear elliptic problems in two and three dimensions.
2. Established a reusable local linear-convergence theorem for nonlinear Hilbert-space iterations.
3. Characterized the required nonlinear Steklov–Poincaré operators and their derivatives.
4. Transferred interface convergence to the corresponding subdomain solutions.
5. Demonstrated mesh-independent finite-element convergence and compared competing nonoverlapping methods.

2025EB

ChatGPT (July 2026)

Summary

Per-iteration bounds for randomized Gauss–Seidel and Kaczmarz methods are often much slower than observed convergence and incorrectly suggest that relaxation can only hurt. This paper instead studies the global asymptotic rate, which is governed almost surely by a Lyapunov exponent of random projector products. The authors bound that exponent through the spectral radius of the covariance-evolution superoperator $\mathcal{A} = \mathbb{E}[(I - \omega P) \otimes (I - \omega P)]$. A new min–max argument, connected to Perron–Frobenius theory for positive maps on noncommutative algebras, produces an asymptotic A-bound using the two smallest eigenvalues of $\mathbb{E}[P]$ and a fourth-order statistic. The bound is strictly sharper than the standard per-iteration B-bound when the spectral gap is positive. Its closed-form minimizing relaxation parameter is provably faster than $\omega = 1$, explaining over-relaxation and resolving the cited open question. Experiments show much tighter predictions across tested matrices.

Contributions

1. Reframed randomized Gauss–Seidel and Kaczmarz convergence through global asymptotic Lyapunov rates.
2. Connected the rate to a covariance superoperator governed by noncommutative Perron–Frobenius theory.
3. Developed a new spectral-radius bound using second- and fourth-order projector statistics.
4. Proved when the asymptotic A-bound improves on the standard per-iteration B-bound.

5. Derived a provably beneficial relaxation parameter and validated the tighter rate predictions numerically.

2025EDLN

ChatGPT (July 2026)

Summary

Unstructured discrete-ordinates transport sweeps form a directed acyclic graph whose wavefront widths and finite-element solve times vary, causing load imbalance at bulk-synchronous barriers. After benchmarking the existing UnSNAP mini-application across current CPUs and GPUs, this work replaces its shared-memory bucket schedule with recursive OpenMP tasks. One initial task per angle processes all energy groups for an element; atomic dependency counters release downwind elements when their final upwind prerequisite completes. Work-first generation limits the number of live tasks, while distributed task queues, affinity, and work stealing improve locality and balance. On 16^3 -element synthetic meshes, the asynchronous many-task implementation matches or outperforms the bulk-synchronous version in almost every tested angle, group, and finite-element-order configuration on AMD Genoa, Intel Sapphire Rapids, and NVIDIA Grace CPUs. Gains reach roughly $2.4\times$ and are strongest at first order; they diminish as higher-order element work dominates scheduling overhead. On Sapphire Rapids, synchronization-wait cycles fall from 33.58% to 9.45% for first-order tests. The evaluation preserves integrated-flux results within floating-point reordering effects. Scope remains single-node CPU execution with parallel-block-Jacobi communication; GPU tasking and other distributed-memory sweep schemes are future work.

Contributions

1. Updated the performance characterization of UnSNAP's bulk-synchronous transport sweep on modern many-core CPUs and GPUs.
2. Designed a recursive work-first task algorithm that respects unstructured sweep dependencies through atomic counters.
3. Combined angle-level concurrency with distributed task queues, affinity, and work stealing to reduce imbalance and synchronization.
4. Demonstrated up to roughly $2.4\times$ speedup across three CPU architectures while preserving the numerical result.
5. Attributed performance gains to sharply reduced synchronization waiting and quantified their decline as finite-element order increases.

2025EGTL

ChatGPT (July 2026)

Summary

Ultra-high-contrast, relativistically intense femtosecond laser pulses were directed at periodic nanowire arrays with silicon cores and titanium-dioxide shells. Relative to planar targets, the arrays produced distinctive backward plasma jets extending as far as 1 mm and emitting K-shell radiation from He-like Ti^{20+} . Electron and proton diagnostics, high-resolution X-ray spectroscopy, source imaging, collisional-radiative calculations, and three-dimensional particle-in-cell simulations jointly constrain the interaction. The nanowire volume reaches approximately solid electron density and keV temperature, while the observed jets are inferred to retain 0.4–1 keV temperatures and 10^{21} – 10^{22} cm^{-3} densities for several nanoseconds. Backward protons above 5 MeV are also produced at laser intensities roughly two orders of magnitude below earlier comparable reports. Simulations predict rapidly formed, kilotesla-scale global magnetic fields. The authors propose that current-driven, Weibel-like dynamics could sustain hundreds of tesla and magnetically confine the jets. That long-time mechanism remains a hypothesis: the experiment establishes the persistent hot, dense plasma, but does not directly measure the confining field or resolve its nanosecond evolution.

Contributions

1. Experimentally observed millimeter-scale, nanosecond-lived hot plasma jets from relativistically irradiated composite nanowire arrays.
2. Combined particle diagnostics, element-specific X-ray spectroscopy, source imaging, FLYCHK modeling, and three-dimensional particle-in-cell simulation in one analysis.
3. Inferred distinct density and temperature regimes for the near-solid-density nanowire plasma and the expanding jets.
4. Demonstrated backward proton acceleration above 5 MeV at substantially lower intensity than comparable earlier experiments.
5. Connected simulated kilotesla field generation to a testable magnetic-confinement hypothesis while explicitly identifying the unresolved long-time mechanism.

2025EHBG

ChatGPT (July 2026)

Summary

Emdee and coauthors use SOLPS-ITER fluid-kinetic edge simulations to explain inboard-outboard recycling asymmetries in a DIII-D H-mode plasma. The simulations compare cases with and without scrape-off-layer drifts and with different core-boundary rotation conditions derived from charge-exchange rotation measurements. Either drifts or rotation can enhance inner-target particle flux when the ion $B \times \nabla B$ drift points toward the X-point, but their combination produces a much stronger high-field-side/low-field-side asymmetry than either effect alone. With realistic rotation, radial transport of parallel momentum changes upstream scrape-off-layer flows, sending more deuterium flux toward the inboard divertor entrance and improving agreement with measured Lyman- α brightness profiles. The modeled inboard-outboard target recycling ratio rises from 1.9 with low momentum transport to 5.8 with high momentum transport. The Letter argues that rotation, viscous momentum coupling, and drifts should be included in 2D edge modeling for detachment-relevant reactor predictions.

Contributions

1. Connected core plasma rotation to divertor recycling asymmetry through SOLPS-ITER edge simulations.
2. Quantified the combined effect of scrape-off-layer drifts and imposed rotation on DIII-D target fluxes.
3. Showed that radial parallel-momentum transport alters upstream scrape-off-layer flow patterns.
4. Improved agreement with experimental inboard-outboard Lyman- α brightness asymmetry.
5. Identified rotation and viscous coupling as important inputs for future divertor and detachment modeling.

2025FKF

ChatGPT (July 2026)

Summary

Classical resistive magnetohydrodynamics predicts that the fastest linear tearing mode is independent of plasma beta, the ratio of thermal to magnetic pressure. This work relaxes strict pressure isotropy using a nonideal, energy-conserving gyrotropic MHD model. Boundary-layer analysis and pseudospectral eigenvalue calculations show that dynamically generated pressure-anisotropy fluctuations suppress tearing even when the equilibrium is isotropic. At high β , the maximum growth rate follows $\sigma_{\max} \tau_a \propto \beta^{-1/4}$, while the fastest-growing mode shifts to longer wavelengths. The full scaling retains the classical Lundquist-number dependence but acquires beta corrections in both growth rate and wavenumber. An imposed anisotropy changes the outcome: $p_{\parallel,0} > p_{\perp,0}$ strengthens suppression, whereas $p_{\parallel,0} < p_{\perp,0}$ can offset it and, at low beta, produce faster growth than isotropic MHD predicts. Viscous effects remain consistent with classical magnetic-Prandtl-number behavior. The result narrows the conditions under which plasmoid-mediated reconnection may begin in high-beta astrophysical plasmas. Its scope is the linear, macroscopic, single-fluid regime; Hall, two-fluid, kinetic, turbulent, and nonlinear reconnection physics remain outside the model.

Contributions

1. Derived a linear tearing theory for an energy-conserving nonideal gyrotropic MHD model with self-consistent pressure dynamics.
2. Established the high-beta scaling $\sigma_{\max}\tau_a \propto \beta^{-1/4}$ for initially isotropic equilibria.
3. Generalized the classical fastest-mode scalings to include coupled Lundquist-number and beta dependence in both growth rate and wavenumber.
4. Verified the analytical predictions by solving the complete linear eigenvalue problem with a converged pseudospectral method.
5. Quantified how the sign of equilibrium pressure anisotropy can either suppress tearing or overcome beta stabilization in low-beta plasmas.

2025G

ChatGPT (July 2026)

Summary

This dissertation develops global bounds for extreme roots of multivariate real-zero polynomials, whose restriction to every line through the origin has only real zeros. Its central tool is a computationally small spectrahedral relaxation of the polynomial's rigidly convex set, constructed from the degree-three truncation and sized by the number of variables. The thesis analyzes why straightforward attempts to strengthen this relaxation by adding higher-degree monomials or variables fail in general: positive semidefiniteness or the relaxation property can be lost. Eulerian polynomials provide a structured exception. Stability-preserving recurrences embed the univariate family into multivariate Eulerian polynomials, and restricting the resulting relaxation along carefully chosen directions yields exponentially improved asymptotic bounds for the leftmost root relative to the direct univariate relaxation. Comparisons with interlacing, Sobolev estimates, and Dandelin–Lobachevski–Gräffe root transformations refine earlier asymptotics and produce a still tighter combined bound. The work also proves real-zero and mirror-reciprocity properties for multivariate Eulerian polynomials and derives descent-set counting identities. The improvements depend on family-specific stability and experimentally chosen linearization directions, so a general extension theory and optimal directions remain open.

Contributions

1. Developed and analyzed a degree-three spectrahedral relaxation for bounding extreme roots of multivariate real-zero polynomials.
2. Identified positive-semidefiniteness and real-zero-preservation obstructions to straightforward matrix and variable extensions.
3. Proved that multivariate Eulerian embeddings yield exponentially better asymptotic root bounds than the direct univariate relaxation.
4. Refined Sobolev and Dandelin–Lobachevski–Gräffe estimates and combined them with the relaxation for a stronger bound.
5. Established new structural results for multivariate Eulerian polynomials, including mirror-reciprocity and descent-set identities.

2025GAST

ChatGPT (July 2026)

Summary

The familiar separation of ideal magnetohydrodynamic waves into incompressible Alfvén waves and compressible magnetosonic waves applies cleanly only to a uniform, unbounded plasma. This review shows how equilibrium non-uniformity couples displacement, compression, pressure perturbations, and vorticity so that a single wave can exhibit different character in different regions. Using the displacement formulation for cylindrical plasmas, it identifies total-pressure perturbation and radial displacement as the variables governing the coupled system and examines special

cases where partial decoupling survives. Resonant layers are especially important: a wave may become locally Alfvénic while retaining pressure variations that classical uniform-plasma terminology would exclude. The authors therefore argue that wave classification should be based on locally dominant physical properties rather than inherited global labels. The framework consolidates results relevant to structured solar and laboratory plasmas and clarifies why “Alfvénic” is often more precise than “Alfvén wave.”

Contributions

1. Explained why the uniform-plasma Alfvén/magnetosonic classification fails in non-uniform equilibria.
2. Organized mixed wave behavior through displacement, compression, pressure perturbation, and parallel vorticity.
3. Showed that pressure variations can accompany locally Alfvénic motion near resonant regions.
4. Identified the exceptional geometries and symmetries in which subsets of wave variables still decouple.
5. Provided a property-based classification framework for interpreting waves in structured plasmas.

2025GKST

ChatGPT (July 2026)

Summary

This experiment demonstrates a single-beam, all-optical inverse-Compton gamma-ray source driven by a petawatt laser. The laser first accelerates a GeV electron beam in a plasma wakefield; after reaching a plasma mirror, its reflected pulse collides head-on with those electrons and upshifts optical photons to multi-MeV energies. The source produced an average of $(1.2 \pm 0.6) \times 10^9$ photons above 1 MeV per shot, about one hundred times the yield of earlier all-optical sources, and reached a peak spectral brightness of $(3.9 \pm 1.5) \times 10^{22}$ photons/mm²/mrad²/s/0.1% bandwidth near 11 MeV. Measurements of the electron beam, laser evolution, and gamma spectrum support a compact model in which the drive pulse remains guided to the mirror. The model reproduces the measured gamma energy and identifies routes to higher conversion efficiency, including plasma-density tailoring and focusing plasma mirrors. Although its average flux remains below conventional facilities, the source’s ultrashort duration and high peak brightness create complementary capabilities for nuclear resonance fluorescence, transmutation, imaging, and fundamental-physics experiments.

Contributions

1. Demonstrated a petawatt-scale single-beam all-optical Compton source producing photons above 10 MeV.
2. Generated more than 10^9 photons above 1 MeV per shot, improving substantially on previous all-optical demonstrations.
3. Measured a peak spectral brightness near 11 MeV of $(3.9 \pm 1.5) \times 10^{22}$ photons/mm²/mrad²/s/0.1% bandwidth.
4. Developed a compact model that quantitatively connects laser and electron properties at the collision to gamma-ray output.
5. Identified experimentally grounded routes toward greater efficiency and higher repetition-rate nuclear-physics sources.

2025GPSH

ChatGPT (July 2026)

Summary

Gerard and coauthors propose a common language for describing flux-surface shape in both tokamaks and stellarators. The method analyzes cross-sections using Fourier shaping modes, extending familiar axisymmetric notions such as elongation, triangularity, and squareness to non-planar and non-axisymmetric surfaces. In stellarators, each shaping mode gains an additional degree of freedom: rotation about the magnetic axis. Applying the method to precisely quasi-axisymmetric and quasi-helical equilibria, and to selected QUASR database configurations, the authors find that quasi-symmetry correlates with a spatial resonance between shape complexity and axial rotation of the shaping modes. The slopes of the shaping-distribution lines track rotational transform and field-period number. The paper

frames flux-surface shaping as a quantitative diagnostic for comparing stellarator geometries and for studying how quasi-symmetry, transform, and higher-order geometric content interact.

Contributions

1. Defined Fourier-based shaping modes for general ideal-MHD flux surfaces.
2. Extended tokamak-style shape descriptors to non-axisymmetric stellarator geometries.
3. Identified axial rotation of shaping modes as the extra stellarator shape degree of freedom.
4. Linked quasi-symmetry to resonant distributions of shape complexity and rotation.
5. Connected shaping spectra to rotational transform and field-period dependence across configurations.

2025GRD

ChatGPT (July 2026)

Summary

The paper develops cell-centered finite-volume schemes with genuinely multidimensional nodal fluxes on unstructured polygonal Voronoi-like meshes. One solver extends Osher–Solomon dissipation by integrating absolute flux Jacobians over a dual simplex and returns the full multidimensional flux tensor. A second reformulates residual-distribution fluctuations as corner fluxes and instantiates them with Roe’s multidimensional N scheme. Both are complete because they use the hyperbolic system’s full eigenstructure. CWENO reconstruction and one-step ADER time discretization provide orders up to four in space and time. Compressible-Euler tests verify designed convergence and show sharper contacts and shear structures than conventional edge-normal solvers. The multidimensional Osher solver strongly suppresses carbuncle behavior without an added fix; the N-based method with its standard shock fix remains carbuncle-free and comparatively low-dissipation. Some low-dissipation tests exhibit reduced angular symmetry or boundary instabilities.

Contributions

1. Provided a corner-flux framework for complete multidimensional Riemann solvers on polygonal meshes.
2. Constructed an Osher-type solver computing an entire flux tensor through dual-simplex Jacobian integration.
3. Related finite-volume corner-flux fluctuations to residual-distribution schemes.
4. Converted the multidimensional N-scheme fluctuation into a practical finite-volume nodal flux.
5. Built high-order CWENO–ADER variants and demonstrated convergence, low dissipation, and carbuncle resistance.

2025GST

ChatGPT (July 2026)

Summary

This paper presents algorithms for computing classical Gaussian quadrature rules, including Gauss–Jacobi, Gauss–Laguerre, and Gauss–Hermite formulas. The methods combine globally convergent fourth-order iteration with asymptotic approximations selected for complementary parameter regimes. This improves speed, accuracy, and usable parameter ranges relative to existing algorithms. For Gauss–Jacobi quadrature, the MATLAB implementation removes earlier upper restrictions such as $\alpha, \beta \leq 5$ and can compute nodes and weights up to degree 10^6 for broad parameter ranges subject to underflow and overflow limits. The paper also gives a specialized efficient approach for Gauss–Legendre quadrature, adds subsampling and scaling strategies for Laguerre and Hermite rules, and treats Gauss–Radau, Gauss–Lobatto, and barycentric weights. Arbitrary-precision Maple algorithms are provided for symmetric cases, supporting high-precision computations with thousands of nodes.

Contributions

1. Developed fast algorithms for classical Gaussian quadrature nodes and weights.
2. Combined fourth-order globally convergent iteration with asymptotic starting approximations.
3. Expanded the practical parameter range for Gauss–Jacobi quadrature.
4. Added efficient Gauss–Legendre, Laguerre, Hermite, Radau, Lobatto, and barycentric-weight computations.
5. Provided arbitrary-precision algorithms for symmetric high-accuracy cases.

2025H

ChatGPT (July 2026)

Summary

Explicit two-stage fourth-order (TSFO) methods are compact but become inefficient for stiff multiscale problems because stability constrains the time step. This paper constructs an implicit TSFO temporal discretization using Taylor expansions and undetermined coefficients. A model-equation analysis with the maximum-modulus principle yields sufficient conditions for L -stability, so strongly damped modes decay as required for stiff integration. Newton iteration supplies a practical nonlinear solution procedure. Classical stiff benchmarks verify fourth-order temporal convergence using only two stages. In the reported comparisons, the method accepts larger stable time steps and produces errors roughly an order of magnitude below a classical fourth-order implicit Runge–Kutta method. The proposed time discretization is designed as a foundation for implicit Lax–Wendroff-type solvers, where temporal derivatives can incorporate transport and stiff sources together and thereby avoid operator-splitting error. That fully coupled temporal-spatial extension, including strong-discontinuity calculations, is presented as subsequent work rather than established here.

Contributions

1. Constructed an implicit two-stage method with fourth-order temporal accuracy.
2. Derived sufficient L -stability conditions from a model-equation analysis and the maximum-modulus principle.
3. Formulated a Newton iteration for solving the method’s implicit stage equations.
4. Verified fourth-order convergence and stiff stability on classical benchmark problems.
5. Established a temporal framework for unsplit Lax–Wendroff-type treatment of flow transport and stiff source terms.

2025HBCC

ChatGPT (July 2026)

Summary

This paper introduces Entity, an open-source particle-in-cell code for plasma astrophysics, and focuses on its curvilinear special-relativistic module. The code targets performance portability across GPU and CPU architectures through Kokkos while preserving an interface intended to be usable and customizable without modifying core internals. The module solves the Vlasov–Maxwell system in general coordinates, enabling coordinate-agnostic simulations in Cartesian, axisymmetric spherical, quasi-spherical, and future geometries. A central numerical choice is to advance particle equations of motion in a global orthonormal Cartesian basis even when fields and grids use generalized coordinates, improving stability. Charge conservation is maintained through conformal current deposition. The implementation is validated on standard Cartesian plasma tests and relativistic magnetosphere simulations in curvilinear axisymmetric geometries. Performance results show portability across AMD, NVIDIA, and Intel GPU platforms, with efficient particle push and current deposition kernels.

Contributions

1. Presented Entity as a hardware-agnostic open-source PIC code for plasma astrophysics.

2. Implemented a general-coordinate special-relativistic Vlasov–Maxwell module.
3. Advanced particles in a global orthonormal basis to improve curvilinear-grid stability.
4. Preserved charge conservation with conformal current deposition.
5. Validated accuracy and performance portability across standard tests and multiple GPU platforms.

2025HBMG

ChatGPT (July 2026)

Summary

NCCL’s GPU-Initiated Networking (GIN) extends device-side communication from intra-node memory access to inter-node RDMA, allowing CUDA kernels to initiate fine-grained transfers without host orchestration. The design has three layers: host APIs create device communicators and register collective memory windows; device APIs expose remote memory operations; and network plugins provide transport-specific execution. A GPUDirect Async Kernel-Initiated backend uses DOCA GPUNetIO for direct GPU-to-NIC commands, while a proxy backend uses lock-free GPU-to-CPU queues to support conventional RDMA hardware with equivalent semantics. GIN shares the NCCL Device API with load/store-accessible and multimemory modes, preserving access to NCCL’s topology, collective algorithms, and deployment infrastructure. Integration with the DeepEP mixture-of-experts communication library demonstrates practical device-driven dispatch and combine operations. Benchmarks characterize latency and show where direct initiation reduces software overhead, while the proxy path trades some latency for portability.

Contributions

1. Added inter-node GPU-initiated RDMA to NCCL’s device-side communication model.
2. Defined host setup, device operations, and transport plugins as a three-layer architecture.
3. Implemented direct GPU-to-NIC execution through DOCA GPUNetIO.
4. Supplied compatible proxy semantics for standard RDMA networks through lock-free queues.
5. Demonstrated GIN in DeepEP for fine-grained mixture-of-experts communication.

2025HBS

ChatGPT (July 2026)

Summary

Resolving particle gyromotion inside an MHD domain creates a severe scale separation. The MHD-gPIC module in Athena++ instead advances energetic particles with guiding-center equations while representing the thermal component as magnetohydrodynamic fluid. Carefully derived momentum and energy source terms couple particle feedback to the fluid and permit a non-negligible nonthermal fraction. A correction to the guiding-center momentum equation restores Galilean invariance, which is important for particle confinement in moving plasmoids, while including perpendicular particle pressure in the fluid pressure lets the Riemann solver suppress gradient noise. The second-order implementation supports static and adaptive mesh refinement, parallel load balancing, and adds about 10–18% per-particle overhead relative to MHD-PIC. Benchmarks validate trajectories, feedback, and wave behavior. Preliminary reconnection simulations find mainly Fermi acceleration, pressure anisotropy, and strong feedback from energetic ions. The guiding-center approximation excludes gyro-resonant and sub-grid scattering effects, defining the method’s intended regime.

Contributions

1. Formulated and implemented an MHD–guiding-center-PIC method in Athena++ for multiscale plasma simulations.
2. Added a Galilean-invariant guiding-center momentum correction that improves confinement in moving structures.
3. Derived backreaction terms valid when energetic particles form a non-negligible plasma fraction.
4. Reduced particle-pressure noise by incorporating perpendicular particle pressure into the fluid Riemann solve.
5. Validated the coupled implementation and demonstrated particle acceleration with feedback in magnetic reconnection.

2025HGR

ChatGPT (July 2026)

Summary

The paper derives a hot-electron-limit closure for the Hermite–Laguerre gyromoment hierarchy by expanding gyroaveraging kernels in $\tau = T_i/T_e \ll 1$ and retaining the minimal $O(\tau)$ terms needed for consistent coupling. The resulting four-field system evolves density, parallel velocity, and parallel and perpendicular temperatures. In Z-pinch geometry it is analytically equivalent to the Ivanov fluid model, after an empirical factor aligns the Dougherty and Landau collision operators. Gyacomo benchmarks show that four gyromoments close in the asymptotic regime and reproduce published linear growth rates, nonlinear heat flux, low-collisionality dynamics, and parallel-domain transport plateaus. Extension to tokamak s – α geometry qualitatively preserves heat-flux dynamics even near $\tau = 1$, but overpredicts growth and substantially weakens or removes the Dimits shift. Higher heat-flux and pressure-tensor moments are therefore essential for zonal-flow amplification and near-threshold transport suppression.

Contributions

1. Derived a controlled hot-electron asymptotic closure directly from the gyromoment hierarchy.
2. Proved equivalence of the reduced Z-pinch equations to the Ivanov model.
3. Established that four gyromoments form a closed set for $\tau \ll 1$.
4. Reproduced linear, nonlinear, collisionality, and parallel-domain Z-pinch benchmarks.
5. Demonstrated that omitted higher moments are crucial for the Dimits shift and zonal-flow regulation.

2025HJP

ChatGPT (July 2026)

Summary

This paper presents a magnetohydrodynamics algorithm for staggered, logically Cartesian meshes that combines constrained transport, cell-based adaptive mesh refinement, and temporal substepping. The method targets simulations where magnetic-field divergence control, local spatial refinement, and local time-step variation must coexist without fragile synchronization between refinement levels. Rather than interpolating or projecting states between nested or neighboring grids, the algorithm handles resolution changes directly on the grid and keeps the constrained-transport update compatible with $\nabla \cdot B = 0$ preservation. This is important for space, astrophysical, and laboratory plasma calculations where localized structures require refinement but global time stepping would be inefficient. The paper describes the update logic for refined cells and staggered electromagnetic quantities, then tests the method on multidimensional benchmark problems. The reported examples show that temporal substepping can be coupled to cell-based refinement while maintaining stable MHD evolution and divergence control. The result is a practical route toward AMR MHD simulations that avoid level-wise interpolation/projection machinery and retain the geometric advantages of staggered constrained transport.

Contributions

1. Developed a staggered-grid MHD algorithm that combines constrained transport with cell-based adaptive mesh refinement.
2. Added temporal substepping so refined cells can advance with local time steps rather than a single global step size.
3. Formulated refinement-boundary updates directly on logically Cartesian grids without interpolation or projection between refinement levels.
4. Preserved the divergence-free magnetic-field constraint within the combined AMR and substepping update.
5. Demonstrated the approach on numerical MHD tests relevant to space, astrophysical, computational, and plasma-physics applications.

2025HL

ChatGPT (July 2026)

Summary

The paper generalizes alternating Anderson acceleration into a periodic scheme, $\text{aAA}(m)[s]\text{--FP}[t]$, that performs t fixed-point steps followed by s Anderson steps with window size m . For linear Richardson iteration, it relates the full-history, single-Anderson-step variant to GMRES, proves convergence when the fixed-point map is contractive, and gives a sufficient condition for diagonalizable noncontractive iteration matrices. Experiments cover ill-conditioned nonsymmetric linear systems, Navier–Stokes Picard solves, regularized logistic-regression gradient descent, and ADMM for nonlinear nonsmooth optimization. Suitable m, s, t choices reduce iterations and runtime relative to unaccelerated iterations, windowed Anderson acceleration, and the earlier $s = 1$ scheme; ADMM and gradient-descent cases show roughly order-of-magnitude acceleration. Performance remains parameter-dependent, and large Anderson windows can make the internal least-squares problem rank deficient.

Contributions

1. Defined the flexible periodic $\text{aAA}(m)[s]\text{--FP}[t]$ family.
2. Established a periodic connection between full-history alternating Anderson acceleration and GMRES.
3. Proved convergence for contractive linear fixed-point maps.
4. Gave a sufficient convergence condition for diagonalizable noncontractive Richardson matrices.
5. Demonstrated acceleration across linear solvers, Navier–Stokes, gradient descent, and ADMM while identifying limitations.

2025HLS

ChatGPT (July 2026)

Summary

Hauck, Laiu, and Schnake combine high-order/low-order acceleration with a micro-macro decomposition for fully implicit time stepping of the BGK kinetic equation. The micro-macro representation splits the distribution into a local Maxwellian and a perturbation, allowing the nonequilibrium component to be compressed when the flow is close to equilibrium. Both HOLO and micro-macro HOLO are implemented with discontinuous Galerkin discretizations in phase space so that the kinetic and moment systems remain consistent. A linearized BGK analysis compares source iteration, HOLO, and micro-macro variants, while numerical tests on Sod shock tube and sudden wall heating problems assess accuracy, convergence, and compression. The results show that micro-macro HOLO preserves the acceleration properties of HOLO and can reduce storage near equilibrium, though compression gains are limited when boundary layers dominate nonequilibrium behavior. Future work targets stronger fluid solvers, hybrid iteration strategies, higher-dimensional problems, and more accurate collision models.

Contributions

1. Combined micro-macro decomposition with HOLO acceleration for fully implicit BGK time stepping.
2. Implemented the coupled kinetic-moment methods with phase-space discontinuous Galerkin discretizations.
3. Analyzed convergence behavior of source iteration, HOLO, and micro-macro HOLO on linear BGK models.
4. Compared accuracy and performance on shock-tube and wall-heating benchmark problems.
5. Quantified when micro-macro compression helps and when boundary-layer nonequilibrium limits storage savings.

2025HMST

ChatGPT (July 2026)

Summary

This paper develops a finite-element scheme for the electrostatic Euler-Poisson equations with a prescribed constant magnetic field. The method targets the magnetic-drift regime, where smooth hydrodynamic motion coexists with plasma and cyclotron oscillations separated by more than ten orders of magnitude in time scale. An operator split treats the compressible Euler subsystem explicitly with a conservative, invariant-domain-preserving graph-viscosity method, while the stiff source subsystem is advanced implicitly. A PDE Schur complement eliminates the coupled momentum and Lorentz-force variables, reducing each source update to one nonsymmetric Poisson-like solve. At the fully discrete level, the authors prove positivity of density and internal energy, a minimum principle for specific entropy, and a total-energy balance. Numerical tests show that the scheme can operate near the magnetic-drift limit without resolving the fastest oscillations. A diocotron-instability calculation reproduces analytical growth rates and illustrates the approach on a demanding multiscale plasma problem. The construction is presented as a step toward similarly robust Euler-Maxwell solvers with self-consistent electric and magnetic fields.

Contributions

1. Constructed a fully discrete finite-element scheme for the magnetic Euler-Poisson system.
2. Proved positivity, an entropy minimum principle, conservation properties, and a total-energy balance.
3. Split explicit invariant-domain Euler dynamics from an implicit treatment of stiff plasma and cyclotron motion.
4. Reduced the coupled source update to a single nonsymmetric Poisson-like problem through a PDE Schur complement.
5. Validated magnetic-drift behavior with a diocotron instability whose growth rates agree with analytical predictions.

2025HS

ChatGPT (July 2026)

Summary

This paper improves error estimates for Gauss-Christoffel quadrature when the integrand has limited endpoint regularity. Classical Chebyshev-coefficient and quadrature bounds require a weighted bounded-variation condition involving $(1 - x^2)^{-1/2}$, which can fail for functions with endpoint singularities. The authors derive a recurrence relation for derivatives of Chebyshev polynomials that permits a weaker assumption: absolute integrability of $f^{(r+1)}$ without the singular weight. This yields new Chebyshev coefficient decay estimates and a Gauss-Christoffel quadrature error bound depending on $U_r = \int_{-1}^1 |f^{(r+1)}(x)| dx$ rather than the classical weighted quantity V_r . The estimates are extended to Gauss-Gegenbauer rules, including Gauss-Legendre and Gauss-Chebyshev cases. Numerical tests show that the new bounds remain finite and useful for endpoint-singular functions where older bounds become extremely large or infinite, and can also be smaller for smooth examples.

Contributions

1. Proved a new recurrence identity for higher derivatives of Chebyshev polynomials.
2. Replaced weighted endpoint-regularity assumptions with an unweighted L^1 derivative condition.
3. Derived improved Chebyshev coefficient decay bounds under reduced regularity.
4. Established corresponding Gauss-Christoffel and Gauss-Gegenbauer quadrature error estimates.
5. Validated the estimates on endpoint-singular and smooth numerical examples.

2025HSBJ

ChatGPT (July 2026)

Summary

This paper reverse-engineers NVIDIA Collective Communication Library (NCCL) 2.19.1 to explain how GPU collectives are executed across devices and nodes. It traces communicator setup, channel partitioning, CUDA block and warp roles, buffer slots, and intra- and inter-node transports, including GPUDirect P2P, shared memory, sockets, InfiniBand verbs, and GPUDirect RDMA. The analysis contrasts Simple, LL, and LL128 protocols: Simple favors large transfers, LL minimizes small-message latency, and LL128 combines flag-based synchronization with 120-byte payloads but requires suitable atomic-write semantics. Ring and tree collectives are decomposed into NCCL communication primitives and classified by whether loop iterations can pipeline. Benchmarks on 16 Grace Hopper nodes show that LL/LL128 and trees favor small messages, while Simple and rings dominate large inter-node transfers; LL128 remains strong within a node over NVLink. These implementation findings underpin ATLAHS, whose NCCL-aware schedules model large-scale training communication with reported simulation errors below 5%.

Contributions

1. Documented NCCL’s channel, buffer, CUDA-thread, and protocol execution hierarchy from its implementation.
2. Explained the synchronization, payload, bandwidth, and hardware trade-offs of Simple, LL, and LL128.
3. Mapped intra- and inter-node paths through P2P, shared-memory, socket, InfiniBand, and GPUDirect RDMA transports.
4. Decomposed ring and tree collectives into communication primitives and pipelined or non-pipelined iteration patterns.
5. Connected the analysis to ATLAHS and validated protocol and algorithm regimes on a 16-node GH200 system.

2025HYSB

ChatGPT (July 2026)

Summary

Meshfree collocation methods approximate differential operators directly on irregular point clouds, but overlapping terminology and independently developed derivations obscure how the methods relate. This overview traces their development from scattered-data interpolation and generalized finite differences through moving least squares, reproducing kernels, radial basis functions, and related particle methods. It formulates their strong-form discrete operators in common vector and matrix notation, then classifies reviewed methods by whether they arise from moment approximation, minimization of approximation error, or minimization of stencil weights. The framework reveals cases that are mathematically identical despite different names and isolates consequential distinctions, including treatment of the zeroth moment, symmetry of the moment matrix, allowable basis functions, and added penalty terms. Those choices can affect computational cost, explicit-time stability, wavenumber dissipation, boundary treatment, and noise robustness. The authors also derive a generalized ℓ^2 weight-minimization principle that permits arbitrary anisotropic basis functions and asymmetric moment matrices. It recovers the same discrete operator as moment cancellation, creating a formal bridge between two derivational families and potentially allowing theory to transfer between them. A tabulated classification and supplementary encyclopedia consolidate the surveyed methods, while application-specific method selection and detailed numerical comparisons remain outside the review’s scope.

Contributions

1. Reconstructed the historical development of meshfree collocation across communities whose differing terminology led to parallel derivations and rediscoveries.
2. Expressed the reviewed methods within one strong-form operator framework using consistent vector and matrix notation.

3. Classified methods by their derivational principle and supplied a consolidated table and supplementary encyclopedia of their operators, bases, and moment treatments.
4. Identified exact equivalences and technically important differences involving zeroth moments, matrix symmetry, basis restrictions, and penalty terms.
5. Introduced generalized ℓ^2 weight minimization and proved that it reproduces the moment-approximation operator while allowing arbitrary anisotropic basis functions.

2025IMCR

ChatGPT (July 2026)

Summary

Two-dimensional particle-in-cell simulations investigate relativistic turbulence and reconnection in plasmas containing electrons, positrons, and protons at a realistic proton-to-electron mass ratio. Varying the proton fraction changes the hierarchy of magnetic islands, current sheets, and plasmoids generated by the turbulent cascade. Proton-rich mixtures suppress the smallest-scale structures but strengthen global dissipation, showing that composition controls both the number of active reconnection sites and their energy-conversion efficiency. To diagnose the mechanism, the authors derive a relativistic multispecies generalization of Ohm's law. Its term balance indicates that divergence of the electron and positron pressure tensors supplies the principal nonideal electric field governing the reconnection rate. The work extends kinetic reconnection studies beyond pair or electron-ion plasmas and connects composition-dependent microphysics to possible particle acceleration and high-energy emission near black holes and neutron stars.

Contributions

1. Simulated relativistic turbulence in a three-species electron-positron-proton plasma.
2. Used a realistic mass ratio while varying composition systematically.
3. Showed that increasing proton fraction removes fine structures but amplifies global dissipation.
4. Derived a relativistic multispecies Ohm's law for reconnection diagnostics.
5. Identified electron and positron pressure-tensor divergence as the main reconnection driver.

2025ISSS

ChatGPT (July 2026)

Summary

This paper argues that Farley-Buneman turbulence in the auroral E region self-organizes near marginal stability rather than behaving only as locally forced turbulence. ICEBEAR radar, CHAIN GPS, Swarm, Arase, and SuperMAG observations connect small-scale electrojet irregularities to magnetospheric forcing. Composite radar-GPS spectra repeatedly peak near a kinetic-Alfvén-like $k^{-8/3}$ slope across roughly four wavenumber decades, while radar phase speeds saturate near the ion-acoustic speed. Above a critical driving threshold, radar echo density scales nearly linearly with magnetospheric wave power, with fitted exponent 0.98 ± 0.09 . The authors formulate a Martin-Siggia-Rose effective field theory in which inertial dynamics become irrelevant under renormalization, leaving an overdamped advective-diffusive fixed point. Saturation electric fields supply anomalous drag, producing a Bohm-limit diffusion coefficient and an Adler-type noise-broadened transition. The resulting closed transport relations offer a proposed parameter-free subgrid closure for anomalous ionospheric currents and heating, although the causal interpretation of matching spectra and the proposed critical-state picture need further substantiation.

Contributions

1. Combined ICEBEAR, CHAIN, Swarm, Arase, and geomagnetic-index observations to relate auroral E-region turbulence to particle precipitation and magnetospheric driving.
2. Resolved a composite turbulence cascade centered near $k^{-8/3}$, with recurring kilometer- and decameter-scale breaks and ion-acoustic phase-speed saturation.

3. Measured a thresholded response in which turbulent-wave density is flat below critical driving and scales as $\Delta E^{0.98 \pm 0.09}$ above it.
4. Derived a Martin–Siggia–Rose renormalized effective theory whose macroscopic limit is an overdamped $z = 2$ advective–diffusive fixed point.
5. Obtained anomalous-resistivity, Bohm-diffusion, and Adler–Ohmic relations proposed as subgrid closures for ionospheric currents and heating.

2025JHJ

ChatGPT (July 2026)

Summary

Relativistic continuum-kinetic simulations require a discrete Maxwell–Jüttner equilibrium whose density, bulk velocity, and temperature match prescribed moments. Direct projection onto a finite momentum grid loses distribution tails and perturbs these moments, which can violate conservation in collision operators. The authors derive the required rest-frame moments from the particle four-flux and stress-energy tensor, then develop robust discontinuous Galerkin operations for Lorentz factors and other nonlinear quantities. Computing with spatial four-velocity avoids superluminal intermediate states, while quadrature and weak multiplication or division preserve the modal representation. An iterative Picard correction adjusts the equilibrium’s input moments until its discrete moments agree with the targets; density rescaling also avoids fragile evaluation of modified Bessel normalizations. On adequately resolved grids, convergence typically takes 3–20 iterations. A test that otherwise needs about 10,000 momentum cells to reach machine precision uses only 32 corrected cells. In a relativistic BGK relaxation test, density, velocity, and temperature remain conserved near machine precision after ten collision times. The method is applicable beyond DG to other grid-based relativistic kinetic discretizations, but cannot recover distributions whose requested moments are unrealizable on the finite grid.

Contributions

1. Developed an iterative projection that matches discrete Maxwell–Jüttner density, velocity, and temperature to machine precision.
2. Derived rest-frame equilibrium moments from lab-frame relativistic flux and stress-energy quantities.
3. Constructed robust DG operations for Lorentz factors using four-velocity and physically bounded fallback projections.
4. Quantified finite-grid temperature and momentum limits that determine whether the correction can converge.
5. Demonstrated near-machine-precision conservation in a relativistic BGK relaxation using a practical 32-cell grid.

2025JM

ChatGPT (September 2026)

Summary

A closed string is a single character or has a nonempty border—a matching proper prefix and suffix—with no internal occurrence. This paper represents all closed substring occurrences of a length- n string using $O(n \log n)$ intervals of admissible lengths, avoiding explicit enumeration of a potentially quadratic output. A maximal right-closed array records substrings that cease to be closed when extended one position to the right, together with their border lengths. Two algorithms construct this array in $O(n \log n)$ time with $O(n)$ auxiliary space: one merges ordered suffix-position sets while scanning suffix and longest-common-prefix arrays; the other modifies Crochemore’s equivalence-class refinement. The stored output may itself require $O(n \log n)$ space. A further scan extracts substrings maximal under both left and right extension. An exact parity-dependent count of maximal closed substring occurrences in Fibonacci words approaches 1.382 times the word length. Experiments favor equivalence-class refinement for Fibonacci and Tribonacci words, and the suffix-array implementation for the other tested sequences. These timings compare the two proposed implementations; no experimental comparison with the earlier suffix-tree algorithm is reported.

Contributions

1. Introduced an interval representation grouping closed occurrences by starting position and consecutive allowable lengths, with an $O(n \log n)$ storage bound.
2. Constructed the maximal right-closed array using longest-common-prefix transitions and ordered-set union operations, proving $O(n \log n)$ time and linear auxiliary space.
3. Adapted Crochemore’s refinement algorithm to obtain the same array directly from the string, using level timestamps to avoid quadratic auxiliary storage.
4. Derived the exact Fibonacci-word count $M(f_k) = F_k + F_{k-2} - 1$ for odd k and $F_k + F_{k-2} - 2$ for even k , for $k \geq 5$ and $F_k = |f_k|$, by classifying singleton occurrences, runs, and maximal gapped repeats.
5. Supplied implementations and experiments separating the algorithms’ behavior on repetitive words, Thue–Morse words, digits of π , biological sequences, and random strings.

2025JMS

ChatGPT (July 2026)

Summary

Block-tridiagonal symmetric positive-definite systems appear repeatedly in state estimation and optimal control, where solution latency can limit real-time operation. The BlockDSS implementation applies recursive Schur-complement reduction, eliminating alternating block rows and producing a hierarchy of smaller independent systems. At each level, factorization and triangular-solve work is expressed with batched BLAS and LAPACK kernels, exposing GPU parallelism while preserving the known block structure. A common implementation targets both NVIDIA and AMD devices. Benchmarks show substantial acceleration over CPU sparse-direct solvers CHOLMOD and HSL MA57 and performance competitive with NVIDIA cuDSS for favorable block sizes. The study also identifies two practical limits: recursion levels still launch batched operations sequentially, and small blocks do not amortize kernel-launch overhead. The method is therefore most effective when the structured problem supplies enough independent, sufficiently large dense blocks at each reduction level.

Contributions

1. Specialized recursive Schur-complement reduction for block-tridiagonal SPD systems on GPUs.
2. Mapped independent factorization and solve tasks to batched BLAS/LAPACK kernels.
3. Delivered a cross-platform implementation for NVIDIA and AMD accelerators.
4. Benchmarked against CHOLMOD, HSL MA57, and NVIDIA cuDSS.
5. Identified sequential recursion levels and kernel-launch amortization as the main performance limits.

2025K

ChatGPT (September 2026)

Summary

Kwon gives a way to turn online regret-minimizing algorithms into iterations for finding fixed points of nonexpansive maps. The central inequality bounds squared fixed-point residuals by terms that online algorithms control through regret, so a regret guarantee becomes a residual guarantee. Converting online gradient descent recovers Krasnoselskii–Mann iteration, including a projected form for maps whose outputs need not lie in the feasible set. Converting AdaGrad-Norm yields best-iterate residual bounds adaptive to an unknown scalar scaling of the map; diagonal and full-matrix AdaGrad variants adapt to unknown preconditioners. The matrix guarantees require bounded iterates for convergence, and the adaptive bounds generally concern the best iterate rather than the last. Illustrative experiments on Markov-chain stationary distributions, total-variation image denoising, and zero-sum games compare the methods with problem-specific baselines and show faster or more robust convergence in some settings. The comparisons concern selected instances and iteration counts, not runtime or broad benchmark averages.

Contributions

1. Converted regret bounds into fixed-point residual bounds using the equivalence between nonexpansive maps and cocoercive residual operators.
2. Recovered Krasnoselskii–Mann iteration from online gradient descent and derived projected iterations for non-self maps.
3. Proved AdaGrad-Norm best-iterate bounds that adapt to an unknown favorable scalar scaling.
4. Extended the analysis to diagonal and full-matrix AdaGrad updates that adapt to unknown positive-definite preconditioners.
5. Tested the resulting iterations on Markov chains, image denoising, and zero-sum games, documenting both gains and limitations against the chosen baselines.

2025KM

Claude (October 2026)

Summary

This article reports extensions to Irksome, the Firedrake library that converts Unified Form Language descriptions of semidiscrete finite element problems into stage-coupled Runge–Kutta (RK) systems solved with PETSc. Strong Dirichlet conditions are now imposed as algebraic equations on the stage approximations, $u^n + \Delta t \sum_j a_{ij} k_j = g(t_n + c_i \Delta t)$ on Γ , solved for the stage boundary values when the Butcher matrix A is invertible; this differential-algebraic (DAE) viewpoint retains the stage order and handles incompatible initial and boundary data, which the earlier rule $k_i = g'(t_n + c_i \Delta t)$ misses. Butcher’s change of variables $w = Ak$, stage-value formulations that skip the mass solve for stiffly accurate methods, and a DIRK path that builds a single-stage solver are selected by TimeStepper keywords. Preconditioners for the stage-coupled Jacobian are generated by re-deriving the coupled form with an approximate Butcher matrix inside firedrake.AuxiliaryOperatorPC, giving block Jacobi, block Gauss–Seidel, and the Rana LD preconditioner, beside stage-coupled monolithic multigrid. Heat-equation, Navier–Stokes cylinder, and Cahn–Hilliard experiments show RadauIIA methods with stage-independent iteration counts, runtimes comparable to DIRKs of equal stage count, and better accuracy per runtime than Alexander’s DIRK and weak-stage-order DIRKs, although stiffness keeps most schemes below their formal order.

Contributions

1. Derived the DAE-type boundary conditions (2.16)–(2.17), $\sum_j a_{ij} k_j = (g(t_n + c_i \Delta t, \cdot) - u^n)/\Delta t$ on Γ , noted that u^{n+1} matches $g(t_{n+1})$ up to projection error for stiffly accurate schemes since $u^{n+1} = u_s$, and showed on the 1D heat equation with $u(0, x) = 0$ and boundary values 1 (ten linear elements, three-stage LobattoIIIC, $\Delta t = 0.05$ to $t = 0.5$) that the ODE-type rule, seeing zero boundary time derivative, returns a steady zero solution while the DAE-type rule captures convergence to $u = 1$ (Fig. 2); the keyword `bc_method` defaults to ‘DAE’.
2. Added the formulation $\sum_j (A^{-1})_{ij} w_j - F(t_n + c_i \Delta t, y^n + \Delta t w_i) = 0$ for $w = Ak$, whose nonlinear Jacobian contribution occupies s rather than s^2 blocks, selected by `splitting = IA` or `AI` (the default); the stage-value form (2.20)–(2.21), `stage_type = ‘value’`, with automatic skipping of the mass matrix update for stiffly accurate tableaux; and `stage_type = ‘dirk’`, which creates one single-stage variational problem updated between stages, with Alexander’s L-stable three-stage DIRK and the weak-stage-order tableaux WSODIRK433 and WSODIRK744 included.
3. Implemented block preconditioners $P_D = D^{-1} \otimes M + \Delta t I \otimes K$, $P_L = (L + D)^{-1} \otimes M + \Delta t I \otimes K$, and $\tilde{P}_{LD} = \tilde{A}_{LD}^{-1} \otimes M + \Delta t I \otimes K$ with $A = LDU$ by overriding the `form` method of firedrake.AuxiliaryOperatorPC to repeat the Irksome transformation with $\tilde{A}$, so the construction applies to nonlinear problems and exposes the block inverses to further PETSc options; monolithic multigrid uses additive Schwarz on patches $V_i \times \dots \times V_i$ via `PCPatch` or `ASMPatchPC`, with the theory of Kirby [2024] that its iteration matrix’s spectral radius does not depend explicitly on the stage count; the Southworth and Leveque preconditioners are not yet supported.
4. For the heat equation with $u = e^{-0.1t} \sin(\pi x) \cos(\pi y)$, serendipity elements of degree 2 and 3, and $\Delta t = M/N$, $M = 1, 4, 8$, found suboptimal L^2 accuracy for RadauIIA(1) at all M and for Alexander’s DIRK at $M = 8$, an L^2 loss for RadauIIA(2) with cubic elements at large steps, and optimal accuracy for RadauIIA($s \geq 3$) and

WSODIRK433 (Figs. 3, 4); with FGMRES at tolerance 10^{-8} and one BoomerAMG sweep per block, P_D and P_L iterations grew with stages to about 47 and 22 at four stages while $\tilde{P}_{LD}$ stayed near 10 and DIRK near 8, with comparable runtimes for the latter two (Fig. 5).

5. For flow past a cylinder with Mardal–Tai–Winther velocities, piecewise constant pressures, a three-level netgen hierarchy of 12,800 cells, Newton–Krylov with Eisenstat–Walker forcing, and V(3,3) cycles with elementwise Vanka patches, found the Rana preconditioner faster for RadauIIA(2) and (3) but monolithic multigrid faster for RadauIIA(4) (Fig. 8); over $[0, 8]$ with Δt from 0.16 to 0.0025, iteration counts were stable in the stage count, DIRKs ran about as fast as RadauIIA with equal stages, and RadauIIA(3) and (4) gave the best final drag and lift error per runtime, with WSODIRK744 accurate but costliest (Figs. 9–11); for Cahn–Hilliard with Bell elements, $f(c) = 50(c^2 - 1)^2$, $\lambda = 10^{-2}$, Nitsche parameter 250, and Δt from 5×10^{-6} to 3.125×10^{-7} , about four Newton iterations per step sufficed, RadauIIA(3) and (4) had the lowest final-energy error, RadauIIA(2) beat Alexander and WSODIRK433 at lower cost, and implicit Euler stayed qualitatively different even at $\Delta t = 6.25 \times 10^{-7}$ (Figs. 13–15).

Comments

- Section 2, Eq. (2.8): the update is written $u^{n+1} = u^n + \sum_i b_i k_i$ without the factor Δt present in (2.3) and in the two-stage LobattoIIIC example $u^{n+1} = u^n + \frac{\Delta t}{2}(k_1 + k_2)$ after (2.10); the factor Δt is evidently intended.
- Sections 2 and 4.2: the semidiscrete Navier–Stokes system (2.12) is called a “degree-2 DAE” in Section 2 and “an index 2 DAE” in Section 4.2; the index is evidently intended.
- Section 3.1: “the linear case discussed in Southworth et al. [2022a] could be easily implemented in Irksome, the nonlinear case in Southworth et al. [2022a] is much more involved”, but in the reference list [2022a] is Part II, Nonlinearities and DAEs, and [2022b] is Part I, The Linear Setting; [2022b] is evidently intended for the linear case.
- Section 4.1 and Fig. 5: the solver comparison is said to run “on a fixed 256×256 mesh, integrating over a time interval $[0, 2]$ ”, whereas the caption of Fig. 5 states that the heat equation was integrated “using cubic serendipity elements on a 64×64 mesh”; unresolved.

2025KMZ

ChatGPT (July 2026)

Summary

The proximal Galerkin method combines a finite-element discretization with a proximal-point algorithm for variational problems subject to pointwise inequality constraints. This paper gives its first abstract a priori error analysis for quadratic optimization problems in Sobolev spaces. The framework treats discretization and iteration errors together rather than analyzing only linearized subproblems. It establishes checkable conditions for existence and uniqueness of the proximal subproblems, monotonic decrease of the objective, best-approximation properties, convergence rates, and asymptotic mesh-independent iteration complexity. The analysis also preserves feasibility at the discrete level and does not require a penalty parameter to approach a singular limit. The authors apply the theory to obstacle and Signorini contact problems using several conforming finite-element pairs and construct the stability and approximation operators needed by the abstract results. For the lowest-order conforming approximations covered by the proof, the resulting estimates attain optimal convergence rates. The work supplies a general mathematical foundation for the method while identifying high-order spaces and nonquadratic energies as important extensions beyond the present theory.

Contributions

1. Presented the first abstract a priori error framework for the proximal Galerkin method.
2. Unified iteration and spatial-discretization errors for pointwise-constrained quadratic variational problems.
3. Proved objective decrease, best-approximation properties, convergence rates, and asymptotic mesh independence.
4. Derived optimal error estimates for lowest-order obstacle and Signorini finite-element discretizations.
5. Established convergence without singular penalty limits while retaining constraint-preserving approximations.

2025KPPT

ChatGPT (July 2026)

Summary

Krylov complexity measures how far a quantum state spreads in the basis generated by its dynamics. This work applies it to two quantum kicked-rotor models and four localization mechanisms: quantum antiresonance, localization induced by regular classical phase-space structures, dynamical localization from quantum interference, and power-law localization from a singular kicking potential. Arnoldi iteration constructs the Floquet Krylov basis, where probability profiles and inverse participation ratios reflect localization strength. Each mechanism produces a distinct pattern of complexity growth, oscillation, or saturation. Long-time averaged complexity and fluctuations of the Arnoldi coefficients further separate classical-induced localization from quantum localization: the former lacks clear scaling with effective Planck's constant, whereas dynamical localization shows systematic power-law scaling. Arnoldi-coefficient variance also detects the transition from integrability to classical chaos even when quantum dynamics remains localized. Conventional level-spacing statistics identify localization but do not distinguish its type. The authors additionally show that the conclusions depend strongly on the initial state; random delocalized states can erase differences among localization mechanisms, limiting Krylov complexity as a standalone chaos diagnostic.

Contributions

1. Characterized four distinct localization mechanisms of quantum kicked rotors in a Floquet Krylov basis.
2. Connected Krylov-space probability profiles and participation ratios to the degree of localization.
3. Identified mechanism-specific growth, oscillation, and saturation signatures in Krylov complexity.
4. Distinguished classical-induced and interference-driven localization through effective-Planck-constant scaling of complexity and Arnoldi fluctuations.
5. Showed that Arnoldi-coefficient variance detects chaos onset while exposing the strong initial-state dependence of Krylov diagnostics.

2025L

ChatGPT (July 2026)

Summary

The paper develops structure-preserving particle-in-cell methods for hybrid plasma models in which ions remain fully kinetic, electrons obey a Boltzmann relation, and a Poisson equation retains space-charge effects. Discretizing either the action integral or the noncanonical Poisson bracket yields equivalent finite-dimensional Hamiltonian systems. Hamiltonian splitting then gives symplectic time integrators, while discrete-gradient methods preserve the discrete energy exactly. Under periodic or homogeneous Neumann boundary conditions, the schemes also retain the model's global neutrality condition. The formulation is extended to an electromagnetic hybrid model with a slowly varying laser envelope. Numerical experiments covering finite-grid instability, Landau damping, and resonantly driven nonlinear ion waves demonstrate the methods' conservation behavior and dynamics; in a reported nonlinear-wave test, the discrete-gradient method maintains energy to near solver tolerance. The analysis focuses mainly on one-dimensional electrostatic examples, while higher-dimensional discretizations, other boundary conditions, and broader electromagnetic applications are left for future work.

Contributions

1. Derived variational and Hamiltonian formulations for a kinetic-ion, Boltzmann-electron plasma model with space charge.
2. Constructed equivalent finite-dimensional Hamiltonian systems by discretizing the action and the Poisson bracket.
3. Developed symplectic Hamiltonian-splitting and energy-preserving discrete-gradient particle-in-cell integrators.
4. Proved preservation of global neutrality under suitable boundary conditions.
5. Extended the geometric formulation to an electromagnetic hybrid model and tested it on three plasma benchmarks.

2025L2

ChatGPT (July 2026)

Summary

This proceedings paper summarizes theoretical modeling of high-energy particle acceleration in magnetized turbulent plasmas. It reviews recent findings from fully kinetic simulations of large-amplitude, semi-relativistic to fully relativistic turbulence and interprets them through a generalized Fermi picture of stochastic acceleration. In this view, acceleration is not spatially uniform: the strongest contributions arise when particles interact with sharp, time-dependent bends of magnetic field lines and with regions of velocity compression. The acceleration-rate distribution is highly intermittent, following a broken power law that extends to large values, which makes relativistic large-amplitude turbulence an efficient extreme accelerator. The paper also discusses consequences for particle transport and for energy spectra over long times, especially when radiative losses are present. Its main role is synthetic rather than algorithmic: it organizes simulation evidence into a physical model that connects turbulent electromagnetic structures, stochastic acceleration rates, transport, and observable high-energy particle spectra in astrophysical plasmas.

Contributions

1. Reviewed fully kinetic simulation evidence for particle acceleration in large-amplitude relativistic turbulence.
2. Framed the results through a generalized Fermi model of stochastic acceleration.
3. Identified dynamic magnetic-field bends and velocity-compression regions as dominant acceleration sites.
4. Emphasized the spatial intermittency and broken-power-law distribution of acceleration rates.
5. Discussed implications for particle transport, radiative losses, and long-time energy spectra in astrophysical plasmas.

2025LCDG

ChatGPT (September 2026)

Summary

This review develops the physics and experimental status of beam-driven plasma-wakefield acceleration, in which a relativistic charged-particle bunch excites a plasma wave that accelerates a trailing bunch at gradients reaching roughly 100 GV/m. It treats linear wakes, nonlinear blowout and suck-in regimes, transformer-ratio enhancement, hollow channels, driver matching and depletion, and the longitudinal and transverse evolution of the accelerated bunch. Beam loading can flatten the accelerating field and combine small energy spread with efficient energy transfer, while ion motion, scattering, hosing, mismatch, and alignment errors threaten emittance. The review compares electron, positron, and self-modulated proton drivers; surveys internal injection, plasma recovery, multistage coupling, numerical methods, facilities, and landmark experiments; and assesses collider and free-electron-laser concepts. Experiments have separately demonstrated large energy gain, substantial efficiency, energy-spread control, and emittance preservation. They have not yet combined application-level beam quality, charge, efficiency, repetition rate, and multistage operation. Positron acceleration, high-average-power plasma sources, compact achromatic staging, instability control, and predictive long-distance simulation therefore remain central challenges.

Contributions

1. Unifies linear and nonlinear wake excitation with driver propagation and trailing-bunch dynamics in one beam-to-beam energy-transfer framework.
2. Relates beam loading, transformer ratio, depletion, matching, and focusing to the competing goals of high efficiency and preserved energy spread and emittance.
3. Compares electron, positron, and proton-driven regimes, including hollow channels, positron-specific plasma response, and self-modulation of long proton bunches.
4. Surveys internal-injection schemes, plasma recovery, multistage coupling, simulation strategies, experimental facilities, and major results across four decades.

5. Quantifies the gap between individually demonstrated performance and the simultaneous beam quality, power, efficiency, stability, and staging required by colliders and free-electron lasers.

2025LD

ChatGPT (July 2026)

Summary

Lucca and Dumbser construct finite-volume discretizations for nonlinear symmetric-hyperbolic and thermodynamically compatible systems, where total energy is a structural conservation law and many models also carry stationary involution constraints. The paper first presents a collocated, cell-centered HTC scheme that is compatible with semi-discrete energy conservation but does not generally preserve discrete vector-calculus identities. Its main contribution is a staggered, vertex-based, semi-implicit scheme written in discrete Godunov form. Compatible mimetic operators make $\nabla \cdot \nabla \times A = 0$ and $\nabla \times \nabla \phi = 0$ hold exactly at the fully discrete level, while the implicit solve reduces to symmetric positive-definite linear systems inside a fixed-point iteration. Tests on nonlinear acoustics, nonlinear Maxwell equations, and nonlinear Maxwell-GLM equations show exact total-energy conservation and exact preservation of the relevant divergence or curl involutions for the staggered method.

Contributions

1. Developed structure-preserving finite-volume schemes for nonlinear SHTC systems.
2. Separated energy-compatible collocated discretization from fully compatible staggered discretization.
3. Built a semi-implicit Godunov-form method leading to symmetric positive-definite linear systems.
4. Preserved discrete vector-calculus identities exactly through compatible mimetic operators.
5. Verified conservation and involution preservation on nonlinear acoustic and Maxwell-type systems.

2025LJHH

ChatGPT (July 2026)

Summary

Explicit Fokker–Planck collision operators impose very small timesteps in collisional gyrokinetic scrape-off-layer simulations. This work implements a Bhatnagar–Gross–Krook (BGK) operator in Gkeyll’s discontinuous Galerkin full- f solver, treating collisionless advection explicitly and collisions with backward Euler. A fixed-point correction adjusts the discrete Maxwellian so density, momentum, and energy moments are conserved to near machine precision; the formulation also supports cross-species collisions. Benchmarks against the more detailed Lenard–Bernstein–Dougherty (LBD) operator cover shock relaxation, anisotropic interspecies equilibration, and a two-dimensional axisymmetric ASDEX Upgrade SOL. At converged resolution, BGK and LBD produce similar steady density, temperature, potential, and heat-flux profiles. The BGK scheme is more robust at lower resolution, yielding a net $56\times$ speedup for the SOL case. Its velocity-independent collision frequency cannot reproduce all transport coefficients, cross-species treatment is only partially implicit, and three-dimensional turbulence tests remain future work.

Contributions

1. Implemented an implicit BGK collision operator for Gkeyll’s discontinuous Galerkin gyrokinetic solver.
2. Enforced discrete density, momentum, and energy conservation through iterative Maxwellian correction.
3. Extended the implicit and conservative formulation to cross-species collisions.
4. Validated the operator against LBD results in relaxation tests and an ASDEX Upgrade SOL simulation.
5. Achieved a $56\times$ end-to-end speedup by combining larger timesteps with lower-resolution convergence.

2025LMST

ChatGPT (July 2026)

Summary

A tree shortcutting adds distance-preserving edges to reduce the number of hops between vertices. Classical constant-hop constructions are globally sparse but contain dense subgraphs. This paper initiates a systematic study of shortcuttings constrained by arboricity and treewidth, two measures of how tree-like a graph remains. It proves upper and lower bounds on the hop-diameter–treewidth tradeoff, including optimal bounds through hop diameter $O(\log \log n)$ and lower bounds for every larger value. Together these imply hop-diameter $\times$ treewidth $= \Omega((\log \log n)^2)$ for all hop diameters, resolving a prior open question even for paths. The constructions also yield constant-hop $(1 + \varepsilon)$ -spanners of doubling metrics with arboricity $O(\log^* n)$. A three-hop fixed-port routing scheme with $O(\log^2 n / \log \log n)$ local bits resolves another open problem. The results connect structural sparsity of tree shortcuttings to geometric spanners and compact routing.

Contributions

1. Formulated tree shortcutting under explicit tree-likeness constraints measured by treewidth and arboricity.
2. Established optimal hop-diameter–treewidth tradeoffs up to $O(\log \log n)$ hops.
3. Proved all-regime lower bounds, including the product bound $\Omega((\log \log n)^2)$ even for paths.
4. Built constant-hop $(1 + \varepsilon)$ -spanners for doubling metrics with arboricity $O(\log^* n)$.
5. Derived a three-hop compact routing scheme with $O(\log^2 n / \log \log n)$ local memory.

2025LWZ

ChatGPT (July 2026)

Summary

Relativistic electron beams counterstreaming through plasma normally develop filamentation that degrades beam uniformity and transport. This paper studies a scissors-shaped geometry of two intersecting beams and a neutralizing return current using linear theory and two-dimensional particle-in-cell simulations. Breaking cylindrical symmetry replaces filamentation with a stratification mode. Collisionless magnetic reconnection rapidly quenches that mode, producing magnetic energy about two orders of magnitude below the one-beam case and maintaining fast-beam transverse nonuniformity near 10% for roughly 8 ps, compared with more than 300% under filamentation. A later bulk-cavitation stage expels background plasma from magnetic vortices; a reduced model predicts magnetic energy growth proportional to $(t - t_0)^4$, consistent with fitted simulation exponents. The intersecting-beam systems also show much less spectral broadening and energy loss. These results establish geometric configuration as a passive instability-control mechanism, but uniform-density, periodic two-dimensional simulations are only a proof of principle; realistic gradients and full three-dimensional transport remain unresolved.

Contributions

1. Introduced a canonical scissors-shaped multivelocity beam-plasma configuration.
2. Identified stratification, rather than filamentation, as its dominant initial instability.
3. Showed reconnection-driven suppression of magnetic energy and preservation of beam uniformity.
4. Modeled bulk cavitation and verified its predicted fourth-power magnetic-energy growth.
5. Quantified reduced beam thermalization and stated testable predictions and dimensional limitations.

2025LZGY

ChatGPT (July 2026)

Summary

This study characterizes turbulence generated internally by three-dimensional resistive magnetohydrodynamic reconnection rather than imposed by external forcing. Simulations with the AMUN code vary explicit resistivity, initial perturbation amplitude, guide field, and isothermal versus adiabatic closure. Power spectra, scale-dependent anisotropy, compressive–solenoidal decomposition, and structure functions describe the resulting cascade. Across the tested cases, velocity follows a Kolmogorov-like $E_v(k) \propto k^{-5/3}$ spectrum, while magnetic energy is steeper and can approach $k^{-8/3}$ in the direction most affected by anisotropy. Resistivity and perturbation amplitude have little effect on these statistical signatures. Most cases develop anisotropic cascades broadly consistent with strong-MHD-turbulence expectations, but a guide field produces an approximately isotropic cascade in these simulations. Adiabatic runs are nearly incompressible and dominated by solenoidal velocity energy. The velocity field is also more intermittent than the magnetic field, reversing the ordering commonly reported for homogeneous MHD turbulence. The conclusions are limited by numerical resolution: small-scale anisotropy may be contaminated by numerical dissipation, and the guide-field result differs from some externally driven studies and requires further high-resolution testing.

Contributions

1. Measured multiple statistical properties of self-driven reconnection turbulence in a common three-dimensional simulation framework.
2. Identified a robust combination of Kolmogorov-like velocity spectra and substantially steeper magnetic spectra.
3. Showed that the tested spectral and anisotropy signatures are insensitive to explicit resistivity and initial perturbation amplitude.
4. Found a guide-field-induced transition to an approximately isotropic cascade within the studied setup.
5. Established quasi-incompressible, solenoidal dynamics in adiabatic runs and stronger velocity than magnetic intermittency.

2025LZL

ChatGPT (July 2026)

Summary

Weibel-generated fields must merge before seeding large-scale cosmic magnetism, but magnetohydrodynamics omits particle-scale kinetics. Using fully kinetic, 2.5-dimensional particle-in-cell simulations of decaying electron–positron plasmas, the paper follows reconnection-driven coalescence of magnetic islands at moderate to high plasma β . Mergers initially transfer magnetic energy to larger scales. They also create pressure anisotropy, however, and falling instability thresholds excite mirror and especially firehose fluctuations. Near firehose marginality, effective magnetic tension vanishes; reconnection and contraction then slow or stop, leaving elongated structures near the particle Larmor scale. Simulations with guide fields show that the outcome is nonmonotonic: an intermediate field produces a frozen, worm-like state, whereas a stronger field prevents firehose onset and permits inverse transfer. Greater separation between island size and Larmor radius also prevents complete arrest because firehose activity saturates below the reconnection scale. A Chew–Goldberger–Low model yields a criterion in initial β and normalized guide-field strength for crossing the firehose boundary, agrees with simulated anisotropy trajectories, and predicts limited scale growth after crossing. The mechanism may confine Weibel seeds to microscopic scales in high- β astrophysical plasmas. Extrapolation remains limited by the simulations’ two-dimensional geometry and pair-plasma composition.

Contributions

1. Identified pressure-anisotropy-driven microinstabilities as a kinetic mechanism that can suppress reconnection-mediated inverse magnetic-energy transfer.

2. Showed that firehose marginality removes magnetic tension at the merger scale and produces arrested, elongated magnetic structures in marginally magnetized plasmas.
3. Demonstrated that stronger guide fields or greater separation between island and Larmor scales restore continued coalescence.
4. Derived and validated an analytical criterion relating initial plasma β and guide-field strength to firehose onset and cascade suppression.
5. Connected the suppression mechanism to the viability of Weibel-generated seed fields for cosmic magnetogenesis while identifying dimensional and plasma-composition limitations.

2025MAC

ChatGPT (July 2026)

Summary

Malleable MPI applications can return or acquire resources while running, but process creation and communicator layout make resizing expensive. Existing approaches either respawn the full application or retain processes that prevent entire nodes from being released. This paper introduces two cooperative spawning strategies in which active and newly created processes participate in parallel process creation before data redistribution. The Hypercube strategy targets homogeneous allocations; Iterative Diffusive spawning also supports allocations with different core counts per node. Both arrange each MPI_COMM_WORLD-derived communicator within a node, enabling fast shrinking by terminating surplus processes instead of leaving zombie ranks or spawning replacements. The methods are integrated into the MaM malleability library and evaluated on two systems. Expansion remains competitive, with worst-case overheads of $1.13\times$ for homogeneous and $1.25\times$ for heterogeneous allocations. Shrinking is at least $1387\times$ and $20\times$ faster in the corresponding scenarios. Heterogeneous hardware load balancing and further synchronization reductions are left open.

Contributions

1. Designed a parallel MPI spawning algorithm for malleable applications on homogeneous allocations.
2. Extended the strategy to allocations with heterogeneous numbers of cores per node.
3. Enabled shrink operations to terminate surplus ranks and release complete nodes without respawning the application.
4. Integrated the new resizing paths into the MaM library while preserving existing processes.
5. Demonstrated competitive expansion and orders-of-magnitude faster shrinking on two systems.

2025MATV

ChatGPT (July 2026)

Summary

This paper reports laser wakefield acceleration experiments using a novel non-symmetric, 3D-printed gas nozzle to improve relativistic electron-beam charge and energy. The target is designed to shape the gas-density profile so that ionization injection and downramp injection reinforce each other. Measurements over backing-pressure scans show a tunable source whose charge per energy per shot increases by at least a factor of ten and whose maximum electron energy doubles relative to the authors' earlier symmetric-nozzle experiments. Particle-in-cell simulations support the interpretation that the non-symmetric profile lowers the injection threshold and promotes additional trapping. The optimal operating range is about 45–55 bar, where higher-energy, more beam-like electrons are obtained, although beam divergence remains relatively large. Lower-pressure bunches still fall in the 6–25 MeV range used by conventional clinical linear accelerators, making them relevant for comparative radiobiology. The work positions tailored gas-target geometry as a practical route toward high-dose very-high-energy electron sources for radiotherapy studies.

Contributions

1. Demonstrated a non-symmetric gas nozzle for increasing laser wakefield electron charge and energy.

2. Measured pressure-dependent beam energy, charge, divergence, and dose profiles for the new target.
3. Achieved at least tenfold charge-per-energy enhancement and a twofold maximum-energy increase relative to symmetric nozzles.
4. Used particle-in-cell simulations to identify combined ionization and downramp injection mechanisms.
5. Assessed the source's relevance to high-dose and radiotherapy-oriented very-high-energy electron applications.

2025MBK

ChatGPT (July 2026)

Summary

Parallel C++ algorithms can slow down when a fixed core count or chunk size is mismatched to the workload. This paper models execution time as ideal parallel work plus a measured constant overhead, then derives the work per core required to target 95% efficiency. The model is implemented in HPX as the `adaptive_core_chunk_size` execution-parameter object. On first use for an application and architecture, it measures per-iteration work and parallel overhead; subsequent calls select a bounded core count and chunk size without changing the algorithm implementation. Experiments cover the memory-bound `adjacent_difference`, a compute-bound synthetic kernel, and a `ChplX forall` loop on Intel, AMD, and RISC-V systems. Smaller inputs benefit from fewer cores, whereas larger inputs use more resources; eight chunks per core performed best in the reported sweep. The adaptive executor consistently matches the workload more effectively than fixed configurations. Memory bandwidth limits `adjacent_difference` to about $10\times$ speedup on 40 cores, while compute-bound tests reach up to $38\times$, $46\times$, and $57\times$ on 40-, 48-, and 64-core machines. The approach reduces tuning burden, although it assumes repeated invocations can amortize the initial measurements.

Contributions

1. Derived an overhead-based model for selecting core counts at a target parallel efficiency.
2. Implemented runtime core and chunk adaptation as an HPX execution-parameter object.
3. Kept performance policy separate from parallel algorithm implementations through the executor API.
4. Evaluated memory-bound and compute-bound workloads across Intel, AMD, and RISC-V systems.
5. Demonstrated workload-dependent resource selection and near-linear speedups for the tested compute-bound kernels.

2025MJLT

ChatGPT (July 2026)

Summary

JAX-in-Cell is a fully electromagnetic, multispecies, relativistic 1D3V particle-in-cell framework implemented entirely in JAX. The code is designed as a bridge between compact educational PIC scripts and larger production frameworks, using just-in-time compilation and automatic vectorization to run efficiently on CPUs, GPUs, and TPUs. It solves the Vlasov–Maxwell system on a staggered Yee lattice with periodic, reflective, or absorbing boundary conditions. Particle motion can be advanced with an explicit Boris pusher or with an implicit Crank–Nicolson method solved by Picard iteration for improved energy behavior. Because the implementation is differentiable end to end, it supports gradient-based optimization and integration with AI workflows. The paper documents the numerical methods, validates the framework against standard plasma benchmarks, and demonstrates automatic-differentiation capabilities. Its main significance is to make PIC modeling more accessible for differentiable physics while retaining electromagnetic, relativistic, multispecies functionality in a modern Python-based software stack.

Contributions

1. Introduced a JAX-based 1D3V electromagnetic PIC framework for relativistic, multispecies plasma simulations.
2. Implemented the Vlasov–Maxwell system on a staggered Yee lattice with multiple boundary-condition options.
3. Provided both explicit Boris and implicit Crank–Nicolson particle advance options.

4. Used JIT compilation and vectorization to target CPU, GPU, and TPU execution from Python.
5. Demonstrated differentiable PIC simulations suitable for gradient-based optimization and AI-assisted plasma modeling.

2025MKK

ChatGPT (July 2026)

Summary

This paper proposes theoretical foundations for GPU-native compilation as a way to reduce latency in AI-assisted code iteration. The premise is that current code generation, compilation, execution, and testing workflows pay large CPU–GPU transfer costs. The authors outline three approaches: adapting traditional compilation passes for parallel GPU execution, using neural sequence-to-sequence compilation with probabilistic verification, and combining the two in hybrid systems. They derive latency and energy bounds that suggest traditional GPU compilation could improve iteration by transfer elimination, while neural compilation could gain more from massive parallelism. The paper also formalizes a probabilistic verification framework in which compilation accuracy can be traded against parallel exploration. The analysis is speculative and systems-oriented rather than an implementation report, but it gives a structured model for evaluating future compilers designed around GPU-resident AI coding loops.

Contributions

1. Identified CPU–GPU transfer latency as a bottleneck in AI code-iteration workflows.
2. Classified traditional, neural, and hybrid approaches to GPU-native compilation.
3. Derived theoretical latency and energy bounds for the proposed approaches.
4. Formalized probabilistic verification for neural compilation candidates.
5. Connected GPU-native compilation to self-improving AI systems and rapid testing loops.

2025MQPF

ChatGPT (July 2026)

Summary

Hall magnetohydrodynamics modifies ideal MHD below the ion skin depth, but its complete wave classification and topology had remained disputed. The authors linearize homogeneous, barotropic Hall MHD and cast it as a Hermitian matrix Schrödinger system. Cardano’s formula gives exact eigenfrequencies, while explicit eigenvectors and symmetry analysis organize the spectrum. They prove that the positive-frequency spectrum has exactly three branches: slow magnetosonic-Hall, shear Alfvén-Hall, and fast magnetosonic-Hall waves. Each branch continuously approaches its ideal-MHD counterpart as the Hall parameter vanishes, so the Hall term changes dispersion and polarization without creating additional branches. Previously proposed circularly polarized Beltrami modes are identified as special parallel-propagation forms of existing branches. The decisive qualitative change is topological. Two branches meet at an isolated Weyl point whose local dispersion forms a Dirac cone. Eigenmode line bundles over a sphere enclosing that point have nonzero Chern numbers, whereas the Weyl point moves to infinity in the ideal-MHD limit. The study isolates bulk topology in a uniform equilibrium and connects these invariants to the possibility of protected boundary waves in inhomogeneous Hall-MHD plasmas.

Contributions

1. Derived a Hermitian Hamiltonian, exact dispersion relations, and eigenvectors for the full linear Hall-MHD wave system.
2. Proved a one-to-one homotopy between the three Hall-MHD and ideal-MHD wave branches.
3. Resolved a literature disagreement by showing that Hall MHD introduces no additional wave branches.
4. Located an isolated Weyl degeneracy and characterized its surrounding Dirac-cone dispersion.
5. Computed nonzero Chern numbers for the eigenmode bundles, establishing topology absent from ideal MHD.

2025MRHH

ChatGPT (July 2026)

Summary

This paper studies impurity transport in three projected SPARC H-mode scenarios using the ASTRA+STRAHL workflow. Neoclassical transport is modeled with FACIT, turbulent transport with TGLF-SAT2, and pedestal boundary conditions with a neural network trained on EPED simulations. The calculations benchmark earlier constant-impurity-fraction studies and then scan uncertain assumptions for a low-current scenario. The main result is robust: turbulent transport dominates neoclassical transport for argon and tungsten impurities over broad variations in pedestal concentrations, toroidal rotation, and temperature and density gradients. Tungsten peaking is nearly insensitive to pedestal tungsten concentration, while argon scans show competing changes in pedestal pressure, dilution, density peaking, and fusion gain that largely cancel. Rotation has little impact in the no-neutral-beam SPARC setting. Separate deuterium and tritium treatment finds maximum fusion power near a 55-45% D-T mixture because isotope effects alter impurity transport. The predictions support low tungsten accumulation in low-collisionality next-step tokamaks.

Contributions

1. Modeled SPARC H-mode impurity transport with coupled ASTRA, STRAHL, FACIT, TGLF-SAT2, and EPED-surrogate tools.
2. Benchmarked variable impurity profiles against earlier constant-fraction SPARC performance predictions.
3. Performed pedestal-source and boundary-condition scans for argon and tungsten impurities.
4. Showed turbulent impurity transport robustly exceeds neoclassical transport across sensitivity studies.
5. Assessed rotation and D-T fuel-composition effects on impurity peaking and fusion performance.

2025MSAK

ChatGPT (July 2026)

Summary

Free-boundary tokamak equilibrium calculations must combine plasma currents, external coil fields, limiter geometry, and numerical control of vertical drift. The paper presents pyIPREQ, a Python implementation that solves the axisymmetric Grad-Shafranov equation using Green's functions for boundary fluxes, finite differences in the interior, and Picard iteration for a prescribed toroidal-current profile. It adds limiter-defined plasma boundaries, derived equilibrium quantities, G-EQDSK output, and artificial feedback coils that stabilize iterations for elongated plasmas. A reconstruction mode constrains the magnetic-axis position inferred from diagnostics by optimizing current-profile parameters; an alternate geometric shift improves the vertical constraint. Comparisons with a published ITER equilibrium, the original IPREQ code, and FREEGS show close agreement in flux surfaces and quantities such as safety factor, plasma beta, and internal inductance. ADITYA-U cases demonstrate position-constrained reconstructions and agreement with magnetic-probe measurements, while SST-1 and divertor-coil cases illustrate predictive use. The method is not a full diagnostic reconstruction: results depend on initial profile guesses, the axis and current-density maxima are approximately aligned, and the unoptimized serial Python code is slower than IPREQ.

Contributions

1. Implemented a free-boundary Grad-Shafranov solver in Python using Green's-function boundary fluxes, finite differences, and self-consistent Picard iteration.
2. Added limiter geometry, equilibrium diagnostics, G-EQDSK export, and feedback-coil stabilization for elongated configurations.
3. Developed an optimization-based method that constrains equilibria to a prescribed magnetic-axis position inferred from measurements.
4. Introduced a vertical current-profile transformation that improves axis alignment when direct parameter optimization is inadequate.

5. Benchmarked the implementation against ITER, IPREQ, and FREEGS results and demonstrated it on ADITYA-U and SST-1 scenarios.

2025NH

ChatGPT (July 2026)

Summary

Nonlocal traffic models let vehicle speed depend on an averaged downstream density, but numerical schemes can lose stability as the interaction horizon ε shrinks toward the local Lighthill–Whitham–Richards conservation law. The paper analyzes a Godunov-type finite-volume discretization with mesh size h , focusing on the convolved density $W_{\varepsilon,h}$ rather than the possibly variation-growing density $\rho_{\varepsilon,h}$. For convex, anisotropic kernels and normalized quadrature weights, a refined argument proves a maximum principle and uniform total-variation-diminishing estimates. A new nonlocal entropy condition yields a discrete Kružkov inequality compatible with the local entropy condition. Compactness and Kuznetsov’s method then show that $W_{\varepsilon,h}$ converges along arbitrary joint limits $\varepsilon, h \rightarrow 0$ to the unique entropy solution, with L^1 error bounded at order $\varepsilon + h + \sqrt{\varepsilon t} + \sqrt{ht}$. For the exponential kernel, an L^1 deviation estimate transfers convergence to $\rho_{\varepsilon,h}$, while the continuous convolved density satisfies an L^1 contraction property. Numerical experiments with shocks, rarefactions, smooth data, several kernels, velocity laws, and limiting paths support the theory. They also show that quadrature normalization is essential: unnormalized weights can cause persistent error, whereas normalized weights recover convergence. Results beyond the hypotheses suggest extensions to constant kernels and broader density convergence. This supplies a robust scheme across both nonlocal and local modeling regimes.

Contributions

1. Proved asymptotic compatibility of a Godunov-type scheme for bounded-variation initial data, general convex downstream kernels, and nonlinear velocity laws, without requiring density to remain bounded away from zero.
2. Established a maximum principle and uniform total-variation-diminishing bounds for the discretized convolved density by exploiting kernel convexity.
3. Introduced a nonlocal entropy condition and a compatible discrete Kružkov inequality that ensure entropy admissibility in the local limit.
4. Derived the explicit L^1 convergence rate $O(\varepsilon + h + \sqrt{\varepsilon t} + \sqrt{ht})$ along arbitrary joint limits of the horizon and mesh parameters.
5. For exponential kernels, extended convergence to the discretized traffic density and proved an L^1 contraction property for the continuous convolved density.

2025P

ChatGPT (July 2026)

Summary

The paper develops multi-order Runge–Kutta methods that act directly on initial-value problems of arbitrary differential order instead of first rewriting them as first-order systems. Starting from Taylor’s theorem with integral remainder and Gauss–Jacobi quadrature, it constructs stage formulas that approximate the solution and its derivatives together. A directed maximum-weight graph records stage dependencies; strongly connected components identify implicit blocks, while independent blocks expose possible parallelism. This graph-based view yields an intrinsic definition of explicitness invariant under stage permutations. The paper defines consistency separately by derivative rank, derives order conditions and convergence results, and studies how reducing, prolonging, or overwriting an initial-value problem affects accuracy. Numerical experiments on a sixth-order linear problem show much smaller local errors than matched classical Runge–Kutta schemes, although unstable large steps can erase that advantage. The stability theory generalizes A- and L-stability, proves a multi-order analogue of the second Dahlquist barrier, and introduces half-line stability. Appendices provide Rust-oriented implementation details and extensive solved order conditions.

Contributions

1. Defined a Runge–Kutta framework that directly advances arbitrary-order initial-value problems and their derivatives.
2. Derived the methods from Taylor integral formulas and Gauss–Jacobi quadrature without differentiating the right-hand side.
3. Introduced weight digraphs, implicit blocks, implicit rank, and parallel blocks to characterize stage-system structure.
4. Established rank-dependent consistency, convergence, rewriting, and order-condition theory.
5. Generalized A- and L-stability, introduced half-line stability, and supplied numerical comparisons and an implementation framework.

2025PE

ChatGPT (July 2026)

Summary

Low-diffusion AUSM-family fluxes preserve stationary contacts and resolve shear and boundary layers accurately, but their weak dissipation can permit shock oscillations. The paper demonstrates that AUSM+, AUSM+-up, LDFSS(2), LDFSS-2001, and SLAU2 can become unstable when shocks are not aligned with unstructured tetrahedral meshes, even at low Mach number; it also exposes carbuncle behavior in AUSM+-up on unstructured meshes. To stabilize AUSM+ and LDFSS, three subsonic-only corrections add dissipation through velocity-jump scaling (2025u), pressure-jump scaling (2025p), or modified Mach splitting (2025M). Each correction vanishes for stationary contacts and retains pressure equilibrium across moving contacts and total-enthalpy preservation. Tests in the Quinoa hydrodynamics solver cover colliding streams, contact and shear waves, expansion, Mach-6 duct shocks, Mach-6 and Mach-20 blunt bodies, and a viscous double-ramp flow. The modifications suppress post-shock oscillations and avoid carbuncles in the tested cases. The velocity variant is most dissipative and robust; the Mach-splitting variant preserves shear layers best but leaves mild oscillations; the pressure variant offers the preferred balance. The most dissipative correction remains less diffusive than Van Leer flux-vector splitting. The analysis focuses on flux corrections and mostly first-order discretizations; grid-structure effects and a definitive carbuncle cause remain unresolved.

Contributions

1. Demonstrated that shock-mesh misalignment can destabilize several low-diffusion AUSM-family schemes independently of shock strength and exposed AUSM+-up carbuncles on uniform unstructured meshes.
2. Introduced velocity-, pressure-, and Mach-splitting-based subsonic dissipation mechanisms for stabilizing AUSM+ and LDFSS fluxes.
3. Established that the modified schemes exactly preserve stationary contacts, pressure equilibrium across moving contacts, and total enthalpy.
4. Validated the modifications across colliding-flow, contact, shear, expansion, duct-shock, blunt-body, and viscous double-ramp benchmarks, with no carbuncles in the shock tests.
5. Quantified the robustness-diffusion trade-off among the three corrections and identified the pressure-based variant as the preferred compromise.

2025PGHS

Claude (September 2026)

Summary

The authors build GyroSwin, a neural surrogate that autoregressively advances the 5D distribution function of nonlinear gyrokinetics, the gyrophase-averaged kinetic description of magnetized-plasma turbulence. In the background theory, quasilinear (QL) models weight per-mode linear simulations by saturation rules fitted to nonlinear fluxes, omitting nonlinear physics such as zonal flows. The data are 255 collisionless electrostatic GKW flux-tube simulations of

ion-temperature-gradient turbulence with adiabatic electrons on a $32 \times 8 \times 16 \times 85 \times 32$ grid in $(v_{\parallel}, \mu, s, x, k_y)$, varying safety factor, magnetic shear, and temperature and density gradients. GyroSwin is a shifted-window (Swin) transformer U-Net on patch tokens of the perturbed ion distribution, with a cross-attention integrator over velocity space giving 3D latents for the electrostatic potential, a heat-flux head, and the zonal ($k_y = 0$) mode as separate channels. Trained on 48 simulations, it gives heat-flux RMSEs of 47.35 (in-distribution) and 59.15 (out-of-distribution), against 89.48 and 95.18 for a QL saturation rule and about 120 for other neural surrogates, whose fluxes decay; with 241 simulations errors fall to 16–26, and the largest model’s rollouts stay above Pearson correlation 0.1 for over 100 snapshots. Stated limitations are error accumulation from ignoring the distributional nature of chaotic turbulence, the omitted linear phase, adiabatic electrons, and incomplete parameter-space coverage.

Contributions

1. Built the potential branch as a 3D Swin decoder without an encoder, taking its latents at every resolution level from a learned query attending over all $v_{\parallel}$ tokens at each spatial location, coupled the δf and ϕ branches by bidirectional cross-attention at the bottleneck, in every decoder stage, and before unpatching, and conditioned every Swin block through FiLM on sinusoidal embeddings of $(t, R/L_T, R/L_n, \hat{s}, q)$.
2. Sampled the four parameters by Latin hypercube in two passes, in which about half of 100 runs over $R/L_T \in [3, 12]$, $R/L_n \in [1, 7]$, $q \in [1, 9]$, $\hat{s} \in [0.5, 5]$ and all 200 runs over the less stabilizing $R/L_T \in [6, 12]$, $R/L_n \in [0, 2]$, $q \in [5, 9]$, $\hat{s} \in [0.5, 2]$ became turbulent, dropped the first 80 of 266 snapshots per run as linear phase, and held out six in-distribution runs inside and five out-of-distribution runs outside the convex hull of the training parameters, plus three for validation.
3. Reported time-averaged Pearson correlations with the ground-truth flux spectrum $Q(k_y)$ of 0.51 ± 0.4 (in-distribution) and 0.58 ± 0.33 (out-of-distribution) for QL, 0.59 ± 0.06 and 0.69 ± 0.06 for GyroSwin on 48 simulations, and 0.87 ± 0.11 and 0.91 ± 0.08 for GyroSwin_{Large}, whose spectra fall off at high k_y , attributed to spectral bias, and whose time-averaged zonal-flow profiles are smoothed, overestimated in amplitude, and on some test cases entirely off.
4. Fitted $L(N, D) = E + AN^{-\alpha} + BD^{-\beta}$ to the one-step validation loss of 11 configurations with 45.5M–893.4M parameters and 37–241 simulations, giving $\alpha = 0.46$ (95% CI [0.22, 0.79]) and $\beta = 0.37$ ([0.32, 0.87]) with the irreducible term E not identifiable, while the single-H100 forward time grew only from 11.8 ms at 90.2M parameters to 15.4 ms at 998.3M, against 569.6 ms, 963.3 ms, and 3103 ms for factorized CNN, factorized FNO, and Transolver.
5. Trained GyroSwin_{Linear}, a learned saturation rule mapping the last snapshot of each linear simulation to the nonlinear time-averaged flux, which with 241 simulations reached RMSEs of 42.85 (in-distribution) and 23.40 (out-of-distribution), against 56.68 and 56.06 for the QuaLiKiz saturation rule with its normalization fitted by least squares to the training fluxes and 36.95 and 25.89 for Gaussian-process regression on the four parameters.

Comments

- Section 3: the core layer performs attention “within 5D windows of size $M := M_{v_{\parallel}} \times M_{v_{\mu}} \times M_s \times M_x \times M_y$ ” and patch merging concatenates “ $2 \times 2 \times 2 \times 2 \times 2$ neighboring patches ... reducing resolution by 2^5 times”, but Appendix D states that “The δf branch operates on a 4D token grid. The μ axis is folded into the channel dimension”, with windows $(4, 2, 9, 4)$ over $(v_{\parallel}, s, x, k_y)$, an $8 \times 4 \times 17 \times 8$ grid of 4,352 tokens, and merging by 2 on each axis with more than two tokens; unresolved.
- Sections 1 and 3, Eq. (8), and Figs. 1–2: GyroSwin is trained with the multitask loss $\mathcal{L} = w_f \mathcal{L}_f(\hat{f}, f) + w_{\phi} \mathcal{L}_{\phi}(\hat{\phi}, \phi) + w_{\bar{Q}} \mathcal{L}_{\bar{Q}}(\hat{Q}, \bar{Q})$ to predict distribution, potential, and fluxes “simultaneously”, with the “heat flux at the next timestep” (Fig. 2 caption), whereas Appendix D states “The total loss is $\mathcal{L} = \mathcal{L}_{\delta f} + w_{\phi}(\tau) \mathcal{L}_{\phi}$ ” and afterwards trains only a flux head attached to “the frozen pretrained backbone”, with an L_1 loss against “the heat flux averaged over the last 80 snapshots of the trajectory”; unresolved.
- Section 4, Appendix D, and Table 6: 266 snapshots per simulation, of which the first 80 are discarded, give 185 one-step pairs per trajectory, matching subsets XS ($37 \times 185 = 6,845$) and S ($63 \times 185 = 11,655$), but the 27,380, 35,705, and 44,400 samples listed for 150, 199, and 241 trajectories equal 148×185 , 193×185 , and 240×185 , whereas $241 \times 185 = 44,585$; unresolved.
- Section 5, “Scalability”: “GyroSwin_{Small} is roughly three orders of magnitude faster than the numerical code GKW (4200 vs. 756 GFLOPs)”, as also claimed in the abstract and Section 1, but the quoted costs differ by a

factor of about 5.6; unresolved.

- Section 5, “5D→1D flux spectrum”: “all neural surrogates exhibit higher correlation than the QL approach” and GyroSwin “performs slightly worse on the ID cases than competitors”, but in Table 3 PointNet (0.49 ± 0.02 , 0.54 ± 0.09) and Transolver (0.50 ± 0.02 , 0.53 ± 0.09) fall below QL (0.51 ± 0.4 , 0.58 ± 0.33) on both test sets, and GyroSwin’s in-distribution 0.59 ± 0.06 is the highest among QL and all 48-simulation surrogates, ViT being next at 0.52 ± 0.05 .
- Table 7: the final design GyroSwin _{$f+\phi+\bar{Q}$} repeats the four Table 2 results of GyroSwin (Ours) (17.67 ± 2.16 , 19.40 ± 1.46 , 47.35 ± 6.20 , 59.15 ± 15.62) but is listed with 91.1M parameters, whereas Table 3 gives 90.2M for GyroSwin and GyroSwin_{Small}; unresolved.
- Fig. 4 (right) and Fig. 11(b): the ground-truth zonal-flow profile of the selected OOD simulation in Fig. 4 coincides with that in the upper-right panel of Fig. 11(b), yet the GyroSwin_{Large} prediction peaks near $x \approx 7$ with its minimum near $x \approx 21$ in Fig. 4, close to the ground truth, but near $x \approx 10$ and $x \approx 26$ in Fig. 11(b), which Appendix G attributes to GyroSwin_{Large}; unresolved.
- Fig. 5 caption: “The sampled parameter space is evenly distributed”, but the plotted histograms concentrate in $\hat{s} \in [0.5, 2]$, $R/L_n \in [0, 2]$, and $q \in [5, 9]$, the second-pass ranges of Appendix C, with the first-pass runs forming what Appendix C calls “the sparse point cloud in Figure 5”.
- Appendix F: Fig. 8 is said to compare predictions “for timesteps {10, 100, 200}” and to show that “the predictions in wavenumber space remain somewhat stable”, but its rows are labeled t_5 , t_{20} , t_{50} , and t_{100} , and its caption places x and y in real space, so no panel is in wavenumber space; unresolved.
- Appendix G: the improvement over QL on $W(k_y)$ “is reflected in the pearson correlation coefficients shown in Table 3”, and wrong zonal-flow profiles explain “the rather high variance observed in the main table”, but Table 3 reports only $Q(k_y)$ correlations and no table reports $W(k_y)$ or zonal-flow metrics.
- Appendices H and I: the ablation results are said to be reported “in Section H”, the section making the statement, and the learned-saturation-rule results “in Equation (5)”, the QL saturation rule of Section 2; the results are in Tables 7 and 8.
- Appendix I: “with increasing training data GyroSwin_{Linear} outperforms tabular regressors GPR and MLP”, but with 241 simulations Table 8 gives GyroSwin_{Linear} an in-distribution RMSE of 42.85 ± 12.65 against 36.95 ± 9.24 for GPR; it beats both only out of distribution (23.40 ± 5.89 against 25.89 ± 8.78 and 34.99 ± 12.80).
- Appendix I.1: $O(\text{MSA}) = 4(xyzhw)^2 d^2 + 2(xyzhw)^2 d$ squares the token count also in the projection term, which windowing leaves unchanged and $O(\text{W-MSA}) = 4xyzhw d^2 + 2M^5 xyzhw d$ gives as $4xyzhw d^2$, the evident intended term; both formulas also use w for the extent named k in the resolution $x \times y \times z \times h \times k$.

2025PGVA

ChatGPT (July 2026)

Summary

This living report develops a reproducible benchmark for constructing multivariate surrogate models from sampled tensors. Fifty functions spanning two to ten variables, polynomial, rational, and non-rational structure, smooth and nonsmooth behavior, and tensors from kilobytes to hundreds of megabytes are evaluated with seven implementations: direct and adaptive multivariate Loewner methods, MDSPACK, Kolmogorov–Arnold networks, parametric AAA variants, and a multilayer perceptron. Models are compared using out-of-sample root-mean-square error, construction time, surrogate complexity, convergence, tuning sensitivity, and usability. The report also derives the multivariate Loewner Framework’s tensor-product barycentric representation, explains recursive null-space construction and variable decoupling, and supplies a MATLAB tutorial. Across the benchmark, Loewner-based methods recover polynomial and rational functions accurately and scale to tensors near one gigabyte, whereas several alternatives encounter memory or runtime limits as dimensionality grows. Non-rational cases favor adaptive methods but expose problem-dependent tuning. The study emphasizes diagnostic comparison rather than a universal ranking and releases code to reproduce and extend the results.

Contributions

1. Curated 50 multivariate test functions varying in dimension, algebraic structure, regularity, symmetry, and tensor size.
2. Defined a common protocol measuring out-of-sample accuracy, construction time, model complexity, convergence, tuning sensitivity, and usability.
3. Consolidated the multivariate Loewner Framework, including tensor barycentric forms, recursive null-space construction, and a MATLAB tutorial.
4. Benchmarked seven rational-approximation and neural-network implementations across the complete test collection.
5. Released reproducible code and identified parameter selection, approximation-order choice, and memory limits as concrete open problems.

2025PHR

ChatGPT (July 2026)

Summary

Landau-damping code tests often compare only late-time decay rates because reconstructing the early-time response from many complex poles is difficult. This paper derives a semi-analytical time-domain solution of the linearized collisionless Vlasov–Maxwell system that avoids pole summation. Positive- and negative-time frequency-domain solutions are combined using time-reversal symmetry; symmetrizing their spectra removes slow Fourier decay and makes numerical inversion efficient. For the distribution function, a real-axis singularity is subtracted and integrated analytically while the regular remainder is evaluated numerically. The method computes both ion-density and full distribution-function responses to an initial electrostatic density perturbation and extends to arbitrary linear initial conditions by separating symmetric and antisymmetric components. A Langmuir-wave example and comparison with a BSL6D semi-Lagrangian simulation illustrate its use as a high-precision reference and expose velocity-dependent interpolation error. The construction is limited to linearized, collisionless dynamics.

Contributions

1. Derived a semi-analytical time-domain reference solution for the linearized Vlasov–Maxwell system.
2. Replaced difficult Landau-pole summation with symmetry-conditioned numerical Fourier inversion.
3. Regularized the distribution-function inversion by analytically isolating its real-axis singular contribution.
4. Extended the construction to arbitrary linear initial perturbations through symmetric and antisymmetric decomposition.
5. Demonstrated distribution-level verification of a six-dimensional semi-Lagrangian Vlasov code.

2025PLGC

ChatGPT (July 2026)

Summary

This paper describes a multigrid Poisson solver for adaptive multiresolution grids, targeting unbounded and semi-unbounded domains common in computational physics. The solver is implemented in the murphy framework for partial differential equations on collocated adaptive grids. Its main design combines multigrid iteration with a Fourier-based direct solve, giving flexibility for boundary-condition combinations that include periodic, unbounded, and semi-unbounded directions. High-order compact stencils are used to improve accuracy while reducing communication compared with wider discretizations. The method is validated against analytical solutions in periodic and unbounded settings. The implementation is also tested at large scale, with reported scalability to 16,384 cores on major European high-performance computing systems. The paper therefore contributes both a numerical method and an implementation strategy for the Poisson equation, often the dominant cost in incompressible-flow and related adaptive-grid simulations.

Contributions

1. Implemented a flexible multigrid Poisson solver in an adaptive multiresolution framework.
2. Combined multigrid iteration with Fourier direct solvers for unbounded boundary conditions.
3. Used high-order compact stencils to improve accuracy and reduce communication.
4. Validated the solver against analytical periodic and unbounded-domain solutions.
5. Demonstrated large-scale parallel scalability to 16,384 cores.

2025PSWR

ChatGPT (July 2026)

Summary

This paper analyzes isotope production as an economic driver for fusion systems before energy breakeven. The central idea is to place selected feedstock materials in a fusion blanket so neutron-driven transmutation can co-produce high-value isotopes and, at higher plasma gain, electricity. The authors calculate the value of this co-generation and formulate an economic breakeven condition based on net present value rather than only energy balance. For low plasma gain, including $Q_{\text{plas}} \lesssim 1$, the analysis indicates that medical-radioisotope production can be viable in watt-to-megawatt fusion systems; one example is $^{102}\text{Ru} \rightarrow ^{99}\text{Mo}$ production in a 3 MW system. For higher gain, $Q_{\text{plas}} \gtrsim 3$, the paper argues that electricity plus isotope co-production can reduce the gain needed for economic viability. It also discusses asymmetric neutron wall loading as a way to enhance transmutation.

Contributions

1. Framed neutron-driven isotope production as an early revenue path for fusion systems.
2. Derived an economic breakeven criterion based on net present value.
3. Quantified low-gain medical-radioisotope production scenarios.
4. Analyzed higher-gain co-production of electricity and valuable isotopes.
5. Identified asymmetric neutron wall loading as a strategy for improving transmutation yield.

2025RAML

ChatGPT (July 2026)

Summary

This paper develops adaptive step-size control for embedded multirate infinitesimal (MRI) integrators. Its Decoupled controllers independently regulate slow and fast errors without estimating accumulated fast error, while H-Tol controllers allocate a requested tolerance between the scales using such an estimate. Both outperform the earlier H-h family on the tested three-scale KPR and stiff Brusselator problems, where H-h often fails to maintain adequate scale separation. H-Tol performs best when slow evaluations dominate cost; Decoupled is preferable when slow and fast costs are comparable. The framework theoretically supports arbitrarily many time scales, although experiments use three. The paper also constructs embeddings for explicit multirate exponential Runge–Kutta methods of orders two through five, including the first fifth-order embedded MRI method, and compares explicit, implicit, and implicit–explicit MRI schemes. Results emphasize that method, controller, and right-hand-side splitting choices remain problem-dependent; large-scale PDE applications are future work.

Contributions

1. Introduced Decoupled multirate controllers that regulate slow and fast errors independently.
2. Developed H-Tol controllers that distribute tolerance using accumulated fast-error estimates.
3. Established a controller design that theoretically extends to an arbitrary number of time scales.
4. Constructed embedded multirate exponential Runge–Kutta formulas of orders two through five, including the first fifth-order embedded MRI scheme.

5. Compared a broad set of embedded MRI methods and controllers and identified cost-dependent selection trade-offs.

2025RGHL

Claude (October 2026)

Summary

The authors accelerate s -stage implicit Runge–Kutta (IRK) time stepping, mainly Gauss–Legendre, for the linear wave equation $u_{tt} - c^2 \Delta u = b$ with Neumann conditions, written as a first-order system and discretized by P_1 finite elements with mass and stiffness matrices M and E . After factoring out $A \otimes I$ and using the eigendecomposition $A^{-T} = WDW^{-1}$, the $2Ns \times 2Ns$ stage system becomes a Sylvester equation equivalent to s shifted systems $(\bar{M} - (h_t/d_i)B)\bar{K}_i = (FW)_i/d_i$, of which conjugate symmetry leaves $\lceil s/2 \rceil$ to solve. Each is solved by GMRES with a block LU preconditioner needing one solve with M (two Gauss–Seidel sweeps) and one with $M + (h_t/d_i)^2 E$, where the complex d_i is replaced by the real average d_{avg} so that one real AMG setup and V-cycle serve all shifts. The systems are solved separately ($\mathcal{P}_{Syl}^I$) or stacked ($\mathcal{P}_{Syl}^{II}$) to batch sparse products. Against block triangular preconditioners $\mathcal{P}_{LD}$, $\mathcal{P}_\kappa$, and a new $\mathcal{P}_{TAI}$ minimizing $\|L^{-1}A - I\|_2$, the Sylvester approach is about twice as fast for $s = 5$ on $h = 2^{-9}$ meshes with fewer V-cycles and scales as $O(sN)$ apart from the W transform. Results cover two-dimensional linear problems; nonlinear and time-dependent coefficients are deferred.

Contributions

1. Derived the Sylvester reformulation (3.1)–(3.3): with $\hat{B} = -h_t \bar{M}^{-1} B$ and $\hat{k} = (A \otimes I_{2N})k$, the system becomes $\hat{B}\hat{K} + \hat{K}A^{-T} = \hat{F}$, which $A^{-T} = WDW^{-1}$ turns into shifted systems $(\bar{M} - (h_t/d_i)B)(\bar{K})_i = (FW)_i/d_i$ with stages recovered by $k = (A^{-1} \otimes I_{2N}) \text{vec}(\bar{K}DW^{-1})$ (3.5); showed $(FW)_{2i} = \overline{(FW)_{2i-1}}$, so solutions for eigenvalues with negative imaginary part follow by conjugation.
2. Replaced $M + (h_t/d_i)^2 E$ by $M + (h_t/d_{avg})^2 E$ with $d_{avg} = (d_1 + \dots + d_s)/s$, which is real because the eigenvalues pair, minimizes $\|D - dI\|_F$, and was found numerically close to the best real shift for the slowest system; the resulting exact preconditioner (4.4) has $(2, 2)$ block $h_t^2(d_i^2 - d_{avg}^2)/(d_i^2 d_{avg}^2) E + M$, and one AMG setup replaces $\lceil s/2 \rceil$ complex ones.
3. Introduced $\mathcal{P}_{TAI}$, the block lower triangular preconditioner $I_s \otimes \bar{M} - h_t L \otimes B$ with L minimizing $\|L^{-1}A - I\|_2$ over lower triangular matrices, computed from s small least-squares problems instead of the Nelder–Mead minimization of $\kappa(L^{-1}A)$ in $\mathcal{P}_\kappa$; for $h = 2^{-3}$ and $s = 3, 5$ its exactly constructed version had the eigenvalues best separated from the origin among $\mathcal{P}_{LD}$, $\mathcal{P}_\kappa$, and $\mathcal{P}_{TAI}$ (Fig. 4.1), although the fields of values for $s = 5$ all contain $(-2, 4) \times (-3i, 3i)$.
4. Reported in Table 4.1 eigenbasis condition numbers of 3.5 to 3.9 for the $\mathcal{P}_{Syl}^I$ systems and $\mathcal{P}_{Syl}^{II}$ with $s = 2$ to 5, against 5.8×10^7 to 1.6×10^8 for the block preconditioners, which fall to 1.1×10^2 to 5.4×10^2 once the single zero-eigenvalue mode of the singular Neumann stiffness matrix is removed, noting that naive numpy.linalg routines give inaccurate values because of the dense eigenvalue clusters.
5. At $h = 2^{-9}$, $\mathcal{P}_{Syl}^I$ needed fewer AMG V-cycles for $s \geq 4$ and about half the runtime of $\mathcal{P}_{TAI}$ at $s = 5$ and throughout mesh refinement (Figs. 4.4, 4.5), with runtime linear in N and, after division by s , overlapping across stage counts, whereas $\mathcal{P}_{TAI}$ shifted as $O(s^2 N)$ (Fig. 4.6); $\mathcal{P}_{Syl}^{II}$ outperformed the others with SuperLU multiple right-hand sides but only for $s \geq 4$ with pyAMG (Fig. 4.7); the space-variable wave and Klein–Gordon variants with $\kappa = 1 + 0.75 \cos(\sqrt{2}x_1 + x_2/\sqrt{3})$ and $\beta = 0.75 + 0.375 \sin(x_1/\sqrt{6} - \sqrt{5}x_2)$ showed $O(h_x^2)$ errors without order reduction (Fig. 4.9).

Comments

- Sections 4.1 and 4.3: Section 4.1 states that $\mathcal{P}_{Syl}^I$ and $\mathcal{P}_{Syl}^{II}$ need “ $\lceil s/2 \rceil$ V-cycles and $2s$ Gauss–Seidel iterations per application”, whereas Section 4.3 counts for $\mathcal{P}_{Syl}^I$ “two V-cycles” and “two iterations of Gauss–Seidel” per GMRES iteration of each of the $\lceil s/2 \rceil$ systems, that is, $2\lceil s/2 \rceil$ of each; unresolved.
- Section 4.3, paragraph on $\mathcal{P}_{Syl}^{II}$: the batched solves are with “ $M - (h_t/d_{avg})^2 E$ ” and the compared system matrices are “ $M - (h_t/d_{avg})^2 E$ and $M - h_t^2 l_{ii}^2 E$ ”, whereas (3.6), Section 3.1, and Section 4.1 define these matrices as $M + (h_t/d_{avg})^2 E$ and $M + h_t^2 l_{ii}^2 E$; the plus sign is evidently intended.

2025RMBC

ChatGPT (July 2026)

Summary

This white paper summarizes UK priorities for magnetic reconnection research over the next decade. It emphasizes reconnection as a universal plasma process that converts magnetic energy into heating and particle acceleration in the heliosphere and beyond. The paper connects reconnection to solar flares, coronal mass ejections, coronal heating, solar wind acceleration, geomagnetic storms, and solar-wind interactions with planetary magnetospheres. It argues that reconnection science is central to multiple future space-mission objectives and that the UK has strengths in in situ measurements, solar observations, numerical simulation, and theory. The document identifies nine priority science objectives, gives recommendations for investment in theory, simulations, and infrastructure, and lists mission priorities and required measurements. As a white paper, its main contribution is agenda-setting rather than a new model or dataset, but it provides a compact roadmap for maintaining and expanding UK capability in reconnection science.

Contributions

1. Positioned magnetic reconnection as a cross-cutting process for heliophysics and plasma astrophysics.
2. Identified nine priority science objectives for reconnection research.
3. Summarized UK strengths in observation, simulation, and theory.
4. Recommended investments in theoretical, computational, and research infrastructure.
5. Defined mission priorities and measurements needed for future reconnection science.

2025RW

ChatGPT (July 2026)

Summary

This MFCS 2025 paper develops lazy B-trees, an external-memory analogue of lazy search trees. Lazy search trees are sorted dictionaries whose cost adapts automatically to the actual mixture of updates and queries, interpolating between priority-queue-like and binary-search-tree-like behavior. The paper transfers this idea to the block-transfer model so that the I/O savings of B-trees over binary search trees extend to the same adaptive regime. A central obstacle is that lazy search trees depend on biased search trees, while a fully satisfactory external-memory biased-search-tree replacement is not available. The authors construct an external variant with the subset of guarantees needed for lazy B-trees, which is also of independent data-structural interest. The resulting structure supports external-memory sorted dictionaries with lazy updates and includes an external-memory priority-queue special case competitive with specialized heaps, with faster decrease-key and insert operations than known alternatives. The work links adaptive indexing, deferred data structures, and external-memory dictionaries.

Contributions

1. Introduced lazy B-trees as external-memory lazy search trees for adaptive sorted dictionaries.
2. Generalized B-tree I/O speedups to workloads that interpolate between priority queues and search trees.
3. Built an external-memory biased-search-tree substitute with guarantees sufficient for the lazy-search-tree framework.
4. Derived an external-memory priority-queue special case competitive with specialized heap structures.
5. Connected lazy updates, external memory, database cracking, and deferred data structures in a unified design.

2025S

ChatGPT (July 2026)

Summary

Worst-case sorting complexity depends only on input size and therefore misses structure in partially sorted inputs. The paper proposes a broader view of universal optimality in which an oblivious algorithm, without knowing a problem-dependent parameter in advance, matches the best asymptotic complexity available when that parameter is known. For sorting, it defines maximal α -sorted subsequences: input-order subsequences whose elements are contiguous in the final sorted order. A stable partition sort that checks whether each recursive subproblem is already sorted exploits maximal subsequences of sizes $n_1, \dots, n_k$ without explicitly discovering them. It uses $O\left(\sum_i n_i \log(n/n_i + 1)\right)$ comparisons, equivalently an entropy-sensitive bound plus $O(n)$. A counting argument gives a matching comparison lower bound for the same family of permutations. The result generalizes multisorting and shows how ordinary partition sorting can adapt to a new presortedness measure, although in-place implementation, practical constants, and other parameterizations remain open.

Contributions

1. Formulated a parameter-sensitive notion of universal optimality for algorithms with implicit input structure.
2. Defined maximal α -sorted subsequences as a new measure of presortedness.
3. Modified stable partition sort to exploit that measure without first computing the subsequences.
4. Proved an entropy-sensitive upper bound based on the maximal-subsequence sizes.
5. Established a matching comparison lower bound and related the result to multisorting and adaptive sorting.

2025S2

ChatGPT (July 2026)

Summary

This technical monograph documents the mathematics and implementation of VMEC++, a C++ reimplement of the widely used VMEC fixed- and free-boundary ideal-MHD equilibrium solver with a Python-facing interface. It derives the toroidal-coordinate and Fourier representations, spectral condensation, radial discretization, force residuals, accelerated-descent iteration, radial preconditioning, and equilibrium outputs used to compute nested-flux-surface stellarator and tokamak equilibria. A major portion reconstructs NESTOR's free-boundary vacuum-field calculation, including singularity subtraction and analytical Fourier coefficients. Later chapters specify inputs and HDF5 outputs and derive post-processing quantities such as Mercier stability, rotational transform, current density, field-line following, inverse coordinate transforms, $\nabla\alpha$, and BNORM data. A Solov'ev tokamak equilibrium supplies analytical cross-checks. The work makes previously scattered legacy numerics reproducible while recording current VMEC++ limitations, notably incomplete support for non-stellarator-symmetric configurations.

Contributions

1. Consolidated the toroidal-coordinate, Fourier-parity, quadrature, and spectral-condensation conventions needed to reproduce VMEC++ behavior.
2. Derived the fixed-boundary VMEC core, including magnetic representation, radial discretization, MHD forces, accelerated descent, and radial preconditioning.
3. Reconstructed the free-boundary NESTOR formulation from Green's identity through singularity cancellation and its numerical linear system.
4. Documented the software-facing numerical contract, including input parameters, unified HDF5 output, and compatibility mappings to legacy VMEC products.
5. Derived downstream quantities and verification paths, including Mercier stability, rotational transform, current density, field-line coordinates, inverse transforms, $\nabla\alpha$, BNORM, and a Solov'ev equilibrium.

2025S3

ChatGPT (July 2026)

Summary

This theoretical paper characterizes residual histories that can be constructed for restarted GMRES and extends the construction to restarted block GMRES. A simplified polynomial-minimization proof shows stagnation mirroring: if the last s iterations of one restart cycle stagnate, the first s iterations of the next cycle must also stagnate; a following cycle of length at most s cannot improve the residual. For nonstagnating cycle transitions, the paper constructs matrices and block right-hand sides realizing prescribed admissible residual norms while independently assigning matrix eigenvalues and per-iteration Ritz values. The block theory treats block vectors as a right vector space over the $*$ -algebra of $p \times p$ matrices and uses matrix-valued residual normalizing quantities with a partial order. Partial stagnation forces at least one zero Ritz value. The results are existential rather than convergence predictions for typical matrices and currently exclude block-Arnoldi breakdown; end-of-cycle stagnation can also make the restarted basis linearly dependent.

Contributions

1. Gave a shorter proof of stagnation mirroring for restarted GMRES.
2. Constructed restarted factorizations realizing admissible residual histories.
3. Extended the construction to block GMRES over a matrix $*$ -algebra.
4. Separated prescribed residual behavior from assigned eigenvalues and Ritz values.
5. Characterized basis freedom and its trade-off with final-residual control.

2025SA

ChatGPT (July 2026)

Summary

The paper proposes a three-dimensional laser–plasma framework that uses fluid evolution in comparatively stable regions and particle-in-cell dynamics where a learned classifier predicts kinetic behavior. Its model combines finite-difference time-domain electromagnetic updates, relativistic fluid and particle equations, blended multiphoton and Ammosov–Delone–Krainov ionization rates, and approximate quantum-electrodynamic radiation-reaction and pair-production terms. SwitchNet classifies fluid versus kinetic regimes, while a residual three-dimensional convolutional network predicts the next normalized plasma state. Physics-based losses penalize energy violations, and Monte Carlo dropout estimates predictive uncertainty. The demonstration comprises 100 internally generated simulations on a 16^3 grid, yielding 500 state transitions split into 400 training and 100 test samples. On this synthetic test set, the paper reports SwitchNet area under the ROC curve of 0.9689, mean state-prediction R^2 of 0.9743, and boundary-concentrated field errors. Differential evolution then searches laser intensity, duration, and spot size for a target energy. The evidence remains proof-of-concept: validation uses small, self-generated data, and the paper provides no experimental comparison or production-scale end-to-end benchmark of switching accuracy, physical fidelity, or speedup.

Contributions

1. Proposed an adaptive fluid–particle-in-cell architecture controlled by a learned plasma-regime classifier.
2. Integrated ionization blending and approximate strong-field quantum-electrodynamic processes into the hybrid workflow.
3. Added physics-regularized state prediction and Monte Carlo-dropout uncertainty estimates.
4. Reported classifier and field-prediction metrics on a documented synthetic simulation dataset.
5. Coupled the surrogate model to differential-evolution optimization of laser pulse parameters.

2025SC

ChatGPT (July 2026)

Summary

This paper describes Armadillo, a C++ linear algebra library intended to make production code close to MATLAB-style prototype notation while retaining efficient BLAS and LAPACK execution. The library uses expression templates and template metaprogramming to analyze linear algebra expressions, avoid unnecessary temporaries, and map operations to optimized kernels. Examples show automatic simplifications for diagonal products, transposes, symmetric products, and expression combinations that would otherwise require manual memory management or verbose BLAS calls. Benchmarks indicate substantial reductions in processing time compared with direct or naive implementations, with little performance penalty for the high-level abstraction. The paper also situates Armadillo as mature scientific software: it supports dense and sparse matrices, real, complex, and integer element types, more than 200 functions, and integration in R, machine-learning, and optimization ecosystems. Planned directions include FP16/BF16 support and similar expression optimizations for GPU linear algebra through Bandicoot.

Contributions

1. Presented Armadillo as a production-oriented C++ framework for high-level numerical linear algebra.
2. Explained expression-template optimizations that remove temporaries and exploit BLAS/LAPACK backends.
3. Demonstrated performance gains on representative matrix expressions.
4. Documented the library's mature API coverage across dense, sparse, real, complex, and integer data.
5. Connected Armadillo to downstream scientific, machine-learning, optimization, and R-package ecosystems.

2025SETS

ChatGPT (July 2026)

Summary

Symmetric time integrators preserve important geometric behavior and can reverse a trajectory without storing every intermediate state, but symmetric Runge–Kutta methods are necessarily implicit. The paper analyzes symmetry through the Hopf algebra of rooted trees underlying B-series. It shows that every B-series method decomposes into symmetric and antisymmetric components and uses the symmetric component to define equivalence classes of Runge–Kutta schemes. New order conditions quantify and suppress the antisymmetric component, yielding Explicit and Effectively Symmetric (EES) methods: explicit schemes that approximate symmetry to a prescribed order. Constructed second-order EES schemes exhibit stability regions competitive with higher-order explicit methods. On inverse-square attraction and galactic-orbit tests, a four-stage EES method gives substantially better reversibility and Hamiltonian preservation than standard explicit Runge–Kutta schemes, with behavior comparable to implicit symmetric methods at lower runtime. The evidence is limited to two Hamiltonian test problems; stochastic, partitioned, adaptive-step, and stiff extensions remain future work.

Contributions

1. Developed a Hopf-algebraic framework for studying symmetry of B-series and Runge–Kutta methods.
2. Proved a decomposition of every B-series method into symmetric and antisymmetric components.
3. Introduced symmetric-component equivalence classes for comparing Runge–Kutta schemes.
4. Derived antisymmetric order conditions and used them to construct explicit EES schemes.
5. Demonstrated strong reversibility, Hamiltonian preservation, stability, and runtime behavior on two geometric-integration benchmarks.

2025SKDF

ChatGPT (July 2026)

Summary

This overview presents Helios, Thea Energy’s preconceptual planar-coil stellarator power-plant design. Helios uses a two-field-period quasi-axisymmetric equilibrium with aspect ratio 4.5, a tokamak-like X-point divertor, and high-temperature-superconducting coils. The magnet set combines 12 large plasma-encircling planar coils with 324 individually controlled planar field-shaping coils, constrained to 20 T on coil. A 1.2 m plasma-to-coil clearance supports shielding, breeding blanket space, and a projected 40-year coil lifetime. The design targets 1.1 GW thermal power and 390 MW net electric output, with helium-cooled blanket and shield systems and a tritium breeding ratio around 1.3. Companion analyses cover MHD, gyrokinetic transport, structural mechanics, neutronics, and control. The central claim is practical integration: quasi-axisymmetric stellarator physics, configurable planar coils, maintainable sectors, and present-day materials are combined into a credible power-plant architecture.

Contributions

1. Summarized a full preconceptual planar-coil stellarator power-plant design.
2. Integrated quasi-axisymmetric confinement with a tokamak-like X-point divertor.
3. Specified an HTS coil architecture with planar, controllable shaping coils.
4. Connected physics performance to shielding, breeding, maintenance, and lifetime constraints.
5. Positioned Helios and the planned Eos facility as staged risk-reduction steps.

2025SPH

ChatGPT (July 2026)

Summary

CASTOR3D linear extended-magnetohydrodynamic calculations examine external kink and peeling-ballooning instabilities in a high- β quasi-axisymmetric stellarator equilibrium. Numerical improvements, increased computing capacity, and a high-fidelity GVEC equilibrium permit substantially higher dominant toroidal harmonics n^* than earlier studies. Low- n^* modes strongly couple the positive- and negative-Fourier branches and can form distinct slow- and fast-growing locked solutions; this coupling weakens as higher- n^* modes localize, making a reduced eigenproblem adequate. Ideal growth rates rise with n^* , whereas parallel viscosity lowers growth and flattens this trend without unlocking modes. Gyroviscosity, ion-diamagnetic drift, $E \times B$ flow, and sufficiently strong externally driven quasi-symmetric flow introduce oscillation, alter mode structure, and can unlock low- n^* modes. External-flow direction can reinforce or oppose drift and viscous stabilization, relock low- n^* modes through frequency compensation, and determine whether an external kink or peeling-ballooning mode is most unstable. The results are configuration-specific but show that omitting these effects can misidentify the controlling instability.

Contributions

1. Extended CASTOR3D stability calculations to high- n^* modes in a three-dimensional quasi-axisymmetric equilibrium.
2. Characterized Fourier-branch coupling, locking, rotation, and localization across the mode spectrum.
3. Quantified how parallel viscosity changes growth rates and dominant mode types without inducing oscillation.
4. Resolved the combined influence of gyroviscosity, diamagnetic drift, $E \times B$ flow, and external flow.
5. Showed that flow magnitude and direction can switch the equilibrium’s most unstable mode family.

2025SRHK

ChatGPT (July 2026)

Summary

Directional semi-Lagrangian splitting makes high-dimensional Vlasov solvers practical, but splitting the magnetic $\mathbf{v} \times \mathbf{B}$ rotation introduces interpolation error and numerical heating. For a constant background magnetic field, this paper transforms the full Vlasov equation to a rotating velocity grid that treats cyclotron rotation exactly and directionally splits only the remaining advections. The authors prove stability and second-order convergence of the rotating-grid Strang scheme under regularity, bounded-density, and consistency assumptions; the one-step flow error is $O(h^3)$. Numerical tests recover the expected second- and fourth-order rates for Strang and composed splittings. In stable and unstable neutralized ion Bernstein-wave problems, the rotating grid resolves more dispersion branches and accurate growth rates over a wider wavenumber range than fixed-grid splittings, while better preserving mass, the L^2 norm, and total energy. High-wavenumber accuracy remains limited by interpolation damping, and the analysis assumes a constant magnetic background.

Contributions

1. Transformed the magnetized Vlasov equation onto a grid rotating with the cyclotron motion.
2. Reduced the number of directional interpolation steps while treating magnetic rotation exactly.
3. Proved stability, local third-order consistency, and global second-order convergence for rotating-grid Strang splitting.
4. Verified the predicted orders numerically, including a fourth-order composed method.
5. Improved ion Bernstein-wave dispersion and growth-rate calculations relative to fixed-grid splittings.

2025SRKH

ChatGPT (July 2026)

Summary

Low-recycling scrape-off layers (SOLs) can improve core confinement through high edge temperature and low density, but lithium targets evaporate under intense heat loads and collisionless SOL dynamics violate fluid assumptions. Using STEP geometry, the authors combine axisymmetric Gkeyll gyrokinetic simulations, EIRENE neutral transport, and comparisons with SOLPS fluid calculations. The simulations suggest that lithium or another low-recycling material can coat lower-flux side walls while a heat-tolerant target remains high recycling, preserving a hot, dilute SOL without a lithium target. Kinetic parallel dynamics also change exhaust predictions. A large divertor-leg potential confines sputtered impurities downstream more strongly than the fluid model predicts. Mirror trapping combined with drifts broadens the mapped heat-flux width from about 3 mm to 6.4 mm and halves the peak heat flux in the studied case. These results support the feasibility of the concept but do not resolve target erosion or helium exhaust; the latter requires future simulations with helium.

Contributions

1. Proposed separating the low-recycling wall function from the high-heat-flux divertor target material.
2. Coupled gyrokinetic plasma and Monte Carlo neutral models for a low-recycling STEP SOL scenario.
3. Demonstrated stronger kinetic confinement of sputtered impurities to the divertor region than in fluid simulations.
4. Showed that mirror trapping and drifts can broaden the heat-flux channel and reduce its peak.
5. Identified helium exhaust and material erosion as unresolved feasibility constraints.

2025SSX

ChatGPT (July 2026)

Summary

Explicitly forming an $n \times n$ kernel matrix costs quadratic time, limiting kernel methods on large datasets. This paper develops subquadratic approximation algorithms that access Gaussian and related kernel matrices through kernel-density-estimation queries instead of individual entries. Its central primitive approximates multiplication by a nonnegative vector with relative ℓ_2 error and improved dependence on the target error ε . Applying this primitive yields stronger Frobenius-error guarantees for multiplication by nonnegative matrices and faster relative-error approximation of nonnegative quadratic forms. A new noisy power-iteration analysis shows that matrix–vector products need only $\Theta(\varepsilon)$ accuracy to approximate the top eigenvalue and a witnessing vector, reducing the Gaussian method’s dependence from about $\varepsilon^{-7.7}$ to $\varepsilon^{-3.2}$. A separate estimator approximates the sum of all kernel entries using $\Theta(\sqrt{n}/\varepsilon^2)$ sampled points, matching a proved sampling lower bound. Strong Exponential Time Hypothesis reductions establish near-quadratic conditional hardness for related mixed-sign, bilinear, and asymmetric-kernel problems, delineating where the positive-vector and symmetric-kernel assumptions matter. The results are primarily theoretical; experiments only validate the noisy power method’s error scaling.

Contributions

1. Improved nonnegative kernel matrix–vector multiplication by more than a factor of $1/\varepsilon$ over the prior bound.
2. Derived matrix–matrix and nonnegative quadratic-form approximations from the new multiplication primitive with stronger relative-error guarantees.
3. Proved that $\Theta(\varepsilon)$ noisy products suffice for power iteration and obtained a substantially faster top-eigenvalue algorithm.
4. Developed a faster kernel-sum estimator whose $\Theta(\sqrt{n}/\varepsilon^2)$ sample count is asymptotically optimal.
5. Established SETH-based near-quadratic lower bounds for mixed-sign and asymmetric variants, clarifying limits of KDE-based acceleration.

2025SVLN

ChatGPT (July 2026)

Summary

Particle-in-cell simulations test whether embedded nanoparticles can make electron injection and charge in laser wakefield accelerators tunable. Scanning elements from lithium to gold and diameters from 50 to 200 nm produces beams spanning roughly 10–600 pC. Material and size initially control injection through the nanoparticle’s ionized electron density and electric field. Above a field-strength saturation threshold, however, the captured charge is governed mainly by the total number of atoms, so different material–size combinations can yield comparable injection. Nanoparticle electrons enter across several plasma periods rather than one localized event; increasing the injected charge consequently broadens the energy distribution. The study converts these trends into experimentally relevant selection rules for nanoparticle composition and size. It establishes a controllable charge-injection mechanism in simulation, while the trade-off between higher charge and larger energy spread remains a central design constraint.

Contributions

1. Systematically scanned nanoparticle material and size for laser-wakefield electron injection.
2. Demonstrated simulated charge control over approximately 10–600 pC.
3. Identified an electric-field saturation threshold separating density- and atom-count-controlled regimes.
4. Resolved injection over multiple plasma periods and its dependence on nanoparticle properties.
5. Quantified the trade-off between increased beam charge and broader energy spread.

2025SWRP

ChatGPT (July 2026)

Summary

Thermal radiation transport spans rapidly changing physical timescales, from photon streaming in optically thin regions to slow diffusion in optically thick material, making fixed timesteps inefficient or inaccurate. This work develops adaptive time integration for deterministic gray and multifrequency transport using a high-order–low-order moment representation. It extends a semi-implicit-explicit formulation to multifrequency problems and introduces a higher-order semi-implicit variant that treats every component except opacity implicitly. Four second-order, asymptotic-preserving implicit-explicit Runge–Kutta schemes receive newly derived embedded methods for local error estimation. Because storing embedded angular and frequency-dependent intensities would be expensive, timestep control uses only material temperature or radiation energy, adding negligible storage and computational overhead. Gray two-dimensional tophat and multifrequency one-dimensional Larsen tests show timesteps varying naturally over four to five orders of magnitude as radiation enters new materials or navigates geometric corners. Radiation-energy estimates generally demand smaller steps and improve accuracy relative to temperature-based estimates. In fixed-step multifrequency tests, the implicit-explicit formulation exhibits order reduction, whereas the semi-implicit formulation recovers the intended convergence behavior and errors two to three orders of magnitude smaller. The study is limited to standalone transport; coupling to radiation hydrodynamics remains future work.

Contributions

1. Extended moment-based semi-implicit-explicit integration from gray to multifrequency thermal radiation transport.
2. Introduced a higher-order semi-implicit formulation that resolves all components implicitly except temperature-dependent opacity.
3. Derived embedded error estimators for four robust, asymptotic-preserving implicit-explicit Runge–Kutta schemes.
4. Reduced adaptive-step overhead by estimating temporal error solely from low-dimensional temperature or radiation-energy moments.
5. Demonstrated four-to-five-order timestep variation on tophat and Larsen benchmarks and quantified the accuracy trade-offs between estimators and integration formulations.

2025SZ

ChatGPT (July 2026)

Summary

This paper studies prefix sums through matrix identities, decomposing triangular all-ones matrices into sums of Kronecker products. The decomposition exposes a recursive structure behind known prefix circuits and shows where Brent-Kung-style constructions waste one output per recursive call. Removing that deficiency leads to a parameterized family of zero-deficiency prefix circuits with constant fan-out per level. For block size s , the circuits have depth bounded by $s\lceil\log_s n\rceil + O(1)$; choosing $s = 3$ gives about $1.893 \log n + O(1)$ depth, below the $2 \log n$ depth of classic zero-deficiency families. The construction is LOGTIME-uniform and retains linear size. The authors then use the prefix circuits for quantum carry-lookahead addition, obtaining quantum adders with $O(n)$ Toffoli gates, $O(n)$ auxiliary qubits, and $1.893 \log n + O(1)$ Toffoli depth. The work connects linear-algebraic identities, parallel prefix circuits, and quantum arithmetic design.

Contributions

1. Derived a Kronecker-product decomposition of triangular prefix-sum matrices.
2. Identified and removed the recursive deficiency in Brent-Kung-style prefix circuits.
3. Constructed LOGTIME-uniform zero-deficiency circuits with constant fan-out per level.
4. Optimized block size to obtain sub- $2 \log n$ prefix depth with linear size.
5. Applied the circuits to quantum adders with improved Toffoli depth and linear resource counts.

2025TBAB

ChatGPT (July 2026)

Summary

This study analyzes how the ion-to-electron temperature ratio T_i/T_e affects core microturbulence and turbulent transport in DIII-D quiescent H-mode plasmas. Using the Gyrokinetic Toroidal Code with a QH-mode equilibrium, the authors perform linear and nonlinear simulations while varying T_i and T_e separately. The results show that reducing T_i/T_e relatively destabilizes trapped-electron modes compared with ion-temperature-gradient modes, with the dominant instability changing across the scanned temperature ratios. Raising T_e at fixed T_i increases saturated transport levels, while lowering T_i at fixed T_e produces a weaker transport increase. The simulations connect these trends to changes in radial correlation length: eddy sizes grow with higher T_e and shrink with lower T_i . The paper also tests impurity effects and finds little change in transport quantities or growth rates relative to impurity-free cases. Overall, the work identifies T_i/T_e as an organizing parameter for turbulence-regime balance and transport levels in QH-mode core plasmas.

Contributions

1. Performed linear and nonlinear gyrokinetic simulations of DIII-D QH-mode core turbulence across controlled T_i/T_e variations.
2. Identified a transition in relative dominance between trapped-electron and ion-temperature-gradient modes as T_i/T_e changes.
3. Distinguished transport effects caused by increasing T_e from those caused by decreasing T_i .
4. Connected saturated transport trends to radial correlation-length changes in turbulent eddies.
5. Assessed impurity addition and found that it does not significantly alter growth rates or transport in the studied cases.

2025TFSF

ChatGPT (July 2026)

Summary

Using 1,251 JET ITER-like-wall H-mode pedestals from the EUROfusion database, this study reconstructs electron-temperature profiles from density profiles and engineering parameters. Global neural networks predict unseen profiles within about 20% of experiment, while local models establish that the task is learnable from local quantities augmented by radial position. Separatrix loss power, plasma current, fuelling rate, and strike-point configuration are the most informative engineering inputs. Physics-motivated reconstructions find $\eta_e = L_{n_e}/L_{T_e} \approx 2$ in the steep-gradient region but a broad overall distribution, inconsistent with a single marginal slab-electron-temperature-gradient threshold. A flexible scaling $R/L_{T_e} = A(R/L_{n_e})^\alpha$ gives a database-wide $\alpha \approx 0.4$; pulse-specific fits place A and α on a common similarity line. Gyrokinetics-inspired heat-flux models improve on database-wide marginal models, particularly when the heat flux scales quadratically with R/L_{T_e} , but retain systematic shape errors and often require an unphysical negative critical η_e . This gap from machine-learning accuracy exposes missing pedestal-transport physics and limits device generalization.

Contributions

1. Demonstrated accurate data-driven temperature-pedestal reconstruction across a large JET-ILW dataset.
2. Ranked the engineering parameters that most improve global and local predictions.
3. Quantified the broad pedestal distribution of η_e and its steep-gradient mean near two.
4. Discovered a pulse-spanning similarity relation between power-law gradient-model parameters.
5. Diagnosed specific successes and physical shortcomings of gyrokinetics-inspired heat-flux closures.

2025TOMT

ChatGPT (July 2026)

Summary

Python’s sparse-computing ecosystem often lacks portable high-performance kernels, advanced iterative solvers, and preconditioners. pyGinkgo provides a lightweight Python interface to the Ginkgo C++ linear-operator framework through Pybind11, exposing Reference, OpenMP, CUDA, HIP, and SYCL executors together with sparse matrices, solvers, and preconditioners. A Python dispatch layer hides C++ template instantiations, while the buffer protocol permits zero-copy NumPy exchange and interfaces support NumPy and PyTorch workflows. Benchmarks use SuiteSparse matrices on Intel Xeon CPUs, NVIDIA A100 GPUs, and AMD MI100 GPUs. For sparse matrix–vector multiplication, pyGinkgo outperforms SciPy, CuPy, PyTorch, and TensorFlow on the tested GPU cases and scales well across CPU threads. For iterative solvers, it beats CuPy for CG and CGS but CuPy is slightly faster for the tested GMRES configuration; CPU solvers also outperform SciPy. Comparison with native Ginkgo shows that relative binding overhead can be substantial for small matrices but falls below 10% for most cases above 10^7 nonzeros on NVIDIA hardware, with small absolute timing differences. The framework therefore brings Ginkgo performance and portability to Python without hiding measured small-problem overheads or solver-specific trade-offs.

Contributions

1. Exposed Ginkgo’s portable sparse operators, executors, iterative solvers, and preconditioners through a Pythonic API.
2. Designed a dispatch layer that maps Python types to pre-instantiated C++ templates without expanding binding complexity unnecessarily.
3. Enabled efficient ecosystem interoperability through the Python buffer protocol and NumPy and PyTorch-compatible interfaces.
4. Benchmarked SpMV and iterative solvers across CPU, NVIDIA, and AMD hardware against major Python libraries.
5. Quantified Pybind11 overhead relative to native Ginkgo, showing low absolute cost and diminishing relative overhead for large sparse matrices.

2025TRBF

ChatGPT (July 2026)

Summary

Adaptive deterministic quadrature recursively refines subdomains according to local error estimates, creating irregular workloads that are difficult to distribute across GPUs. This paper formulates multidimensional integration as hierarchical domain decomposition: each MPI rank owns one GPU and independently advances a collection of subregions using a Genz–Malik cubature rule. A decentralized cyclic round-robin policy periodically moves bounded batches of subregions between ranks through nonblocking, CUDA-aware MPI communication that partially overlaps evaluation. On one GPU, the method is competitive with the PAGANI package and is more robust for oscillatory and discontinuous integrands at tight tolerances. Distributing memory makes cases through dimension 11 feasible and produces speedups of up to an order of magnitude over PAGANI in the reported tests. Strong scaling is limited: performance generally improves from two to four GPUs but can flatten or regress at eight and twelve because global convergence checks, slowly propagated imbalance, and unfinished communication create idle time. The study therefore demonstrates that multi-GPU adaptivity extends the attainable dimension and accuracy, while showing that better initial partitioning and more informed load redistribution are needed for scalable efficiency.

Contributions

1. Cast adaptive multidimensional quadrature as a distributed hierarchical domain-decomposition problem for multi-GPU systems.

2. Designed decentralized cyclic load redistribution using bounded, nonblocking CUDA-aware MPI transfers.
3. Improved robustness over PAGANI for difficult oscillatory and discontinuous integrands at tight tolerances.
4. Extended deterministic adaptive integration to reported cases as high as dimension 11 with up to order-of-magnitude speedups.
5. Quantified the synchronization, load-imbalance, and communication bottlenecks that limit strong scaling beyond four GPUs.

2025UN

ChatGPT (July 2026)

Summary

This paper introduces nonlinear Gaussian process tomography for plasma diagnostics when the reconstructed physical quantity must be non-negative. Instead of enforcing positivity through sampling-based posteriors in earlier Gaussian process tomography, the method uses a logarithmic Gaussian process and applies a Laplace approximation. This makes the constraint natural while keeping inference faster than fully sampling-based alternatives. The authors demonstrate the method on optical-diagnostic tomography for the Ring Trap 1 device, where emissivity is the constrained positive quantity. In that case study, the logarithmic Gaussian process reconstruction is more accurate than standard Gaussian process tomography and minimum Fisher information reconstruction. The work therefore frames positivity not as an after-the-fact correction, but as part of the statistical model used for the inverse problem. Its main technical value is a practical constrained Bayesian reconstruction method for diagnostics where negative inferred densities, emissivities, or related quantities are physically meaningless.

Contributions

1. Formulated nonlinear Gaussian process tomography for positive-valued plasma diagnostic quantities.
2. Used a logarithmic Gaussian process to encode non-negativity directly in the reconstruction model.
3. Applied a Laplace approximation to reduce the cost relative to sampling-based constrained posteriors.
4. Demonstrated the method on Ring Trap 1 optical-diagnostic tomography.
5. Showed improved reconstruction accuracy against standard Gaussian process tomography and minimum Fisher information methods.

2025VCCC

ChatGPT (July 2026)

Summary

The paper constructs projection-based reduced-order models for finite-element Lagrangian hydrodynamics and hyper-reduces their nonlinear force evaluations with the empirical quadrature procedure (EQP). Proper-orthogonal-decomposition bases reduce position, velocity, and internal-energy states, while time windowing limits basis growth in advection-dominated flows. The central contribution is a conservative EQP variant that enriches the energy basis with the constant-energy mode and uses one shared reduced quadrature rule for velocity and energy forces. These conditions preserve the pressure-work cancellation underlying discrete total-energy conservation. Implementations in Laghos and libROM are tested on Sedov blast, Gresho vortex, triple-point, and Taylor–Green vortex benchmarks. Conservative EQP maintains accuracy and useful speedups comparable to basic EQP while reducing total-energy drift to approximately 10^{-13} – 10^{-14} . The experiments reproduce training trajectories rather than testing out-of-sample prediction.

Contributions

1. Derived two projection-based reduced formulations from a high-order finite-element Lagrangian Euler discretization.
2. Applied EQP to replace full nonlinear spatial integration with sparse, nonnegative reduced quadrature rules.
3. Established sufficient conditions for strong discrete total-energy conservation in the hyper-reduced RK2A model.

4. Realized those conditions through zero offsets, energy-basis enrichment, and a coupled velocity–energy quadrature rule.
5. Validated near-machine-precision energy conservation on four two- and three-dimensional hydrodynamics benchmarks.

2025VGRB

ChatGPT (July 2026)

Summary

This 26-part roadmap surveys warm dense matter, where thermal, Fermi, Coulomb-correlation, and binding energies compete and neither conventional condensed-matter nor ideal-plasma approximations are uniformly reliable. It connects first-principles and reduced theories—path-integral Monte Carlo, finite-temperature density-functional theory, response and Green-function methods, radiation hydrodynamics, molecular dynamics, and average-atom models—to experimental creation and diagnosis using diamond-anvil cells, lasers, particle beams, shocks, X-ray scattering and absorption, and optical techniques. Material-focused chapters treat hydrogen, planetary mixtures, rocky-planet matter, compact stars, inertial-confinement fusion, and high-pressure synthesis. The roadmap identifies cross-cutting bottlenecks: the fermion sign problem, exchange-correlation and equation-of-state uncertainty, theory-dependent thermometry, sparse validation data, and multiscale coupling. It closes with machine learning, interoperable databases, reproducible open software, and future high-repetition-rate facilities.

Contributions

1. Integrated the principal warm-dense-matter theories and identified their validity regimes and unresolved errors.
2. Compared complementary creation and diagnostic platforms, emphasizing multi-diagnostic, model-independent temperature and density inference.
3. Organized current knowledge across hydrogen, planetary mixtures, compact stars, and inertial-confinement-fusion capsules.
4. Connected warm-dense-matter capabilities to material synthesis while identifying missing equation-of-state, transport, opacity, and phase-boundary data.
5. Set an infrastructure agenda for physics-aware machine learning, uncertainty detection, standardized metadata, interoperable databases, provenance-aware workflows, and open software.

2025VKCK

ChatGPT (July 2026)

Summary

Vandecasteele, Karris, Cloninger, and Kevrekidis extend matrix-free timestepper methods for steady states from deterministic systems to stochastic particle simulations. Instead of treating a stochastic timestepper as a map on individual particle realizations, the paper reformulates it as an operator on probability measures using optimal transport. This allows steady states to be computed through smooth distributional representations, including cumulative- and inverse-cumulative-distribution-function timesteppers in one dimension and CDF-related or sliced-Wasserstein representations in higher dimensions. The authors analyze why naive particle-level Newton–Krylov finite differences fail under stochastic noise: Jacobian-vector estimates acquire a noise-dependent bias-variance tradeoff scaling with the perturbation size. Smooth distributional timesteppers reduce microscopic variability and restore robust Newton–Krylov convergence. Numerical examples recover steady-state distributions in noisy particle systems and clarify that representation smoothness is the key condition for fast matrix-free solvers.

Contributions

1. Recast stochastic particle timesteppers as operators on probability measures.
2. Connected steady-state computation with optimal-transport residual formulations.

3. Analyzed stochastic noise in finite-difference Jacobian-vector products.
4. Introduced smooth ICDF, CDF, and sliced-Wasserstein timestepper representations.
5. Demonstrated Newton–Krylov steady-state computation for noisy particle distributions.

2025VR

ChatGPT (July 2026)

Summary

This paper develops two high-order methods for integrating functions known on scattered surface point clouds without triangulation, prescribed point layouts, or closed-form basis-function moments. Both methods use norm-minimizing Hermite–Birkhoff interpolation and inherit its super-algebraic convergence. Method 1 determines the scaling that makes a Poisson problem solvable and thereby computes ratios of integrals, including surface averages; it applies directly to closed surfaces and surfaces with boundary. Method 2 uses the divergence theorem to reduce a surface integral to boundary line integrals without imposing boundary conditions. It evaluates integrals and surface areas directly, but closed surfaces must first be partitioned into patches with boundary. Numerical tests on irregular point distributions and a genus-two surface show substantially higher accuracy than Monte Carlo or linear triangulation. The framework is also augmented with known singular functions, retaining rapid convergence for logarithmic and inverse-distance singular integrands without refining the point cloud near the singularity.

Contributions

1. Developed two fully meshfree high-order quadrature methods for arbitrary piecewise-smooth surfaces.
2. Formulated a Poisson-solvability method for ratios of surface integrals and average values.
3. Derived a divergence-theorem method for direct integrals and accurate surface-area estimation.
4. Preserved super-algebraic convergence for scattered, nonuniform point clouds without triangulation.
5. Extended norm-minimizing interpolation to singular integrands without local point refinement.

2025WAFG

ChatGPT (July 2026)

Summary

This work uses PICNIC, a fully kinetic, exactly energy-conserving particle-in-cell Monte Carlo collision code, to simulate the burn phase of National Ignition Facility shot N210808 in one-dimensional spherical geometry. Initialized from HYDRA profiles about 150 ps before bang time, the model evolves the hot spot, deuterium–tritium ice, and carbon liner while treating radiation, fusion, Coulomb collisions, and nuclear elastic scattering. Large-angle Rutherford and nuclear elastic collisions with fusion α particles generate suprathermal fuel ions and reproduce the measured extent and relative yield of the 15.5–18 MeV α knock-on neutron signal. Adding these processes, however, neither materially changes total yield nor produces the experiment’s anomalous greater-than-50-keV primary-neutron spectral shift. PICNIC qualitatively captures HYDRA’s burn-wave dynamics but predicts a delayed bang time and higher peak hot-spot temperature; its 2.35–2.45 MJ yields also exceed both HYDRA’s 2.1 MJ result and the experimental 1.37 MJ yield. The authors therefore limit the negative conclusion to the modeled one-dimensional collision physics and identify multidimensional magnetic effects, impurity kinetics, neutron scattering, and transport-model differences as unresolved directions.

Contributions

1. Applies a fully kinetic, energy-conserving PIC-MCC model to the complete burn phase of an ignition-regime inertial-confinement-fusion capsule.
2. Integrates large-angle Rutherford scattering, nuclear elastic scattering, radiation transport, and anisotropic fusion within a self-consistent spherical simulation.

3. Reproduces the experimentally identified extent and relative yield of the 15.5–18 MeV α knock-on neutron spectrum.
4. Shows that the modeled large-angle collision physics does not explain N210808’s anomalously large primary-neutron spectral shift.
5. Benchmarks PICNIC against HYDRA’s burn evolution and quantifies yield, bang-time, and peak-temperature discrepancies requiring further transport and multidimensional study.

2025WFTJ

ChatGPT (July 2026)

Summary

This paper studies how plasma-ion motion affects resonantly driven wakefield accelerators, especially schemes that use a train of drivers or self-modulation of a long beam to excite a plasma wave over many periods. Because these concepts rely on persistent electron plasma waves, ion motion can disrupt the beam–plasma resonance and limit wakefield growth. Simulations show two related suppression mechanisms. First, ion motion shifts the plasma response so the drive beam loses resonance with the wave it excites. Second, transverse wavebreaking causes phase mixing, degrading coherent wake excitation. Earlier work emphasized the phase-mixing mechanism; this study shows that resonance loss is a distinct contribution and that both effects share the same scaling with ion mass. The results are directly relevant to recent self-modulation experiments and to the choice of plasma species in resonant wakefield accelerator design. The paper clarifies why heavier ions can be advantageous and how ion dynamics constrain long interaction lengths in multi-period plasma-wave excitation.

Contributions

1. Simulated ion-motion effects in resonantly driven plasma wakefield acceleration and long-beam self-modulation.
2. Identified resonance loss between the drive beam and plasma wave as a wakefield-suppression mechanism.
3. Distinguished resonance loss from transverse-wavebreaking phase mixing.
4. Demonstrated that both suppression mechanisms scale similarly with ion mass.
5. Connected ion-motion limits to plasma-species choices in recent and future wakefield acceleration experiments.

2025WH

ChatGPT (July 2026)

Summary

This update extends a 2022 cross-concept assessment of fusion progress using Lawson parameters, ion temperature, triple product, and scientific energy gain. It incorporates peer-reviewed results from magnetic, inertial, and magneto-inertial experiments, including C-2W, FuZE, JET, MagLIF, NIF, OMEGA, PCS, PI3, and ST40. Revised plots compare Lawson parameter and triple product against temperature and track record triple products over time; a new plot reports measured scientific gain directly. The paper clarifies that Lawson criteria specify both temperature and density–confinement requirements and that the target gain or ignition definition must be stated. For inertial confinement, it reframes the hot-spot ignition threshold using measured stagnation time, raising the plotted idealized threshold by a factor of three. It also explains why triple product does not uniquely determine gain and becomes misleading beyond inertial-confinement ignition, where propagating burn shortens the measured burn duration. Updated tables document source data and highlight NIF’s scientific breakeven, later gain increases, and JET’s record fusion energies. The compilation is transparent and reproducible through released plotting code, but several values require stated inference assumptions and the authors disclose financial interests in some represented companies.

Contributions

1. Updated a cross-concept Lawson-parameter and triple-product dataset with experimental results published since 2022.

2. Added a direct time history of measured scientific energy gain for deuterium–tritium experiments.
3. Reframed inertial-confinement ignition criteria using experimentally measured stagnation time and a corrected threshold.
4. Clarified the limits of triple product as a proxy for gain, especially near and beyond inertial-confinement ignition.
5. Released revised tables and plotting code that make assumptions, corrections, and source provenance explicit.

2025WMKK

ChatGPT (July 2026)

Summary

Traditional inter-node MPI communication traverses network protocols and device software stacks even when nodes share a Compute Express Link memory pool. cMPI instead maps one-sided and two-sided point-to-point operations onto loads, stores, and copies in shared CXL memory. A shared-memory arena gives named-object allocation over the inflexible devdax interface and supports MPI remote-memory-access windows. Per-process single-producer/single-consumer queues coordinate messages without cross-node atomic operations, while cache flushes, fences, and non-temporal stores provide software coherence where hardware coherence is absent. Measurements on a real pooled-memory platform show raw CXL access with flushing has 7.2–8.1 times lower latency than TCP-based links. For small MPI messages, cMPI reaches up to 49 times lower latency and 72 times higher bandwidth than evaluated Ethernet and SmartNIC paths. Conjugate-gradient and miniAMR tests show that low latency benefits small-scale communication, whereas conventional high-bandwidth networking regains advantage as message sizes and node counts grow. The study establishes a working MPI design for CXL sharing, but the physical platform contains only two nodes, and cache maintenance plus CPU-driven copies limit large-message bandwidth and demonstrated scalability.

Contributions

1. Implemented MPI one-sided and two-sided inter-node communication over a real CXL pooled-memory platform.
2. Designed a named-object shared-memory arena that makes devdax storage usable for MPI buffers and windows.
3. Replaced unsupported cross-node atomics with per-process single-producer/single-consumer message queues.
4. Provided software cache coherence using targeted flushes, fences, and non-temporal memory operations.
5. Quantified latency and bandwidth gains with microbenchmarks and application kernels while identifying large-message and scaling limits.

2025WWTG

ChatGPT (July 2026)

Summary

This paper quantitatively compares scrape-off-layer filament properties from real-size GBS fluid turbulence simulations with experiments from the TCV-X21 validation case. The comparison is enabled by extending the open TCV-X21 dataset with two-dimensional gas-puff imaging measurements at the outboard midplane and divertor region. To make the simulation output comparable to the diagnostic, the authors analyze GBS data with a synthetic gas-puff imaging model that includes helium-plasma emission physics and line-integration effects. Filament radial and poloidal velocities agree well between simulation and experiment, while the simulations overpredict filament sizes and underpredict relative fluctuation levels. The simulations indicate that scrape-off-layer filaments are mainly density perturbations rather than temperature perturbations, supporting assumptions used in experimental transport analysis. The paper also examines the relationship between filament poloidal velocity and the time-averaged $E \times B$ direction, finding agreement at the outboard midplane and X-point region but not in the divertor leg.

Contributions

1. Extended the TCV-X21 validation comparison with two-dimensional gas-puff imaging filament data.
2. Built a synthetic gas-puff imaging diagnostic for GBS turbulence simulations.

3. Compared filament sizes, velocities, and fluctuation levels between simulation and experiment.
4. Identified strong velocity agreement but systematic size and fluctuation discrepancies.
5. Clarified the density-dominated character and $E \times B$ alignment of simulated filaments.

2025WZLW

ChatGPT (July 2026)

Summary

This paper uses particle-in-cell simulations and multi-objective Bayesian optimization to improve the efficiency and beam quality of laser wakefield acceleration. The authors focus on moderately short, sub-petawatt laser pulses and laser defocusing injection, which can create nanocoulomb electron bunches with an initially negative chirp. During subsequent acceleration the bunch undergoes an automatic two-step dechirping process, producing small energy spread without sacrificing charge. Optimization over laser and plasma parameters identifies a broad operating region with 10–30% laser-to-beam energy-transfer efficiency and percent-level energy spread. A representative case produces a roughly 420 MeV, 5.5 nC beam with 2% RMS energy spread using an 8.3 J, 7.2 fs laser pulse. The work argues that short-pulse operation and automated multi-objective search can jointly address efficiency, charge, and energy-spread constraints for compact accelerator applications.

Contributions

1. Demonstrated high-efficiency laser wakefield acceleration with moderately short laser pulses.
2. Identified laser defocusing injection as a route to nanocoulomb, initially chirped electron beams.
3. Explained the two-step dechirping mechanism that narrows the final energy spread.
4. Used multi-objective Bayesian optimization to search the coupled laser-plasma design space.
5. Reported operating points with high charge, percent-level energy spread, and up to 28% transfer efficiency.

2025YCT

ChatGPT (July 2026)

Summary

CARMA is a server-scale resource manager that collocates multiple deep-learning training tasks on a GPU while controlling out-of-memory failures and performance interference. It combines per-second GPU telemetry, warm-up-aware bookkeeping that waits for a task's first kernel, risk filters over compute and memory activity, and placement policies based on available memory or utilization. Optional Horus, PyTorch FakeTensor, and GPUMemNet estimators predict task memory demand; an OOM recovery queue relaunches failed tasks on an idle GPU. Experiments on two production-trace-derived workloads running on four NVIDIA A100 GPUs show that the best recovery-based policy reduces makespan by about 35%. Reported aggregate gains include 54% higher streaming-multiprocessor activity, 61% higher occupancy, 62% higher memory use, and roughly 15% lower GPU energy. Memory estimators reduce or eliminate OOMs but can overpredict demand and suppress useful collocation. CARMA targets one server and exposes a throughput-versus-per-task-latency trade-off; cluster-scale scheduling, fairness, and interference-aware task matching remain future work.

Contributions

1. Designed a task-level collocation manager integrating GPU monitoring, risk analysis, placement, memory estimation, and OOM recovery.
2. Introduced time-to-first-kernel bookkeeping that prevents premature collocation while a newly launched task is still warming up.
3. Evaluated memory- and utilization-aware policies with three distinct GPU-memory estimators and recovery-only baselines.
4. Implemented lightweight recovery that detects an OOM-failed task and relaunches it on a fully idle GPU.

5. Demonstrated approximately 35% lower makespan and 15% lower GPU energy for the best policy on a trace-derived training workload.

2025YHR

ChatGPT (July 2026)

Summary

Particle-in-cell methods efficiently follow collisionless plasma characteristics, but common Monte Carlo collision operators add noise and usually provide only first-order temporal accuracy. This paper develops a deterministic particle treatment of the nonlinear Lenard–Bernstein–Dougherty Fokker–Planck operator and couples it to Vlasov field dynamics. Kernel regularization turns the collision term into a particle velocity field, while a local optimization enforces discrete conservation of mass, momentum, and kinetic energy. A fully explicit second-order integrator then applies an accuracy-preserving velocity correction so that particle plus field energy is conserved at the fully discrete level. The construction covers electrostatic Ampere and electromagnetic Maxwell formulations and provides a more robust correction variant for rare particles whose nominal factor is not real. Cell lists remove negligible distant interactions, and GPU evaluation accelerates the otherwise quadratic collision calculation. Homogeneous relaxation tests recover near second-order velocity accuracy and Maxwellian relaxation. Collisional linear and nonlinear Landau damping and two-stream benchmarks preserve energy substantially better than forward-Euler or Verlet comparisons while reproducing expected collisional smoothing. Remaining work includes a convergence proof and broader electromagnetic validation.

Contributions

1. Constructed a deterministic particle discretization of the nonlinear Fokker–Planck collision operator.
2. Enforced discrete mass, momentum, and kinetic-energy conservation through a local optimization procedure.
3. Developed a fully explicit second-order correction that conserves total particle–field energy exactly at the discrete level.
4. Extended the energy-conserving construction to both electrostatic and electromagnetic plasma systems.
5. Demonstrated accuracy and robustness on relaxation, Landau-damping, and two-stream benchmarks with cell-list and GPU acceleration.

2025Z

ChatGPT (July 2026)

Summary

This paper analyzes distributed-memory algorithms for estimating the variance of a sample mean by bootstrap resampling. It compares two naive MPI baselines with two proposed strategies under explicit dataset size, resample count, process count, bandwidth, and per-process memory models. Data Broadcast and Statistic Aggregation computes local sample means and transmits only their first and second moments; this removes the resample count from the dominant communication term, reducing it from $O(DN)$ to $O(D)$ for dataset size D and N resamples. When no process can store all data, Distributed Data and RNG Synchronization partitions the dataset and gives every process the same pseudorandom index stream. Each process contributes partial sums for indices in its partition, using $O(D/P)$ memory across P processes at an $O(NP)$ communication cost. The analysis therefore exposes a direct trade-off: aggregation minimizes communication when replicated data fit in memory, while synchronized generation bounds memory when they do not. Python/mmpi4py sketches illustrate both designs. The work is theoretical rather than an empirical scaling study; latency, nonblocking communication, load imbalance, and heterogeneous systems remain open.

Contributions

1. Formulated explicit communication, computation, and memory models for four MPI bootstrap strategies.
2. Replaced full-sample returns with sufficient-statistic aggregation for variance estimation.
3. Reduced the aggregation strategy’s dominant communication complexity from $O(DN)$ to $O(D)$.

4. Designed synchronized pseudorandom resampling for datasets partitioned across process memories.
5. Characterized the $O(D)$ -communication versus $O(D/P)$ -memory trade-off between the proposed methods.

2025ZCSV

ChatGPT (July 2026)

Summary

This paper studies electrostatic instabilities of one-dimensional Bernstein–Greene–Kruskal equilibria using high-precision Vlasov–Poisson simulations with controlled collisionality and resolution. The initial state is produced by driving initially thermalized electrons with an external electric field until high-amplitude electron acoustic waves create a stationary two-hole phase-space configuration. After the driver is removed, large-scale density perturbations trigger an instability in which vortices merge into a one-hole state. By varying the short-range collision rate and grid resolution, the simulations separate physical collisional effects from numerical phase-space dissipation. The growth rate follows the Landau-theory expectation that it is controlled by the distribution-function slope in the resonant region, while the instability onset time is delayed by collisions. Long-time simulations also characterize the saturation stage through Hermite spectral analysis. Collisionless, well-resolved runs exhibit a power-law Hermite spectrum associated with constant enstrophy flux, whereas significant collisional effects introduce a high-mode cutoff and decay. The work clarifies how BGK equilibria lose stability and how collisions modify the timing and spectral structure without simply changing the resonant growth mechanism.

Contributions

1. Constructed unstable BGK equilibria through self-consistent Vlasov–Poisson simulations driven by high-amplitude electron acoustic waves.
2. Quantified the transition from a two-hole to one-hole phase-space state under large-scale density perturbations.
3. Separated effects of physical collisionality and grid resolution on instability onset and growth.
4. Verified that the growth rate is governed by the resonant distribution-function slope, consistent with weakly inhomogeneous Landau theory.
5. Used Hermite spectra to distinguish collisionless enstrophy-flux behavior from collisional high-mode cutoff and decay.

2025ZS

ChatGPT (July 2026)

Summary

This paper analyzes a fully discrete finite-element scheme for two-phase diffuse-interface magnetohydrodynamics, coupling Cahn–Hilliard, Navier–Stokes, and magnetic-field equations in two or three dimensions. A semi-implicit backward-Euler convex splitting treats the phase evolution, while pressure correction separates velocity and pressure and the remaining subproblems are decoupled. The authors prove mass conservation and unconditional discrete energy stability. Their central analysis uses elliptic Ritz and Stokes quasi-projections, together with a Maxwell projection, to prevent lower-order phase-field approximation errors from contaminating velocity and magnetic-field estimates. This yields optimal spatial and first-order temporal error bounds for the phase field and, for the matched Taylor–Hood or Mini-type choices considered, the fluid and magnetic variables. FreeFem experiments recover the predicted 2D and 3D convergence rates. Spinodal-decomposition and Kelvin–Helmholtz simulations further demonstrate energy behavior, phase coarsening, interface roll-up, and changes caused by stronger magnetic effects. The analysis does not cover lower-order magnetic-field elements.

Contributions

1. Formulated a fully discrete convex-splitting finite-element scheme that decouples phase, magnetic, velocity, and pressure solves.

2. Proved mass conservation and unconditional discrete energy stability.
3. Introduced Ritz, Stokes, and Maxwell projection tools that isolate coupled approximation errors.
4. Derived optimal temporal and spatial error estimates for matched Mini- and Taylor–Hood-type discretizations.
5. Confirmed the theory in 2D/3D convergence tests and simulated spinodal decomposition and Kelvin–Helmholtz instability.

2025ZSHC

ChatGPT (July 2026)

Summary

Grid-based Vlasov–Poisson solvers become prohibitively expensive in six-dimensional phase space. This paper extends the semi-Lagrangian adaptive-rank method from matrix representations to hierarchical Tucker tensors, enabling simulations through three spatial and three velocity dimensions. A compact quadratic reconstruction evaluates the distribution at characteristic feet with third-order space-time accuracy and $O(d^2)$ stencil complexity, while semi-Lagrangian tracing permits large time steps without dimensional splitting. A recursive hierarchical Tucker adaptive cross approximation samples selected tensor entries and adjusts ranks along a dimension tree. The self-consistent electric field is computed by an FFT-based low-rank Poisson solver that avoids rank-inflating Hadamard products. The resulting worst-case cost is $O(d^4 N r^{3+\lceil \log_2 d \rceil})$, linear in the per-dimension grid size N for fixed dimension and rank. Landau-damping and two-stream tests in up to 3D3V recover expected electric-energy dynamics and fine phase-space structures while compressing storage. Rank growth tracks nonlinear complexity, but mass, momentum, and energy errors rise in late strongly nonlinear stages, motivating the authors’ proposed extension of locally macroscopic conservation techniques.

Contributions

1. Extended semi-Lagrangian adaptive-rank integration from matrices to general hierarchical Tucker tensors for up to 3D3V dynamics.
2. Developed a compact $O(d^2)$ reconstruction with third-order space-time accuracy and no dimensional splitting.
3. Introduced recursive hierarchical Tucker adaptive cross approximation using only selected tensor entries.
4. Constructed an FFT-based low-rank Poisson solver that avoids rank-inflating tensor products.
5. Demonstrated adaptive compression and resolved kinetic structures in high-dimensional Landau-damping and two-stream benchmarks.

2025ZSRR

ChatGPT (July 2026)

Summary

This paper models the first DIII-D divertor-detachment experiments in negative-triangularity plasmas with the two-dimensional multifluid edge code UEDGE, including cross-field particle drifts. Density scans reproduce the observed outer-target ion-saturation-current rollover and compare forward and reverse toroidal-field configurations. The simulations find that forward field requires roughly 40% higher density for detachment onset than reverse field, while reverse field fails to reach deep detachment because $E \times B$ drift drives plasma into the outer far scrape-off layer and prevents density buildup at the outer strike point. In forward field, high-field-side radiation can move above the X-point before outer-target detachment, explaining experimentally observed confinement degradation during density ramps. Comparison with positive-triangularity Ohmic discharges shows that negative triangularity needs much higher separatrix density, near or above the Greenwald limit, to detach. The modeling attributes this difficulty mainly to shorter divertor-leg and connection lengths and weaker cross-field transport.

Contributions

1. Modeled DIII-D negative-triangularity detachment experiments with drift-inclusive UEDGE simulations.
2. Reproduced the field-direction dependence of detachment onset and shallow-detachment behavior.

3. Explained reverse-field detachment saturation through drift-driven outer far-SOL plasma transport.
4. Linked forward-field confinement degradation to high-field-side radiation above the X-point.
5. Compared positive and negative triangularity to isolate geometric and transport limits on detachment.

2025ZZS

ChatGPT (July 2026)

Summary

This work extends diagonalization-based parallel-in-time integration to nonlinear evolution equations containing first-, second-, or third-order time derivatives. A hybrid finite-difference formula produces an all-at-once time matrix B . For nonlinear problems, a simplified quasi-Newton iteration replaces the time-dependent block Jacobians by their average, preserving a Kronecker structure. Diagonalizing $B = VDV^{-1}$ then decouples each iteration into independent shifted spatial systems that can be solved concurrently across time modes; higher-order equations are reduced to equivalent systems involving powers of B . By relating the characteristic equation to Chebyshev polynomials, the authors derive explicit eigenvalues and eigenvectors, prove $\text{cond}_2(V) = \mathcal{O}(n^3)$, and construct an $\mathcal{O}(n^2)$ algorithm for V^{-1} . At $n = 8192$, the reported spectral-decomposition implementation is about ten times faster than MATLAB's eig-based route, although the reconstruction residual worsens with n because V is increasingly ill-conditioned. Four two-dimensional second- and third-order examples show decreasing runtime through 20 CPU cores, with diminishing returns from communication. The experiments establish feasibility and parallel speedup, but do not compare solution time or robustness against other modern parallel-in-time methods.

Contributions

1. Extended a direct diagonalization-based parallel-in-time formulation to nonlinear equations with time derivatives through third order.
2. Preserved separability for nonlinear problems through an average-Jacobian simplified quasi-Newton iteration.
3. Derived explicit spectral factors of the all-at-once time matrix using Chebyshev-polynomial identities.
4. Proved cubic growth of the eigenvector condition number and developed an $\mathcal{O}(n^2)$ inverse-eigenvector algorithm.
5. Demonstrated spectral-decomposition and multicore speedups on second- and third-order partial-differential-equation examples.

2024AK

ChatGPT (July 2026)

Summary

Arya and Kalyanaraman construct an implicit leapfrog family of arbitrary even temporal order R for a three-field (p, E, H) formulation of Maxwell's equations, where the auxiliary pressure enforces the electric divergence constraint. The method, denoted LF_R , uses staggered time grids and operator corrections generated from Taylor expansions; a specialized half-step initialization preserves the designed order. Coupling LF_R to compatible simplicial finite elements from a discrete de Rham sequence maintains the differential structure among scalar, electric, and magnetic fields. The analysis proves exact conservation of the weighted discrete energy for sufficiently small fixed time steps. Under regularity assumptions, the time-semidiscrete error is $\mathcal{O}(\Delta t^R)$, while the fully discrete error is bounded by $\mathcal{O}(\Delta t^R + h^r + h^r \Delta t^R)$ for Whitney-form polynomial order r . Two two-dimensional manufactured-solution experiments test the sixth-order member with linear and quadratic Whitney elements, including a case with nonzero pressure and nonhomogeneous boundary data. The work generalizes the authors' earlier second- and fourth-order schemes into one structure-preserving framework.

Contributions

1. Defined an arbitrary-even-order implicit staggered leapfrog family LF_R for the three-field Maxwell system, including consistent startup equations.

2. Proved exact conservation of the weighted discrete p - E - H energy for the time-semidiscrete method.
3. Established R th-order temporal convergence under stated regularity assumptions.
4. Coupled the scheme to compatible simplicial de Rham finite elements and proved a full space-time error bound.
5. Validated the sixth-order method on two analytical Maxwell problems with linear and quadratic Whitney bases.

2024BC

ChatGPT (July 2026)

Summary

This paper analyzes implicit–explicit Runge–Kutta discretizations of the ellipsoidal-statistical BGK kinetic model in the small-Knudsen-number regime. Through a discrete Chapman–Enskog expansion, the authors extend Navier–Stokes-level asymptotic analysis from a special class of schemes to general Type I and Type II IMEX-RK methods. They derive conditions under which the numerical distribution and its moments approximate the compressible Navier–Stokes limit without resolving collision times proportional to the Knudsen number. The analysis also identifies order-reduction mechanisms and additional stage-order and coupling conditions needed for uniform accuracy across kinetic and fluid regimes. Among examined methods, a Type II, asymptotically accurate IMEX-II-ISA3 construction satisfies the extra conditions and avoids the observed Navier–Stokes-level order loss. Relaxation, smooth-flow, and shock-tube experiments support the truncation-error results and show agreement of stress and heat flux with their Navier–Stokes closures. The guarantees depend on well-prepared data and the stated regularity and coefficient conditions; arbitrary IMEX schemes do not automatically retain their nominal order in the fluid limit.

Contributions

1. Extended Navier–Stokes asymptotic analysis to general Type I and Type II IMEX-RK schemes.
2. Derived discrete Chapman–Enskog expansions and local truncation errors for the ES-BGK model.
3. Identified algebraic conditions controlling asymptotic accuracy and fluid-limit order reduction.
4. Highlighted an IMEX-II-ISA3 scheme that maintains uniform accuracy under those conditions.
5. Verified the theory with relaxation, smooth-flow, and shock-tube numerical experiments.

2024G

ChatGPT (September 2026)

Summary

Plasma kinetic theory is reformulated using exterior calculus, which describes geometric quantities independently of coordinates. Distribution functions become differential forms transported by Hamiltonian flow; their Liouville equation is expressed with a Lie derivative. Integrating out particles yields a geometric BBGKY hierarchy coupling successive reduced distributions. Canonical-coordinate expressions and closure assumptions recover Vlasov, Boltzmann, master, Fokker–Planck, fluid, and magnetohydrodynamic descriptions. The collisional reductions distinguish a thermal bath assumption from the additional small-jump approximation needed for diffusion in action space. A generalized variational reaction-rate construction bounds steady flux through surfaces invariant under the unperturbed Hamiltonian motion, using an assumed upper bound on the distribution. Applied to three-body recombination in a strong magnetic field, it separates axial bounce motion from perpendicular drift and estimates their contributions for a stationary ion. Finite ion mass and motion motivate a two-bottleneck scaling model, with field ionization suppressing recombination in specified parameter regimes relevant to antihydrogen formation. The framework supplies geometric interpretation and rate estimates; its weak-correlation closures and approximate finite-mass model do not establish a general strongly coupled plasma theory.

Contributions

1. Expressed Liouville evolution and the reduced-distribution hierarchy through differential forms and Lie derivatives for Hamiltonians with pairwise interactions, explicitly identifying the required vanishing boundary fluxes.

2. Formulated a reaction-rate upper bound by minimizing flux across invariant dividing surfaces, making the dependence on distribution bounds and the chosen surface family explicit.
3. Organized canonical-coordinate kinetic reductions by their closure assumptions, separating collision kernels, thermal-bath master equations, and small-action-jump diffusion approximations.
4. Evaluated axial and perpendicular-drift contributions to the stationary-ion recombination estimate, including an impact-parameter cutoff and comparison with an earlier Monte Carlo rate.
5. Developed an approximate finite-ion-mass recombination model that distinguishes guiding-center atoms, drifting pairs, field ionization, and a second kinetic bottleneck associated with ion gyromotion.

2024HM

ChatGPT (July 2026)

Summary

This paper examines which magnetohydrodynamic invariants remain dynamically relevant in weakly collisional solar-wind-like turbulence. A two-dimensional hybrid simulation treats protons kinetically and electrons as a fluid, starts with strong cross helicity but negligible kinetic and magnetic helicities, and is analyzed through K’arm’an–Howarth–Monin scale-by-scale balances. Combined kinetic-plus-magnetic energy decays at large scales, cascades through MHD nonlinearities and then Hall coupling, and transfers to internal energy through resistive and pressure–strain effects. Although Hall MHD suggests a mixed combination of cross and kinetic helicities, the simulation instead finds that cross helicity most closely parallels energy: it decays, cascades through joint MHD–Hall transfer, and is dissipated mainly by its pressure–strain analogue. Hall effects influence cross-helicity transfer even above ion scales. Magnetic helicity is generated only weakly and does not cascade. Spectral-transfer and coarse-graining analyses reproduce these conclusions. The evidence comes from one two-dimensional parameter set, so three-dimensional, fully kinetic, and different-helicity regimes remain open.

Contributions

1. Derived and evaluated hybrid K’arm’an–Howarth–Monin balances for energy and several helicities.
2. Resolved the MHD-to-Hall continuation and small-scale conversion of the combined-energy cascade.
3. Identified cross helicity, rather than mixed helicity, as the energy-like rugged quantity in the simulated regime.
4. Showed that Hall and pressure–strain physics alter cross-helicity transfer across a broad scale range.
5. Confirmed the scale-by-scale findings with independent spectral-transfer and coarse-graining diagnostics.

2024IMSC

ChatGPT (July 2026)

Summary

High-resolution two-dimensional particle-in-cell simulations reveal that kinetic plasma turbulence self-organizes into long-lived, magnetically dominated vortices rather than remaining entirely disordered. The structures persist for many dynamical times, merge, and generate larger vortices with similar profiles, making them active mediators of the turbulent evolution. To explain them, the authors construct a self-consistent cylindrical kinetic equilibrium from an explicit particle velocity distribution inspired by the Harris pinch. Its macroscopic limit reduces to a modified force-free Gold–Hoyle configuration, connecting the measured current and magnetic-field profiles to an analytic model. The work therefore links coherent structures observed in fully kinetic turbulence to exact metastable equilibria rather than transient eddies alone. Because the simulations are 2.5D, extrapolation to fully three-dimensional systems remains open, but the structures may inform models of heliospheric coherence and plasmoid formation near compact astrophysical objects.

Contributions

1. Observed persistent force-free vortices in fully kinetic plasma-turbulence simulations.
2. Showed that vortex mergers produce self-similar, larger metastable structures.

3. Constructed a cylindrical kinetic equilibrium from an explicit velocity distribution.
4. Connected the kinetic solution to a modified Gold–Hoyle force-free state.
5. Identified a possible coherent-structure mechanism relevant to heliospheric and compact-object plasmas.

2024K

ChatGPT (July 2026)

Summary

The paper studies accuracy-aware floating-point compression for standard $\mathcal{H}$ -, uniform $\mathcal{H}$ -, and $\mathcal{H}^2$ -matrix representations and its effect on bandwidth-bound matrix–vector multiplication. Variable-accuracy low-rank compression assigns precision according to singular-value contributions while keeping the total perturbation near the requested tolerance. Compression is fused into multiplication so coefficients are decompressed only when consumed. Experiments show the largest memory savings for $\mathcal{H}$ -matrices, followed by uniform $\mathcal{H}$ - and $\mathcal{H}^2$ -matrices; adaptive floating-point compression outperforms a fixed-exponent alternative. Reduced bandwidth demand yields roughly $2\text{--}3\times$ speedup for $\mathcal{H}$ -matrix multiplication and $1.5\text{--}2.5\times$ for uniform $\mathcal{H}$ -matrices, with smaller gains for $\mathcal{H}^2$. Benefits grow with problem size but decline at tighter accuracy.

Contributions

1. Extended floating-point compression analysis across three major hierarchical-matrix formats.
2. Adapted variable-accuracy low-rank compression to shared and nested cluster bases with controlled error.
3. Designed fused matrix–vector kernels that decompress coefficients only on access.
4. Quantified format-dependent storage savings and explained them through low-rank and cluster-basis structure.
5. Demonstrated bandwidth-driven speedups and evaluated their distance from roofline limits.

2024LJKL

ChatGPT (July 2026)

Summary

This paper develops a GPU backend for the adaptive finite element library Gascoigne 3d, targeting geometric multigrid solvers on locally refined meshes. Rather than redesigning the discretization around GPU-specific kernels, the implementation maps existing dense and sparse linear algebra operations to cuBLAS and cuSPARSE and wraps the library’s data interfaces to limit host-device transfers. Custom CUDA kernels are used only where they remove bottlenecks, such as right-hand-side assembly. The approach is tested on transport-diffusion, linear elasticity, and instationary Navier-Stokes problems, showing substantial speedups, in some cases up to about $20\times$ relative to multicore CPU runs. The experiments also identify limits: sparse matrix operations already reach high memory bandwidth, thousands of small kernel calls constrain further gains, and CPU-side matrix assembly restricts nonlinear applications. The paper argues that a conservative linear-algebra-based GPU port can deliver useful acceleration while keeping the CPU and GPU code paths maintainable.

Contributions

1. Added GPU acceleration to an adaptive finite element multigrid code with limited code-path divergence.
2. Mapped finite element solver operations mainly to cuBLAS and cuSPARSE primitives.
3. Encapsulated data movement through custom Gascoigne 3d interface implementations.
4. Benchmarked transport-diffusion, elasticity, and Navier-Stokes examples on adaptive meshes.
5. Identified remaining bottlenecks from CPU assembly, sparse-kernel memory bandwidth, and kernel-call granularity.

2024LLW

ChatGPT (July 2026)

Summary

This paper develops deterministic and uncertainty-quantification solvers for the Vlasov–Poisson system with massless electrons near the quasineutral limit. Because the nonlinear Poisson equation contains e^ϕ , classical particle-in-cell discretizations inherit restrictive Debye-length and plasma-frequency scales. The authors differentiate and reformulate that field equation, then construct an explicit asymptotic-preserving particle-in-cell method whose spatial and temporal resolution need not track the small Debye length. Numerical tests show stable long-time behavior and convergence toward the quasineutral model as $\varepsilon \rightarrow 0$ on a fixed mesh. For random initial data, the high-fidelity kinetic solver is paired with an Euler–Poisson low-fidelity model in a stochastic-collocation bi-fidelity procedure. Low-fidelity snapshots select a reduced basis, while a small number of high-fidelity runs supplies its coefficients. Tests with five-dimensional random inputs and a Karhunen–Loève expansion report roughly 10^{-3} errors with as few as five high-fidelity runs in one case, with accurate means and standard deviations across $\varepsilon = 0.01$ to 1.

Contributions

1. Reformulated the nonlinear Vlasov–Poisson field equation to remain suitable in the quasineutral limit.
2. Constructed an explicit asymptotic-preserving particle-in-cell discretization for massless electrons.
3. Demonstrated fixed-mesh convergence to the quasineutral system over short and long simulations.
4. Coupled the kinetic solver to an Euler–Poisson surrogate through a reduced bi-fidelity stochastic-collocation method.
5. Validated efficient uncertainty estimates for multiple random-data models and plasma regimes.

2024O

ChatGPT (July 2026)

Summary

This paper explains how representation theory can expose parallelism hidden in discretized partial differential equations. When a differential operator and its discretization commute with a finite symmetry group, a basis adapted to the group’s irreducible representations block-diagonalizes the resulting matrix. Schur’s lemma then guarantees that couplings between inequivalent symmetry classes vanish, converting one large solve into smaller independent systems. The paper develops the required representation theory through concrete square and cube symmetry groups and gives both explicit and numerical procedures for constructing symmetry-adapted polynomial bases. Schrödinger equations illustrate the method: octahedral symmetry on a cube yields 20 independent systems, while permutation symmetry yields eight, each smaller than systems obtained from reflection parity alone. For high-dimensional many-particle problems, bosonic and spinless-fermionic symmetry can reduce the relevant subspaces dramatically, although constructing those discretizations without prohibitive preprocessing remains unresolved. The approach is especially promising for global spectral methods, where multigrid is less readily available, and extends in principle to vector-valued and equivariant operators such as Maxwell systems.

Contributions

1. Formulated symmetry-based PDE parallelization directly in terms of irreducible group representations.
2. Used Schur’s lemma to prove block sparsity and decoupling in symmetry-adapted discretizations.
3. Gave practical explicit and numerical constructions for symmetry-adapted bases on squares and cubes.
4. Demonstrated decompositions into 20 systems for octahedral symmetry and eight for permutation symmetry.
5. Analyzed growing dimension reduction for high-dimensional bosonic and spinless-fermionic Schrödinger problems.

2024PS

ChatGPT (July 2026)

Summary

This paper proposes a mesh-free collocation solver for path-dependent partial differential equations using signature kernels, universal kernels for sequential data. The PPDE solution is approximated by the minimum-norm element of a signature-kernel reproducing kernel Hilbert space that satisfies the differential equation and boundary data at finitely many paths. For linear PPDEs, the representer theorem reduces this infinite-dimensional optimal-recovery problem to a finite linear system with a unique closed-form solution. The authors establish differentiability and universality properties for the kernel and, under compactness, regularity, uniqueness, and dense-collocation assumptions, prove consistency as the number of collocation points grows. Tests on fractional-Brownian path-dependent heat equations accurately recover analytic conditional expectations. Rough-Bergomi option-pricing experiments obtain errors comparable to Monte Carlo for matched numbers of paths and collocation points. Unlike time-discretized neural PPDE solvers, the method operates on continuous paths, exposes derivatives through kernel differentiation, and solves a convex optimization problem. Error rates versus collocation count and nonlinear PPDE extensions remain open.

Contributions

1. Formulated PPDE solution as minimum-norm signature-kernel collocation in path space.
2. Reduced linear PPDE recovery to a uniquely solvable finite linear system.
3. Proved kernel regularity, universality, and conditional consistency of the collocation approximation.
4. Recovered analytic fractional-Brownian prices and matched Monte Carlo accuracy in rough-Bergomi tests.
5. Delivered a continuous-path, differentiable, convex alternative to neural and Monte Carlo PPDE solvers.

2024RBIW

ChatGPT (July 2026)

Summary

This paper proposes C++ language annotations that let HPC developers specify compact memory layouts for structs while preserving ordinary assignment semantics and avoiding external-library dependencies. The attributes expose information unavailable to the compiler, such as integer ranges, boolean packing, reduced floating-point precision, and willingness to relax member-order guarantees. The authors implement the extensions in LLVM and add complementary annotations for MPI datatype construction, so communication descriptions can track compressed struct layouts and include only fields that actually change. Smoothed-particle hydrodynamics benchmarks show that the approach can reduce memory footprint enough to run larger simulations and can improve throughput for cache- and bandwidth-limited kernels, although compression may introduce overhead when arithmetic intensity or access patterns are unfavorable. The paper stresses that compression is an additional performance-engineering choice rather than an automatic speedup. Its main value is incremental, source-level memory modesty that remains syntactically standard C++ and can be ignored by compilers that do not implement the attributes.

Contributions

1. Designed C++ attributes for range-aware integer packing, reduced-precision floating point, and padding control.
2. Implemented the memory-layout extensions as LLVM compiler support.
3. Added MPI-oriented annotations that reduce datatype-maintenance burden under layout changes.
4. Evaluated memory-footprint and throughput effects in smoothed-particle hydrodynamics kernels.
5. Characterized when compact layouts help bandwidth-limited codes and when conversion overhead can dominate.

2024SV

ChatGPT (July 2026)

Summary

Runge–Kutta methods commute with affine changes of variables, but applications such as Lie–Poisson reduction use quadratic maps and seek to evolve only the reduced variables. This paper characterizes when a Runge–Kutta discretization of a quadratic-projectable vector field descends to a closed method in those projected variables. Algebraic conditions on the Butcher coefficients eliminate cross-stage quadratic terms. The main theorem proves that the nontrivial methods satisfying these conditions and symplecticity are precisely symplectic diagonally implicit Runge–Kutta (SyDIRK) methods, up to stage permutation. The descended schemes are generally not Runge–Kutta methods themselves. Higher than second order requires at least one negative composition coefficient. The framework recovers matrix Lie–Poisson and isospectral integrators, extends through Jordan operator algebras and momentum maps, and also applies to nonconservative dynamics. For the Navier–Stokes–Zeitlin system, the resulting method dissipates energy and enstrophy monotonically up to $O(h^3)$ when its weights are nonnegative, while preserving enstrophy exactly in the inviscid case.

Contributions

1. Defined algebraic conditions under which Runge–Kutta stages project through quadratic transformations.
2. Proved that quadratic-projectable symplectic Runge–Kutta methods are exactly SyDIRK methods up to stage ordering.
3. Derived closed integration formulas expressed entirely in the projected variables.
4. Connected the construction to matrix Lie–Poisson, isospectral, Jordan-algebra, and momentum-map reductions.
5. Established dissipation and exact inviscid enstrophy preservation for a Navier–Stokes–Zeitlin application.

2024T

ChatGPT (July 2026)

Summary

These lecture notes develop parallel computing from analytical foundations through practical programming on shared- and distributed-memory systems. They distinguish parallel computing from concurrent and distributed computing, introduce machine and programming models such as RAM, PRAM, shared memory, and message passing, and define the metrics needed to judge algorithms: time, work, cost, speedup, efficiency, load balance, and scalability. Classical limits, including Amdahl’s law and memory-bandwidth constraints, are connected to empirical benchmarking. The notes then build reusable design patterns—task graphs, pipelines, stencils, work pools, domain decomposition, reductions, scans, and collective communication—and analyze algorithms for merging, prefix sums, searching, sorting, and linear algebra. Shared-memory programming is treated through C, POSIX threads, and OpenMP, including synchronization, races, memory consistency, scheduling, and NUMA effects. Distributed-memory programming receives detailed MPI coverage: communicators, datatypes, point-to-point, nonblocking, persistent and one-sided communication, collectives, topology, and performance models. Correct C-like examples and extensive exercises turn the theory into implementation and benchmarking practice. GPUs and randomized algorithms are deliberately excluded, but the principles of locality, bandwidth, overhead, and scalable work remain applicable across parallel architectures.

Contributions

1. Unifies theoretical algorithm analysis with practical shared- and distributed-memory programming in a course-length reference.
2. Develops a consistent framework for reasoning about parallel time, work, cost, speedup, efficiency, load balance, and scalability.
3. Organizes reusable parallel patterns and analyzes representative algorithms for merging, prefix sums, searching, sorting, and linear algebra.

4. Provides detailed, concept-driven coverage of POSIX threads, OpenMP, and MPI, including synchronization, advanced communication modes, and performance implications.
5. Supplies correct C-style examples and extensive exercises designed for implementation projects, benchmarking, self-study, and classroom assessment.

2024UTIS

ChatGPT (July 2026)

Summary

This survey organizes GPU-centric communication: mechanisms that reduce CPU participation in the critical path of multi-GPU data movement and synchronization. It introduces separate intra-node and inter-node taxonomies based on where APIs execute and whether data traverse device, host, or network paths, distinguishing host-native, host-controlled, device-native, and fallback modes. The paper traces vendor mechanisms across NVIDIA, AMD, and Intel, including peer access, unified or virtual memory, GPUDirect RDMA, and GPU-initiated networking. It then compares higher-level ecosystems such as GPU-aware MPI, NCCL/RCCL/oneCCL, NVSHMEM/ROC_SHMEM/Intel SHMEM, UCX, and related research libraries, emphasizing that API placement alone does not determine the data path. Research directions include CPU-free networking, topology-aware collective scheduling, communication-computation overlap, debugging and profiling for multi-GPU races, fault tolerance, and compression-accelerated communication. The survey also identifies practical gaps: vendor fragmentation, inconsistent terminology, limited cross-vendor portability, and insufficient tools for fine-grained device-native transfers and multi-GPU race detection.

Contributions

1. Defined GPU-centric communication in terms of reduced CPU involvement rather than a single API or transport technology.
2. Developed operation-based taxonomies for intra-node and inter-node communication control and data paths.
3. Mapped NVIDIA, AMD, and Intel hardware, runtime, memory-management, and networking mechanisms into a common framework.
4. Compared major collective, one-sided, MPI, and middleware libraries across capabilities, limitations, and reported performance behavior.
5. Synthesized open problems in CPU-free networking, topology awareness, overlap, debugging, resilience, portability, and compressed communication.

2023ATF

ChatGPT (August 2026)

Summary

This paper supplies the theoretical and numerical temporal-accuracy analysis missing from the original FastRK3 work. FastRK3 replaces the intermediate pressure projections of standard three-stage RK3 with a two-level linear extrapolation of pseudo-pressure and retains one pressure Poisson solve per step. By comparing stage errors with standard RK3, the authors show that final velocity retains third-order global accuracy for non-stiff flows when the RK coefficients satisfy a product condition and the extrapolation errors cancel. They derive the general s -stage constraint and the α - β relation; for the chosen RK3 scheme it becomes $\beta = -2\alpha/3 + 5/6$. Taylor–Green vortex tests confirm that midpoint and standard-approximation extrapolations give third-order velocity and second-order pressure with periodic or unsteady Dirichlet boundaries, whereas endpoint extrapolation reduces those orders to two and one. In a stiff lid-driven polar cavity, FastRK3 and standard RK3 exhibit the same order reduction: second order for $0.0625 < CFL_\nu < 1$ and first order below 0.0625. The authors recommend midpoint extrapolation because the standard-approximation coefficients may be flow-dependent.

Contributions

1. Derived FastRK3's stage-wise local error and the conditions under which its global velocity error remains third order for non-stiff flows.
2. Generalized the required Runge–Kutta coefficient condition to an arbitrary s -stage scheme.
3. Established the pressure-extrapolation relation and obtained $\beta = -2\alpha/3 + 5/6$ for the selected RK3 coefficients.
4. Distinguished endpoint, midpoint, and standard-approximation extrapolations using Taylor–Green vortices with periodic and unsteady Dirichlet boundaries.
5. Showed that FastRK3 matches standard RK3's reduced temporal order in a stiff wall-bounded cavity, with no additional penalty from intermediate pressure extrapolation.

2023SKYI

ChatGPT (July 2026)

Summary

This work connects Boost Odeint to OpenFPM by defining custom distributed state types, an algebra for elementwise operations and global norms, and CUDA/HIP operations for GPU backends. Odeint's explicit multistage, multistep, symplectic, and adaptive steppers can thereby act on OpenFPM vectors while MPI halo exchange and spatial discretization remain inside a reusable system functor. The separation lets users change time-stepping schemes independently of the right-hand side, spatial method, and target architecture, while retaining OpenFPM's expression language. Accuracy tests for exponential and sigmoidal dynamics recover expected orders through eighth order and give identical results across process counts. A three-dimensional Gray–Scott reaction–diffusion benchmark attains about 80% strong-scaling efficiency through 512 CPU cores; adaptive Dormand–Prince stepping roughly halves runtime without a significant scalability penalty. In a smoothed-particle-hydrodynamics dam-break case, the interface adds at most 6.6% CPU overhead, adds no significant GPU overhead, and runs about five times faster on one RTX 4090 than on 32 CPU cores without handwritten CUDA. A 60-line Gray–Scott example illustrates compactness. Current limitations include explicit methods only, state dimension at most six, and no support for stiff or implicit Odeint integrators.

Contributions

1. Implemented interoperable OpenFPM state types and distributed Odeint algebras for MPI-parallel CPU execution.
2. Added CUDA and HIP operation backends so the same time-integration interface targets single- and multi-GPU systems.
3. Separated temporal integration from model right-hand sides, spatial operators, communication, and hardware through template expressions and system functors.
4. Verified the designed convergence orders of fixed-step, adaptive, and multistep schemes independently of process count.
5. Demonstrated low interface overhead, 80% strong-scaling efficiency on 512 CPU cores, and approximately fivefold GPU speedup in representative PDE and particle simulations.

2023T

ChatGPT (July 2026)

Summary

Terekhov addresses the linear systems produced by vector finite-element discretizations of time-harmonic Maxwell equations. These systems are indefinite, ill conditioned, and can have nontrivial null spaces, making direct Gaussian elimination memory-intensive and iterative convergence difficult. The proposed preconditioner uses an integral Laguerre transform in time to convert the frequency-domain problem into auxiliary sign-definite systems that can be handled by multigrid methods. Numerical tests compare GMRES with and without the preconditioner on homogeneous and heterogeneous two-dimensional models with impedance and perfect-conductor boundaries. The approach is not competitive with direct methods in the reported two-dimensional cases, but its purpose is reduced memory use rather

than immediate two-dimensional speed. The paper argues that the method should be more valuable in three dimensions, where direct solvers become prohibitive and multigrid complexity and memory demands are more favorable.

Contributions

1. Formulated a Laguerre-transform preconditioner for frequency-domain Maxwell systems.
2. Targeted indefinite Nedelec finite-element matrices with large direct-solver memory costs.
3. Converted the preconditioning problem into sign-definite systems suitable for multigrid.
4. Demonstrated nonstagnating GMRES behavior on homogeneous and heterogeneous examples.
5. Clarified that the expected advantage is three-dimensional memory reduction rather than two-dimensional speed.

2022NH

ChatGPT (July 2026)

Summary

The paper uses multi-species gyrokinetic Vlasov simulations to evaluate steady-burning profile conditions in an ITER-like deuterium–tritium–helium plasma. The model evolves deuterium, tritium, thermalized helium ash, and real-mass kinetic electrons with interspecies collisions, capturing ion-temperature-gradient and trapped-electron-mode turbulence. Unlike zero-dimensional confinement-time balances, the analysis directly evaluates turbulent energy and particle fluxes needed for helium-ash exhaust and inward fuel transport. Scans over fuel composition and helium concentration reveal that deuterium and tritium can have strongly imbalanced particle fluxes even at equal fractions; an effective single-ion model cannot reproduce this behavior. Density- and temperature-gradient scans identify regimes satisfying Reiter’s steady-burning inequalities. For the thermal-helium case studied, relatively flat density profiles favor inward fuel pinch, including a regime with $R_{ax}/L_n \leq 1.2$. Suppressing zonal flows removes inward fuel transport, while hotter, nonthermal helium can reduce or reverse helium exhaust. Species-resolved gyrokinetics is therefore necessary for predictive fuel-control and ash-exhaust assessment.

Contributions

1. Formulated a gyrokinetic flux-based evaluation of Reiter’s steady-burning condition.
2. Simulated deuterium, tritium, helium ash, and real-mass electrons with interspecies collisions in an ITER-like plasma.
3. Revealed deuterium–tritium particle-flux imbalance and showed why effective single-ion models miss it.
4. Identified density- and temperature-profile regimes combining helium exhaust with inward fuel transport.
5. Demonstrated that zonal flows and nonthermal helium materially alter whether those steady-burning regimes persist.

2022SKP

Claude (October 2026)

Summary

This companion paper treats nonlinear method-of-lines systems $Mu' = \mathcal{N}(u, t)$ and differential-algebraic equations (DAEs) integrated by s -stage fully implicit Runge–Kutta (IRK) or DG-in-time methods. After $A_0 \otimes I$ is pulled out of the linearized stage system, the real Schur decomposition $A_0^{-1} = Q_0 R_0 Q_0^T$ makes the simplified-Newton operator block upper triangular. For stage-dependent linearizations $\hat{\mathcal{L}}_i$, the rotated operator $(Q_0^T \otimes I) \text{diag}(\hat{\mathcal{L}}_i) (Q_0 \otimes I)$ has off-diagonal blocks whose coefficients sum to zero, motivating three Newton-like approximations: lumping each diagonal block onto its dominant stage operator, keeping the weighted diagonal blocks, or truncating to the pattern of R_0 . The 2×2 blocks of conjugate eigenvalue pairs $\eta \pm i\beta$ receive block triangular preconditioners whose Schur complement $\eta I - \hat{\mathcal{L}}_2 + \beta^2(\eta I - \hat{\mathcal{L}}_1)^{-1}$ is replaced by $\gamma_* I - \hat{\mathcal{L}}_2$, $\gamma_* = \eta + \beta^2/\eta$; under field-of-values assumptions its condition number is at most $1 + \beta^2/(2\eta^2)$ for equal operators and at most twice that otherwise, so backward-Euler-type

preconditioners suffice. MFEM tests on an isentropic Euler vortex, Reynolds 40,000 flow over a NACA0012 airfoil, and a vorticity–streamfunction double shear layer (an index-1 DAE) reach tenth order; fourth-order Gauss needs about half the preconditioner applications of SDIRK.

Contributions

1. Cast Newton, Picard, and other fixed-point schemes as linearly preconditioned nonlinear Richardson iterations $x_{k+1} = x_k + \mathcal{P}^{-1}\mathcal{F}(x_k)$ and, using $\sum_i (d_{k,\ell})_i = \delta_{k\ell}$ for $d_{k,\ell}$ the elementwise product of the k th row of Q_0^T with the ℓ th column of Q_0 , showed that two-stage Gauss has a block-diagonal rotated operator, so exact Newton costs no more than simplified Newton, while two-stage Radau IIA has diagonal blocks $0.985\hat{\mathcal{L}}_1 + 0.015\hat{\mathcal{L}}_2$ and off-diagonal blocks $0.121(\hat{\mathcal{L}}_1 - \hat{\mathcal{L}}_2)$ (Eq. (13)).
2. Proved Theorem 4.4: if $W(\hat{\mathcal{L}}_1), W(\hat{\mathcal{L}}_2) \leq 0$ and $\langle \hat{\mathcal{L}}_1 w, \hat{\mathcal{L}}_2 w \rangle \geq 0$, right preconditioning the Schur complement by $(\gamma_* I - \hat{\mathcal{L}}_2)^{-1}$ gives condition number at most $2 + \beta^2/\eta^2$, by choosing γ_* so that the identity perturbation $(\eta^2 + \beta^2 - \gamma\eta)I$ in the minimum singular value vanishes; for equal operators Table 1 gives bounds 1.17 to 2.18 (Gauss), 1.25 to 2.60 (Radau IIA), and 1.50 to 3.44 (Lobatto IIIC) for two to five stages, and Remark 4.3 gives the tight bound $\frac{1}{2}(1 + \sqrt{1 + \beta^2/\eta^2})$ for symmetric negative semidefinite operators.
3. Extended the framework to DAEs $Mu_t = \mathcal{N}(u, w, t)$, $0 = \mathcal{G}(u, w, t)$, where the Schur-decomposed blocks become the 4×4 systems (34); for the Picard-linearized vorticity–streamfunction form with $\mathcal{L}_u = K(\psi^{(i)})$, $\mathcal{L}_w = 0$, $\mathcal{G}_u = M_{mix}$, and $\mathcal{G}_w = A$, reordered (41) into the block triangular (42), so the 2×2 vorticity block is solved first, with AIR for advection–diffusion blocks, and two independent Poisson problems preconditioned by AMG follow.
4. For the isentropic Euler vortex (degree-4 DG, 532 triangles, Roe flux) observed orders 2 to 7 against an explicit fourth-order Runge–Kutta reference with $\delta t = 5 \times 10^{-5}$ (Table 2); at $\delta t = 2 \times 10^{-2}$ with elementwise block ILU, Gauss(2) and Gauss(4) needed 15 and 36 preconditioner applications against 26 and 59 for second- and fourth-order SDIRK, Radau(3) 44 against 45, and Lobatto methods 66 to 292 (Table 3); γ_* instead of η reduced average Krylov iterations per 2×2 system, e.g. from 35.2 to 11.2 for Lobatto(8) (Table 4).
5. For the NACA0012 case at $\delta t = 5 \times 10^{-2}$ (degree-3 compact DG, 3154 triangles), Gauss(2), Radau(3), and Gauss(4) needed 128, 314, and 244 preconditioner applications against 200, 359, and 481 for SDIRK of orders 2 to 4 (Table 5), and Newton-like(3) lowered Gauss(4) runtime from 33.8 s with simplified Newton (9 Newton iterations, 148 matrix-vector products) to 21.6 s (4 iterations, 76 products) (Fig. 2); for the double shear layer with $p = 3$, $h = 0.0025$, $\delta t = 10^{-2}$ on 288 cores, Gauss(10) needed 333 preconditioner applications against 365 for SDIRK(4) at $\text{Re} = 10$ and 194 against 177 at $\text{Re} = 25,000$ (Tables 6, 7), the reordered preconditioner (42) needing up to 1064 AIR and 1204 AMG iterations for Gauss(10).

Comments

- Section 3.2: “extract the real Schur decomposition as in subsection 3.2” refers to the subsection in which the sentence appears; the decomposition $A_0^{-1} = Q_0 R_0 Q_0^T$ is introduced in subsection 3.1, the evident intended reference.
- Table 2, Lobatto 4: the errors 2.39×10^{-5} at $\delta t = 1.25 \times 10^{-2}$ and 2.39×10^{-6} at $\delta t = 6.25 \times 10^{-3}$ give the rate $\log_2 10 \approx 3.32$, not the listed 3.67; unresolved.
- Section 7.2 and Table 8: the long-time accuracy study states “We choose a Reynolds number of 100”, whereas the caption of Table 8 reads “double shear layer problem with $\text{Re} = 10$ ”; unresolved.
- Section 7.2: the reference solution is computed “by applying explicit sixth-order SDIRK integration”, but SDIRK denotes singly diagonally implicit Runge–Kutta methods (Section 2.1), which are not explicit; whether an explicit sixth-order method or an SDIRK method was used is unresolved.
- Appendix A, Eqs. (45) and (51): (45) gives $\|\mathcal{P}_\gamma\| \leq 1 + \beta^2/(\gamma\eta)$, which for $\gamma_* = (\eta^2 + \beta^2)/\eta$ is $1 + \beta^2/(\eta^2 + \beta^2)$, but (51) uses the factor $1 + \eta^2/(\eta^2 + \beta^2)$ to reach $2 + \beta^2/\eta^2$, the bound of Theorem 4.4, Eq. (25); combining (45) with (50) instead gives $1 + 2\beta^2/\eta^2$, which exceeds $2 + \beta^2/\eta^2$ whenever $\beta > \eta$; unresolved.

Summary

The authors solve the stage equations of s -stage fully implicit Runge–Kutta (IRK) methods, and of equivalent discontinuous Galerkin (DG) in time discretizations, for linear method-of-lines systems $Mu' = \mathcal{L}u + f$. Pulling $A_0 \otimes I$ out of the stage system leaves $A_0^{-1} \otimes I - I \otimes \hat{\mathcal{L}}, \hat{\mathcal{L}} = \delta t M^{-1} \mathcal{L}$, whose adjugate-based inverse reduces the update to inverting the characteristic polynomial $P_s(\hat{\mathcal{L}})$ of A_0^{-1} . Conjugate eigenvalue pairs $\eta \pm i\beta$ of A_0^{-1} give real quadratic factors $(\eta I - \hat{\mathcal{L}})^2 + \beta^2 I$, each preconditioned by $(\gamma_* I - \hat{\mathcal{L}})^{-2}$ with $\gamma_* = \sqrt{\eta^2 + \beta^2}$. If A_0 has eigenvalues with positive real part and the field of values of $\mathcal{L}$ lies in the closed left half-plane, the preconditioned condition number is at most $\sqrt{1 + \beta^2/\eta^2}$, tight, optimal in γ_* , independent of mesh and time step, and below 2.5 for Gauss, Radau IIA, and Lobatto IIIC methods with up to five stages. Backward-Euler-type preconditioners suffice, and CG or MINRES applies for symmetric operators. MFEM tests on finite-difference and DG advection–diffusion and matrix-free diffusion reach tenth order; fourth-order Gauss needs about half the preconditioner applications of SDIRK schemes, and diffusion-dominated cases need extra inner iterations.

Contributions

1. Showed (Lemma 5, Remark 6) that $\mathcal{M}_s = A_0^{-1} \otimes I - I \otimes \hat{\mathcal{L}}$ has inverse $(I_s \otimes P_s(\hat{\mathcal{L}})^{-1}) \text{adj}(\mathcal{M}_s)$, with diagonal adjugate entries monic of degree $s - 1$ and off-diagonal entries of degree $s - 2$, so that stiffly accurate methods, for which $b_0^T A_0^{-1} = (0, \dots, 0, 1)$, need only the last adjugate row at $(s - 1)^2$ matrix-vector products, against $(s - 1)s$ for Gauss methods, and Algorithm 1 forms the right-hand side by Horner evaluation of precomputed polynomials $R_i(\hat{\mathcal{L}})$ applied to $M^{-1}f_i$.
2. Proved Theorem 11: for the factored preconditioner $(\delta I - \hat{\mathcal{L}})^{-1}(\gamma I - \hat{\mathcal{L}})^{-1}$ of $(\eta I - \hat{\mathcal{L}})^2 + \beta^2 I$, the choice $\gamma_* = (\eta^2 + \beta^2)/\delta$ gives $\kappa \leq (\delta + (\eta^2 + \beta^2)/\delta)/(2\eta)$, attained by operators with eigenvalues $\{0, \pm i\sqrt{\delta\gamma_*}\}$ and minimal over γ , by bounding the Rayleigh-type ratio (25) termwise through the constants $c_0 = 1 \geq c_1 = c_2 = c_3 = 2\eta/(\delta + \gamma_*)$; Table 1 lists the resulting bounds, 1.83 for five-stage Gauss and 2.42 for five-stage Lobatto IIIC, while the naive $\gamma = \eta$ squares the bound for SPD operators and raises it to the power $3/2$ for skew-symmetric ones.
3. Iterated on $M\mathcal{Q}_\eta = (\eta M - \delta t \mathcal{L})M^{-1}(\eta M - \delta t \mathcal{L}) + \beta^2 M$ with preconditioner $\mathcal{P}M^{-1}\mathcal{P}$, $\mathcal{P} \approx (\gamma_* M - \delta t \mathcal{L})^{-1}$ (Remark 9, Algorithm 1), halving the mass-inverse applications per iteration and keeping operator and preconditioner SPD for Hermitian M and $\mathcal{L}$, so that three-term Krylov recursions apply (Remark 8); each conjugate pair is resolved before the next to avoid compounding condition numbers, and inexact inner solves are wrapped in flexible outer Krylov iterations (Remark 10).
4. On the periodic finite-difference advection–diffusion problem (18) with $\delta t = 2h$, fourth- and eighth-order stencils and one hypr AMG V-cycle per factor, observed orders 3, 4, 7, and 8 with AMG iterations per step roughly constant under refinement, fourth-order Gauss needing about half the AMG iterations of the five-stage L-stable SDIRK(4), and the block preconditioners GSL [51] and LD [42] taking 1.16 to 2.97 times the runtime of the new method (Table 2); on DG advection–diffusion with weakly imposed time-dependent Dirichlet data, L-stable SDIRK(4) reduced to order 2.7 in ℓ^2 and 2 in broken $\mathcal{H}^1$, whereas Gauss(4) kept fourth order (Fig. 2).
5. For the purely advective DG problem on 288 cores with AIR, γ_* instead of η cut inner iterations by up to 5.8 times (five-stage Lobatto IIIC, 175 to 30) and never increased them (Table 3); for $\varepsilon = 0.01$ one AIR iteration per preconditioner application failed to converge in 1000 FGMRES iterations, whereas three gave 150 (fixed-point) or 112 (inner GMRES) total AIR iterations per step (Fig. 5); on 256 cores the new solver had the smallest wallclock time in 11 of 18 cases, e.g. 4.40 s against 24.9 (GSL), 11.5 (LD), and 12.7 (ILU) for Gauss(10) at $\varepsilon = 0$, while LD was faster at $\varepsilon = 0.01$ (Table 4); for matrix-free diffusion with low-order-refined preconditioning, inner CG reduced preconditioner applications by about 1.5 times, and second- and fourth-order Gauss needed 1.3 to 2 times fewer than equal-order DIRK (Fig. 7).

Comments

- Section 3.4, discussion of line 4 of Algorithm 1: “ R_i is a polynomial of degree s ” whose Horner evaluation “only requires computing the action of $\hat{\mathcal{L}}$ s times”, but Remark 6 states that each element of $(b_0^T A_0^{-1} \otimes I) \text{adj}(\mathcal{M}_s)$ is a

polynomial in $\hat{\mathcal{L}}$ of degree $s - 1$, as the cofactors of an $s \times s$ matrix are; the evident intended degree is $s - 1$, with $s - 1$ applications of $\hat{\mathcal{L}}$.

- Section 4.2.4 and Table 4: the text states that the method “is faster than ILU in all cases but one (in which the times are very close) while offering as much as a $7\times$ speedup in one example (Radau(9) $\epsilon = 0.01$)”, but Table 4 lists 113 s for ILU against 5.41 s for IRK with Lobatto(8) at $\epsilon = 0$, a ratio of about 21, and 220 s against 33.4 s for Radau(9) at $\epsilon = 0.01$, a ratio of about 6.6; unresolved.
- Section 4.2.4: “2,359,296 DOFs” on “256 cores” is said to give “9,360 DOFs/core”, but $2,359,296/256 = 9,216$.
- Section 2.1 and Table 4: “the block ILU approach in [40]” and the column heading “ILU [40]” cite Pazner’s low-order refined preconditioners paper, whereas Section 2.1 (“block ILU preconditioners were successfully applied to a transformed version of (3) in [41]”) and Section 4.2.4 (“a block ILU preconditioner as in [41]”) cite Pazner and Persson; the evident intended reference is [41].

2021EMCW

ChatGPT (July 2026)

Summary

This Nuclear Fusion paper experimentally investigates how magnetic islands affect plasma flow and turbulence in the Wendelstein 7-X stellarator. The study examines magnetic configurations where the 5/5 island is placed at the plasma edge inside the last closed flux surface. A V-band Doppler reflectometer provides spatially resolved measurements of perpendicular plasma flow and density fluctuations. The data reveal a clear island signature in the perpendicular-flow profile. Comparing the measured flow with neoclassically driven $E \times B$ flow shows that the island contribution is strongest near island boundaries and nearly vanishes near the island O-point. The density fluctuation level is also reduced close to the O-point. By relating these observations to results from other devices and gyrokinetic simulations, the paper provides experimental evidence that island geometry can structure both flow shear and turbulence intensity in stellarator edge plasmas. The work helps connect W7-X island-divertor physics with broader magnetic-island turbulence studies.

Contributions

1. Measured magnetic-island effects on perpendicular flow and density fluctuations in W7-X.
2. Used V-band Doppler reflectometry to obtain spatially resolved flow and turbulence profiles.
3. Identified a characteristic 5/5 island signature in the perpendicular-flow profile.
4. Showed that island-driven flow contributions peak near boundaries and vanish near the O-point.
5. Observed reduced density fluctuation levels near the island O-point and compared the behavior with other devices and simulations.

2021FKM

Claude (October 2026)

Summary

Irksome is a Python library that turns Unified Form Language (UFL) descriptions of semidiscrete finite element problems, extended by a time-derivative operator Dt , into UFL for the stage-coupled equations of any Runge–Kutta (RK) method given by its Butcher tableau, solved through Firedrake and PETSc. For a form $G(t, u, u_t; v) = 0$, stage i replaces u_t by k_i , u by $u^n + \Delta t \sum_j a_{ij} k_j$, v by the i th test component, and t by $t_n + c_i \Delta t$; strongly enforced Dirichlet data g become $k_i|_\Gamma = \partial_t g(t_n + c_i \Delta t)$. A TimeStepper class and tableaux for Gauss–Legendre, RadauIIA, LobattoIIIA, and LobattoIIIC of arbitrary order, and other schemes, are provided. Since the system stays symbolic, block diagonal, block triangular, and monolithic multigrid preconditioners follow from PETSc options, and DIRKs reduce to block forward substitution through field splits. Heat-equation tests show stage-order reduction and a monolithic vertex-star multigrid whose iteration counts are independent of stages, mesh, and time step; the mixed wave equation confirms exact energy conservation for Gauss–Legendre methods; Navier–Stokes drag and lift, Benjamin–Bona–Mahony invariants,

and three-dimensional Gross–Pitaevskii dynamics cover nonlinear, DAE, and Sobolev-type problems. The monolithic preconditioner lacks theory, and explicit and IMEX methods are not efficiently supported.

Contributions

1. Built the transformation on `getForm`, which takes the semidiscrete UFL form, a Butcher tableau, the time and time step as UFL Constants, and the solution Coefficient, and returns the coupled stage form on the s -fold product space together with stage boundary conditions; a pre-order variant of `ufl.replace` was needed because the post-order traversal substitutes u before $u[i]$ and so indexes the wrong piece of the stage vector, and the implementation assumes Dt acts only on unknown fields, so one writes $2tu + t^{**2}*Dt(u)$ rather than $Dt(t^{**2}*u)$.
2. Verified accuracy on the heat equation with $u = e^{-t} \sin(\pi x) \cos(\pi y)$ and cubic serendipity elements on $N \times N$ meshes, $N = 8$ to 128: two-stage Gauss–Legendre reaches fourth order in L^2 only for small time steps and degenerates to stage order two, LobattoIIIC(3) loses L^2 but not H^1 order, RadauIIA(2) loses order only in L^2 , and RadauIIA(3) keeps full accuracy; with third-order Raviart–Thomas elements for the mixed form, Gauss–Legendre(2) again shows order reduction while RadauIIA(3) does not (Figs. 1, 2).
3. Compared, for Q_2 elements on a 128×128 mesh with $\Delta t = 0.078125$ and RadauIIA(s), $s = 1, 2, 3$, the block diagonal and block triangular preconditioners (2.35), (2.36) with one geometric multigrid V-cycle per block against a monolithic multigrid whose Chebyshev-accelerated vertex-star smoother couples all stages: block diagonal makes RadauIIA(2) about four times and RadauIIA(3) about eight times as expensive as backward Euler, whereas the monolithic variant keeps about five FGMRES iterations per step for every stage count, costing about two and four times the single-stage time, and needs four or five iterations for $N = 16$ to 128 and $\Delta t = c/N$, $c = 1, 4, 16$ (Table 1); on a 32×32 mesh RadauIIA(2) with $\Delta t = 1/8$ reaches error 2.9×10^{-4} in 2.01 s against 1.1×10^{-3} in 7.75 s for RadauIIA(1) with $\Delta t = 1/128$ (Table 2).
4. Showed for the mixed wave equation on a 10×10 mesh with $RT_2 \times DG_1$ and $\Delta t = 0.1, 0.5, 1.0$ to $t = 10$ that one- and two-stage Gauss–Legendre keep the energy ratio at 1 while LobattoIIIC(2) decays to 3.79×10^{-1} , 9.75×10^{-18} , and 1.17×10^{-20} and RadauIIA(2) to 9.00×10^{-1} , 7.10×10^{-4} , and 3.77×10^{-7} (Table 3), and that the symplectic two-stage Qin–Zhang DIRK, solved exactly by a multiplicative field split blocking the u and σ unknowns of each stage (Listing 5), reproduces the Gauss–Legendre energy behavior.
5. Reproduced the Schäfer–Turek cylinder drag and lift at Reynolds number 100 with a curvilinear mesh of 14,640 cells, Scott–Vogelius P_4 /discontinuous P_3 elements (about 382k unknowns), and two-stage RadauIIA with $\Delta t = 0.00125$, against a reference using about 667k unknowns and $\Delta t = 0.000625$, with discrete divergence between 10^{-5} and 10^{-3} ; for the Benjamin–Bona–Mahony solitary wave with $c = 1/2$ on $[0, 100]$, $h = 0.1$, the two-stage Gauss–Legendre method with $\Delta t = 10h$ gave 0.14% error against 0.15% for the implicit midpoint rule with $\Delta t = h$ and over 10% with $\Delta t = 10h$, conserving I_1 and I_2 to 10^{-16} with cubic-invariant drift 9.6×10^{-7} versus 5.1×10^{-3} for one stage (Table 4); for the Gross–Pitaevskii equation on $[-6, 6]^3$ with cubic Lagrange elements, Gauss–Legendre with one to three stages and $\Delta t = 0.05$ to 0.25 needed 4.86 to 6.79 Krylov iterations per Newton step with monolithic multigrid, and the energy drift fell from 7.56×10^{-2} to 5.45×10^{-8} (Table 5).

Comments

- Section 2.1, Eqs. (2.3)–(2.4): the stages are said to satisfy (2.4) “for all $1 \leq i \leq n$ ”, the coefficients are indexed “for $1 \leq i, j \leq n$ ”, and the sum in (2.4) runs over $i = 1, \dots, s$, although n is the dimension of y and (2.3) and (2.14) index the s stages by i and sum over j ; the evident intended ranges are $1 \leq i, j \leq s$ with the sum over j .
- Section 2.2.2, Eq. (2.23): the stage equations for the wave equation, introduced as $u_{tt} - \Delta u = 0$ with the homogeneous system (2.22), contain the forcing term $-(f(t_n + c_i \Delta t, \cdot), v_i)$ carried over from the mixed heat equation (2.21); the evident intended form has no forcing term.
- Section 4.1: after discussing the order reduction of the two-stage Gauss–Legendre method in Fig. 2(a), the text concludes that “the extra formal order of accuracy for Gauss–Lobatto methods does not materialize unless one also takes a very small time step”, although no Gauss–Lobatto method appears in Fig. 2; Gauss–Legendre is evidently intended.
- Section 4.2, Listing 4, and Table 1: the solver option sets are cited as “Figure 2”, “Figure 3”, and “Figure 4”, whereas they are Listings 2–4 and Figs. 2–4 show the mixed heat-equation errors, the solver performance, and the cylinder domain; the caption of Listing 4 refers to “the monolithic multi-grid method described in Equation (2)” and that of Table 1 to “the Monolithic Multi-grid Options in Figure 4”; Listings 2 and 4 are evidently intended.

2020AF

ChatGPT (August 2026)

Summary

FastRK3 is an explicit pressure-correction projection method for incompressible Navier–Stokes flows on orthogonal curvilinear grids. It combines a three-stage Runge–Kutta update with extrapolated intermediate pressure gradients, so the pressure Poisson problem is solved only at the final stage. Expressing the operators in orthogonal coordinates removes cross-derivatives and yields Cartesian-like seven-point stencils. When the metric factors are invariant in one direction, FastPoc applies an FFT in that direction and direct sparse factorization to independent two-dimensional systems. Verification against standard RK3, Blasius and Jeffery–Hamel solutions shows near-identical results and approximately second-order spatial accuracy; lid-driven polar-cavity results agree with experiments. On the tested grids, FastPoc is 30–60 times faster than hypre SMG at divergence tolerances from 10^{-6} to 10^{-12} , and the complete solver is 4–7 times faster. Applications reproduce attached and separated laminar flow over ramp and Gaussian-bump geometries. FastPoc requires orthogonal grids with metric coefficients invariant in one coordinate, and its direct factorizations can consume substantial memory.

Contributions

1. Formulated the velocity-based incompressible Navier–Stokes equations in general orthogonal curvilinear coordinates, including curved-wall and outflow boundary treatments.
2. Introduced FastRK3, which extrapolates intermediate pressure gradients and requires only one pressure Poisson solve per time step.
3. Developed FastPoc, an FFT/direct sparse solver for orthogonal grids whose metric coefficients are invariant in one coordinate direction.
4. Verified approximately second-order spatial accuracy using analytical solutions and comparisons with standard RK3 and polar-cavity experiments.
5. Demonstrated 30–60-fold Poisson-solver and 4–7-fold full-solver speedups on the tested systems, then applied the method to ramp and bump flows.

2020FBMH

ChatGPT (July 2026)

Summary

The authors construct a conservative discontinuous Galerkin discretization of a gyroaveraged Lenard–Bernstein–Dougherty collision operator for full- f , long-wavelength gyrokinetics. The continuous model is an advective–diffusive Fokker–Planck operator that conserves particles, momentum, and energy while satisfying an H -theorem. At the discrete level, weak equality supplies consistent primitive moments, while a recovery DG treatment handles velocity-space diffusion. This design preserves the three moments exactly at any resolution, including with piecewise-linear bases when projected quadratic terms are treated carefully. An explicit SSP-RK3 update is accompanied by a conservative stability bound. Implemented in Gkeyll, the operator is tested through relaxation toward a discrete maximum-entropy state, collisional Landau damping, and five-dimensional turbulence in a helical open-field-line model of the Texas Helimak. Low Hermite moments follow analytic relaxation behavior and higher moments converge with resolution. The implementation is limited to like-species collisions; positivity, discrete self-adjointness, a discrete H -theorem, and efficient treatment of high collisionality remain future work.

Contributions

1. Derived a gyroaveraged Dougherty collision operator suitable for full- f long-wavelength gyrokinetic simulations.
2. Introduced a weak-equality and recovery-DG discretization that exactly conserves particles, momentum, and energy.
3. Established practical explicit time-step stability bounds for the combined drag and diffusion operator.

4. Verified relaxation and collisional Landau-damping behavior against analytic moment and dispersion predictions.
5. Integrated the operator into Gkeyll and exercised it in a five-dimensional Texas Helimak turbulence calculation.

2020MHF

ChatGPT (July 2026)

Summary

This work presents an energy-conserving continuum discontinuous Galerkin scheme for the long-wavelength, electromagnetic full- f gyrokinetic system. In a symplectic formulation using parallel velocity, the inductive term $\partial A_{\parallel}/\partial t$ enters the kinetic equation explicitly; the method computes it directly from a generalized Ohm's law. The phase-space DG discretization conserves particles and, when paired with continuous finite-element field solves so that the discrete Hamiltonian is continuous, conserves energy. Kinetic Alfvén-wave and local kinetic-ballooning-mode benchmarks agree with analytic results and show that the method avoids the Ampère cancellation problem. A nonlinear Gkeyll simulation of helical open field lines, used as a simplified tokamak scrape-off-layer model, provides the first published nonlinear electromagnetic gyrokinetic turbulence calculation on open field lines. Relative to an electrostatic run, it shows less radial transport, shallower density and temperature profiles, and larger, more intermittent density fluctuations. The geometry omits realistic magnetic shear, X-points, gyroaveraging, and a complete positivity treatment, so the nonlinear comparison is demonstrative rather than predictive.

Contributions

1. Formulated a particle- and energy-conserving DG scheme for electromagnetic full- f gyrokinetics.
2. Derived a direct generalized-Ohm-law update for the inductive parallel electric field.
3. Demonstrated avoidance of the Ampère cancellation problem in two linear electromagnetic benchmarks.
4. Produced the first published nonlinear electromagnetic gyrokinetic turbulence simulation on open field lines.
5. Quantified transport, profile, and intermittency differences between electromagnetic and electrostatic scrape-off-layer simulations.

2018MDL

ChatGPT (July 2026)

Summary

This paper projects the flux-tube gyrokinetic equation onto Laguerre polynomials in magnetic moment and Hermite polynomials in parallel velocity. The resulting pseudo-spectral moment hierarchy smoothly connects low-dimensional gyrofluid models to conventional five-dimensional gyrokinetics as velocity resolution increases. Its formulation preserves density, momentum, energy, and free energy even at low resolution. A sparse model collision operator includes pitch-angle scattering, energy diffusion, slowing down, and classical diffusion while satisfying conservation laws and an H theorem. The authors analyze closures generated by parallel streaming, magnetic drifts, and nonlinear phase mixing, emphasizing that sharp velocity-space structure can require high resolution. Linear ion-temperature-gradient and zonal-flow tests agree with GS2 when sufficiently resolved and expose where low-order gyrofluid closure fails. The approach underlies the GPU-oriented GX code, offering a small-memory, parallel representation and the possibility of adapting fidelity between gyrofluid efficiency and gyrokinetic accuracy.

Contributions

1. Derived a Laguerre–Hermite pseudo-spectral representation of nonlinear flux-tube gyrokinetics.
2. Unified gyrofluid and gyrokinetic velocity resolutions within one moment hierarchy.
3. Preserved key particle, energy, and free-energy invariants at low spectral resolution.
4. Constructed a sparse conservative collision model satisfying an H theorem.
5. Verified linear instabilities and zonal-flow dynamics while identifying closure and phase-mixing limits.

2017NNSW

ChatGPT (July 2026)

Summary

This paper identifies a mechanism by which heavier hydrogen isotopes can reduce trapped-electron-mode (TEM) turbulent transport, contrary to conventional gyro-Bohm mass scaling. Linear and nonlinear multi-species gyrokinetic simulations use real-mass kinetic electrons, finite collisions, and realistic helical and tokamak geometries. The ratio of electron-ion collision frequency to ion transit frequency grows with ion mass, so collisions stabilize TEMs more strongly in deuterium and tritium than in hydrogen. Near marginal stability, the reduced TEM drive also makes steady zonal-flow shear more effective. In the Large Helical Device geometry, deuterium lowers the nonlinear heat flux to 48% of the hydrogen value and raises the relative zonal-flow energy; in a Cyclone-base-case-like tokamak, the corresponding heat-flux ratio is 43% at the examined collisionality. Suppressing zonal components increases tokamak TEM transport by more than fourfold, confirming their nonlinear role. The mechanism appears across axisymmetric and non-axisymmetric toroidal configurations and is relevant to density- or electron-temperature-gradient-driven outer-core and pedestal-top plasmas.

Contributions

1. Performed the first nonlinear LHD TEM simulations with hydrogen isotopes and real-mass kinetic electrons.
2. Identified ion-mass-dependent collisional TEM stabilization through the electron-ion-collision to ion-transit frequency ratio.
3. Demonstrated transport scaling opposite to gyro-Bohm expectations in finite-collisionality TEM turbulence.
4. Quantified how near-marginal steady zonal flows amplify the deuterium transport reduction.
5. Reproduced the mechanism in both helical LHD and axisymmetric tokamak configurations.

2015SHH

ChatGPT (July 2026)

Summary

This paper develops and benchmarks a one-dimensional electrostatic gyrokinetic model for parallel propagation of a JET-like edge-localized-mode heat pulse through the scrape-off layer to divertor plates. Gyrokinetic quasineutrality and logical-sheath boundary conditions remove the need to resolve the electron Debye length and plasma frequency while retaining kinetic electrons and ions. To make explicit calculations tractable, the authors modify the ion-polarization term with a minimum perpendicular wavenumber and derive the corresponding Hamiltonian correction required for energy conservation. A continuum discontinuous Galerkin implementation in Gkeyll uses a flux-balanced sheath cutoff and specified benchmark initial conditions. Its divertor heat-flux history agrees well with earlier Vlasov and full-orbit particle-in-cell results: fast electrons cause only an early modest signal because the rising sheath potential confines most electrons, and the bulk ELM energy reaches the target on the slower ion transit timescale. A Boltzmann-electron reduction reproduces the main heat pulse at much lower cost but omits early kinetic-electron effects.

Contributions

1. Formulated an energy-consistent electrostatic gyrokinetic scrape-off-layer model with a modified polarization term.
2. Implemented logical-sheath boundary conditions that enforce zero net wall current without resolving the Debye sheath.
3. Supplied reproducible JET-like edge-localized-mode source and initial-condition specifications for kinetic-code benchmarks.
4. Matched prior Vlasov and full-orbit PIC heat-flux results and explained the ion-timescale bulk energy arrival.
5. Developed a faster Boltzmann-electron model and identified the kinetic-electron effects that it omits.

2013LE

ChatGPT (July 2026)

Summary

This paper develops a collocation discretization for the semi-infinite speed coordinate in continuum kinetic calculations. Its nonclassical orthogonal polynomials use the Maxwellian weight $\exp(-x^2)$ on $[0, \infty)$, producing nodes and operators that accurately integrate and differentiate Maxwellian-like functions with very few speed points. For the Fokker–Planck field term, whose Rosenbluth potentials have algebraic rather than Maxwellian tails, the authors solve the potential equations on a separate fine grid and eliminate that grid when assembling the kinetic matrix. Careful large-speed extrapolation preserves accuracy for collisions between species of disparate mass. Tests of plasma resistivity, tokamak bootstrap current, and deuterium–molybdenum flows show rapid convergence; several neoclassical quantities reach percent-level or two-digit accuracy with roughly four to six speed nodes. The method therefore reduces velocity-space cost without sacrificing the integral, differential, and collision-operator accuracy needed by continuum plasma solvers.

Contributions

1. Introduced Maxwellian-weighted nonclassical orthogonal polynomials for spectral speed-space collocation.
2. Constructed accurate quadrature, differentiation, and interpolation operators on the semi-infinite speed domain.
3. Separated the fine Rosenbluth-potential grid from the coarse distribution-function grid in the assembled collision operator.
4. Developed large-speed extrapolation needed for accurate interspecies collisions with disparate masses.
5. Demonstrated rapid convergence for resistivity, bootstrap-current, and multispecies neoclassical-flow calculations.

2012NWS

ChatGPT (July 2026)

Summary

This paper uses five-dimensional nonlinear gyrokinetic Vlasov simulations to resolve how zonal flows exchange entropy with ion-temperature-gradient (ITG) and electron-temperature-gradient (ETG) turbulence. Separate entropy balances for zonal and nonzonal modes are coupled by a transfer function that kinetically generalizes Reynolds-stress production. A symmetrized triad transfer diagnostic identifies both participating Fourier modes and transfer direction. During ITG saturation, substantial entropy flows from nonzonal fluctuations into zonal modes and builds strong zonal flows. In the statistically steady state, direct transfer into the zonal component becomes weak; established zonal flows instead mediate transfer from heat-carrying, low-radial-wavenumber nonzonal modes toward higher-radial-wavenumber modes that contribute less transport, broadening the spectrum and regulating heat flux. Toroidal ETG turbulence behaves differently: interactions among low-wavenumber nonzonal modes dominate both saturation and steady transport, while zonal mediation remains weak for the parameters studied. The authors note that ETG regimes with weaker magnetic shear and stronger zonal flows may behave differently.

Contributions

1. Derived coupled gyrokinetic entropy balances for zonal and nonzonal fluctuations.
2. Introduced a direction-resolving symmetrized triad entropy-transfer diagnostic.
3. Identified transfer into zonal modes as a principal mechanism saturating the simulated ITG instability.
4. Showed that steady ITG zonal flows redirect transport-carrying entropy toward higher radial wavenumbers.
5. Distinguished this ITG pathway from nonzonal-dominated entropy transfer in the simulated ETG regime.

2009GKS

ChatGPT (August 2026)

Summary

This review updates SSP theory for nonlinear hyperbolic PDEs with discontinuities. An SSP method transfers any forward-Euler stability bound in a norm, seminorm, or convex functional to a higher-order time discretization under a computable step restriction. Burgers, linear-advection/WENO, and near-vacuum Euler tests show that A-, L-, or linearly stable non-SSP integrators can develop oscillations or lose positivity even when SSP schemes remain well behaved at comparable cost. The paper identifies the SSP coefficient with the radius of absolute monotonicity, making it a necessary as well as sufficient worst-case bound, and summarizes stage and order barriers for Runge–Kutta, multistep, and general linear methods. It catalogs optimal or best-known explicit and implicit schemes, emphasizing low-storage explicit Runge–Kutta methods and multistep methods with controlled starts; implicit SSP methods rarely justify their solve cost. Spectral deferred-correction schemes are recast as Runge–Kutta methods, simplifying their optimization and showing that high-order SSP variants require downwinding. The review closes by identifying higher-order general linear and implicit–explicit SSP methods as open directions.

Contributions

1. Demonstrated practical SSP necessity through examples where conventional linear or implicit stability does not prevent oscillations or loss of positivity.
2. Established the optimal SSP coefficient as the radius of absolute monotonicity, making the bound necessary and sufficient in the stated worst-case setting.
3. Synthesized stage, order, and step-size barriers for explicit and implicit Runge–Kutta, multistep, and general linear methods.
4. Collected optimal or best-known schemes and highlighted efficient low-storage constructions, including a ten-stage fourth-order method with $c = 6$.
5. Recast spectral deferred-correction formulas as Runge–Kutta methods and derived downwinding requirements for higher-order variants.

2007BH

ChatGPT (July 2026)

Summary

This review develops a unified foundation for nonlinear gyrokinetics, the reduced kinetic description of low-frequency turbulence in strongly magnetized plasmas. It organizes the theory around three mutually consistent elements: a gyrokinetic Vlasov equation with a gyrocenter Hamiltonian containing quadratic ponderomotive-like terms; gyrokinetic Poisson–Ampere equations containing polarization and magnetization contributions; and an exact energy conservation law retaining the required linear and nonlinear couplings. Phase-space Lagrangian and Hamiltonian Lie-transform methods systematically separate fast gyromotion from slower dynamics through guiding-center and gyrocenter transformations, with push-forward and pull-back operations connecting reduced and particle variables. Simplified equations are presented for electrostatic, shear-Alfvenic, and compressional magnetic fluctuations, and velocity moments connect the kinetic formulation to reduced fluid models. A variational formulation produces the self-consistent Vlasov–Maxwell system and its conserved energy through Noether’s method. The framework accommodates nonuniform plasmas and arbitrary magnetic geometry, clarifies the approximations behind commonly used models, and discusses extensions including strong $E \times B$ flow shear and bounce-center kinetics.

Contributions

1. Organized modern nonlinear gyrokinetic theory around consistent kinetic, field, and exact-energy-conservation equations.

2. Presented Lie-transform perturbation theory as a systematic method for removing gyrophase dependence while preserving Hamiltonian structure.
3. Identified the polarization, magnetization, and quadratic Hamiltonian terms required for self-consistent nonlinear coupling.
4. Derived the gyrokinetic Vlasov–Maxwell system and its exact conserved energy from a variational principle.
5. Connected the general theory to electrostatic, electromagnetic, reduced-fluid, strong-flow-shear, and bounce-averaged applications.

2007CHPG

ChatGPT (August 2026)

Summary

This paper develops a unified design framework for MPI collective communication on distributed-memory machines. Using a latency–bandwidth–computation model with message cost $\alpha + n\beta$ and reduction cost γ , it derives lower bounds for broadcast, reduce, scatter, gather, allgather, reduce-scatter, and allreduce. It presents dual families of minimum-spanning-tree, recursive-doubling or halving, bidirectional-exchange, bucket, and simple algorithms for linear arrays, multidimensional meshes, hypercubes, and fully connected networks. Short messages favor latency-minimizing tree algorithms; long messages favor bandwidth-efficient composed algorithms. For intermediate sizes, the authors organize algorithms into parameterized hybrids obtained by factoring processors into logical mesh dimensions and choosing kernels per dimension. They prove a restricted optimality result for hypercubes and derive a practical heuristic for arbitrary factorizations. MPI point-to-point implementations in the InterCol library are benchmarked on a Myrinet-connected Cray-Dell cluster; broadcast and reduce-scatter trends largely match the model and are often competitive with or faster than the tested MPICH-GM collectives. The model and measurements assume a uniform, historical cluster architecture, so performance rankings do not directly generalize to modern systems.

Contributions

1. Derived latency, bandwidth, and computation lower bounds for seven principal collective communication operations under one cost model.
2. Presented collective algorithms as dual, reusable families, including tree, exchange, doubling or halving, and bucket constructions.
3. Generalized one-dimensional algorithms systematically to meshes, hypercubes, and logically fully connected architectures.
4. Constructed parameterized hybrids for intermediate messages, proved restricted hypercube optimality, and proposed a factorization-based heuristic for general processor counts.
5. Implemented the designs with MPI point-to-point operations and compared broadcast and reduce-scatter results with the model and MPICH-GM.

2007GSAB

ChatGPT (July 2026)

Summary

GYSELA evolves the full ion distribution function in a global five-dimensional, electrostatic gyrokinetic model of ion-temperature-gradient turbulence in simplified toroidal geometry. Because a full- f calculation evolves equilibrium and fluctuations together, initialization is consequential. A canonical Maxwellian balances curvature-driven vertical charge separation through parallel flow, whereas a local Maxwellian generates polarization, large-scale electric fields, and sheared flows. Those artificial flows reduce the effective linear growth rate and delay turbulence, although the late nonlinear diffusivities become similar. A nonlinear comparison with the δf particle-in-cell code ORB5 gives broadly consistent transport, with remaining differences attributed to boundary conditions, profile relaxation, and self-consistent zonal flows. A scan over ρ_* shows that transport scaling is not a function of system size alone: correlation properties are

consistent with gyroBohm behavior at small ρ_* well above threshold and Bohm-like behavior at larger ρ_* near threshold. Finally, a reduced three-dimensional flux-driven model with a constant heat source reaches a statistical steady state while retaining bursty relaxation, large convection cells, zonal flows, and measurable non-Maxwellian moments. These results motivate fully flux-driven five-dimensional studies.

Contributions

1. Demonstrated global full- f gyrokinetic evolution of the complete ion distribution with the semi-Lagrangian GYSELA code.
2. Showed how non-equilibrium initialization creates sheared flows that suppress linear growth and delay turbulence onset.
3. Nonlinearly benchmarked GYSELA against the independently formulated ORB5 δf particle-in-cell code.
4. Found that ρ_* transport scaling depends jointly on system size and distance from the instability threshold.
5. Achieved a flux-driven statistical steady state in a reduced model and characterized its bursty, non-Maxwellian dynamics.

2005G

ChatGPT (August 2026)

Summary

Strong stability preserving (SSP) time integrators are designed for method-of-lines systems $\dot{u} = L(u)$ whose forward Euler step is nonlinearly stable in a norm, seminorm, or convex functional under $\Delta t \leq \Delta t_{\text{FE}}$. This review explains how a Runge–Kutta or multistep method written as convex combinations of forward Euler steps inherits that property under $\Delta t \leq c\Delta t_{\text{FE}}$, with downwind spatial operators needed when coefficients are negative. It catalogs efficient explicit SSP Runge–Kutta schemes for nonlinear and linear constant-coefficient problems, including low-storage implementations, order barriers, effective CFL measures, and the need for stable intermediate stages. Explicit and implicit multistep methods are assessed; higher-order methods generally face restrictive coefficients, while suitable starting procedures can relax boundedness requirements. The paper also shows why higher-order implicit SSP methods are not unconditionally stable and are often not cost-effective. Finally, it connects the optimal SSP CFL coefficient of a Runge–Kutta method to its radius of absolute monotonicity, providing a constructive route from Butcher coefficients to an optimal Shu–Osher representation.

Contributions

1. Presented the Shu–Osher convex-decomposition argument that transfers forward-Euler strong stability to higher-order Runge–Kutta methods under an explicit CFL coefficient.
2. Collected optimal nonlinear SSP schemes, including low-storage variants and a five-stage fourth-order method with coefficient $c = 1.508$.
3. Treated linear constant-coefficient problems separately, giving higher-order method families, effective CFL comparisons, and coefficient bounds.
4. Demonstrated why SSP behavior is needed at intermediate stages and surveyed restrictive results for explicit and implicit multistep and Runge–Kutta methods.
5. Connected the optimal SSP coefficient to the radius of absolute monotonicity and constructed an optimal Shu–Osher representation from Butcher data.

2003LW

ChatGPT (July 2026)

Summary

Eight explicit finite-difference schemes for the compressible Euler equations are compared across demanding one- and two-dimensional tests. The selection spans central and upwind-biased methods, dimensional splitting, Riemann

solvers, flux limiting, hybridization, and Runge–Kutta time stepping; every scheme is at least formally second-order accurate. Seven one-dimensional Riemann problems and the Woodward–Colella blast-wave problem permit quantitative comparisons, including relative L^1 errors, while the two-dimensional suite probes smooth-flow accuracy, interacting Riemann waves, strong shocks, unstable interfaces, symmetry preservation, odd–even decoupling, and vortex advection. Smooth periodic tests recover the superior formal accuracy of WENO and PPM, but discontinuous and multidimensional problems expose method-dependent trade-offs in dissipation, shock and contact resolution, oscillations, grid-aligned artifacts, and symmetry errors. CPU timings are reported but treated cautiously because the implementations required different adaptations to a common framework. No scheme is uniformly best: rankings change with the flow and diagnostic. The work therefore supplies a comparative benchmark rather than a universal recommendation, with complete results and animations provided separately.

Contributions

1. Compared eight representative high-resolution finite-difference schemes within one computational framework.
2. Assembled a broad benchmark suite spanning exact-solution Riemann tests and difficult multidimensional flows.
3. Quantified one-dimensional errors and smooth-problem convergence while also reporting implementation-qualified timing data.
4. Documented scheme-specific failures, including excessive diffusion, oscillations, symmetry loss, grid alignment, and odd–even decoupling.
5. Established that method quality is problem-dependent and released a larger body of results for continued code assessment.

2002S

ChatGPT (August 2026)

Summary

This review recasts adaptive time-stepping for ordinary differential and differential-algebraic equations as a feedback-control problem. After logarithmic transformation, the relation between step size and estimated local error becomes an approximately linear discrete process, making control-theoretic stability analysis applicable. Elementary controllers can generate irregular steps or poor tolerance response; proportional–integral controllers instead yield smoother step sequences, predictable tolerance proportionality, and robust work–precision behavior. Predictive controllers extend this approach to implicit and stiff integration by estimating changes in the principal error function. The paper also coordinates step-size selection with modified Newton iterations: measured convergence rates determine when to reuse a Jacobian, refactor a matrix, or reject and retry a step. This replaces disconnected heuristics with interacting controllers whose dynamics can be analyzed. Computational stability and tolerance proportionality are proposed as software-quality criteria alongside accuracy and efficiency. The framework is broadly applicable, but its detailed recommendations depend on reliable local-error and iteration-convergence estimates; multistep controller dynamics and differential-algebraic error estimation remain incompletely understood.

Contributions

1. Formulated adaptive time-stepping as a discrete feedback-control problem through logarithmic models of step size and local error.
2. Analyzed elementary controllers and advocated proportional–integral designs that stabilize step sequences and improve tolerance proportionality.
3. Developed predictive controllers for implicit and stiff solvers by estimating changes in the principal error function.
4. Coordinated time-step control with modified Newton convergence monitoring, Jacobian reevaluation, and matrix refactorization.
5. Introduced computational stability and tolerance proportionality as practical measures of adaptive solver quality and identified unresolved controller problems.

1999HW

Claude (September 2026)

Summary

The authors trace the development of Radau IIA implicit Runge–Kutta methods for stiff ordinary differential equations, in which nearby rapid solutions perturb numerical approximations of slow smooth ones, and describe RADAU, a new variable-order code. In the background part, the s -stage Radau IIA methods are collocation methods of order $2s - 1$ at the zeros of $\frac{d^{s-1}}{dx^{s-1}}(x^{s-1}(x-1)^s)$, whose stability functions, the subdiagonal Padé approximations $R_{s-1,s}$ of e^z , vanish at infinity; their A-stability (stability domain $|R(z)| \leq 1$ containing the left half-plane), order stars, B-stability, stiff accuracy, and convergence for singular perturbation, differential-algebraic, and parabolic problems are reviewed. The implementation solves $My' = f(x, y)$, with constant, possibly singular M , by simplified Newton iterations decoupled by transforming the coefficient matrix. Fixed-order runs at orders 5, 9, and 13 on Robertson's chemical-kinetics problem show high orders becoming costlier at loose tolerances, because large proposed steps make the Newton iteration converge slowly or fail. Since cost factors and optimal step sizes are hard to predict for implicit Runge–Kutta methods, RADAU instead changes the order according to the observed Newton contraction. On Robertson's problem it matches the best fixed-order code at every tolerance, the best achievable in the authors' view; further tests use Van der Pol, linear, and circuit problems.

Contributions

1. Traced stiff problems to Crank and Nicolson's heat equation with nonlinear heat generation, semi-discretized by Hartree's 'method II' (the method of lines), to Curtiss and Hirschfelder, who called $dy/dx = [y - G(x)]/a(x, y)$ stiff when $|a(x, y)/\Delta x| \ll 1$, introduced BDF, and took $a(x, y)$ positive, thus seeking stable numerical solutions of unstable problems, and to the 'building-up errors' of Fox and Goodwin's trapezoidal rule with $h = 0.2$ on $y' = -10y + 6z$, $z' = 13.5y - 10z$, and recounted that Butcher's implicit Runge–Kutta paper was accepted by a 'computational' journal only with an appendix on fixed-point iteration.
2. Estimated the local error by $\text{err} = \|(M - h\gamma_0 J)^{-1}(\hat{y}_1 - y_1)\|$, with an order- s solution $\hat{y}_1$ combining $hf(x_0, y_0)$ and the stage values and with γ_0 chosen so that $M - h\gamma_0 J$ is already factored, and took the smaller of the standard step $\text{fac} \cdot h_n(1/\text{err}_{n+1})^{1/(s+1)}$ and the step size control with memory, based on $C_{n+1}/C_n \approx C_n/C_{n-1}$, which multiplies it by $(h_n/h_{n-1})(\text{err}_n/\text{err}_{n+1})^{1/(s+1)}$ and is selected where steps decrease, avoiding many rejections.
3. Defined the contractivity factor as the last Ψ_k of a step, with $\Theta_k = \|\Delta W^k\|/\|\Delta W^{k-1}\|$, $\Psi_1 = \Theta_1$, and $\Psi_k = \sqrt{\Theta_k \Theta_{k-1}}$, and changed the order by 4 within $\{5, 9, 13\}$, up when the factor is ≤ 0.002 and down when it is ≥ 0.8 or the iteration fails to converge, starting at low order (for example 5) with no change during the first 10 steps and no increase during 10 steps after a decrease.
4. Ran the fixed-order codes on Robertson's problem over $[0, 10^{11}]$ with $\text{Rtol} = 10^{-2-m/4}$, $m = 0, \dots, 40$, and $\text{Atol} = 10^{-6}\text{Rtol}$, taking the error as the maximum at $x = 1, 10, \dots, 10^{11}$; at $\text{Rtol} = 10^{-2}, \dots, 10^{-12}$ RADAU took only order-5 steps (87 to 195) down to 10^{-5} , 10 order-5 then 98 to 138 order-9 steps for 10^{-6} to 10^{-8} , and 10, 12 to 14, and 90 to 134 steps of orders 5, 9, and 13 for 10^{-9} to 10^{-12} , never lowering the order.
5. For the Van der Pol equation with $\varepsilon = 10^{-6}$ on $[0, 11]$, RADAU alternated between order 5 in stiff phases and 9 in transients for $\text{Rtol} > 10^{-5}$ and between 9 and 13 for tighter tolerances, staying at 13 for $\text{Rtol} \leq 10^{-8}$; on linear constant-coefficient problems simplified Newton is exact after one iteration, so the code switches to order 13, the most efficient for problem B5 (eigenvalues $-10 \pm 100i$, -4 , -1 , -0.5 , -0.1) even at loose tolerances, and it picks a nearly optimal order on the 15-dimensional ring modulator with $C_S = 2 \times 10^{-12}$ on $[0, 10^{-3}]$, whose Jacobian has eigenvalues near $-2 \times 10^4 \pm 3 \times 10^7 i$, 10^5 , and -10^{11} .

Comments

- Section 6: the standard strategy is said "to choose the order p in such a way that the error per unit step is minimal, i.e., $C^{(p)}/h^{(p)} \rightarrow \min$ ", but the same sentence defines $C^{(p)}$ and $h^{(p)}$ as "the cost factors and the proposed step sizes for the method of order p ", and the next sentence concerns estimating these cost factors, so the minimized quantity is the cost per unit step, which is evidently intended.